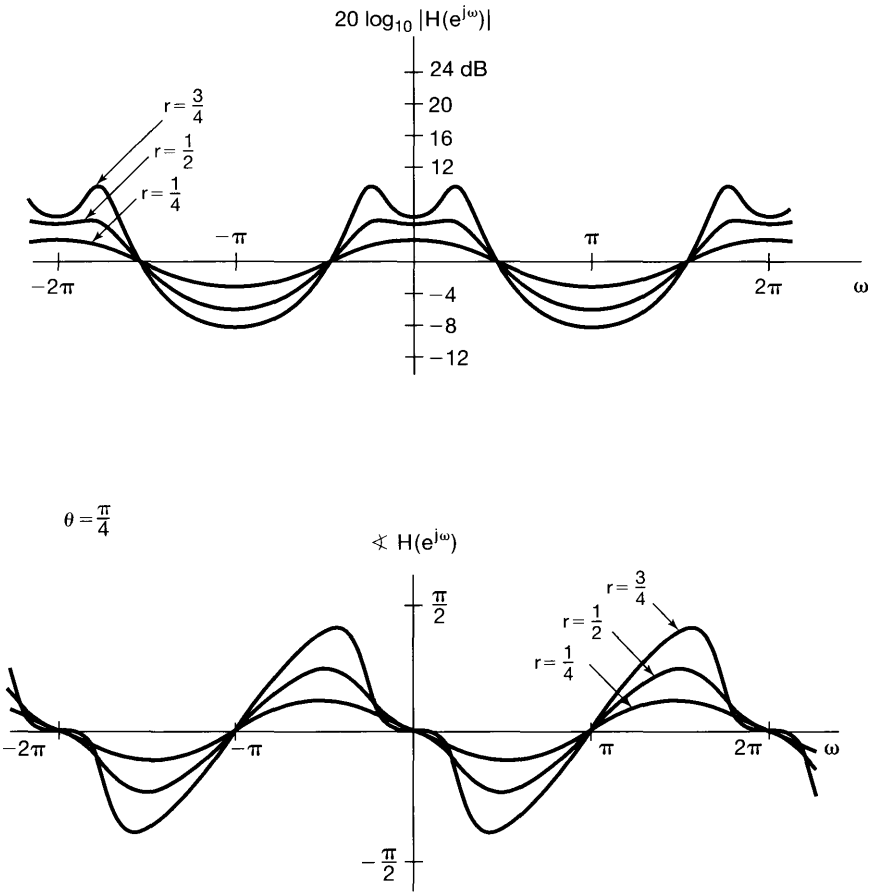


(b)

Figure 6.31 *Continued*

$$y[n] - (d_1 + d_2)y[n-1] + d_1 d_2 y[n-2] = x[n]. \tag{6.71}$$

In this case,

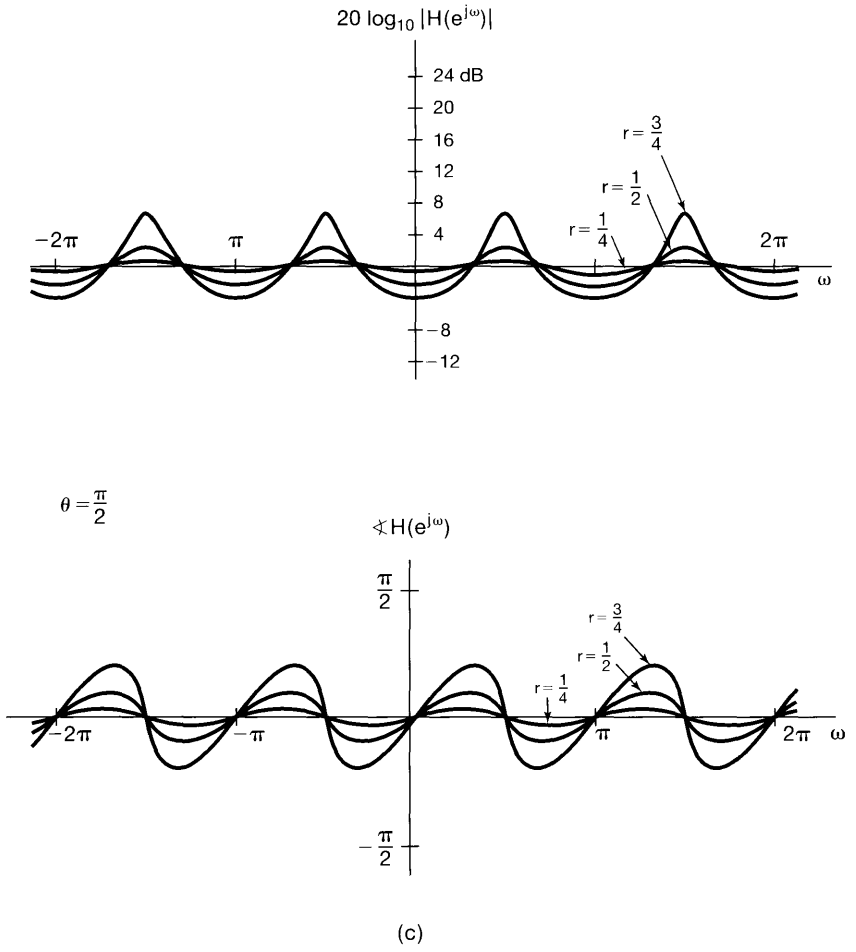
$$H(e^{j\omega}) = \frac{A}{1 - d_1 e^{-j\omega}} + \frac{B}{1 - d_2 e^{-j\omega}}, \tag{6.72}$$

where

$$A = \frac{d_1}{d_1 - d_2}, \quad B = \frac{d_2}{d_2 - d_1}. \tag{6.73}$$

Thus,

$$h[n] = [Ad_1^n + Bd_2^n]u[n], \tag{6.74}$$



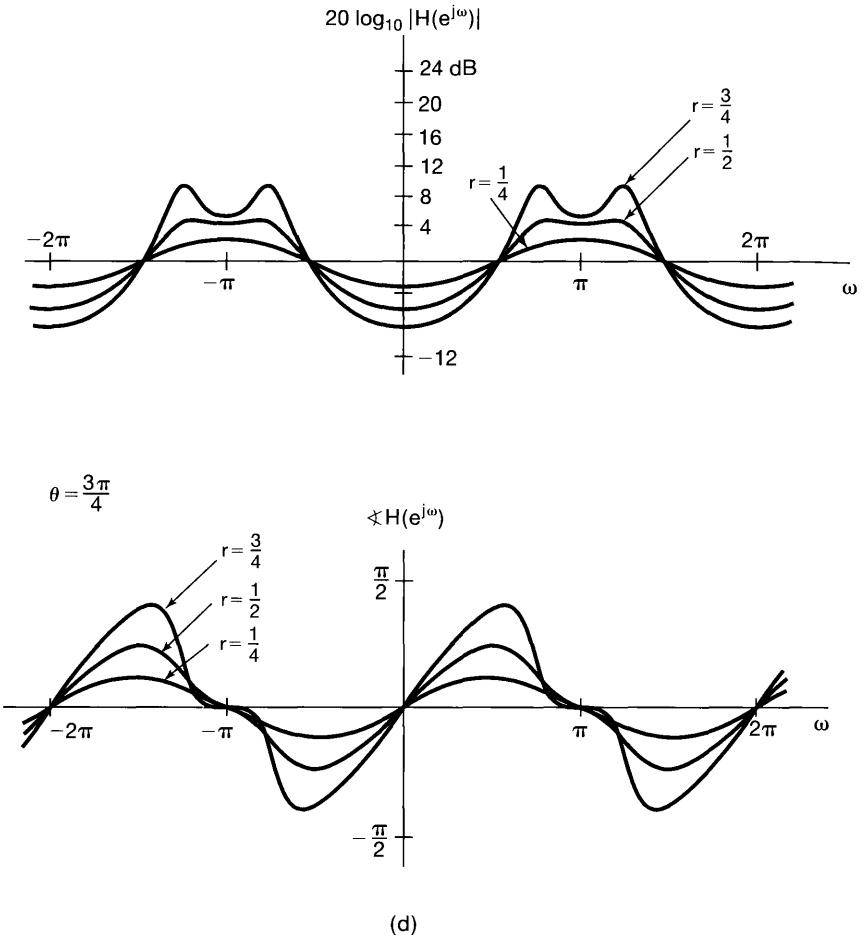
(c)

Figure 6.31 *Continued*

which is the sum of two decaying real exponentials. Also,

$$s[n] = \left[A\left(\frac{1 - d_1^{n+1}}{1 - d_1}\right) + B\left(\frac{1 - d_2^{n+1}}{1 - d_2}\right)\right]u[n]. \tag{6.75}$$

The system with frequency response given by eq. (6.70) corresponds to the cascade of two first-order systems. Therefore, we can deduce most of its properties from our understanding of the first-order case. For example, the log-magnitude and phase plots for eq. (6.70) can be obtained by adding together the plots for each of the two first-order terms. Also, as we saw for first-order systems, the response of the system is fast if $|d_1|$ and $|d_2|$ are small, but the system has a long settling time if either of these magnitudes is near 1. Furthermore, if d_1 and d_2 are negative, the response is oscillatory. The case when both d_1 and d_2 are positive is the counterpart of the overdamped case in continuous time, with the impulse and step responses settling without oscillation.



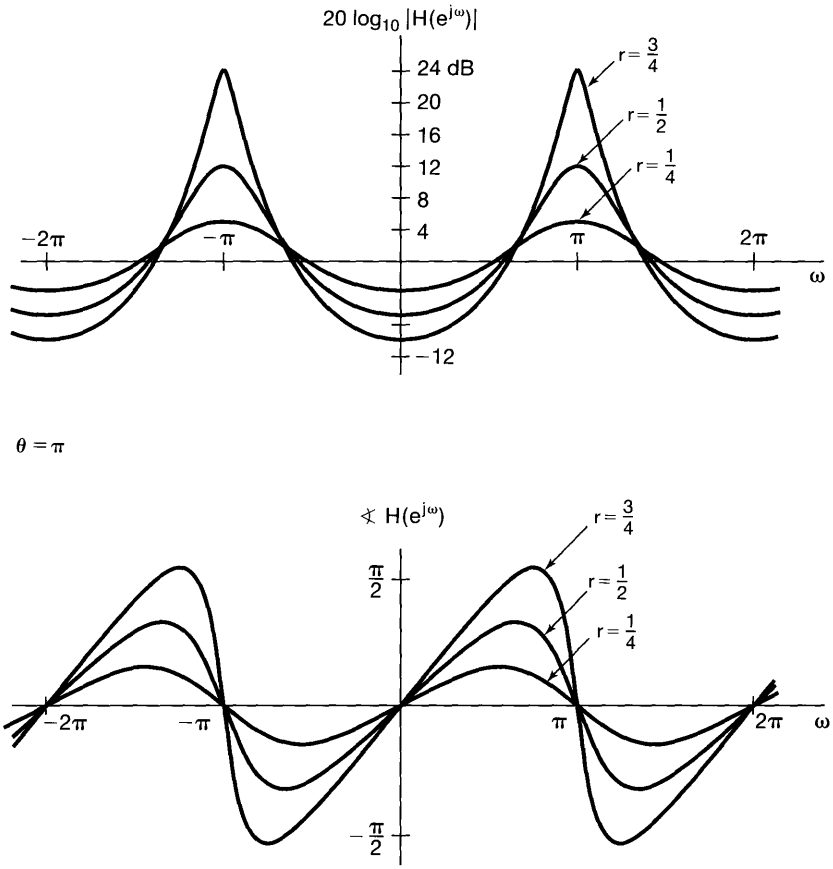
(d)

Figure 6.31 *Continued*

In this section, we have restricted attention to those causal first- and second-order systems that are stable and for which the frequency response can be defined. In particular, the causal system described by eq. (6.51) is unstable for $|a| \geq 1$. Also, the causal system described by eq. (6.56) is unstable if $r \geq 1$, and that described by eq. (6.71) is unstable if either $|d_1|$ or $|d_2|$ exceeds 1.

6.7 EXAMPLES OF TIME- AND FREQUENCY-DOMAIN ANALYSIS OF SYSTEMS

Throughout this chapter, we have illustrated the importance of viewing systems in both the time domain and the frequency domain and the importance of being aware of trade-offs in the behavior between the two domains. In this section, we illustrate some of these issues further. In Section 6.7.1, we discuss these trade-offs for continuous time in the context of an automobile suspension system. In Section 6.7.2, we discuss an important class of discrete-time filters referred to as moving-average or nonrecursive systems.



(e)

Figure 6.31 *Continued*

6.7.1 Analysis of an Automobile Suspension System

A number of the points that we have made concerning the characteristics and trade-offs in continuous-time systems can be illustrated in the interpretation of an automobile suspension system as a lowpass filter. Figure 6.32 shows a diagrammatic representation of a simple suspension system comprised of a spring and dashpot (shock absorber). The road surface can be thought of as a superposition of rapid small-amplitude changes in elevation (high frequencies), representing the roughness of the road surface, and gradual changes in elevation (low frequencies) due to the general topography. The automobile suspension system is generally intended to filter out rapid variations in the ride caused by the road surface (i.e., the system acts as a lowpass filter).

The basic purpose of the suspension system is to provide a smooth ride, and there is no sharp, natural division between the frequencies to be passed and those to be rejected. Thus, it is reasonable to accept and, in fact, prefer a lowpass filter that has a gradual

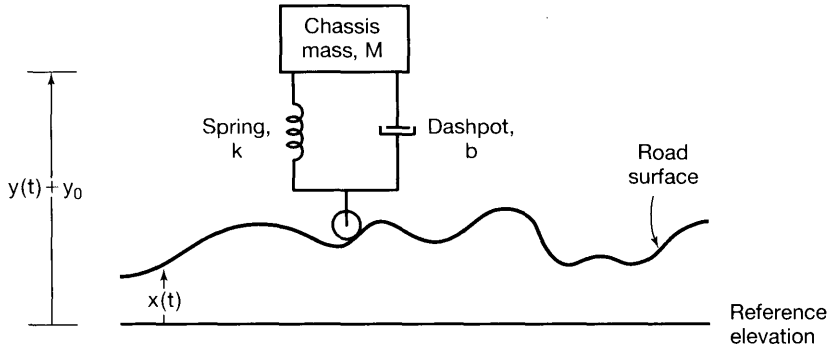


Figure 6.32 Diagrammatic representation of an automotive suspension system. Here, y_0 represents the distance between the chassis and the road surface when the automobile is at rest, $y(t) + y_0$ the position of the chassis above the reference elevation, and $x(t)$ the elevation of the road above the reference elevation.

transition from passband to stopband. Furthermore, the time-domain characteristics of the system are important. If the impulse response or step response of the suspension system exhibits ringing, then a large bump in the road (modeled as an impulse input) or a curb (modeled as a step input) will result in an uncomfortable oscillatory response. In fact, a common test for a suspension system is to introduce an excitation by depressing and then releasing the chassis. If the response exhibits ringing, it is an indication that the shock absorbers need to be replaced.

Cost and ease of implementation also play an important role in the design of automobile suspension systems. Many studies have been carried out to determine the most desirable frequency-response characteristics for suspension systems from the point of view of passenger comfort. In situations where the cost may be warranted, such as for passenger railway cars, intricate and costly suspension systems are used. For the automotive industry, cost is an important factor, and simple, less costly suspension systems are generally used. A typical automotive suspension system consists simply of the chassis connected to the wheels through a spring and a dashpot.

In the diagrammatic representation in Figure 6.32, y_0 represents the distance between the chassis and the road surface when the automobile is at rest, $y(t) + y_0$ the position of the chassis above the reference elevation, and $x(t)$ the elevation of the road above the reference elevation. The differential equation governing the motion of the chassis is then

$$M\frac{d^2y(t)}{dt^2} + b\frac{dy(t)}{dt} + ky(t) = kx(t) + b\frac{dx(t)}{dt}, \tag{6.76}$$

where M is the mass of the chassis and k and b are the spring and shock absorber constants, respectively. The frequency response of the system is

$$H(j\omega) = \frac{k + bj\omega}{(j\omega)^2M + b(j\omega) + k},$$

or

$$H(j\omega) = \frac{\omega_n^2 + 2\zeta\omega_n(j\omega)}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}, \tag{6.77}$$

where

$$\omega_n = \sqrt{\frac{k}{M}} \quad \text{and} \quad 2\zeta\omega_n = \frac{b}{M}.$$

As in Section 6.5.2, the parameter ω_n is referred to as the undamped natural frequency and ζ as the damping ratio. A Bode plot of the log magnitude of the frequency response in eq. (6.77) can be constructed by using first-order and second-order Bode plots. The Bode plot for eq. (6.77) is sketched in Figure 6.33 for several different values of the damping ratio. Figure 6.34 illustrates the step response for several different values of the damping ratio.

As we saw in Section 6.5.2, the filter cutoff frequency is controlled primarily through ω_n, or equivalently for a chassis with a fixed mass, by an appropriate choice of spring constant k. For a given ω_n, the damping ratio is then adjusted through the damping factor b associated with the shock asorbers. As the natural frequency ω_n is decreased, the suspension will tend to filter out slower road variations, thus providing a smoother ride. On the other hand, we see from Figure 6.34 that the rise time of the system increases, and thus the system will feel more sluggish. On the one hand, it would be desirable to keep ω_n small to improve the lowpass filtering; on the other hand, it would be desirable to have ω_n large for a rapid time response. These, of course, are conflicting requirements and illustrate the need for a trade-off between time-domain and frequency-domain characteristics. Typically, a suspension system with a low value of ω_n, so that the rise time is long, is characterized as "soft" and one with a high value of ω_n, so that the rise time is short, is characterized as "hard." From Figures 6.33 and 6.34, we observe also that, as the damping ratio decreases, the frequency response of the system cuts off more sharply, but the overshoot and ringing in the step response tend to increase, another trade-off between the time and frequency

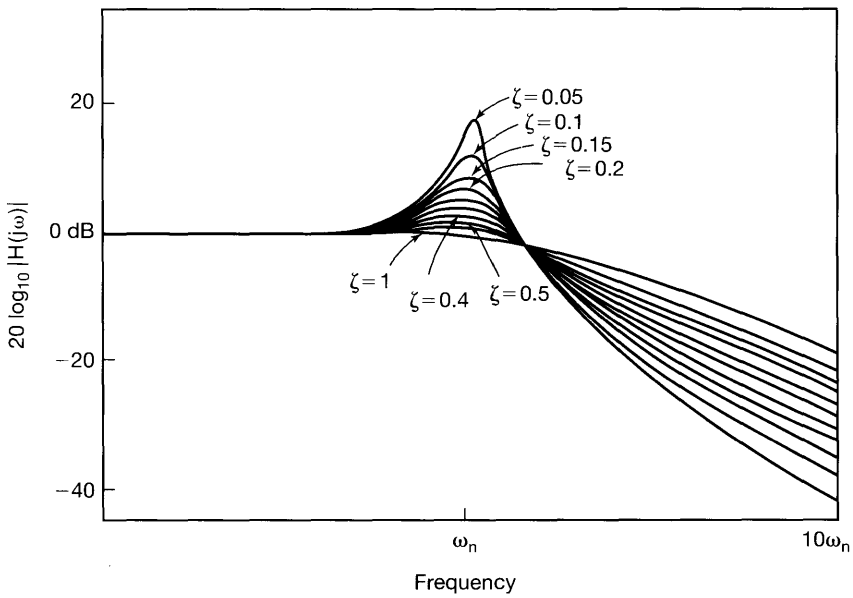


Figure 6.33 Bode plot for the magnitude of the frequency response of the automobile suspension system for several values of the damping ratio.

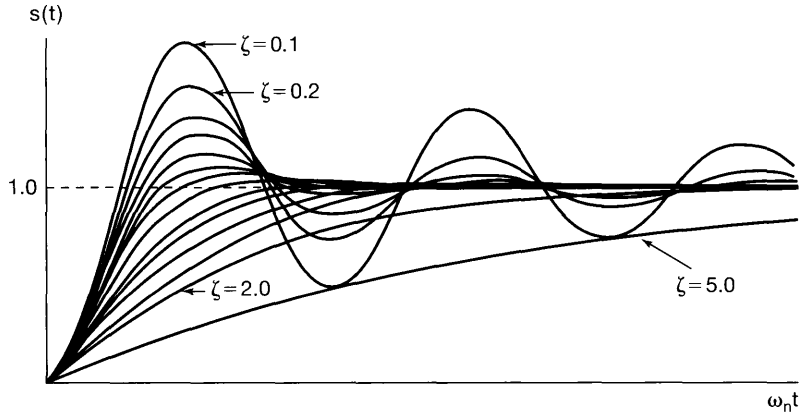


Figure 6.34 Step response of the automotive suspension system for various values of the damping ratio (ζ = 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.2, 1.5, 2.0, 5.0).

domains. Generally, the shock absorber damping is chosen to have a rapid rise time and yet avoid overshoot and ringing. This choice corresponds to the critically damped case, with ζ = 1.0, considered in Section 6.5.2.

6.7.2 Examples of Discrete-Time Nonrecursive Filters

In Section 3.11, we introduced the two basic classes of LTI filters described by difference equations, namely, recursive or infinite impulse response (IIR) filters and nonrecursive or finite impulse response (FIR) filters. Both of these classes of filters are of considerable importance in practice and have their own advantages and disadvantages. For example, recursive filters implemented as interconnections of the first- and second-order systems described in Section 6.6 provide a flexible class of filters that can be easily and efficiently implemented and whose characteristics can be adjusted by varying the number and the parameters of each of the component first- and second-order subsystems. On the other hand, as shown in Problem 6.64, it is not possible to design a causal, recursive filter with exactly linear phase, a property that we have seen is often desirable since, in that case, the effect of the phase on the output signal is a simple time delay. In contrast, as we show in this section, nonrecursive filters *can* have exactly linear phase. However, it is generally true that the same filter specifications require a higher order equation and hence more coefficients and delays when implemented with a nonrecursive equation, compared with a recursive difference equation. Consequently, for FIR filters, one of the principal trade-offs between the time and frequency domains is that increasing the flexibility in specifying the frequency domain characteristics of the filter, including, for example, achieving a higher degree of frequency selectivity, requires an FIR filter with an impulse response of longer duration.

One of the most basic nonrecursive filters, introduced in Section 3.11.2, is the moving-average filter. For this class of filters, the output is the average of the values of the input over a finite window:

$$y[n] = \frac{1}{N+M+1} \sum_{k=-N}^{M} x[n-k]. \tag{6.78}$$

The corresponding impulse response is a rectangular pulse, and the frequency response is

$$H(e^{j\omega}) = \frac{1}{N+M+1} e^{j\omega[(N-M)/2]} \frac{\sin[\omega(M+N+1)/2]}{\sin(\omega/2)}. \tag{6.79}$$

In Figure 6.35, we show the log magnitude for $M+N+1 = 33$ and $M+N+1 = 65$. The main, center lobe of each of these frequency responses corresponds to the effective passband of the corresponding filter. Note that, as the impulse response increases in length, the width of the main lobe of the magnitude of the frequency response decreases. This provides another example of the trade-off between the time and frequency domains. Specifically, in order to have a narrower passband, the filter in eqs. (6.78) and (6.79) must have a longer impulse response. Since the length of the impulse response of an FIR filter has a direct impact on the complexity of its implementation, this implies a trade-off between frequency selectivity and the complexity of the filter, a topic of central concern in filter design.

Moving-average filters are commonly applied in economic analysis in order to attenuate the short-term fluctuations in a variety of economic indicators in relation to longer term trends. In Figure 6.36, we illustrate the use of a moving-average filter of the form of eq. (6.78) on the weekly Dow Jones stock market index for a 10-year period. The weekly Dow Jones index is shown in Figure 6.36(a). Figure 6.36(b) is a 51-day moving average (i.e., $N = M = 25$) applied to that index, and Figure 6.36(c) is a 201-day moving average (i.e., $N = M = 100$) applied to the index. Both moving averages are considered useful, with the 51-day average tracking cyclical (i.e., periodic) trends that occur during the course of the year and the 201-day average primarily emphasizing trends over a longer time frame.

The more general form of a discrete-time nonrecursive filter is

$$y[n] = \sum_{k=-N}^{M} b_k x[n-k], \tag{6.80}$$

so that the output of this filter can be thought of as a weighted average of $N + M + 1$ neighboring points. The simple moving-average filter in eq. (6.78) then corresponds to setting all of these weights to the same value, namely, $1/(N+M+1)$. However, by choosing these coefficients in other ways, we have considerable flexibility in adjusting the filter's frequency response.

There are, in fact, a variety of techniques available for choosing the coefficients in eq. (6.80) so as to meet certain specifications on the filter, such as sharpening the transition band as much as possible for a filter of a given length (i.e., for N+M+1 fixed). These procedures are discussed in detail in a number of texts,[3] and although we do not discuss the procedures here, it is worth emphasizing that they rely heavily on the basic concepts and tools developed in this book. To illustrate how adjustment of the coefficients can influence

[3]See, for example, R. W. Hamming, *Digital Filters*, 3rd ed. (Englewood Cliffs, NJ: Prentice-Hall, Inc., 1989); A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing* (Englewood Cliffs, NJ: Prentice-Hall, Inc., 1989); and L. R. Rabiner and B. Gold, *Theory and Application of Digital Signal Processing* (Englewood Cliffs, NJ: Prentice-Hall, Inc., 1975).

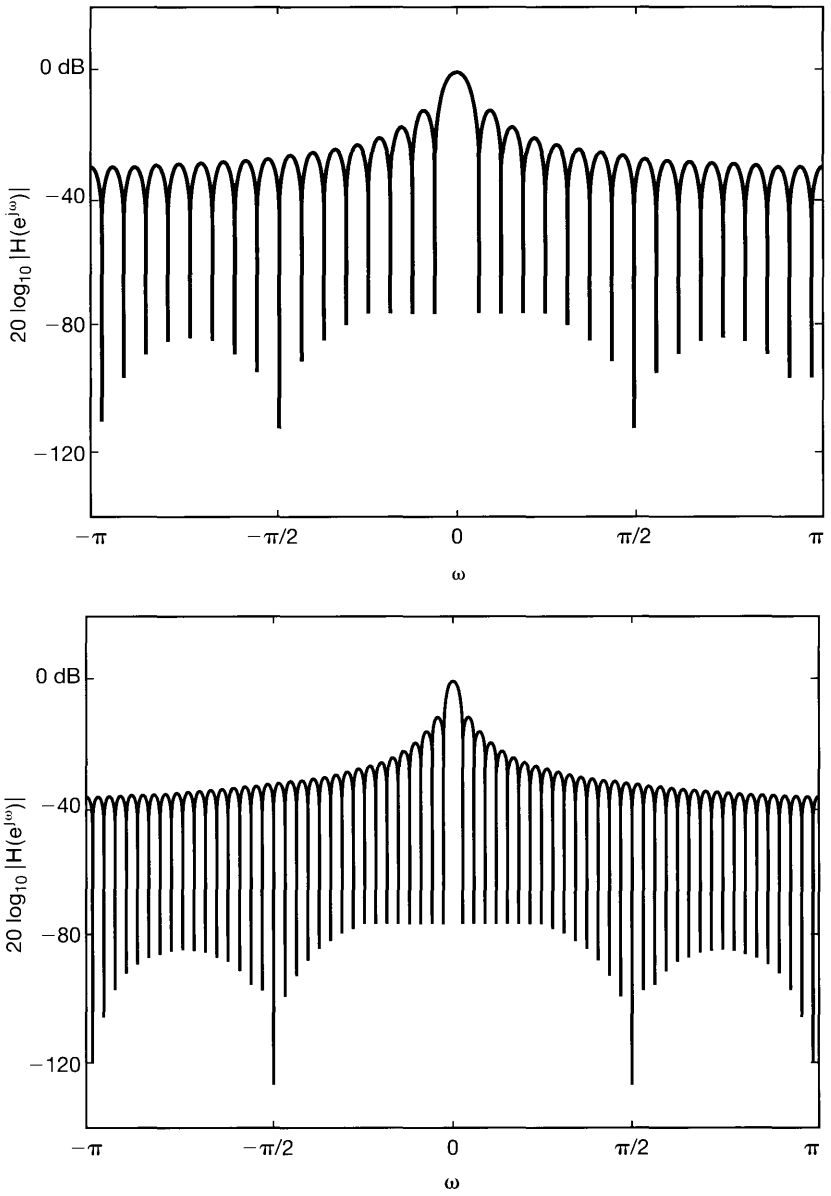


Figure 6.35 Log-magnitude plots for the moving-average filter of eqs. (6.78) and (6.79) for (a) $M + N + 1 = 33$ and (b) $M + N + 1 = 65$.

the response of the filter, let us consider a filter of the form of eq. (6.80), with $N = M = 16$ and the filter coefficients chosen to be

$$b_k = \begin{cases} \dfrac{\sin(2\pi k/33)}{\pi k}, & |k| \leq 32 \\ 0, & |k| > 32 \end{cases}. \tag{6.81}$$

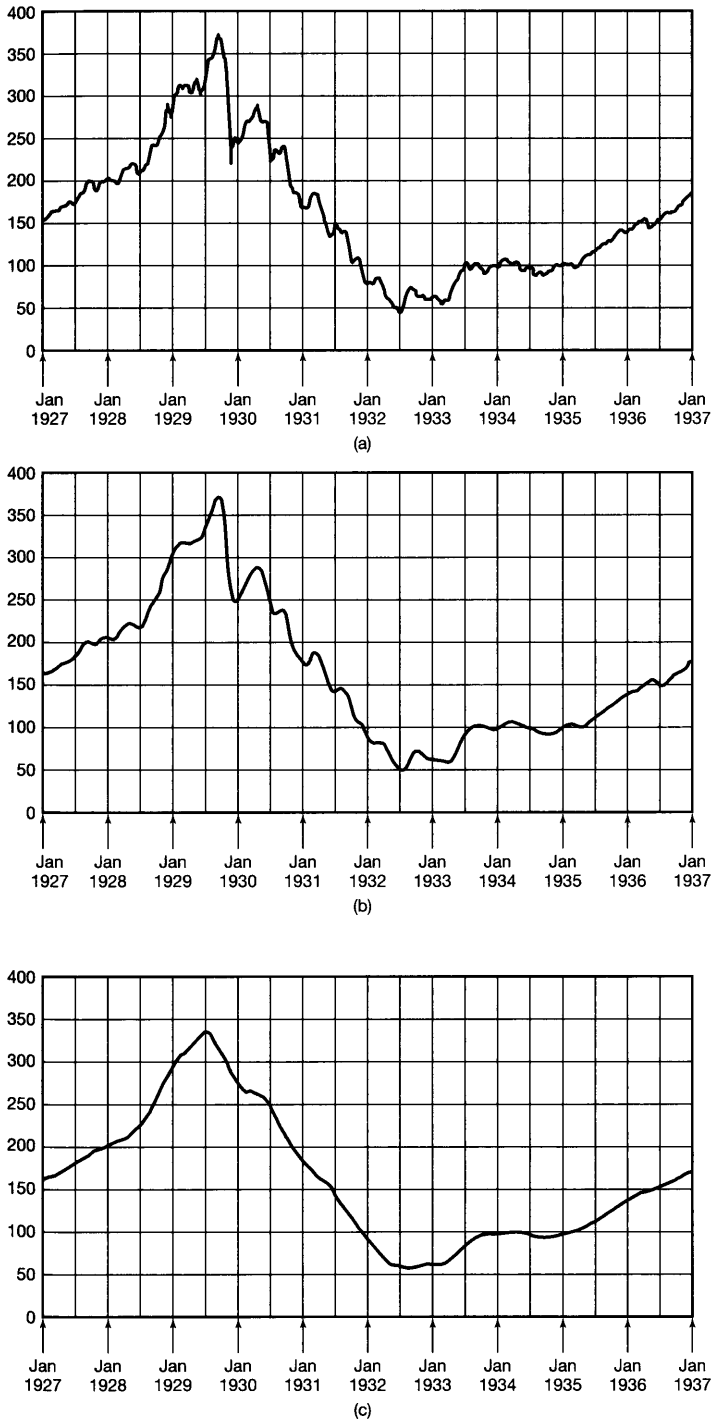


Figure 6.36 Effect of lowpass filtering on the Dow Jones weekly stock market index over a 10-year period using moving-average filters: (a) weekly index; (b) 51-day moving average applied to (a); (c) 201-day moving average applied to (a). The weekly stock market index and the two moving averages are discrete-time sequences. For clarity in the graphical display, the three sequences are shown here with their individual values connected by straight lines to form a continuous curve.

The impulse response of the filter is

$$h[n] = \begin{cases} \dfrac{\sin(2\pi n/33)}{\pi n}, & |n| \leq 32 \\ 0, & |n| > 32 \end{cases}. \tag{6.82}$$

Comparing this impulse response with eq. (6.20), we see that eq. (6.82) corresponds to truncating, for $|n| > 32$, the impulse response for the ideal lowpass filter with cutoff frequency $\omega_c = 2\pi/33$.

In general, the coefficients b_k can be adjusted so that the cutoff is at a desired frequency. For the example shown in Figure 6.37, the cutoff frequency was chosen to match approximately the cutoff frequency of Figure 6.35 for $N = M = 16$. Figure 6.37(a) shows the impulse response of the filter, and Figure 6.37(b) shows the log magnitude of the frequency response in dB. Comparing this frequency response to Figure 6.35, we observe that the passband of the filter has approximately the same width, but that the transition to

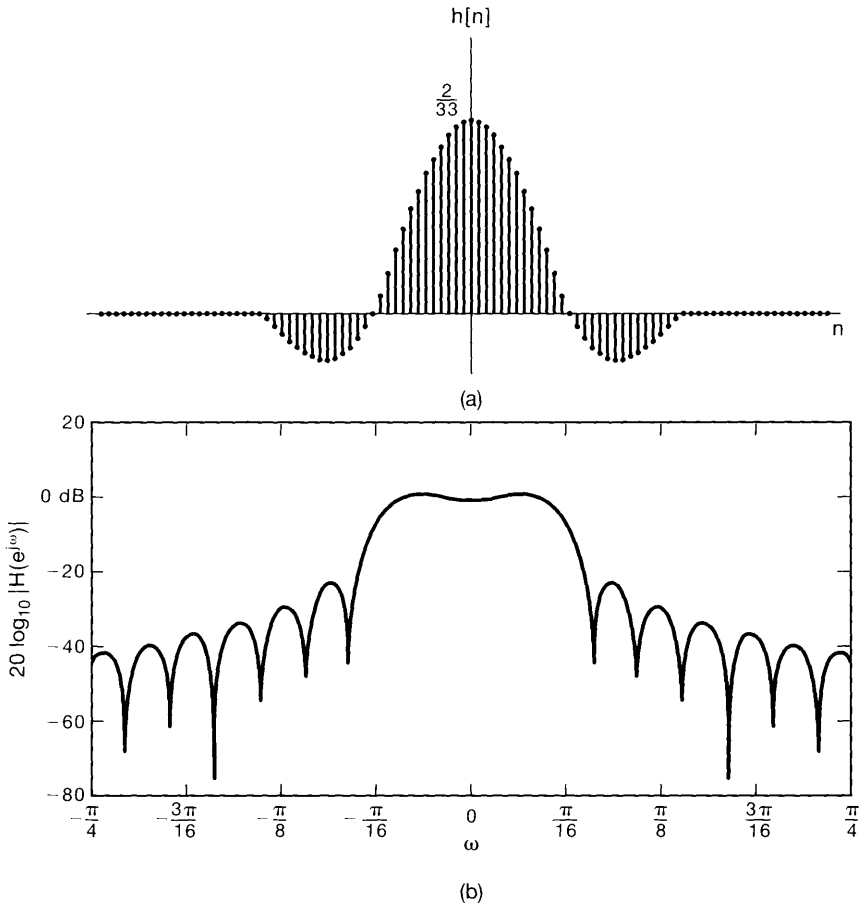


Figure 6.37 (a) Impulse response for the nonrecursive filter of eq. (6.82); (b) log magnitude of the frequency response of the filter.

the stopband is sharper. In Figures 6.38(a) and (b), the magnitudes (on a linear amplitude scale) of the two filters are shown for comparison. It should be clear from the comparison of the two examples that, by the intelligent choice of the weighting coefficients, the transition band can be sharpened. An example of a higher order lowpass filter ($N = M = 125$), with the coefficients determined through a numerical algorithm referred to as the Parks-McClellan algorithm,[4] is shown in Figure 6.39. This again illustrates the trade-off between the time and frequency domains: If we increase the length $N + M + 1$ of a filter, then, by a judicious choice of the filter coefficients in eq. (6.80), we can achieve sharper transition band behavior and a greater degree of frequency selectivity.

An important property of the examples we have given is that they all have zero or linear phase characteristics. For example, the phase of the moving-average filter of eq. (6.79) is $\omega[(N - M)/2]$. Also, since the impulse response in eq. (6.82) is real and even, the impulse response of the filter described by that equation is real and even, and thus has zero phase. From the symmetry properties of the Fourier transform of real signals, we know that *any* nonrecursive filter with an impulse response that is real and even will have a frequency response $H(e^{j\omega})$ that is real and even and, consequently, has zero phase. Such a filter, of course, is noncausal, since its impulse response $h[n]$ has nonzero values for $n < 0$. However, if a causal filter is required, then a simple change in the impulse response can achieve this, resulting in a system with *linear* phase. Specifically, since $h[n]$ is the impulse response of an FIR filter, it is identically zero outside a range of values centered at the origin

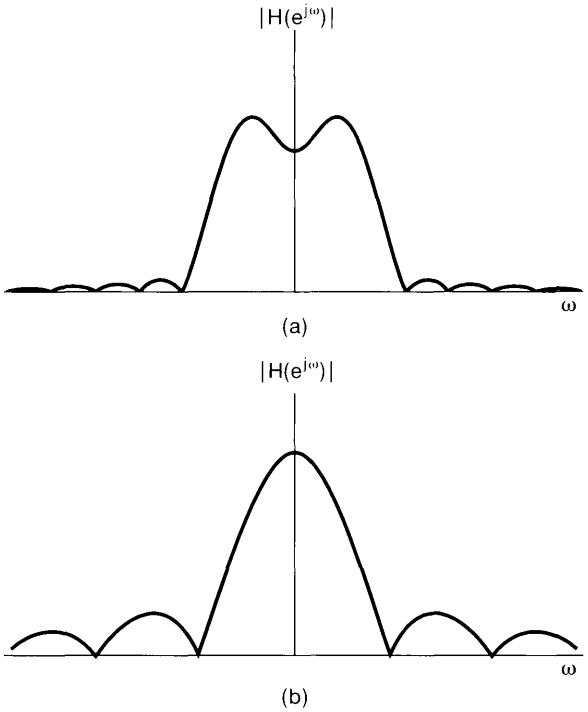


Figure 6.38 Comparison, on a linear amplitude scale, of the frequency responses of (a) Figure 6.37 and (b) Figure 6.35.

[4]A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing* (Englewood Cliffs, NJ: Prentice-Hall, Inc., 1989), Chap. 7.

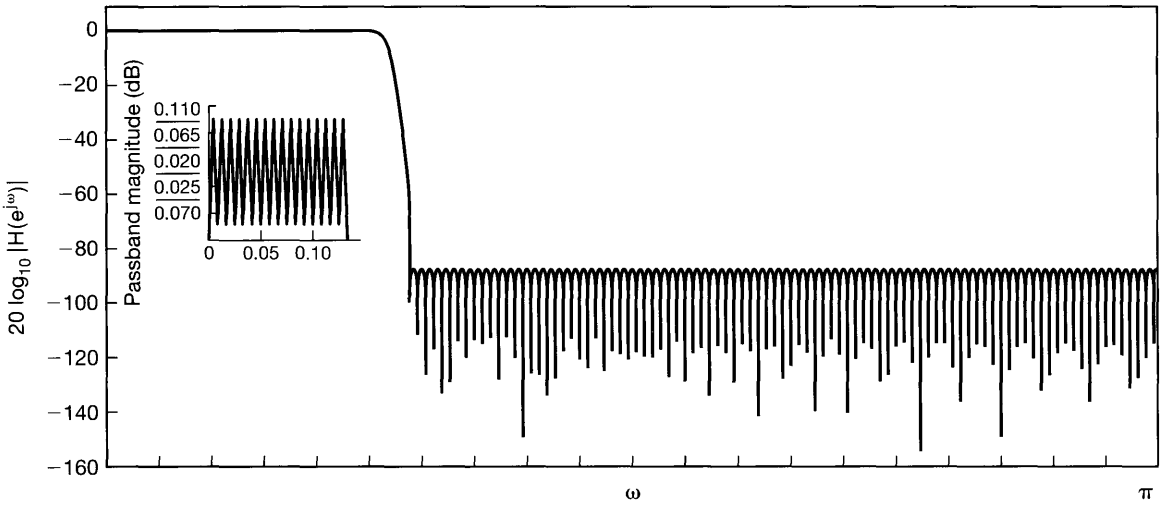


Figure 6.39 Lowpass nonrecursive filter with 251 coefficients designed to obtain the sharpest possible cutoff.

(i.e., $h[n] = 0$ for all $|n| > N$). If we now define the nonrecursive LTI system resulting from a simple N-step delay of $h[n]$, i.e.,

$$h_1[n] = h[n - N], \tag{6.83}$$

then $h_1[n] = 0$ for all $n < 0$, so that this LTI system is causal. Furthermore, from the time-shift property for discrete-time Fourier transforms, we see that the frequency response of the system is

$$H_1(e^{j\omega}) = H(e^{j\omega})e^{-j\omega N}. \tag{6.84}$$

Since $H(e^{j\omega})$ has zero phase, $H_1(e^{j\omega})$ does indeed have linear phase.

6.8 SUMMARY

In this chapter, we have built on the foundation of Fourier analysis of signals and systems developed in Chapters 3–5 in order to examine in more detail the characteristics of LTI systems and the effects they have on signals. In particular, we have taken a careful look at the magnitude and phase characteristics of signals and systems, and we have introduced log-magnitude and Bode plots for LTI systems. We have also discussed the impact of phase and phase distortion on signals and systems. This examination led us to understand the special role played by linear phase characteristics, which impart a constant delay at all frequencies and which, in turn, led to the concept of nonconstant group delay and dispersion associated with systems having nonlinear phase characteristics. Using these tools and insights, we took another look at frequency-selective filters and the time-frequency trade-offs involved. We examined the properties of both ideal and non-ideal frequency-selective filters and saw that time-frequency considerations, causality constraints, and implementation issues frequently make non-ideal filters, with transition bands and tolerance limits in the passbands and stopbands, the preferred choice.

We also examined in detail the time-frequency characteristics of first- and second-order systems in both continuous and discrete time. We noted in particular the trade-off between the response time of these systems and the frequency-domain bandwidth. Since first- and second-order systems are the building blocks for more complex, higher order LTI systems, the insights developed for those basic systems are of considerable use in practice.

Finally, we presented several examples of LTI systems in order to illustrate many of the points developed in the chapter. In particular, we examined a simple model of an automobile suspension system to provide a concrete example of the time-response–frequency-response concerns that drive system design in practice. We also considered several examples of discrete-time nonrecursive filters, ranging from simple moving-average filters to higher order FIR filters designed to have enhanced frequency selectivity. We saw, in addition, that FIR filters can be designed so as to have exactly linear phase. These examples, the development of the tools of Fourier analysis that preceded them, and the insights those tools provide illustrate the considerable value of the methods of Fourier analysis in analyzing and designing LTI systems.

Chapter 6 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining two sections contain problems belonging to the basic and advanced categories, respectively.

BASIC PROBLEMS WITH ANSWERS

6.1. Consider a continuous-time LTI system with frequency response $H(j\omega) = |H(j\omega)|\, e^{j\measuredangle H(j\omega)}$ and real impulse response $h(t)$. Suppose that we apply an input $x(t) = \cos(\omega_0 t + \phi_0)$ to this system. The resulting output can be shown to be of the form

$$y(t) = Ax(t - t_0),$$

where A is a nonnegative real number representing an *amplitude-scaling* factor and t_0 is a time delay.

(a) Express A in terms of $|H(j\omega_0)|$.

(b) Express t_0 in terms of $\measuredangle H(j\omega_0)$.

6.2. Consider a discrete-time LTI system with frequency response $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\measuredangle H(e^{j\omega})}$ and real impulse response $h[n]$. Suppose that we apply the input $x[n] = \sin(\omega_0 n + \phi_0)$ to this system. The resulting output can be shown to be of the form

$$y[n] = |H(e^{j\omega_0})|x[n - n_0],$$

provided that $\measuredangle H(e^{j\omega_0})$ and ω_0 are related in a particular way. Determine this relationship.

6.3. Consider the following frequency response for a causal and stable LTI system:

$$H(j\omega) = \frac{1 - j\omega}{1 + j\omega}.$$

(a) Show that $|H(j\omega)| = A$, and determine the value of A.

(b) Determine which of the following statements is true about $\tau(\omega)$, the group delay of the system. (*Note:* $\tau(\omega) = -d(\measuredangle H(j\omega))/d\omega$, where $\measuredangle H(j\omega)$ is expressed in a form that does not contain any discontinuities.)

1. $\tau(\omega) = 0$ for $\omega > 0$

2. $\tau(\omega) > 0$ for $\omega > 0$

3. $\tau(\omega) < 0$ for $\omega > 0$

6.4. Consider a linear-phase discrete-time LTI system with frequency response $H(e^{j\omega})$ and real impulse response $h[n]$. The group delay function for such a system is defined as

$$\tau(\omega) = -\frac{d}{d\omega}\measuredangle H(e^{j\omega}),$$

where $\measuredangle H(e^{j\omega})$ has no discontinuities. Suppose that, for this system,

$$|H(e^{j\pi/2})| = 2, \quad \measuredangle H(e^{j0}) = 0, \quad \text{and} \quad \tau\left(\frac{\pi}{2}\right) = 2.$$

Determine the output of the system for each of the following inputs:

(a) $\cos(\frac{\pi}{2}n)$ **(b)** $\sin(\frac{7\pi}{2}n + \frac{\pi}{4})$

6.5. Consider a continuous-time ideal bandpass filter whose frequency response is

$$H(j\omega) = \begin{cases} 1, & \omega_c \le |\omega| \le 3\omega_c \\ 0, & \text{elsewhere} \end{cases}.$$

(a) If $h(t)$ is the impulse response of this filter, determine a function $g(t)$ such that

$$h(t) = \left(\frac{\sin \omega_c t}{\pi t}\right) g(t).$$

(b) As ω_c is increased, does the impulse response of the filter get more concentrated or less concentrated about the origin?

6.6. Consider a discrete-time ideal highpass filter whose frequency response is specified as

$$H(e^{j\omega}) = \begin{cases} 1, & \pi - \omega_c \le |\omega| \le \pi \\ 0, & |\omega| < \pi - \omega_c \end{cases}$$

(a) If $h[n]$ is the impulse response of this filter, determine a function $g[n]$ such that

$$h[n] = \left(\frac{\sin \omega_c n}{\pi n}\right) g[n].$$

(b) As ω_c is increased, does the impulse response of the filter get more concentrated or less concentrated about the origin?

6.7. A continuous-time lowpass filter has been designed with a passband frequency of 1,000 Hz, a stopband frequency of 1,200 Hz, passband ripple of 0.1, and stopband ripple of 0.05. Let the impulse response of this lowpass filter be denoted by $h(t)$. We wish to convert the filter into a bandpass filter with impulse response

$$g(t) = 2h(t)\cos(4{,}000\pi t).$$

Assuming that $|H(j\omega)|$ is negligible for $|\omega| > 4{,}000\pi$, answer the following questions:

(a) If the passband ripple for the bandpass filter is constrained to be 0.1, what are the two passband frequencies associated with the bandpass filter?

(b) If the stopband ripple for the bandpass filter is constrained to be 0.05, what are the two stopband frequencies associated with the bandpass filter?

6.8. A causal, nonideal lowpass filter is designed with frequency response $H(e^{j\omega})$. The difference equation relating the input $x[n]$ and output $y[n]$ for this filter is specified as

$$y[n] = \sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k].$$

The filter also satisfies the following specifications for the magnitude of its frequency response:

$$\begin{aligned} \text{passband frequency} &= \omega_p, \\ \text{passband tolerance} &= \delta_p, \\ \text{stopband frequency} &= \omega_s, \\ \text{stopband tolerance} &= \delta_s. \end{aligned}$$

Now consider a causal LTI system whose input and output are related by the difference equation

$$y[n] = \sum_{k=1}^{N} (-1)^k a_k y[n-k] + \sum_{k=0}^{M} (-1)^k b_k x[n-k].$$

Show that this filter has a passband with a tolerance of δ_p, and specify the corresponding location of the passband.

6.9. Consider a continuous-time causal and stable LTI system whose input $x(t)$ and output $y(t)$ are related by the differential equation

$$\frac{dy(t)}{dt} + 5y(t) = 2x(t).$$

What is the final value $s(\infty)$ of the step response $s(t)$ of this filter? Also, determine the value of t_0 for which

$$s(t_0) = s(\infty)\left[1 - \frac{1}{e^2}\right].$$

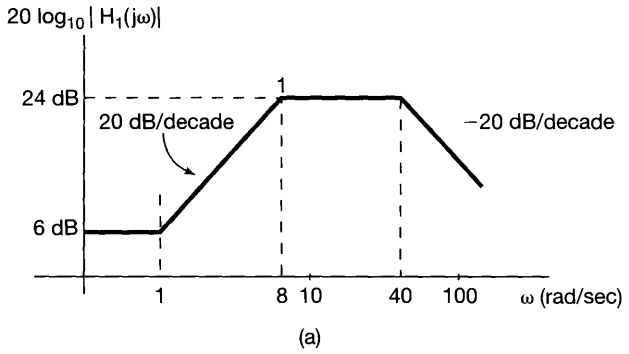
6.10. For each first-order system whose frequency response is as follows, specify the straight-line approximation of the Bode magnitude plot:

(a) $40\left(\frac{j\omega+0.1}{j\omega+40}\right)$ **(b)** $0.04\left(\frac{j\omega+50}{j\omega+0.2}\right)$

6.11. For each second-order system whose frequency response is as follows, specify the straight-line approximation of the Bode magnitude plot:

(a) $\frac{250}{(j\omega)^2+50.5j\omega+25}$ **(b)** $0.02\frac{j\omega+50}{(j\omega)^2+0.2j\omega+1}$

6.12. A continuous-time LTI system S with frequency response $H(j\omega)$ is constructed by cascading two continuous-time LTI systems with frequency responses $H_1(j\omega)$ and $H_2(j\omega)$, respectively. Figures P6.12(a) and P6.12(b) show the straight-line approximations of the Bode magnitude plots of $H_1(j\omega)$ and $H(j\omega)$, respectively. Specify $H_2(j\omega)$.



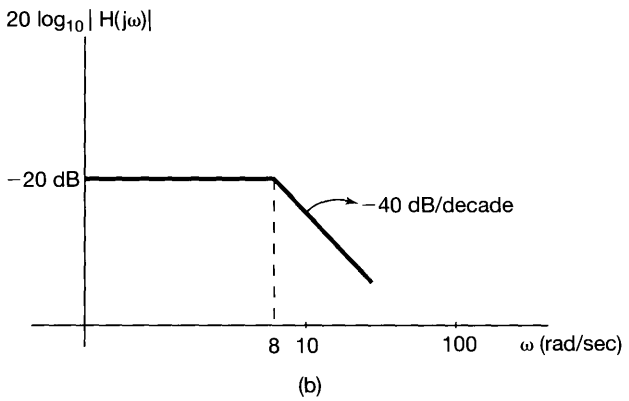


Figure P6.12

6.13. The straight-line approximation of the Bode magnitude plot of a second-order continuous-time LTI system S is shown in Figure P6.13. S may be constructed by

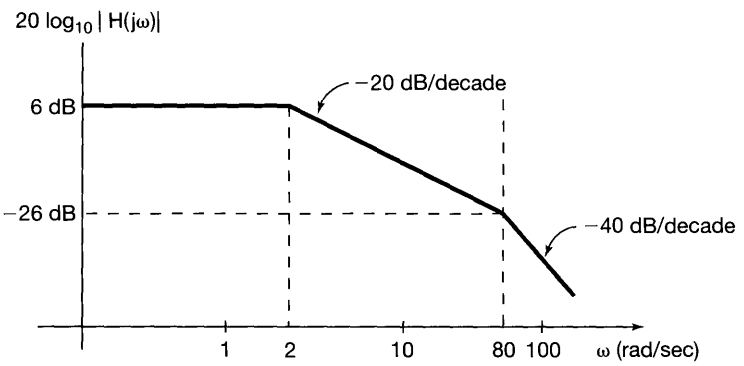


Figure P6.13

either connecting two first-order systems S_1 and S_2 in cascade or two first-order systems S_3 and S_4 in parallel. Determine which, if any, of the following statements are true or false. Justify your answers.

(a) The frequency responses of S_1 and S_2 may be determined uniquely.

(b) The frequency responses of S_3 and S_4 may be determined uniquely.

6.14. The straight-line approximation of the Bode magnitude plot of a causal and stable continuous-time LTI system S is shown in Figure P6.14. Specify the frequency response of a system that is the inverse of S.

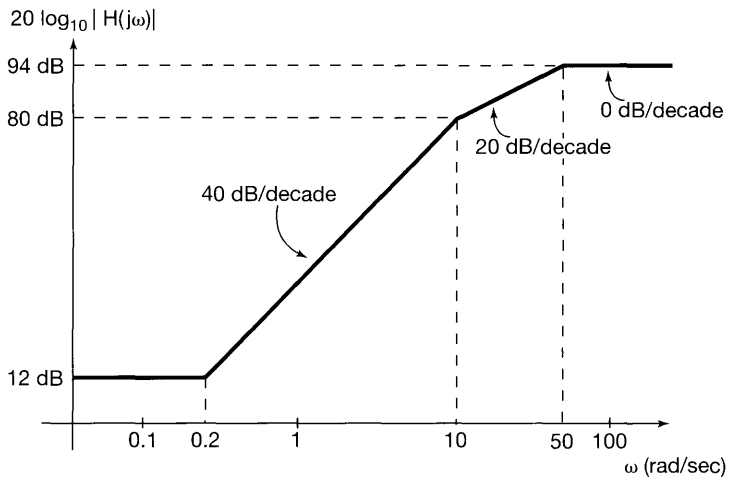


Figure P6.14

6.15. For each of the following second-order differential equations for causal and stable LTI systems, determine whether the corresponding impulse response is underdamped, overdamped, or critically damped:

(a) $\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 4y(t) = x(t)$

(b) $5\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 5y(t) = 7x(t)$

(c) $\frac{d^2y(t)}{dt^2} + 20\frac{dy(t)}{dt} + y(t) = x(t)$

(d) $5\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 5y(t) = 7x(t) + \frac{1}{3}\frac{dx(t)}{dt}$

6.16. A particular first-order causal and stable discrete-time LTI system has a step response whose maximum overshoot is 50% of its final value. If the final value is 1, determine a difference equation relating the input $x[n]$ and output $y[n]$ of this filter.

6.17. For each of the following second-order difference equations for causal and stable LTI systems, determine whether or not the step response of the system is oscillatory:

(a) $y[n] + y[n-1] + \frac{1}{4}y[n-2] = x[n]$

(b) $y[n] - y[n-1] + \frac{1}{4}y[n-2] = x[n]$

6.18. Consider the continuous-time LTI system implemented as the RC circuit shown in Figure P6.18. The voltage source $x(t)$ is considered the input to this system. The voltage $y(t)$ across the capacitor is considered the system output. Is it possible for the step response of the system to have an oscillatory behavior?

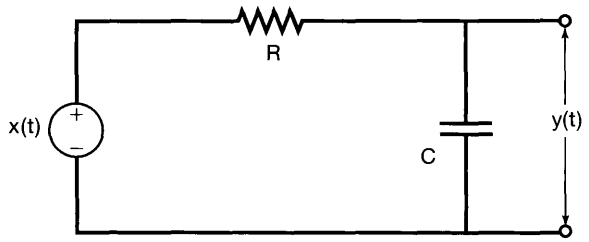


Figure P6.18

6.19. Consider the continuous-time LTI system implemented as the *RLC* circuit shown in Figure P6.19. The voltage source $x(t)$ is considered the input to this system. The voltage $y(t)$ across the capacitor is considered the system output. How should R, L, and C be related so that there is no oscillation in the step response?

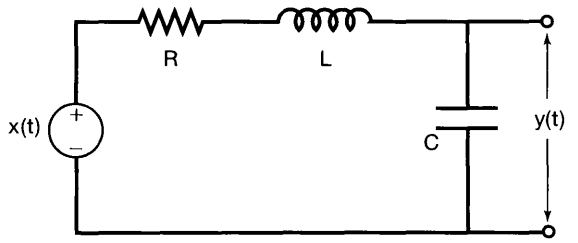


Figure P6.19

6.20. Consider a nonrecursive filter with the impulse response shown in Figure P6.20. What is the group delay as a function of frequency for this filter?

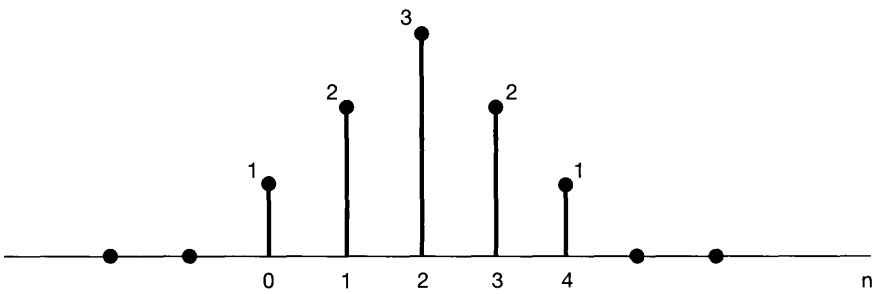


Figure P6.20

BASIC PROBLEMS

6.21. A causal LTI filter has the frequency response $H(j\omega)$ shown in Figure P6.21. For each of the input signals given below, determine the filtered output signal $y(t)$.

(a) $x(t) = e^{jt}$ **(b)** $x(t) = (\sin \omega_0 t)u(t)$

(c) $X(j\omega) = \frac{1}{(j\omega)(6+j\omega)}$ **(d)** $X(j\omega) = \frac{1}{2+j\omega}$

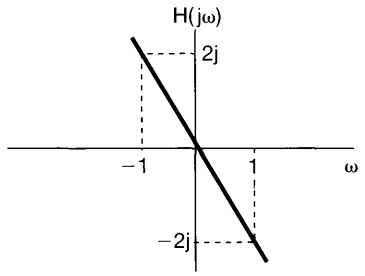


Figure P6.21

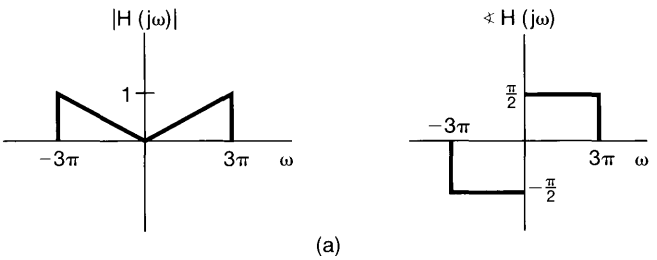
6.22. Shown in Figure P6.22(a) is the frequency response $H(j\omega)$ of a continuous-time filter referred to as a lowpass differentiator. For each of the input signals $x(t)$ below, determine the filtered output signal $y(t)$.

(a) $x(t) = \cos(2\pi t + \theta)$

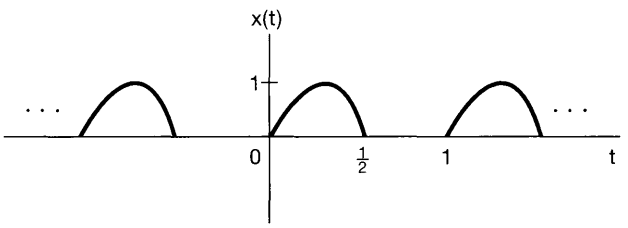
(b) $x(t) = \cos(4\pi t + \theta)$

(c) $x(t)$ is a half-wave rectified sine wave of period, as sketched in Figure P6.22(b).

$$x(t) = \begin{cases} \sin 2\pi t, & m \le t \le (m + \frac{1}{2}) \\ 0, & (m + \frac{1}{2}) \le t \le m \text{ for any integer } m \end{cases}$$



(a)



(b)

Figure P6.22

6.23. Shown in Figure P6.23 is $|H(j\omega)|$ for a lowpass filter. Determine and sketch the impulse response of the filter for each of the following phase characteristics:

(a) $\sphericalangle H(j\omega) = 0$

(b) $\sphericalangle H(j\omega) = \omega T$, where T is a constant

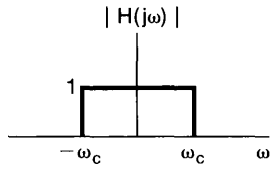


Figure P6.23

(c) $\measuredangle H(j\omega) = \begin{cases} \frac{\pi}{2}, & \omega > 0 \\ \frac{-\pi}{2}, & \omega < 0 \end{cases}$

6.24. Consider a continuous-time lowpass filter whose impulse response $h(t)$ is known to be real and whose frequency response magnitude is given as:

$$|H(j\omega)| = \begin{cases} 1, & |\omega| \leq 200\pi \\ 0, & \text{otherwise} \end{cases}.$$

(a) Determine and sketch the real-valued impulse response $h(t)$ for this filter when the corresponding group delay function is specified as:
(i) $\tau(\omega) = 5$ **(ii)** $\tau(\omega) = \frac{5}{2}$ **(iii)** $\tau(\omega) = -\frac{5}{2}$

(b) If the impulse response $h(t)$ had not been specified to be real, would knowledge of $|H(j\omega)|$ and $\tau(\omega)$ be sufficient to determine $h(t)$ uniquely? Justify your answer.

6.25. By computing the group delay at two selected frequencies, verify that each of the following frequency responses has nonlinear phase.
(a) $H(j\omega) = \frac{1}{j\omega+1}$ **(b)** $H(j\omega) = \frac{1}{(j\omega+1)^2}$ **(c)** $H(j\omega) = \frac{1}{(j\omega+1)(j\omega+2)}$

6.26. Consider an ideal highpass filter whose frequency response is specified as

$$H(j\omega) = \begin{cases} 1, & |\omega| > \omega_c \\ 0, & \text{otherwise} \end{cases}.$$

(a) Determine the impulse response $h(t)$ for this filter.
(b) As ω_c is increased, does $h(t)$ get more or less concentrated about the origin?
(c) Determine $s(0)$ and $s(\infty)$, where $s(t)$ is the step response of the filter.

6.27. The output $y(t)$ of a causal LTI system is related to the input $x(t)$ by the differential equation

$$\frac{dy(t)}{dt} + 2y(t) = x(t).$$

(a) Determine the frequency response

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

of the system, and sketch its Bode plot.
(b) Specify, as a function of frequency, the group delay associated with this system.
(c) If $x(t) = e^{-t}u(t)$, determine $Y(j\omega)$, the Fourier transform of the output.

(d) Using the technique of partial-fraction expansion, determine the output $y(t)$ for the input $x(t)$ in part (c).

(e) Repeat parts (c) and (d), first if the input has as its Fourier transform

(i) $X(j\omega) = \frac{1+j\omega}{2+j\omega}$,

then if

(ii) $X(j\omega) = \frac{2+j\omega}{1+j\omega}$,

and finally, if

(iii) $X(j\omega) = \frac{1}{(2+j\omega)(1+j\omega)}$

6.28. **(a)** Sketch the Bode plots for the following frequency responses:

(i) $1 + (j\omega/10)$ **(ii)** $1 - (j\omega/10)$

(iii) $\frac{16}{(j\omega+2)^4}$ **(iv)** $\frac{1-(j\omega/10)}{1+j\omega}$

(v) $\frac{(j\omega/10)-1}{1+j\omega}$ **(vi)** $\frac{1+(j\omega/10)}{1+j\omega}$

(vii) $\frac{1-(j\omega/10)}{(j\omega)^2+(j\omega)+1}$ **(viii)** $\frac{10+5j\omega+10(j\omega)^2}{1+(j\omega/10)}$

(ix) $1 + j\omega + (j\omega)^2$ **(x)** $1 - j\omega + (j\omega)^2$

(xi) $\frac{(j\omega+10)(10j\omega+1)}{[(j\omega/100+1)][((j\omega)^2+j\omega+1)]}$

(b) Determine and sketch the impulse response and the step response for the system with frequency response (iv). Do the same for the system with frequency response (vi).

The system given in (iv) is often referred to as a non-minimum-phase system, while the system specified in (vi) is referred to as being a minimum phase. The corresponding impulse responses of (iv) and (vi) are referred to as a non-minimum-phase signal and a minimum-phase signal, respectively. By comparing the Bode plots of these two frequency responses, we can see that they have identical magnitudes; however, the magnitude of the *phase* of the system of (iv) is larger than for the system of (vi).

We can also note differences in the time-domain behavior of the two systems. For example, the impulse response of the minimum-phase system has more of its energy concentrated near $t = 0$ than does the impulse response of the non-minimum-phase system. In addition, the step response of (iv) initially has the opposite sign from its asymptotic value as $t \to \infty$, while this is not the case for the system of (vi).

The important concept of minimum- and non-minimum-phase systems can be extended to more general LTI systems than the simple first-order systems we have treated here, and the distinguishing characteristics of these systems can be described far more thoroughly than we have done.

6.29. An LTI system is said to have *phase lead* at a particular frequency $\omega = \omega_0$ if $\measuredangle H(j\omega_0) > 0$. The terminology stems from the fact that if $e^{j\omega_0 t}$ is the input to this system, then the phase of the output will exceed, or lead, the phase of the input. Similarly, if $\measuredangle H(j\omega_0) < 0$, the system is said to have *phase lag* at this frequency. Note that the system with frequency response

$$\frac{1}{1 + j\omega\tau}$$

has phase lag for all $\omega > 0$, while the system with frequency response

$$1 + j\omega\tau$$

has phase lead for all $\omega > 0$.

(a) Construct the Bode plots for the following two systems. Which has phase lead and which phase lag? Also, which one amplifies signals at certain frequencies?

(i) $\frac{1+(j\omega/10)}{1+10j\omega}$ **(ii)** $\frac{1+10j\omega}{1+(j\omega/10)}$

(b) Repeat part (a) for the following three frequency responses:

(i) $\frac{(1+(j\omega/10))^2}{(1+10j\omega)^3}$ **(ii)** $\frac{1+j\omega/10}{100(j\omega)^2+10j\omega+1}$ **(iii)** $\frac{1+10j\omega}{0.01(j\omega)^2+0.2j\omega+1}$

6.30. Let $h(t)$ have the Bode plot depicted in Figure P6.30. The dashed lines in the figure represent straight-line approximations. Sketch the Bode plots for $10h(10t)$.

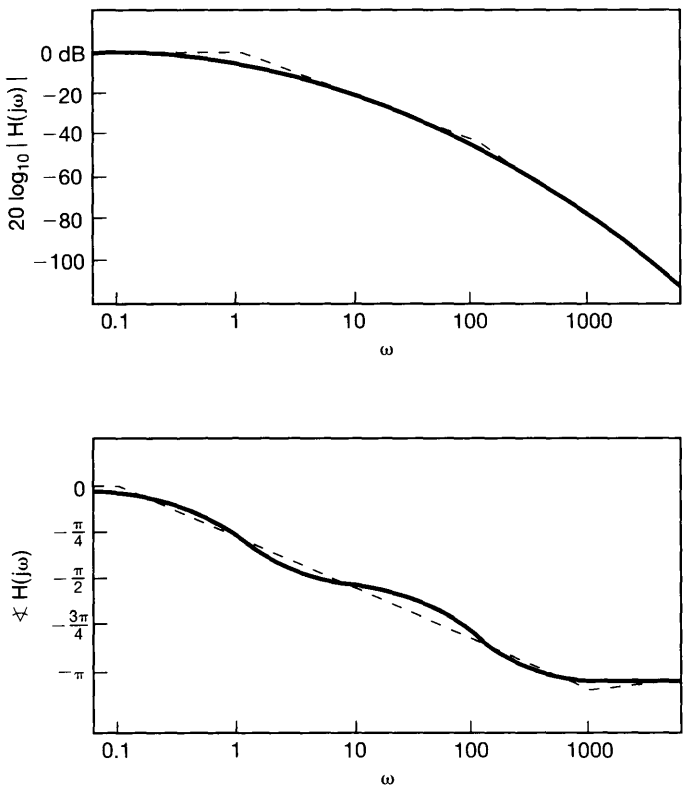


Figure P6.30

6.31. An integrator has as its frequency response

$$H(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega),$$

where the impulse at $\omega = 0$ is a result of the fact that the integration of a constant input from $t = -\infty$ results in an infinite output. Thus, if we avoid inputs that are

constant, or equivalently, only examine $H(j\omega)$ for $\omega > 0$, we see that

$$20\log|H(j\omega)| = -20\log(\omega),$$

$$\measuredangle H(j\omega) = \frac{-\pi}{2}.$$

In other words, the Bode plot for an integrator, as illustrated in Figure P6.31, consists of two straight-line plots. These plots reflect the principal characteristics of an integrator: a phase shift of $-90°$ at all positive values of frequency and the amplification of low frequencies.

(a) A useful, simple model of an electric motor is an LTI system with input equal to the applied voltage and output given by the motor shaft angle. This system can be visualized as the cascade of a stable LTI system (with the voltage as input and shaft angular velocity as output) and an integrator (representing the integration of the angular velocity). Often, a model of first-order system is used for the first part of the cascade. Assuming, for example, that this first-order system has a time constant of 0.1 second, we obtain an overall motor frequency response of

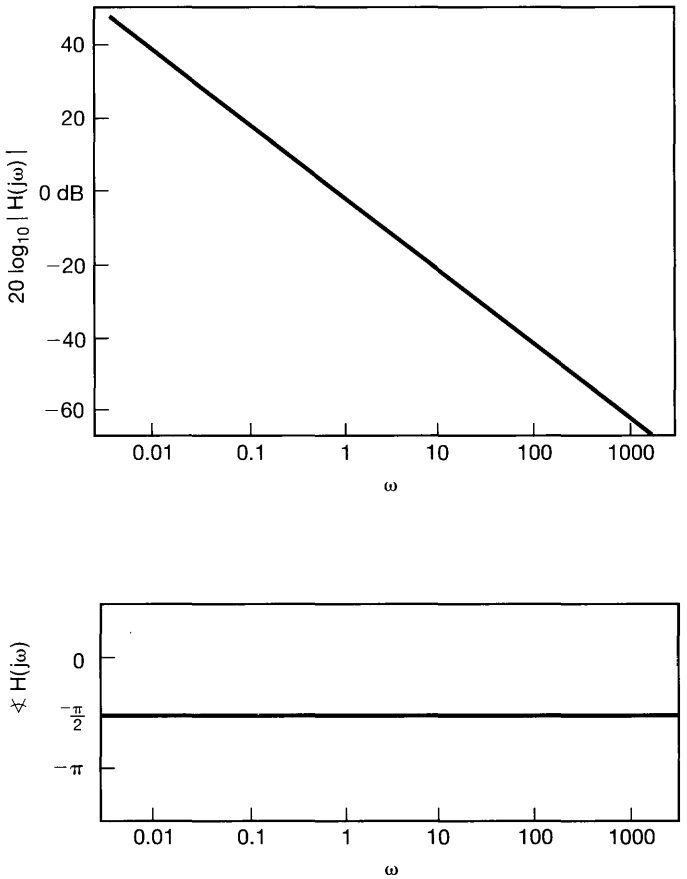


Figure P6.31

the form

$$H(j\omega) = \frac{1}{j\omega(1 + j\omega/10)} + \pi\delta(\omega).$$

Sketch the Bode plot for the system for $\omega > 0.001$.

(b) Sketch the Bode plot for a differentiator.

(c) Do the same for systems with the following frequency responses:

(i) $H(j\omega) = \frac{j\omega}{1+j\omega/100}$

(ii) $H(j\omega) = \frac{j\omega}{(1+(j\omega)/10+(j\omega)^2/100)}$

6.32. Consider the system depicted in Figure P6.32. This "compensator" box is a continuous-time LTI system.

(a) Suppose that it is desired to choose the frequency response of the compensator so that the overall frequency response $H(j\omega)$ of the cascade satisfies the following two conditions:

1. The log magnitude of $H(j\omega)$ has a slope of -40 dB/decade beyond $\omega = 1{,}000$.
2. For $0 < \omega < 1{,}000$, the log magnitude of $H(j\omega)$ should be between -10 dB and 10 dB.

Design a suitable compensator (that is, determine a frequency response for a compensator that meets the preceding requirements), and draw the Bode plot for the resulting $H(j\omega)$.

(b) Repeat (a) if the specifications on the log magnitude of $H(j\omega)$ are as follows:

1. It should have a slope of $+20$ dB/decade for $0 < \omega < 10$.
2. It should be between $+10$ and $+30$ dB for $10 < \omega < 100$.
3. It should have a slope of -20 dB/decade for $100 < \omega < 1{,}000$.
4. It should have a slope of -40 dB/decade for $\omega > 1{,}000$.

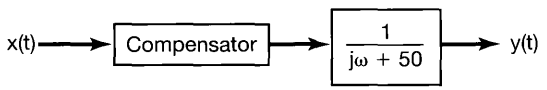


Figure P6.32

6.33. Figure P6.33 shows a system commonly used to obtain a highpass filter from a lowpass filter and vice versa.

(a) Show that, if $H(j\omega)$ is a lowpass filter with cutoff frequency ω_{lp}, the overall system corresponds to an ideal highpass filter. Determine the system's cutoff frequency and sketch its impulse response.

(b) Show that, if $H(j\omega)$ is an ideal highpass filter with cutoff frequency ω_{hp}, the overall system corresponds to an ideal lowpass filter, and determine the cutoff frequency of the system.

(c) If the interconnection of Figure P6.33 is applied to an ideal discrete-time lowpass filter, will the resulting system be an ideal discrete-time highpass filter?

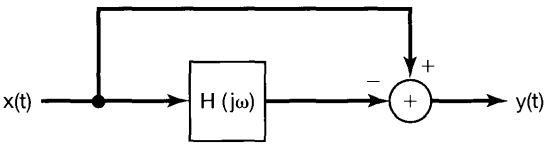


Figure P6.33

6.34. In Problem 6.33, we considered a system commonly used to obtain a highpass filter from a lowpass filter and vice versa. In this problem, we explore the system further and, in particular, consider a potential difficulty if the phase of $H(j\omega)$ is not properly chosen.

(a) Referring to Figure P6.33, let us assume that $H(j\omega)$ is real and as shown in Figure P6.34. Then

$$1 - \delta_1 < H(j\omega) < 1 + \delta_1, \quad 0 \leq \omega \leq \omega_1,$$

$$-\delta_2 < H(j\omega) < +\delta_2, \quad \omega_2 < \omega.$$

Determine and sketch the resulting frequency response of the overall system of Figure P6.33. Does the resulting system correspond to an approximation to a highpass filter?

(b) Now let $H(j\omega)$ in Figure P6.33 be of the form

$$H(j\omega) = H_1(j\omega)e^{j\theta(\omega)}, \tag{P6.34-1}$$

where $H_1(j\omega)$ is identical to Figure P6.34 and $\theta(\omega)$ is an unspecified phase characteristic. With $H(j\omega)$ in this more general form, does it still correspond to an approximation to a lowpass filter?

(c) Without making any assumptions about $\theta(\omega)$, determine and sketch the tolerance limits on the magnitude of the frequency response of the overall system of Figure P6.33.

(d) If $H(j\omega)$ in Figure P6.33 is an approximation to a lowpass filter with unspecified phase characteristics, will the overall system in that figure necessarily correspond to an approximation to a highpass filter?

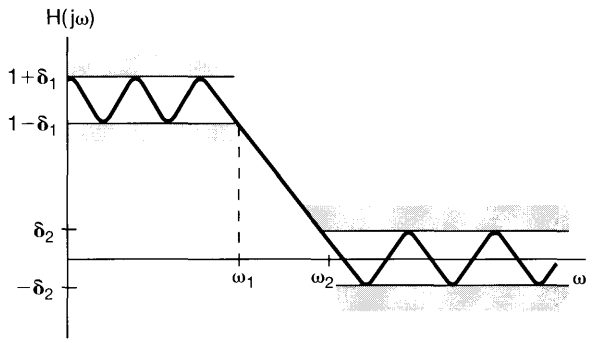


Figure P6.34

6.35. Shown in Figure P6.35 is the frequency response $H(e^{j\omega})$ of a discrete-time differentiator. Determine the output signal $y[n]$ as a function of ω_0 if the input $x[n]$ is

$$x[n] = \cos[\omega_0 n + \theta].$$

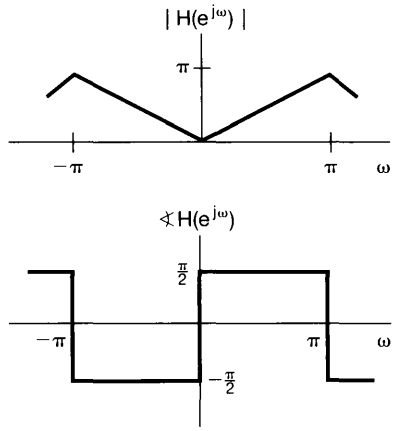


Figure P6.35

6.36. Consider a discrete-time lowpass filter whose impulse response $h[n]$ is known to be real and whose frequency response magnitude in the region $-\pi \leq \omega \leq \pi$ is given as:

$$|H(e^{j\omega})| = \begin{cases} 1, & |\omega| \leq \frac{\pi}{4} \\ 0, & \text{otherwise} \end{cases}.$$

Determine and sketch the real-valued impulse response $h[n]$ for this filter when the corresponding group delay function is specified as:

(a) $\tau(\omega) = 5$ **(b)** $\tau(\omega) = \frac{5}{2}$ **(c)** $\tau(\omega) = -\frac{5}{2}$

6.37. Consider a causal LTI system whose frequency response is given as:

$$H(e^{j\omega}) = e^{-j\omega}\frac{1 - \frac{1}{2}e^{j\omega}}{1 - \frac{1}{2}e^{-j\omega}}.$$

(a) Show that $|H(e^{j\omega})|$ is unity at all frequencies.

(b) Show that

$$\measuredangle H(e^{j\omega}) = -\omega - 2\tan^{-1}\left(\frac{\frac{1}{2}\sin\omega}{1 - \frac{1}{2}\cos\omega}\right).$$

(c) Show that the group delay for this filter is given by

$$\tau(\omega) = \frac{\frac{3}{4}}{\frac{5}{4} - \cos\omega}.$$

Sketch $\tau(\omega)$.

(d) What is the output of this filter when the input is $\cos(\frac{\pi}{3}n)$?

6.38. Consider an ideal bandpass filter whose frequency response in the region $-\pi \leq \omega \leq \pi$ is specified as

$$H(e^{j\omega}) = \begin{cases} 1, & \frac{\pi}{2} - \omega_c \leq |\omega| \leq \frac{\pi}{2} + \omega_c \\ 0, & \text{otherwise} \end{cases}.$$

Determine and sketch the impulse response $h[n]$ for this filter when

(a) $\omega_c = \frac{\pi}{5}$

(b) $\omega_c = \frac{\pi}{4}$

(c) $\omega_c = \frac{\pi}{3}$

As ω_c is increased, does $h[n]$ get more or less concentrated about the origin?

6.39. Sketch the log magnitude and phase of each of the following frequency responses.

(a) $1 + \frac{1}{2}e^{-j\omega}$ **(b)** $1 + 2e^{-j\omega}$

(c) $1 - 2e^{-j\omega}$ **(d)** $1 + 2e^{-j2\omega}$

(e) $\frac{1}{(1+\frac{1}{2}e^{-j\omega})^3}$ **(f)** $\frac{1+\frac{1}{2}e^{-j\omega}}{1-\frac{1}{2}e^{-j\omega}}$

(g) $\frac{1+2e^{-j\omega}}{1+\frac{1}{2}e^{-j\omega}}$ **(h)** $\frac{1-2e^{-j\omega}}{1+\frac{1}{2}e^{-j\omega}}$

(i) $\frac{1}{(1-\frac{1}{4}e^{-j\omega})(1-\frac{3}{4}e^{-j\omega})}$ **(j)** $\frac{1}{(1-\frac{1}{4}e^{-j\omega})(1+\frac{3}{4}e^{-j\omega})}$

(k) $\frac{1+2e^{-2j\omega}}{(1-\frac{1}{2}e^{-j\omega})^2}$

6.40. Consider an ideal discrete-time lowpass filter with impulse response $h[n]$ and for which the frequency response $H(e^{j\omega})$ is that shown in Figure P6.40. Let us consider obtaining a new filter with impulse response $h_1[n]$ and frequency response $H_1(e^{j\omega})$ as follows:

$$h_1[n] = \begin{cases} h[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

This corresponds to inserting a sequence value of zero between each sequence value of $h[n]$. Determine and sketch $H_1(e^{j\omega})$ and state the class of ideal filters to which it belongs (e.g., lowpass, highpass, bandpass, multiband, etc.).

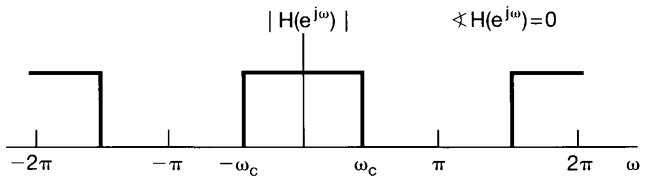


Figure P6.40

6.41. A particular causal LTI system is described by the difference equation

$$y[n] - \frac{\sqrt{2}}{2}y[n-1] + \frac{1}{4}y[n-2] = x[n] - x[n-1].$$

(a) Find the impulse response of this system.

(b) Sketch the log magnitude and the phase of the frequency response of the system.

6.42. **(a)** Consider two LTI systems with the following frequency responses:

$$H_1(e^{j\omega}) = \frac{1 + \frac{1}{2}e^{-j\omega}}{1 + \frac{1}{4}e^{-j\omega}},$$

$$H_2(e^{j\omega}) = \frac{\frac{1}{2} + e^{-j\omega}}{1 + \frac{1}{4}e^{-j\omega}}.$$

Show that both of these frequency responses have the same magnitude function [i.e., $|H_1(e^{j\omega})| = |H_2(e^{j\omega})|$], but the group delay of $H_2(e^{j\omega})$ is greater than the group delay of $H_1(e^{j\omega})$ for $\omega > 0$.

(b) Determine and sketch the impulse and step responses of the two systems.

(c) Show that

$$H_2(e^{j\omega}) = G(e^{j\omega})H_1(e^{j\omega}),$$

where $G(e^{j\omega})$ is an *all-pass system* [i.e., $|G(e^{j\omega})| = 1$ for all ω].

6.43. When designing filters with highpass or bandpass characteristics, it is often convenient first to design a lowpass filter with the desired passband and stopband specifications and then to transform this prototype filter to the desired highpass or bandpass filter. Such transformations are called lowpass-to-highpass or highpass-to-lowpass transformations. Designing filters in this manner is convenient because it requires us only to formulate our filter design algorithms for the class of filters with lowpass characteristics. As one example of such a procedure, consider a discrete-time lowpass filter with impulse response $h_{lp}[n]$ and frequency response $H_{lp}(e^{j\omega})$, as sketched in Figure P6.43. Suppose the impulse response is modulated with the sequence $(-1)^n$ to obtain $h_{hp}[n] = (-1)^n h_{lp}[n]$.

(a) Determine and sketch $H_{hp}(e^{j\omega})$ in terms of $H_{lp}(e^{j\omega})$. Show in particular that, for $H_{lp}(e^{j\omega})$ as shown in Figure P6.43, $H_{hp}(e^{j\omega})$ corresponds to a highpass filter.

(b) Show that modulation of the impulse response of a discrete-time highpass filter by $(-1)^n$ will transform it to a lowpass filter.

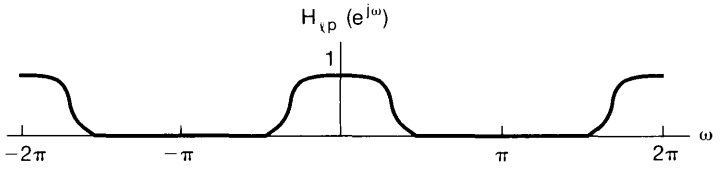


Figure P6.43

6.44. A discrete-time system is implemented as shown in Figure P6.44. The system S shown in the figure is an LTI system with impulse response $h_{lp}[n]$.

(a) Show that the overall system is time invariant.

(b) If $h_{lp}[n]$ is a lowpass filter, what type of filter does the system of the figure implement?

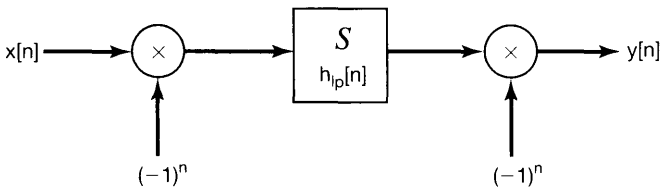


Figure P6.44

6.45. Consider the following three frequency responses for causal and stable third-order LTI systems. By utilizing the properties of first- and second-order systems discussed in Section 6.6, determine whether or not the impulse response of each of the third-order systems is oscillatory. (*Note:* You should be able to answer this question without taking the inverse Fourier transforms of the frequency responses of the third-order systems.)

$$H_1(e^{j\omega}) = \frac{1}{(1-\frac{1}{2}e^{-j\omega})(1-\frac{1}{3}e^{-j\omega})(1-\frac{1}{4}e^{-j\omega})},$$

$$H_2(e^{j\omega}) = \frac{1}{(1+\frac{1}{2}e^{-j\omega})(1-\frac{1}{3}e^{-j\omega})(1-\frac{1}{4}e^{-j\omega})},$$

$$H_3(e^{j\omega}) = \frac{1}{(1-\frac{1}{2}e^{-j\omega})(1-\frac{3}{4}e^{-j\omega}+\frac{9}{16}e^{-j2\omega})}.$$

6.46. Consider a causal, nonrecursive (FIR) filter whose real-valued impulse response $h[n]$ is zero for $n \geq N$.

(a) Assuming that N is odd, show that if $h[n]$ is symmetric about $(N-1)/2$ (i.e., if $h[(N-1)/2+n] = h[(N-1)/2-n]$), then

$$H(e^{j\omega}) = A(\omega)e^{-j[(N-1)/2]\omega},$$

where $A(\omega)$ is a real-valued function of ω. We conclude that the filter has linear phase.

(b) Give an example of the impulse response $h[n]$ of a causal, linear-phase FIR filter such that $h[n] = 0$ for $n \geq 5$ and $h[n] \neq 0$ for $0 \leq n \leq 4$.

(c) Assuming that N is even, show that if $h[n]$ is symmetric about $(N-1)/2$ (i.e., if $h[(N/2)+n] = h[N/2-n-1]$), then

$$H(e^{j\omega}) = A(\omega)e^{-j[(N-1)/2]\omega},$$

where $A(\omega)$ is a real-valued function of ω.

(d) Give an example of the impulse response $h[n]$ of a causal, linear-phase FIR filter such that $h[n] = 0$ for $n \geq 4$ and $h[n] \neq 0$ for $0 \leq n \leq 3$.

6.47. A three-point symmetric moving average, referred to as a weighted moving average, is of the form

$$y[n] = b\{ax[n-1] + x[n] + ax[n+1]\}. \qquad \text{(P6.47–1)}$$

(a) Determine, as a function of a and b, the frequency response $H(e^{j\omega})$ of the three-point moving average in eq. (P6.47–1).

(b) Determine the scaling factor b such that $H(e^{j\omega})$ has unity gain at zero frequency.

(c) In many time-series analysis problems, a common choice for the coefficient a in the weighted moving average in eq. (P6.47–1) is $a = 1/2$. Determine and sketch the frequency response of the resulting filter.

6.48. Consider a four-point, moving-average, discrete-time filter for which the difference equation is

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] + b_3x[n-2].$$

Determine and sketch the magnitude of the frequency response for each of the following cases:

(a) $b_0 = b_3 = 0, b_1 = b_2$

(b) $b_1 = b_2 = 0, b_0 = b_3$

(c) $b_0 = b_1 = b_2 = b_3$

(d) $b_0 = -b_1 = b_2 = -b_3$

ADVANCED PROBLEMS

6.49. The time constant provides a measure of how fast a first-order system responds to inputs. The idea of measuring the speed of response of a system is also important for higher order systems, and in this problem we investigate the extension of the time constant to such systems.

(a) Recall that the time constant of a first-order system with impulse response

$$h(t) = ae^{-at}u(t), \quad a > 0,$$

is $1/a$, which is the amount of time from $t = 0$ that it takes the system step response $s(t)$ to settle within $1/e$ of its final value [i.e., $s(\infty) = \lim_{t\to\infty} s(t)$]. Using this same quantitative definition, find the equation that must be solved in order to determine the time constant of the causal LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 10y(t) = 9x(t). \tag{P6.49–1}$$

(b) As can be seen from part (a), if we use the precise definition of the time constant set forth there, we obtain a simple expression for the time constant of a first-order system, but the calculations are decidedly more complex for the system of eq. (P6.49–1). However, show that this system can be viewed as the parallel interconnection of two first-order systems. Thus, we usually think of the system of eq. (P6.49–1) as having *two* time constants, corresponding to the two first-order factors. What are the two time constants for this system?

(c) The discussion given in part (b) can be directly generalized to all systems with impulse responses that are linear combinations of decaying exponentials. In any system of this type, one can identify the *dominant* time constants of the system, which are simply the largest of the time constants. These represent the slowest parts of the system response, and consequently, they have the dominant effect on how fast the system as a whole can respond. What is the dominant time constant of the system of eq. (P6.49–1)? Substitute this time constant into the equation determined in part (a). Although the number will not satisfy the equation exactly, you should see that it nearly does, which is an indication that it is very close to the time constant defined in part (a). Thus, the approach we have outlined in part (b) and here is of value in providing insight into the speed of response of LTI systems without requiring excessive calculation.

(d) One important use of the concept of dominant time constants is in the reduction of the order of LTI systems. This is of great practical significance in problems

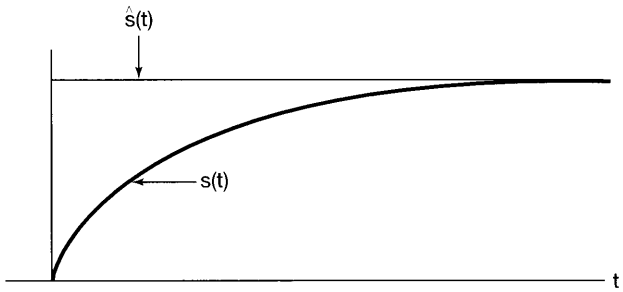


Figure P6.49

involving the analysis of complex systems having a few dominant time constants and other very small time constants. In order to reduce the complexity of the model of the system to be analyzed, one often can simplify the fast parts of the system. That is, suppose we regard a complex system as a parallel interconnection of first- and second-order systems. Suppose also that one of these subsystems, with impulse response $h(t)$ and step response $s(t)$, is fast—that is, that $s(t)$ settles to its final value $s(\infty)$ very quickly. Then we can approximate this subsystem by the subsystem that settles to the same final value *instantaneously*. That is, if $\hat{s}(t)$ is the step response to our approximation, then

$$\hat{s}(t) = s(\infty)u(t).$$

This is illustrated in Figure P6.49. Note that the impulse response of the approximate system is then

$$\hat{h}(t) = s(\infty)\delta(t),$$

which indicates that the approximate system is *memoryless*.

Consider again the causal LTI system described by eq. (P6.49–1) and, in particular, the representation of it as a parallel interconnection of two first-order systems, as described in part (b). Use the method just outlined to replace the faster of the two subsystems by a memoryless system. What is the differential equation that then describes the resulting overall system? What is the frequency response of this system? Sketch $|H(j\omega)|$ (not $\log|H(j\omega)|$) and $\measuredangle H(j\omega)$ for both the original and approximate systems. Over what range of frequencies are these frequency responses nearly equal? Sketch the step responses for both systems. Over what range of time are the step responses nearly equal? From your plots, you will see some of the similarities *and* differences between the original system and its approximation. The utility of an approximation such as this depends upon the specific application. In particular, one must take into account both how widely separated the different time constants are and also the nature of the inputs to be considered. As you will see from your answers in this part of the problem, the frequency response of the approximate system is essentially the same as the frequency response of the original system at low frequencies. That is, when the fast parts of the system are sufficiently fast compared to the rate of fluctuation of the input, the approximation becomes useful.

6.50. The concepts associated with frequency-selective filtering are often used to separate two signals that have been added together. If the spectra of the two signals do not overlap, ideal frequency-selective filters are desirable. However, if the spectra overlap, it is often preferable to design the filter to have a gradual transition between passband and stopband. In this problem, we explore one approach for determining the frequency response of a filter to be used for separating signals with overlapping spectra. Let $x(t)$ denote a composite continuous-time signal consisting of the sum of two signals $s(t) + w(t)$. As indicated in Figure P6.50(a), we would like to design an LTI filter to recover $s(t)$ from $x(t)$. The filter's frequency response $H(j\omega)$ is to be chosen so that, in some sense, $y(t)$ is a "good" approximation to $s(t)$.

Let us define a measure of the error between $y(t)$ and $s(t)$ at each frequency ω as

$$\epsilon(\omega) \triangleq |S(j\omega) - Y(j\omega)|^2,$$

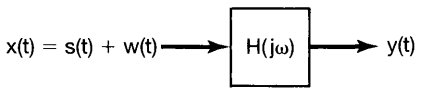
where $S(j\omega)$ and $Y(j\omega)$ are the Fourier transforms of $s(t)$ and $y(t)$, respectively.

(a) Express $\epsilon(\omega)$ in terms of $S(j\omega)$, $H(j\omega)$, and $W(j\omega)$, where $W(j\omega)$ is the Fourier transform of $w(t)$.

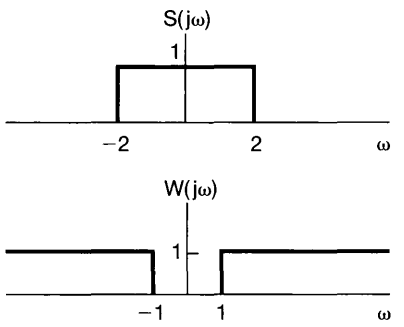
(b) Let us restrict $H(j\omega)$ to be real, so that $H(j\omega) = H^*(j\omega)$. By setting the derivative of $\epsilon(\omega)$ with respect to $H(j\omega)$ to be zero, determine the $H(j\omega)$ required to minimize the error $\epsilon(\omega)$.

(c) Show that if the spectra of $S(j\omega)$ and $W(j\omega)$ are non-overlapping, the result in part (b) reduces to an ideal frequency-selective filter.

(d) From your result in part (b), determine and sketch $H(j\omega)$ if $S(j\omega)$ and $W(j\omega)$ are as shown in Figure P6.50(b).



(a)



(b)

Figure P6.50

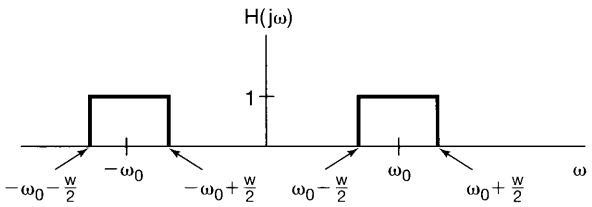
6.51. An ideal bandpass filter is a bandpass filter that passes only a range of frequencies, without any change in amplitude or phase. As shown in Figure P6.51(a), let the passband be

$$\omega_0 - \frac{w}{2} \leq |\omega| \leq \omega_0 + \frac{w}{2}.$$

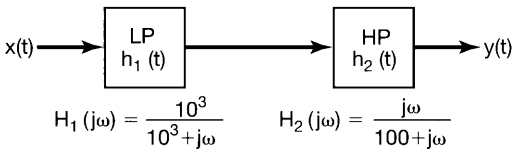
(a) What is the impulse response $h(t)$ of this filter?

(b) We can approximate an ideal bandpass filter by cascading a first-order lowpass and a first-order highpass filter, as shown in Figure P6.51(b). Sketch the Bode diagrams for each of the two filters $H_1(j\omega)$ and $H_2(j\omega)$.

(c) Determine the Bode diagram for the overall bandpass filter in terms of your results from part (b).



(a)



(b)

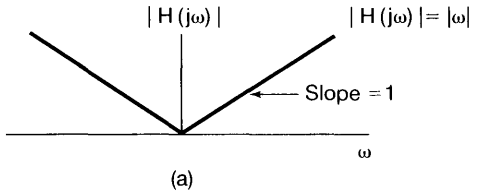
Figure P6.51

6.52. In Figure P6.52(a), we show the magnitude of the frequency response for an ideal continuous-time differentiator. A nonideal differentiator would have a frequency response that is some approximation to the frequency response in the figure.

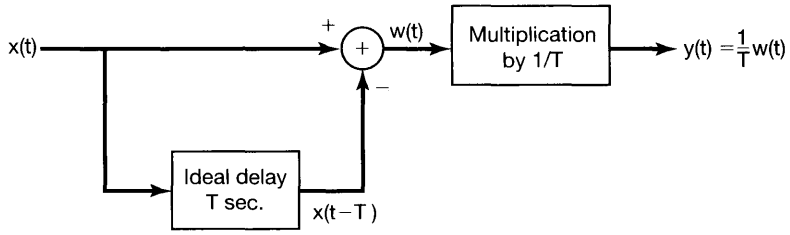
(a) Consider a nonideal differentiator with frequency response $G(j\omega)$ for which $|G(j\omega)|$ is constrained to be within $\pm 10\%$ of the magnitude of the frequency response of the ideal differentiator at all frequencies; that is,

$$-0.1|H(j\omega)| \leq [|G(j\omega)| - |H(j\omega)|] \leq 0.1|H(j\omega)|.$$

Sketch the region in a plot of $G(j\omega)$ vs. ω where $|G(j\omega)|$ must be confined to meet this specification.



(a)



(b)

Figure P6.52

(b) The system in Figure P6.52(b), incorporating an ideal delay of T seconds, is sometimes used to approximate a continuous-time differentiator. For $T = 10^{-2}$ second, determine the frequency range over which the magnitude of the frequency response of the system in the figure is within $\pm 10\%$ of that for an ideal differentiator.

6.53. In many filtering applications, it is often undesirable for the step response of a filter to overshoot its final value. In processing pictures, for example, the overshoot in the step response of a linear filter may produce flare—that is, an increase in intensity—at sharp boundaries. It is possible, however, to eliminate overshoot by requiring that the impulse response of the filter be positive for all time.

Show that if $h(t)$, the impulse response of a continuous-time LTI filter, is always greater than or equal to zero, the step response of the filter is a monotonically nondecreasing function and therefore will not have overshoot.

6.54. By means of a specific filter design procedure, a nonideal continuous-time lowpass filter with frequency response $H_0(j\omega)$, impulse response $h_0(t)$, and step response $s_0(t)$ has been designed. The cutoff frequency of the filter is at $\omega = 2\pi \times 10^2$ rad/sec, and the step response rise time, defined as the time required for the step response to go from 10% of its final value to 90% of its final value, is $\tau_r = 10^{-2}$ second. From this design, we can obtain a new filter with an arbitrary cutoff frequency ω_c by the use of frequency scaling. The frequency response of the resulting filter is then of the form

$$H_{lp}(j\omega) = H_0(ja\omega),$$

where a is an appropriate scale factor.

(a) Determine the scale factor a such that $H_{lp}(j\omega)$ has a cutoff frequency of ω_c.

(b) Determine the impulse response $h_{lp}(t)$ of the new filter in terms of ω_c and $h_0(t)$.

(c) Determine the step response $s_{lp}(t)$ of the new filter in terms of ω_c and $s_0(t)$.

(d) Determine and sketch the rise time of the new filter as a function of its cutoff frequency ω_c.

This is one illustration of the trade-off between time-domain and frequency-domain characteristics. In particular, as the cutoff frequency decreases, the rise time tends to increase.

6.55. The square of the magnitude of the frequency response of a class of continuous-time lowpass filters, known as Butterworth filters, is

$$|B(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2N}}.$$

Let us define the passband edge frequency ω_p as the frequency below which $|B(j\omega)|^2$ is greater than one-half of its value at $\omega = 0$; that is,

$$|B(j\omega)|^2 \geq \frac{1}{2}|B(j0)|^2, \quad |\omega| < \omega_p.$$

Now let us define the stopband edge frequency ω_s as the frequency above which $|B(j\omega)|^2$ is less than 10^{-2} of its value at $\omega = 0$; that is,

$$|B(j\omega)|^2 \leq 10^{-2}|B(j0)|^2, \quad |\omega| > \omega_s.$$

The transition band is then the frequency range between ω_p and ω_s. The ratio ω_s/ω_p is referred to as the transition ratio.

For fixed ω_p, and making reasonable approximations, determine and sketch the transition ratio as a function of N for the class of Butterworth filters.

6.56. In this problem, we explore some of the filtering issues involved in the commercial version of a typical system that is used in most modern cassette tape decks to reduce noise. The primary source of noise is the high-frequency hiss in the tape playback process, which, in some part, is due to the friction between the tape and the playback head. Let us assume that the noise hiss that is added to the signal upon playback has the spectrum of Figure P6.56(a) when measured in decibels, with 0 dB equal to the signal level at 100 Hz. The spectrum $S(j\omega)$ of the signal has the shape shown in Figure P6.56(b).

The system that we analyze has a filter $H_1(j\omega)$ which conditions the signal $s(t)$ before it is recorded. Upon playback, the hiss $n(t)$ is added to the signal. The system is represented schematically in Figure P6.56(c).

Suppose we would like our overall system to have a signal-to-noise ratio of 40 dB over the frequency range 50 Hz $< \omega/2\pi <$ 20 kHz.

(a) Determine the transfer characteristic of the filter $H_1(j\omega)$. Sketch the Bode plot of $H_1(j\omega)$.

(b) If we were to listen to the signal $p(t)$, assuming that the playback process does nothing more than add hiss to the signal, how do you think it would sound?

(c) What should the Bode plot and transfer characteristic of the filter $H_2(j\omega)$ be in order for the signal $\hat{s}(t)$ to sound similar to $s(t)$?

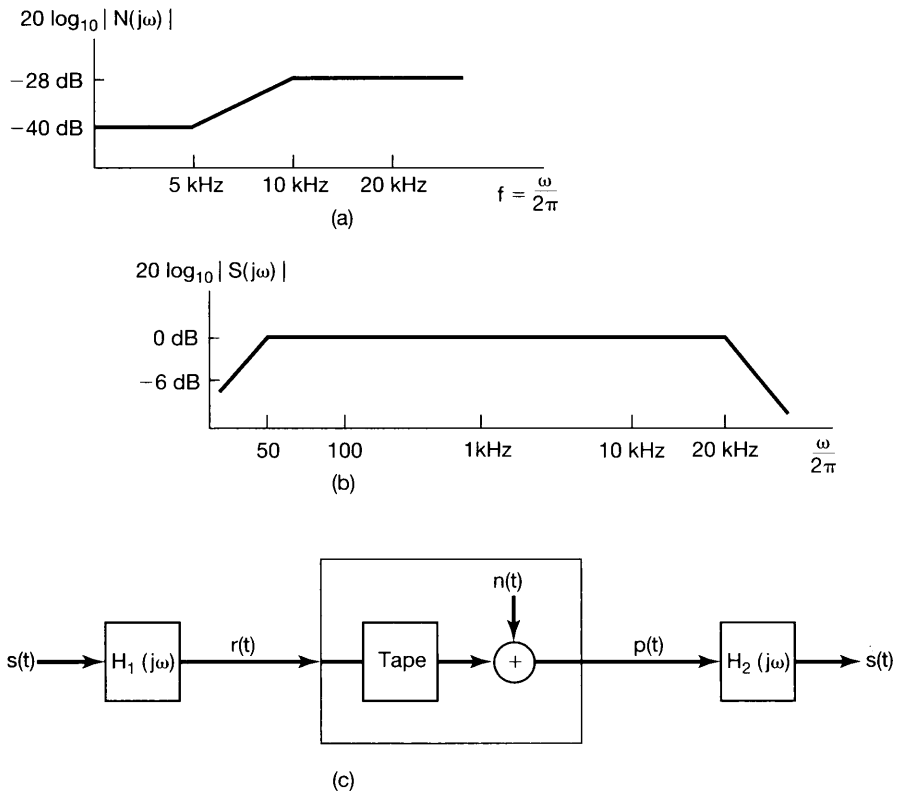


Figure P6.56

6.57. Show that if $h[n]$, the impulse response of a discrete-time LTI filter, is always greater than or equal to zero, the step response of the filter is a monotonically nondecreasing function and therefore will not have overshoot.

6.58. In the design of either analog or digital filters, we often approximate a specified magnitude characteristic without particular regard to the phase. For example, standard design techniques for lowpass and bandpass filters are typically derived from a consideration of the magnitude characteristics only.

In many filtering problems, one would ideally like the phase characteristics to be zero or linear. For causal filters, it is impossible to have zero phase. However, for many digital filtering applications, it is not necessary that the unit sample response of the filter be zero for $n < 0$ if the processing is not to be carried out in real time.

One technique commonly used in digital filtering when the data to be filtered are of finite duration and stored, for example, on a disc or magnetic tape is to process the data forward and then backward through the same filter.

Let $h[n]$ be the unit sample response of a causal filter with an arbitrary phase characteristic. Assume that $h[n]$ is real, and denote its Fourier transform by $H(e^{j\omega})$. Let $x[n]$ be the data that we want to filter. The filtering operation is performed as follows:

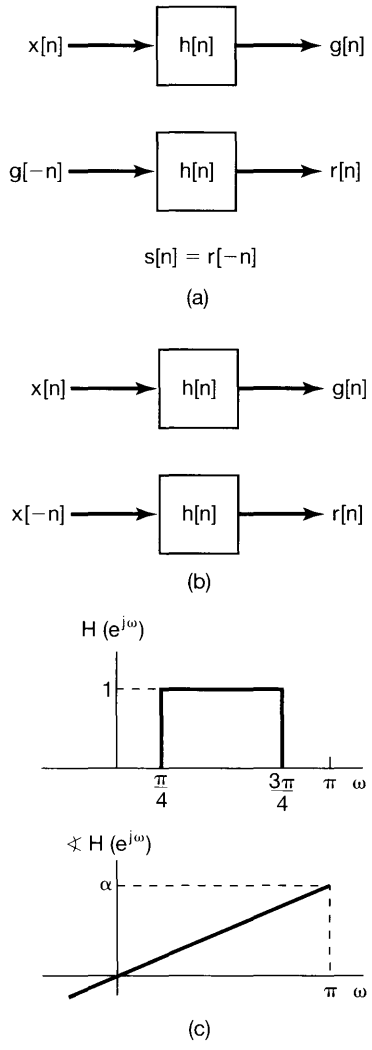


Figure P6.58

(a) *Method A:* Process $x[n]$ to get $s[n]$, as indicated in Figure P6.58(a).
   1. Determine the overall unit sample response $h_1[n]$ that relates $x[n]$ and $s[n]$, and show that it has zero phase characteristic.
   2. Determine $|H_1(e^{j\omega})|$ and express it in terms of $|H(e^{j\omega})|$ and $\measuredangle H(e^{j\omega})$.

(b) *Method B:* Process $x[n]$ through the filter $h[n]$ to get $g[n]$ [Figure P6.58(b)]. Also, process $x[n]$ backward through $h[n]$ to get $r[n]$. The output $y[n]$ is taken to be the sum of $g[n]$ and $r[-n]$. The composite set of operations can be represented by a filter with input $x[n]$, output $y[n]$, and unit sample response $h_2[n]$.
   1. Show that the composite filter $h_2[n]$ has zero phase characteristic.
   2. Determine $|H_2(e^{j\omega})|$, and express it in terms of $|H(e^{j\omega})|$ and $\measuredangle H(e^{j\omega})$.

(c) Suppose that we are given a sequence of finite duration on which we would like to perform bandpass, zero-phase filtering. Furthermore, assume that we

are given the bandpass filter $h[n]$ with frequency response as specified in Figure P6.58(c) and with magnitude characteristic that we desire, but with linear phase. To achieve zero phase, we could use either of the preceding methods, A or B. Determine and sketch $|H_1(e^{j\omega})|$ and $|H_2(e^{j\omega})|$. From these results, which method would you use to achieve the desired bandpass filtering operation? Explain why. More generally, if $h[n]$ has the desired magnitude, but a nonlinear phase characteristic, which method is preferable to achieve a zero phase characteristic?

6.59. Let $h_d[n]$ denote the unit sample response of a desired ideal system with frequency response $H_d(e^{j\omega})$, and let $h[n]$ denote the unit sample response for an FIR system of length N and with frequency response $H(e^{j\omega})$. In this problem, we show that a rectangular window of length N samples applied to $h_d[n]$ will produce a unit sample response $h[n]$ such that the mean square error

$$\epsilon^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |H_d(e^{j\omega}) - H(e^{j\omega})|^2 \, d\omega$$

is minimized.

(a) The error function $E(e^{j\omega}) = H_d(e^{j\omega}) - H(e^{j\omega})$ can be expressed as the power series

$$E(e^{j\omega}) = \sum_{n=-\infty}^{\infty} e[n]e^{-j\omega n}.$$

Find the coefficients $e[n]$ in terms of $h_d[n]$ and $h[n]$.

(b) Using Parseval's relation, express the mean square error ϵ^2 in terms of the coefficients $e[n]$.

(c) Show that for a unit sample response $h[n]$ of length N samples, ϵ^2 is minimized when

$$h[n] = \begin{cases} h_d[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}.$$

That is, simple truncation gives the best mean square approximation to a desired frequency response for a fixed value of N.

6.60. In Problem 6.50, we considered one specific criterion for determining the frequency response of a continuous-time filter that would recover a signal from the sum of two signals when their spectra overlapped in frequency. For the discrete-time case, develop the result corresponding to that obtained in part (b) of Problem 6.50.

6.61. In many situations we have available an analog or digital filter module, such as a basic hardware element or computer subroutine. By using the module repetitively or by combining identical modules, it is possible to implement a new filter with improved passband or stopband characteristics. In this and the next problem, we consider two procedures for doing just that. Although the discussion is phrased in terms of discrete-time filters, much of it applies directly to continuous-time filters as well.

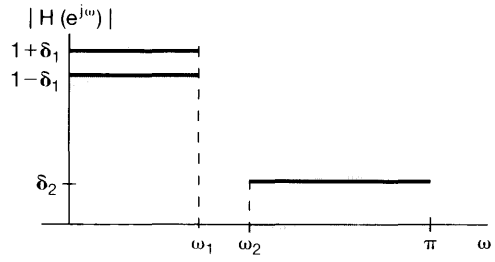


Figure P6.61

Consider a lowpass filter with frequency response $H(e^{j\omega})$ for which $|H(e^{j\omega})|$ falls within the tolerance limits shown in Figure P6.61; that is,

$$1 - \delta_1 \leq |H(e^{j\omega})| \leq 1 + \delta_1, \quad 0 \leq \omega \leq \omega_1,$$
$$0 \leq |H(e^{j\omega})| \leq \delta_2, \quad \omega_2 \leq \omega \leq \pi.$$

A new filter with frequency response $G(e^{j\omega})$ is formed by cascading two identical filters, both with frequency response $H(e^{j\omega})$.

(a) Determine the tolerance limits on $|G(e^{j\omega})|$.

(b) Assuming that $H(e^{j\omega})$ is a good approximation to a lowpass filter, so that $\delta_1 << 1$ and $\delta_2 << 1$, determine whether the passband ripple for $G(e^{j\omega})$ is larger or smaller than the passband ripple for $H(e^{j\omega})$. Also, determine whether the stopband ripple for $G(e^{j\omega})$ is larger or smaller than the stopband ripple for $H(e^{j\omega})$.

(c) If N identical filters with frequency response $H(e^{j\omega})$ are cascaded to obtain a new frequency response $G(e^{j\omega})$, then, again assuming that $\delta_1 << 1$ and $\delta_2 << 1$, determine the approximate tolerance limits on $|G(e^{j\omega})|$.

6.62. In Problem 6.61, we considered one method for using a basic filter module repetitively to implement a new filter with improved characteristics. Let us now consider an alternative approach, proposed by J. W. Tukey in the book, *Exploratory Data Analysis* (Reading, MA: Addison-Wesley Publishing Co., Inc., 1976). The procedure is shown in block diagram form in Figure P6.62(a).

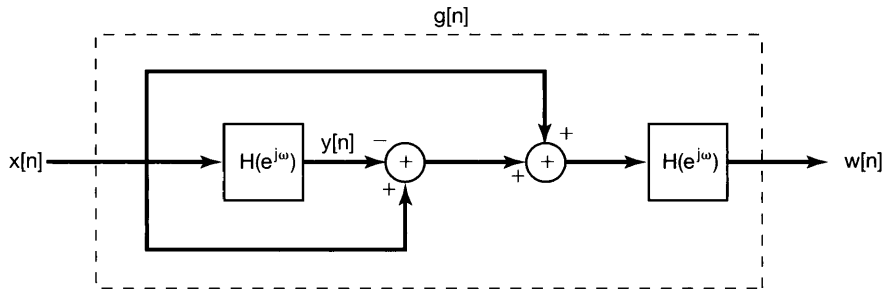
(a) Suppose that $H(e^{j\omega})$ is real and has a passband ripple of $\pm\delta_1$ and a stopband ripple of $\pm\delta_2$ (i.e., $H(e^{j\omega})$ falls within the tolerance limits indicated in Figure P6.62(b)). The frequency response $G(e^{j\omega})$ of the overall system in Figure P6.62(a) falls within the tolerance limits indicated in Figure P6.62(c). Determine A, B, C, and D in terms of δ_1 and δ_2.

(b) If $\delta_1 << 1$ and $\delta_2 << 1$, what is the approximate passband ripple and stopband ripple associated with $G(e^{j\omega})$? Indicate in particular whether the passband ripple for $G(e^{j\omega})$ is larger or smaller than the passband ripple for $H(e^{j\omega})$. Also, indicate whether the stopband ripple for $G(e^{j\omega})$ is larger or smaller than the stopband ripple for $H(e^{j\omega})$.

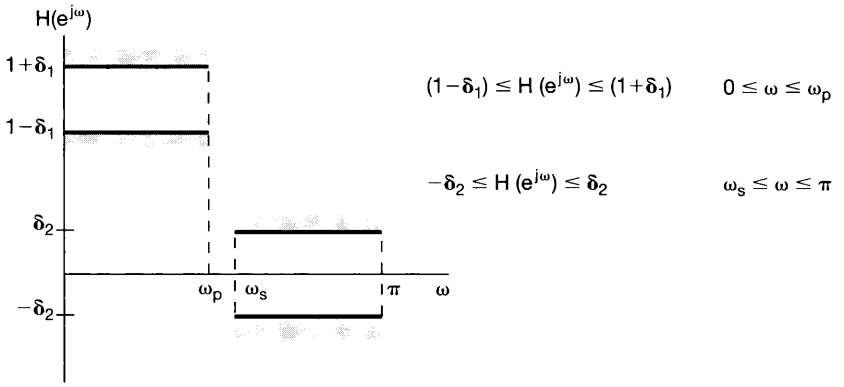
(c) In parts (a) and (b), we assumed that $H(e^{j\omega})$ is real. Now consider $H(e^{j\omega})$ to have the more general form

$$H(e^{j\omega}) = H_1(e^{j\omega})e^{j\theta(\omega)},$$

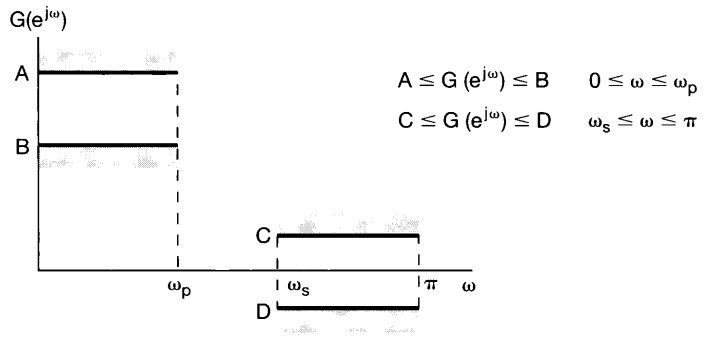
where $H_1(e^{j\omega})$ is real and $\theta(\omega)$ is an unspecified phase characteristic. If $|H(e^{j\omega})|$



(a)



(b)



(c)

Figure P6.62

is a reasonable approximation to an ideal lowpass filter, will $|G(e^{j\omega})|$ necessarily be a reasonable approximation to an ideal lowpass filter?

(d) Now assume that $H(e^{j\omega})$ is an FIR linear-phase lowpass filter, so that

$$H(e^{j\omega}) = H_1(e^{j\omega})e^{jM\omega},$$

where $H_1(e^{j\omega})$ is real and M is an integer. Show how to modify the system in Figure P6.62(a) so that the overall system will approximate a lowpass filter.

6.63. In the design of digital filters, we often choose a filter with a specified magnitude characteristic that has the shortest duration. That is, the impulse response, which is the inverse Fourier transform of the complex frequency spectrum, should be as narrow as possible. Assuming that $h[n]$ is real, we wish to show that if the phase $\theta(\omega)$ associated with the frequency response $H(e^{j\omega})$ is zero, the duration of the impulse response is minimal. Let the frequency response be expressed as

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)},$$

and let us consider the quantity

$$D = \sum_{n=-\infty}^{\infty} n^2 h^2[n] = \sum_{n=-\infty}^{\infty} (nh[n])^2$$

to be a measure of the duration of the associated impulse response $h[n]$.

(a) Using the derivative property of the Fourier transform and Parseval's relation, express D in terms of $H(e^{j\omega})$.

(b) By expressing $H(e^{j\omega})$ in terms of its magnitude $|H(e^{j\omega})|$ and phase $\theta(\omega)$, use your result from part (a) to show that D is minimized when $\theta(\omega) = 0$.

6.64. For a discrete-time filter to be *causal* and to have exactly linear phase, its impulse response must be of finite length and consequently the difference equation must be nonrecursive. To focus on the insight behind this statement, we consider a particular case, that of a linear phase characteristic for which the slope of the phase is an integer. Thus, the frequency response is assumed to be of the form

$$H(e^{j\omega}) = H_r(e^{j\omega})e^{-jM\omega}, \quad -\pi < \omega < \pi \tag{P6.64–1}$$

where $H_r(e^{j\omega})$ is real and even.

Let $h[n]$ denote the impulse response of the filter with frequency response $H(e^{j\omega})$ and let $h_r[n]$ denote the impulse response of the filter with frequency response $H_r(e^{j\omega})$.

(a) By using the appropriate properties in Table 5.1, show that:

1. $h_r[n] = h_r[-n]$ (i.e., $h_r[n]$ is symmetric about $n = 0$).

2. $h[n] = h_r[n - M]$.

(b) Using your result in part (a), show that with $H(e^{j\omega})$ of the form shown in eq. (P6.64–1), $h[n]$ is symmetric about $n = M$, that is,

$$h[M + n] = h[M - n]. \tag{P6.64–2}$$

(c) According to the result in part (b), the linear phase characteristic in eq. (P6.64–1) imposes a symmetry in the impulse response. Show that if $h[n]$ is causal and has the symmetry in eq. (P6.64–2), then

$$h[n] = 0, \qquad n < 0 \text{ and } n > 2M$$

(i.e., it must be of finite length).

6.65. For a class of discrete-time lowpass filters, known as Butterworth filters, the squared magnitude of the frequency response is given by

$$|B(e^{j\omega})|^2 = \frac{1}{1 + \left(\frac{\tan(\omega/2)}{\tan(\omega_c/2)}\right)^{2N}},$$

where ω_c is the cutoff frequency (which we shall take to be $\pi/2$) and N is the order of the filter (which we shall consider to be $N = 1$). Thus, we have

$$|B(e^{j\omega})|^2 = \frac{1}{1 + \tan^2(\omega/2)}.$$

(a) Using trigonometric identities, show that $|B(e^{j\omega})|^2 = \cos^2(\omega/2)$.

(b) Let $B(e^{j\omega}) = a\cos(\omega/2)$. For what complex values of a is $|B(e^{j\omega})|^2$ the same as in part (a)?

(c) Show that $B(e^{j\omega})$ from part (b) is the transfer function corresponding to a difference equation of the form

$$y[n] = \alpha x[n] + \beta x[n - \gamma].$$

Determine α, β, and γ.

6.66. In Figure P6.66(a) we show a discrete-time system consisting of a parallel combination of N LTI filters with impulse response $h_k[n]$, $k = 0, 1, \cdots, N - 1$. For any k, $h_k[n]$ is related to $h_0[n]$ by the expression

$$h_k[n] = e^{j(2\pi nk/N)}h_0[n].$$

(a) If $h_0[n]$ is an ideal discrete-time lowpass filter with frequency response $H_0(e^{j\omega})$ as shown in Figure P6.66(b), sketch the Fourier transforms of $h_1[n]$ and $h_{N-1}[n]$ for ω in the range $-\pi < \omega \leq +\pi$.

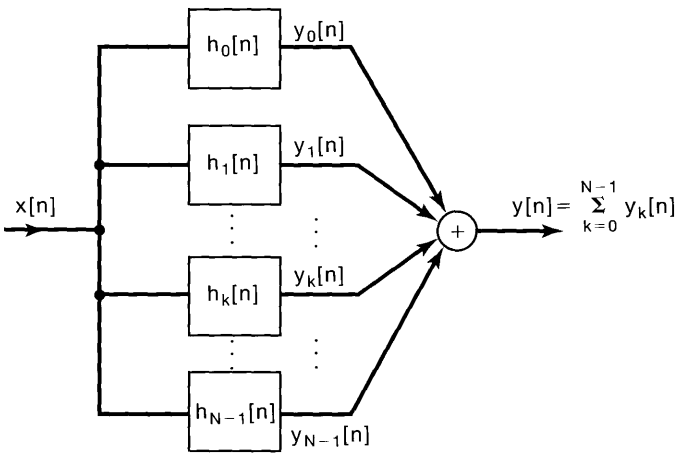


Figure P6.66a

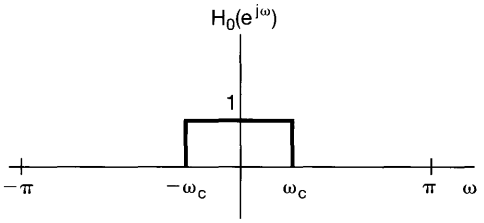


Figure P6.66b

(b) Determine the value of the cutoff frequency ω_c in Figure P6.66(b) in terms of N ($0 < \omega_c \leq \pi$) such that the system of Figure P6.66(a) is an identity system; that is, $y[n] = x[n]$ for all n and any input $x[n]$.

(c) Suppose that $h[n]$ is no longer restricted to be an ideal lowpass filter. If $h[n]$ denotes the impulse response of the entire system in Figure P6.66(a) with input $x[n]$ and output $y[n]$, then $h[n]$ can be expressed in the form

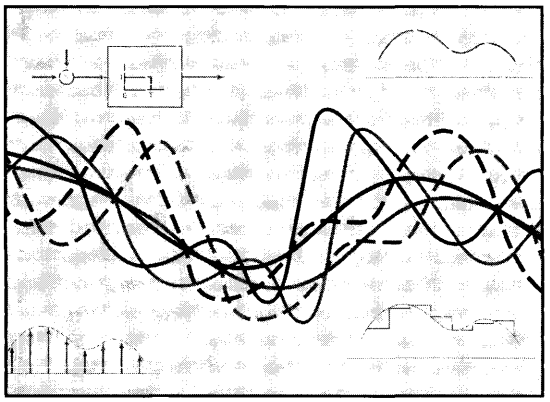
$$h[n] = r[n]h_0[n].$$

Determine and sketch $r[n]$.

(d) From your result of part (c), determine a necessary and sufficient condition on $h_0[n]$ to ensure that the overall system will be an identity system (i.e., such that for any input $x[n]$, the output $y[n]$ will be identical to $x[n]$). Your answer should not contain any sums.

7

SAMPLING



7.0 INTRODUCTION

Under certain conditions, a continuous-time signal can be completely represented by and recoverable from knowledge of its values, or *samples,* at points equally spaced in time. This somewhat surprising property follows from a basic result that is referred to as the *sampling theorem.* This theorem is extremely important and useful. It is exploited, for example, in moving pictures, which consist of a sequence of individual frames, each of which represents an instantaneous view (i.e., a sample in time) of a continuously changing scene. When these samples are viewed in sequence at a sufficiently fast rate, we perceive an accurate representation of the original continuously moving scene. As another example, printed pictures typically consist of a very fine grid of points, each corresponding to a sample of the spatially continuous scene represented in the picture. If the samples are sufficiently close together, the picture appears to be spatially continuous, although under a magnifying glass its representation in terms of samples becomes evident.

Much of the importance of the sampling theorem also lies in its role as a bridge between continuous-time signals and discrete-time signals. As we will see in this chapter, the fact that under certain conditions a continuous-time signal can be completely recovered from a sequence of its samples provides a mechanism for representing a continuous-time signal by a discrete-time signal. In many contexts, processing discrete-time signals is more flexible and is often preferable to processing continuous-time signals. This is due in large part to the dramatic development of digital technology over the past few decades, resulting in the availability of inexpensive, lightweight, programmable, and easily reproducible discrete-time systems. The concept of sampling, then, suggests an extremely attractive and widely employed method for using discrete-time system technology to implement continuous-time systems and process continuous-time signals: We exploit sampling to

convert a continuous-time signal to a discrete-time signal, process the discrete-time signal using a discrete-time system, and then convert back to continuous time.

In the following discussion, we introduce and develop the concept of sampling and the process of reconstructing a continuous-time signal from its samples. In this discussion, we both identify the conditions under which a continuous-time signal can be exactly reconstructed from its samples and examine the consequences when these conditions are not satisfied. Following this, we explore the processing of continuous-time signals that have been converted to discrete-time signals through sampling. Finally, we examine the sampling of discrete-time signals and the related concepts of decimation and interpolation.

7.1 REPRESENTATION OF A CONTINUOUS-TIME SIGNAL BY ITS SAMPLES: THE SAMPLING THEOREM

In general, in the absence of any additional conditions or information, we would not expect that a signal could be uniquely specified by a sequence of equally spaced samples. For example, in Figure 7.1 we illustrate three different continuous-time signals, all of which have identical values at integer multiples of T; that is,

$$x_1(kT) = x_2(kT) = x_3(kT).$$

Clearly, an infinite number of signals can generate a given set of samples. As we will see, however, if a signal is band limited—i.e., if its Fourier transform is zero outside a finite band of frequencies—and if the samples are taken sufficiently close together in relation to the highest frequency present in the signal, then the samples *uniquely* specify the signal, and we can reconstruct it perfectly. This result, known as the *sampling theorem,* is of profound importance in the practical application of the methods of signal and system analysis.

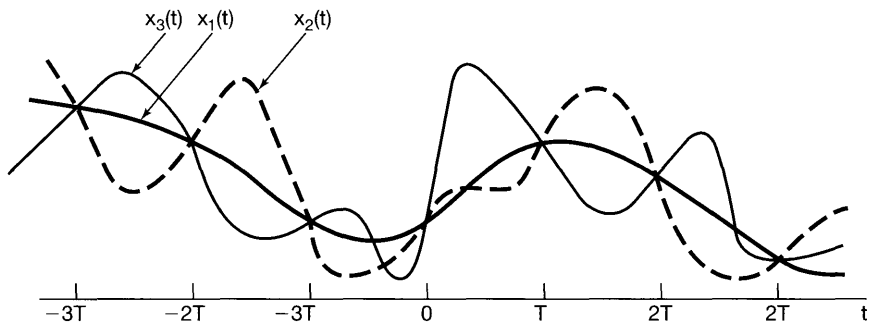


Figure 7.1 Three continuous-time signals with identical values at integer multiples of T.

7.1.1 Impulse-Train Sampling

In order to develop the sampling theorem, we need a convenient way in which to represent the sampling of a continuous-time signal at regular intervals. A useful way to do this is through the use of a periodic impulse train multiplied by the continuous-time signal $x(t)$ that we wish to sample. This mechanism, known as *impulse-train sampling*, is depicted in Figure 7.2. The periodic impulse train $p(t)$ is referred to as the *sampling function*, the period T as the *sampling period*, and the fundamental frequency of $p(t)$, $\omega_s = 2\pi/T$, as the *sampling frequency*. In the time domain,

$$x_p(t) = x(t)p(t), \tag{7.1}$$

where

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT). \tag{7.2}$$

Because of the sampling property of the unit impulse discussed in Section 1.4.2, we know that multiplying $x(t)$ by a unit impulse samples the value of the signal at the point at which the impulse is located; i.e., $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$. Applying this to eq. (7.1), we see, as illustrated in Figure 7.2, that $x_p(t)$ is an impulse train with the amplitudes of

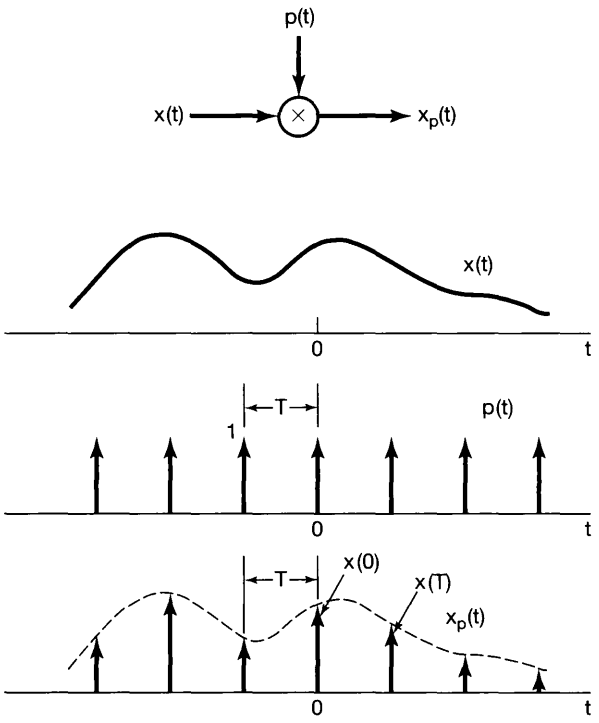


Figure 7.2 Impulse-train sampling.

the impulses equal to the samples of $x(t)$ at intervals spaced by T; that is,

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT). \tag{7.3}$$

From the multiplication property (Section 4.5), we know that

$$X_p(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\theta)P(j(\omega - \theta))d\theta. \tag{7.4}$$

and from Example 4.8,

$$P(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s). \tag{7.5}$$

Since convolution with an impulse simply shifts a signal [i.e., $X(j\omega) * \delta(\omega - \omega_0) = X(j(\omega - \omega_0))$], it follows that

$$X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s)). \tag{7.6}$$

That is, $X_p(j\omega)$ is a periodic function of ω consisting of a superposition of shifted replicas of $X(j\omega)$, scaled by $1/T$, as illustrated in Figure 7.3. In Figure 7.3(c), $\omega_M < (\omega_s - \omega_M)$, or equivalently, $\omega_s > 2\omega_M$, and thus there is no overlap between the shifted replicas of $X(j\omega)$, whereas in Figure 7.3(d), with $\omega_s < 2\omega_M$, there is overlap. For the case illustrated in Figure 7.3(c), $X(j\omega)$ is faithfully reproduced at integer multiples of the sampling frequency. Consequently, if $\omega_s > 2\omega_M$, $x(t)$ can be recovered exactly from $x_p(t)$ by means of

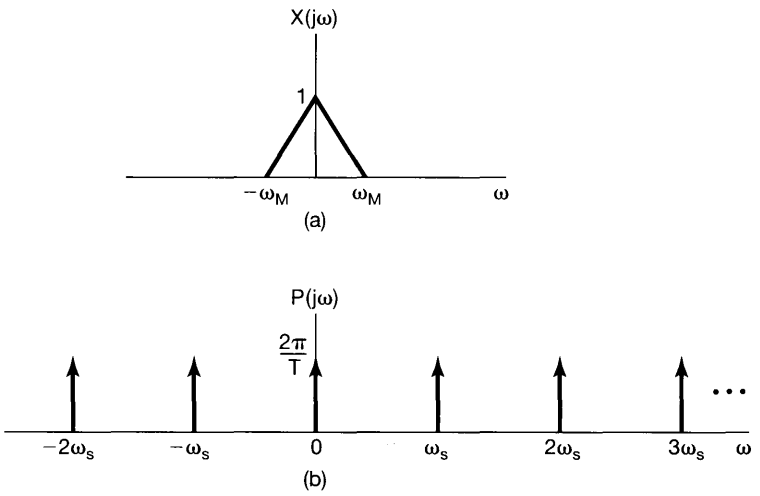


Figure 7.3 Effect in the frequency domain of sampling in the time domain: (a) spectrum of original signal; (b) spectrum of sampling function;

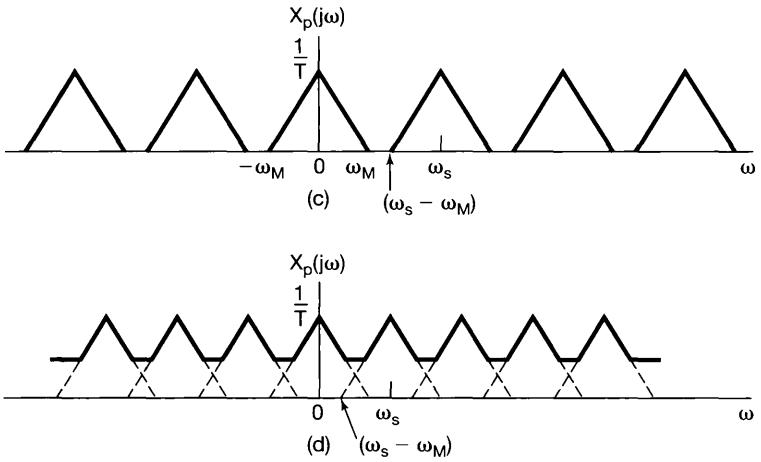


Figure 7.3 *Continued* (c) spectrum of sampled signal with $\omega_s > 2\omega_M$; (d) spectrum of sampled signal with $\omega_s < 2\omega_M$.

a lowpass filter with gain T and a cutoff frequency greater than ω_M and less than $\omega_s - \omega_M$, as indicated in Figure 7.4. This basic result, referred to as the *sampling theorem,* can be stated as follows: [1]

Sampling Theorem:
Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$, if

$$\omega_s > 2\omega_M,$$

where

$$\omega_s = \frac{2\pi}{T}.$$

Given these samples, we can reconstruct $x(t)$ by generating a periodic impulse train in which successive impulses have amplitudes that are successive sample values. This impulse train is then processed through an ideal lowpass filter with gain T and cutoff frequency greater than ω_M and less than $\omega_s - \omega_M$. The resulting output signal will exactly equal $x(t)$.

[1]The important and elegant sampling theorem was available for many years in a variety of forms in the mathematics literature. See, for example, J. M. Whittaker, "Interpolatory Function Theory," (New York: Stecher-Hafner Service Agency, 1964), chap. 4. It did not appear explicitly in the literature of communication theory until the publication in 1949 of the classic paper by Shannon entitled "Communication in the Presence of Noise" (*Proceedings of the IRE,* January 1949, pp. 10–21). However, H. Nyquist in 1928 and D. Gabor in 1946 had pointed out, based on the use of the Fourier Series, that $2TW$ numbers are sufficient to represent a function of duration T and highest frequency W. [H. Nyquist, "Certain Topics in Telegraph Transmission Theory," *AIEE Transactions,* 1928, p. 617; D. Gabor, "Theory of Communication,"*Journal of IEE* 93, no. 26 (1946), p. 429.]

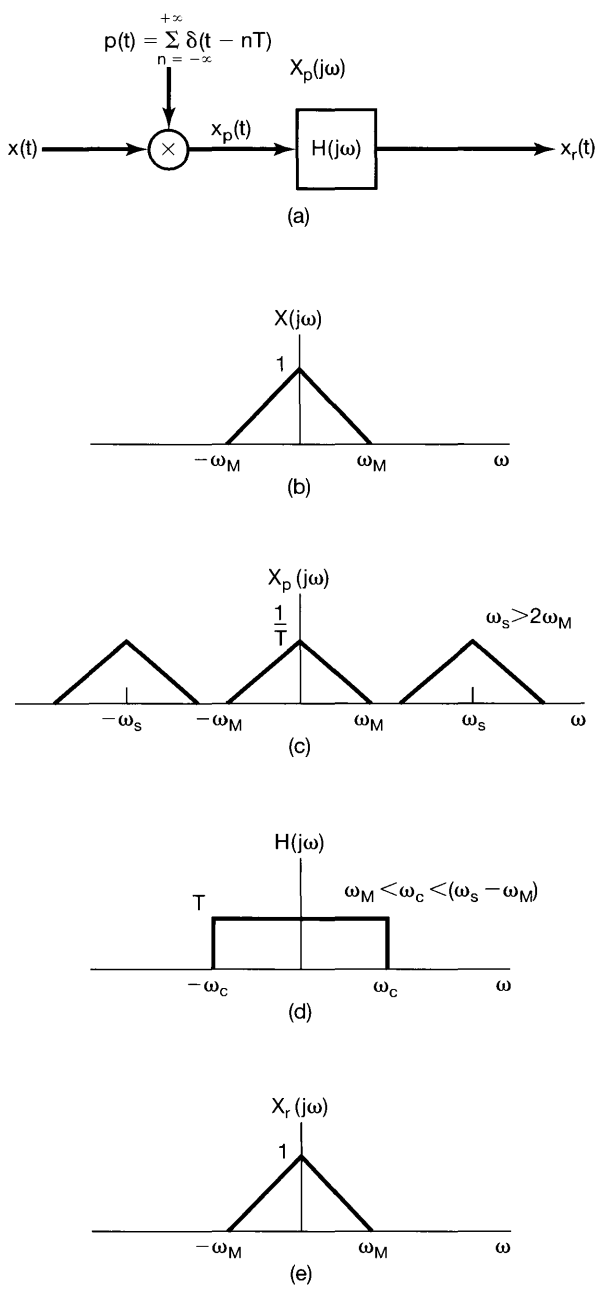


Figure 7.4 Exact recovery of a continuous-time signal from its samples using an ideal lowpass filter: (a) system for sampling and reconstruction; (b) representative spectrum for $x(t)$; (c) corresponding spectrum for $x_p(t)$; (d) ideal lowpass filter to recover $X(j\omega)$ from $X_p(j\omega)$; (e) spectrum of $x_r(t)$.

The frequency $2\omega_M$, which, under the sampling theorem, must be exceeded by the sampling frequency, is commonly referred to as the *Nyquist rate.*[2]

As discussed in Chapter 6, ideal filters are generally not used in practice for a variety of reasons. In any practical application, the ideal lowpass filter in Figure 7.4 would be

[2]The frequency ω_M corresponding to one-half the Nyquist rate is often referred to as the *Nyquist frequency.*

replaced by a nonideal filter $H(j\omega)$ that approximated the desired frequency characteristic accurately enough for the problem of interest (i.e., $H(j\omega) \simeq 1$ for $|\omega| < \omega_M$, and $H(j\omega) \simeq 0$ for $|\omega| > \omega_S - \omega_M$). Obviously, any such approximation in the lowpass filtering stage will lead to some discrepancy between $x(t)$ and $x_r(t)$ in Figure 7.4 or, equivalently, between $X(j\omega)$ and $X_r(j\omega)$. The particular choice of nonideal filter is then dictated by the acceptable level of distortion for the application under consideration. For convenience and to emphasize basic principles such as the sampling theorem, we will regularly assume the availability and use of ideal filters throughout this and the next chapter, with the understanding that in practice such a filter must be replaced by a nonideal filter designed to approximate the ideal characteristics accurately enough for the problem at hand.

7.1.2 Sampling with a Zero-Order Hold

The sampling theorem, which is most easily explained in terms of impulse-train sampling, establishes the fact that a band-limited signal is uniquely represented by its samples. In practice, however, narrow, large-amplitude pulses, which approximate impulses, are also relatively difficult to generate and transmit, and it is often more convenient to generate the sampled signal in a form referred to as a *zero-order hold.* Such a system samples $x(t)$ at a given instant and holds that value until the next instant at which a sample is taken, as illustrated in Figure 7.5. The reconstruction of $x(t)$ from the output of a zero-order hold can again be carried out by lowpass filtering. However, in this case, the required filter no longer has constant gain in the passband. To develop the required filter characteristic, we first note that the output $x_0(t)$ of the zero-order hold can in principle be generated by impulse-train sampling followed by an LTI system with a rectangular impulse response, as depicted in Figure 7.6. To reconstruct $x(t)$ from $x_0(t)$, we consider processing $x_0(t)$ with an LTI system with impulse response $h_r(t)$ and frequency response $H_r(j\omega)$. The cascade of this system with the system of Figure 7.6 is shown in Figure 7.7, where we wish to specify $H_r(j\omega)$ so that $r(t) = x(t)$. Comparing the system in Figure 7.7 with that in Figure 7.4, we see that $r(t) = x(t)$ if the cascade combination of $h_0(t)$ and $h_r(t)$ is the ideal lowpass filter $H(j\omega)$ used in Figure 7.4. Since, from Example 4.4 and the time-shifting property in Section 4.3.2,

$$H_0(j\omega) = e^{-j\omega T/2}\left[\frac{2\sin(\omega T/2)}{\omega}\right], \tag{7.7}$$

this requires that

$$H_r(j\omega) = \frac{e^{j\omega T/2}H(j\omega)}{\dfrac{2\sin(\omega T/2)}{\omega}}. \tag{7.8}$$

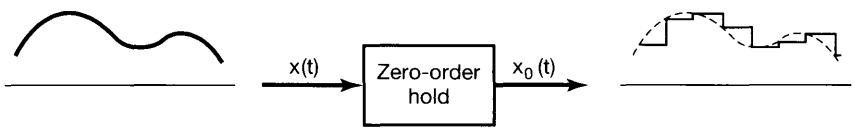


Figure 7.5 Sampling utilizing a zero-order hold.

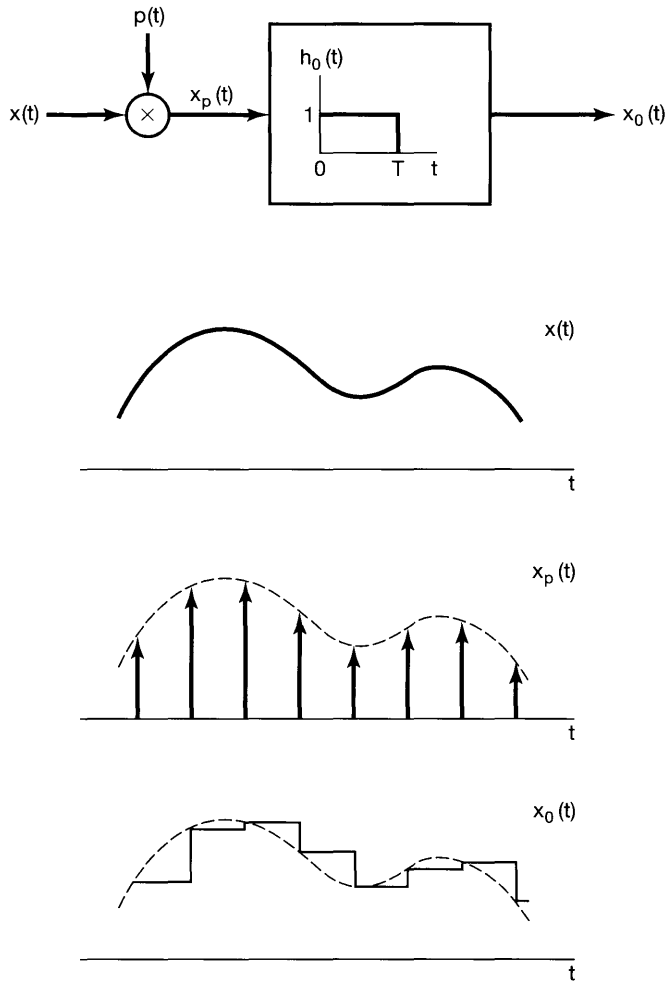


Figure 7.6 Zero-order hold as impulse-train sampling followed by an LTI system with a rectangular impulse response.

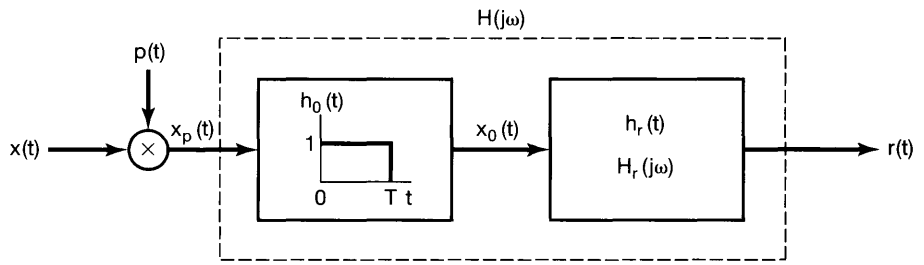


Figure 7.7 Cascade of the representation of a zero-order hold (Figure 7.6) with a reconstruction filter.

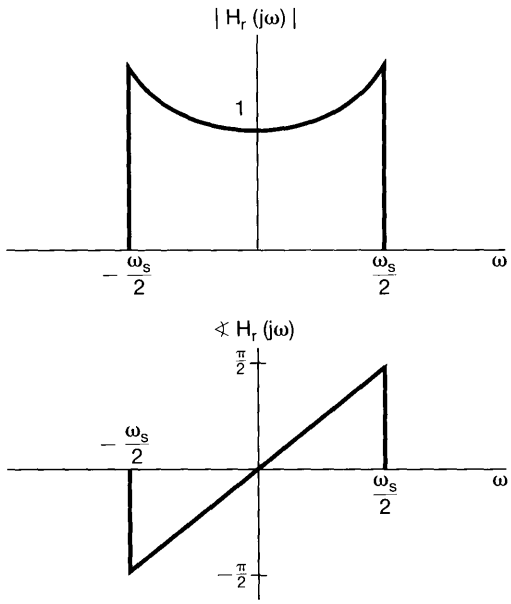


Figure 7.8 Magnitude and phase for the reconstruction filter for a zero-order hold.

For example, with the cutoff frequency of $H(j\omega)$ equal to $\omega_s/2$, the ideal magnitude and phase for the reconstruction filter following a zero-order hold is that shown in Figure 7.8.

Once again, in practice the frequency response in eq. (7.8) cannot be exactly realized, and thus an adequate approximation to it must be designed. In fact, in many situations, the output of the zero-order hold is considered an adequate approximation to the original signal by itself, without any additional lowpass filtering, and in essence represents a possible, although admittedly very coarse, interpolation between the sample values. Alternatively, in some applications, we may wish to perform some smoother interpolation between sample values. In the next section, we explore in more detail the general concept of interpreting the reconstruction of a signal from its samples as a process of interpolation.

7.2 RECONSTRUCTION OF A SIGNAL FROM ITS SAMPLES USING INTERPOLATION

Interpolation, that is, the fitting of a continuous signal to a set of sample values, is a commonly used procedure for reconstructing a function, either approximately or exactly, from samples. One simple interpolation procedure is the zero-order hold discussed in Section 7.1. Another useful form of interpolation is *linear interpolation,* whereby adjacent sample points are connected by a straight line, as illustrated in Figure 7.9. In more

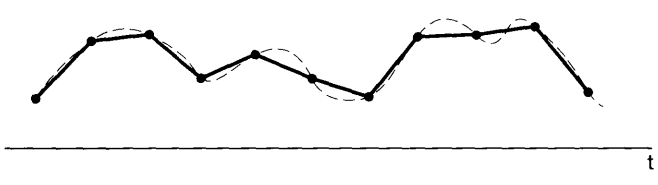


Figure 7.9 Linear interpolation between sample points. The dashed curve represents the original signal and the solid curve the linear interpolation.

complicated interpolation formulas, sample points may be connected by higher order polynomials or other mathematical functions.

As we have seen in Section 7.1, for a band-limited signal, if the sampling instants are sufficiently close, then the signal can be reconstructed exactly; i.e., through the use of a lowpass filter, exact interpolation can be carried out between the sample points. The interpretation of the reconstruction of $x(t)$ as a process of interpolation becomes evident when we consider the effect in the time domain of the lowpass filter in Figure 7.4. In particular, the output is

$$x_r(t) = x_p(t) * h(t)$$

or, with $x_p(t)$ given by eq. (7.3),

$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT)h(t - nT). \tag{7.9}$$

Equation (7.9) describes how to fit a continuous curve between the sample points $x(nT)$ and consequently represents an interpolation formula. For the ideal lowpass filter $H(j\omega)$ in Figure 7.4,

$$h(t) = \frac{\omega_c T \sin(\omega_c t)}{\pi \omega_c t}, \tag{7.10}$$

so that

$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT)\frac{\omega_c T}{\pi}\frac{\sin(\omega_c(t - nT))}{\omega_c(t - nT)}. \tag{7.11}$$

The reconstruction according to eq. (7.11) with $\omega_c = \omega_s/2$ is illustrated in Figure 7.10. Figure 7.10(a) represents the original band-limited signal $x(t)$, and Figure 7.10(b) represents $x_p(t)$, the impulse train of samples. In Figure 7.10(c), the superposition of the individual terms in eq. (7.11) is illustrated.

Interpolation using the impulse response of an ideal lowpass filter as in eq. (7.11) is commonly referred to as *band-limited interpolation,* since it implements exact reconstruction if $x(t)$ is band limited and the sampling frequency satisfies the conditions of the sampling theorem. As we have indicated, in many cases it is preferable to use a less accurate, but simpler, filter or, equivalently, a simpler interpolating function than the function in eq. (7.10). For example, the zero-order hold can be viewed as a form of interpolation between sample values in which the interpolating function $h(t)$ is the impulse response $h_0(t)$ depicted in Figure 7.6. In that sense, with $x_0(t)$ in the figure corresponding to the approximation to $x(t)$, the system $h_0(t)$ represents an approximation to the ideal lowpass filter required for the exact interpolation. Figure 7.11 shows the magnitude of the transfer function of the zero-order-hold interpolating filter, superimposed on the desired transfer function of the exact interpolating filter.

Both from Figure 7.11 and from Figure 7.6, we see that the zero-order hold is a very rough approximation, although in some cases it is sufficient. For example, if additional

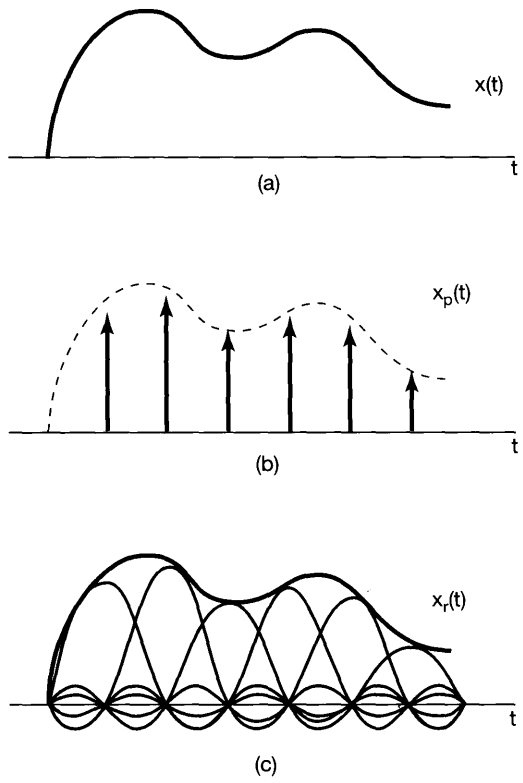


Figure 7.10 Ideal band-limited interpolation using the sinc function: (a) band-limited signal $x(t)$; (b) impulse train of samples of $x(t)$; (c) ideal band-limited interpolation in which the impulse train is replaced by a superposition of sinc functions [eq. (7.11)].

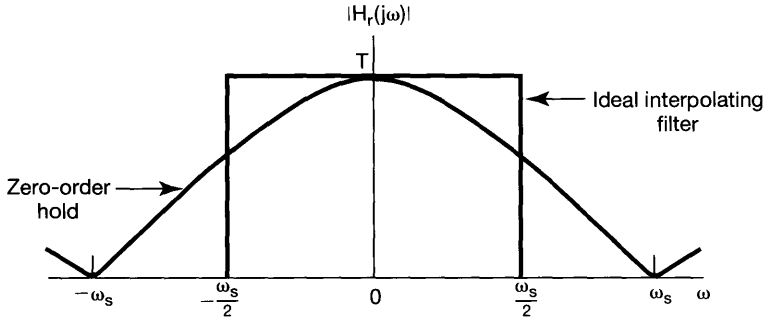


Figure 7.11 Transfer function for the zero-order hold and for the ideal interpolating filter.

lowpass filtering is naturally applied in a given application, it will tend to improve the overall interpolation. This is illustrated in the case of pictures in Figure 7.12. Figure 7.12(a) shows pictures with impulse sampling (i.e., sampling with spatially narrow pulses). Figure 7.12(b) is the result of applying a two-dimensional zero-order hold to Figure 7.12(a), with a resulting mosaic effect. However, the human visual system inherently imposes lowpass filtering, and consequently, when viewed at a distance, the discontinuities in the mosaic are smoothed. For example, in Figure 7.12(c) a

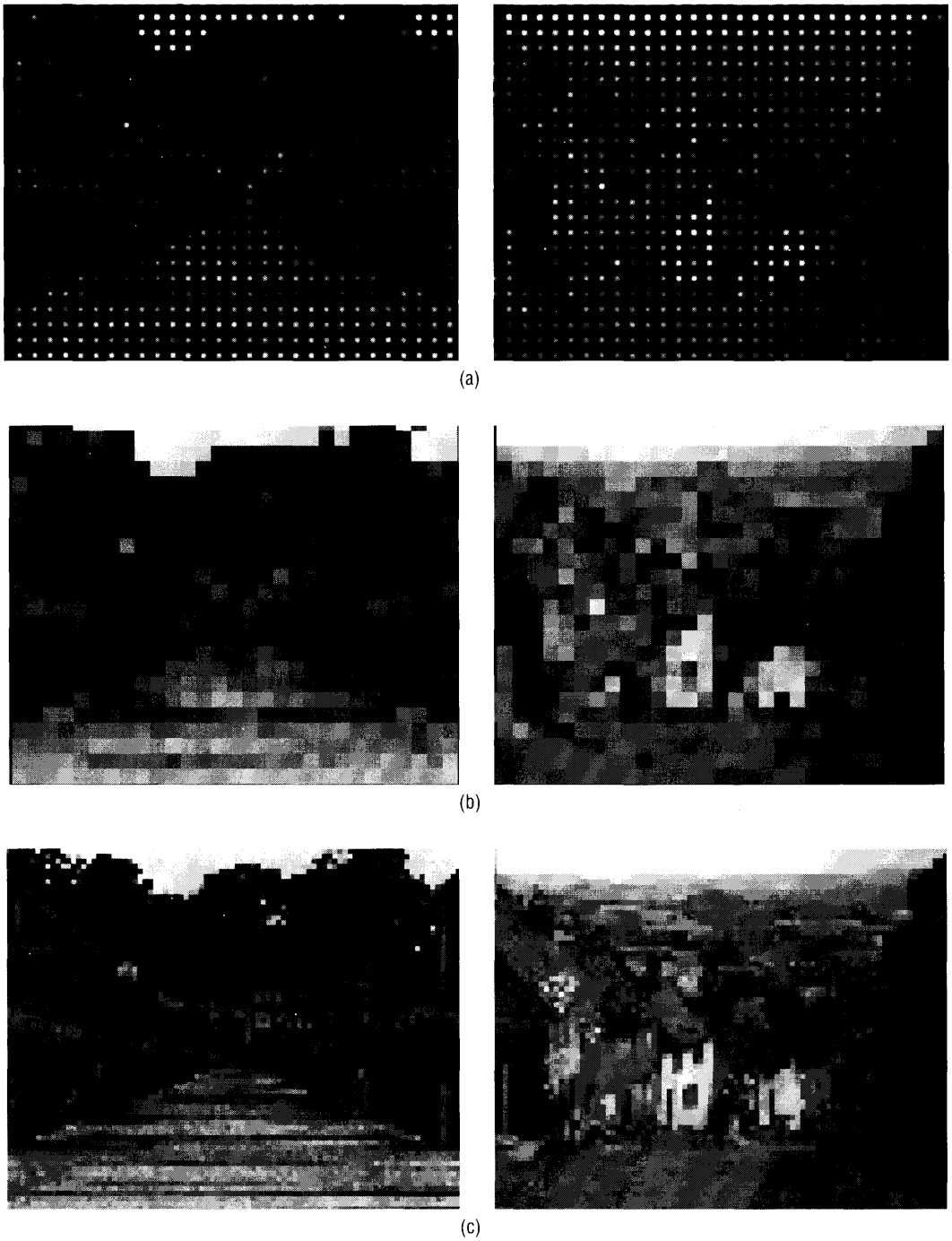


Figure 7.12 (a) The original pictures of Figures 6.2(a) and (g) with impulse sampling; (b) zero-order hold applied to the pictures in (a). The visual system naturally introduces lowpass filtering with a cutoff frequency that decreases with distance. Thus, when viewed at a distance, the discontinuities in the mosaic in Figure 7.12(b) are smoothed; (c) result of applying a zero-order hold after impulse sampling with one-fourth the horizontal and vertical spacing used in (a) and (b).

zero-order hold is again used, but here the sample spacing in each direction is one-fourth that in Figure 7.12(a). With normal viewing, considerable lowpass filtering is naturally applied, although the mosaic effect is still evident.

If the crude interpolation provided by the zero-order hold is insufficient, we can use a variety of smoother interpolation strategies, some of which are known collectively as *higher order holds.* In particular, the zero-order hold produces an output signal, as in Figure 7.5, that is discontinuous. In contrast, linear interpolation, as illustrated in Figure 7.9, yields reconstructions that are continuous, although with discontinous derivatives due to the changes in slope at the sample points. Linear interpolation, which is sometimes referred to as a first-order hold, can also be viewed as interpolation in the form of Figure 7.4 and eq. (7.9) with $h(t)$ triangular, as illustrated in Figure 7.13. The associated transfer function is also shown in the figure and is

$$H(j\omega) = \frac{1}{T}\left[\frac{\sin(\omega T/2)}{\omega/2}\right]^2. \tag{7.12}$$

The transfer function of the first-order hold is shown superimposed on the transfer function for the ideal interpolating filter. Figure 7.14 corresponds to the same pictures as those in Figure 7.12(b), but with a first-order hold applied to the sampled picture. In an analogous fashion, we can define second- and higher order holds that produce reconstructions with a higher degree of smoothness. For example, the output of a second-order hold provides an interpolation of the sample values that is continuous and has a continuous first derivative and discontinuous second derivative.

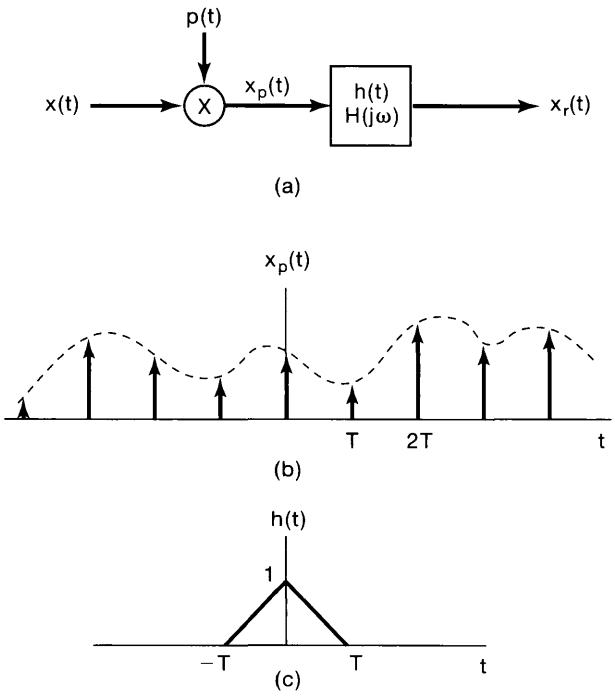


Figure 7.13 Linear interpolation (first-order hold) as impulse-train sampling followed by convolution with a triangular impulse response: (a) system for sampling and reconstruction; (b) impulse train of samples; (c) impulse response representing a first-order hold;

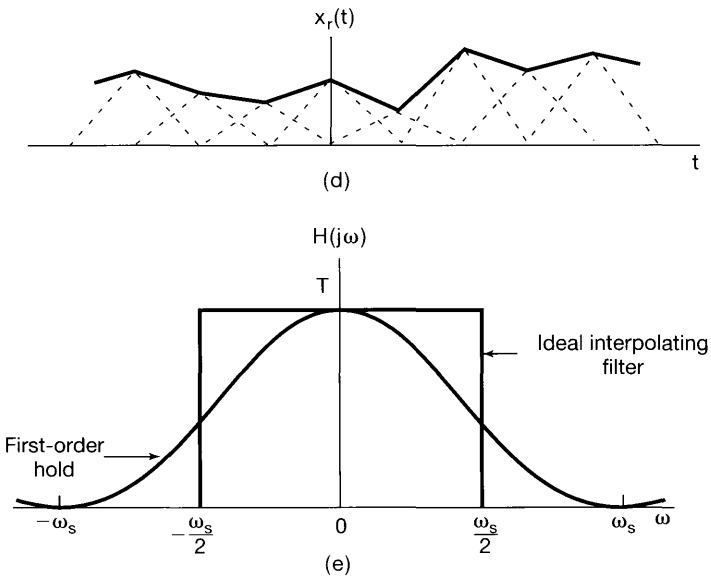


Figure 7.13 *Continued* (d) first-order hold applied to the sampled signal; (e) comparison of transfer function of ideal interpolating filter and first-order hold.

7.3 THE EFFECT OF UNDERSAMPLING: ALIASING

In previous sections in this chapter, it was assumed that the sampling frequency was sufficiently high that the conditions of the sampling theorem were met. As illustrated in Figure 7.3, with $\omega_s > 2\omega_M$, the spectrum of the sampled signal consists of scaled replications of the spectrum of $x(t)$, and this forms the basis for the sampling theorem. When

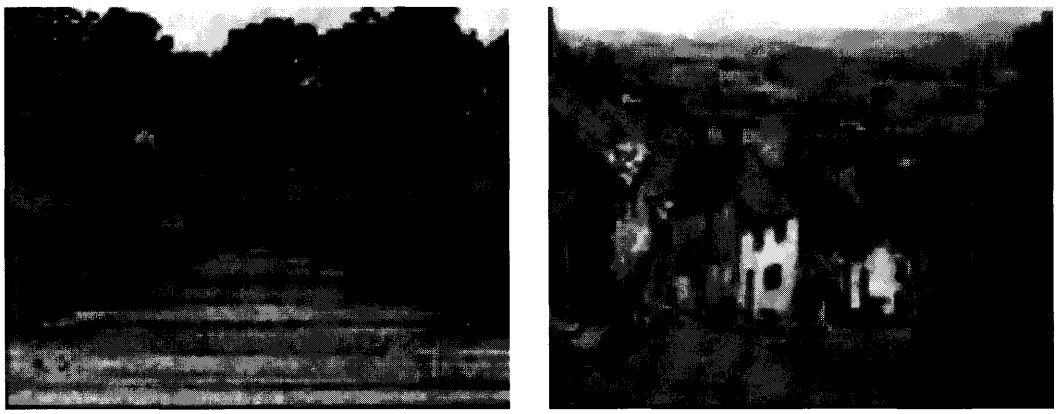


Figure 7.14 Result of applying a first-order hold rather than a zero-order hold after impulse sampling with one-third the horizontal and vertical spacing used in Figures 7.12(a) and (b).

$\omega_s < 2\omega_M$, $X(j\omega)$, the spectrum of $x(t)$, is no longer replicated in $X_p(j\omega)$ and thus is no longer recoverable by lowpass filtering. This effect, in which the individual terms in eq. (7.6) overlap, is referred to as *aliasing,* and in this section we explore its effect and consequences.

Clearly, if the system of Figure 7.4 is applied to a signal with $\omega_s < 2\omega_M$, the reconstructed signal $x_r(t)$ will no longer be equal to $x(t)$. However, as explored in Problem 7.25, the original signal and the signal $x_r(t)$ that is reconstructed using bandlimited interpolation will always be equal at the sampling instants; that is, for any choice of ω_s,

$$x_r(nT) = x(nT), \qquad n = 0, \pm 1, \pm 2, \ldots. \tag{7.13}$$

Some insight into the relationship between $x(t)$ and $x_r(t)$ when $\omega_s < 2\omega_M$ is provided by considering in more detail the comparatively simple case of a sinusoidal signal. Thus, let

$$x(t) = \cos \omega_0 t, \tag{7.14}$$

with Fourier transform $X(j\omega)$ as indicated in Figure 7.15(a). In this figure, we have graphically distinguished the impulse at ω_0 from that at $-\omega_0$ for convenience. Let us consider $X_p(j\omega)$, the spectrum of the sampled signal, and focus in particular on the effect of a change in the frequency ω_0 with the sampling frequency ω_s fixed. In Figures 7.15(b)–(e), we illustrate $X_p(j\omega)$ for several values of ω_0. Also indicated by a dashed line is the passband of the lowpass filter of Figure 7.4 with $\omega_c = \omega_s/2$. Note that no aliasing occurs in (b) and (c), since $\omega_0 < \omega_s/2$, whereas aliasing does occur in (d) and (e). For each of the four cases, the lowpass filtered output $x_r(t)$ is given as follows:

(a) $\omega_0 = \dfrac{\omega_s}{6}$; $\qquad x_r(t) = \cos \omega_0 t = x(t)$

(b) $\omega_0 = \dfrac{2\omega_s}{6}$; $\qquad x_r(t) = \cos \omega_0 t = x(t)$

(c) $\omega_0 = \dfrac{4\omega_s}{6}$; $\qquad x_r(t) = \cos(\omega_s - \omega_0)t \neq x(t)$

(d) $\omega_0 = \dfrac{5\omega_s}{6}$; $\qquad x_r(t) = \cos(\omega_s - \omega_0)t \neq x(t)$.

When aliasing occurs, the original frequency ω_0 takes on the identity of a lower frequency, $\omega_s - \omega_0$. For $\omega_s/2 < \omega_0 < \omega_s$, as ω_0 increases relative to ω_s, the output frequency $\omega_s - \omega_0$ decreases. When $\omega_s = \omega_0$, for example, the reconstructed signal is a constant. This is consistent with the fact that, when sampling once per cycle, the samples are all equal and would be identical to those obtained by sampling a constant signal ($\omega_0 = 0$). In Figure 7.16, we have depicted, for each of the four cases in Figure 7.15, the signal $x(t)$, its samples, and the reconstructed signal $x_r(t)$. From the figure, we can see how the lowpass

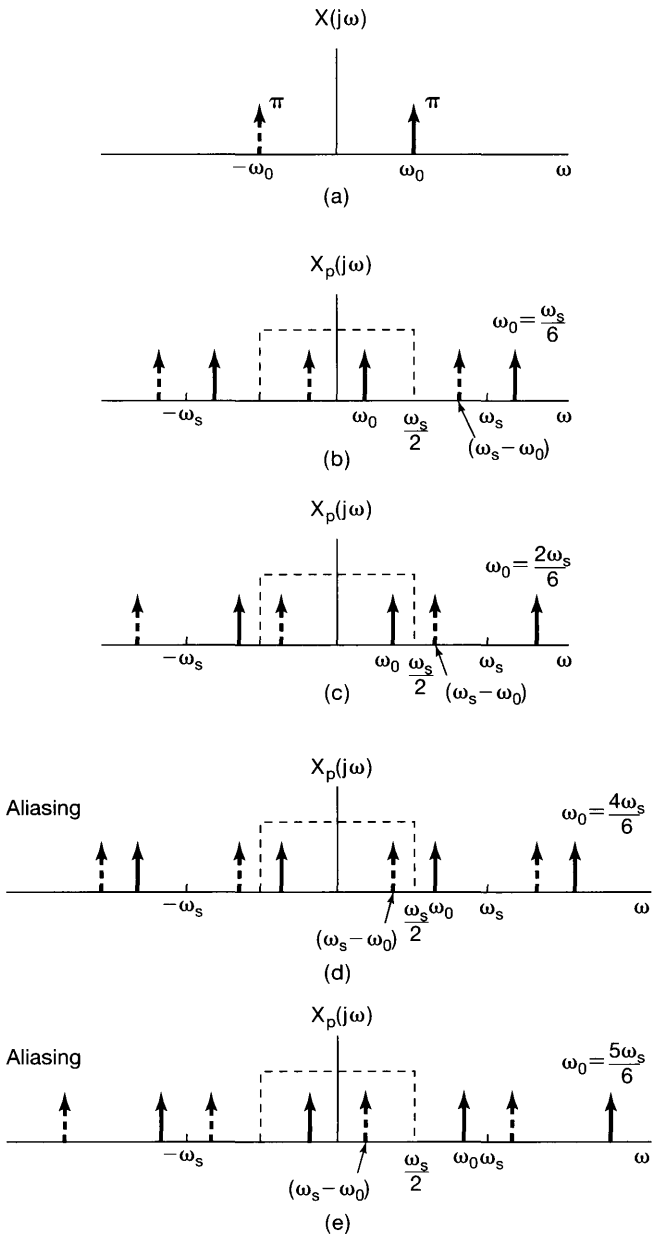


Figure 7.15 Effect in the frequency domain of oversampling and undersampling: (a) spectrum of original sinusoidal signal; (b), (c) spectrum of sampled signal with $\omega_s > 2\omega_0$; (d), (e) spectrum of sampled signal with $\omega_s < 2\omega_0$. As we increase ω_0 in moving from (b) through (d), the impulses drawn with solid lines move to the right, while the impulses drawn with dashed lines move to the left. In (d) and (e), these impulses have moved sufficiently that there is a change in the ones falling within the passband of the ideal lowpass filter.

filter interpolates between the samples, in particular always fitting a sinusoid of frequency less than $\omega_s/2$ to the samples of $x(t)$.

As a variation on the preceding examples, consider the signal

$$x(t) = \cos(\omega_0 t + \phi). \tag{7.15}$$

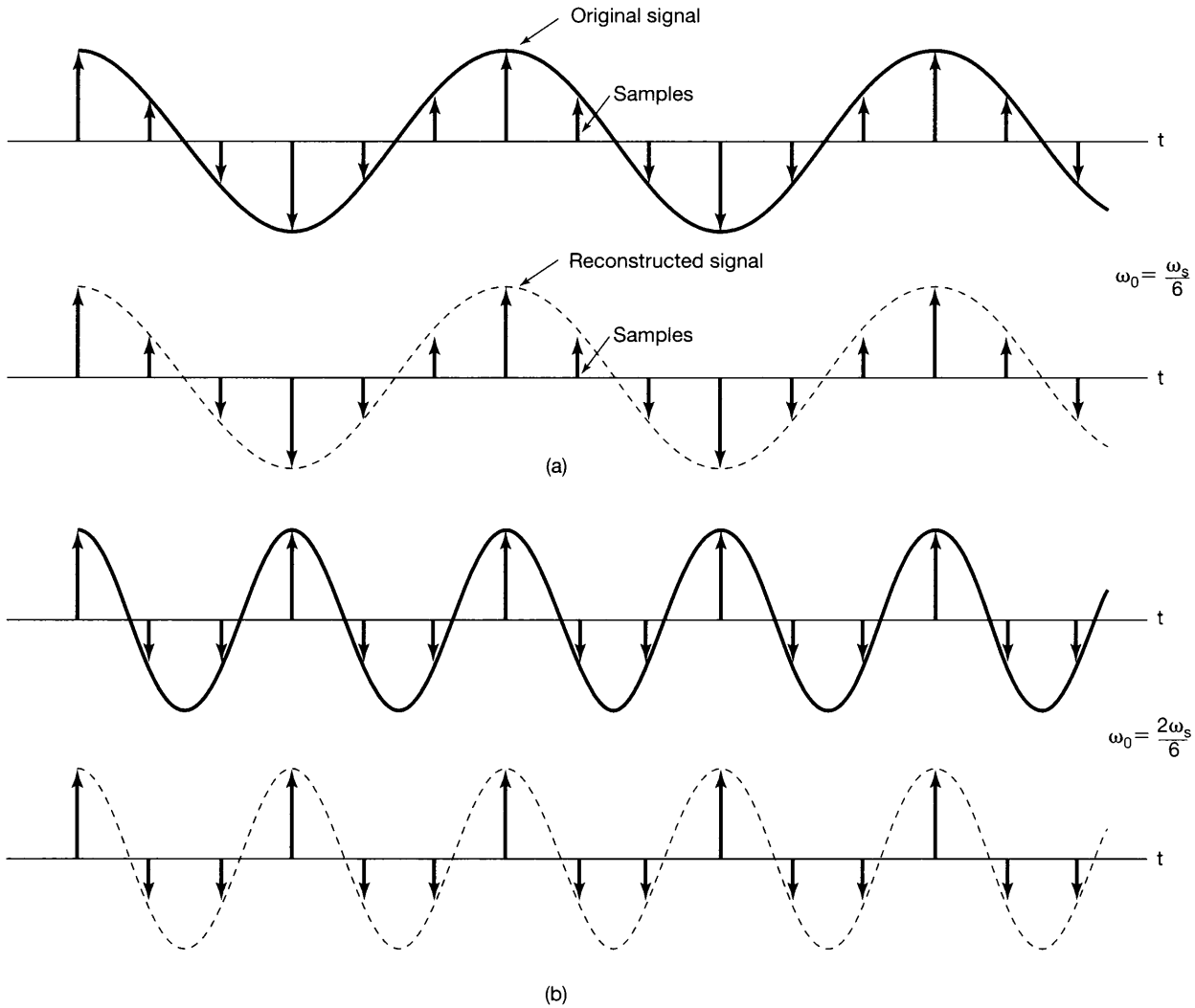


Figure 7.16 Effect of aliasing on a sinusoidal signal. For each of four values of ω_0, the original sinusoidal signal (solid curve), its samples, and the reconstructed signal (dashed curve) are illustrated: (a) $\omega_0 = \omega_s/6$; (b) $\omega_0 = 2\omega_s/6$; (c) $\omega_0 = 4\omega_s/6$; (d) $\omega_0 = 5\omega_s/6$. In (a) and (b) no aliasing occurs, whereas in (c) and (d) there is aliasing.

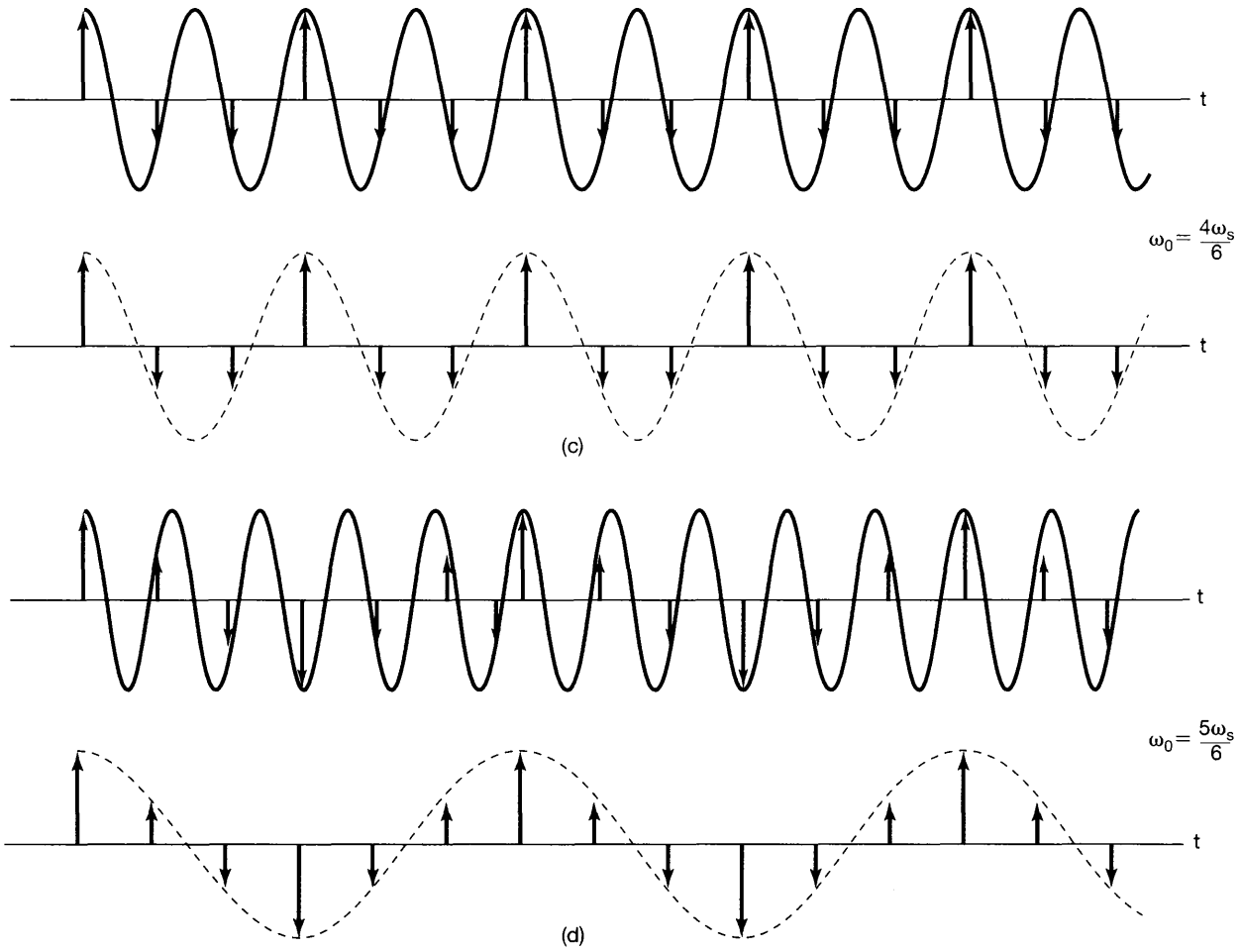


Figure 7.16 *Continued*

In this case, the Fourier transform of $x(t)$ is essentially the same as Figure 7.15(a), except that the impulse indicated with a solid line now has amplitude $\pi e^{j\phi}$, while the impulse indicated with a dashed line has amplitude with the opposite phase, namely, $\pi e^{-j\phi}$. If we now consider the same set of choices for ω_0 as in Figure 7.15, the resulting spectra for the sampled versions of $\cos(\omega_0 t + \phi)$ are exactly as in the figure, with all solid impulses having amplitude $\pi e^{j\phi}$ and all dashed ones having amplitude $\pi e^{-j\phi}$. Again, in cases (b) and (c) the condition of the sampling theorem is met, so that $x_r(t) = \cos(\omega_0 t + \phi) = x(t)$, while in (d) and (e) we again have aliasing. However, we now see that there has been a reversal in the solid and dashed impulses appearing in the passband of the lowpass filter. As a result, we find that in these cases, $x_r(t) = \cos[(\omega_s - \omega_0)t - \phi]$, where we have a change in the sign of the phase ϕ, i.e., a *phase reversal.*

It is important to note that the sampling theorem explicity requires that the sampling frequency be *greater than* twice the highest frequency in the signal, rather than greater than or equal to twice the highest frequency. The next example illustrates that sampling a sinusoidal signal at exactly twice its frequency (i.e., exactly two samples per cycle) is not sufficient.

Example 7.1

Consider the sinusoidal signal

$$x(t) = \cos\left(\frac{\omega_s}{2}t + \phi\right),$$

and suppose that this signal is sampled, using impulse sampling, at exactly twice the frequency of the sinusoid, i.e., at sampling frequency ω_s. As shown in Problem 7.39, if this impulse-sampled signal is applied as the input to an ideal lowpass filter with cutoff frequency $\omega_s/2$, the resulting output is

$$x_r(t) = (\cos\phi)\cos\left(\frac{\omega_s}{2}t\right).$$

As a consequence, we see that perfect reconstruction of $x(t)$ occurs only in the case in which the phase ϕ is zero (or an integer multiple of 2π). Otherwise, the signal $x_r(t)$ does not equal $x(t)$.

As an extreme example, consider the case in which $\phi = -\pi/2$, so that

$$x(t) = \sin\left(\frac{\omega_s}{2}t\right).$$

This signal is sketched in Figure 7.17. We observe that the values of the signal at integer multiples of the sampling period $2\pi/\omega_s$ are zero. Consequently, sampling at this rate produces a signal that is identically zero, and when this zero input is applied to the ideal lowpass filter, the resulting output $x_r(t)$ is also identically zero.

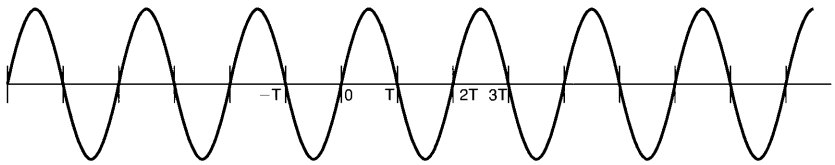


Figure 7.17 Sinusoidal signal for Example 7.1.

The effect of undersampling, whereby higher frequencies are reflected into lower frequencies, is the principle on which the stroboscopic effect is based. Consider, for example, the situation depicted in Figure 7.18, in which we have a disc rotating at a constant rate with a single radial line marked on the disc. The flashing strobe acts as a sampling system, since it illuminates the disc for extremely brief time intervals at a periodic rate. When the strobe frequency is much higher than the rotational speed of the disc, the speed of rotation of the disc is perceived correctly. When the strobe frequency becomes less than twice the rotational frequency of the disc, the rotation appears to be at a lower frequency than is actually the case. Furthermore, because of phase reversal, the disc will appear to be rotating in the wrong direction! Roughly speaking, if we track the position of a fixed line on the disc at successive samples, then when $\omega_0 < \omega_s < 2\omega_0$, so that we sample somewhat more frequently than once per revolution, samples of the disc will show the fixed line in positions that are successively displaced in a counterclockwise direction, opposite to the clockwise rotation of the disc itself. At one flash per revolution, corresponding to $\omega_s = \omega_0$, the radial line appears stationary (i.e., the rotational frequency of the disc and its harmonics have been aliased to zero frequency). A similar effect is commonly observed in Western movies,

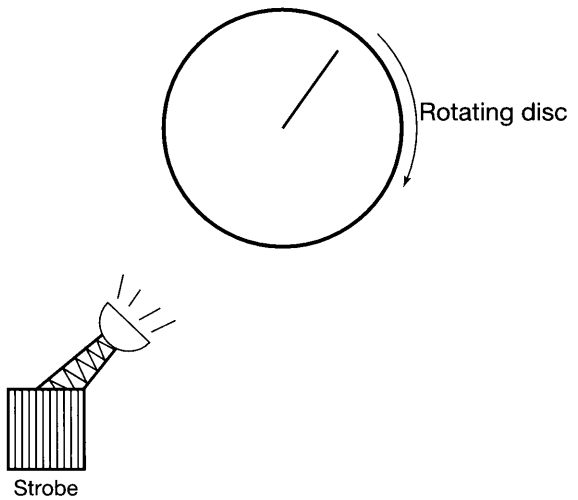


Figure 7.18 Strobe effect.

where the wheels of a stagecoach appear to be rotating more slowly than would be consistent with the coach's forward motion, and sometimes in the wrong direction. In this case, the sampling process corresponds to the fact that moving pictures are a sequence of individual frames with a rate (usually between 18 and 24 frames per second) corresponding to the sampling frequency.

The preceding discussion suggests interpreting the stroboscopic effect as an example of a useful application of aliasing due to undersampling. Another practical application of aliasing arises in a measuring instrument referred to as a *sampling oscilloscope*. This instrument is intended for observing very high-frequency waveforms and exploits the principles of sampling to alias these frequencies into ones that are more easily displayed. The sampling oscilloscope is explored in more detail in Problem 7.38.

7.4 DISCRETE-TIME PROCESSING OF CONTINUOUS-TIME SIGNALS

In many applications, there is a significant advantage offered in processing a continuous-time signal by first converting it to a discrete-time signal and, after discrete-time processing, converting back to a continuous-time signal. The discrete-time signal processing can be implemented with a general- or special-purpose computer, with microprocessors, or with any of the variety of devices that are specifically oriented toward discrete-time signal processing.

In broad terms, this approach to continuous-time signal processing can be viewed as the cascade of three operations, as indicated in Figure 7.19, where $x_c(t)$ and $y_c(t)$ are continuous-time signals and $x_d[n]$ and $y_d[n]$ are the discrete-time signals corresponding to $x_c(t)$ and $y_c(t)$. The overall system is, of course, a continuous-time system in the sense that its input and output are both continuous-time signals. The theoretical basis for converting a continuous-time signal to a discrete-time signal and reconstructing a continuous-time signal from its discrete-time representation lies in the sampling theorem, as discussed in Section 7.1. Through the process of periodic sampling with the sampling frequency consistent with the conditions of the sampling theorem, the continuous-time signal $x_c(t)$ is exactly represented by a sequence of instantaneous sample values $x_c(nT)$; that is, the discrete-time sequence $x_d[n]$ is related to $x_c(t)$ by

$$x_d[n] = x_c(nT). \tag{7.16}$$

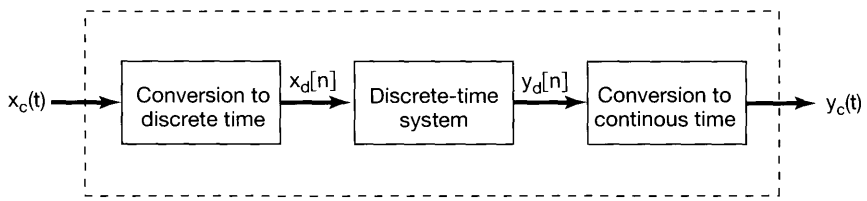


Figure 7.19 Discrete-time processing of continuous-time signals.

The transformation of $x_c(t)$ to $x_d[n]$ corresponding to the first system in Figure 7.19 will be referred to as *continuous-to-discrete-time conversion* and will be abbreviated C/D. The reverse operation corresponding to the third system in Figure 7.19 will be abbreviated D/C, representing *discrete-time to continuous-time conversion*. The D/C operation performs an interpolation between the sample values provided to it as input. That is, the D/C operation produces a continuous-time signal $y_c(t)$ which is related to the discrete-time signal $y_d[n]$ by

$$y_d[n] = y_c(nT).$$

This notation is made explicit in Figure 7.20. In systems such as digital computers and digital systems for which the discrete-time signal is represented in digital form, the device commonly used to implement the C/D conversion is referred to as an *analog-to-digital* (A-to-D) converter, and the device used to implement the D/C conversion is referred to as a *digital-to-analog* (D-to-A) converter.

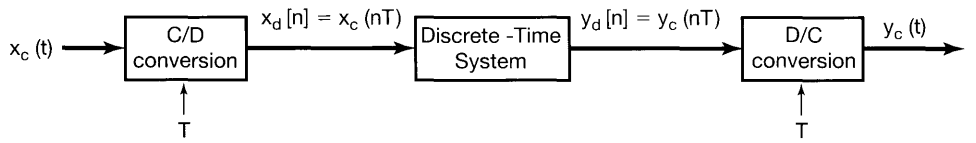


Figure 7.20 Notation for continuous-to-discrete-time conversion and discrete-to-continuous-time conversion. T represents the sampling period.

To understand further the relationship between the continuous-time signal $x_c(t)$ and its discrete-time representation $x_d[n]$, it is helpful to represent C/D as a process of periodic sampling followed by a mapping of the impulse train to a sequence. These two steps are illustrated in Figure 7.21. In the first step, representing the sampling process, the impulse train $x_p(t)$ corresponds to a sequence of impulses with amplitudes corresponding to the samples of $x_c(t)$ and with a time spacing equal to the sampling period T. In the conversion from the impulse train to the discrete-time sequence, we obtain $x_d[n]$, corresponding to the same sequence of samples of $x_c(t)$, but with unity spacing in terms of the new independent variable n. Thus, in effect, the conversion from the impulse train sequence of samples to the discrete-time sequence of samples can be thought of as a normalization in time. This normalization in converting $x_p(t)$ to $x_d[n]$ is evident in Figures 7.21(b) and (c), in which $x_p(t)$ and $x_d[n]$ are respectively illustrated for sampling rates of $T = T_1$ and $T = 2T_1$.

It is also instructive to examine the processing stages in Figure 7.19 in the frequency domain. Since we will be dealing with Fourier transforms in both continuous and discrete time, in this section only we distinguish the continuous-time and discrete-time frequency variables by using ω in continuous time and Ω in discrete time. For example, the continuous-time Fourier transforms of $x_c(t)$ and $y_c(t)$ are $X_c(j\omega)$ and $Y_c(j\omega)$, respectively, while the discrete-time Fourier transforms of $x_d[n]$ and $y_d[n]$ are $X_d(e^{j\Omega})$ and $Y_d(e^{j\Omega})$, respectively.

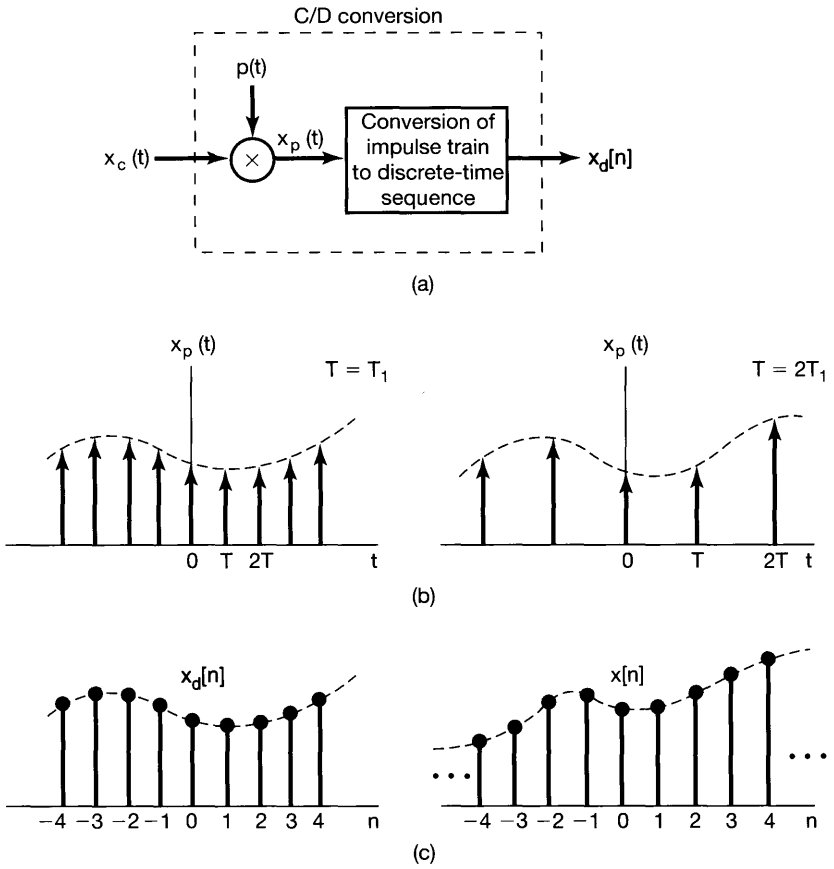


Figure 7.21 Sampling with a periodic impulse train followed by conversion to a discrete-time sequence: (a) overall system; (b) $x_p(t)$ for two sampling rates. The dashed envelope represents $x_c(t)$; (c) the output sequence for the two different sampling rates.

To begin let us express $X_p(j\omega)$, the continuous-time Fourier transform of $x_p(t)$, in terms of the sample values of $x_c(t)$ by applying the Fourier transform to eq. (7.3). Since

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT), \tag{7.17}$$

and since the transform of $\delta(t - nT)$ is $e^{-j\omega nT}$, it follows that

$$X_p(j\omega) = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\omega nT}. \tag{7.18}$$

Now consider the discrete-time Fourier transform of $x_d[n]$, that is,

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n]e^{-j\Omega n}, \tag{7.19}$$

or, using eq. (7.16),

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\Omega n}. \tag{7.20}$$

Comparing eqs. (7.18) and (7.20), we see that $X_d(e^{j\Omega})$ and $X_p(j\omega)$ are related through

$$X_d(e^{j\Omega}) = X_p(j\Omega/T). \tag{7.21}$$

Also, recall that, as developed in eq. (7.6) and illustrated in Figure 7.3,

$$X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s)). \tag{7.22}$$

Consequently,

$$X_d(e^{j\Omega}) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\Omega - 2\pi k)/T). \tag{7.23}$$

The relationship among $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ is illustrated in Figure 7.22 for two different sampling rates. Here, $X_d(e^{j\Omega})$ is a frequency-scaled version of $X_p(j\omega)$

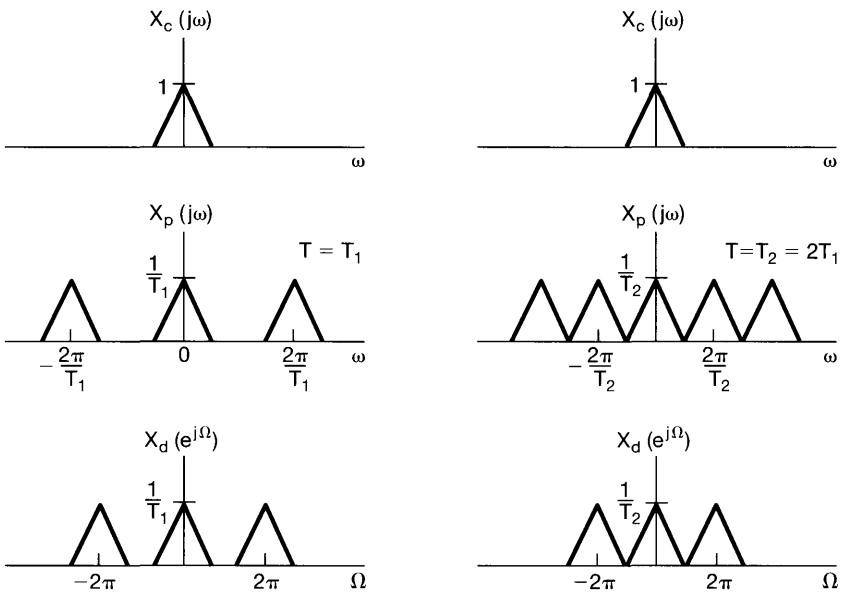


Figure 7.22 Relationship between $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ for two different sampling rates.

and, in particular, is periodic in Ω with period 2π. This periodicity is, of course, characteristic of any discrete-time Fourier transform. The spectrum of $x_d[n]$ is related to that of $x_c(t)$ through periodic replication, represented by eq. (7.22), followed by linear frequency scaling, represented by eq. (7.21). The periodic replication is a consequence of the first step in the conversion process in Figure 7.21, namely, the impulse-train sampling. The linear frequency scaling in eq. (7.21) can be thought of informally as a consequence of the normalization in time introduced by converting from the impulse train $x_p(t)$ to the discrete-time sequence $x_d[n]$. From the time-scaling property of the Fourier transform in Section 4.3.5, scaling of the time axis by $1/T$ will introduce a scaling of the frequency axis by T. Thus, the relationship $\Omega = \omega T$ is consistent with the notion that, in converting from $x_p(t)$ to $x_d[n]$, the time axis is scaled by $1/T$.

In the overall system of Figure 7.19, after processing with a discrete-time system, the resulting sequence is converted back to a continuous-time signal. This process is the reverse of the steps in Figure 7.21. Specifically, from the sequence $y_d[n]$, a continuous-time impulse train $y_p(t)$ can be generated. Recovery of the continuous-time signal $y_c(t)$ from this impulse train is then accomplished by means of lowpass filtering, as illustrated in Figure 7.23.

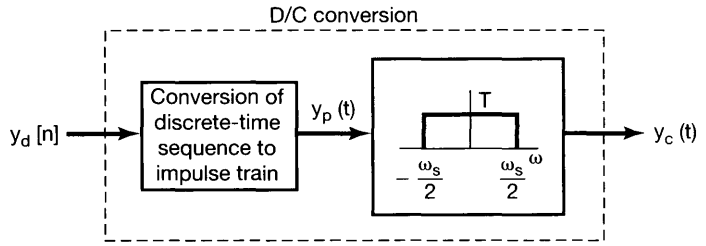


Figure 7.23 Conversion of a discrete-time sequence to a continuous-time signal.

Now let us consider the overall system of Figure 7.19, represented as shown in Figure 7.24. Clearly, if the discrete-time system is an identity system (i.e., $x_d[n] = y_d[n]$), then, assuming that the conditions of the sampling theorem are met, the overall system will be an identity system. The characteristics of the overall system with a more general frequency response $H_d(e^{j\Omega})$ are perhaps best understood by examining the representative example depicted in Figure 7.25. On the left-hand side of the figure are the representative

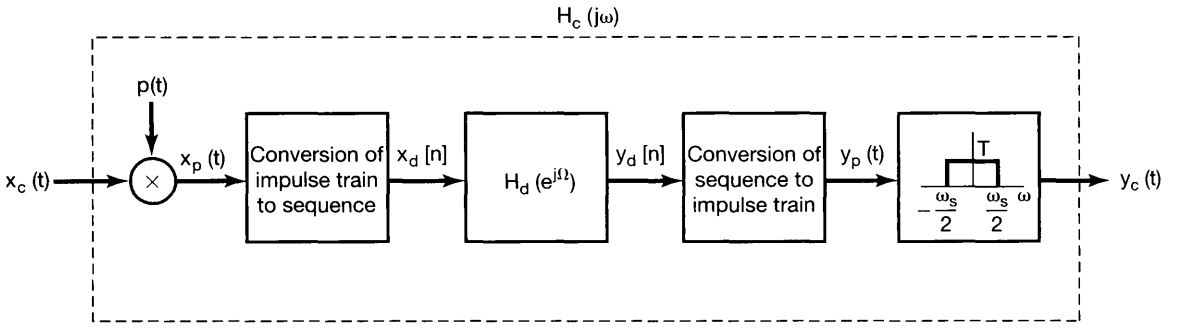


Figure 7.24 Overall system for filtering a continuous-time signal using a discrete-time filter.

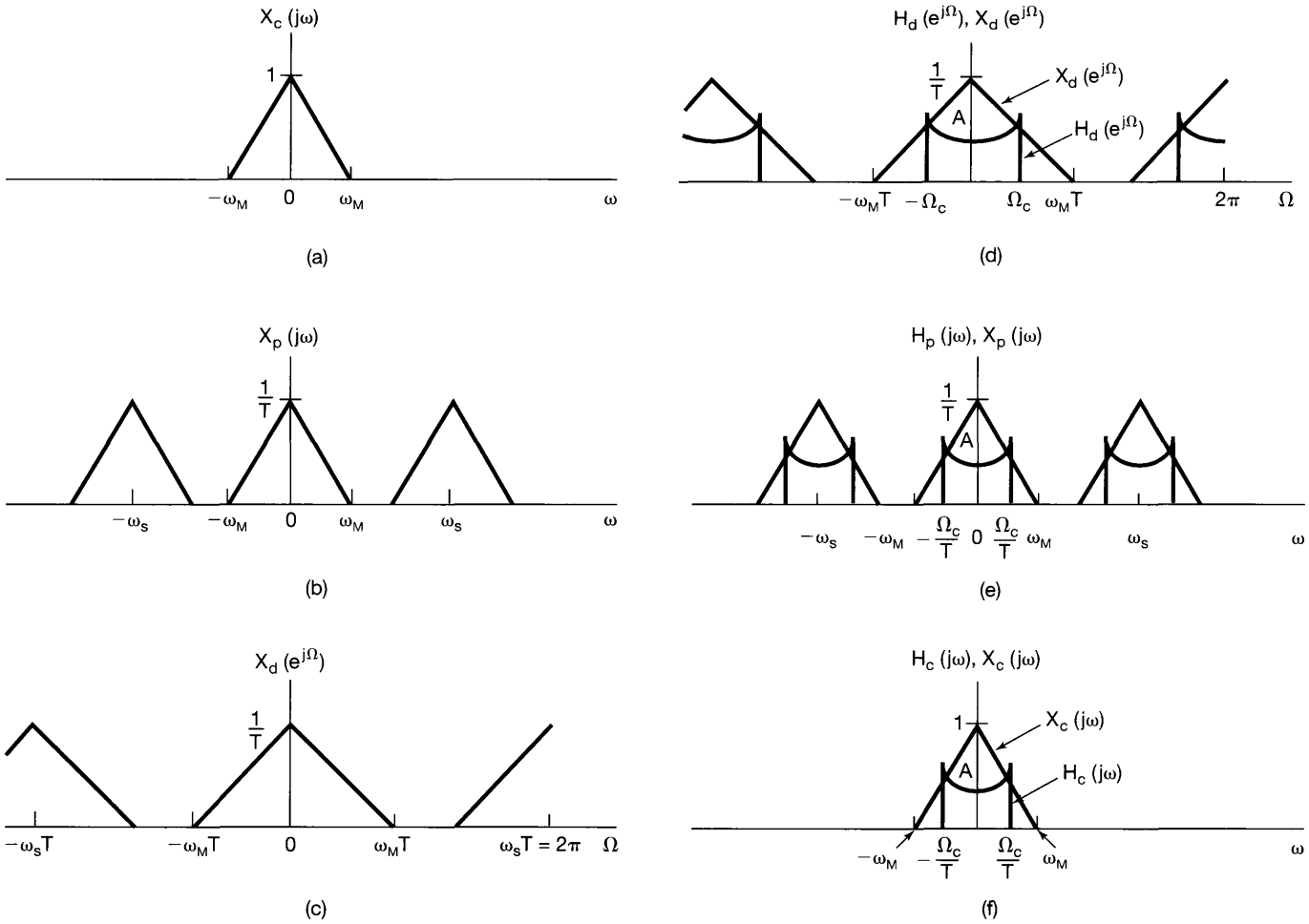


Figure 7.25 Frequency-domain illustration of the system of Figure 7.24: (a) continuous-time spectrum $X_c(j\omega)$; (b) spectrum after impulse-train sampling; (c) spectrum of discrete-time sequence $x_d[n]$; (d) $H_d(e^{j\Omega})$ and $X_d(e^{j\Omega})$ that are multiplied to form $Y_d(e^{j\Omega})$; (e) spectra that are multiplied to form $Y_p(j\omega)$; (f) spectra that are multiplied to form $Y_c(j\omega)$.

spectra $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$, where we assume that $\omega_M < \omega_s/2$, so that there is no aliasing. The spectrum $Y_d(e^{j\Omega})$ corresponding to the output of the discrete-time filter is the product of $X_d(e^{j\Omega})$ and $H_d(e^{j\Omega})$, and this is depicted in Figure 7.25(d) by overlaying $H_d(e^{j\Omega})$ and $X_d(e^{j\Omega})$. The transformation to $Y_c(j\omega)$ then corresponds to applying a frequency scaling and lowpass filtering, resulting in the spectra indicated in Figure 7.25(e) and (f). Since $Y_d(e^{j\Omega})$ is the product of the two overlaid spectra in Figure 7.25(d), the frequency scaling and lowpass filtering are applied to both. In comparing Figures 7.25(a) and (f), we see that

$$Y_c(j\omega) = X_c(j\omega)H_d(e^{j\omega T}). \tag{7.24}$$

Consequently, for inputs that are sufficiently band limited, so that the sampling theorem is satisfied, the overall system of Figure 7.24 is, in fact, equivalent to a continuous-time LTI system with frequency response $H_c(j\omega)$ which is related to the discrete-time frequency response $H_d(e^{j\Omega})$ through

$$H_c(j\omega) = \begin{cases} H_d(e^{j\omega T}), & |\omega| < \omega_s/2 \\ 0, & |\omega| > \omega_s/2 \end{cases}. \tag{7.25}$$

The equivalent frequency response for this continuous-time filter is one period of the frequency response of the discrete-time filter with a linear scale change applied to the frequency axis. This relationship between the discrete-time frequency response and the equivalent continuous-time frequency response is illustrated in Figure 7.26.

The equivalence of the overall system of Figure 7.24 to an LTI system is somewhat surprising in view of the fact that multiplication by an impulse train is *not* a time-invariant operation. In fact, the overall system of Figure 7.24 is not time invariant for arbitrary inputs. For example, if $x_c(t)$ was a narrow rectangular pulse of duration less than T, then a time shift of $x_c(t)$ could generate a sequence $x[n]$ that either had all zero values or had one nonzero value, depending on the alignment of the rectangular pulse relative to the

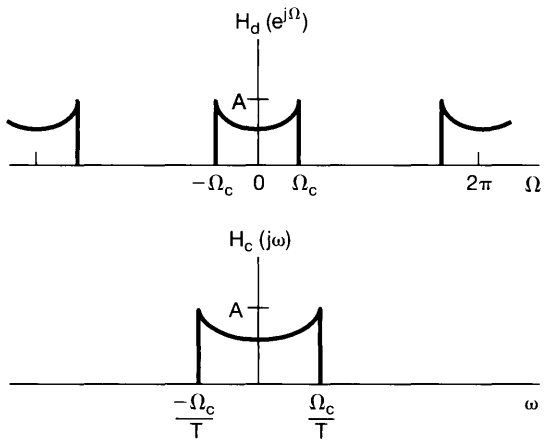


Figure 7.26 Discrete-time frequency response and the equivalent continuous-time frequency response for the system of Figure 7.24.

sampling impulse train. However, as suggested by the spectra of Figure 7.25, for *band-limited input signals* with a sampling rate sufficiently high so as to avoid aliasing, the system of Figure 7.24 *is* equivalent to a continuous-time LTI system. For such inputs, Figure 7.24 and eq. (7.25) provide the conceptual basis for continuous-time processing using discrete-time filters. This is now explored further in the context of some examples.

7.4.1 Digital Differentiator

Consider the discrete-time implementation of a continuous-time band-limited differentiating filter. As discussed in Section 3.9.1, the frequency response of a continuous-time differentiating filter is

$$H_c(j\omega) = j\omega, \tag{7.26}$$

and that of a band-limited differentiator with cutoff frequency ω_c is

$$H_c(j\omega) = \begin{cases} j\omega, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}, \tag{7.27}$$

as sketched in Figure 7.27. Using eq. (7.25) with a sampling frequency $\omega_s = 2\omega_c$, we see that the corresponding discrete-time transfer function is

$$H_d(e^{j\Omega}) = j\left(\frac{\Omega}{T}\right), \quad |\Omega| < \pi, \tag{7.28}$$

as sketched in Figure 7.28. With this discrete-time transfer function, $y_c(t)$ in Figure 7.24 will be the derivative of $x_c(t)$, assuming that there is no aliasing in sampling $x_c(t)$.

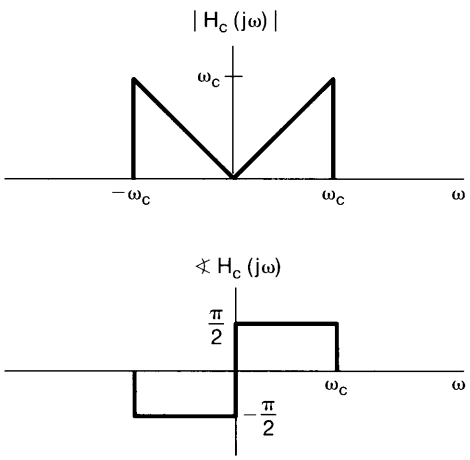


Figure 7.27 Frequency response of a continuous-time ideal band-limited differentiator $H_c(j\omega) = j\omega$, $|\omega| < \omega_c$.

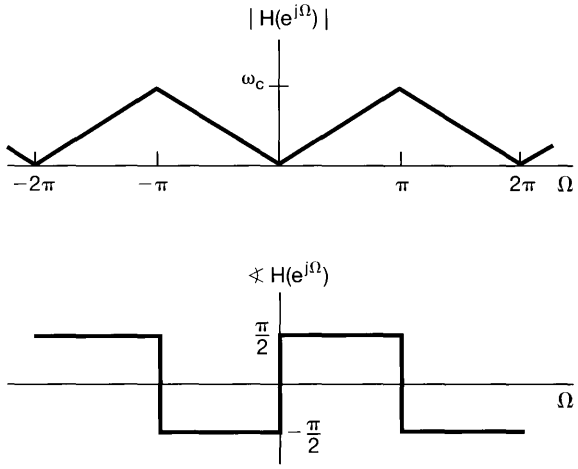


Figure 7.28 Frequency response of discrete-time filter used to implement a continuous-time band-limited differentiator.

Example 7.2

By considering the output of the digital differentiator for a continuous-time sinc input, we may conveniently determine the impulse response $h_d[n]$ of the discrete-time filter in the implementation of the digital differentiator. With reference to Figure 7.24, let

$$x_c(t) = \frac{\sin(\pi t/T)}{\pi t}, \tag{7.29}$$

where T is the sampling period. Then

$$X_c(j\omega) = \begin{cases} 1, & |\omega| < \pi/T \\ 0, & \text{otherwise} \end{cases},$$

which is sufficiently band limited to ensure that sampling $x_c(t)$ at frequency $\omega_s = 2\pi/T$ does not give rise to any aliasing. It follows that the output of the digital differentiator is

$$y_c(t) = \frac{d}{dt}x_c(t) = \frac{\cos(\pi t/T)}{Tt} - \frac{\sin(\pi t/T)}{\pi t^2}. \tag{7.30}$$

For $x_c(t)$ as given by eq. (7.29), the corresponding signal $x_d[n]$ in Figure 7.24 may be expressed as

$$x_d[n] = x_c(nT) = \frac{1}{T}\delta[n]. \tag{7.31}$$

That is, for $n \neq 0$, $x_c(nT) = 0$, while

$$x_d[0] = x_c(0) = \frac{1}{T}$$

which can be verified by l'Hôpital's rule. We can similarly evaluate $y_d[n]$ in Figure 7.24 corresponding to $y_c(t)$ in eq. (7.30). Specifically

$$y_d[n] = y_c(nT) = \begin{cases} \dfrac{(-1)^n}{nT^2}, & n \neq 0 \\ 0 & n = 0 \end{cases}. \tag{7.32}$$

which can be verified for $n \neq 0$ by direct substitution into eq. (7.30) and for $n = 0$ by application of l'Hôpital's rule.

Thus when the input to the discrete-time filter given by eq. (7.28) is the scaled unit impulse in eq. (7.31), the resulting output is given by eq. (7.32). We then conclude that the impulse response of this filter is given by

$$h_d[n] = \begin{cases} \dfrac{(-1)^n}{nT}, & n \neq 0 \\ 0, & n = 0 \end{cases}$$

7.4.2 Half-Sample Delay

In this section, we consider the implementation of a time shift (delay) of a continuous-time signal through the use of a system in the form of Figure 7.19. Thus, we require that the input and output of the overall system be related by

$$y_c(t) = x_c(t - \Delta) \tag{7.33}$$

when the input $x_c(t)$ is band limited and the sampling rate is high enough to avoid aliasing and where Δ represents the delay time. From the time-shifting property derived in Section 4.3.2

$$Y_c(j\omega) = e^{-j\omega\Delta} X_c(j\omega).$$

From eq. (7.25), the equivalent continuous-time system to be implemented must be band limited. Therefore, we take

$$H_c(j\omega) = \begin{cases} e^{-j\omega\Delta}, & |\omega| < \omega_c \\ 0, & \text{otherwise} \end{cases}, \tag{7.34}$$

where ω_c is the cutoff frequency of the continuous-time filter. That is, $H_c(j\omega)$ corresponds to a time shift as in eq. (7.33) for band-limited signals and rejects all frequencies greater than ω_c. The magnitude and phase of the frequency response are shown in Figure 7.29(a). With the sampling frequency ω_s taken as $\omega_s = 2\omega_c$, the corresponding discrete-time

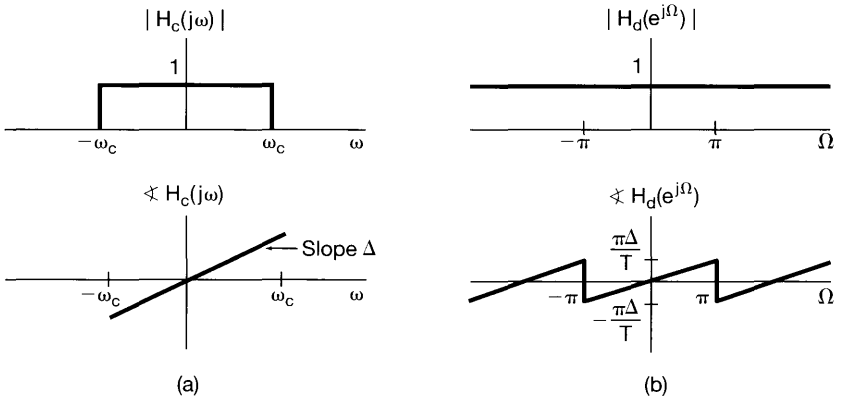


Figure 7.29 (a) Magnitude and phase of the frequency response for a continuous-time delay; (b) magnitude and phase of the frequency response for the corresponding discrete-time delay.

frequency response is

$$H_d(e^{j\Omega}) = e^{-j\Omega\Delta/T}, \quad |\Omega| < \pi, \tag{7.35}$$

and is shown in Figure 7.29(b).

For appropriately band-limited inputs, the output of the system of Figure 7.24 with $H_d(e^{j\Omega})$ as in eq. (7.35) is a delayed replica of the input. For Δ/T an integer, the sequence $y_d[n]$ is a delayed replica of $x_d[n]$; that is,

$$y_d[n] = x_d\left[n - \frac{\Delta}{T}\right]. \tag{7.36}$$

For Δ/T not an integer, eq. (7.36), as written, has no meaning, since sequences are defined only at integer values of the index. However, we can interpret the relationship between $x_d[n]$ and $y_d[n]$ in these cases in terms of band-limited interpolation. The signals $x_c(t)$ and $x_d[n]$ are related through sampling and band-limited interpolation, as are $y_c(t)$ and $y_d[n]$. With $H_d(e^{j\Omega})$ in eq. (7.35), $y_d[n]$ is equal to samples of a shifted version of the band-limited interpolation of the sequence $x_d[n]$. This is illustrated in Figure 7.30 with $\Delta/T = 1/2$, which is sometimes referred to as a half-sample delay.

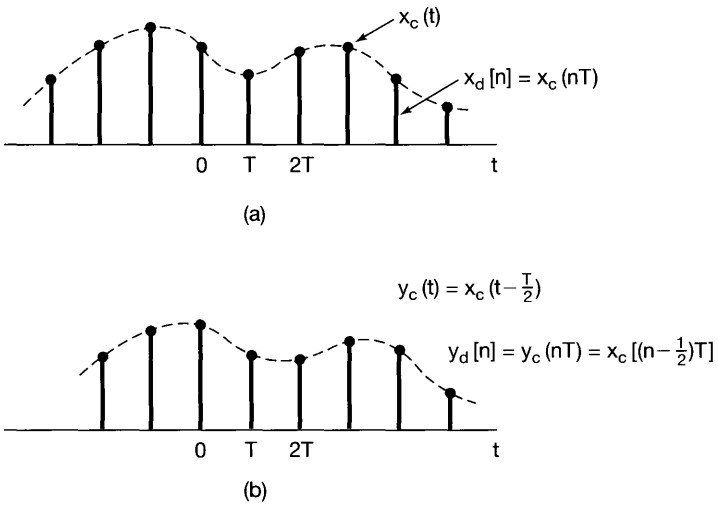


Figure 7.30 (a) Sequence of samples of a continuous-time signal $x_c(t)$; (b) sequence in (a) with a half-sample delay.

Example 7.3

The approach in Example 7.2 is also applicable to determining the impulse response $h_d[n]$ of the discrete-time filter in the half-sample delay system. With reference to Figure 7.24, let

$$x_c(t) = \frac{\sin(\pi t/T)}{\pi t}. \tag{7.37}$$

It follows from Example 7.2 that

$$x_d[n] = x_c(nT) = \frac{1}{T}\delta[n].$$

Also, since there is no aliasing for the band-limited input in eq. (7.37), the output of the half-sample delay system is

$$y_c(t) = x_c(t - T/2) = \frac{\sin(\pi(t - T/2)/T)}{\pi(t - T/2)},$$

and the sequence $y_d[n]$ in Figure 7.24 is

$$y_d[n] = y_c(nT) = \frac{\sin(\pi(n - \frac{1}{2}))}{T\pi(n - \frac{1}{2})}.$$

We conclude that

$$h[n] = \frac{\sin(\pi(n - \frac{1}{2}))}{\pi(n - \frac{1}{2})}.$$

7.5 SAMPLING OF DISCRETE-TIME SIGNALS

Thus far in this chapter, we have considered the sampling of continuous-time signals, and in addition to developing the analysis necessary to understand continuous-time sampling, we have introduced a number of its applications. As we will see in this section, a very similar set of properties and results with a number of important applications can be developed for sampling of discrete-time signals.

7.5.1 Impulse-Train Sampling

In analogy with continuous-time sampling as carried out using the system of Figure 7.2, sampling of a discrete-time signal can be represented as shown in Figure 7.31. Here, the new sequence $x_p[n]$ resulting from the sampling process is equal to the original sequence $x[n]$ at integer multiples of the sampling period N and is zero at the intermediate samples; that is,

$$x_p[n] = \begin{cases} x[n], & \text{if } n = \text{an integer multiple of } N \\ 0, & \text{otherwise} \end{cases}. \tag{7.38}$$

As with continuous-time sampling in Section 7.1, the effect in the frequency domain of discrete-time sampling is seen by using the multiplication property developed in Section 5.5. Thus, with

$$x_p[n] = x[n]p[n] = \sum_{k=-\infty}^{+\infty} x[kN]\delta[n - kN], \tag{7.39}$$

we have, in the frequency domain,

$$X_p(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} P(e^{j\theta})X(e^{j(\omega-\theta)})d\theta. \tag{7.40}$$

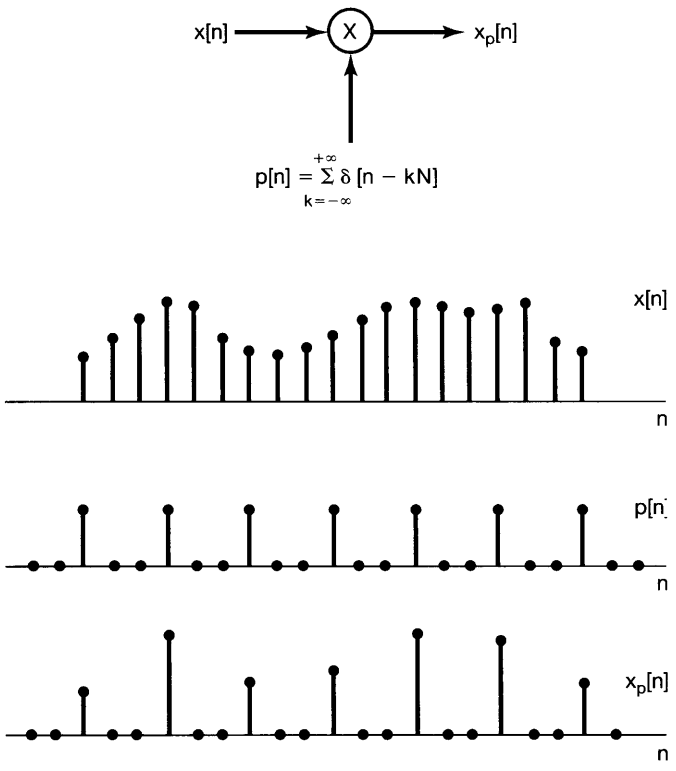


Figure 7.31 Discrete-time sampling.

As in Example 5.6, the Fourier transform of the sampling sequence $p[n]$ is

$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s), \tag{7.41}$$

where ω_s, the sampling frequency, equals $2\pi/N$. Combining eqs. (7.40) and (7.41), we have

$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=0}^{N-1} X\left(e^{j(\omega - k\omega_s)}\right). \tag{7.42}$$

Equation (7.42) is the counterpart for discrete-time sampling of eq. (7.6) for continuous-time sampling and is illustrated in Figure 7.32. In Figure 7.32(c), with $\omega_s - \omega_M > \omega_M$, or equivalently, $\omega_s > 2\omega_M$, there is no aliasing [i.e., the nonzero portions of the replicas of $X(e^{j\omega})$ do not overlap], whereas with $\omega_s < 2\omega_M$, as in Figure 7.32(d), frequency-domain aliasing results. In the absence of aliasing, $X(e^{j\omega})$ is faithfully reproduced around $\omega = 0$ and integer multiples of 2π. Consequently, $x[n]$ can be recovered from $x_p[n]$ by means of a lowpass filter with gain N and a cutoff frequency greater than

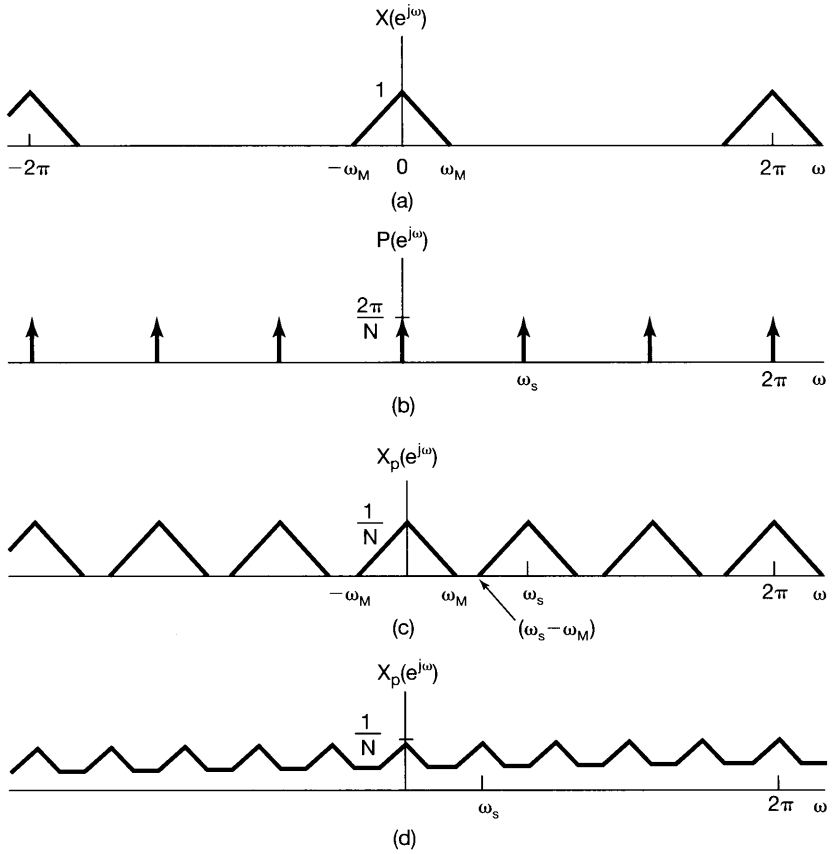


Figure 7.32 Effect in the frequency domain of impulse-train sampling of a discrete-time signal: (a) spectrum of original signal; (b) spectrum of sampling sequence; (c) spectrum of sampled signal with $\omega_S > 2\omega_M$; (d) spectrum of sampled signal with $\omega_S < 2\omega_M$. Note that aliasing occurs.

ω_M and less than $\omega_s - \omega_M$, as illustrated in Figure 7.33, where we have specified the cutoff frequency of the lowpass filter as $\omega_s/2$. If the overall system of Figure 7.33(a) is applied to a sequence for which $\omega_s < 2\omega_M$, so that aliasing results, $x_r[n]$ will no longer be equal to $x[n]$. However, as with continuous-time sampling, the two sequences *will* be equal at multiples of the sampling period; that is, corresponding to eq. (7.13), we have

$$x_r[kN] = x[kN], \quad k = 0, \pm 1, \pm 2, \ldots, \tag{7.43}$$

independently of whether aliasing occurs. (See Problem 7.46.)

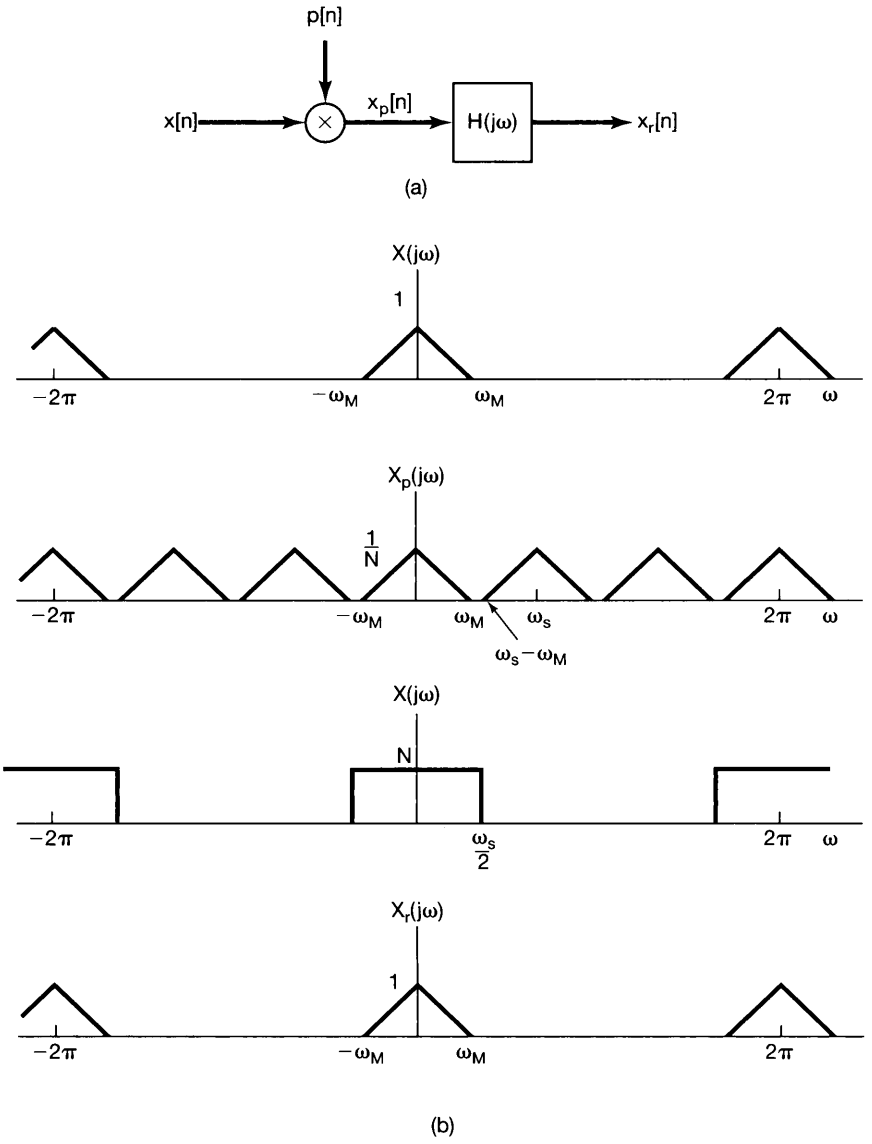


Figure 7.33 Exact recovery of a discrete-time signal from its samples using an ideal lowpass filter: (a) block diagram for sampling and reconstruction of a band-limited signal from its samples; (b) spectrum of the signal $x[n]$; (c) spectrum of $x_p[n]$; (d) frequency response of an ideal lowpass filter with cutoff frequency $\omega_s/2$; (e) spectrum of the reconstructed signal $x_r[n]$. For the example depicted here $\omega_s > 2\omega_M$ so that no aliasing occurs and consequently $x_r[n] = x[n]$.

Example 7.4

Consider a sequence $x[n]$ whose Fourier transform $X(e^{j\omega})$ has the property that

$$X(e^{j\omega}) = 0 \qquad \text{for} \quad 2\pi/9 \leq |\omega| \leq \pi.$$

To determine the lowest rate at which $x[n]$ may be sampled without the possibility of aliasing, we must find the largest N such that

$$\frac{2\pi}{N} \geq 2\left(\frac{2\pi}{9}\right) \Longrightarrow N \leq 9/2.$$

We conclude that $N_{\max} = 4$, and the corresponding sampling frequency is $2\pi/4 = \pi/2$.

The reconstruction of $x[n]$ through the use of a lowpass filter applied to $x_p[n]$ can be interpreted in the time domain as an interpolation formula similar to eq. (7.11). With $h[n]$ denoting the impulse response of the lowpass filter, we have

$$h[n] = \frac{N\omega_c}{\pi} \frac{\sin \omega_c n}{\omega_c n}. \tag{7.44}$$

The reconstructed sequence is then

$$x_r[n] = x_p[n] * h[n], \tag{7.45}$$

or equivalently,

$$x_r[n] = \sum_{k=-\infty}^{+\infty} x[kN] \frac{N\omega_c}{\pi} \frac{\sin \omega_c (n - kN)}{\omega_c (n - kN)}. \tag{7.46}$$

Equation (7.46) represents ideal band-limited interpolation and requires the implementation of an ideal lowpass filter. In typical applications a suitable approximation for the lowpass filter in Figure 7.33 is used, in which case the equivalent interpolation formula is of the form

$$x_r[n] = \sum_{k=-\infty}^{+\infty} x[kN] h_r[n - kN], \tag{7.47}$$

where $h_r[n]$ is the impulse response of the interpolating filter. Some specific examples, including the discrete-time counterparts of the zero-order hold and first-order hold discussed in Section 7.2 for continuous-time interpolation, are considered in Problem 7.50.

7.5.2 Discrete-Time Decimation and Interpolation

There are a variety of important applications of the principles of discrete-time sampling, such as in filter design and implementation or in communication applications. In many of these applications it is inefficient to represent, transmit, or store the sampled sequence $x_p[n]$ directly in the form depicted in Figure 7.31, since, in between the sampling instants, $x_p[n]$ is *known* to be zero. Thus, the sampled sequence is typically replaced by a new sequence $x_b[n]$, which is simply every Nth value of $x_p[n]$; that is,

$$x_b[n] = x_p[nN]. \tag{7.48}$$

Also, equivalently,

$$x_b[n] = x[nN], \tag{7.49}$$

since $x_p[n]$ and $x[n]$ are equal at integer multiples of N. The operation of extracting every Nth sample is commonly referred to as *decimation.*[3] The relationship between $x[n]$, $x_p[n]$, and $x_b[n]$ is illustrated in Figure 7.34.

To determine the effect in the frequency domain of decimation, we wish to determine the relationship between $X_b(e^{j\omega})$—the Fourier transform of $x_b[n]$—and $X(e^{j\omega})$. To this end, we note that

$$X_b(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} x_b[k]e^{-j\omega k}, \tag{7.50}$$

or, using eq. (7.48),

$$X_b(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} x_p[kN]e^{-j\omega k}. \tag{7.51}$$

If we let $n = kN$, or equivalently $k = n/N$, we can write

$$X_b(e^{j\omega}) = \sum_{\substack{n=\text{integer} \\ \text{multiple of } N}} x_p[n]e^{-j\omega n/N},$$

and since $x_p[n] = 0$ when n is not an integer multiple of N, we can also write

$$X_b(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x_p[n]e^{-j\omega n/N}. \tag{7.52}$$

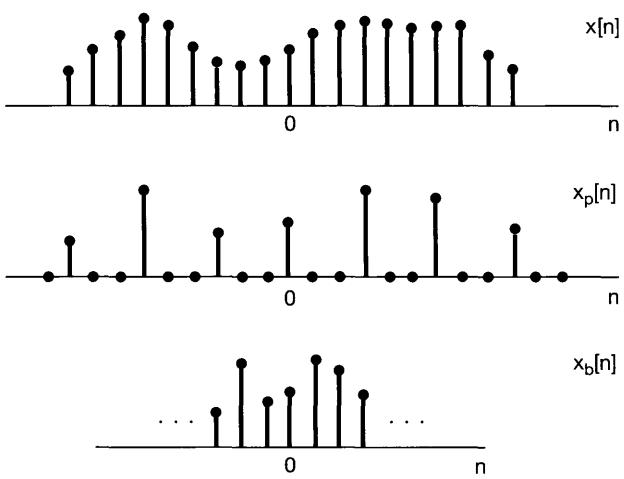


Figure 7.34 Relationship between $x_p[n]$ corresponding to sampling and $x_b[n]$ corresponding to decimation.

[3]Technically, decimation would correspond to extracting every *tenth* sample. However, it has become common terminology to refer to the operation as decimation even when N is not equal to 10.

Furthermore, we recognize the right-hand side of eq. (7.52) as the Fourier transform of $x_p[n]$; that is,

$$\sum_{n=-\infty}^{+\infty} x_p[n]e^{-j\omega n/N} = X_p(e^{j\omega/N}). \tag{7.53}$$

Thus, from eqs. (7.52) and (7.53), we conclude that

$$X_b(e^{j\omega}) = X_p(e^{j\omega/N}). \tag{7.54}$$

This relationship is illustrated in Figure 7.35, and from it, we observe that the spectra for the sampled sequence and the decimated sequence differ only in a frequency scaling or normalization. If the original spectrum $X(e^{j\omega})$ is appropriately band limited, so that there is no aliasing present in $X_p(e^{j\omega})$, then, as shown in the figure, the effect of decimation is to spread the spectrum of the original sequence over a larger portion of the frequency band.

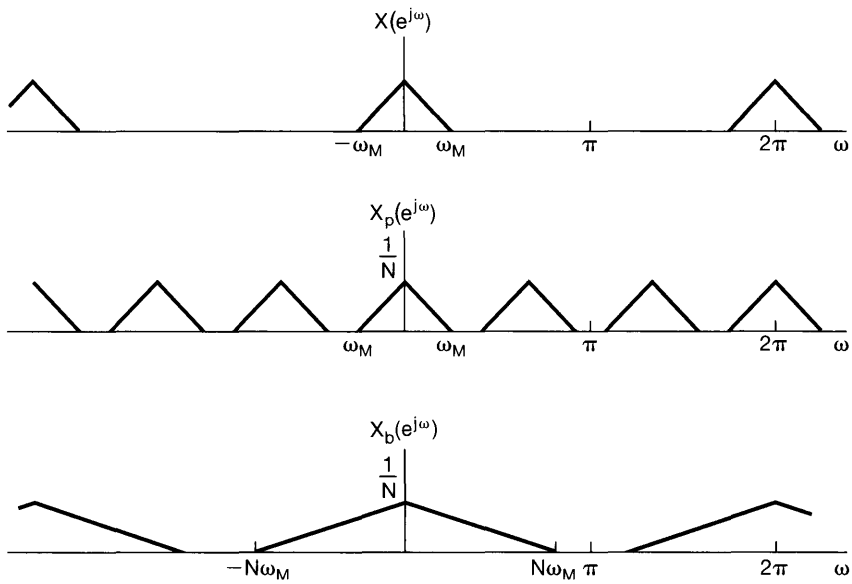


Figure 7.35 Frequency-domain illustration of the relationship between sampling and decimation.

If the original sequence $x[n]$ is obtained by sampling a continuous-time signal, the process of decimation can be viewed as reducing the sampling rate on the signal by a factor of N. To avoid aliasing, $X(e^{j\omega})$ cannot occupy the full frequency band. In other words, if the signal can be decimated without introducing aliasing, then the original continuous-time signal was oversampled, and thus, the sampling rate can be reduced without aliasing. With the interpretation of the sequence $x[n]$ as samples of a continuous-time signal, the process of decimation is often referred to as *downsampling*.

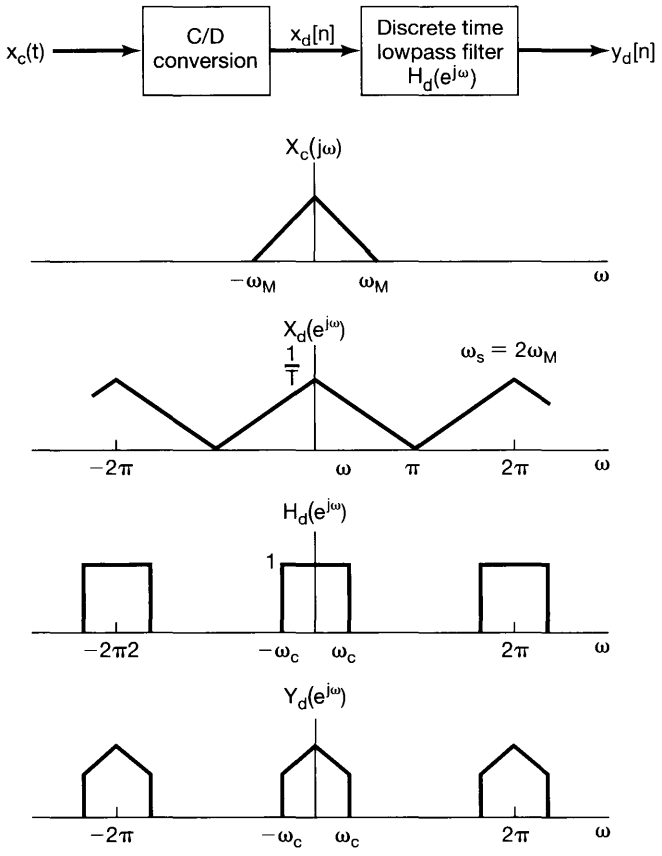


Figure 7.36 Continuous-time signal that was originally sampled at the Nyquist rate. After discrete-time filtering, the resulting sequence can be further downsampled. Here $X_c(j\omega)$ is the continuous-time Fourier transform of $x_c(t)$, $X_d(e^{j\omega})$ and $Y_d(e^{j\omega})$ are the discrete-time Fourier transforms of $x_d[n]$ and $y_d[n]$ respectively, and $H_d(e^{j\omega})$ is the frequency response of the discrete-time lowpass filter depicted in the block diagram.

In some applications in which a sequence is obtained by sampling a continuous-time signal, the original sampling rate may be as low as possible without introducing aliasing, but after additional processing and filtering, the bandwidth of the sequence may be reduced. An example of such a situation is shown in Figure 7.36. Since the output of the discrete-time filter is band limited, downsampling or decimation can be applied.

Just as in some applications it is useful to downsample, there are situations in which it is useful to convert a sequence to a *higher* equivalent sampling rate, a process referred to as *upsampling* or *interpolation.* Upsampling is basically the reverse of decimation or downsampling. As illustrated in Figures 7.34 and 7.35, in decimation we first sample and then retain only the sequence values at the sampling instants. To upsample, we reverse the process. For example, referring to Figure 7.34, we consider upsampling the sequence $x_b[n]$ to obtain $x[n]$. From $x_b[n]$, we form the sequence $x_p[n]$ by inserting $N - 1$ points with zero amplitude between each of the values in $x_b[n]$. The interpolated sequence $x[n]$ is then obtained from $x_p[n]$ by lowpass filtering. The overall procedure is summarized in Figure 7.37.

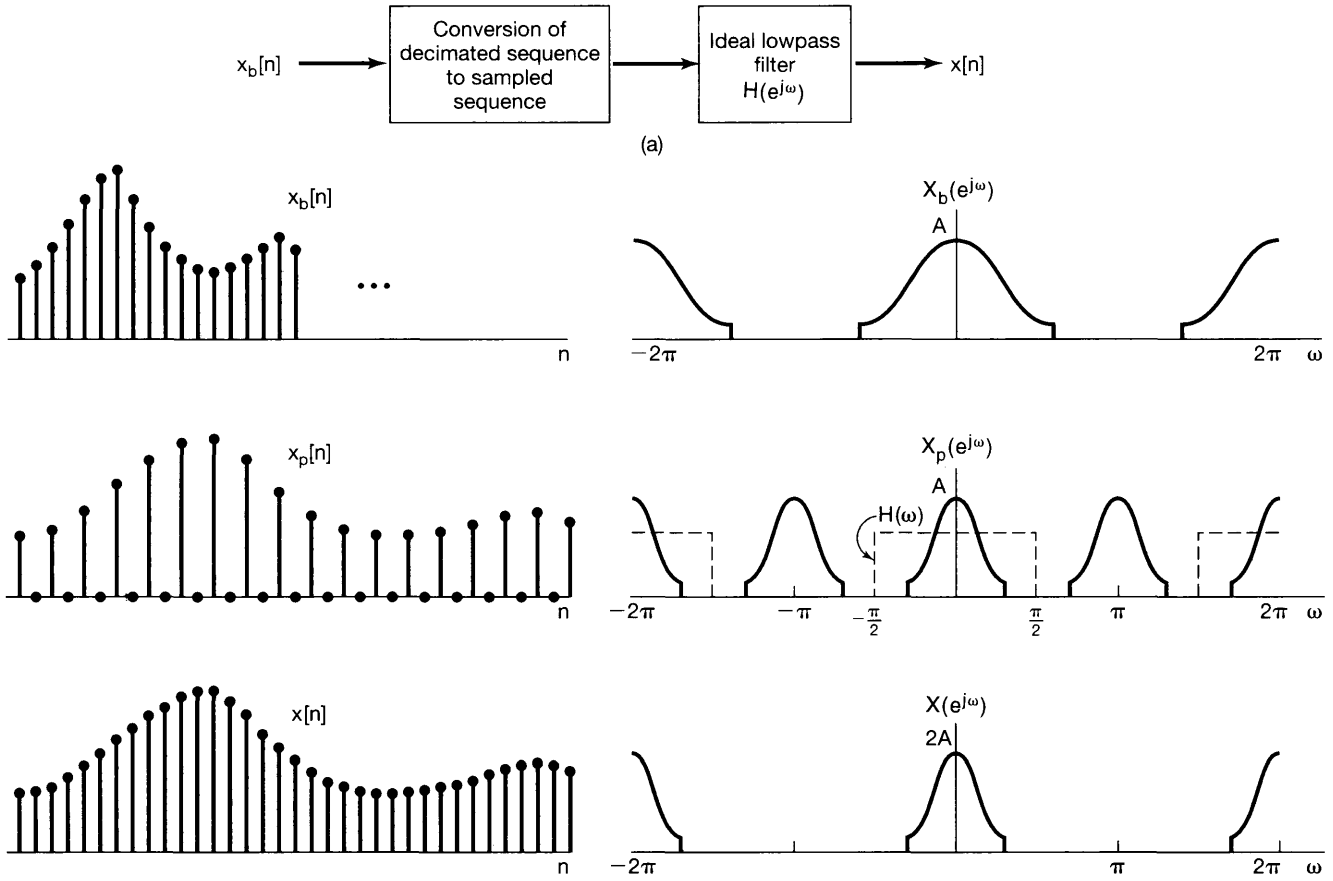


Figure 7.37 Upsampling: (a) overall system; (b) associated sequences and spectra for upsampling by a factor of 2.

Example 7.5

In this example, we illustrate how a combination of interpolation and decimation may be used to further downsample a sequence without incurring aliasing. It should be noted that maximum possible downsampling is achieved once the non-zero portion of one period of the discrete-time spectrum has expanded to fill the entire band from $-\pi$ to π.

Consider the sequence $x[n]$ whose Fourier transform $X(e^{j\omega})$ is illustrated in Figure 7.38(a). As discussed in Example 7.4, the lowest rate at which impulse-train sampling may be used on this sequence without incurring aliasing is $2\pi/4$. This corresponds to

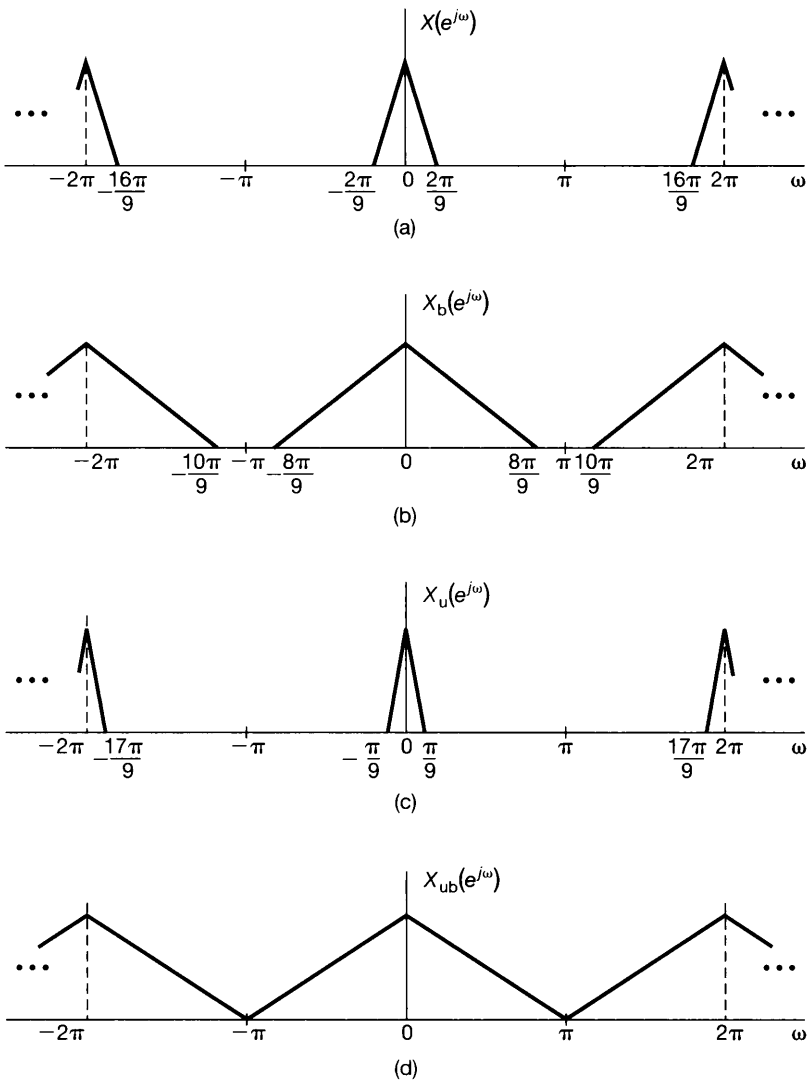


Figure 7.38 Spectra associated with Example 7.5. (a) Spectrum of $x[n]$; (b) spectrum after downsampling by 4; (c) spectrum after upsampling $x[n]$ by a factor of 2; (d) spectrum after upsampling $x[n]$ by 2 and then downsampling by 9.

sampling every 4th value of $x[n]$. If the result of such sampling is decimated by a factor of 4, we obtain a sequence $x_b[n]$ whose spectrum is shown in Figure 7.38(b). Clearly, there is still no aliasing of the original spectrum. However, this spectrum is zero for $8\pi/9 \leq |\omega| \leq \pi$, which suggests there is room for further downsampling.

Specifically, examining Figure 7.38(a) we see that if we could scale frequency by a factor of 9/2, the resulting spectrum would have nonzero values over the entire frequency interval from $-\pi$ to π. However, since 9/2 is not an integer, we can't achieve this purely by downsampling. Rather we must first upsample $x[n]$ by a factor of 2 and then downsample by a factor of 9. In particular, the spectrum of the signal $x_u[n]$ obtained when $x[n]$ is upsampled by a factor of 2, is displayed in Figure 7.38(c). When $x_u[n]$ is then downsampled by a factor of 9, the spectrum of the resulting sequence $x_{ub}[n]$ is as shown in Figure 7.38(d). This combined result effectively corresponds to downsampling $x[n]$ by a noninteger amount, 9/2. Assuming that $x[n]$ represents unaliased samples of a continuous-time signal $x_c(t)$, our interpolated and decimated sequence represents the maximum possible (aliasing-free) downsampling of $x_c(t)$.

7.6 SUMMARY

In this chapter we have developed the concept of sampling, whereby a continuous-time or discrete-time signal is represented by a sequence of equally spaced samples. The conditions under which the signal is exactly recoverable from the samples is embodied in the sampling theorem. For exact reconstruction, this theorem requires that the signal to be sampled be band limited and that the sampling frequency be greater than twice the highest frequency in the signal to be sampled. Under these conditions, exact reconstruction of the original signal is carried out by means of ideal lowpass filtering. The time-domain interpretation of this ideal reconstruction procedure is often referred to as ideal band-limited interpolation. In practical implementations, the lowpass filter is approximated and the interpolation in the time domain is no longer exact. In some instances, simple interpolation procedures such as a zero-order hold or linear interpolation (a first-order hold) suffice.

If a signal is undersampled (i.e., if the sampling frequency is less than that required by the sampling theorem), then the signal reconstructed by ideal band-limited interpolation will be related to the original signal through a form of distortion referred to as aliasing. In many instances, it is important to choose the sampling rate so as to avoid aliasing. However, there are a variety of important examples, such as the stroboscope, in which aliasing is exploited.

Sampling has a number of important applications. One particularly significant set of applications relates to using sampling to process continuous-time signals with discrete-time systems, by means of minicomputers, microprocessors, or any of a variety of devices specifically oriented toward discrete-time signal processing.

The basic theory of sampling is similar for both continuous-time and discrete-time signals. In the discrete-time case there is the closely related concept of decimation, whereby the decimated sequence is obtained by extracting values of the original sequence at equally spaced intervals. The difference between sampling and decimation lies in the fact that, for the sampled sequence, values of zero lie in between the sample values, whereas in the decimated sequence these zero values are discarded, thereby compressing the sequence in time. The inverse of decimation is interpolation. The ideas of decima-

tion and interpolation arise in a variety of important practical applications of signals and systems, including communication systems, digital audio, high-definition television, and many other applications.

Chapter 7 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining two sections contain problems belonging to the basic and advanced categories, respectively.

BASIC PROBLEMS WITH ANSWERS

7.1. A real-valued signal $x(t)$ is known to be uniquely determined by its samples when the sampling frequency is $\omega_s = 10{,}000\pi$. For what values of ω is $X(j\omega)$ guaranteed to be zero?

7.2. A continuous-time signal $x(t)$ is obtained at the output of an ideal lowpass filter with cutoff frequency $\omega_c = 1{,}000\pi$. If impulse-train sampling is performed on $x(t)$, which of the following sampling periods would guarantee that $x(t)$ can be recovered from its sampled version using an appropriate lowpass filter?

(a) $T = 0.5 \times 10^{-3}$

(b) $T = 2 \times 10^{-3}$

(c) $T = 10^{-4}$

7.3. The frequency which, under the sampling theorem, must be exceeded by the sampling frequency is called the *Nyquist rate*. Determine the Nyquist rate corresponding to each of the following signals:

(a) $x(t) = 1 + \cos(2{,}000\pi t) + \sin(4{,}000\pi t)$

(b) $x(t) = \frac{\sin(4{,}000\pi t)}{\pi t}$

(c) $x(t) = \left(\frac{\sin(4{,}000\pi t)}{\pi t}\right)^2$

7.4. Let $x(t)$ be a signal with Nyquist rate ω_0. Determine the Nyquist rate for each of the following signals:

(a) $x(t) + x(t-1)$

(b) $\frac{dx(t)}{dt}$

(c) $x^2(t)$

(d) $x(t)\cos\omega_0 t$

7.5. Let $x(t)$ be a signal with Nyquist rate ω_0. Also, let

$$y(t) = x(t)p(t-1),$$

where

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT), \text{ and } T < \frac{2\pi}{\omega_0}.$$

Specify the constraints on the magnitude and phase of the frequency response of a filter that gives $x(t)$ as its output when $y(t)$ is the input.

7.6. In the system shown in Figure P7.6, two functions of time, $x_1(t)$ and $x_2(t)$, are multiplied together, and the product $w(t)$ is sampled by a periodic impulse train. $x_1(t)$ is band limited to ω_1, and $x_2(t)$ is band limited to ω_2; that is,

$$X_1(j\omega) = 0, |\omega| \geq \omega_1,$$

$$X_2(j\omega) = 0, |\omega| \geq \omega_2.$$

Determine the *maximum* sampling interval T such that $w(t)$ is recoverable from $w_p(t)$ through the use of an ideal lowpass filter.

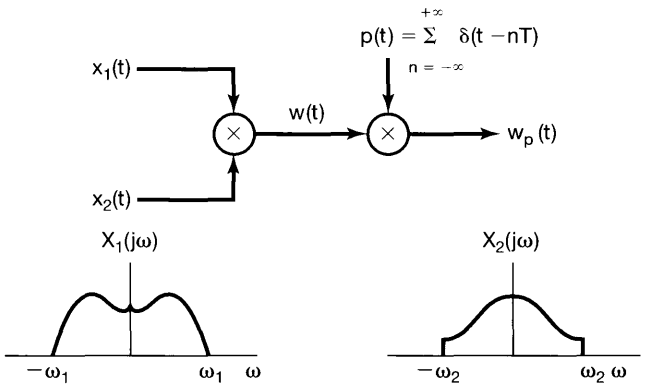


Figure P7.6

7.7. A signal $x(t)$ undergoes a zero-order hold operation with an effective sampling period T to produce a signal $x_0(t)$. Let $x_1(t)$ denote the result of a first-order hold operation on the samples of $x(t)$; i.e.,

$$x_1(t) = \sum_{n=-\infty}^{\infty} x(nT)h_1(t - nT),$$

where $h_1(t)$ is the function shown in Figure P7.7. Specify the frequency response of a filter that produces $x_1(t)$ as its output when $x_0(t)$ is the input.

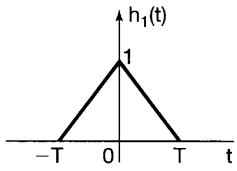


Figure P7.7

7.8. Consider a real, odd, and periodic signal $x(t)$ whose Fourier series representation may be expressed as

$$x(t) = \sum_{k=0}^{5} \left(\frac{1}{2}\right)^k \sin(k\pi t).$$

Let $\hat{x}(t)$ represent the signal obtained by performing impulse-train sampling on $x(t)$ using a sampling period of $T = 0.2$.

(a) Does aliasing occur when this impulse-train sampling is performed on $x(t)$?

(b) If $\hat{x}(t)$ is passed through an ideal lowpass filter with cutoff frequency π/T and passband gain T, determine the Fourier series representation of the output signal $g(t)$.

7.9. Consider the signal

$$x(t) = \left(\frac{\sin 50\pi t}{\pi t}\right)^2,$$

which we wish to sample with a sampling frequency of $\omega_s = 150\pi$ to obtain a signal $g(t)$ with Fourier transform $G(j\omega)$. Determine the maximum value of ω_0 for which it is guaranteed that

$$G(j\omega) = 75X(j\omega) \text{ for } |\omega| \leq \omega_0,$$

where $X(j\omega)$ is the Fourier transform of $x(t)$.

7.10. Determine whether each of the following statements is true or false:

(a) The signal $x(t) = u(t + T_0) - u(t - T_0)$ can undergo impulse-train sampling without aliasing, provided that the sampling period $T < 2T_0$.

(b) The signal $x(t)$ with Fourier transform $X(j\omega) = u(\omega + \omega_0) - u(\omega - \omega_0)$ can undergo impulse-train sampling without aliasing, provided that the sampling period $T < \pi/\omega_0$.

(c) The signal $x(t)$ with Fourier transform $X(j\omega) = u(\omega) - u(\omega - \omega_0)$ can undergo impulse-train sampling without aliasing, provided that the sampling period $T < 2\pi/\omega_0$.

7.11. Let $x_c(t)$ be a continuous-time signal whose Fourier transform has the property that $X_c(j\omega) = 0$ for $|\omega| \geq 2{,}000\pi$. A discrete-time signal

$$x_d[n] = x_c(n(0.5 \times 10^{-3}))$$

is obtained. For each of the following constraints on the Fourier transform $X_d(e^{j\omega})$ of $x_d[n]$, determine the corresponding constraint on $X_c(j\omega)$:

(a) $X_d(e^{j\omega})$ is real.

(b) The maximum value of $X_d(e^{j\omega})$ over all ω is 1.

(c) $X_d(e^{j\omega}) = 0$ for $\frac{3\pi}{4} \leq |\omega| \leq \pi$.

(d) $X_d(e^{j\omega}) = X_d(e^{j(\omega-\pi)})$.

7.12. A discrete-time signal $x_d[n]$ has a Fourier transform $X_d(e^{j\omega})$ with the property that $X_d(e^{j\omega}) = 0$ for $3\pi/4 \leq |\omega| \leq \pi$. The signal is converted into a continuous-time signal

$$x_c(t) = T \sum_{n=-\infty}^{\infty} x_d[n] \frac{\sin \frac{\pi}{T}(t - nT)}{\pi(t - nT)},$$

where $T = 10^{-3}$. Determine the values of ω for which the Fourier transform $X_c(j\omega)$ of $x_c(t)$ is guaranteed to be zero.

7.13. With reference to the filtering approach illustrated in Figure 7.24, assume that the sampling period used is T and the input $x_c(t)$ is band limited, so that $X_c(j\omega) = 0$ for $|\omega| \geq \pi/T$. If the overall system has the property that $y_c(t) = x_c(t-2T)$, determine the impulse response $h[n]$ of the discrete-time filter in Figure 7.24.

7.14. Repeat the previous problem, except this time assume that

$$y_c(t) = \frac{d}{dt} x_c\left(t - \frac{T}{2}\right).$$

7.15. Impulse-train sampling of $x[n]$ is used to obtain

$$g[n] = \sum_{k=-\infty}^{\infty} x[n]\delta[n - kN].$$

If $X(e^{j\omega}) = 0$ for $3\pi/7 \leq |\omega| \leq \pi$, determine the largest value for the sampling interval N which ensures that no aliasing takes place while sampling $x[n]$.

7.16. The following facts are given about the signal $x[n]$ and its Fourier transform:

1. $x[n]$ is real.
2. $X(e^{j\omega}) \neq 0$ for $0 < \omega < \pi$.
3. $x[n]\sum_{k=-\infty}^{\infty}\delta[n - 2k] = \delta[n]$.

Determine $x[n]$. You may find it useful to note that the signal $(\sin \frac{\pi}{2}n)/(\pi n)$ satisfies two of these conditions.

7.17. Consider an ideal discrete-time bandstop filter with impulse response $h[n]$ for which the frequency response in the interval $-\pi \leq \omega \leq \pi$ is

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{4} \text{ and } |\omega| \geq \frac{3\pi}{4} \\ 0, & \text{elsewhere} \end{cases}.$$

Determine the frequency response of the filter whose impulse response is $h[2n]$.

7.18. Suppose the impulse response of an ideal discrete-time lowpass filter with cutoff frequency $\pi/2$ is interpolated (in accordance with Figure 7.37) to obtain an upsampling by a factor of 2. What is the frequency response corresponding to this upsampled impulse response?

7.19. Consider the system shown in Figure P7.19, with input $x[n]$ and the corresponding output $y[n]$. The zero-insertion system inserts two points with zero amplitude between each of the sequence values in $x[n]$. The decimation is defined by

$$y[n] = w[5n],$$

where $w[n]$ is the input sequence for the decimation system. If the input is of the form

$$x[n] = \frac{\sin \omega_1 n}{\pi n},$$

determine the output $y[n]$ for the following values of ω_1:

(a) $\omega_1 \leq \frac{3\pi}{5}$

(b) $\omega_1 > \frac{3\pi}{5}$

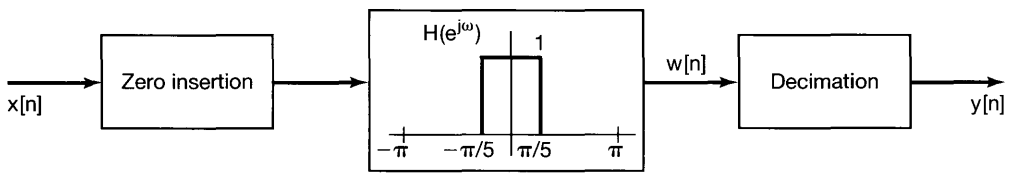


Figure P7.19

7.20. Two discrete-time systems S_1 and S_2 are proposed for implementing an ideal lowpass filter with cutoff frequency $\pi/4$. System S_1 is depicted in Figure P7.20(a). System S_2 is depicted in Figure P7.20(b). In these figures, S_A corresponds to a zero-insertion system that inserts one zero after every input sample, while S_B corresponds to a decimation system that extracts every second sample of its input.

(a) Does the proposed system S_1 correspond to the desired ideal lowpass filter?

(b) Does the proposed system S_2 correspond to the desired ideal lowpass filter?

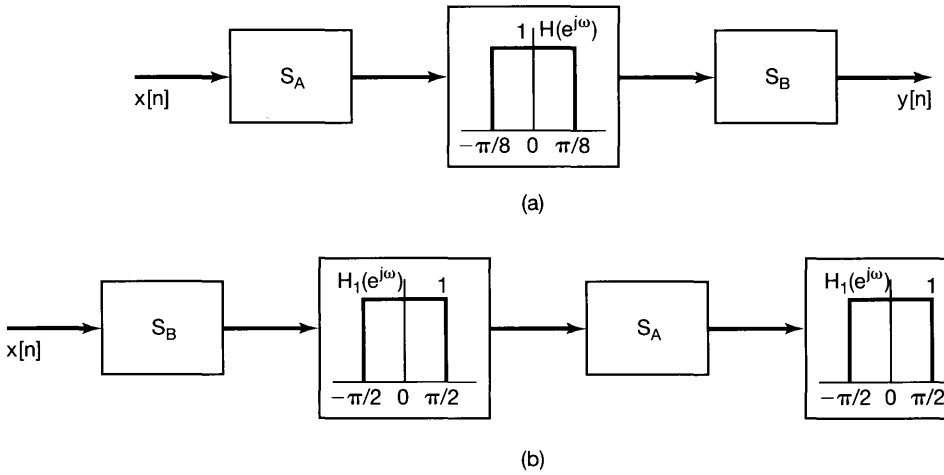


Figure P7.20

BASIC PROBLEMS

7.21. A signal $x(t)$ with Fourier transform $X(j\omega)$ undergoes impulse-train sampling to generate

$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t - nT)$$

where $T = 10^{-4}$. For each of the following sets of constraints on $x(t)$ and/or $X(j\omega)$, does the sampling theorem (see Section 7.1.1) guarantee that $x(t)$ can be recovered exactly from $x_p(t)$?

(a) $X(j\omega) = 0$ for $|\omega| > 5000\pi$
(b) $X(j\omega) = 0$ for $|\omega| > 15000\pi$
(c) $\mathcal{R}e\{X(j\omega)\} = 0$ for $|\omega| > 5000\pi$
(d) $x(t)$ real and $X(j\omega) = 0$ for $\omega > 5000\pi$
(e) $x(t)$ real and $X(j\omega) = 0$ for $\omega < -15000\pi$
(f) $X(j\omega) * X(j\omega) = 0$ for $|\omega| > 15000\pi$
(g) $|X(j\omega)| = 0$ for $\omega > 5000\pi$

7.22. The signal $y(t)$ is generated by convolving a band-limited signal $x_1(t)$ with another band-limited signal $x_2(t)$, that is,

$$y(t) = x_1(t) * x_2(t)$$

where

$$X_1(j\omega) = 0 \qquad \text{for } |\omega| > 1000\pi$$
$$X_2(j\omega) = 0 \qquad \text{for } |\omega| > 2000\pi.$$

Impulse-train sampling is performed on $y(t)$ to obtain

$$y_p(t) = \sum_{n=-\infty}^{+\infty} y(nT)\delta(t - nT).$$

Specify the range of values for the sampling period T which ensures that $y(t)$ is recoverable from $y_p(t)$.

7.23. Shown in Figure P7.23 is a system in which the sampling signal is an impulse train with alternating sign. The Fourier transform of the input signal is as indicated in the figure.

(a) For $\Delta < \pi/(2\omega_M)$, sketch the Fourier transform of $x_p(t)$ and $y(t)$.

(b) For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $x_p(t)$.

(c) For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $y(t)$.

(d) What is the *maximum* value of Δ in relation to ω_M for which $x(t)$ can be recovered from either $x_p(t)$ or $y(t)$?

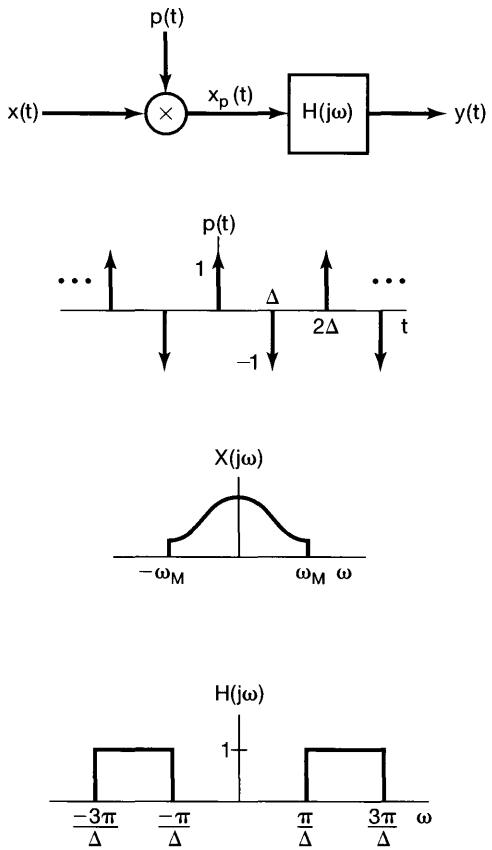


Figure P7.23

7.24. Shown in Figure P7.24 is a system in which the input signal is multiplied by a periodic square wave. The period of $s(t)$ is T. The input signal is band limited with $|X(j\omega)| = 0$ for $|\omega| \geq \omega_M$.

(a) For $\Delta = T/3$, determine, in terms of ω_M, the maximum value of T for which there is no aliasing among the replicas of $X(j\omega)$ in $W(j\omega)$.

(b) For $\Delta = T/4$, determine, in terms of ω_M, the maximum value of T for which there is no aliasing among the replicas of $X(j\omega)$ in $W(j\omega)$.

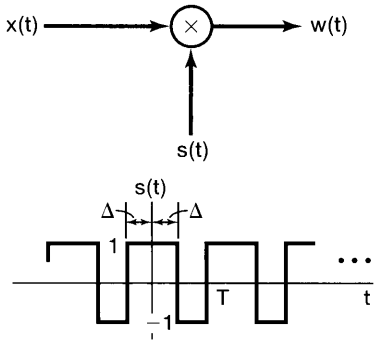


Figure P7.24

7.25. In Figure P7.25 is a sampler, followed by an ideal lowpass filter, for reconstruction of $x(t)$ from its samples $x_p(t)$. From the sampling theorem, we know that if $\omega_s = 2\pi/T$ is greater than twice the highest frequency present in $x(t)$ and $\omega_c = \omega_s/2$, then the reconstructed signal $x_r(t)$ will exactly equal $x(t)$. If this condition on the bandwidth of $x(t)$ is violated, then $x_r(t)$ will *not* equal $x(t)$. We seek to show in this problem that if $\omega_c = \omega_s/2$, then for any choice of T, $x_r(t)$ and $x(t)$ will always be equal at the sampling instants; that is,

$$x_r(kT) = x(kT), k = 0, \pm 1, \pm 2, \ldots.$$

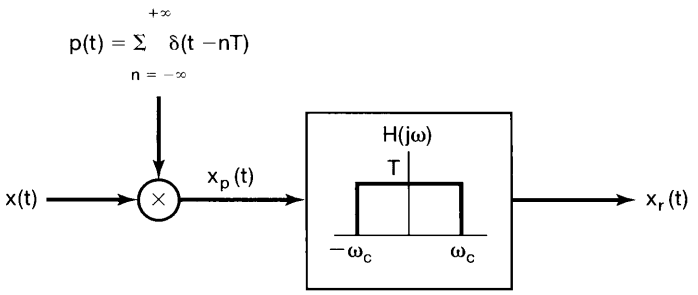


Figure P7.25

To obtain this result, consider eq. (7.11), which expresses $x_r(t)$ in terms of the samples of $x(t)$:

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT)T\frac{\omega_c}{\pi}\frac{\sin[\omega_c(t-nT)]}{\omega_c(t-nT)}.$$

With $\omega_c = \omega_s/2$, this becomes

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT)\frac{\sin\left[\frac{\pi}{T}(t-nT)\right]}{\frac{\pi}{T}(t-nT)}. \qquad \text{(P7.25–1)}$$

By considering the values of α for which $[\sin(\alpha)]/\alpha = 0$, show from eq. (P7.25–1) that, without any restrictions on $x(t)$, $x_r(kT) = x(kT)$ for any integer value of k.

7.26. The sampling theorem, as we have derived it, states that a signal $x(t)$ must be sampled at a rate greater than its bandwidth (or equivalently, a rate greater than twice its highest frequency). This implies that if $x(t)$ has a spectrum as indicated in Figure P7.26(a) then $x(t)$ must be sampled at a rate greater than $2\omega_2$. However, since the signal has most of its energy concentrated in a narrow band, it would seem reasonable to expect that a sampling rate lower than twice the highest frequency could be used. A signal whose energy is concentrated in a frequency band is often referred to as a *bandpass signal*. There are a variety of techniques for sampling such signals, generally referred to as *bandpass-sampling* techniques.

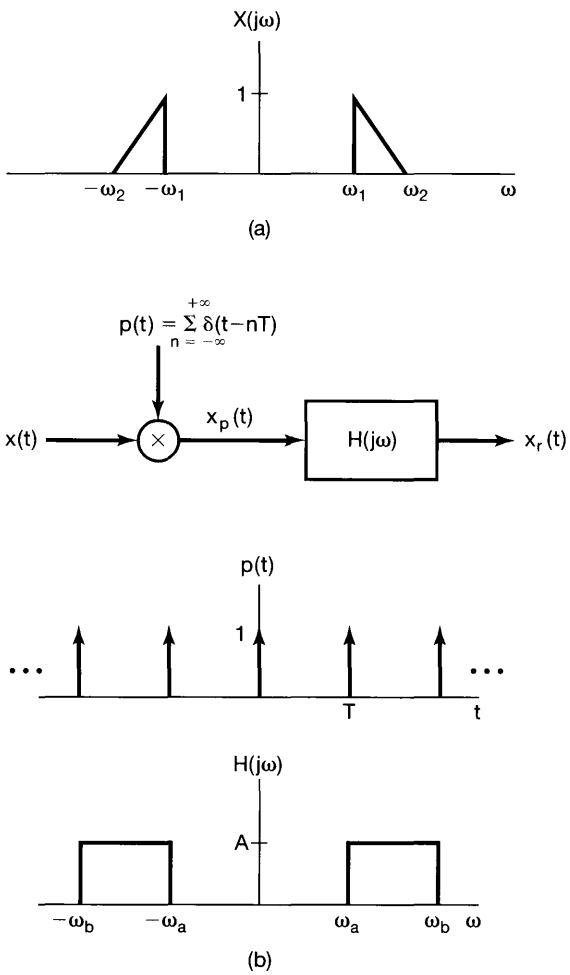


Figure P7.26

To examine the possibility of sampling a bandpass signal as a rate less than the total bandwidth, consider the system shown in Figure P7.26(b). Assuming that $\omega_1 > \omega_2 - \omega_1$, find the maximum value of T and the values of the constants A, ω_a, and ω_b such that $x_r(t) = x(t)$.

7.27. In Problem 7.26, we considered one procedure for bandpass sampling and reconstruction. Another procedure, used when $x(t)$ is real, consists of multiplying $x(t)$ by a complex-exponential and then sampling the product. The sampling system is shown in Figure P7.27(a). With $x(t)$ real and with $X(j\omega)$ nonzero only for $\omega_1 < |\omega| < \omega_2$, the frequency is chosen to be $\omega_0 = (1/2)(\omega_1 + \omega_2)$, and the lowpass filter $H_1(j\omega)$ has cutoff frequency $(1/2)(\omega_2 - \omega_1)$.

(a) For $X(j\omega)$ as shown in Figure P7.27(b), sketch $X_p(j\omega)$.

(b) Determine the maximum sampling period T such that $x(t)$ is recoverable from $x_p(t)$.

(c) Determine a system to recover $x(t)$ from $x_p(t)$.

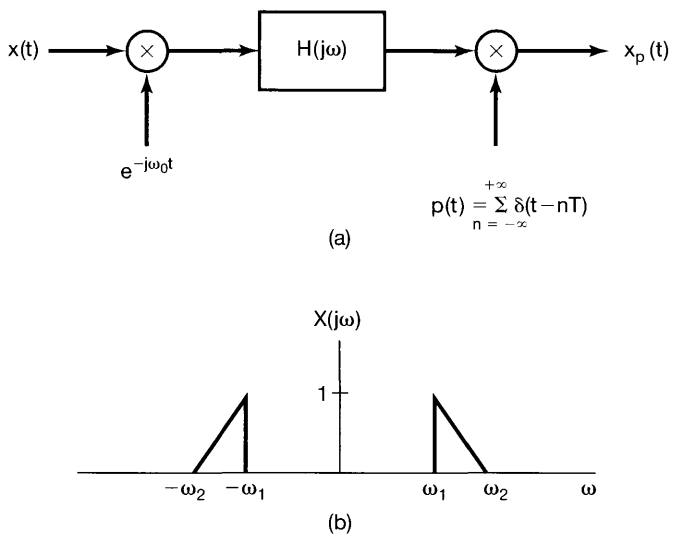


Figure P7.27

7.28. Figure P7.28(a) shows a system that converts a continuous-time signal to a discrete-time signal. The input $x(t)$ is periodic with a period of 0.1 second. The Fourier series coefficients of $x(t)$ are

$$a_k = \left(\frac{1}{2}\right)^{|k|}, \quad -\infty < k < +\infty.$$

The lowpass filter $H(j\omega)$ has the frequency response shown in Figure P7.28(b). The sampling period $T = 5 \times 10^{-3}$ second.

(a) Show that $x[n]$ is a periodic sequence, and determine its period.

(b) Determine the Fourier series coefficients of $x[n]$.

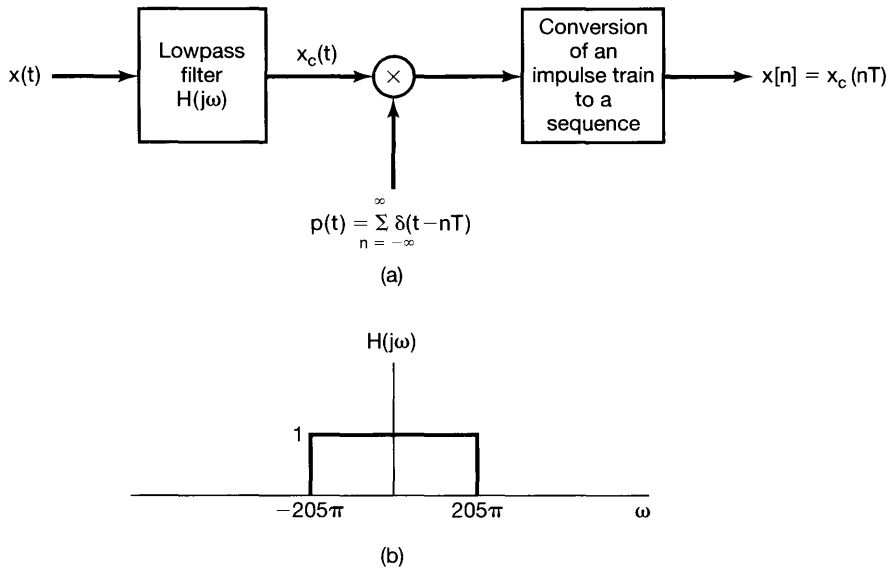


Figure P7.28

7.29. Figure P7.29(a) shows the overall system for filtering a continuous-time signal using a discrete-time filter. If $X_c(j\omega)$ and $H(e^{j\omega})$ are as shown in Figure P7.29(b), with $1/T = 20$ kHz, sketch $X_p(j\omega)$, $X(e^{j\omega})$, $Y(e^{j\omega})$, $Y_p(j\omega)$, and $Y_c(j\omega)$.

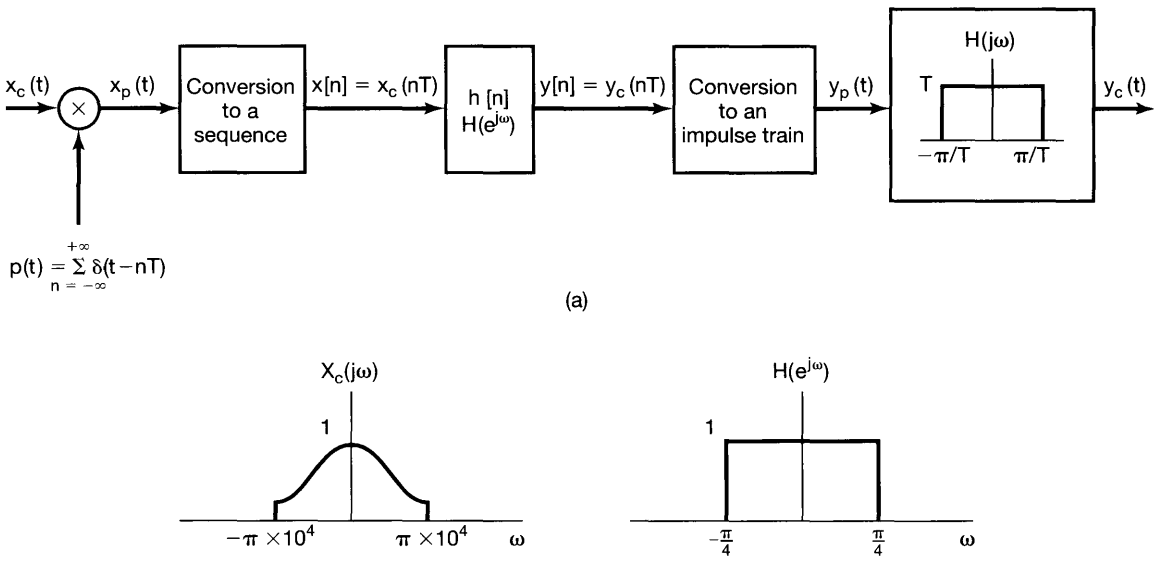


Figure P7.29

7.30. Figure P7.30 shows a system consisting of a continuous-time LTI system followed by a sampler, conversion to a sequence, and an LTI discrete-time system. The continuous-time LTI system is causal and satisfies the linear, constant-coefficient differential equation

$$\frac{dy_c(t)}{dt} + y_c(t) = x_c(t).$$

The input $x_c(t)$ is a unit impulse $\delta(t)$.

(a) Determine $y_c(t)$.

(b) Determine the frequency response $H(e^{j\omega})$ and the impulse response $h[n]$ such that $w[n] = \delta[n]$.

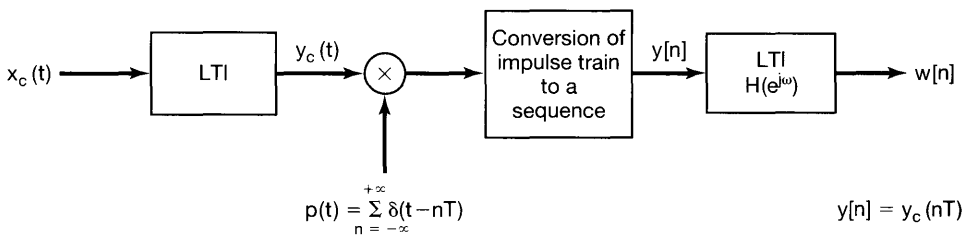


Figure P7.30

7.31. Shown in Figure P7.31 is a system that processes continuous-time signals using a digital filter $h[n]$ that is linear and causal with difference equation

$$y[n] = \frac{1}{2}y[n-1] + x[n].$$

For input signals that are band limited such that $X_c(j\omega) = 0$ for $|\omega| > \pi/T$, the system in the figure is equivalent to a continuous-time LTI system.

Determine the frequency response $H_c(j\omega)$ of the equivalent overall system with input $x_c(t)$ and output $y_c(t)$.

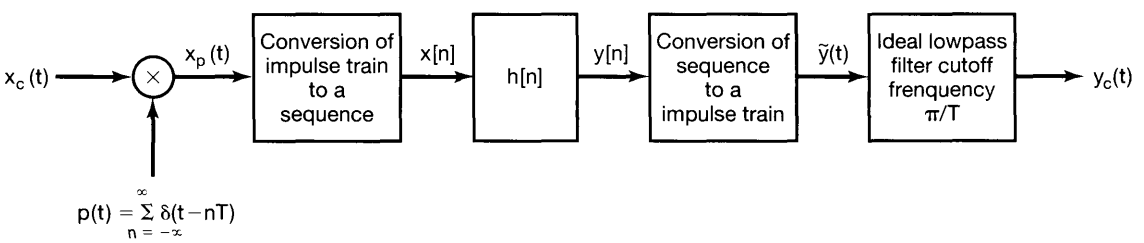


Figure P7.31

7.32. A signal $x[n]$ has a Fourier transform $X(e^{j\omega})$ that is zero for $(\pi/4) \leq |\omega| \leq \pi$. Another signal

$$g[n] = x[n] \sum_{k=-\infty}^{\infty} \delta[n - 1 - 4k]$$

is generated. Specify the frequency response $H(e^{j\omega})$ of a lowpass filter that produces $x[n]$ as output when $g[n]$ is the input.

7.33. A signal $x[n]$ with Fourier transform $X(e^{j\omega})$ has the property that

$$\left(x[n] \sum_{k=-\infty}^{\infty} \delta[n - 3k]\right) * \left(\frac{\sin \frac{\pi}{3}n}{\frac{\pi}{3}n}\right) = x[n].$$

For what values of ω is it guaranteed that $X(e^{j\omega}) = 0$?

7.34. A real-valued discrete-time signal $x[n]$ has a Fourier transform $X(e^{j\omega})$ that is zero for $3\pi/14 \leq |\omega| \leq \pi$. The nonzero portion of the Fourier transform of one period of $X(e^{j\omega})$ can be made to occupy the region $|\omega| < \pi$ by first performing upsampling by a factor of L and then performing downsampling by a factor of M. Specify the values of L and M.

7.35. Consider a discrete-time sequence $x[n]$ from which we form two new sequences, $x_p[n]$ and $x_d[n]$, where $x_p[n]$ corresponds to sampling $x[n]$ with a sampling period of 2 and $x_d[n]$ corresponds to decimating $x[n]$ by a factor of 2, so that

$$x_p[n] = \begin{cases} x[n], & n = 0, \pm 2, \pm 4, \ldots \\ 0, & n = \pm 1, \pm 3, \ldots \end{cases}$$

and

$$x_d[n] = x[2n].$$

(a) If $x[n]$ is as illustrated in Figure P7.35(a), sketch the sequences $x_p[n]$ and $x_d[n]$.
(b) If $X(e^{j\omega})$ is as shown in Figure P7.35(b), sketch $X_p(e^{j\omega})$ and $X_d(e^{j\omega})$.

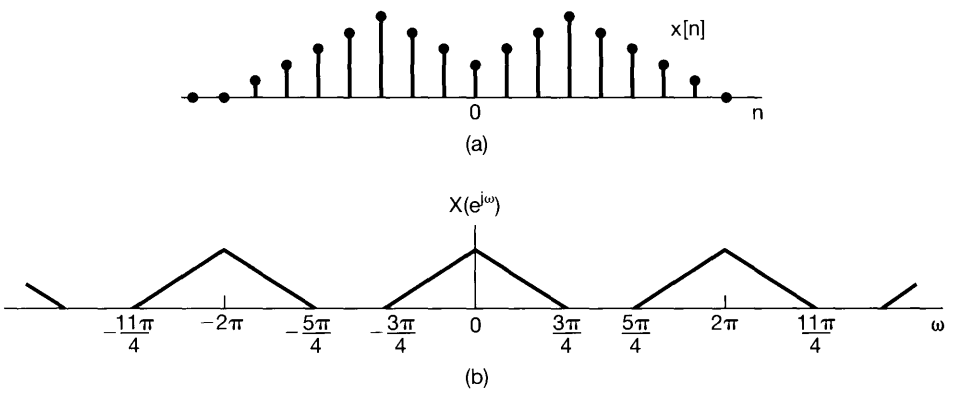


Figure P7.35

ADVANCED PROBLEMS

7.36 Let $x(t)$ be a band-limited signal such that $X(j\omega) = 0$ for $|\omega| \geq \frac{\pi}{T}$.
(a) If $x(t)$ is sampled using a sampling period T, determine an interpolating function

$g(t)$ such that

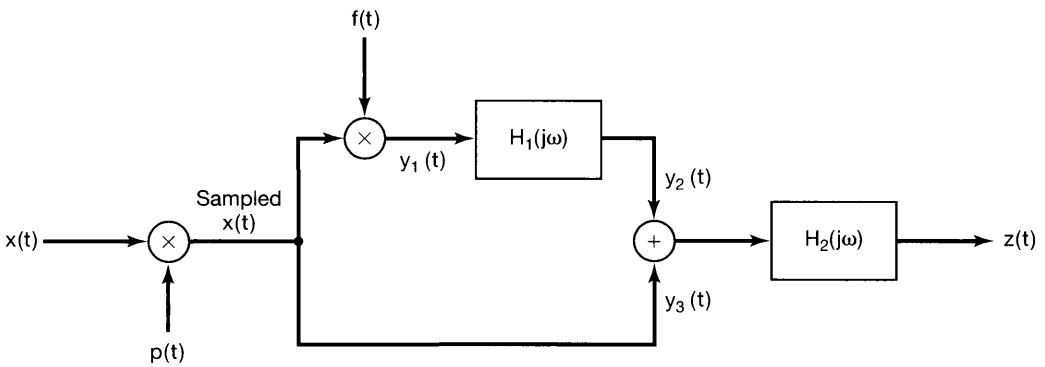
$$\frac{dx(t)}{dt} = \sum_{n=-\infty}^{\infty} x(nT)g(t - nT).$$

(b) Is the function $g(t)$ unique?

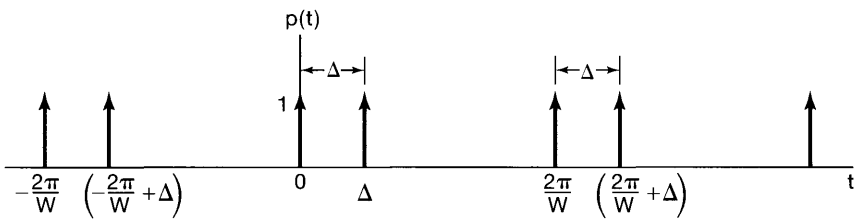
7.37. A signal limited in bandwidth to $|\omega| < W$ can be recovered from nonuniformly spaced samples as long as the average sample density is $2(W/2\pi)$ samples per second. This problem illustrates a particular example of nonuniform sampling. Assume that in Figure P7.37(a):

1. $x(t)$ is band limited; $X(j\omega) = 0, |\omega| > W$.
2. $p(t)$ is a nonuniformly spaced periodic pulse train, as shown in Figure P7.37(b).
3. $f(t)$ is a periodic waveform with period $T = 2\pi/W$. Since $f(t)$ multiplies an impulse train, only its values $f(0) = a$ and $f(\Delta) = b$ at $t = 0$ and $t = \Delta$, respectively, are significant.
4. $H_1(j\omega)$ is a 90° phase shifter; that is,

$$H_1(j\omega) = \begin{cases} j, & \omega > 0 \\ -j, & \omega < 0 \end{cases}.$$



(a)



(b)

Figure P7.37

5. $H_2(j\omega)$ is an ideal lowpass filter; that is,

$$H_2(j\omega) = \begin{cases} K, & 0 < \omega < W \\ K^*, & -W < \omega < 0 \\ 0, & |\omega| > W \end{cases},$$

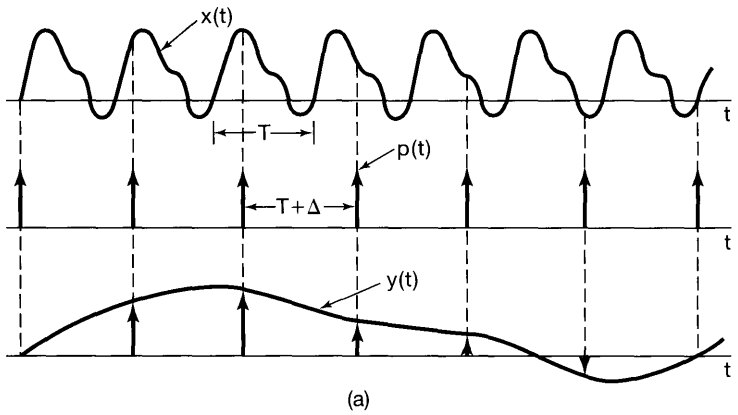
where K is a (possibly complex) constant.

(a) Find the Fourier transforms of $p(t)$, $y_1(t)$, $y_2(t)$, and $y_3(t)$.

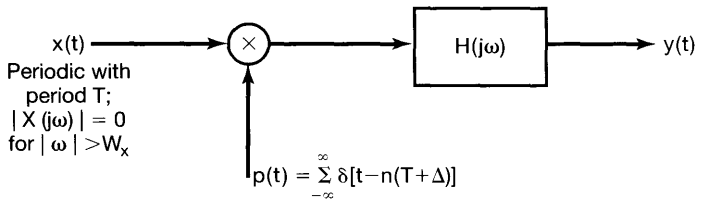
(b) Specify the values of a, b, and K as functions of Δ such that $z(t) = x(t)$ for any band-limited $x(t)$ and any Δ such that $0 < \Delta < \pi/W$.

7.38. It is frequently necessary to display on an oscilloscope screen waveforms having very short time structures—for example, on the scale of thousandths of a nanosecond. Since the rise time of the fastest oscilloscope is longer than this, such displays cannot be achieved directly. If however, the waveform is periodic, the desired result can be obtained indirectly by using an instrument called a sampling oscilloscope.

The idea, as shown in Figure P7.38(a), is to sample the fast waveform $x(t)$ once each period, but at successively later points in successive periods. The increment Δ should be an appropriately chosen sampling interval in relation to the bandwidth of $x(t)$. If the resulting impulse train is then passed through an appropriate interpolat-



(a)



$$H(j\omega) = \begin{cases} 1, & |\omega| < \dfrac{1}{2(T+\Delta)} \\ 0, & \text{elsewhere} \end{cases}$$

(b)

Figure P7.38

ing lowpass filter, the output $y(t)$ will be proportional to the original fast waveform slowed down or stretched out in time [i.e., $y(t)$ is proportional to $x(at)$, where $a < 1$].

For $x(t) = A + B\cos[(2\pi/T)t + \theta]$, find a range of values of Δ such that $y(t)$ in Figure P7.38(b) is proportional to $x(at)$ with $a < 1$. Also, determine the value of a in terms of T and Δ.

7.39 A signal $x_p(t)$ is obtained through impule-train sampling of a sinusoidal signal $x(t)$ whose frequency is equal to half the sampling frequency ω_s.

$$x(t) = \cos\left(\frac{\omega_s}{2}t + \phi\right)$$

and

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT)$$

where $T = 2\pi/\omega_s$.

(a) Find $g(t)$ such that

$$x(t) = \cos(\phi)\cos\left(\frac{\omega_s}{2}t\right) + g(t).$$

(b) Show that

$$g(nT) = 0 \qquad \text{for } n = 0, \pm 1, \pm 2, \cdots$$

(c) Using the results of the previous two parts, show that if $x_p(t)$ is applied as the input to an ideal lowpass filter with cutoff frequency $\omega_s/2$, the resulting output is

$$y(t) = \cos(\phi)\cos\left(\frac{\omega_s}{2}t\right).$$

7.40. Consider a disc on which four cycles of a sinusoid are painted. The disc is rotated at approximately 15 revolutions per second, so that the sinusoid, when viewed through a narrow slit, has a frequency of 60 Hz.

The arrangement is indicated in Figure P7.40. Let $v(t)$ denote the position of the line seen through the slit. Then

$$v(t) = A\cos(\omega_0 t + \phi), \omega_0 = 120\pi.$$

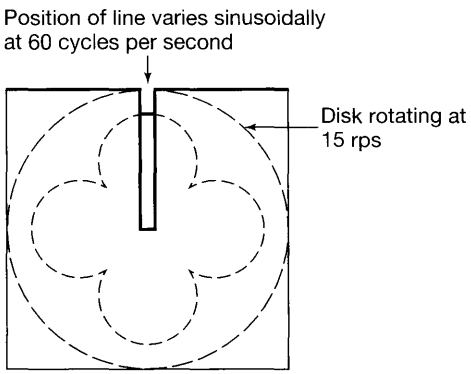


Figure P7.40

For notational convenience, we will normalize $v(t)$ so that $A = 1$. At 60 Hz, the eye is not able to follow $v(t)$, and we will assume that this effect can be explained by modeling the eye as an ideal lowpass filter with cutoff frequency 20 Hz.

Sampling of the sinusoid can be accomplished by illuminating the disc with a strobe light. Thus, the illumination can be represented by an impulse train; that is,

$$i(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT),$$

where $1/T$ is the strobe frequency in hertz. The resulting sampled signal is the product $r(t) = v(t)i(t)$. Let $R(j\omega)$, $V(j\omega)$, and $I(j\omega)$ denote the Fourier transforms of $r(t)$, $v(t)$, and $i(t)$, respectively.

(a) Sketch $V(j\omega)$, indicating clearly the effect of the parameters ϕ and ω_0.

(b) Sketch $I(j\omega)$, indicating the effect of T.

(c) According to the sampling theorem, there is a maximum value for T in terms of ω_0 such that $v(t)$ can be recovered from $r(t)$ using a lowpass filter. Determine this value of T and the cutoff frequency of the lowpass filter. Sketch $R(j\omega)$ when T is slightly less than the maximum value.

If the sampling period T is made greater than the value determined in part (c), aliasing of the spectrum occurs. As a result of this aliasing, we perceive a lower frequency sinusoid.

(d) Suppose that $2\pi/T = \omega_0 + 20\pi$. Sketch $R(j\omega)$ for $|\omega| < 40\pi$. Denote by $v_a(t)$ the apparent position of the line as we perceive it. Assuming that the eye behaves as an ideal lowpass filter with 20-Hz cutoff and unity gain, express $v_a(t)$ in the form

$$v_a(t) = A_a \cos(\omega_a + \phi_a),$$

where A_a is the apparent amplitude, ω_a the apparent frequency, and ϕ_a the apparent phase of $v_a(t)$.

(e) Repeat part (d) for $2\pi/T = \omega_0 - 20\pi$.

7.41. In many practical situations a signal is recorded in the presence of an echo, which we would like to remove by appropriate processing. For example, in Figure P7.41(a), we illustrate a system in which a receiver simultaneously receives a signal $x(t)$ and an echo represented by an attenuated delayed replication of $x(t)$. Thus, the receiver output is $s(t) = x(t) + \alpha x(t - T_0)$, where $|\alpha| < 1$. This output is to be processed to recover $x(t)$ by first converting to a sequence and then using an appropriate digital filter $h[n]$, as indicated in Figure P7.41(b).

Assume that $x(t)$ is band limited [i.e., $X(j\omega) = 0$ for $|\omega| > \omega_M$] and that $|\alpha| < 1$.

(a) If $T_0 < \pi/\omega_M$, and the sampling period is taken to be equal to T_0 (i.e., $T = T_0$), determine the difference equation for the digital filter $h[n]$ such that $y_c(t)$ is proportional to $x(t)$.

(b) With the assumptions of part (a), specify the gain A of the ideal lowpass filter such that $y_c(t) = x(t)$.

(c) Now suppose that $\pi/\omega_M < T_0 < 2\pi/\omega_M$. Determine a choice for the sampling period T, the lowpass filter gain A, and the frequency response for the digital filter $h[n]$ such that $y_c(t)$ is proportional to $x(t)$.

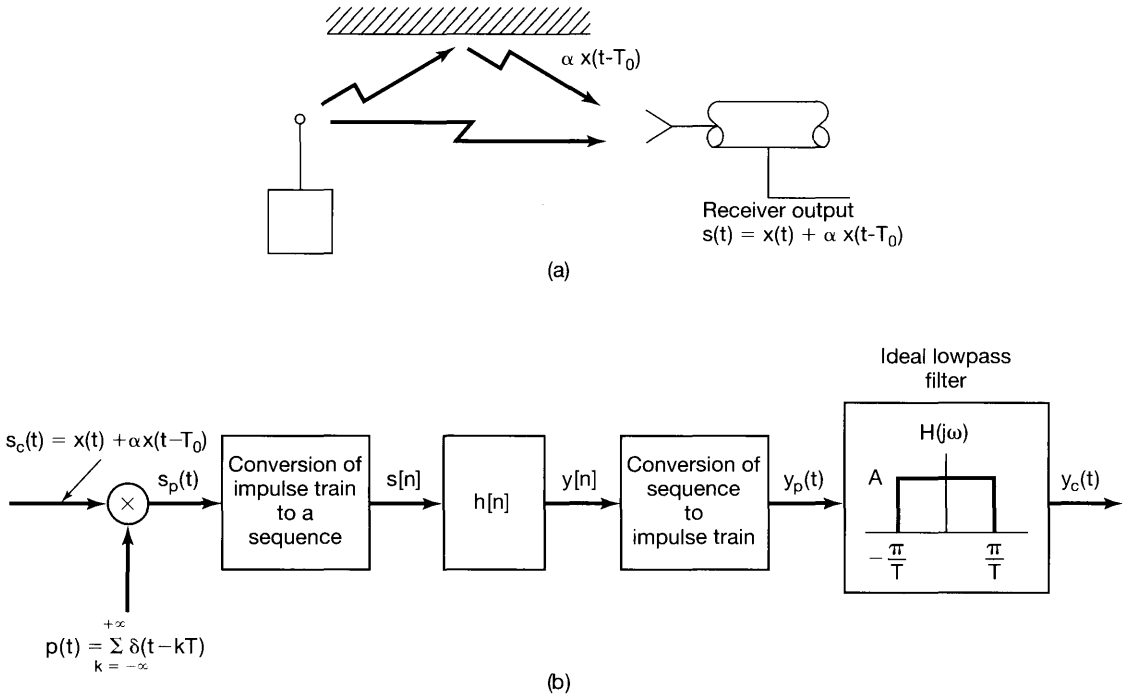


Figure P7.41

7.42. Consider a band-limited signal $x_c(t)$ that is sampled at a rate higher than the Nyquist rate. The samples, spaced T seconds apart, are then converted to a sequence $x[n]$, as indicated in Figure P7.42.

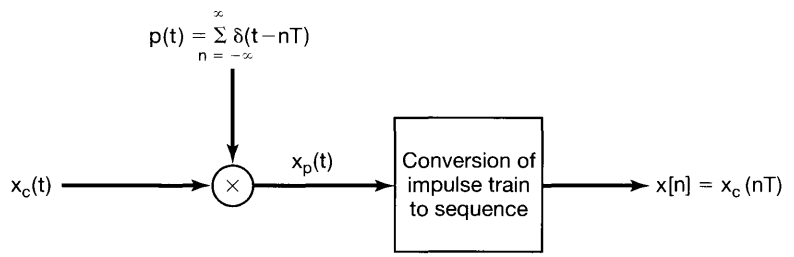


Figure P7.42

Determine the relation between the energy E_d of the sequence, the energy E_c of the original signal, and the sampling interval T. The energy of a sequence $x[n]$ is defined as

$$E_d = \sum_{n=-\infty}^{\infty} |x[n]|^2,$$

and the energy in a continuous-time function $x_c(t)$ is defined as

$$E_c = \int_{-\infty}^{+\infty} |x_c(t)|^2 \, dt.$$

7.43. Figure P7.43(a) depicts a system for which the input and output are discrete-time signals. The discrete-time input $x[n]$ is converted to a continuous-time impulse train $x_p(t)$. The continuous-time signal $x_p(t)$ is then filtered by an LTI system to produce the output $y_c(t)$, which is then converted to the discrete-time signal $y[n]$. The LTI system with input $x_c(t)$ and output $y_c(t)$ is causal and is characterized by the linear constant-coefficient differential equation

$$\frac{d^2y_c(t)}{dt^2} + 4\frac{dy_c(t)}{dt} + 3y_c(t) = x_c(t).$$

The overall system is equivalent to a causal discrete-time LTI system, as indicated in Figure P7.43(b).

Determine the frequency response $H(e^{j\omega})$ and the unit sample response $h[n]$ of the equivalent LTI system.

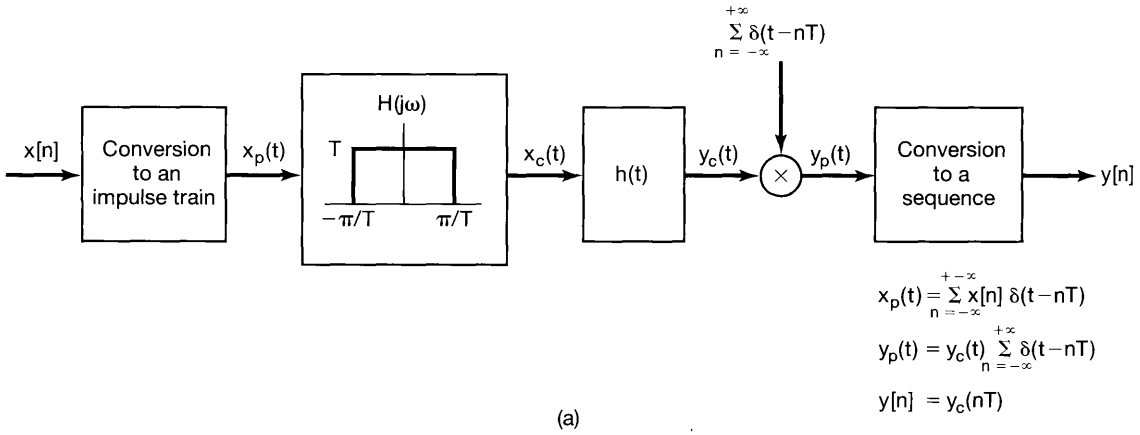


Figure P7.43

7.44. Suppose we wish to design a continuous-time generator that is capable of producing sinusoidal signals at any frequency satisfying

$$\omega_1 \leq \omega \leq \omega_2,$$

where ω_1 and ω_2 are given positive numbers.

Our design is to take the following form: We have stored a discrete-time cosine wave of period N; that is, we have stored $x[0], \ldots, x[N-1]$, where

$$x[k] = \cos\left(\frac{2\pi k}{N}\right).$$

Every T seconds we output an impulse weighted by a value of $x[k]$, where we proceed through the values of $k = 0, 1, \ldots, N-1$ in a cyclic fashion. That is,

$$y_p(kT) = x(k \text{ modulo } N),$$

or equivalently,

$$y_p(kT) = \cos\left(\frac{2\pi k}{N}\right),$$

and

$$y_p(t) = \sum_{k=-\infty}^{+\infty} \cos\left(\frac{2\pi k}{N}\right)\delta(t - kT).$$

(a) Show that by adjusting T, we can adjust the frequency of the cosine signal being sampled. That is, show that

$$y_p(t) = (\cos\omega_0 t)\sum_{k=-\infty}^{+\infty}\delta(t - kT),$$

where $\omega_0 = 2\pi/NT$. Determine a range of values for T such that $y_p(t)$ can represent samples of a cosine signal with a frequency that is variable over the full range

$$\omega_1 \leq \omega \leq \omega_2.$$

(b) Sketch $Y_p(j\omega)$.

The overall system for generating a continuous-time sinusoid is depicted in Figure P7.44(a). $H(j\omega)$ is an ideal lowpass filter with unity gain in its passband; that is,

$$H(j\omega) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \text{otherwise} \end{cases}.$$

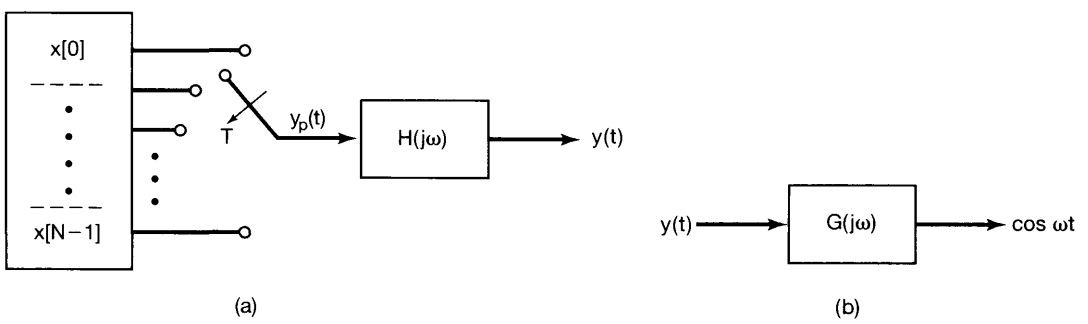


Figure P7.44

The parameter ω_c is to be determined so that $y(t)$ is a continuous-time cosine signal in the desired frequency band.

(c) Consider any value of T in the range determined in part (a). Determine the minimum value of N and some value for ω_c such that $y(t)$ is a cosine signal in the range $\omega_1 \leq \omega \leq \omega_2$.

(d) The amplitude of $y(t)$ will vary, depending upon the value of ω chosen between ω_1 and ω_2. Thus, we must design a system $G(j\omega)$ that normalizes the signal as shown in Figure P7.44(b). Find such a $G(j\omega)$.

7.45. In the system shown in Figure P7.45, the input $x_c(t)$ is band limited with $X_c(j\omega) = 0$, $|\omega| > 2\pi \times 10^4$. The digital filter $h[n]$ is described by the input-output relation

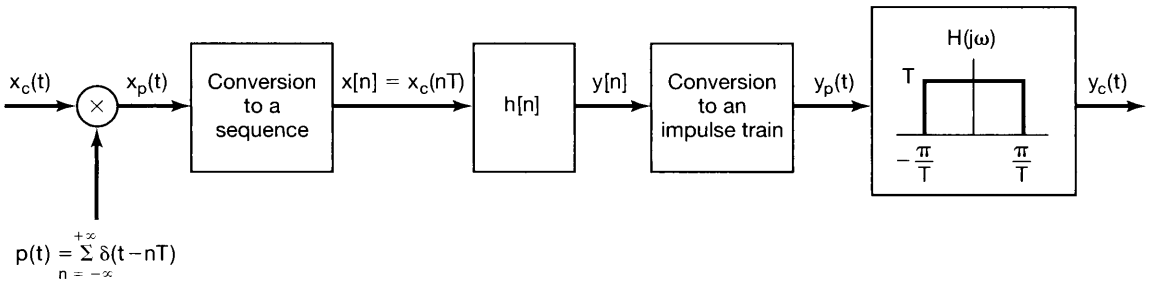


Figure P7.45

$$y[n] = T \sum_{k=-\infty}^{n} x[k]. \tag{P7.45–1}$$

(a) What is the maximum value of T allowed if aliasing is to be avoided in the transformation from $x_c(t)$ to $x_p(t)$.

(b) With the discrete-time LTI system specified through eq. (P7.45–1), determine its impulse response $h[n]$.

(c) Determine whether there is any value of T for which

$$\lim_{n\to\infty} y[n] = \lim_{t\to\infty} \int_{-\infty}^{t} x_c(\tau)\, d\tau. \tag{P7.45–2}$$

If so, determine the *maximum* value. If not, explain and specify how T would be chosen so that the equality in eq. (P7.45–2) is best approximated. (Think carefully about this part; it is easy to jump to the wrong conclusion!)

7.46 A signal $x[n]$ is sampled in discrete time as shown in Figure P7.46. $h_r[n]$ is an ideal lowpass filter with frequency response

$$H_r(e^{j\omega}) = \begin{cases} 1, & |\omega| < \frac{\pi}{N} \\ 0, & \frac{\pi}{N} < |\omega| < \pi \end{cases}$$

From eqs. (7.46) and (7.47), the filter output is expressible as

$$x_r[n] = \sum_{k=-\infty}^{+\infty} x[kN]h_r[n - kN] = \sum_{k=-\infty}^{+\infty} x[kN]\frac{N\omega_c}{\pi}\frac{\sin\omega_c(n - kN)}{\omega_c(n - kN)}$$

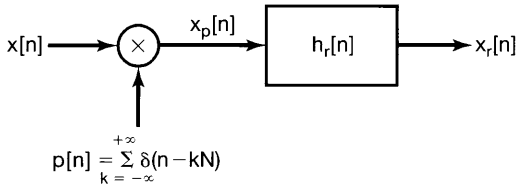


Figure P7.46

where $\omega_c = 2\pi/N$. Show that independent of whether the sequence $x[n]$ is sampled above or below the Nyquist rate, $x_r[mN] = x[mN]$, where m is any positive or negative integer.

7.47. Suppose $x[n]$ has a Fourier transform that is zero for $\pi/3 \leq |\omega| \leq \pi$. Show that

$$x[n] = \sum_{k=-\infty}^{\infty} x[3k]\left(\frac{\sin(\frac{\pi}{3}(n-3k))}{\frac{\pi}{3}(n-3k)}\right).$$

7.48. If $x[n] = \cos(\frac{\pi}{4}n+\phi_0)$ with $0 \leq \phi_0 < 2\pi$ and $g[n] = x[n]\sum_{k=-\infty}^{\infty}\delta[n-4k]$, what additional constraints must be imposed on ϕ_0 to ensure that

$$g[n] * \left(\frac{\sin\frac{\pi}{4}n}{\frac{\pi}{4}n}\right) = x[n]?$$

7.49. As discussed in Section 7.5 and illustrated in Figure 7.37, the procedure for interpolation or upsampling by an integer factor N can be thought of as the cascade of two operations. The first operation, involving system A, corresponds to inserting $N-1$ zero-sequence values between each sequence value of $x[n]$, so that

$$x_p[n] = \begin{cases} x_d\left[\frac{n}{N}\right], & n = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}.$$

For exact band-limited interpolation, $H(e^{j\omega})$ is an ideal lowpass filter.

(a) Determine whether or not system A is linear.

(b) Determine whether or not system A is time invariant.

(c) For $X_d(e^{j\omega})$ as sketched in Figure P7.49 and with $N = 3$, sketch $X_p(e^{j\omega})$.

(d) For $N = 3$, $X_d(e^{j\omega})$ as in Figure P7.49, and $H(e^{j\omega})$ appropriately chosen for exact band-limited interpolation, sketch $X(e^{j\omega})$.

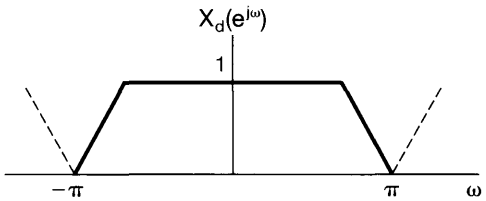


Figure P7.49

7.50. In this problem, we consider the discrete-time counterparts of the zero-order hold and first-order hold, which were discussed for continuous time in Sections 7.1.2 and 7.2.

Let $x[n]$ be a sequence to which discrete-time sampling, as illustrated in Figure 7.31, has been applied. Suppose the conditions of the discrete-time sampling theorem are satisfied; that is, $\omega_s > 2\omega_M$, where ω_s is the sampling frequency and $X(e^{j\omega}) = 0$, $\omega_M < |\omega| \leq \pi$. The original signal $x[n]$ is then exactly recoverable from $x_p[n]$ by ideal lowpass filtering, which, as discussed in Section 7.5, corresponds to band-limited interpolation.

The zero-order hold represents an approximate interpolation whereby every sample value is repeated (or held) $N - 1$ successive times, as illustrated in Figure P7.50(a) for the case of $N = 3$. The first-order hold represents a linear interpolation between samples, as illustrated in the same figure.

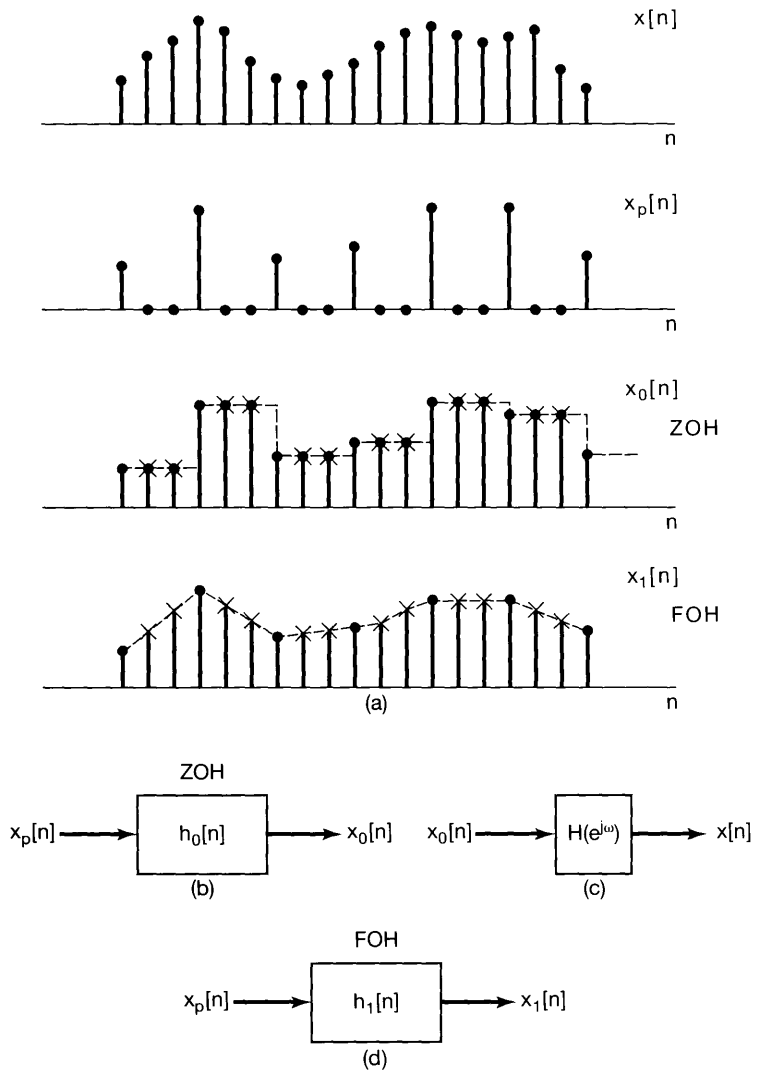


Figure P7.50

(a) The zero-order hold can be represented as an interpolation in the form of eq. (7.47) or, equivalently, the system in Figure P7.50(b). Determine and sketch $h_0[n]$ for the general case of a sampling period N.

(b) $x[n]$ can be exactly recovered from the zero-order-hold sequence $x_0[n]$ using an appropriate LTI filter $H(e^{j\omega})$, as indicated in Figure P7.50(c). Determine and sketch $H(e^{j\omega})$.

(c) The first-order-hold (linear interpolation) can be represented as an interpolation in the form of eq. (7.47) or, equivalently, the system in Figure P7.50(d). Determine and sketch $h_1[n]$ for the general case of a sampling period N.

(d) $x[n]$ can be exactly recovered from the first-order-hold sequence $x_1[n]$ using an appropriate LTI filter with frequency response $H(e^{j\omega})$. Determine and sketch $H(e^{j\omega})$.

7.51. As shown in Figure 7.37 and discussed in Section 7.5.2, the procedure for interpolation or upsampling by an integer factor N can be thought of as a cascade of two operations. For exact band-limited interpolation, the filter $H(e^{j\omega})$ in Figure 7.37 is an ideal lowpass filter. In any specific application, it would be necessary to implement an approximate lowpass filter. In this problem, we explore some useful constraints that are often imposed on the design of these approximate lowpass filters.

(a) Suppose that $H(e^{j\omega})$ is approximated by a zero-phase FIR filter. The filter is to be designed with the constraint that the original sequence values $x_d[n]$ get reproduced *exactly;* that is,

$$x[n] = x_d\left[\frac{n}{L}\right], n = 0, \pm L, \pm 2L, \ldots. \tag{P7.51–1}$$

This guarantees that, even though the interpolation between the original sequence values may not be exact, the original values are reproduced exactly in the interpolation. Determine the constraint on the impulse response $h[n]$ of the lowpass filter which guarantees that eq. (P7.51–1) will hold exactly for any sequence $x_d[n]$.

(b) Now suppose that the interpolation is to be carried out with a *linear-phase, causal, symmetric* FIR filter of length N; that is

$$h[n] = 0, n < 0, n > N - 1, \tag{P7.51–2}$$

$$H(e^{j\omega}) = H_R(e^{j\omega})e^{-j\alpha\omega}, \tag{P7.51–3}$$

where $H_R(e^{j\omega})$ is real. The filter is to be designed with the constraint that the original sequence values $x_d[n]$ get reproduced *exactly,* but with an integer delay α, where α is the negative of the slope of the phase of $H(e^{j\omega})$; that is,

$$x[n] = x_d\left[\frac{n-\alpha}{L}\right], n - \alpha = 0, \pm L, \pm 2L, \ldots \tag{P7.51–4}$$

Determine whether this imposes any constraint on whether the filter length N is odd or even.

(c) Again, suppose that the interpolation is to be carried out with a linear-phase, causal, symmetric FIR filter, so that

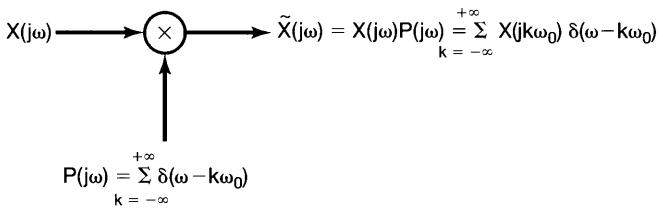
$$H(e^{j\omega}) = H_R(e^{j\omega})e^{-j\beta\omega},$$

where $H_R(e^{j\omega})$ is real. The filter is to be designed with the constraint that the original sequence values $x_d[n]$ get reproduced exactly, but with a delay M that is not necessarily equal to the slope of the phase; that is,

$$x[n] = x_d\left[\frac{n-\alpha}{L}\right], n - M = 0, \pm L, \pm 2L, \ldots.$$

Determine whether this imposes any constraint on whether the filter length N is odd or even.

7.52 In this problem we develop the dual to the time-domain sampling theorem, whereby a time-limited signal can be reconstructed from *frequency-domain* samples. To develop this result, consider the frequency-domain sampling operation in Figure P7.52.



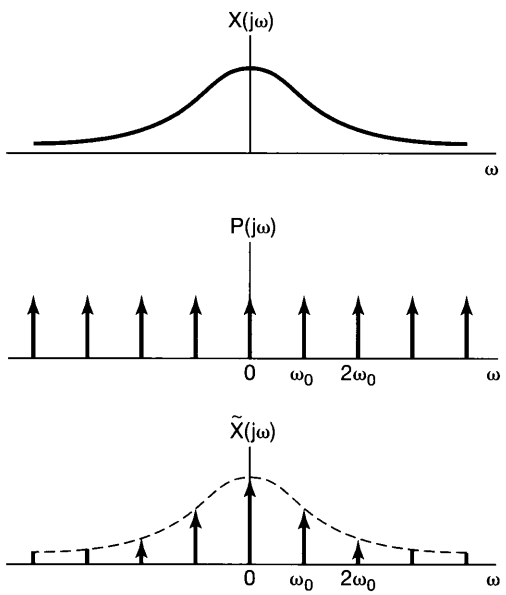


Figure P7.52

(a) Show that

$$\tilde{x}(t) = x(t) * p(t)$$

where $\tilde{x}(t)$, $x(t)$, and $p(t)$ are the inverse Fourier transforms of $\tilde{X}(j\omega)$, $X(j\omega)$, and $P(j\omega)$, respectively.

(b) Assuming that $x(t)$ is time-limited so that $x(t) = 0$ for $|t| \geq \frac{\pi}{\omega_o}$, show that $x(t)$ can be obtained from $\tilde{x}(t)$ through a "low-time windowing" operation. That is,

$$x(t) = \tilde{x}(t)w(t)$$

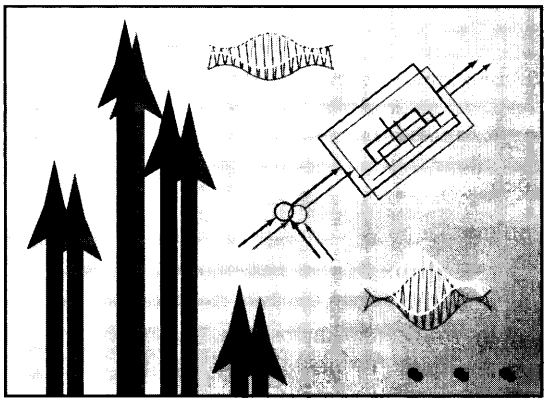
where

$$w(t) = \begin{cases} \omega_0, & |t| \leq \frac{\pi}{\omega_0} \\ 0, & |t| > -\frac{\pi}{\omega_0} \end{cases}$$

(c) Show that $x(t)$ is not recoverable from $\tilde{x}(t)$ if $x(t)$ is not constrained to be zero for $|t| \geq \frac{\pi}{\omega_0}$.

8

COMMUNICATION SYSTEMS



8.0 INTRODUCTION

Communication systems play a key role in our modern world in transmitting information between people, systems, and computers. In general terms, in all communication systems the information at the source is first processed by a transmitter or modulator to change it into a form suitable for transmission over the communication channel. At the receiver, the signal is then recovered through appropriate processing. This processing is required for a variety of reasons. In particular, quite typically, any specific communication channel has associated with it a frequency range over which it is best suited for transmitting a signal and outside of which communication is severely degraded or impossible. For example, the atmosphere will rapidly attenuate signals in the audible frequency range (10 Hz to 20 kHz), whereas it will propagate signals at a higher frequency range over longer distances. Thus, in transmitting audio signals such as speech or music over a communication channel that relies on propagation through the atmosphere, the transmitter first embeds the signal through an appropriate process into another, higher frequency signal.

Many of the concepts and techniques we have developed in the earlier chapters of this text play a central role in the analysis and design of communication systems. As with any concept that is closely tied to a wide variety of important applications, there are a large number of detailed issues to be considered, and, as indicated in the bibliography, there are many excellent texts on the subject. While a full and detailed analysis of communication systems is well beyond the scope of our discussions here, with the background of the previous chapters we are now in a position to introduce some of the basic principles and issues encountered in the design and analysis of these systems.

The general process of embedding an information-bearing signal into a second signal is typically referred to as *modulation.* Extracting the information-bearing signal

is known as *demodulation.* As we will see, modulation techniques not only allow us to embed information into signals that can be transmitted effectively, but also make possible the simultaneous transmission of more than one signal with overlapping spectra over the same channel, through a concept referred to as *multiplexing.*

There are a wide variety of modulation methods used in practice, and in this chapter we examine several of the most important of these. One large class of modulation methods relies on the concept of *amplitude modulation* or AM in which the signal we wish to transmit is used to modulate the amplitude of another signal. A very common form of amplitude modulation is *sinusoidal amplitude modulation,* which we explore in some detail in Sections 8.1–8.4 together with the related concepts of frequency-division multiplexing. Another important class of AM systems involves the modulation of the amplitude of a pulsed signal, and in Sections 8.5 and 8.6 we examine this form of modulation as well as the concept of time-division multiplexing. In Section 8.7 we then examine a different form of modulation, namely *sinusoidal frequency modulation* in which the information-bearing signal is used to vary the frequency of a sinusoidal signal.

All of the discussion up through Section 8.7 focuses attention on continuous-time signals, since most transmission media, such as the atmosphere, are best thought of as continuous-time phenomena. Nevertheless, not only is it possible to develop analogous techniques for discrete-time signals, but it is of considerable practical importance to consider modulation concepts involving such signals, and in Section 8.8 we examine some of the basic ideas behind the communication of discrete-time signals.

8.1 COMPLEX EXPONENTIAL AND SINUSOIDAL AMPLITUDE MODULATION

Many communication systems rely on the concept of sinusoidal amplitude modulation, in which a complex exponential or sinusoidal signal $c(t)$ has its amplitude multiplied (modulated) by the information-bearing signal $x(t)$. The signal $x(t)$ is typically referred to as the *modulating signal* and the signal $c(t)$ as the *carrier signal.* The modulated signal $y(t)$ is then the product of these two signals:

$$y(t) = x(t)c(t)$$

As we discussed in Section 8.0, an important objective in modulation is to produce a signal whose frequency range is suitable for transmission over the communication channel to be used. In telephone transmission systems, for example, long-distance transmission is often accomplished over microwave or satellite links. The individual voice signals are in the frequency range 200 Hz to 4 kHz, whereas a microwave link requires signals in the range 300 megahertz (MHz) to 300 gigahertz (GHz), and communication satellite links operate in the frequency range from a few hundred MHz to over 40 GHz. Thus, for transmission over these channels, the information in a voice signal must be shifted into these higher ranges of frequency. As we will see in this section, sinusoidal amplitude modulation achieves such a shift in frequency in a very simple manner.

8.1.1 Amplitude Modulation with a Complex Exponential Carrier

There are two common forms of sinusoidal amplitude modulation, one in which the carrier signal is a complex exponential of the form

$$c(t) = e^{j(\omega_c t + \theta_c)} \tag{8.1}$$

and the second in which the carrier signal is sinusoidal and of the form

$$c(t) = \cos(\omega_c t + \theta_c). \tag{8.2}$$

In both cases, the frequency ω_c is referred to as the *carrier frequency.* Let us consider first the case of a complex exponential carrier, and for convenience, let us choose $\theta_c = 0$, so that the modulated signal is

$$y(t) = x(t)e^{j\omega_c t}. \tag{8.3}$$

From the multiplication property (Section 4.5), and with $X(j\omega)$, $Y(j\omega)$, and $C(j\omega)$ denoting the Fourier transforms of $x(t)$, $y(t)$, and $c(t)$, respectively,

$$Y(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\theta)C(j(\omega - \theta))d\theta. \tag{8.4}$$

For $c(t)$ a complex exponential as given in eq. (8.1),

$$C(j\omega) = 2\pi\delta(\omega - \omega_c), \tag{8.5}$$

and hence,

$$Y(j\omega) = X(j\omega - j\omega_c). \tag{8.6}$$

Thus, the spectrum of the modulated output $y(t)$ is simply that of the input, shifted in frequency by an amount equal to the carrier frequency ω_c. For example, with $X(j\omega)$ band limited with highest frequency ω_M (and bandwidth $2\omega_M$), as depicted in Figure 8.1(a), the output spectrum $Y(j\omega)$ is that shown in Figure 8.1(c).

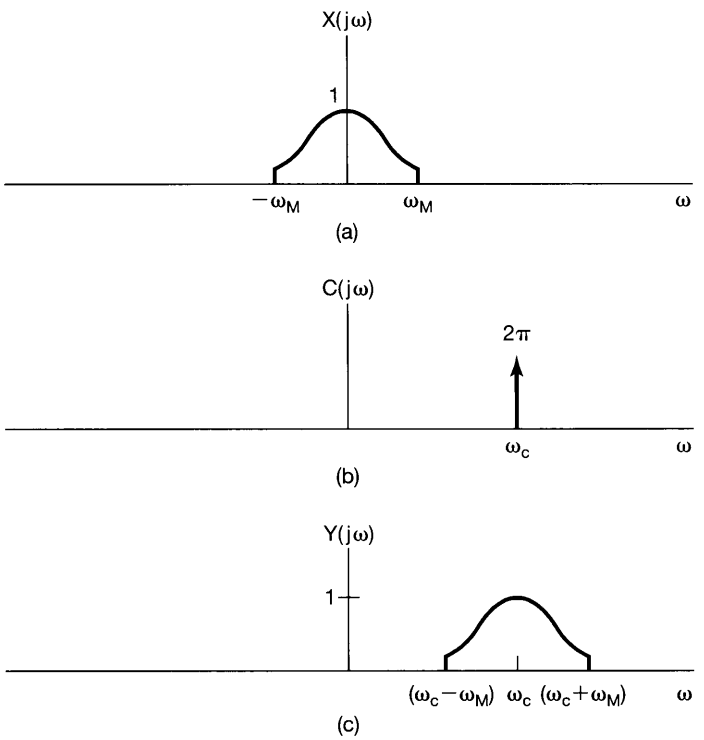


Figure 8.1 Effect in the frequency domain of amplitude modulation with a complex exponential carrier: (a) spectrum of modulating signal $x(t)$; (b) spectrum of carrier $c(t) = e^{j\omega_c t}$; (c) spectrum of amplitude-modulated signal $y(t) = x(t)e^{j\omega_c t}$.

From eq. (8.3), it is clear that $x(t)$ can be recovered from the modulated signal $y(t)$ by multiplying by the complex exponential $e^{-j\omega_c t}$; that is,

$$x(t) = y(t)e^{-j\omega_c t}. \tag{8.7}$$

In the frequency domain, this has the effect of shifting the spectrum of the modulated signal back to its original position on the frequency axis. The process of recovering the original signal from the modulated signal is referred to as *demodulation,* a topic we discuss at more length in Section 8.2.

Since $e^{j\omega_c t}$ is a complex signal, eq. (8.3) can be rewritten as

$$y(t) = x(t)\cos\omega_c t + jx(t)\sin\omega_c t. \tag{8.8}$$

Implementation of eq. (8.7) or (8.8) with $x(t)$ real utilizes two separate multipliers and two sinusoidal carrier signals that have a phase difference of $\pi/2$, as depicted in Figure 8.2 for $c(t)$ given by eq. (8.1). In Section 8.4 we give an example of one of the applications in which there are particular advantages to using a system, such as in Figure 8.2, employing two sinusoidal carriers with a phase difference of $\pi/2$.

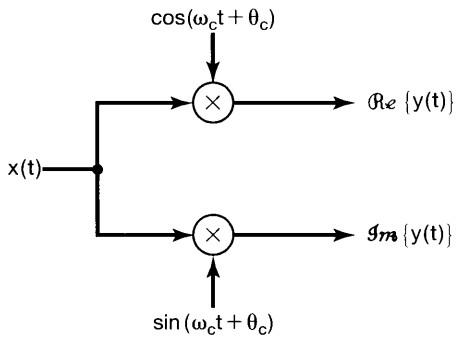


Figure 8.2 Implementation of amplitude modulation with a complex exponential carrier $c(t) = e^{j(\omega_c t+\theta_c)}$.

8.1.2 Amplitude Modulation with a Sinusoidal Carrier

In many situations, using a sinusoidal carrier of the form of eq. (8.2) is often simpler than and equally as effective as using a complex exponential carrier. In effect, using a sinusoidal carrier corresponds to retaining only the real or imaginary part of the output of Figure 8.2. A system that uses a sinusoidal carrier is depicted in Figure 8.3.

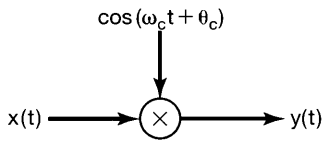


Figure 8.3 Amplitude modulation with a sinusoidal carrier.

The effect of amplitude modulation with a sinusoidal carrier in the form of eq. (8.2) can be analyzed in a manner identical to that in the preceding subsection. Again, for

convenience we choose $\theta_c = 0$. In this case, the spectrum of the carrier signal is

$$C(j\omega) = \pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)], \tag{8.9}$$

and thus, from eq. (8.4),

$$Y(j\omega) = \frac{1}{2}[X(j\omega - j\omega_c) + X(j\omega + j\omega_c)]. \tag{8.10}$$

With $X(j\omega)$ as depicted in Figure 8.4(a), the spectrum of $y(t)$ is that shown in Figure 8.4(c). Note that there is now a replication of the spectrum of the original signal, centered around both $+\omega_c$ and $-\omega_c$. As a consequence, $x(t)$ is recoverable from $y(t)$ only if $\omega_c > \omega_M$, since otherwise the two replications will overlap in frequency. This is in contrast to the case of a complex exponential carrier, for which a replication of the spectrum of the original signal is centered only around ω_c. Specifically, as we saw in Section 8.1.1, in the case of amplitude modulation with a complex exponential carrier, $x(t)$ can always be recovered from $y(t)$ for any choice of ω_c by shifting the spectrum back to its original location by multiplying by $e^{-j\omega_c t}$, as in eq. (8.7). With a sinusoidal carrier, on the other hand, as we see from Figure 8.4, if $\omega_c < \omega_M$, then there will be an overlap between the two replications of $X(j\omega)$. For example, Figure 8.5 depicts $Y(j\omega)$ for $\omega_c = \omega_M/2$. Clearly, the spectrum of $x(t)$ is no longer replicated in $Y(j\omega)$, and thus, it may no longer be possible to recover $x(t)$ from $y(t)$.

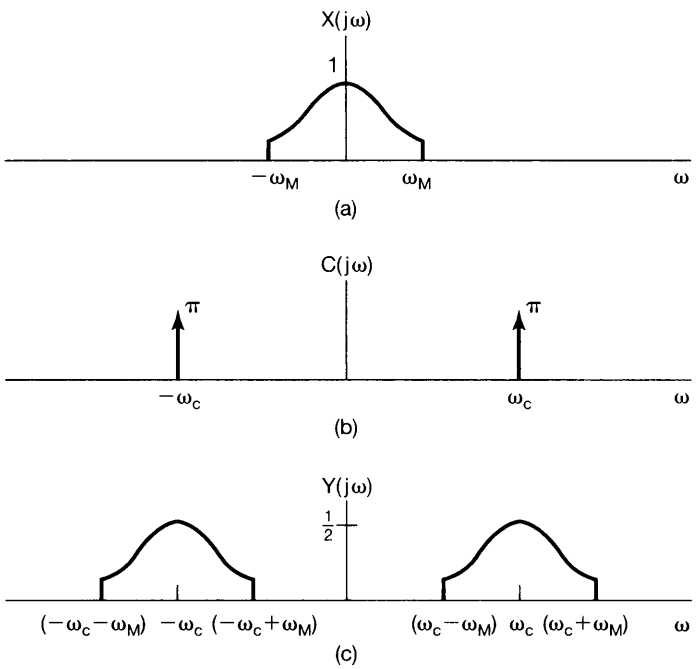


Figure 8.4 Effect in the frequency domain of amplitude modulation with a sinusoidal carrier: (a) spectrum of modulating signal $x(t)$; (b) spectrum of carrier $c(t) = \cos \omega_c t$; (c) spectrum of amplitude-modulated signal.

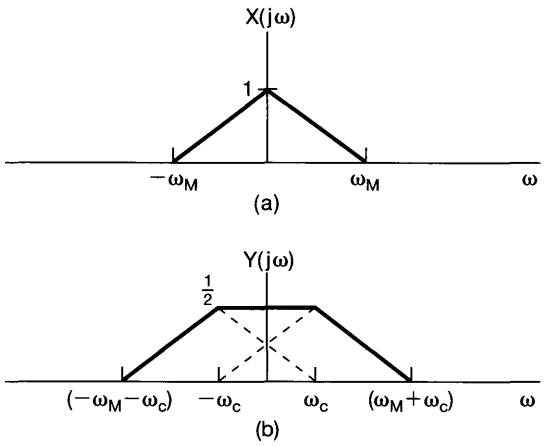


Figure 8.5 Sinusoidal amplitude modulation with carrier $\cos \omega_c t$ for which $\omega_c = \omega_M/2$: (a) spectrum of modulating signal; (b) spectrum of modulated signal.

8.2 DEMODULATION FOR SINUSOIDAL AM

At the receiver in a communication system, the information-bearing signal $x(t)$ is recovered through demodulation. In this section, we examine the process of demodulation for sinusoidal amplitude modulation, as introduced in the previous section. There are two commonly used methods for demodulation, each with its own advantages and disadvantages. In Section 8.2.1 we discuss the first of these, a process referred to as *synchronous demodulation,* in which the transmitter and receiver are synchronized in phase. In Section 8.2.2, we describe an alternative method referred to as *asynchronous demodulation.*

8.2.1 Synchronous Demodulation

Assuming that $\omega_c > \omega_M$, demodulation of a signal that was modulated with a sinusoidal carrier is relatively straightforward. Specifically, consider the signal

$$y(t) = x(t) \cos \omega_c t. \tag{8.11}$$

As was suggested in Example 4.21, the original signal can be recovered by modulating $y(t)$ with the same sinusoidal carrier and applying a lowpass filter to the result. To see this, consider

$$w(t) = y(t) \cos \omega_c t. \tag{8.12}$$

Figure 8.6 shows the spectra of $y(t)$ and $w(t)$, and we observe that $x(t)$ can be recovered from $w(t)$ by applying an ideal lowpass filter with a gain of 2 and a cutoff frequency that is greater than ω_M and less than $2\omega_c - \omega_M$. The frequency response of the lowpass filter is indicated by the dashed line in Figure 8.6(c).

The basis for using eq. (8.12) and a lowpass filter to demodulate $y(t)$ can also be seen algebraically. From eqs. (8.11) and (8.12), it follows that

$$w(t) = x(t) \cos^2 \omega_c t,$$

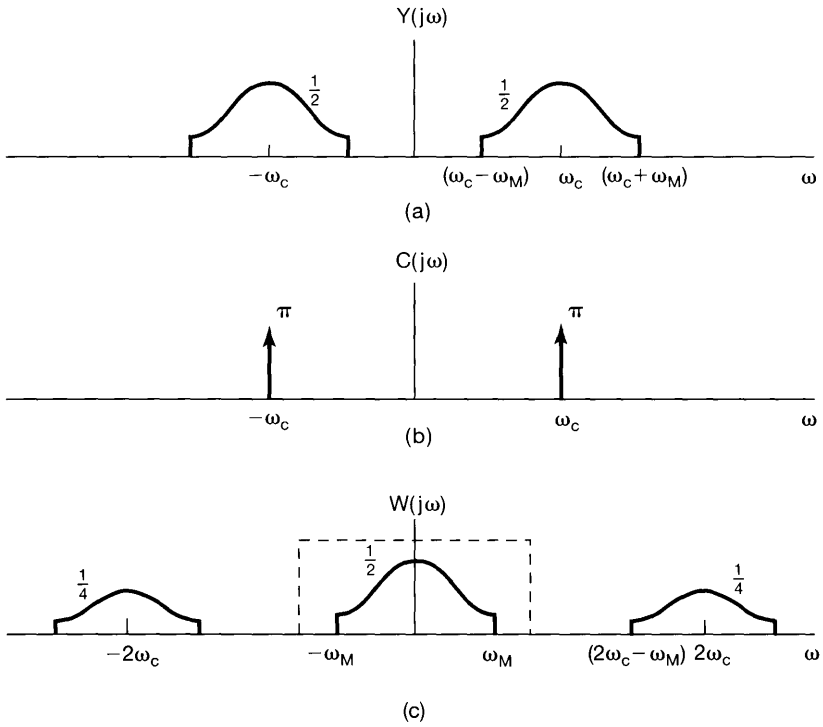


Figure 8.6 Demodulation of an amplitude-modulated signal with a sinusoidal carrier: (a) spectrum of modulated signal; (b) spectrum of carrier signal; (c) spectrum of modulated signal multiplied by the carrier. The dashed line indicates the frequency response of a lowpass filter used to extract the demodulated signal.

or, using the trigonometric identity

$$\cos^2 \omega_c t = \frac{1}{2} + \frac{1}{2}\cos 2\omega_c t,$$

we can rewrite $w(t)$ as

$$w(t) = \frac{1}{2}x(t) + \frac{1}{2}x(t)\cos 2\omega_c t. \tag{8.13}$$

Thus, $w(t)$ consists of the sum of two terms, namely one-half the original signal and one-half the original signal modulated with a sinusoidal carrier at twice the original carrier frequency ω_c. Both of these terms are apparent in the spectrum shown in Figure 8.6(c). Applying the lowpass filter to $w(t)$ corresponds to retaining the first term on the right-hand side of eq. (8.13) and eliminating the second term.

The overall system for amplitude modulation and demodulation using a complex exponential carrier is depicted in Figure 8.7, and the overall system for modulation and demodulation using a sinusoidal carrier is depicted in Figure 8.8. In these figures, we have indicated the more general case in which, for both the complex exponential and the sinusoidal carrier, a carrier phase θ_c is included. The modification of the preceding analysis so as to include θ_c is straightforward and is considered in Problem 8.21.

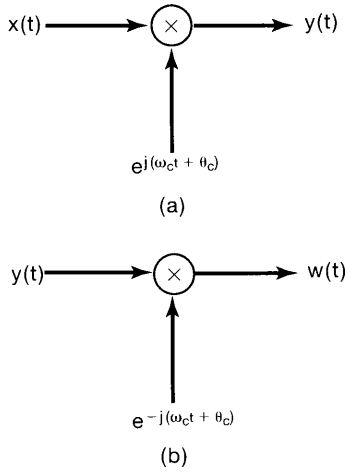


Figure 8.7 System for amplitude modulation and demodulation using a complex exponential carrier: (a) modulation; (b) demodulation.

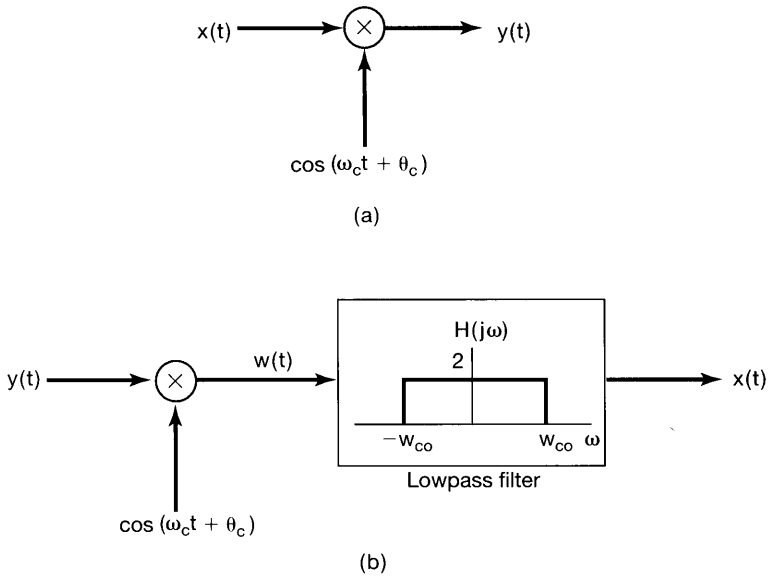


Figure 8.8 Amplitude modulation and demodulation with a sinusoidal carrier: (a) modulation system; (b) demodulation system. The lowpass filter cutoff frequency w_{co} is greater than ω_M and less than $2\omega_c - \omega_M$.

In the systems of Figures 8.7 and 8.8, the demodulating signal is assumed to be synchronized in phase with the modulating signal, and consequently the process is referred to as *synchronous demodulation.* Suppose, however, that the modulator and demodulator are not synchronized in phase. For the case of the complex exponential carrier, with θ_c denoting the phase of the modulating carrier and ϕ_c the phase of the demodulating carrier,

$$y(t) = e^{j(\omega_c t + \theta_c)} x(t), \tag{8.14}$$

$$w(t) = e^{-j(\omega_c t + \phi_c)} y(t), \tag{8.15}$$

and consequently,

$$w(t) = e^{j(\theta_c - \phi_c)} x(t). \tag{8.16}$$

Thus, if $\theta_c \neq \phi_c$, $w(t)$ will have a complex amplitude factor. For the particular case in which $x(t)$ is positive, $x(t) = |w(t)|$, and thus $x(t)$ can be recovered by taking the magnitude of the demodulated signal.

For the sinusoidal carrier, again let θ_c and ϕ_c denote the phases of the modulating and demodulating carriers, respectively, as indicated in Figure 8.9. The input to the lowpass filter is now

$$w(t) = x(t)\cos(\omega_c t + \theta_c)\cos(\omega_c t + \phi_c), \tag{8.17}$$

or, using the trigonometric identity

$$\cos(\omega_c t + \theta_c)\cos(\omega_c t + \phi_c) = \frac{1}{2}\cos(\theta_c - \phi_c) + \frac{1}{2}\cos(2\omega_c t + \theta_c + \phi_c), \tag{8.18}$$

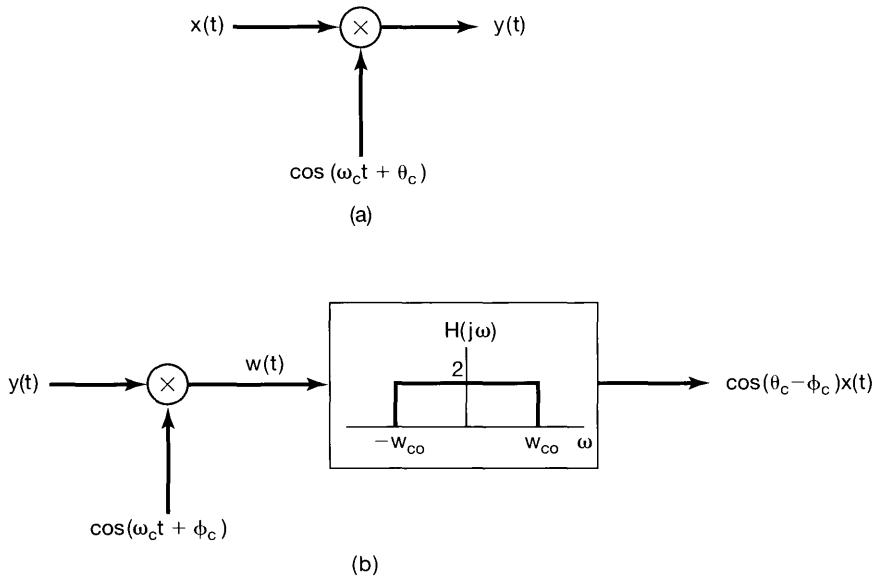


Figure 8.9 Sinusoidal amplitude modulation and demodulation system for which the carrier signals and the modulator and demodulator are not synchronized: (a) modulator; (b) demodulator.

we have

$$w(t) = \frac{1}{2}\cos(\theta_c - \phi_c)x(t) + \frac{1}{2}x(t)\cos(2\omega_c t + \theta_c + \phi_c), \tag{8.19}$$

and the output of the lowpass filter is then $x(t)$ multiplied by the amplitude factor $\cos(\theta_c - \phi_c)$. If the oscillators in the modulator and demodulator are in phase, $\theta_c = \phi_c$, and the output of the lowpass filter is $x(t)$. On the other hand, if these oscillators have a phase difference of $\pi/2$, the output will be zero. In general, for a maximum output signal, the oscillators should be in phase. Of even more importance, the phase relation between the two oscillators must be maintained over time, so that the amplitude factor $\cos(\theta_c - \phi_c)$ does not vary. This requires careful synchronization between the modulator and the demodulator, which is often difficult, particularly when they are geographically separated, as is typical in a communication system. The corresponding effects of, and the need for, synchronization not only between the phase of the modulator and demodulator, but between the frequencies of the carrier signals used in both, are explored in detail in Problem 8.23.

8.2.2 Asynchronous Demodulation

In many systems that employ sinusoidal amplitude modulation, an alternative demodulation procedure referred to as *asynchronous demodulation* is commonly used. Asynchronous demodulation avoids the need for synchronization between the modulator and demodulator. In particular, suppose that $x(t)$ is always positive and that the carrier frequency ω_c is much higher than ω_M, the highest frequency in the modulating signal. The modulated signal $y(t)$ will then have the general form illustrated in Figure 8.10.

In particular, the *envelope* of $y(t)$—that is, a smooth curve connecting the peaks in $y(t)$—would appear to be a reasonable approximation to $x(t)$. Thus, $x(t)$ could be approximately recovered through the use of a system that tracks these peaks to extract the envelope. Such a system is referred to as an *envelope detector.* One example of a simple circuit that acts as an envelope detector is shown in Figure 8.11(a). This circuit is generally followed by a lowpass filter to reduce the variations at the carrier frequency, which are evident in Figure 8.11(b) and which will generally be present in the output of an envelope detector of the type indicated in Figure 8.11(a).

The two basic assumptions required for asynchronous demodulation are that $x(t)$ be positive and that $x(t)$ vary slowly compared to ω_c, so that the envelope is easily tracked. The second condition is satisfied, for example, in audio transmission over a radio-frequency (RF) channel, where the highest frequency present in $x(t)$ is typically 15 to 20 kHz and $\omega_c/2\pi$ is in the range 500 kHz to 2 MHz. The first condition, that $x(t)$ be positive, can be satisfied by simply adding an appropriate constant value to $x(t)$ or, equivalently, by a simple change in the modulator, as shown in Figure 8.12. The output of the envelope detector then approximates $x(t) + A$, from which $x(t)$ is easily obtained.

To use the envelope detector for demodulation, we require that A be sufficiently large so that $x(t) + A$ is positive. Let K denote the maximum amplitude of $x(t)$; that is, $|x(t)| \leq K$. For $x(t) + A$ to be positive, we require that $A > K$. The ratio K/A is commonly referred to as the *modulation index m.* Expressed in percent, it is referred to as the *percent modulation.* An illustration of the output of the modulator of Figure 8.12 for $x(t)$ sinusoidal and for $m = 0.5$ (50% modulation) and $m = 1.0$ (100% modulation), is shown in Figure 8.13.

In Figure 8.14, we show a comparison of the spectra associated with the modulated signal when synchronous demodulation and when asynchronous demodulation are used. We note in particular that the output of the modulator for the asynchronous system in Figure 8.12 has an additional component $A \cos \omega_c t$ that is neither present nor necessary in the synchronous system. This is represented in the spectrum of Figure 8.14(c) by the presence of impulses at $+\omega_c$ and $-\omega_c$. For a fixed maximum amplitude K of the modulating signal, as A is decreased the relative amount of carrier present in the modulated output decreases. Since the carrier component in the output contains no information, its presence

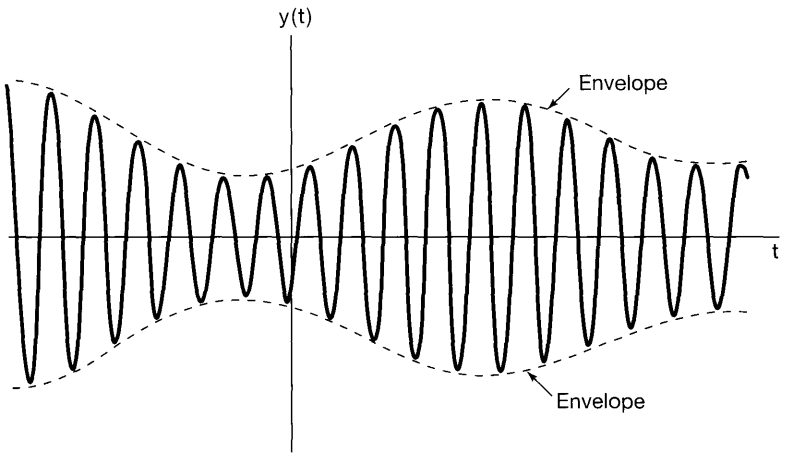
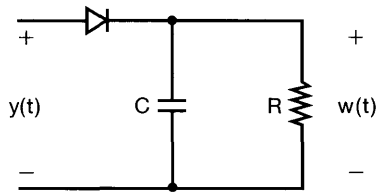
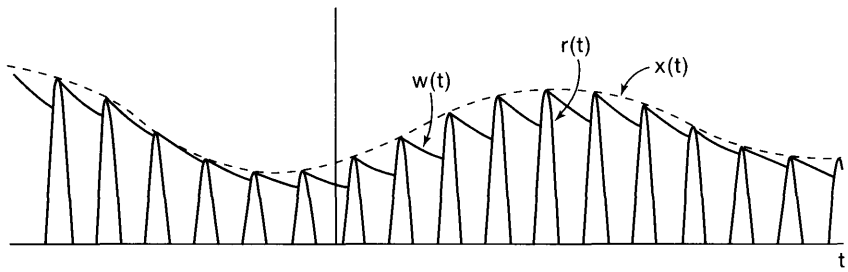


Figure 8.10 Amplitude-modulated signal for which the modulating signal is positive. The dashed curve represents the envelope of the modulated signal.



(a)



(b)

Figure 8.11 Demodulation by envelope detection: (a) circuit for envelope detection using half-wave rectification; (b) waveforms associated with the envelope detector in (a): $r(t)$ is the half-wave rectified signal, $x(t)$ is the true envelope, and $w(t)$ is the envelope obtained from the circuit in (a). The relationship between $x(t)$ and $w(t)$ has been exaggerated in (b) for purposes of illustration. In a practical asynchronous demodulation system, $w(t)$ would typically be a much closer approximation to $x(t)$ than depicted here.

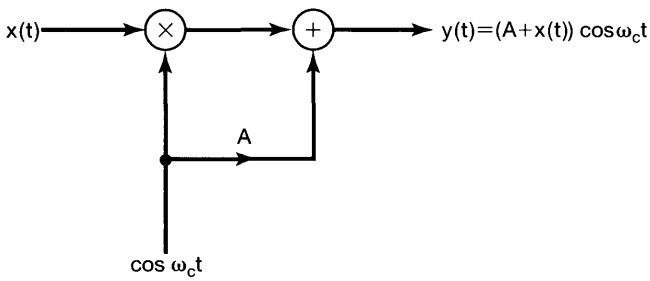


Figure 8.12 Modulator for an asynchronous modulation-demodulation system.

represents an inefficiency—for example, in the amount of power required to transmit the modulated signal—and thus, in one sense it is desirable to make the ratio K/A—i.e., the modulation index m—as large as possible. On the other hand, the ability of a simple envelope detector such as that in Figure 8.11 to follow the envelope and thus extract $x(t)$ improves as the modulation index decreases. Hence, there is a trade-off between the effi-

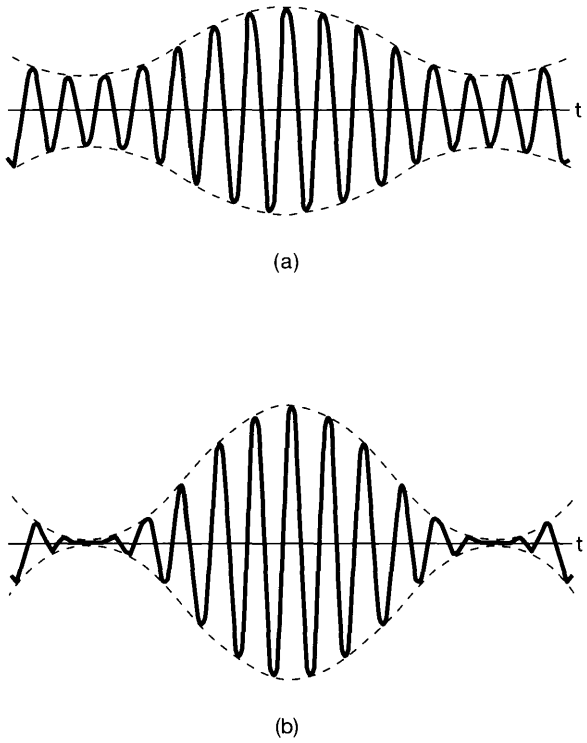


Figure 8.13 Output of the amplitude modulation system of Figure 8.12: (a) modulation index $m = 0.5$; (b) modulation index $m = 1.0$.

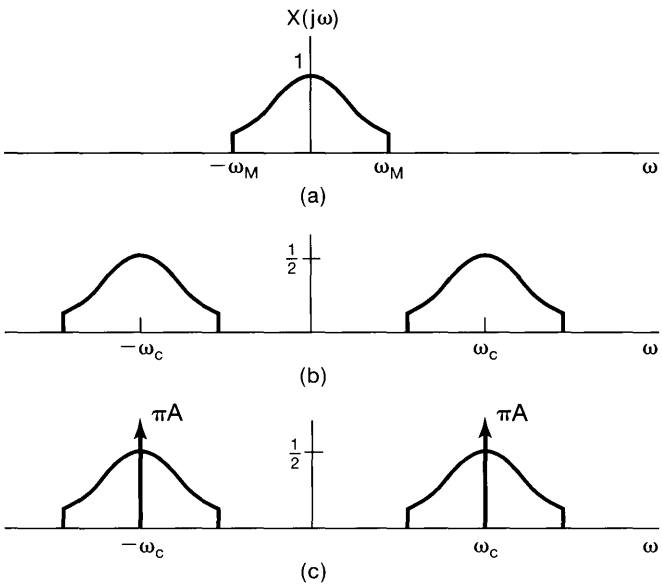


Figure 8.14 Comparison of spectra for synchronous and asynchronous sinusoidal amplitude modulation systems: (a) spectrum of modulating signal; (b) spectrum of $x(t)\cos\omega_c t$ representing modulated signal in a synchronous system; (c) spectrum of $[x(t) + A]\cos\omega_c t$ representing modulated signal in an asynchronous system.

ciency of the system in terms of the power in the output of the modulator and the quality of the demodulated signal.

There are a number of advantages and disadvantages to the asynchronous modulation-demodulation system of Figures 8.11 and 8.12, compared with the synchronous system of Figure 8.8. The synchronous system requires a more sophisticated demodulator because the oscillator in the demodulator must be synchronized with the oscillator in the modulator, both in phase and in frequency. On the other hand, the asynchronous modulator in general requires transmitting more power than the synchronous modulator, since, for the envelope detector to operate properly, the envelope must be positive, or equivalently, there must be a carrier component present in the transmitted signal. This is often preferable in cases such as that associated with public radio broadcasting, in which it is desirable to mass-produce large numbers of receivers (demodulators) at moderate cost. The additional cost in transmitted power is then offset by the savings in cost for the receiver. On the other hand, in situations in which transmitter power requirements are at a premium, as in satellite communication, the cost of implementing a more sophisticated synchronous receiver is warranted.

8.3 FREQUENCY-DIVISION MULTIPLEXING

Many systems used for transmitting signals provide more bandwidth than is required for any one signal. For example, a typical microwave link has a total bandwidth of several gigahertz, which is considerably greater than the bandwidth required for one voice channel. If the individual voice signals, which are overlapping in frequency, have their frequency content shifted by means of sinusoidal amplitude modulation so that the spectra of the modulated signals no longer overlap, they can be transmitted *simultaneously* over a single wideband channel. The resulting concept is referred to as *frequency-division multiplexing* (FDM). Frequency-division multiplexing using a sinusoidal carrier is illustrated in Figure 8.15. The individual signals to be transmitted are assumed to be band limited and

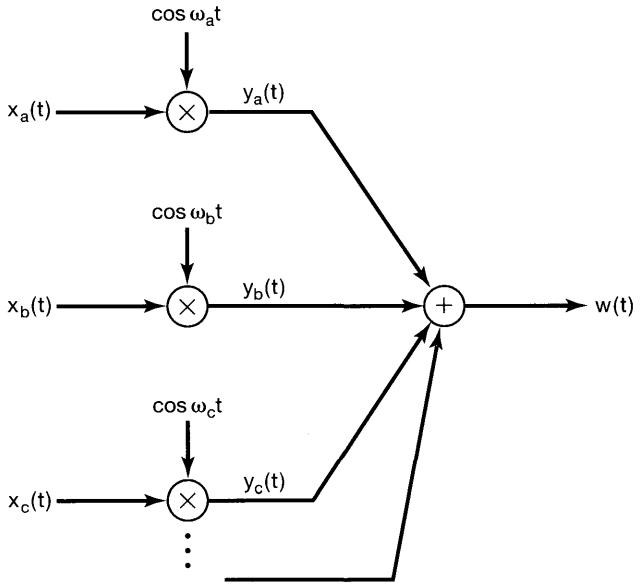


Figure 8.15 Frequency-division multiplexing using sinusoidal amplitude modulation.

are modulated with different carrier frequencies. The modulated signals are then summed and transmitted simultaneously over the same communication channel. The spectra of the individual subchannels and the composite multiplexed signal are illustrated in Figure 8.16. Through this multiplexing process, the individual input signals are allocated distinct segments of the frequency band. To recover the individual channels in the demultiplexing process requires two basic steps: bandpass filtering to extract the modulated signal corresponding to a specific channel, followed by demodulation to recover the original signal. This is illustrated in Figure 8.17 to recover channel a, where, for purposes of illustration, synchronous demodulation is assumed.

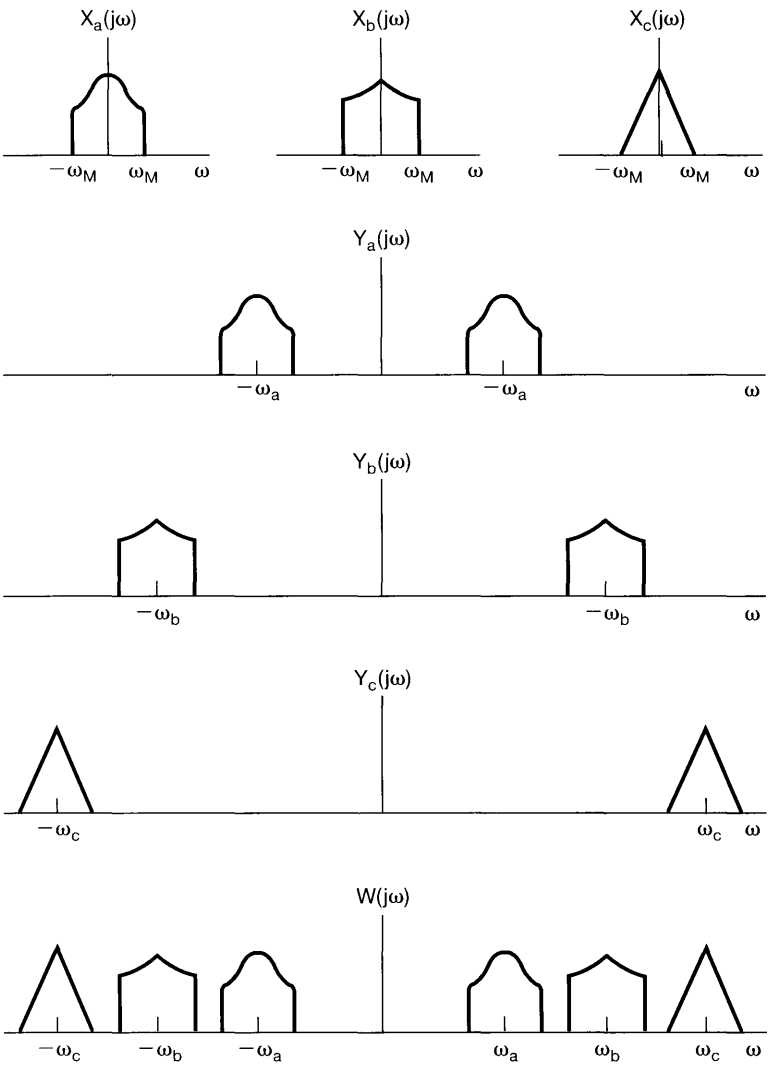


Figure 8.16 Spectra associated with the frequency-division multiplexing system of Figure 8.15.

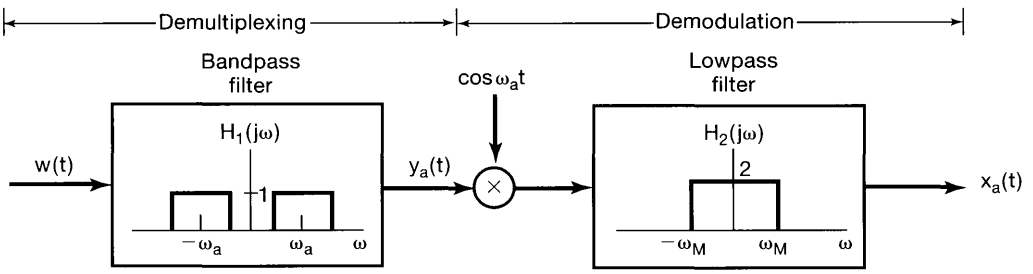


Figure 8.17 Demultiplexing and demodulation for a frequency-division multiplexed signal.

Telephone communication is one important application of frequency-division multiplexing. Another is the transmission of signals through the atmosphere in the RF band. In the United States, the use of radio frequencies for transmitting signals over the range 10 kHz to 275 GHz is controlled by the Federal Communications Commission, and different portions of the range are allocated for different purposes. The current allocation of frequencies is shown in Figure 8.18. As indicated, the frequency range in the neighborhood of 1 MHz is assigned to the AM broadcast band, where *AM* refers specifically to the use of sinusoidal amplitude modulation. Individual AM radio stations are assigned specific frequencies within the AM band, and thus, many stations can broadcast simultaneously through this use of frequency-division multiplexing. In principle, at the receiver, an individual radio station can be selected by demultiplexing and demodulating, as illustrated in Figure 8.17. The tuning dial on the receiver would then control both the center frequency of the bandpass filter and the frequency of the demodulating oscillator. In fact, for public broadcasting, asynchronous modulation and demodulation are used to simplify the receiver and reduce its cost. Furthermore, the demultiplexing in Figure 8.17 requires a sharp cutoff bandpass filter with variable center frequency. Variable frequency-selective filters are difficult to implement, and consequently, a fixed filter is implemented instead, and an intermediate stage of modulation and filtering [referred to in a radio receiver as the *intermediate-frequency* (IF) stage] is used. The use of modulation to slide the spectrum of the signal past a fixed bandpass filter replaces the use of a variable bandpass filter in a manner similar to the procedure discussed in Section 4.5.1. This basic procedure is incorporated into typical home AM radio receivers. Some of the more detailed issues involved are considered in Problem 8.36.

As illustrated in Figure 8.16, in the frequency-division multiplexing system of Figure 8.15 the spectrum of each individual signal is replicated at both positive and negative frequencies, and thus the modulated signal occupies twice the bandwidth of the original. This represents an inefficient use of bandwidth. In the next section we consider an alternative form of sinusoidal amplitude modulation, which leads to more efficient use of bandwidth at the cost of a more complicated modulation system.

Frequency range	Designation	Typical uses	Propagation method	Channel features
30–300 Hz	ELF (extremely low frequency)	Macrowave, submarine communication	Megametric waves	Penetration of conducting earth and seawater
0.3–3 kHz	VF (voice frequency)	Data terminals, telephony	Copper wire	
3–30 kHz	VLF (very low frequency)	Navigation, telephone, telegraph, frequency and timing standards	Surface ducting (ground wave)	Low attenuation, little fading, extremely stable phase and frequency, large antennas
30–300 kHz	LF (low frequency)	Industrial (power line) communication, aeronautical and maritime long-range navigation, radio beacons	Mostly surface ducting	Slight fading, high atmospheric pulse
0.3–3 MHz	MF (medium frequency)	Mobile, AM broadcasting, amateur, public safety	Ducting and ionospheric reflection (sky wave)	Increased fading, but reliable
3–30 MHz	HF (high frequency)	Military communication, aeronautical mobile, international fixed, amateur and citizen's band, industrial	Ionospheric reflecting sky wave, 50–400 km layer altitudes	Intermittent and frequency-selective fading, multipath
30–300 MHz	VHF (very high frequency)	FM and TV broadcast, land transportation (taxis, buses, railroad)	Sky wave (ionospheric and tropospheric scatter)	Fading, scattering, and multipath
0.3–3 GHz	UHF (ultra high frequency)	UHF TV, space telemetry, radar, military	Transhorizon tropospheric scatter and line-of-sight relaying	
3–30 GHz	SHF (super high frequency)	Satellite and space communication, common carrier (CC), microwave	Line-of-sight ionosphere penetration	Ionospheric penetration, extraterrestrial noise, high directly
30–300 GHz	EHF (extremely high frequency)	Experimental, government, radio astronomy	Line of sight	Water vapor and oxygen absorption
10^3–10^7 GHz	Infrared, visible light, ultraviolet	Optical communications	Line of sight	

Figure 8.18 Allocation of frequencies in the RF spectrum.

8.4 SINGLE-SIDEBAND SINUSOIDAL AMPLITUDE MODULATION

For the sinusoidal amplitude modulation systems discussed in Section 8.1, the total bandwidth of the original signal $x(t)$ is $2\omega_M$, including both positive and negative frequencies, where ω_M is the highest frequency present in $x(t)$. With the use of a complex exponential carrier, the spectrum is translated to ω_c, and the total width of the frequency band over which there is energy from the signal is still $2\omega_M$, although the modulated signal is now complex. With a sinusoidal carrier, on the other hand, the spectrum of the signal is shifted to $+\omega_c$ *and* $-\omega_c$, and thus, twice the bandwidth is required. This suggests that there is a basic redundancy in the modulated signal with a sinusoidal carrier. Using a technique referred to as *single-sideband modulation,* we can remove the redundancy.

The spectrum of $x(t)$ is illustrated in Figure 8.19(a), in which we have shaded the positive and negative frequency components differently to distinguish them. The spectrum

in Figure 8.19(b) results from modulation with a sinusoidal carrier, where we identify an upper and lower sideband for the portion of the spectrum centered at $+\omega_c$ and that centered at $-\omega_c$. Comparing Figures 8.19(a) and (b), we see that $X(j\omega)$ can be recovered if only the upper sidebands at positive and negative frequencies are retained, or alternatively, if only the lower sidebands at positive and negative frequencies are retained. The resulting spectrum if only the upper sidebands are retained is shown in Figure 8.19(c), and the resulting spectrum if only the lower sidebands are retained is shown in Figure 8.19(d). The conversion of $x(t)$ to the form corresponding to Figure 8.19(c) or (d) is referred to as *single-sideband modulation* (SSB), in contrast to the *double-sideband modulation* (DSB) of Figure 8.19(b), in which both sidebands are retained.

There are several methods by which the single-sideband signal can be obtained. One is to apply a sharp cutoff bandpass or highpass filter to the double-sideband signal of Figure 8.19(b), as illustrated in Figure 8.20, to remove the unwanted sideband. Another is to use a procedure that utilizes phase shifting. Figure 8.21 depicts a system designed

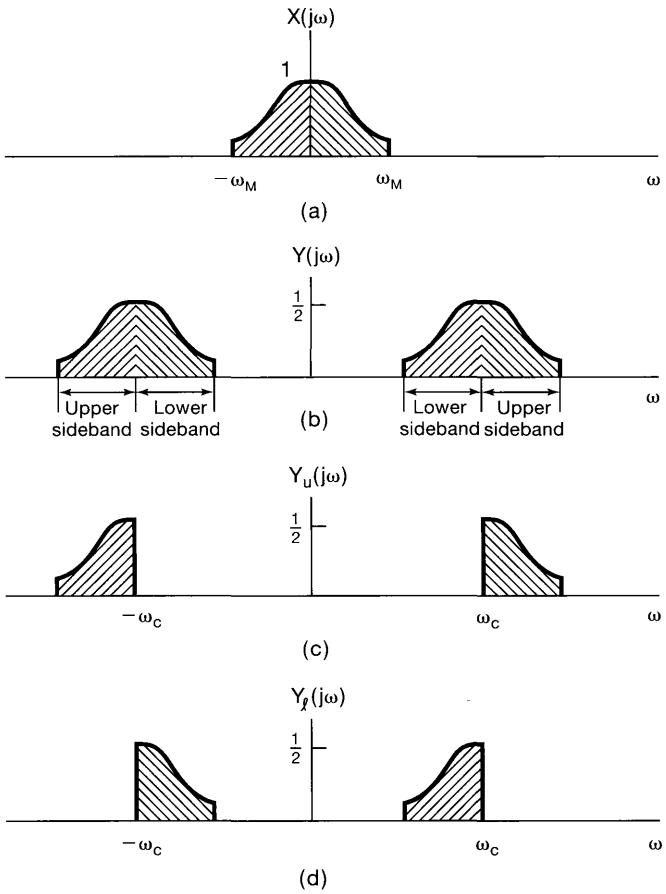


Figure 8.19 Double- and single-sideband modulation: (a) spectrum of modulating signal; (b) spectrum after modulation with a sinusoidal carrier; (c) spectrum with only the upper sidebands; (d) spectrum with only the lower sidebands.

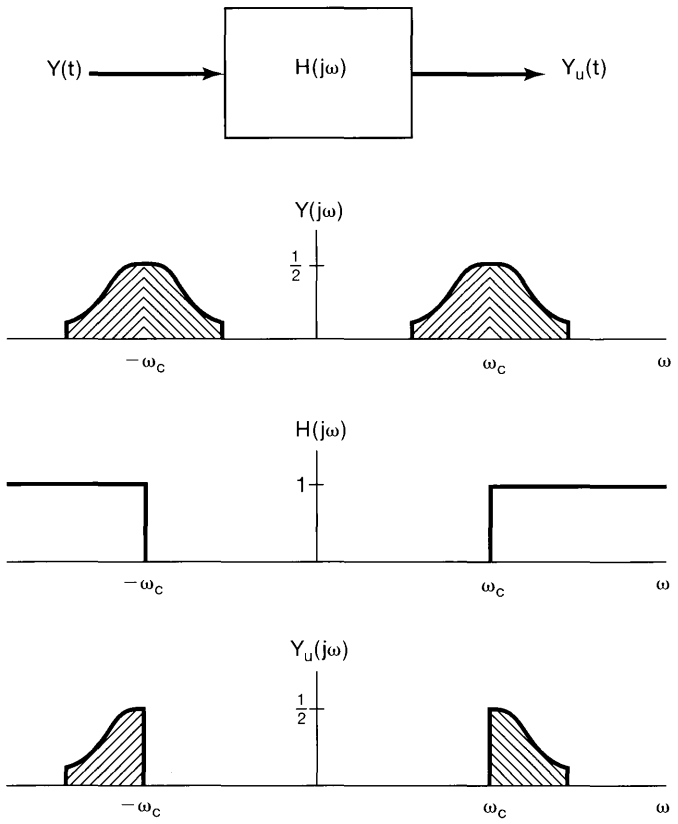


Figure 8.20 System for retaining the upper sidebands using ideal high-pass filtering.

to retain the lower sidebands. The system $H(j\omega)$ in the figure is referred to as a "90° phase-shift network," for which the frequency response is of the form

$$H(j\omega) = \begin{cases} -j, & \omega > 0 \\ j, & \omega < 0 \end{cases}. \tag{8.20}$$

The spectra of $x(t)$, $y_1(t) = x(t)\cos\omega_c t$, $y_2(t) = x_p(t)\sin\omega_c t$, and $y(t)$ are illustrated in Figure 8.22. As is examined in Problem 8.28, to retain the upper sidebands instead of the lower sidebands, the phase characteristic of $H(j\omega)$ is reversed so that

$$H(j\omega) = \begin{cases} j, & \omega > 0 \\ -j, & \omega < 0 \end{cases}. \tag{8.21}$$

As is explored in Problem 8.29, synchronous demodulation of single-sideband systems can be accomplished in a manner identical to synchronous demodulation of double-sideband systems. The price paid for the increased efficiency of single-sideband systems is added complexity in the modulator.

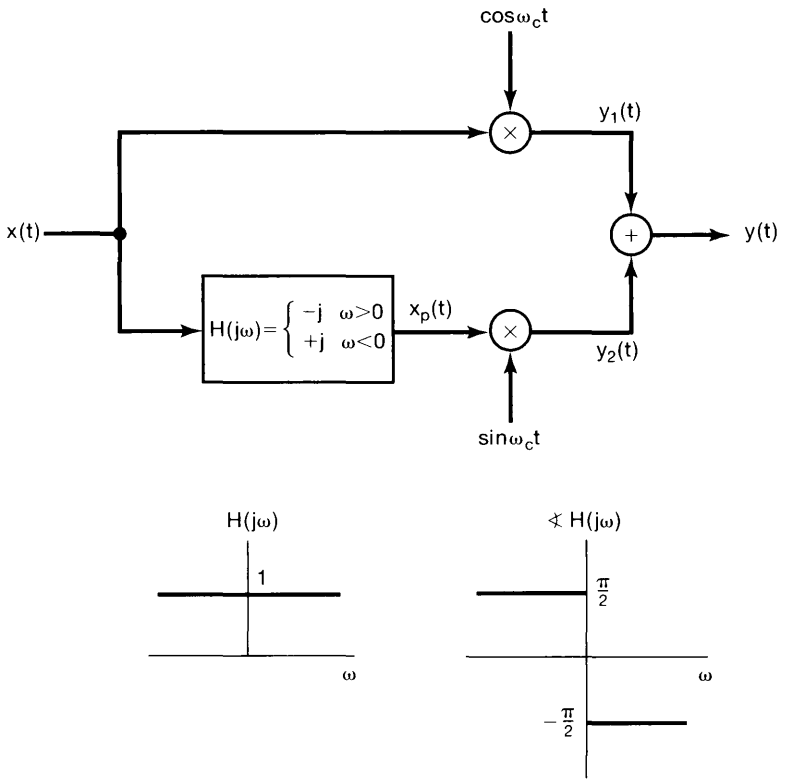


Figure 8.21 System for single-sideband amplitude modulation, using a 90° phase-shift network, in which only the lower sidebands are retained.

In summary, in Sections 8.1 through 8.4 we have seen a number of variations of complex exponential and sinusoidal amplitude modulation. With asynchronous demodulation, discussed in Section 8.2.2, a constant must be added to the modulating signal so that it is positive. This results in the presence of the carrier signal as a component in the modulated output, requiring more power for transmission, but resulting in a simpler demodulator than is required in a synchronous system. Alternatively, only the upper or lower sidebands in the modulated output may be retained, which makes more efficient use of bandwidth and transmitter power, but requires a more sophisticated modulator. Sinusoidal amplitude modulation with both sidebands and the presence of a carrier is typically abbreviated as AM-DSB/WC (amplitude modulation, double sideband/with carrier) and, when the carrier is suppressed or absent, as AM-DSB/SC (amplitude modulation, double-sideband/suppressed carrier). The corresponding single-sideband systems are abbreviated AM-SSB/WC and AM-SSB/SC.

Sections 8.1 through 8.4 are intended to provide an introduction to many of the basic concepts associated with sinusoidal amplitude modulation. There are many variations in details and implementation, and the reader is referred to the bibliography for an indication of the numerous excellent books that explore this topic further.

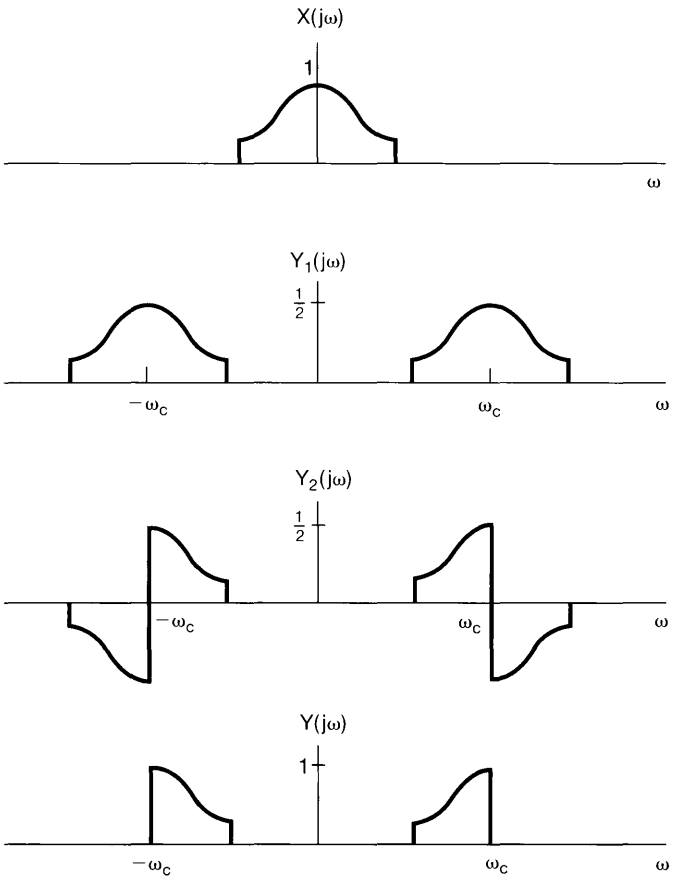


Figure 8.22 Spectra associated with the single-sideband system of Figure 8.21.

8.5 AMPLITUDE MODULATION WITH A PULSE-TRAIN CARRIER

8.5.1 Modulation of a Pulse-Train Carrier

In previous sections, we examined amplitude modulation with a sinusoidal carrier. Another important class of amplitude modulation techniques corresponds to the use of a carrier signal that is a pulse train, as illustrated in Figure 8.23; amplitude modulation of this type effectively corresponds to transmitting equally spaced time slices of $x(t)$. In general, we would not expect that an arbitrary signal could be recovered from such a set of time slices. However, our examination of the concept of sampling in Chapter 7 suggests that this should be possible if $x(t)$ is band limited and the pulse repetition frequency is high enough.

From Figure 8.23,

$$y(t) = x(t)c(t); \tag{8.22}$$

i.e., the modulated signal $y(t)$ is the product of $x(t)$ and the carrier $c(t)$. With $Y(j\omega)$, $X(j\omega)$, and $C(j\omega)$ representing the Fourier transforms of each of these signals, it follows from

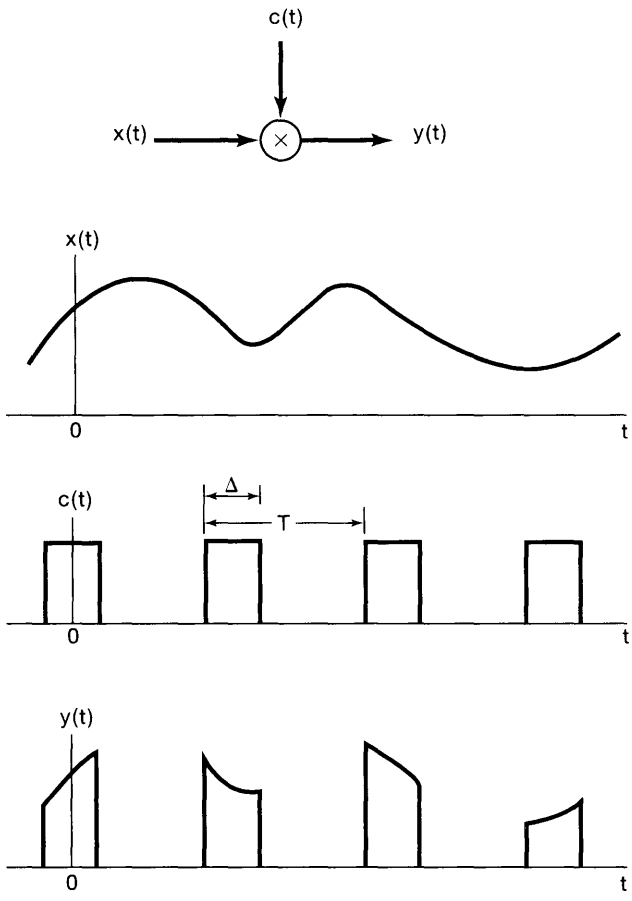


Figure 8.23 Amplitude modulation of a pulse train.

the multiplication property that

$$Y(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\theta)C(j(\omega - \theta))d\theta. \tag{8.23}$$

Since $c(t)$ is periodic with period T, $C(j\omega)$ consists of impulses in frequency spaced by $2\pi/T$; that is,

$$C(j\omega) = 2\pi \sum_{k=-\infty}^{-\infty} a_k \delta(\omega - k\omega_c), \tag{8.24}$$

where $\omega_c = 2\pi/T$ and the coefficients a_k are the Fourier series coefficients of $c(t)$, which, from Example 3.5, are

$$a_k = \frac{\sin(k\omega_c \Delta/2)}{\pi k}. \tag{8.25}$$

The spectrum of $c(t)$ is shown in Figure 8.24(b). With the spectrum of $x(t)$ as illustrated in Figure 8.24(a), the resulting spectrum of the modulated signal $y(t)$ is shown in Figure 8.24(c). From eqs. (8.23) and (8.24), $Y(j\omega)$ is a sum of scaled and shifted replicas of $X(j\omega)$:

$$Y(j\omega) = \sum_{k=-\infty}^{+\infty} a_k X(j(\omega - k\omega_c)). \tag{8.26}$$

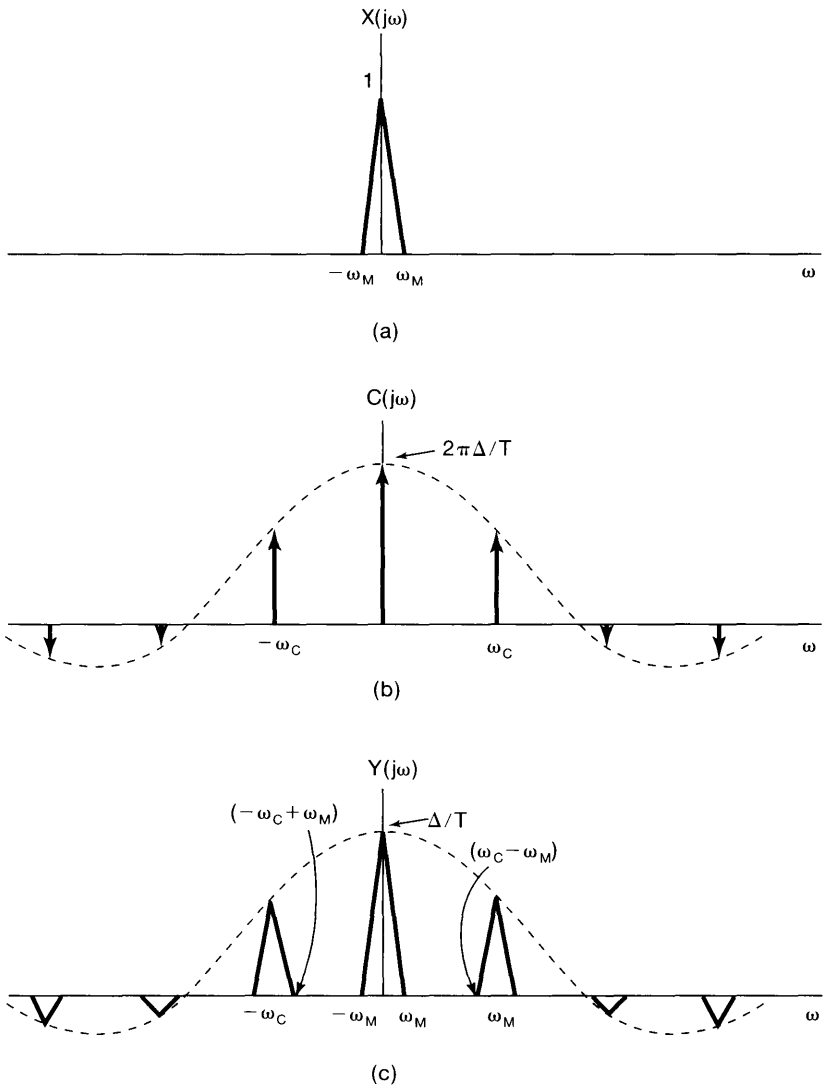


Figure 8.24 Spectra associated with amplitude modulation of a pulse train: (a) spectrum of a bandlimited-signal $x(t)$; (b) spectrum of the pulse carrier signal $c(t)$ in Figure 8.23; (c) spectrum of the modulated pulse train $y(t)$.

Comparing eq. (8.26) with eq. (7.6) and Figure 8.24 with Figure 7.3(c), we see that the spectrum of $y(t)$ is very similar in form to the spectrum resulting from sampling with a periodic impulse train, the only difference being the values of the Fourier coefficients of the pulse train. For the periodic impulse train used in Chapter 7, all of the Fourier coefficients are equal to $1/T$ in value, while for the pulse train $c(t)$ in Figure 8.23, the Fourier coefficients are given by eq. (8.25). Consequently, the replicas of $X(j\omega)$ do not overlap as long as $\omega_c > 2\omega_M$, which corresponds to the condition of the Nyquist sampling theorem. If this constraint is satisfied, then, as with impulse-train sampling, $x(t)$ can be recovered from $y(t)$ through the use of a lowpass filter with cutoff frequency greater than ω_M and less than $\omega_c - \omega_M$.

Note that the same conclusion holds for a wide variety of other pulselike carrier waveforms: If $c(t)$ is *any* periodic signal with Fourier transform as in eq. (8.24) for some set of Fourier coefficients a_k, then $Y(j\omega)$ is given by eq. (8.26). Then, as long as $\omega_c = 2\pi/T > 2\omega_M$, the replicas of $X(j\omega)$ do not overlap, allowing us to recover $x(t)$ by lowpass filtering, provided that the DC Fourier coefficient a_0 is nonzero. As shown in Problem 8.11, if a_0 is zero or unacceptably small, then, by using a bandpass filter to select one of the shifted replicas of $X(j\omega)$ with a larger value of a_k, we obtain a sinusoidal AM signal with a scaled version of $x(t)$ as the modulating signal. Using the demodulation methods described in Section 8.2, we can then recover $x(t)$.

8.5.2 Time-Division Multiplexing

Amplitude modulation with a pulse-train carrier is often used to transmit several signals over a single channel. As indicated in Figure 8.23, the modulated output signal $y(t)$ is nonzero only when the carrier signal $c(t)$ is on (i.e., is nonzero). During the intervals in which $c(t)$ is off, other similarly modulated signals can be transmitted. Two equivalent representations of this process are shown in Figure 8.25. In this technique for transmitting several signals over a single channel, each signal is in effect assigned a set of time slots of duration Δ that repeat every T seconds and that do not overlap with the slots assigned to other signals. The smaller the ratio Δ/T, the larger the number of signals that can be transmitted over the channel. This procedure is referred to as *time-division multiplexing* (TDM). Whereas frequency-division multiplexing, as discussed in Section 8.3, assigns different *frequency* intervals to individual signals, time-division multiplexing assigns different *time* intervals to individual signals. Demultiplexing the individual signals from the composite signal in Figure 8.25 is accomplished by time gating, to select the particular time slots associated with each individual signal.

8.6 PULSE-AMPLITUDE MODULATION

8.6.1 Pulse-Amplitude Modulated Signals

In Section 8.5 we described a modulation system in which a continuous-time signal $x(t)$ modulates a periodic pulse train, corresponding to transmitting time slices of $x(t)$ of duration Δ seconds every T seconds. As we saw both in that discussion and in our investigation of sampling in Chapter 7, our ability to recover $x(t)$ from these time slices depends not on their duration Δ, but rather on their frequency $2\pi/T$, which must exceed the Nyquist rate

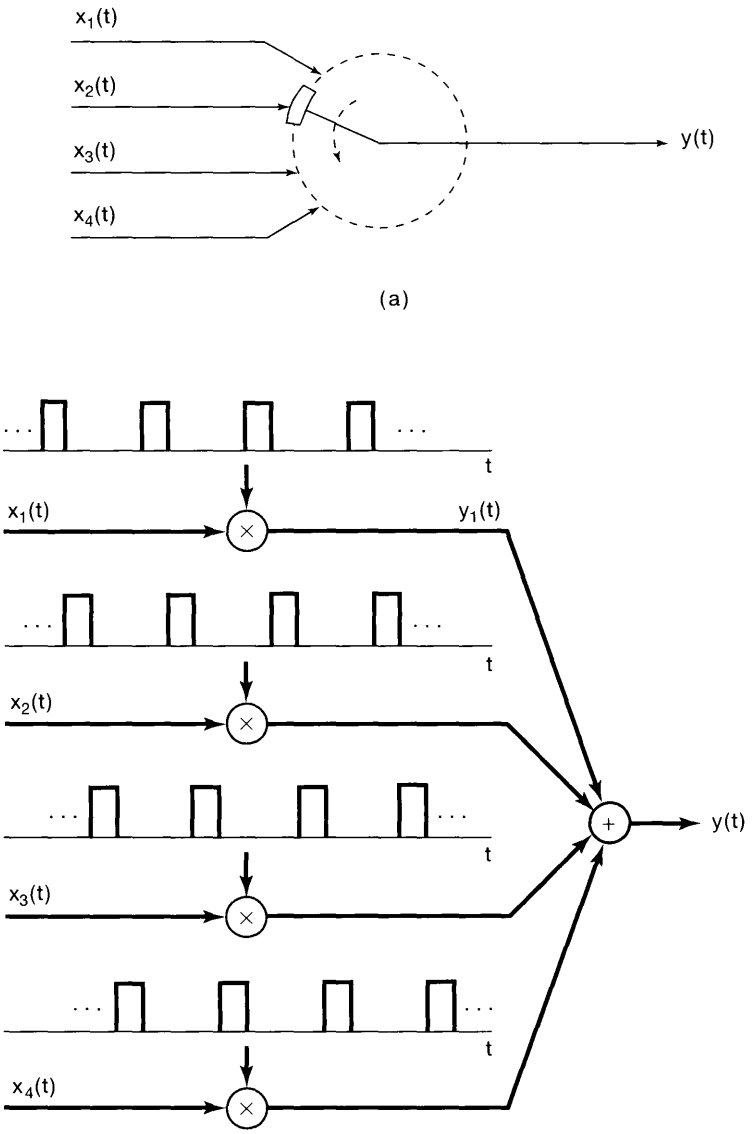


Figure 8.25 Time-division multiplexing.

in order to ensure an alias-free reconstruction of $x(t)$. That is, in principle, we need only transmit the samples $x(nT)$ of the signal $x(t)$.

In fact, in modern communication systems, sampled values of the information-bearing signal $x(t)$, rather than time slices are more typically transmitted. For practical reasons, there are limitations on the maximum amplitude that can be transmitted over a communication channel, so that transmitting impulse-sampled versions of $x(t)$ is not practical. Instead, the samples $x(nT)$ are used to modulate the amplitude of a sequence of pulses, resulting in what is referred to as a *pulse-amplitude modulation* (PAM) system.

The use of rectangular pulses corresponds to a sample-and-hold strategy in which pulses of duration Δ and amplitude proportional to the instantaneous sample values of $x(t)$ are transmitted. The resulting waveform for a single PAM channel of this type is illustrated in Figure 8.26. In the figure, the dotted curve represents the signal $x(t)$. As with the modulation scheme in Section 8.5, PAM signals can be time multiplexed. This is illustrated in Figure 8.27, which depicts the transmitted waveform with three time-multiplexed channels. The pulses associated with each channel are distinguished by shading, as well as by the channel number above each pulse. For a given pulse-repetition period T, as the pulse width decreases, more time-multiplexed channels can be transmitted over the same communication channel or medium. However, as the pulse width decreases, it is typically necessary to increase the amplitude of the transmitted pulses so that a reasonable amount of energy is transmitted in each pulse.

In addition to energy considerations, a number of other issues must be addressed in designing a PAM signal. In particular, as long as the sampling frequency exceeds the Nyquist rate, we know that $x(t)$ can be reconstructed exactly from its samples, and con-

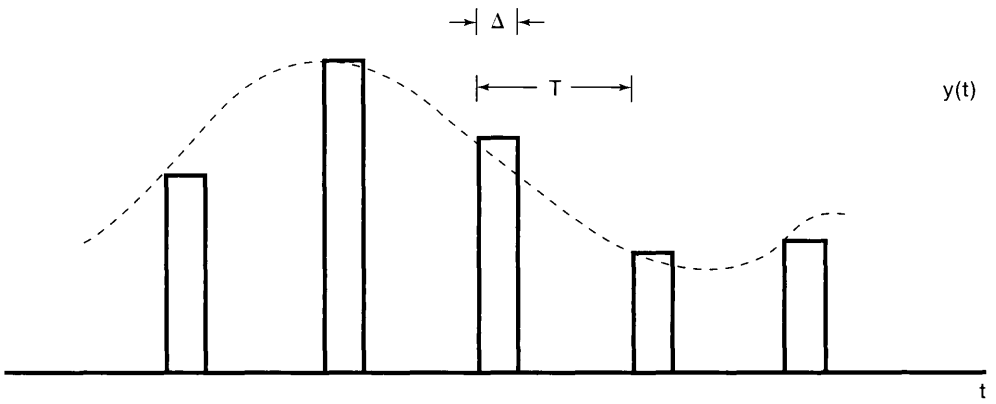


Figure 8.26 Transmitted waveform for a single PAM channel. The dotted curve represents the signal $x(t)$.

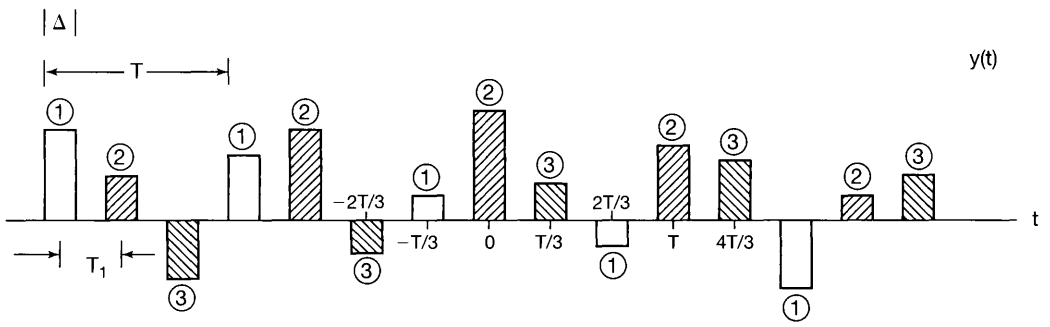


Figure 8.27 Transmitted waveform with three time-multiplexed PAM channels. The pulses associated with each channel are distinguished by shading, as well as by the channel number above each pulse. Here, the intersymbol spacing is $T_1 = T/3$.

sequently we can use these samples to modulate the amplitude of a sequence of pulses of any shape. The choice of pulse shape is dictated by considerations such as the frequency selectivity of the communication medium being used and the problem of intersymbol interference, which we discuss next.

8.6.2 Intersymbol Interference in PAM Systems

In the TDM pulse-amplitude modulation system just described, the receiver can, in principle, separate the channels by sampling the time-multiplexed waveform at appropriate times. For example, consider the time-multiplexed signal in Figure 8.27, which consists of pulse-amplitude-modulated versions of three signals $x_1(t)$, $x_2(t)$, and $x_3(t)$. If we sample $y(t)$ at appropriate times, corresponding, for example, to the midpoints of each pulse, we can separate the samples of the three signals. That is,

$$\begin{aligned} y(t) &= Ax_1(t), \qquad t = 0, \pm 3T_1, \pm 6T_1, \ldots, \\ y(t) &= Ax_2(t), \qquad t = T_1, T_1 \pm 3T_1, T_1 \pm 6T_1, \ldots, \\ y(t) &= Ax_3(t), \qquad t = 2T_1, 2T_1 \pm 3T_1, T_1 \pm 6T_1, \ldots, \end{aligned} \tag{8.27}$$

where T_1 is the intersymbol spacing, here equal to $T/3$, and where A is the appropriate proportionality constant. In other words, samples of $x_1(t)$, $x_2(t)$, and $x_3(t)$ can be obtained by appropriate sampling of the received time-multiplexed PAM signal.

The strategy indicated in the preceding paragraph assumes that the transmitted pulses remain distinct as they propagate over the communication channel. In transmission through any realistic channel, however, the pulses can be expected to be distorted through effects such as additive noise and filtering. Additive noise in the channel will, of course, introduce amplitude errors at the sampling times. Filtering due to the nonideal frequency response of a channel causes a smearing of the individual pulses that can cause the received pulses to overlap in time. This interference is illustrated in Figure 8.28 and is referred to as *intersymbol interference.*

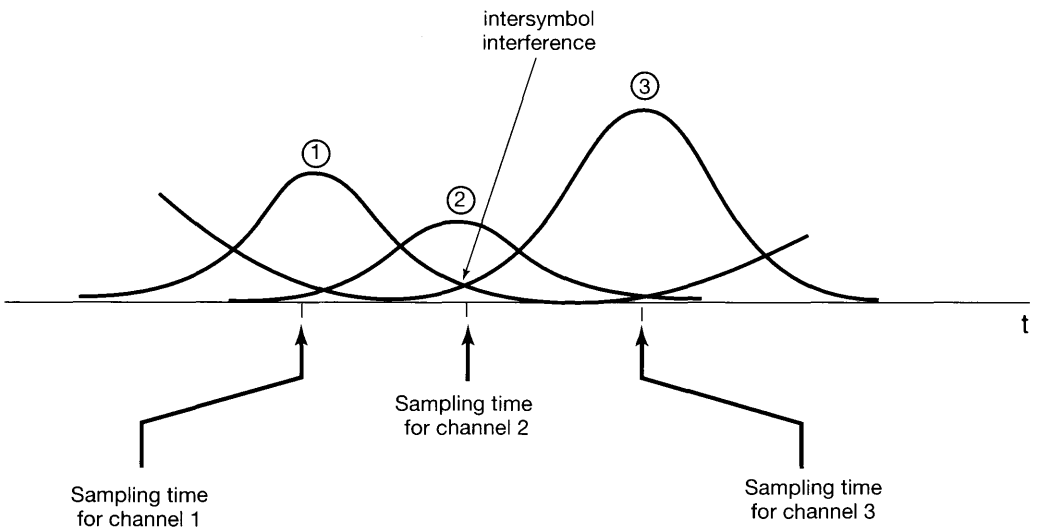


Figure 8.28 Intersymbol interference.

The smearing over time of the idealized pulses in Figure 8.27 can result from the bandwidth constraints of the channel or from phase dispersion caused by nonconstant group delay, as was discussed in Section 6.2.2. (See in particular, Example 6.1.) If the intersymbol interference is due only to the limited bandwidth of the channel, an approach is to use a pulse shape $p(t)$ that is itself band limited and therefore not affected (or only minimally affected) by the restricted bandwidth of the channel. In particular, if the channel has a frequency response $H(j\omega)$ that has no distortion over a specified frequency band (e.g., if $H(j\omega) = 1$ for $|\omega| < W$), then if the pulse that is used is band limited (i.e., if $P(j\omega) = 0$ for $|\omega| \geq W$), each PAM signal will be received without distortion. On the other hand, by using such a pulse, we no longer have pulses without overlap as in Figure 8.27. Nevertheless, intersymbol interference can be avoided in the time domain, even with a band-limited pulse, if the pulse shape is constrained to have zero-crossings at the other sampling times [so that eq. (8.27) continues to hold]. For example, consider the sinc pulse

$$p(t) = \frac{T_1 \sin(\pi t/T_1)}{\pi t}$$

and its corresponding spectrum displayed in Figure 8.29. Since the pulse is zero at integer multiples of the symbol spacing T_1, as indicated in Figure 8.30, there will be no intersymbol interference at these instants. That is, if we sample the received signal at $t = kT_1$, then the contributions to this sampled value from all of the other pulses, i.e., from $p(t - mT_1)$ for $m \neq k$, will be identically zero. Of course, avoiding interference from adjacent symbols

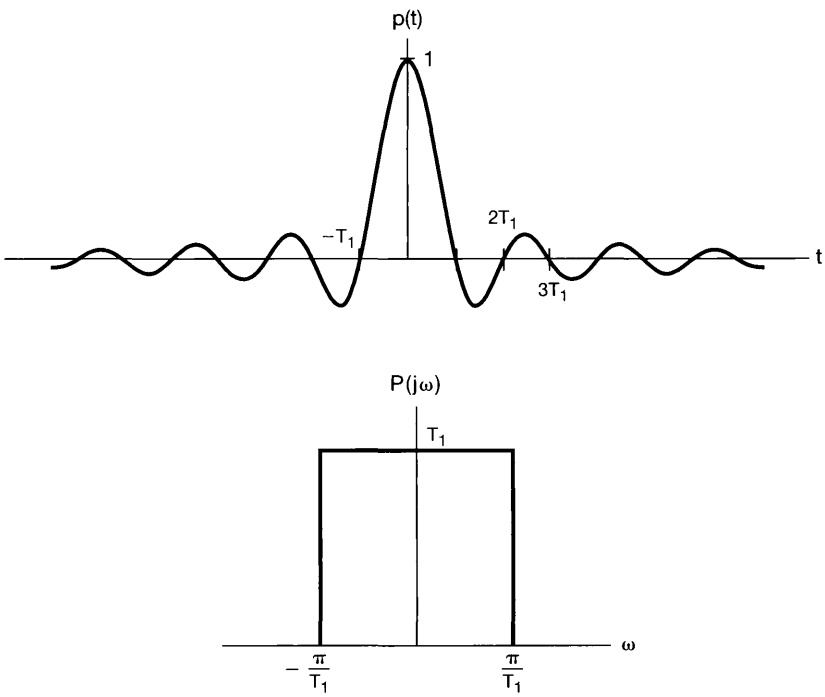


Figure 8.29 A sinc pulse and its corresponding spectrum.

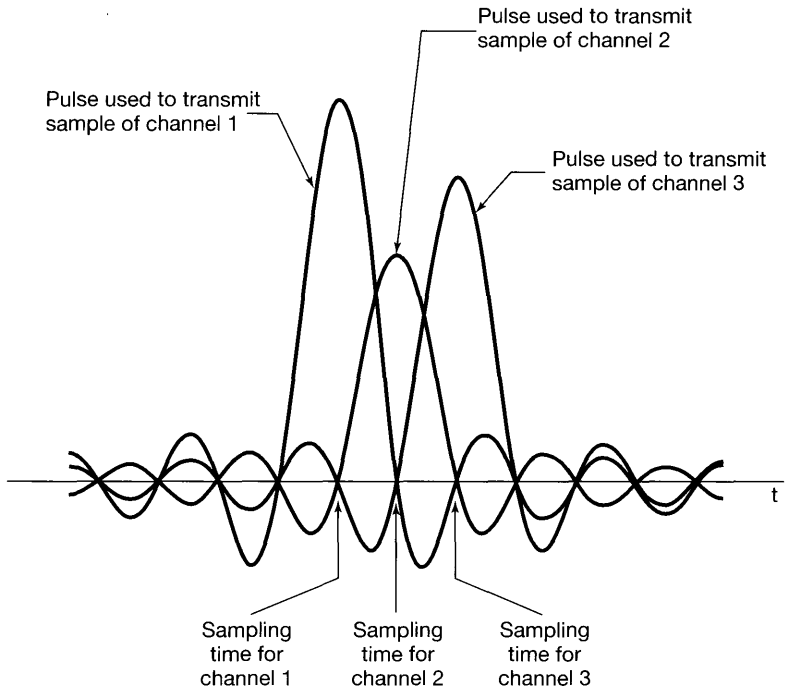


Figure 8.30 Absence of intersymbol interference when sinc pulses with correctly chosen zero-crossings are used.

requires high accuracy in the sampling times, so that sampling occurs at the zero-crossings of the adjacent symbols.

The sinc pulse is only one of many band-limited pulses with time-domain zero-crossings at $\pm T_1$, $\pm 2T_1$, etc. More generally, consider a pulse $p(t)$ with spectrum of the form

$$P(j\omega) = \begin{cases} 1 + P_1(j\omega), & |\omega| \leq \frac{\pi}{T_1}, \\ P_1(j\omega), & \frac{\pi}{T_1} < |\omega| \leq \frac{2\pi}{T_1} \\ 0, & \text{otherwise} \end{cases} \tag{8.28}$$

and with $P_1(j\omega)$ having odd symmetry around π/T_1, so that

$$P_1\left(-j\omega + j\frac{\pi}{T_1}\right) = -P_1\left(j\omega + j\frac{\pi}{T_1}\right) \quad 0 \leq \omega \leq \frac{\pi}{T_1}, \tag{8.29}$$

as illustrated in Figure 8.31. If $P_1(j\omega) = 0$, $p(t)$ is the sinc pulse itself. More generally, as explored in Problem 8.42, for any $P(j\omega)$ satisfying the conditions in eqs. (8.28) and (8.29), $p(t)$ will have zero-crossing at $\pm T_1$, $\pm 2T_1, \ldots$.

While signals satisfying eqs. (8.28) and (8.29) allow us to overcome the problem of limited channel bandwidth, other channel distortions may occur that require a different choice of pulse waveform or some additional processing of the received signal prior to the separation of the different TDM signals. In particular, if $|H(j\omega)|$ is not constant over the passband, there may be a need to perform *channel equalization*—i.e., filtering of

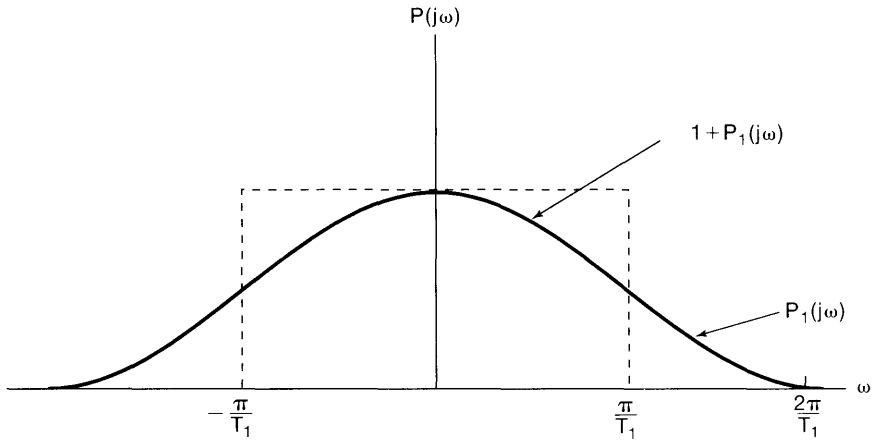


Figure 8.31 Odd symmetry around π/T_1 as defined in eq. (8.29).

the received signal to correct for the nonconstant channel gain. Also, if the channel has nonlinear phase, distortion can result that leads to intersymbol interference, unless compensating signal processing is performed. Problems 8.43 and 8.44 provide illustrations of these effects.

8.6.3 Digital Pulse-Amplitude and Pulse-Code Modulation

The PAM system described in the preceding subsections involves the use of a discrete set of samples to modulate a sequence of pulses. This set of samples can be thought of as a discrete-time signal $x[n]$, and in many applications $x[n]$ is in fact stored in or generated by a digital system. In such cases, the limited word length of a digital system implies that $x[n]$ can take on only a finite, quantized set of values, resulting in only a finite set of possible amplitudes for the modulated pulses.

In fact, in many cases this quantized form of digital PAM is reduced to a system using only a few—typically, only two—amplitude values. In particular, if each sample of $x[n]$ is represented as a binary number (i.e., a finite string of 0's and 1's), then a pulse with one of two possible values (one value corresponding to a 0 and one value to a 1) can be set for each binary digit, or *bit,* in the string. More generally, in order to protect against transmission errors or provide secure communication, the sequence of binary digits representing $x[n]$ might first be transformed or *encoded* into another sequence of 0's and 1's before transmission. For example, a very simple error detection mechanism is to transmit one additional modulated pulse for each sample of $x[n]$, representing a *parity* check. That is, this additional bit would be set to 1 if the binary representation of $x[n]$ has an odd number of 1's in it and to 0 if there is an even number of 1's. The receiver can then check the received parity bit against the other received bits in order to detect inconsistencies. More complex coding and error correction schemes can certainly be employed, and the design of codes with particular desirable properties is an important component of communication system design. For obvious reasons, a PAM system modulated by an encoded sequence of 0's and 1's is referred to as a *pulse-code modulation* (PCM) system.

8.7 SINUSOIDAL FREQUENCY MODULATION

In the preceding sections, we discussed a number of specific amplitude modulation systems in which the modulating signal was used to vary the amplitude of a sinusoidal or a pulse carrier. As we have seen, such systems are amenable to detailed analysis using the frequency-domain techniques we developed in preceding chapters. In another very important class of modulation techniques referred to as *frequency modulation (FM),* the modulating signal is used to control the frequency of a sinusoidal carrier. Modulation systems of this type have a number of advantages over amplitude modulation systems. As suggested by Figure 8.10, with sinusoidal amplitude modulation the peak amplitude of the envelope of the carrier is directly dependent on the amplitude of the modulating signal $x(t)$, which can have a large dynamic range—i.e., can vary significantly. With frequency modulation, the envelope of the carrier is constant. Consequently, an FM transmitter can always operate at peak power. In addition, in FM systems, amplitude variations introduced over a transmission channel due to additive disturbances or fading can, to a large extent, be eliminated at the receiver. For this reason, in public broadcasting and a variety of other contexts, FM reception is typically better than AM reception. On the other hand, as we will see, frequency modulation generally requires greater bandwidth than does sinusoidal amplitude modulation.

Frequency modulation systems are highly nonlinear and, consequently, are not as straightforward to analyze as are the amplitude modulation systems discussed in the preceding sections. However, the methods we have developed in earlier chapters do allow us to gain some understanding of the nature and operation of these systems.

We begin by introducing the general notion of *angle modulation.* Consider a sinusoidal carrier expressed in the form

$$c(t) = A\cos(\omega_c t + \theta_c) = A\cos\theta(t), \tag{8.30}$$

where $\theta(t) = \omega_c t + \theta_c$ and where ω_c is the frequency and θ_c the phase of the carrier. Angle modulation, in general, corresponds to using the modulating signal to change or vary the angle $\theta(t)$. One form that this sometimes takes is to use the modulating signal $x(t)$ to vary the phase θ_c so that the modulated signal takes the form

$$y(t) = A\cos[\omega_c t + \theta_c(t)], \tag{8.31}$$

where θ_c is now a function of time, specifically of the form

$$\theta_c(t) = \theta_0 + k_p x(t). \tag{8.32}$$

If $x(t)$ is, for example, constant, the phase of $y(t)$ will be constant and proportional to the amplitude of $x(t)$. Angle modulation of the form of eq. (8.31) is referred to as *phase modulation.* Another form of angle modulation corresponds to varying the *derivative* of the angle proportionally with the modulating signal; that is,

$$y(t) = A\cos\theta(t), \tag{8.33}$$

where

$$\frac{d\theta(t)}{dt} = \omega_c + k_f x(t). \tag{8.34}$$

For $x(t)$ constant, $y(t)$ is sinusoidal with a frequency that is offset from the carrier frequency ω_c by an amount proportional to the amplitude of $x(t)$. For that reason, angle modulation of the form of eqs. (8.33) and (8.34) is commonly referred to as *frequency modulation.*

Although phase modulation and frequency modulation are different forms of angle modulation, they can be easily related. From eqs. (8.31) and (8.32), for phase modulation,

$$\frac{d\theta(t)}{dt} = \omega_c + k_p \frac{dx(t)}{dt}, \tag{8.35}$$

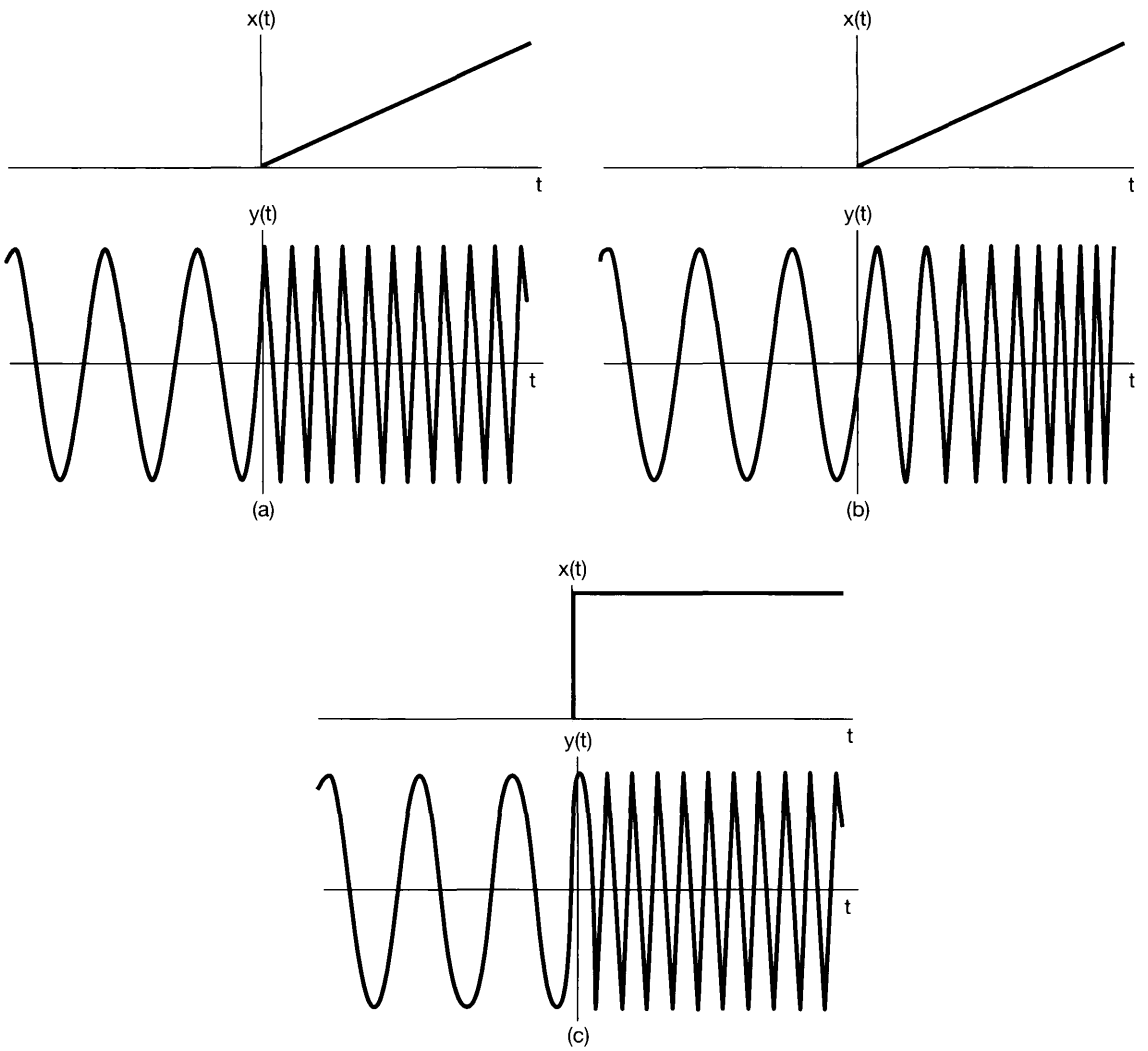


Figure 8.32 Phase modulation, frequency modulation, and their relationship: (a) phase modulation with a ramp as the modulating signal; (b) frequency modulation with a ramp as the modulating signal; (c) frequency modulation with a step (the derivative of a ramp) as the modulating signal.

and thus, comparing eqs. (8.34) and (8.35), we see that phase modulating with $x(t)$ is identical to frequency modulating with the derivative of $x(t)$. Likewise, frequency modulating with $x(t)$ is identical to phase modulating with the integral of $x(t)$. An illustration of phase modulation and frequency modulation is shown in Figures 8.32(a) and (b). In both cases, the modulating signal is $x(t) = tu(t)$ (i.e., a ramp signal increasing linearly with time for $t > 0$). In Figure 8.32(c), an example of frequency modulation is shown with a step (the derivative of a ramp) as the modulating signal [i.e., $x(t) = u(t)$]. The correspondence between Figures 8.32(a) and (c) should be evident.

Frequency modulation with a step corresponds to the frequency of the sinusoidal carrier changing instantaneously from one value to another when $x(t)$ changes value at $t = 0$, much as the frequency of a sinusoidal oscillator changes when the frequency setting is switched instantaneously. When the frequency modulation is a ramp, as in Figure 8.32(b), the frequency changes linearly with time. This notion of a time-varying frequency is often best expressed in terms of the concept of *instantaneous frequency*. For

$$y(t) = A\cos\theta(t), \tag{8.36}$$

the instantaneous frequency of the sinusoid is defined as

$$\omega_i(t) = \frac{d\theta(t)}{dt}. \tag{8.37}$$

Thus, for $y(t)$ truly sinusoidal [i.e., $\theta(t) = (\omega_c t + \theta_0)$], the instantaneous frequency is ω_c, as we would expect. For phase modulation as expressed in eqs. (8.31) and (8.32), the instantaneous frequency is $\omega_c + k_p(dx(t)/dt)$, and for frequency modulation as expressed in eqs. (8.33) and (8.34), the instantaneous frequency is $\omega_c + k_f x(t)$.

Since frequency modulation and phase modulation are easily related, we will phrase the remaining discussion in terms of frequency modulation alone. To gain some insight into how the spectrum of the frequency-modulated signal is affected by the modulating signal $x(t)$, it is useful to consider two cases in which the modulating signal is sufficiently simple so that some of the essential properties of frequency modulation become evident.

8.7.1 Narrowband Frequency Modulation

Consider the case of frequency modulation with

$$x(t) = A\cos\omega_m t. \tag{8.38}$$

From eqs. (8.34) and (8.37), the instantaneous frequency is

$$\omega_i(t) = \omega_c + k_f A\cos\omega_m t, \tag{8.39}$$

which varies sinusoidally between $\omega_c + k_f A$ and $\omega_c - k_f A$. With

$$\Delta\omega = k_f A,$$

we have

$$\omega_i(t) = \omega_c + \Delta\omega\cos\omega_m t,$$

and

$$\begin{aligned} y(t) &= \cos[\omega_c t + \int x(t)dt] \\ &= \cos\left(\omega_c t + \frac{\Delta\omega}{\omega_m}\sin\omega_m t + \theta_0\right), \end{aligned} \tag{8.40}$$

where θ_0 is a constant of integration. For convenience we will choose $\theta_0 = 0$, so that

$$y(t) = \cos\left[\omega_c t + \frac{\Delta\omega}{\omega_m}\sin\omega_m t\right]. \tag{8.41}$$

The factor $\Delta\omega/\omega_m$, which we denote by m, is defined as the *modulation index* for frequency modulation. The properties of FM systems tend to be different, depending on whether the modulation index m is small or large. The case in which m is small is referred to as *narrowband FM*. In general, we can rewrite eq. (8.41) as

$$y(t) = \cos(\omega_c t + m\sin\omega_m t) \tag{8.42}$$

or

$$y(t) = \cos\omega_c t\cos(m\sin\omega_m t) - \sin\omega_c t\sin(m\sin\omega_m t). \tag{8.43}$$

When m is sufficiently small ($\ll \pi/2$), we can make the approximations

$$\cos(m\sin\omega_m t) \simeq 1, \tag{8.44}$$

$$\sin(m\sin\omega_m t) \simeq m\sin\omega_m t, \tag{8.45}$$

so that eq. (8.42) becomes

$$y(t) \simeq \cos\omega_c t - m(\sin\omega_m t)(\sin\omega_c t). \tag{8.46}$$

The spectrum of $y(t)$ based on this approximation is shown in Figure 8.33. We note that it has a similarity to AM-DSB/WC in that the carrier frequency is present in the spectrum and there are sidebands representing the spectrum of the modulating signal in eq. (8.38). However, in AM-DSB/WC the additional carrier injected is in phase with the modulated carrier, whereas, as we see in eq. (8.46) for the case of the narrowband FM, the carrier signal has a phase difference of $\pi/2$ in relation to the amplitude-modulated carrier. The waveforms corresponding to AM-DSB/WC and FM are also very different. Figure 8.34(a) illustrates the narrowband FM waveform corresponding to eq. (8.46). For comparison, Figure 8.34(b) shows the AM-DSB/WC signal

$$y_2(t) = \cos\omega_c t + m(\cos\omega_m t)(\cos\omega_c t). \tag{8.47}$$

For the narrowband FM signal of eq. (8.46), the bandwidth of the sidebands is equal to the bandwidth of the modulating signal, and in particular, although the approximation in the equation is based on assuming that $m \ll \pi/2$, the bandwidth of the sidebands is otherwise independent of the modulation index m (i.e., it depends only on the bandwidth of the modulating signal, not on its amplitude). A similar statement applies for narrowband FM with a more general modulating signal.

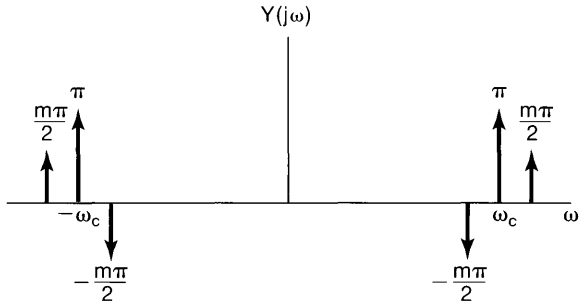
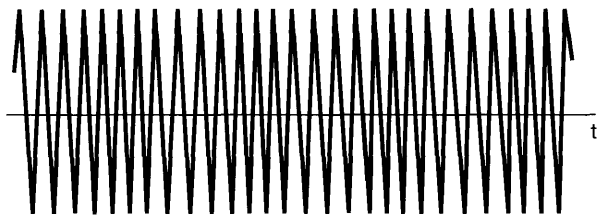
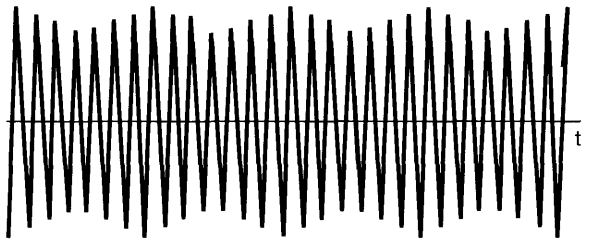


Figure 8.33 Approximate spectrum for narrowband FM.



(a)



(b)

Figure 8.34 Comparison of narrowband FM and AM-DSB/WC: (a) narrowband FM; (b) AM-DSB/WC.

8.7.2 Wideband Frequency Modulation

When m is large, the approximation leading to eq. (8.46) no longer applies, and the spectrum of $y(t)$ depends on both the amplitude and the spectrum of the modulating signal $x(t)$. With $y(t)$ expressed in the form of eq. (8.43), we note that the terms $\cos[m \sin \omega_m t]$ and $\sin[m \sin \omega_m t]$ are periodic signals with fundamental frequency ω_m. Thus, the Fourier transform of each of these signals is an impulse train with impulses at integer multiples of ω_m and amplitudes proportional to the Fourier series coefficients. The coefficients for these two periodic signals involve a class of functions referred to as Bessel functions of the first kind. The first term in eq. (8.43) corresponds to a sinusoidal carrier of the form $\cos \omega_c t$ amplitude modulated by the periodic signal $\cos[m \sin \omega_m t]$ and the second term to a sinusoidal carrier $\sin \omega_c t$ amplitude modulated by the periodic signal $\sin[m \sin \omega_m t]$. Multiplication by the carrier signals has the effect in the frequency domain of translating the spectrum of eq. (8.43) to the carrier frequency, so that it is centered at plus and minus

ω_c. In Figures 8.35(a) and (b) we illustrate, for $\omega > 0$, the the magnitude of the spectra of the two individual terms in eq. (8.43), and in Figure 8.35(c) the magnitude of the combined spectrum representing the modulated signal $y(t)$. The spectrum of $y(t)$ consists of impulses at frequencies $\pm\omega_c + n\omega_m$, $n = 0, \pm 1, \pm 2, \ldots$, and is not, strictly speaking, band limited around $\pm\omega_c$. However, the behavior of the Fourier series coefficients of $\cos[m \sin \omega_m t]$ and $\sin[m \sin \omega_m t]$ are such that the amplitude of the nth harmonic for $|n| > m$ can be considered negligible, and thus, the total bandwidth B of each sideband centered around $+\omega_c$ and $-\omega_c$ is effectively limited to $2m\omega_m$. That is,

$$B \simeq 2m\omega_m, \tag{8.48}$$

or, since $m = k_f A/\omega_m = \Delta\omega/\omega_m$,

$$B \simeq 2k_f A = 2\Delta\omega. \tag{8.49}$$

Comparing eqs. (8.39) and (8.49), we note that the effective bandwidth of each sideband is equal to the total excursion of the instantaneous frequency around the carrier

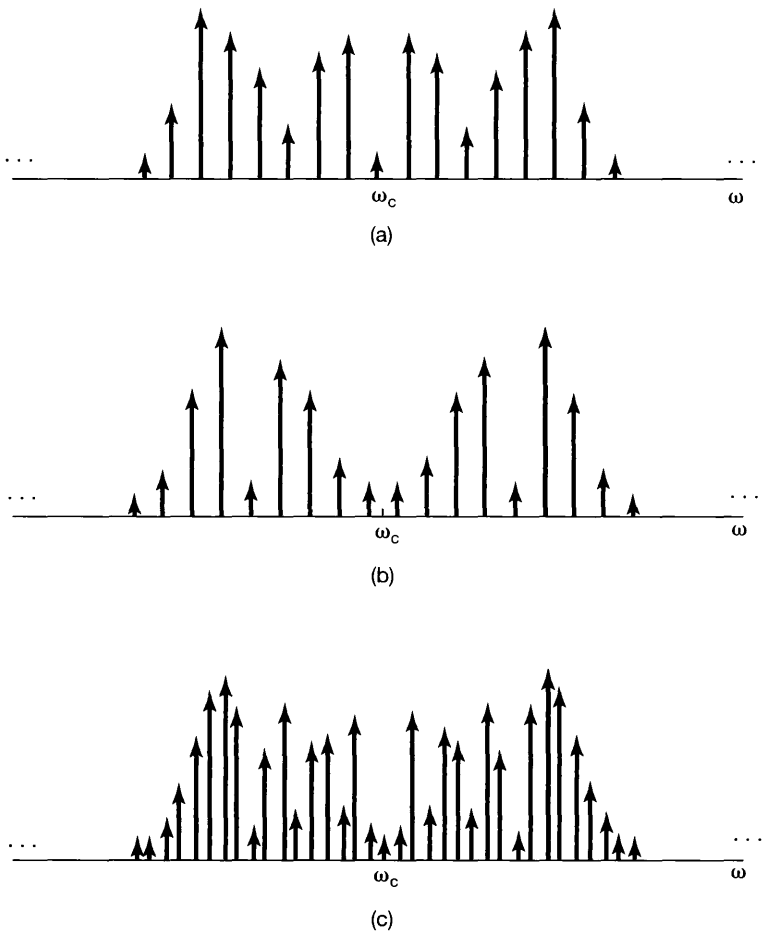


Figure 8.35 Magnitude of spectrum of wideband frequency modulation with $m = 12$: (a) magnitude of spectrum of $\cos \omega_c t \cos[m \sin \omega_m t]$; (b) magnitude of spectrum of $\sin \omega_c t \sin[m \sin \omega_m t]$; (c) combined spectral magnitude of $\cos[\omega_c t + m \sin \omega_m t]$.

frequency. Therefore, for wideband FM, since we assume that m is large, the bandwidth of the modulated signal is much larger than the bandwidth of the modulating signal, and in contrast to the narrowband case, the bandwidth of the transmitted signal in wideband FM is directly proportional to amplitude A of the modulating signal and the gain factor k_f.

8.7.3 Periodic Square-Wave Modulating Signal

Another example that lends insight into the properties of frequency modulation is that of a modulating signal which is a periodic square wave. Referring to eq. (8.39), let $k_f = 1$ so that $\Delta\omega = A$, and let $x(t)$ be given by Figure 8.36. The modulated signal $y(t)$ is illustrated in Figure 8.37. The instantaneous frequency is $\omega_c + \Delta\omega$ when $x(t)$ is positive and $\omega_c - \Delta\omega$ when $x(t)$ is negative. Thus, $y(t)$ can also be written as

$$y(t) = r(t)\cos[(\omega_c + \Delta\omega)t] + r\left(t - \frac{T}{2}\right)\cos[(\omega_c - \Delta\omega)t], \tag{8.50}$$

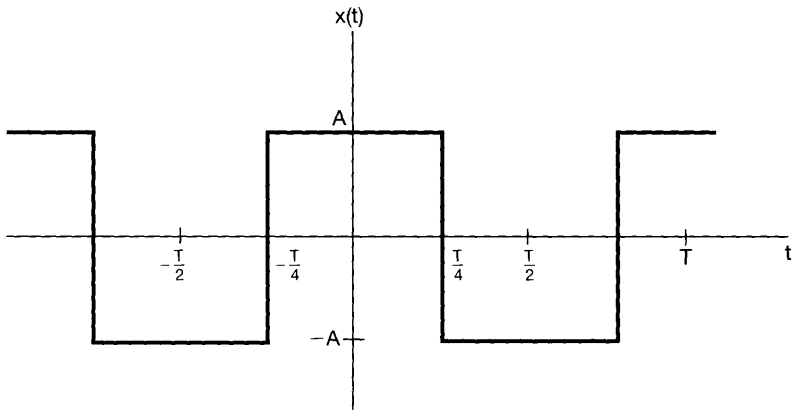


Figure 8.36 Symmetric periodic square wave.

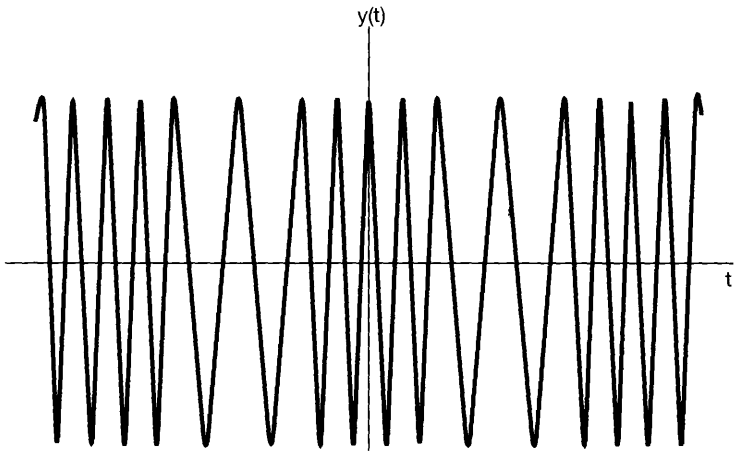


Figure 8.37 Frequency modulation with a periodic square-wave modulating signal.

where $r(t)$ is the symmetric square wave shown in Figure 8.38. Thus, for this particular modulating signal, we are able to recast the problem of determining the spectrum of the FM signal $y(t)$ as the determination of the spectrum of the sum of the two AM signals in eq. (8.50). Specifically,

$$Y(j\omega) = \frac{1}{2}[R(j\omega + j\omega_c + j\Delta\omega) + R(j\omega - j\omega_c - j\Delta\omega)] + \frac{1}{2}[R_T(j\omega + j\omega_c - j\Delta\omega) + R_T(j\omega - j\omega_c + j\Delta\omega)], \tag{8.51}$$

where $R(j\omega)$ is the Fourier transform of the periodic square wave $r(t)$ in Figure 8.38 and $R_T(j\omega)$ is the Fourier transform of $r(t - T/2)$. From Example 4.6, with $T = 4T_1$,

$$R(j\omega) = \sum_{k=-\infty}^{\infty} \frac{2}{2k+1}(-1)^k \delta\left[\omega - \frac{2\pi(2k+1)}{T}\right] + \pi\delta(\omega) \tag{8.52}$$

and

$$R_T(j\omega) = R(j\omega)e^{-j\omega T/2}. \tag{8.53}$$

The magnitude of the spectrum of $Y(j\omega)$ is illustrated in Figure 8.39. As with wideband FM, the spectrum has the general appearance of two sidebands, centered around $\omega_c \pm \Delta\omega$, that decay for $\omega < \omega_c - \Delta\omega$ and $\omega > \omega_c + \Delta\omega$.

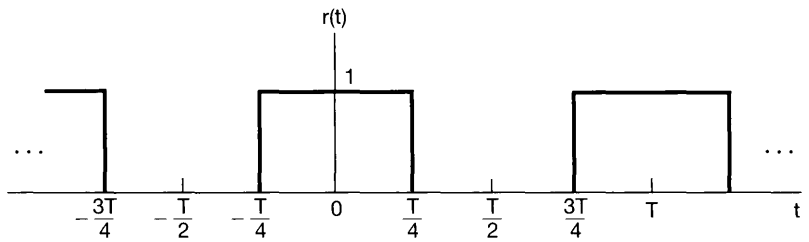


Figure 8.38 Symmetric square wave $r(t)$ in eq. (8.50).

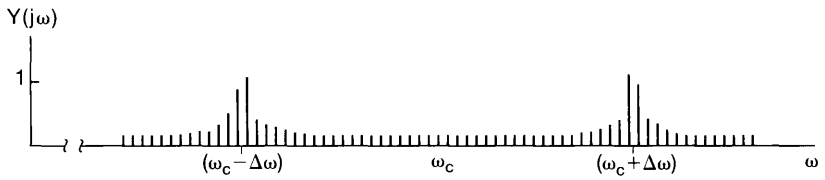


Figure 8.39 Magnitude of the spectrum for $\omega > 0$ corresponding to frequency modulation with a periodic square-wave modulating signal. Each of the vertical lines in the figure represents an impulse of area proportional to the height of the line.

Systems for the demodulation of FM signals typically are of two types. One type of demodulation system corresponds to converting the FM signal to an AM signal through differentiation, while demodulation systems of the second type directly track the phase or frequency of the modulated signal. The foregoing discussion provides only a brief introduction to the characteristics of frequency modulation, and we have again seen how the basic techniques developed in the earlier chapters can be exploited to analyze and gain an insight into an important class of systems.

8.8 DISCRETE-TIME MODULATION

8.8.1 Discrete-Time Sinusoidal Amplitude Modulation

A discrete-time amplitude modulation system is depicted in Figure 8.40, in which $c[n]$ is the carrier and $x[n]$ the modulating signal. The basis for our analysis of continuous-time amplitude modulation was the multiplication property for Fourier transforms—specifically, the fact that multiplication in the time domain corresponds to convolution in the frequency domain. As we discussed in Section 5.5, there is a corresponding property for discrete-time signals which we can use to analyze discrete-time amplitude modulation. Specifically, consider

$$y[n] = x[n]c[n].$$

With $X(e^{j\omega})$, $Y(e^{j\omega})$, and $C(e^{j\omega})$ denoting the Fourier transforms of $x[n]$, $y[n]$, and $c[n]$, respectively, $Y(e^{j\omega})$ is proportional to the periodic convolution of $X(e^{j\omega})$ and $C(e^{j\omega})$; that is,

$$Y(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})C(e^{j(\omega-\theta)})d\theta. \tag{8.54}$$

Since $X(e^{j\omega})$ and $C(e^{j\omega})$ are periodic with a period of 2π, the integration can be performed over any frequency interval of length 2π.

Let us first consider sinusoidal amplitude modulation with a complex exponential carrier, so that

$$c[n] = e^{j\omega_c n}. \tag{8.55}$$

As we saw in Section 5.2, the Fourier transform of $c[n]$ is a periodic impulse train; that is,

$$C(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} 2\pi\delta(\omega - \omega_c + k2\pi), \tag{8.56}$$

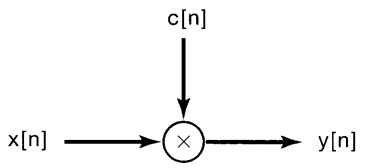


Figure 8.40 Discrete-time amplitude modulation.

which is sketched in Figure 8.41(b). With $X(e^{j\omega})$ as illustrated in Figure 8.41(a), the spectrum of the modulated signal is that shown in Figure 8.41(c). In particular, we note that $Y(e^{j\omega}) = X(e^{j(\omega-\omega_c)})$. This is the discrete-time counterpart to Figure 8.1, and here again, with $x[n]$ real, the modulated signal will be complex. Demodulation is accomplished by multiplying by $e^{-j\omega_c n}$ to translate the spectrum back to its original location on the frequency axis, so that

$$x[n] = y[n]e^{-j\omega_c n}. \tag{8.57}$$

As explored in Problem 6.43, if $\omega_c = \pi$ so that $c[n] = (-1)^n$, the result of modulation in the time domain is to change the algebraic sign of $x[n]$ for odd values of n, while in the frequency domain the consequence is the interchanging of high and low frequencies. Problem 6.44 explores the use of this type of modulation in utilizing a lowpass filter to achieve highpass filtering and vice versa.

As an alternative to a complex exponential carrier, we can use a sinusoidal carrier, in which case, with $x[n]$ real, the modulated signal $y[n]$ will also be real. With $c[n] = \cos\omega_c n$, the spectrum of the carrier consists of periodically repeated pairs of impulses at

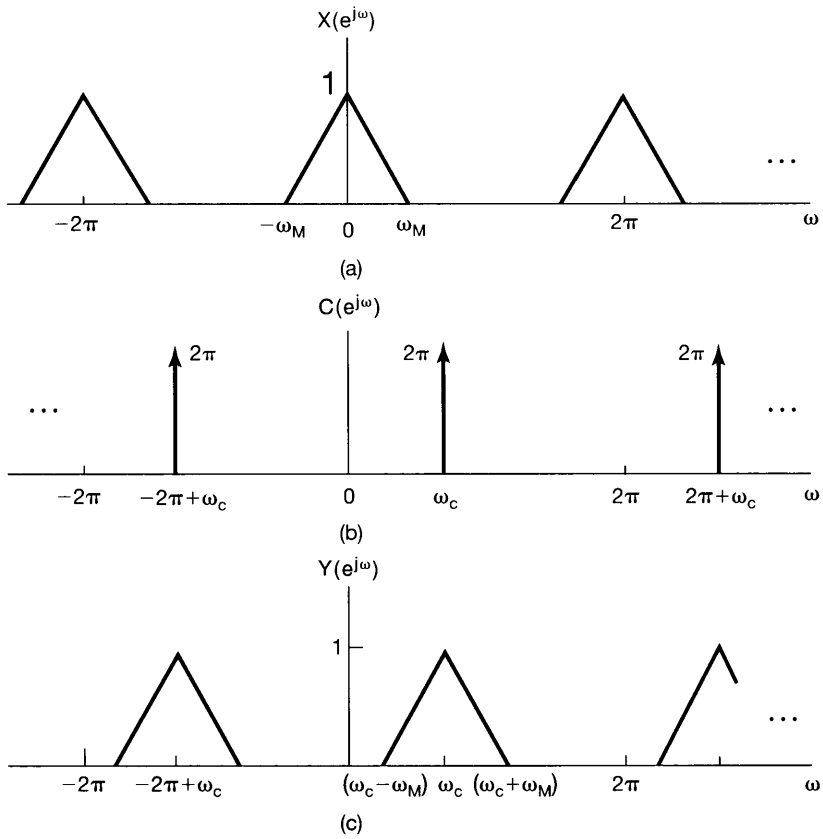


Figure 8.41 (a) Spectrum of $x[n]$; (b) spectrum of $c[n] = e^{j\omega_c n}$; (c) spectrum of $y[n] = x[n]c[n]$.

$\omega = \pm\omega_c + k2\pi$, as illustrated in Figure 8.42(b). With $X(e^{j\omega})$ as shown in Figure 8.42(a), the resulting spectrum for the modulated signal is shown in Figure 8.42(c) and corresponds to replicating $X(e^{j\omega})$ at the frequencies $\omega = \pm\omega_c + k2\pi$. In order that the individual replications of $X(e^{j\omega})$ do not overlap, we require that

$$\omega_c > \omega_M \tag{8.58}$$

and

$$2\pi - \omega_c - \omega_M > \omega_c + \omega_M$$

or, equivalently,

$$\omega_c < \pi - \omega_M. \tag{8.59}$$

The first condition is identical to that in Section 8.2 for continous-time sinusoidal amplitude modulation, while the second results from the inherent periodicity of discrete-time

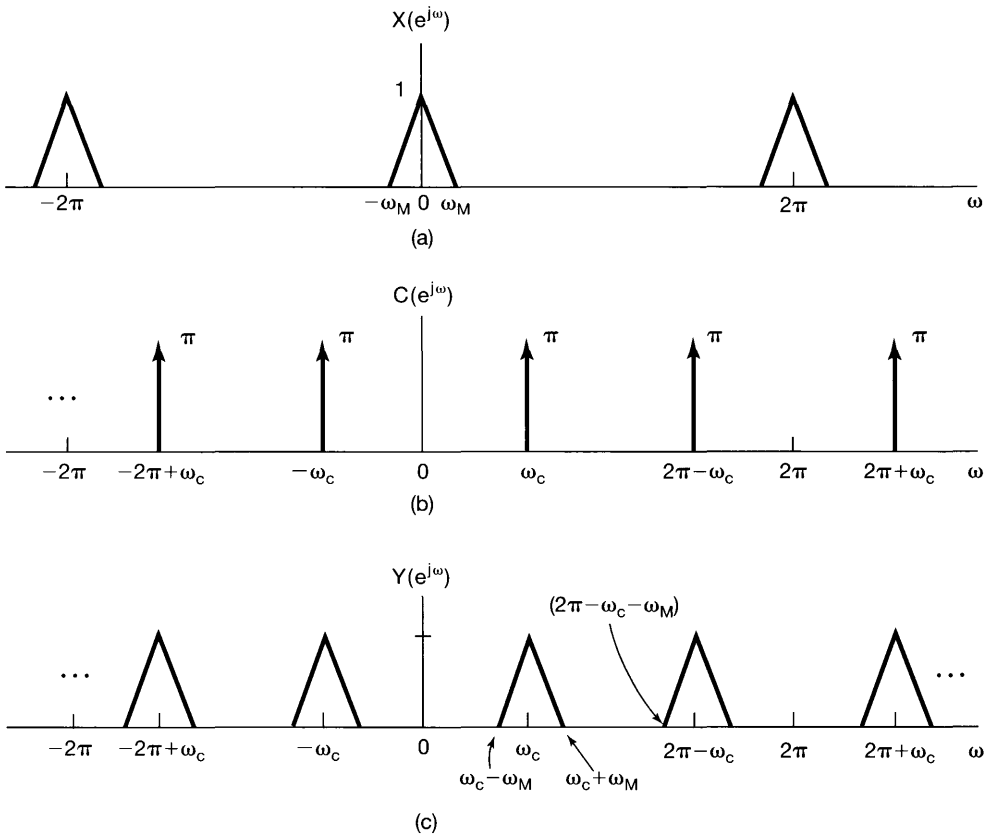


Figure 8.42 Spectra associated with discrete-time modulation using a sinusoidal carrier: (a) spectrum of a bandlimited-signal $x[n]$; (b) spectrum of a sinusoidal carrier signal $c[n] = \cos\omega_c n$; (c) spectrum of the modulated signal $y[n] = x[n]c[n]$.

spectra. Combining eqs. (8.58) and (8.59), we see that for amplitude modulation with a sinusoidal carrier, we must restrict ω_c so that

$$\omega_M < \omega_c < \pi - \omega_M. \tag{8.60}$$

Demodulation can be accomplished in a manner similar to that employed in continuous time. As illustrated in Figure 8.43, multiplication of $y[n]$ with the same carrier used in the modulator results in several replications of the spectrum of the original signal, one of which is centered about $\omega = 0$. By lowpass filtering to eliminate the unwanted replications of $X(e^{j\omega})$, the demodulated signal is obtained.

As should be evident from the foregoing discussion, analysis of discrete-time amplitude modulation proceeds in a manner similar to that of continuous-time amplitude modulation, with only slight differences. For example, as explored in Problem 8.47, in the synchronous modulation and demodulation system, the effect of a phase difference or a frequency difference between the sinusoidal carriers in the modulator and demodulator is identical in both discrete and continuous time. In addition, just as in continuous time, we can use discrete-time sinusoidal AM as the basis for frequency-division multiplexing in

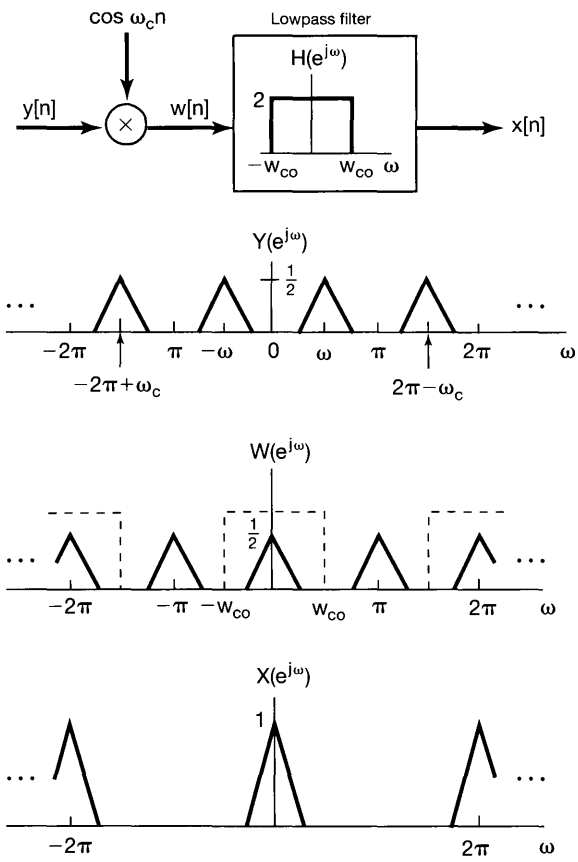


Figure 8.43 System and associated spectra for discrete-time synchronous demodulation.

discrete time. Furthermore, as explored in Problem 8.48, we can also consider using a discrete-time signal to modulate a pulse train, leading to time-division multiplexing of discrete-time signals.

The implementation of discrete-time multiplexing systems provides an excellent example of the flexibility of discrete-time processing in general and the importance of the operation of upsampling (see Section 7.5.2) in particular. Consider a discrete-time FDM system with M sequences that we wish to frequency-division multiplex. With M channels, it is required that the spectral energy for each input channel $x_i[n]$ be band limited; that is,

$$X_i(\omega) = 0, \qquad \frac{\pi}{M} < |\omega| < \pi. \tag{8.61}$$

If the sequences originally occupied the entire frequency band corresponding, for example, to having sampled a set of continuous-time signals at the Nyquist rate, then they would first have to be converted to a higher sampling rate (i.e., upsampled) before frequency-division multiplexing. This idea is explored further in Problem 8.33.

8.8.2 Discrete-Time Transmodulation

One context in which discrete-time modulation is widely used, together with the operations of decimation, upsampling, and interpolation introduced in Chapter 7, is digital communication systems. Typically, in such systems continuous-time signals are transmitted over communication channels in the form of discrete-time signals, obtained by sampling. The continous-time signals are often in the form of time-division-multiplexed (TDM) or frequency-division-multiplexed (FDM) signals. The signals are then converted to discrete-time sequences whose values are represented digitally, for storage or long-distance transmission. In some systems, because of different constraints or requirements at the transmitting end and the receiving end, or because sets of signals that have been individually multiplexed by different methods are then multiplexed together, there is often the requirement for converting from sequences representing TDM signals to sequences representing FDM signals or vice versa. This conversion from one modulation or multiplexing scheme to another is referred to as *transmodulation* or *transmultiplexing*. In the context of digital communication systems, one obvious way of implementing transmultiplexing is to convert back to continuous-time signals, demultiplex and demodulate, and then modulate and multiplex as required. However, if the new signal is then to be converted back to a discrete-time signal, it is clearly more efficient for the entire process to be carried out directly in the discrete-time domain. Figure 8.44 shows, in block diagram form, the steps involved in converting a discrete-time TDM signal to a discrete-time FDM signal. Note that, after demultiplexing the TDM signal, each channel must be upsampled in preparation for frequency-division multiplexing.

8.9 SUMMARY

In this chapter, we have examined a number of the basic concepts associated with communication systems. In particular, we have examined the concept of modulation, in which a signal we wish to communicate is used to modulate a second signal referred to as the

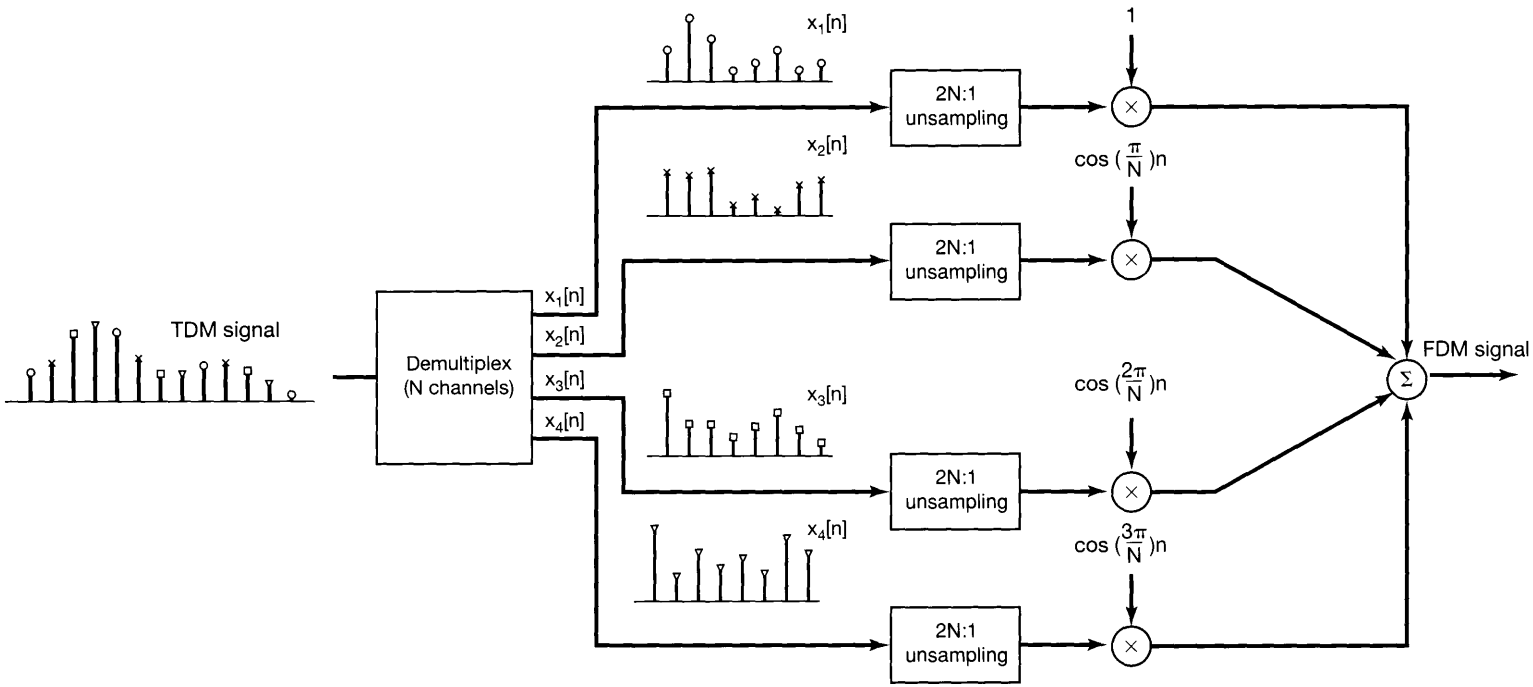


Figure 8.44 Block diagram for TDM-to-FDM transmultiplexing.

carrier, and we have looked in detail at the concept of amplitude modulation. The properties of amplitude modulation are most easily interpreted in the frequency domain through the multiplication property of the Fourier transform. Amplitude modulation with a complex exponential or sinusoidal carrier is typically used to shift the spectrum of a signal in frequency and is applied, for example, in communication systems to place the spectrum in a frequency range suitable for transmission and to permit frequency-division multiplexing. Variations of sinusoidal amplitude modulation, such as the insertion of a carrier signal for asynchronous systems and single- and double-sideband systems, were discussed.

We also examined several other forms of modulation-based communication. In this regard, we briefly introduced the concepts of frequency and phase modulation. Although these forms of modulation are more difficult to analyze in detail, it is possible to gain significant insight into their characteristics through the frequency domain.

We further examined in some detail amplitude modulation of a pulsed signal, which led us to the concepts of time-division multiplexing and pulse-amplitude modulation, in which successive samples of a discrete-time signal are used to modulate the amplitude of a sequence of pulses. This led in turn to an examination of discrete-time modulation and digital communication, in which the flexibility of discrete-time processing facilitates the design and implementation of more sophisticated communication systems involving concepts such as pulse-code modulation and transmodulation.

Chapter 8 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining two sections contain problems belonging to the basic and advanced categories, respectively.

BASIC PROBLEMS WITH ANSWERS

8.1. Let $x(t)$ be a signal for which $X(j\omega) = 0$ when $|\omega| > \omega_M$. Another signal $y(t)$ is specified as having the Fourier transform $Y(j\omega) = 2X(j(\omega - \omega_c))$. Determine a signal $m(t)$ such that

$$x(t) = y(t)m(t).$$

8.2. Let $x(t)$ be a real-valued signal for which $X(j\omega) = 0$ when $|\omega| > 1{,}000\pi$. Supposing that $y(t) = e^{j\omega_c t}x(t)$, answer the following questions:

(a) What constraint should be placed on ω_c to ensure that $x(t)$ is recoverable from $y(t)$?

(b) What constraint should be placed on ω_c to ensure that $x(t)$ is recoverable from $\mathcal{R}e\{y(t)\}$?

8.3. Let $x(t)$ be a real-valued signal for which $X(j\omega) = 0$ when $|\omega| > 2{,}000\pi$. Amplitude modulation is performed to produce the signal

$$g(t) = x(t)\sin(2{,}000\pi t).$$

A proposed demodulation technique is illustrated in Figure P8.3 where $g(t)$ is the input, $y(t)$ is the output, and the ideal lowpass filter has cutoff frequency $2{,}000\pi$ and passband gain of 2. Determine $y(t)$.

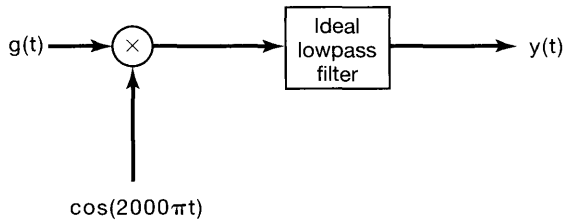


Figure P8.3

8.4. Suppose

$$x(t) = \sin 200\pi t + 2\sin 400\pi t$$

and

$$g(t) = x(t)\sin 400\pi t.$$

If the product $g(t)(\sin 400\pi t)$ is passed through an ideal lowpass filter with cutoff frequency 400π and passband gain of 2, determine the signal obtained at the output of the lowpass filter.

8.5. Suppose we wish to transmit the signal

$$x(t) = \frac{\sin 1{,}000\pi t}{\pi t}$$

using a modulator that creates the signal

$$w(t) = (x(t) + A)\cos(10{,}000\pi t).$$

Determine the largest permissible value of the modulation index m that would allow asynchronous demodulation to be used to recover $x(t)$ from $w(t)$. For this problem, you should assume that the maximum magnitude taken on by a side lobe of a sinc function occurs at the instant of time that is exactly halfway between the two zero-crossings enclosing the side lobe.

8.6. Assume that $x(t)$ is a signal whose Fourier transform $X(j\omega)$ is zero for $|\omega| > \omega_M$. The signal $g(t)$ may be expressed in terms of $x(t)$ as

$$g(t) = x(t)\cos\omega_c t - \left\{ x(t)\cos\omega_c t * \left(\frac{\sin\omega_c t}{\pi t}\right)\right\},$$

where $*$ denotes convolution and $\omega_c > \omega_M$. Determine the value of the constant A such that

$$x(t) = (g(t)\cos\omega_c t) * \frac{A\sin\omega_M t}{\pi t}.$$

8.7. An AM-SSB/SC system is applied to a signal $x(t)$ whose Fourier transform $X(j\omega)$ is zero for $|\omega| > \omega_M$. The carrier frequency ω_c used in the system is greater than ω_M. Let $g(t)$ denote the output of the system, assuming that only the *upper* sidebands are retained. Let $q(t)$ denote the output of the system, assuming that only the *lower* sidebands are retained. The system in Figure P8.7 is proposed for converting $g(t)$ into $q(t)$. How should the parameter ω_0 in the figure be related to ω_c? What should be the value of passband gain A?

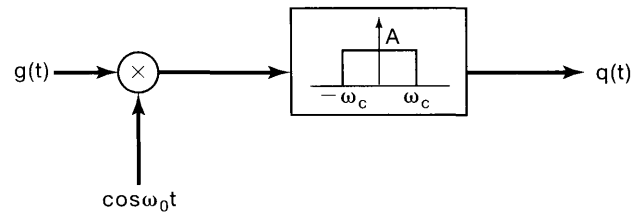


Figure P8.7

8.8. Consider the modulation system shown in Figure P8.8. The input signal $x(t)$ has a Fourier transform $X(j\omega)$ that is zero for $|\omega| > \omega_M$. Assuming that $\omega_c > \omega_M$, answer the following questions:

(a) Is $y(t)$ guaranteed to be real if $x(t)$ is real?

(b) Can $x(t)$ be recovered from $y(t)$?

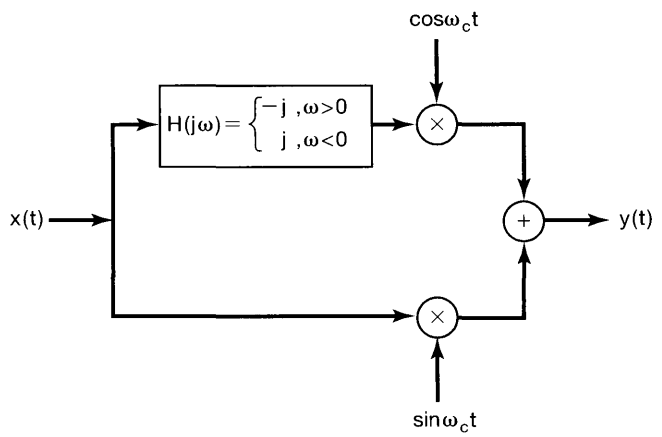


Figure P8.8

8.9. Two signals $x_1(t)$ and $x_2(t)$, each with a Fourier transform that is zero for $|\omega| > \omega_c$, are to be combined using frequency-division multiplexing. The AM-SSB/SC technique of Figure 8.21 is applied to each signal in a manner that retains the lower sidebands. The carrier frequencies used for $x_1(t)$ and $x_2(t)$ are ω_c and $2\omega_c$, respectively. The two modulated signals are then summed together to obtain the FDM signal $y(t)$.

(a) For what values of ω is $Y(j\omega)$ guaranteed to be zero?

(b) Specify the values of A and ω_0 so that

$$x_1(t) = \left[\left\{ y(t) * \frac{\sin \omega_0 t}{\pi t} \right\} \cos \omega_0 t\right] * \frac{A \sin \omega_c t}{\pi t},$$

where $*$ denotes convolution.

8.10. A signal $x(t)$ is multiplied by the rectangular pulse train $c(t)$ shown in Figure P8.10.

(a) What constraint should be placed on $X(j\omega)$ to ensure that $x(t)$ can be recovered from the product $x(t)c(t)$ by using an ideal lowpass filter?

(b) Specify the cutoff frequency ω_c and the passband gain A of the ideal lowpass filter needed to recover $x(t)$ from $x(t)c(t)$. [Assume that $X(j\omega)$ satisfies the constraint determined in part (a).]

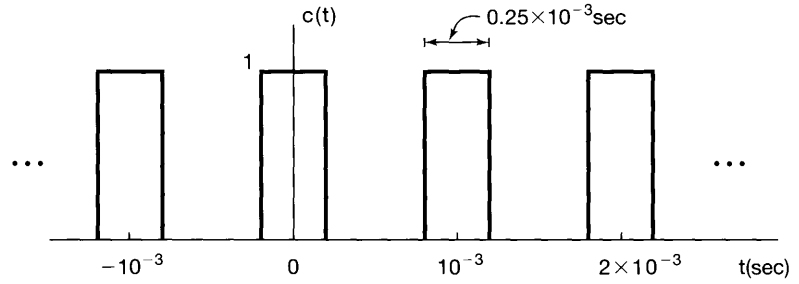


Figure P8.10

8.11. Let

$$c(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_c t},$$

where $a_0 = 0$ and $a_1 \neq 0$, be a real-valued periodic signal. Also, let $x(t)$ be a signal with $X(j\omega) = 0$ for $|\omega| \geq \omega_c/2$. The signal $x(t)$ is used to modulate the carrier $c(t)$ to obtain

$$y(t) = x(t)c(t).$$

(a) Specify the passband and the passband gain of an ideal bandpass filter so that, with input $y(t)$, the output of the filter is

$$g(t) = (a_1 e^{j\omega_c t} + a_1^* e^{-j\omega_c t})x(t).$$

(b) If $a_1 = |a_1|e^{j\sphericalangle a_1}$, show that

$$g(t) = A \cos(\omega_c t + \phi)x(t),$$

and express A and ϕ in terms of $|a_1|$ and $\sphericalangle a_1$.

8.12. Consider a set of 10 signals $x_i(t)$, $i = 1, 2, 3, \ldots, 10$. Assume that each $x_i(t)$ has Fourier transform such that $X_i(j\omega) = 0$ for $|\omega| \geq 2{,}000\pi$. All 10 signals are to be time-division multiplexed after each is multiplied by a carrier $c(t)$ shown in Figure P8.12. If the period T of $c(t)$ is chosen to have the maximum allowable value, what is the largest value of Δ such that all 10 signals can be time-division multiplexed?

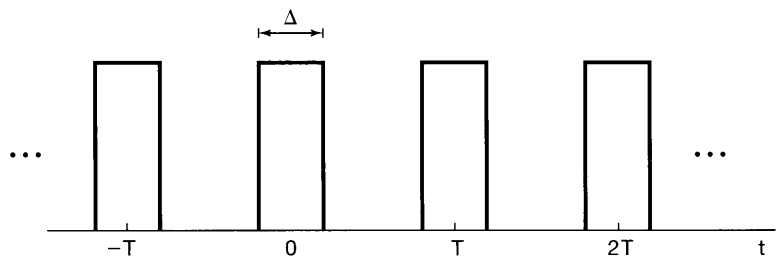


Figure P8.12

8.13. A class of popularly used pulses in PAM are those which have a *raised cosine* frequency response. The frequency response of one of the members of this class is

$$P(j\omega) = \begin{cases} \frac{1}{2}\left(1 + \cos\frac{\omega T_1}{2}\right), & 0 \le |\omega| \le \frac{2\pi}{T_1} \\ 0, & \text{elsewhere} \end{cases},$$

where T_1 is the intersymbol spacing.

(a) Determine $p(0)$.

(b) Determine $p(kT_1)$, where $k = \pm 1, \pm 2, \ldots$.

8.14. Consider the frequency-modulated signal

$$y(t) = \cos(\omega_c t + m\cos\omega_m t),$$

where $\omega_c >> \omega_m$ and $m << \pi/2$. Specify an approximation to $Y(j\omega)$ for $\omega > 0$.

8.15. For what values of ω_0 in the range $-\pi < \omega_0 \le \pi$ is amplitude modulation with carrier $e^{j\omega_0 n}$ equivalent to amplitude modulation with carrier $\cos\omega_0 n$?

8.16. Suppose $x[n]$ is a real-valued discrete-time signal whose Fourier transform $X(e^{j\omega})$ has the property that

$$X(e^{j\omega}) = 0 \quad \text{for } \frac{\pi}{8} \le \omega \le \pi.$$

We use $x[n]$ to modulate a sinusoidal carrier $c[n] = \sin(5\pi/2)n$ to produce

$$y[n] = x[n]c[n].$$

Determine the values of ω in the range $0 \le \omega \le \pi$ for which $Y(e^{j\omega})$ is guaranteed to be zero.

8.17. Consider an arbitrary finite-duration signal $x[n]$ with Fourier transform $X(e^{j\omega})$. We generate a signal $g[n]$ through insertion of zero-valued samples:

$$g[n] = x_{(4)}[n] = \begin{cases} x[n/4], & n = 0, \pm 4, \pm 8, \pm 12, \ldots \\ 0, & \text{otherwise} \end{cases}.$$

The signal $g[n]$ is passed through an ideal lowpass filter with cutoff frequency $\pi/4$ and passband gain of unity to produce a signal $q[n]$. Finally, we obtain

$$y[n] = q[n]\cos\left(\frac{3\pi}{4}n\right).$$

For what values of ω is $Y(e^{j\omega})$ guaranteed to be zero?

8.18. Let $x[n]$ be a real-valued discrete-time signal whose Fourier transform $X(e^{j\omega})$ is zero for $\omega \geq \pi/4$. We wish to obtain a signal $y[n]$ whose Fourier transform has the property that, in the interval $-\pi < \omega \leq \pi$,

$$Y(e^{j\omega}) = \begin{cases} X(e^{j(\omega-\frac{\pi}{2})}), & \frac{\pi}{2} < \omega \leq \frac{3\pi}{4} \\ X(e^{j(\omega+\frac{\pi}{2})}), & -\frac{3\pi}{4} < \omega \leq -\frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases}.$$

The system in Figure P8.18 is proposed for obtaining $y[n]$ from $x[n]$. Determine constraints that the frequency response $H(e^{j\omega})$ of the filter in the figure must satisfy for the proposed system to work.

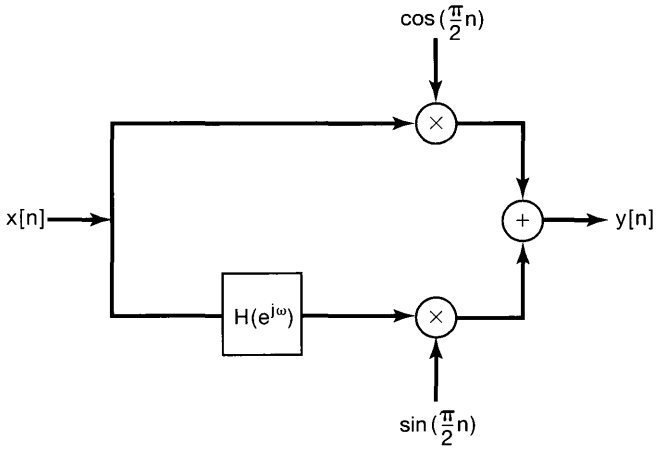


Figure P8.18

8.19. Consider 10 arbitrary real-valued signals $x_i[n]$, $i = 1, 2, \ldots, 10$. Suppose each $x_i[n]$ is upsampled by a factor of N, and then sinusoidal amplitude modulation is applied to it with carrier frequency $\omega_i = i\pi/10$. Determine the value of N which would guarantee that all 10 modulated signals can be summed together to yield an FDM signal $y[n]$ from which each $x_i[n]$ can be recovered.

8.20. Let $v_1[n]$ and $v_2[n]$ be two discrete-time signals obtained through the sampling (without aliasing) of continuous-time signals. Let

$$y[n] = \hat{v}_1[n] + \hat{v}_2[n-1]$$

be a TDM signal, where, for $i = 1, 2$,

$$\hat{v}_i[n] = \begin{cases} v_i\left[\frac{n}{2}\right], & n = 0, \pm 2, \pm 4, \pm 6, \ldots \\ 0, & \text{otherwise} \end{cases}.$$

The signal $y[n]$ is processed by the system S depicted in Figure P8.20 to obtain a signal $g[n]$. For the two filters used in S,

$$H_0(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < \omega \leq \pi \end{cases}.$$

Determine the signal $p[n]$ used in S such that $g[n]$ represents frequency-division multiplexing of $v_1[n]$ and $v_2[n]$.

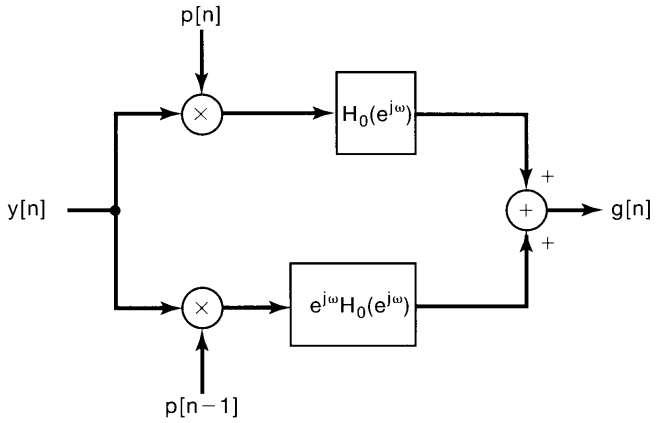


Figure P8.20

BASIC PROBLEMS

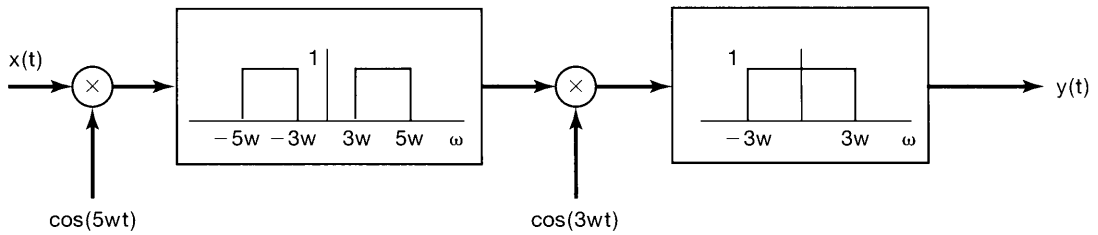
8.21. In Sections 8.1 and 8.2, we analyzed the sinusoidal amplitude modulation and demodulation system of Figure 8.8, assuming that the phase θ_c of the carrier signal was zero.

(a) For the more general case of arbitrary phase θ_c in the figure, show that the signal in the demodulation system can be expressed as

$$w(t) = \frac{1}{2}x(t) + \frac{1}{2}x(t)\cos(2\omega_c t + 2\theta_c).$$

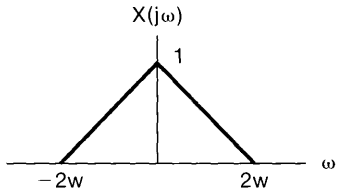
(b) If $x(t)$ has a spectrum that is zero for $|\omega| \geq \omega_M$, determine the relationships required among w_{co} [the cutoff frequency of the ideal lowpass filter in Figure 8.8(b)], ω_c (the carrier frequency), and ω_M so that the output of the lowpass filter is proportional to $x(t)$. Does your answer depend on the carrier phase θ_c?

8.22. In Figure P8.22(a), a system is shown with input signal $x(t)$ and output signal $y(t)$. The input signal has the Fourier transform $X(j\omega)$ shown in Figure P8.22(b). Determine and sketch $Y(j\omega)$, the spectrum of $y(t)$.



(a)

Figure P8.22



(b)

Figure P8.22 *Continued*

8.23. In Section 8.2, we discussed the effect of a loss of synchronization in phase between the carrier signals in the modulator and demodulator in sinusoidal amplitude modulation. We showed that the output of the demodulation is attenuated by the cosine of the phase difference, and in particular, when the modulator and demodulator have a phase difference of $\pi/2$, the demodulator output is zero. As we demonstrate in this problem, it is also important to have *frequency* synchronization between the modulator and demodulator.

Consider the amplitude modulation and demodulation systems in Figure 8.8 with $\theta_c = 0$ and with a change in the *frequency* of the demodulator carrier so that

$$w(t) = y(t)\cos\omega_d t,$$

where

$$y(t) = x(t)\cos\omega_c t.$$

Let us denote the difference in frequency between the modulator and demodulator as $\Delta\omega$ (i.e., $\omega_d - \omega_c = \Delta\omega$). Also, assume that $x(t)$ is band limited with $X(j\omega) = 0$ for $|\omega| \geq \omega_M$, and assume that the cutoff frequency w_{co} of the lowpass filter in the demodulator satisfies the inequality

$$\omega_M + \Delta\omega < \mathrm{w}_{co} < 2\omega_c + \Delta\omega - \omega_M.$$

(a) Show that the output of the lowpass filter in the demodulator is proportional to $x(t)\cos(\Delta\omega t)$.

(b) If the spectrum of $x(t)$ is that shown in Figure P8.23, sketch the spectrum of the output of the demodulator.

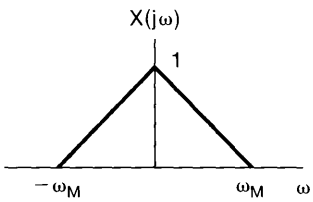


Figure P8.23

8.24. Figure P8.24 shows a system to be used for sinusoidal amplitude modulation, where $x(t)$ is band limited with maximum frequency ω_M, so that $X(j\omega) = 0$, $|\omega| > \omega_M$.

As indicated, the signal $s(t)$ is a periodic impulse train with period T and with an offset from $t = 0$ of Δ. The system $H(j\omega)$ is a bandpass filter.

(a) With $\Delta = 0$, $\omega_M = \pi/2T$, $\omega_l = \pi/T$, and $\omega_h = 3\pi/T$, show that $y(t)$ is proportional to $x(t)\cos\omega_c t$, where $\omega_c = 2\pi/T$.

(b) If ω_M, ω_l, and ω_h are the same as given in part (a), but Δ is not necessarily zero, show that $y(t)$ is proportional to $x(t)\cos(\omega_c t + \theta_c)$, and determine ω_c and θ_c as a function of T and Δ.

(c) Determine the maximum allowable value of ω_M relative to T such that $y(t)$ is proportional to $x(t)\cos(\omega_c t + \theta_c)$.

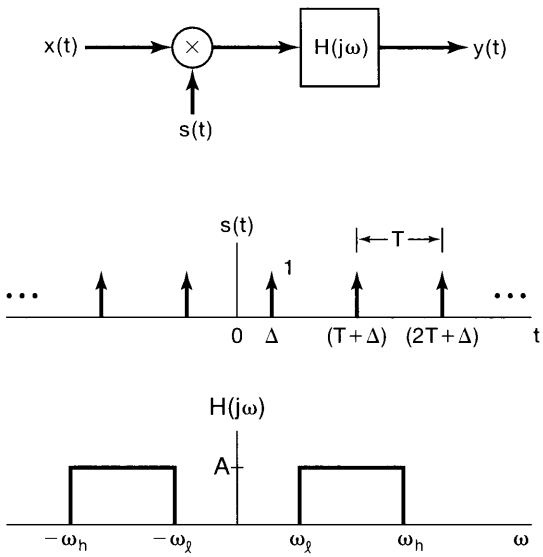


Figure P8.24

8.25. A commonly used system to maintain privacy in voice communication is a *speech scrambler.* As illustrated in Figure P8.25(a), the input to the system is a normal speech signal $x(t)$ and the output is the scrambled version $y(t)$. The signal $y(t)$ is transmitted and then unscrambled at the receiver.

We assume that all inputs to the scrambler are real and band limited to the frequency ω_M; that is, $X(j\omega) = 0$ for $|\omega| > \omega_M$. Given any such input, our proposed scrambler permutes different bands of the spectrum of the input signal. In addition, the output signal is real and band limited to the same frequency band; that is, $Y(j\omega) = 0$ for $|\omega| > \omega_M$. The specific algorithm for the scrambler is

$$Y(j\omega) = X(j(\omega - \omega_M)), \quad \omega > 0,$$
$$Y(j\omega) = X(j(\omega + \omega_M)), \quad \omega < 0.$$

(a) If $X(j\omega)$ is given by the spectrum shown in Figure P8.25(b), sketch the spectrum of the scrambled signal $y(t)$.

(b) Using amplifiers, multipliers, adders, oscillators, and whatever ideal filters you find necessary, draw the block diagram for such an ideal scrambler.

(c) Again using amplifiers, multipliers, adders, oscillators, and ideal filters, draw a block diagram for the associated unscrambler.

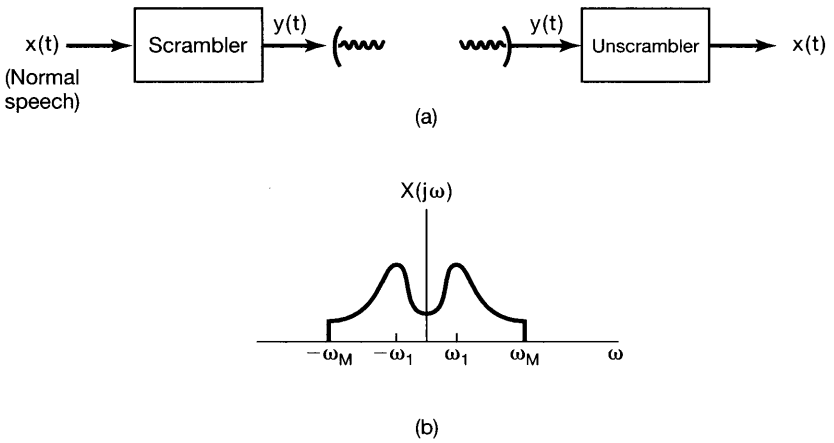


Figure P8.25

8.26. In Section 8.2.2, we discussed the use of an envelope detector for asynchronous demodulation of an AM signal of the form $y(t) = [x(t) + A]\cos(\omega_c t + \theta_c)$. An alternative demodulation system, which also does not require phase synchronization, but does require frequency synchronization, is shown in block diagram form in Figure P8.26. The lowpass filters both have a cutoff frequency of ω_c. The signal $y(t) = [x(t) + A]\cos(\omega_c t + \theta_c)$, with θ_c constant but unknown. The signal $x(t)$ is band limited with $X(j\omega) = 0$, $|\omega| > \omega_M$, and with $\omega_M < \omega_c$. As we required for the use of the envelope detector, $x(t) + A > 0$ for all t.

Show that the system in Figure P8.26 can be used to recover $x(t)$ from $y(t)$ without knowledge of the modulator phase θ_c.

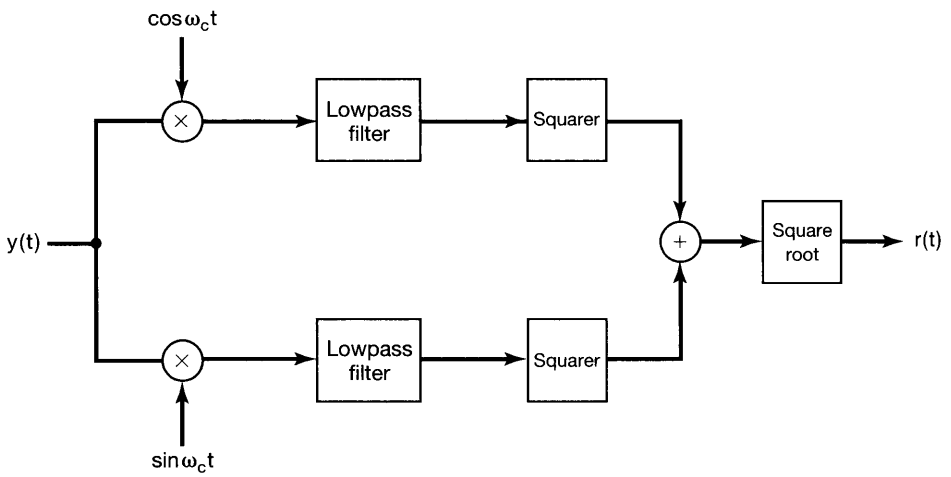


Figure P8.26

8.27. As discussed in Section 8.2.2, asynchronous modulation-demodulation requires the injection of the carrier signal so that the modulated signal is of the form

$$y(t) = [A + x(t)]\cos(\omega_c t + \theta_c), \qquad \text{(P8.27–1)}$$

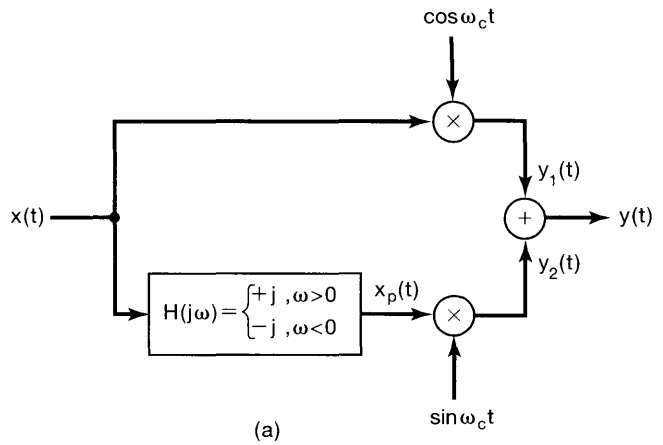
where $A + x(t) > 0$ for all t. The presence of the carrier means that more transmitter power is required, representing an inefficiency.

(a) Let $x(t) = \cos\omega_M t$ with $\omega_M < \omega_c$ and $A + x(t) > 0$. For a periodic signal $y(t)$ with period T, the average power over time is defined as $P_y = (1/T)\int_T y^2(t)\,dt$. Determine and sketch P_y for $y(t)$ in eq. (P8.27–1). Express your answer as a function of the modulation index m, defined as the maximum absolute value of $x(t)$ divided by A.

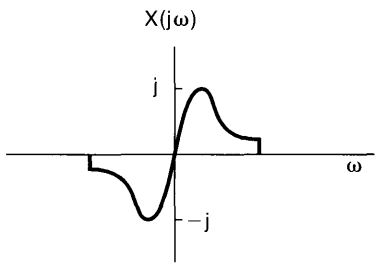
(b) The efficiency of transmission of an amplitude-modulated signal is defined to be the ratio of the power in the sidebands of the signal to the total power in the signal. With $x(t) = \cos\omega_M t$, and with $\omega_M < \omega_c$ and $A + x(t) > 0$, determine and sketch the efficiency of the modulated signal as a function of the modulation index m.

8.28. In Section 8.4 we discussed the implementation of single-sideband modulation using 90° phase-shift networks, and in Figures 8.21 and 8.22 we specifically illustrated the system and associated spectra required to retain the lower sidebands.

Figure P8.28(a) shows the corresponding system required to retain the upper sidebands.



(a)



(b)

Figure P8.28

(a) With the same $X(j\omega)$ illustrated in Figure 8.22, sketch $Y_1(j\omega)$, $Y_2(j\omega)$, and $Y(j\omega)$ for the system in Figure P8.28(a), and demonstrate that only the upper sidebands are retained.

(b) For $X(j\omega)$ imaginary, as illustrated in Figure P8.28(b), sketch $Y_1(j\omega)$, $Y_2(j\omega)$, and $Y(j\omega)$ for the system in Figure P8.28(a), and demonstrate that, for this case also, only the upper sidebands are retained.

8.29. Single-sideband modulation is commonly used in point-to-point voice communication. It offers many advantages, including effective use of available power, conservation of bandwidth, and insensitivity to some forms of random fading in the channel. In double-sideband suppressed carrier (DSB/SC) systems the spectrum of the modulating signal appears in its entirety in two places in the transmitted spectrum. Single-sideband modulation eliminates this redundancy, thus conserving bandwidth and increasing the signal-to-noise ratio within the remaining portion of the spectrum that is transmitted.

In Figure P8.29(a), two systems for generating an amplitude-modulated single-sideband signal are shown. The system on the top can be used to generate a single-sideband signal for which the lower sideband is retained, and the system on the bottom can produce a single-sideband signal for which the upper sideband is retained.

(a) For $X(j\omega)$ as shown in Figure P8.29(b), determine and sketch $S(j\omega)$, the Fourier transform of the lower sideband modulated signal, and $R(j\omega)$, the Fourier transform of the upper sideband modulated signal. Assume that $\omega_c > \omega_3$.

The upper sideband modulation scheme is particularly useful with voice communication, as any real filter has a finite transition region for the cutoff (i.e., near ω_c). This region can be accommodated with negligible distortion, since the voice signal does not have any significant energy near $\omega = 0$ (i.e., for $|\omega| < \omega_1 = 2\pi \times 40$ Hz).

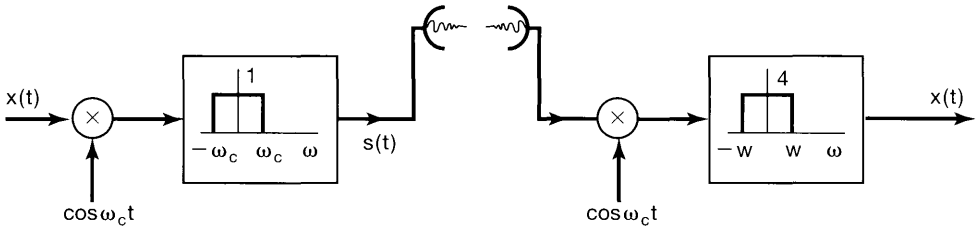
(b) Another procedure for generating a single-sideband signal is termed the *phase-shift method* and is illustrated in Figure P8.29(c). Show that the single-sideband signal generated is proportional to that generated by the lower sideband modulation scheme of Figure P8.29(a) [i.e., $p(t)$ is proportional to $s(t)$].

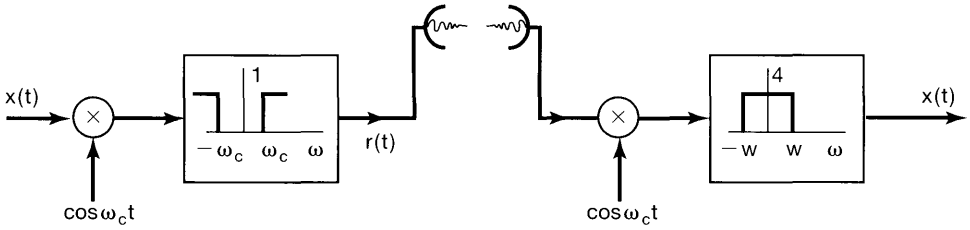
(c) All three AM-SSB signals can be demodulated using the scheme shown on the right-hand side of Figure P8.29(a). Show that, whether the received signal is $s(t)$, $r(t)$, or $p(t)$, as long as the oscillator at the receiver is in phase with oscillators at the transmitter, and $w = \omega_c$, the output of the demodulator is $x(t)$.

The distortion that results when the oscillator is not in phase with the transmitter, called *quadrature distortion,* can be particularly troublesome in data communication.

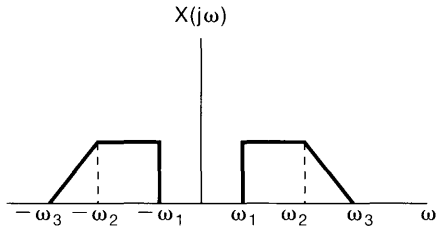
8.30. Amplitude modulation with a pulse-train carrier may be modeled as in Figure P8.30(a). The output of the system is $q(t)$.

(a) Let $x(t)$ be a band-limited signal [i.e., $X(j\omega) = 0$, $|\omega| \geq \pi/T$], as shown in Figure P8.30(b). Determine and sketch $R(j\omega)$ and $Q(j\omega)$.

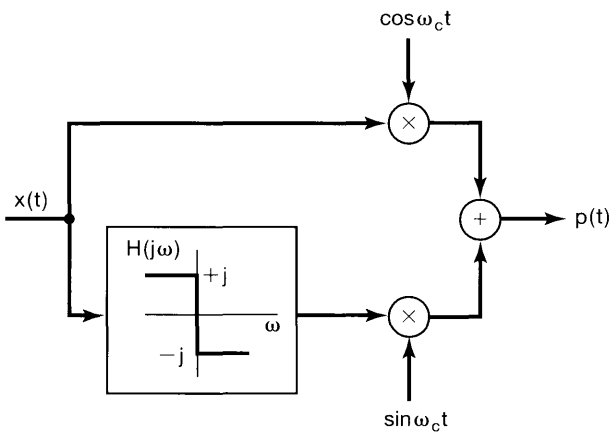




(a)



(b)



(c)

Figure P8.29

(b) Find the maximum value of Δ such that $w(t) = x(t)$ with an appropriate filter $M(j\omega)$.

(c) Determine and sketch the compensating filter $M(j\omega)$ such that $w(t) = x(t)$.

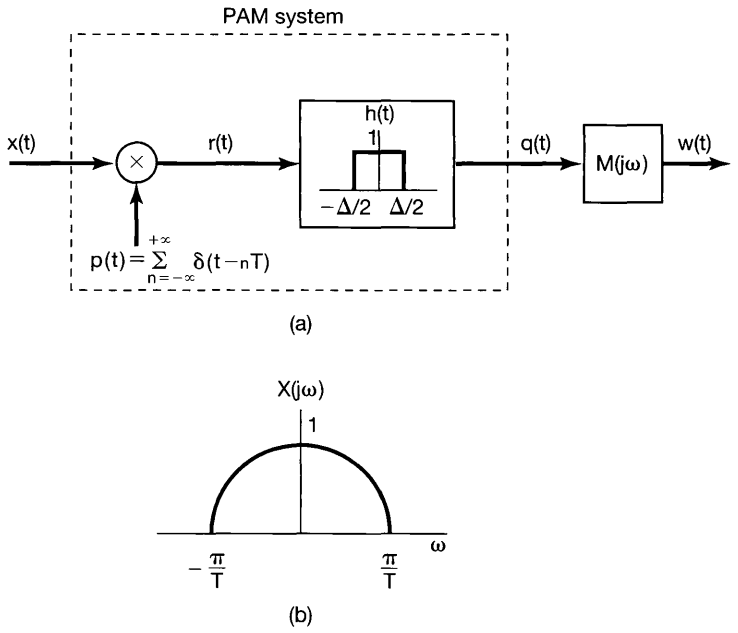


Figure P8.30

8.31. Let $x[n]$ be a discrete-time signal with spectrum $X(e^{j\omega})$, and let $p(t)$ be a continuous-time pulse function with spectrum $P(j\omega)$. We form the signal

$$y(t) = \sum_{n=-\infty}^{+\infty} x[n]p(t - n).$$

(a) Determine the spectrum $Y(j\omega)$ in terms of $X(e^{j\omega})$ and $P(j\omega)$.

(b) If

$$p(t) = \begin{cases} \cos 8\pi t, & 0 \le t \le 1 \\ 0, & \text{elsewhere} \end{cases},$$

determine $P(j\omega)$ and $Y(j\omega)$.

8.32. Consider a discrete-time signal $x[n]$ with Fourier transform shown in Figure P8.32(a). The signal is amplitude modulated by a sinusoidal sequence, as indicated in Figure P8.32(b).

(a) Determine and sketch $Y(e^{j\omega})$, the Fourier transform of $y[n]$.

(b) A proposed demodulation system is shown in Figure P8.32(c). For what value of θ_c, ω_{lp}, and G will $\hat{x}[n] = x[n]$? Are any restrictions on ω_c and ω_{lp} necessary to guarantee that $x[n]$ is recoverable from $y[n]$?

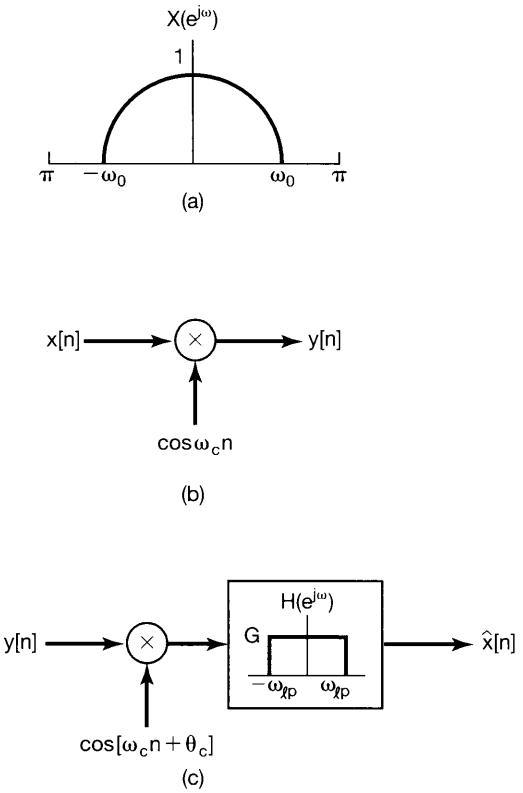


Figure P8.32

8.33. Let us consider the frequency-division multiplexing of discrete-time signals $x_i[n]$, $i = 0, 1, 2, 3$. Furthermore, each $x_i[n]$ potentially occupies the entire frequency band $(-\pi < \omega < \pi)$. The sinusoidal modulation of upsampled versions of each of these signals may be carried out by using either double-sideband techniques or single-sideband techniques.

(a) Suppose each signal $x_i[n]$ is appropriately upsampled and then modulated with $\cos[i(\pi/4)n]$. What is the minimum amount of upsampling that must be carried out on each $x_i[n]$ in order to ensure that the spectrum of the FDM signal does not have any aliasing?

(b) If the upsampling of each $x_i[n]$ is restricted to be by a factor of 4, how would you use single-sideband techniques to ensure that the FDM signal does not have any aliasing? *Hint:* See Problem 8.17.

ADVANCED PROBLEMS

8.34. In discussing amplitude modulation systems, modulation and demodulation were carried out through the use of a multiplier. Since multipliers are often difficult to implement, many practical systems use a nonlinear element. In this problem, we illustrate the basic concept.

In Figure P8.34, we show one such nonlinear system for amplitude modulation. The system consists of squaring the *sum* of the modulating signal and the carrier and then bandpass filtering to obtain the amplitude-modulated signal.

Assume that $x(t)$ is band limited, so that $X(j\omega) = 0$, $|\omega| > \omega_M$. Determine the bandpass filter parameters A, ω_l, and ω_h such that $y(t)$ is an amplitude-modulated version of $x(t)$ [i.e., such that $y(t) = x(t)\cos\omega_c t$]. Specify the necessary constraints, if any, on ω_c and ω_M.

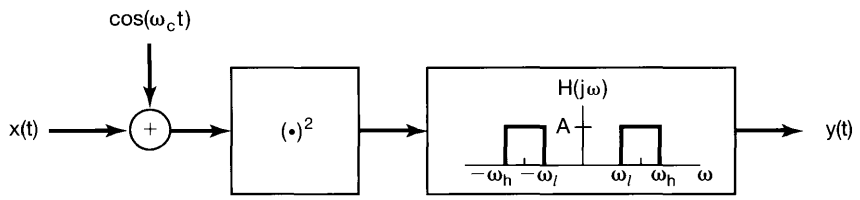
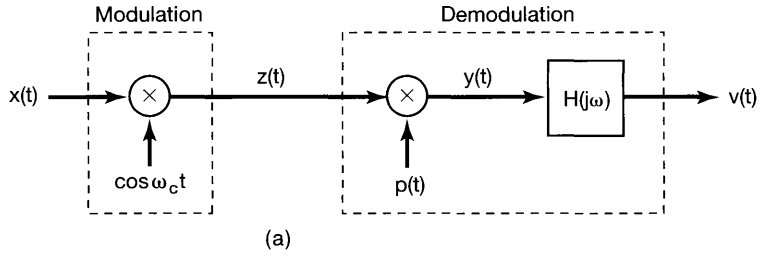


Figure P8.34

8.35. The modulation-demodulation scheme proposed in this problem is similar to sinusoidal amplitude modulation, except that the demodulation is done with a square wave with the same zero-crossings as $\cos\omega_c t$. The system is shown in Figure P8.35(a); the relation between $\cos\omega_c t$ and $p(t)$ is shown in Figure P8.35(b). Let the input signal $x(t)$ be a band-limited signal with maximum frequency $\omega_M < \omega_c$, as shown in Figure P8.35(c).

(a) Sketch and dimension the real and imaginary parts of $Z(j\omega)$, $P(j\omega)$, and $Y(j\omega)$, the Fourier transforms of $z(t)$, $p(t)$, and $y(t)$, respectively.

(b) Sketch and dimension a filter $H(j\omega)$ so that $v(t) = x(t)$.



(a)

Figure P8.35

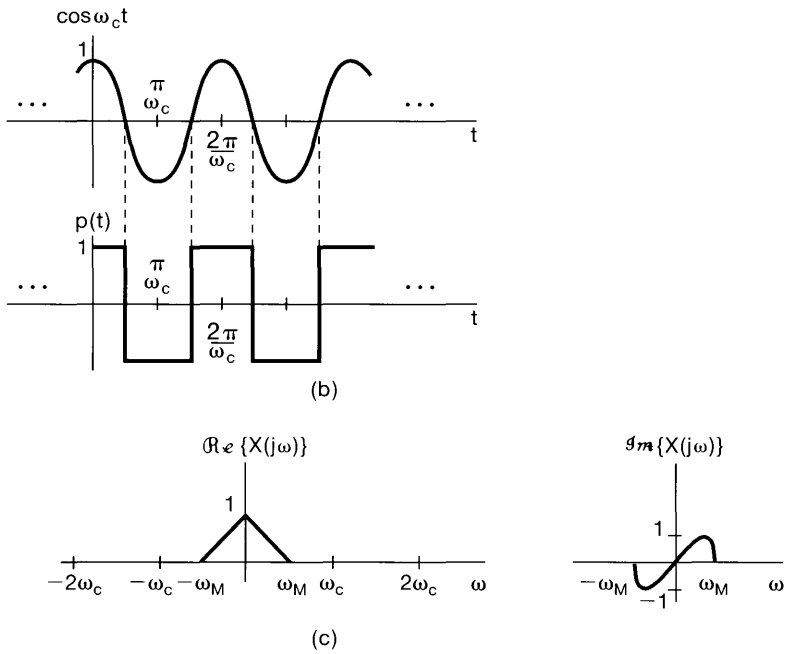


Figure P8.35 *Continued*

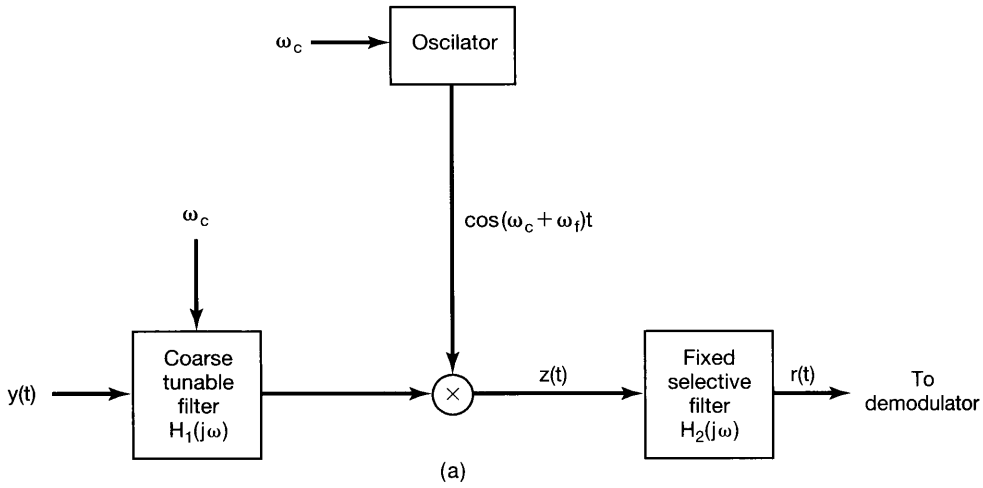
8.36. The accurate demultiplexing-demodulation of radio and television signals is generally performed using a system called the superheterodyne receiver, which is equivalent to a tunable filter. The basic system is shown in Figure P8.36(a).

(a) The input signal $y(t)$ consists of the superposition of many amplitude-modulated signals that have been multiplexed using frequency-division multiplexing, so that each signal occupies a different frequency channel. Let us consider one such channel that contains the amplitude-modulated signal $y_1(t) = x_1(t)\cos\omega_c t$, with spectrum $Y_1(j\omega)$ as depicted at the top of Figure P8.36(b). We want to demultiplex and demodulate $y_1(t)$ to recover the modulating signal $x_1(t)$, using the system of Figure P8.36(a). The coarse tunable filter has the spectrum $H_1(j\omega)$ shown at the bottom of Figure P8.36(b). Determine the spectrum $Z(j\omega)$ of the input signal to the fixed selective filter $H_2(j\omega)$. Sketch and label $Z(j\omega)$ for $\omega > 0$.

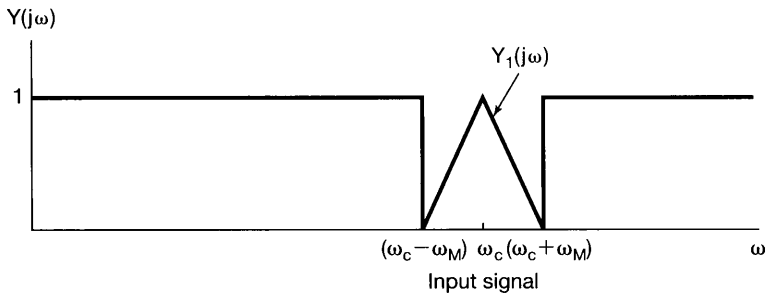
(b) The fixed frequency-selective filter is a bandpass type centered around the fixed frequency ω_f, as shown in Figure P8.36(c). We would like the output of the filter with spectrum $H_2(j\omega)$ to be $r(t) = x_1(t)\cos\omega_f t$. In terms of ω_c and ω_M, what constraint must ω_T satisfy to guarantee that an undistorted spectrum of $x_1(t)$ is centered around $\omega = \omega_f$?

(c) What must G, α, and β be in Figure P8.36(c) so that $r(t) = x_1(t)\cos\omega_f t$?

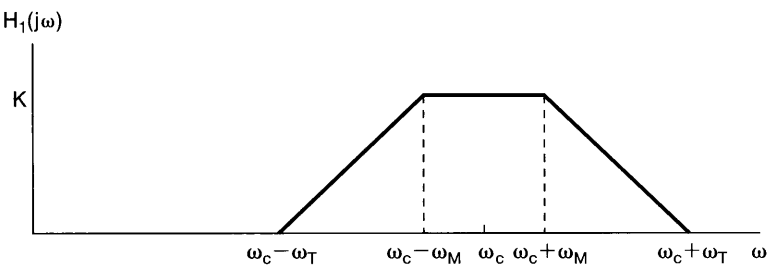
8.37. The following scheme has been proposed to perform amplitude modulation: The input signal $x(t)$ is added to the carrier signal $\cos\omega_c t$ and then put through a nonlinear



(a)

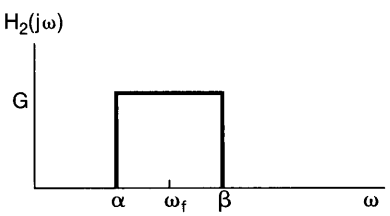


Input signal



Coarse tunable filter

(b)



(c)

Figure P8.36

device, so that the output $z(t)$ is related to the input by

$$z(t) = e^{y(t)} - 1,$$
$$y(t) = x(t) + \cos \omega_c t.$$

This is illustrated in Figure P8.37(a). Such a nonlinear relation can be implemented through the current-voltage characteristics of a diode, where, with $i(t)$ and $v(t)$ the diode and current and voltage, respectively,

$$i(t) = I_0 e^{av(t)} - 1 \quad (a \text{ real}).$$

To study the effects of nonlinearity, we can examine the spectrum of $z(t)$ and how it relates to $X(j\omega)$ and ω_c. To accomplish this, we use the power series for e^y, which is

$$e^y = 1 + y + \frac{1}{2}y^2 + \frac{1}{6}y^3 + \cdots.$$

(a) If the spectrum of $x(t)$ is given by Figure P8.37(b), and if $\omega_c = 100\omega_1$, sketch and label $Z(j\omega)$, the spectrum of $z(t)$, using the first four terms in the power series for e^y.

(b) The bandpass filter has parameters as shown in Figure P8.37(c). Determine the range of α and the range of β such that $r(t)$ is an amplitude-modulated version of $x(t)$.

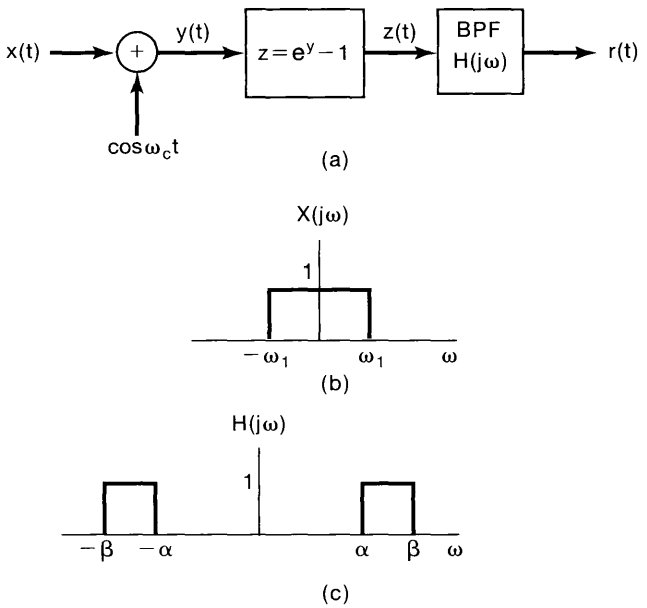


Figure P8.37

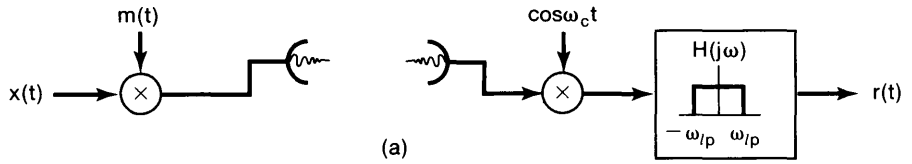
8.38. In Figure P8.38(a), a communication system is shown that transmits a band-limited signal $x(t)$ as periodic bursts of high-frequency energy. Assume that $X(j\omega) = 0$ for $|\omega| > \omega_M$. Two possible choices, $m_1(t)$ and $m_2(t)$, are considered for the modulating signal $m(t)$. $m_1(t)$ is a periodic train of sinusoidal pulses, each of duration D, as shown in Figure P8.38(b). That is,

$$m_1(t) = \sum_{k=-\infty}^{\infty} p(t - kT),$$

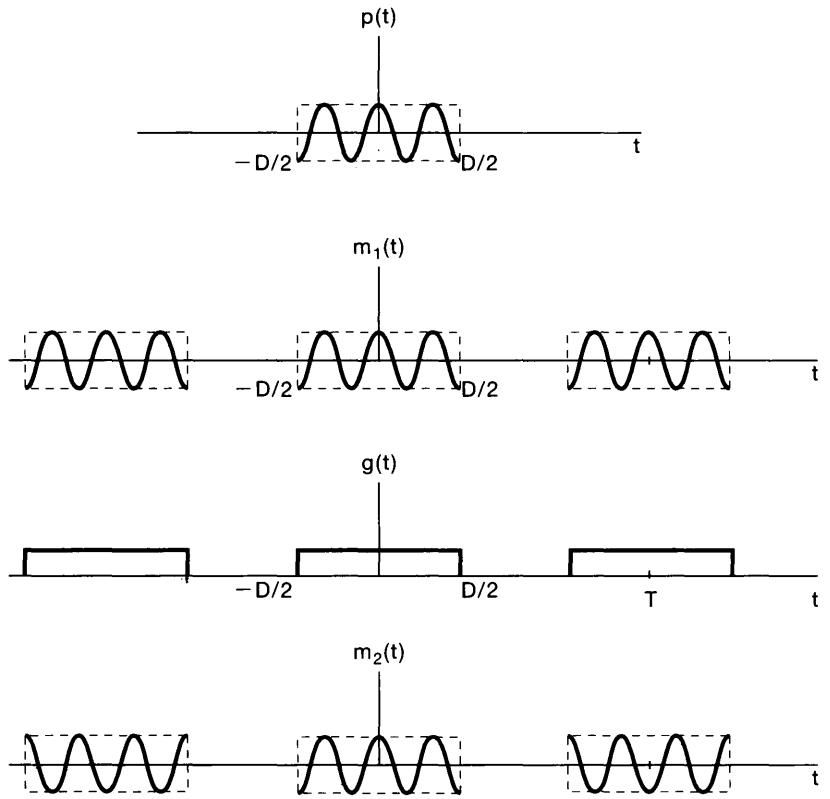
where

$$p(t) = \begin{cases} \cos \omega_c t, & |t| < (D/2) \\ 0, & |t| > (D/2) \end{cases}.$$

$m_2(t)$ is $\cos \omega_c t$ periodically blanked or gated; that is, $m_2(t) = g(t) \cos \omega_c t$, where $g(t)$ is as shown in Figure P8.38(b).



(a)



(b)

Figure P8.38

The following relationships between the parameters T, D, ω_c, and ω_M are assumed:

$$D < T,$$

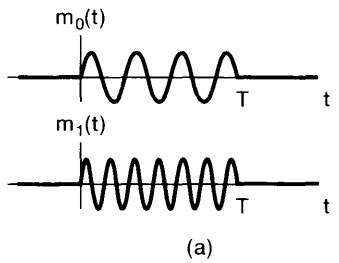
$$\omega_c >> \frac{2\pi}{D},$$

$$\frac{2\pi}{T} > 2\omega_M.$$

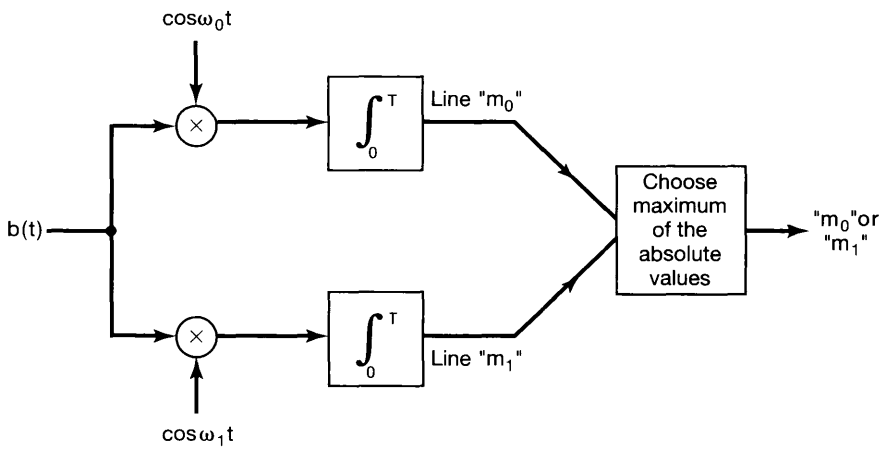
Also, assume that $[\sin(x)]/x$ is negligible for $x >> 1$.

Determine whether, for some choice of ω_{lp}, either $m_1(t)$ or $m_2(t)$ will result in a demodulated signal $x(t)$. For each case in which your answer is yes, determine an acceptable range for ω_{lp}.

8.39. Suppose we wish to communicate one of two possible messages: message m_0 or message m_1. To do so, we will send a burst of one of two frequencies over a time interval of length T. Note that T is independent of which message is being transmitted. For message m_0 we will send $\cos\omega_0 t$, and for message m_1 we will send $\cos\omega_1 t$. Thus, a burst $b(t)$ will look as shown in Figure P8.39(a). Such a communi-



(a)



(b)

Figure P8.39

cation system is called *frequency shift keying* (FSK). When the burst of frequency $b(t)$ is received, we wish to determine whether it represents message m_0 or message m_1. To accomplish this, we do as illustrated in Figure P8.39(b).

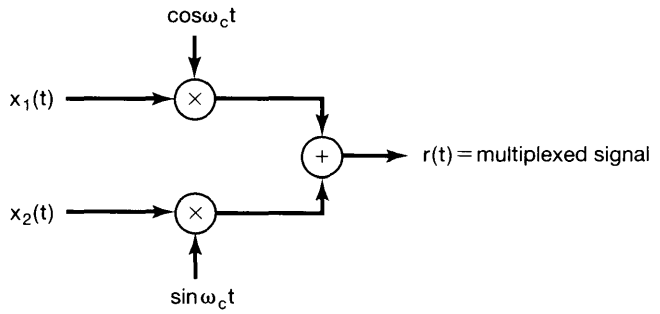
(a) Show that the maximum difference between the absolute values of the two lines in Figure P8.39(b) occurs when $\cos \omega_0 t$ and $\cos \omega_1 t$ have the relationship

$$\int_0^T \cos \omega_0 t \cos \omega_1 t \, dt = 0.$$

(b) Is it possible to choose ω_0 and ω_1 such that there is *no* interval of length T for which

$$\int_0^T \cos \omega_0 t \cos \omega_1 t \, dt = 0?$$

8.40. In Section 8.3, we discussed the use of sinusoidal modulation for frequency-division multiplexing whereby several signals are shifted into different frequency bands and then summed for simultaneous transmission. In the current problem, we explore another multiplexing concept referred to as *quadrature multiplexing*. In this multiplexing procedure, two signals can be transmitted simultaneously in the same frequency band if the two carrier signals are 90° out of phase. The multiplexing system is shown in Figure P8.40(a) and the demultiplexing system in Figure P8.40(b). $x_1(t)$ and $x_2(t)$ are both assumed to be band limited with maximum frequency ω_M, so that $X_1(j\omega) = X_2(j\omega) = 0$ for $|\omega| > \omega_M$. The carrier frequency ω_c is assumed to be greater than ω_M. Show that $y_1(t) = x_1(t)$ and $y_2(t) = x_2(t)$.

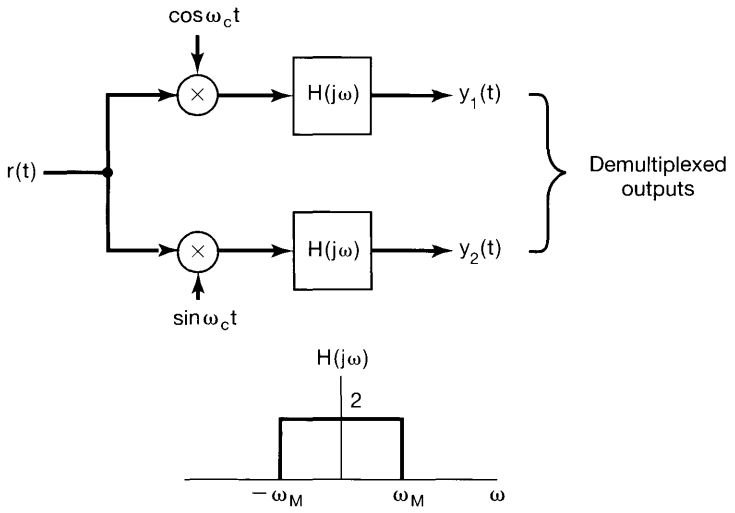


(a)

Figure P8.40

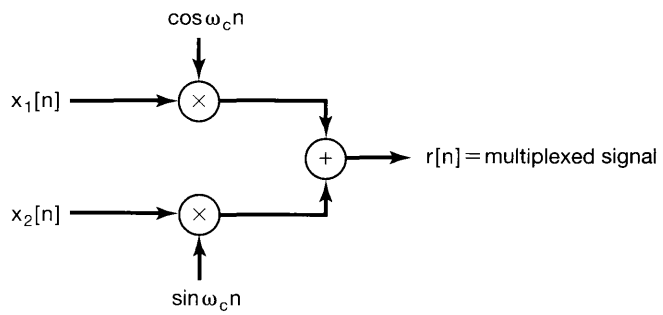
8.41. In Problem 8.40, we introduced the concept of quadrature multiplexing, whereby two signals are summed after each has been modulated with carrier signals of identical frequency, but with a phase difference of 90°. The corresponding discrete-time multiplexer and demultiplexer are shown in Figure P8.41. The signals $x_1[n]$ and $x_2[n]$ are both assumed to be band limited with maximum frequency ω_M, so that

$$X_1(e^{j\omega}) = X_2(e^{j\omega}) = 0 \text{ for } \omega_M < \omega < 2\pi - \omega_M.$$

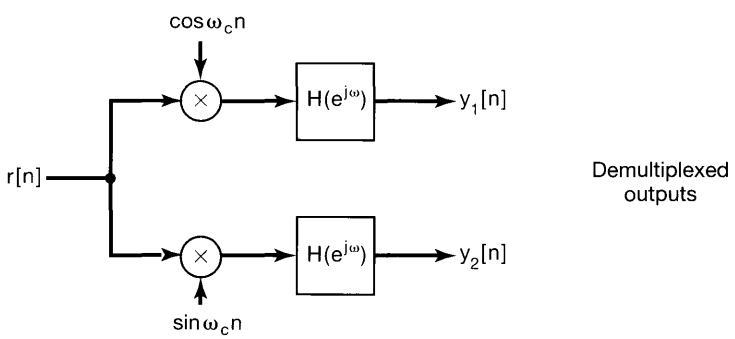


(b)

Figure P8.40 *Continued*



(a)



(b)

Figure P8.41

(a) Determine the range of values for ω_c so that $x_1[n]$ and $x_2[n]$ can be recovered from $r[n]$.

(b) With ω_c satisfying the conditions in part (a), determine $H(e^{j\omega})$ so that $y_1[n] = x_1[n]$ and $y_2[n] = x_2[n]$.

8.42. In order to avoid intersymbol interference, pulses used in PAM systems are designed to be zero at integer multiples of the symbol spacing T_1. In this problem, we develop a class of pulses which are zero at $t = kT_1$, $k = \pm 1, \pm 2, \pm 3, \ldots$.

Consider a pulse $p_1(t)$ that is real and even and that has a Fourier transform $P_1(j\omega)$. Also, assume that

$$P_1\left(-j\omega + j\frac{\pi}{T_1}\right) = -P_1\left(j\omega + j\frac{\pi}{T_1}\right), \quad 0 \le \omega \le \frac{\pi}{T_1}.$$

(a) Define a periodic sequence $\tilde{p}_1(t)$ with Fourier transform

$$\tilde{P}_1(j\omega) = \sum_{m=-\infty}^{\infty} P_1\left(j\omega - jm\frac{4\pi}{T_1}\right),$$

and show that

$$\tilde{P}_1(j\omega) = -\tilde{P}_1\left(j\omega - j\frac{2\pi}{T_1}\right).$$

(b) Use the result of the previous part to show that for some T

$$\tilde{p}_1(t) = 0, \; t = kT, \; k = 0, \pm 2, \pm 4, \ldots.$$

(c) Use the result of the previous part to show that

$$p_1(kT_1) = 0, \; k = \pm 1, \pm 2, \pm 3, \ldots.$$

(d) Show that a pulse $p(t)$ with Fourier transform

$$P(j\omega) = \begin{cases} 1 + P_1(j\omega), & |\omega| \le \frac{\pi}{T_1} \\ P_1(j\omega), & \frac{\pi}{T_1} \le |\omega| \le \frac{2\pi}{T_1} \\ 0, & \text{otherwise} \end{cases}$$

also has the property that

$$p(kT_1) = 0, \quad k = \pm 1, \pm 2, \pm 3, \ldots.$$

8.43. The impulse response of a channel used for PAM communication is specified by

$$h(t) = 10{,}000e^{-1{,}000t}u(t).$$

It is assumed that the phase response of the channel is approximately linear in the bandwidth of the channel. A pulse that is received after passing through the channel is processed using an LTI system S with impulse response $g(t)$ in order to compensate for the nonuniform gain over the channel bandwidth.

(a) Verify that if $g(t)$ has the Fourier transform

$$G(j\omega) = A + jB\omega,$$

where A and B are real constants, then $g(t)$ can compensate for the nonuniform gain over the channel bandwidth. Determine the values of A and B.

(b) It is proposed that S be implemented with the system shown in Figure P8.43. Determine the values of the gain factors α and β in this system.

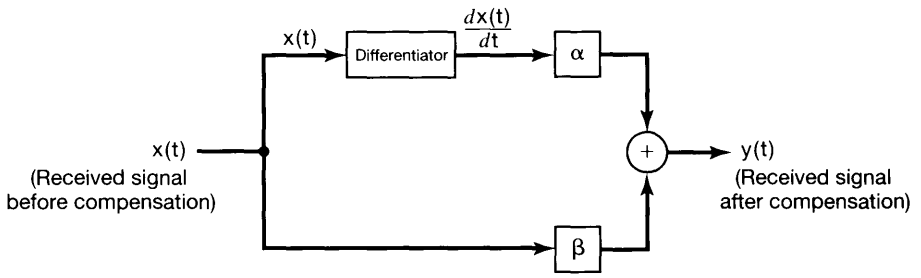


Figure P8.43

8.44. In this problem, we explore an *equalization* method used to avoid intersymbol interference caused in PAM systems by the channel having nonlinear phase over its bandwidth.

When a PAM pulse with zero-crossings at integer multiples of the symbol spacing T_1 is passed through a channel with nonlinear phase, the received pulse may no longer have zero-crossings at times that are integer multiples of T_1. Therefore, in order to avoid intersymbol interference, the received pulse is passed through a *zero-forcing equalizer,* which forces the pulse to have zero-crossings at integer multiples of T_1. This equalizer generates a new pulse $y(t)$ by summing up weighted and shifted versions of the received pulse $x(t)$. The pulse $y(t)$ is given by

$$y(t) = \sum_{l=-N}^{N} a_l x(t - lT_1), \tag{P8.44–1}$$

where the a_l are all real and are chosen such that

$$y(kT_1) = \begin{cases} 1, & k = 0 \\ 0, & k = \pm 1, \pm 2, \pm 3, \ldots, \pm N \end{cases}.$$

(a) Show that the equalizer is a filter and determine its impulse response.

(b) To illustrate the selection of the weights a_l, let us consider an example. If $x(0T_1) = 0.0$, $x(-T_1) = 0.2$, $x(T_1) = -0.2$, and $x(kT_1) = 0$ for $|k| > 1$, determine the values of a_0, a_1, and a_{-1} such that $y(\pm T_1) = 0$.

8.45. A band-limited signal $x(t)$ is to be transmitted using narrowband FM techniques. That is, the modulation index m, as defined in Section 8.7, is much less than $\pi/2$. Before $x(t)$ is transmitted to the modulator, it is processed so that $X(j\omega)|_{\omega=0} = 0$ and $|x(t)| < 1$. This normalized $x(t)$ is now used to angle-modulate a carrier to form the FM signal

$$y(t) = \cos\left(\omega_c t + m\int_{-\infty}^{t} x(\tau)\,d\tau\right).$$

(a) Determine the instantaneous frequency ω_i.

(b) Using eqs. (8.44) and (8.45), the narrowband assumption ($m << \pi/2$), and the preceding normalization conditions, show that

$$y(t) \approx \cos\omega_c t - \left(m\int_{-\infty}^{t} x(\tau)\,d\tau\right)\sin\omega_c t.$$

(c) What is the relationship among the bandwidth of $y(t)$, the bandwidth of $x(t)$, and the carrier frequency ω_c?

8.46. Consider the complex exponential function of time,

$$s(t) = e^{j\theta(t)}, \tag{P8.46-1}$$

where $\theta(t) = \omega_0 t^2/2$.

Since the instantaneous frequency $\omega_i = d\theta/dt$ is also a function of time, the signal $s(t)$ may be regarded as an FM signal. In particular, since the signal sweeps linearly through the frequency spectrum with time, it is often called a frequency "chirp" or "chirp signal."

(a) Determine the instantaneous frequency.

(b) Determine and sketch the magnitude and phase of the Fourier transform of the "chirp signal." To evaluate the Fourier transform integral, you may find it helpful to complete the square in the exponent in the integrand and to use the relation

$$\int_{-\infty}^{+\infty} e^{jz^2}\,dz = \sqrt{\frac{\pi}{2}}(1 + j).$$

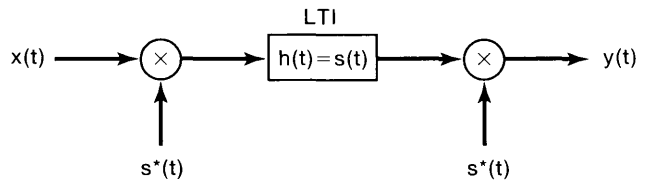


Figure P8.46

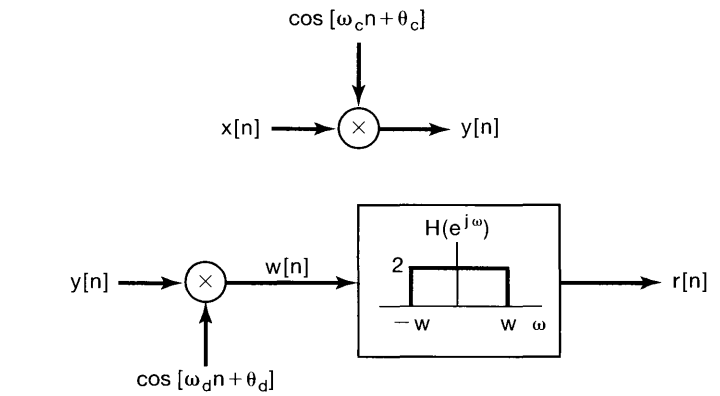
(c) Consider the system in Figure P8.46, in which $s(t)$ is the "chirp signal" in eq. (P8.46–1). Show that $y(t) = X(j\omega_0 t)$, where $X(j\omega)$ is the Fourier transform of $x(t)$.
(Note: The system in Figure P8.46 is referred to as the *"chirp" transform algorithm* and is often used in practice to obtain the Fourier transform of a signal.)

8.47. In Section 8.8 we considered synchronous discrete-time modulation and demodulation with a sinusoidal carrier. In this problem we want to consider the effect of a loss in synchronization in phase and/or frequency. The modulation and demodulation systems are shown in Figure P8.47(a), where both a phase and frequency difference between the modulator and demodulator carriers is indicated. Let the frequency difference $\omega_d - \omega_c$ be denoted as $\Delta\omega$ and the phase difference $\theta_d - \theta_c$ as $\Delta\theta$.

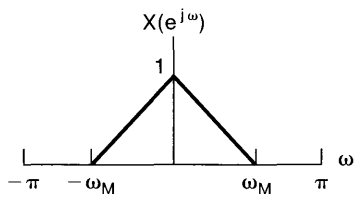
(a) If the spectrum of $x[n]$ is that shown in Figure P8.47(b), sketch the spectrum of $w[n]$, assuming $\Delta\omega = 0$.

(b) If $\Delta\omega = 0$, show that w can be chosen so that the output $r[n]$ is $r[n] = x[n]\cos\Delta\theta$. In particular, what is $r[n]$ if $\Delta\theta = \pi/2$?

(c) For $\Delta\theta = 0$, and $w = \omega_M + \Delta\omega$, show that the output $r[n] = x[n]\cos[\Delta\omega n]$ (assume that $\Delta\omega$ is small).



(a)



(b)

Figure P8.47

8.48. In this problem, we consider the analysis of discrete-time amplitude modulation of a pulse-train carrier. The system to be considered is shown in Figure P8.48(a).

(a) Determine and sketch the discrete-time Fourier transform of the periodic square-wave signal $p[n]$ in Figure P8.48(a).

(b) Assume that $x[n]$ has the spectrum shown in Figure P8.48(b). With $\omega_M = \pi/2N$ and with $M = 1$ in Figure P8.48(a), sketch $Y(e^{j\omega})$, the Fourier transform of $y[n]$.

(c) Now assume that $X(e^{j\omega})$ is known to be band limited with $X(e^{j\omega}) = 0$, $\omega_M < \omega < 2\pi - \omega_M$, but is otherwise unspecified. For the system of Figure P8.48(a), determine, as a function of N, the maximum allowable value of ω_M that will permit $x[n]$ to be recovered from $y[n]$. Indicate whether your result depends on M.

(d) With ω_M and N satisfying the condition determined in part (c), state or show in block diagram form how to recover $x[n]$ from $y[n]$.

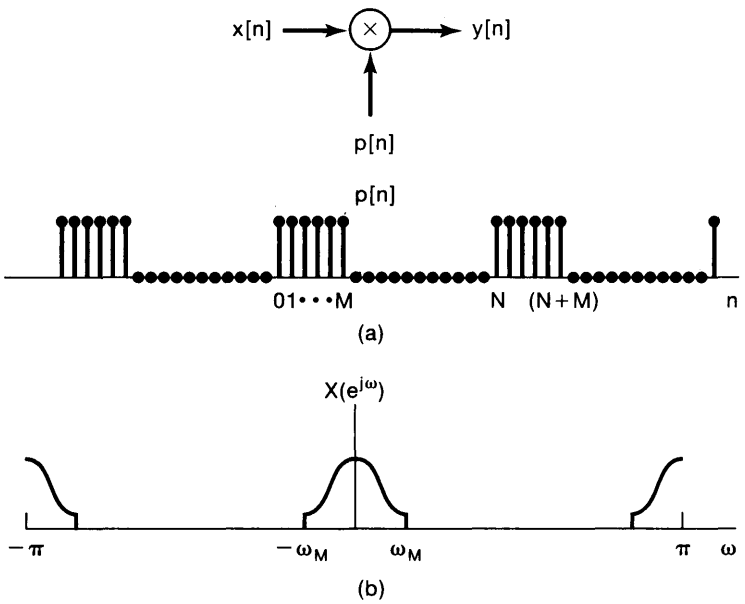


Figure P8.48

8.49. In practice it is often very difficult to build an amplifier at very low frequencies. Consequently, low-frequency amplifiers typically exploit the principles of amplitude modulation to shift the signal into a higher-frequency band. Such an amplifier is referred to as a *chopper amplifier* and is illustrated in the block-diagram form in Figure P8.49.

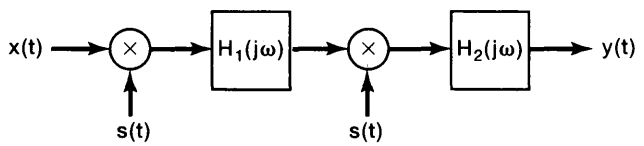


Figure P8.49

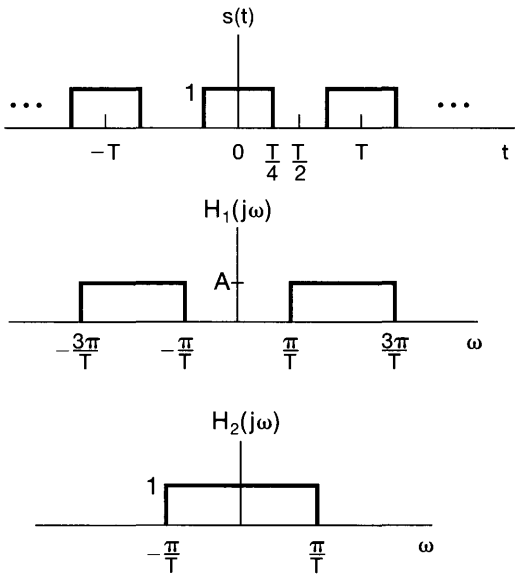


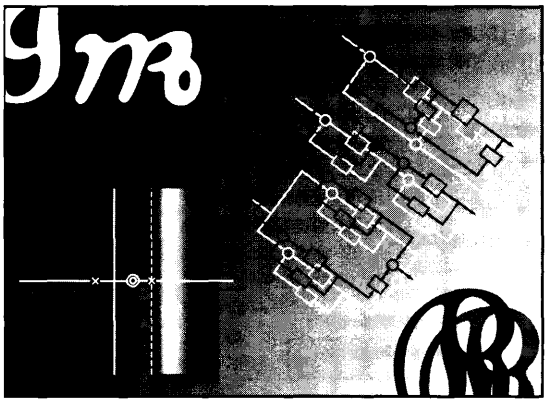
Figure P8.49 *Continued*

(a) Determine in terms of T the highest allowable frequency present in $x(t)$, if $y(t)$ is to be proportional to $x(t)$ (i.e., if the overall system is to be equivalent to an amplifier).

(b) With $x(t)$ bandlimited as specified in part (a), determine the gain of the overall system in Figure P8.49 in terms of A and T.

9

THE LAPLACE TRANSFORM



9.0 INTRODUCTION

In the preceding chapters, we have seen that the tools of Fourier analysis are extremely useful in the study of many problems of practical importance involving signals and LTI systems. This is due in large part to the fact that broad classes of signals can be represented as linear combinations of periodic complex exponentials and that complex exponentials are eigenfunctions of LTI systems. The continuous-time Fourier transform provides us with a representation for signals as linear combinations of complex exponentials of the form e^{st} with $s = j\omega$. However the eigenfunction property introduced in Section 3.2 and many of its consequences apply as well for arbitrary values of s and not only those values that are purely imaginary. This observation leads to a generalization of the continuous-time Fourier transform, known as the Laplace transform, which we develop in this chapter. In the next chapter we develop the corresponding discrete-time generalization known as the z-transform.

As we will see, the Laplace and z-transforms have many of the properties that make Fourier analysis so useful. Moreover, not only do these transforms provide additional tools and insights for signals and systems that can be analyzed using the Fourier transform, but they also can be applied in some very important contexts in which Fourier transforms cannot. For example Laplace and z-transforms can be applied to the analysis of many unstable systems and consequently play an important role in the investigation of the stability or instability of systems. This fact, combined with the algebraic properties that Laplace and z-transforms share with Fourier transforms, leads to a very important set of tools for system analysis and in particular for the analysis of feedback systems, which we develop in Chapter 11.

9.1 THE LAPLACE TRANSFORM

In Chapter 3, we saw that the response of a linear time-invariant system with impulse response $h(t)$ to a complex exponential input of the form e^{st} is

$$y(t) = H(s)e^{st}, \tag{9.1}$$

where

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}\,dt. \tag{9.2}$$

For s imaginary (i.e., $s = j\omega$), the integral in eq. (9.2) corresponds to the Fourier transform of $h(t)$. For general values of the complex variable s, it is referred to as the *Laplace transform* of the impulse response $h(t)$.

The Laplace transform of a general signal $x(t)$ is defined as[1]

$$\boxed{X(s) \triangleq \int_{-\infty}^{+\infty} x(t)e^{-st}\,dt,} \tag{9.3}$$

and we note in particular that it is a function of the independent variable s corresponding to the complex variable in the exponent of e^{-st}. The complex variable s can be written as $s = \sigma + j\omega$, with σ and ω the real and imaginary parts, respectively. For convenience, we will sometimes denote the Laplace transform in operator form as $\mathfrak{L}\{x(t)\}$ and denote the transform relationship between $x(t)$ and $X(s)$ as

$$x(t) \overset{\mathfrak{L}}{\longleftrightarrow} X(s). \tag{9.4}$$

When $s = j\omega$, eq. (9.3) becomes

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}\,dt, \tag{9.5}$$

which corresponds to the *Fourier transform* of $x(t)$; that is,

$$X(s)|_{s=j\omega} = \mathfrak{F}\{x(t)\}. \tag{9.6}$$

The Laplace transform also bears a straightforward relationship to the Fourier transform when the complex variable s is not purely imaginary. To see this relationship, consider $X(s)$ as specified in eq. (9.3) with s expressed as $s = \sigma + j\omega$, so that

$$X(\sigma + j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-(\sigma+j\omega)t}\,dt, \tag{9.7}$$

[1]The transform defined by eq. (9.3) is often called the *bilateral Laplace transform*, to distinguish it from the unilateral Laplace transform, which we discuss in Section 9.9. The bilateral transform in eq. (9.3) involves an integration from $-\infty$ to $+\infty$, while the unilateral transform has a form similar to that in eq. (9.3), but with limits of integration from 0 to $+\infty$. As we are primarily concerned with the bilateral transform, we will omit the word "bilateral," except where it is needed in Section 9.9 to avoid ambiguity.

or

$$X(\sigma + j\omega) = \int_{-\infty}^{+\infty} [x(t)e^{-\sigma t}]e^{-j\omega t}\,dt. \tag{9.8}$$

We recognize the right-hand side of eq. (9.8) as the Fourier transform of $x(t)e^{-\sigma t}$; that is, the Laplace transform of $x(t)$ can be interpreted as the Fourier transform of $x(t)$ after multiplication by a real exponential signal. The real exponential $e^{-\sigma t}$ may be decaying or growing in time, depending on whether σ is positive or negative.

To illustrate the Laplace transform and its relationship to the Fourier transform, let us consider the following example:

Example 9.1

Let the signal $x(t) = e^{-at}u(t)$. From Example 4.1, the Fourier transform $X(j\omega)$ converges for $a > 0$ and is given by

$$X(j\omega) = \int_{-\infty}^{+\infty} e^{-at}u(t)e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-at}e^{-j\omega t}\,dt = \frac{1}{j\omega + a}, \quad a > 0. \tag{9.9}$$

From eq. (9.3), the Laplace transform is

$$X(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st}dt = \int_{0}^{\infty} e^{-(s+a)t}dt, \tag{9.10}$$

or, with $s = \sigma + j\omega$,

$$X(\sigma + j\omega) = \int_{0}^{\infty} e^{-(\sigma+a)t}e^{-j\omega t}\,dt. \tag{9.11}$$

By comparison with eq. (9.9) we recognize eq. (9.11) as the Fourier transform of $e^{-(\sigma+a)t}u(t)$, and thus,

$$X(\sigma + j\omega) = \frac{1}{(\sigma + a) + j\omega}, \quad \sigma + a > 0, \tag{9.12}$$

or equivalently, since $s = \sigma + j\omega$ and $\sigma = \mathcal{R}e\{s\}$,

$$X(s) = \frac{1}{s + a}, \quad \mathcal{R}e\{s\} > -a. \tag{9.13}$$

That is,

$$e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s + a}, \quad \mathcal{R}e\{s\} > -a. \tag{9.14}$$

For example, for $a = 0$, $x(t)$ is the unit step with Laplace transform $X(s) = 1/s$, $\mathcal{R}e\{s\} > 0$.

We note, in particular, that just as the Fourier transform does not converge for all signals, the Laplace transform may converge for some values of $\mathcal{R}e\{s\}$ and not for others. In eq. (9.13), the Laplace transform converges only for $\sigma = \mathcal{R}e\{s\} > -a$. If a is positive,

then $X(s)$ can be evaluated at $\sigma = 0$ to obtain

$$X(0 + j\omega) = \frac{1}{j\omega + a}. \tag{9.15}$$

As indicated in eq. (9.6), for $\sigma = 0$ the Laplace transform is equal to the Fourier transform, as is evident in the preceding example by comparing eqs. (9.9) and (9.15). If a is negative or zero, the Laplace transform still exists, but the Fourier transform does not.

Example 9.2

For comparison with Example 9.1, let us consider as a second example the signal

$$x(t) = -e^{-at}u(-t). \tag{9.16}$$

Then

$$\begin{aligned} X(s) &= -\int_{-\infty}^{\infty} e^{-at}e^{-st}u(-t)\,dt \\ &= -\int_{-\infty}^{0} e^{-(s+a)t}\,dt, \end{aligned} \tag{9.17}$$

or

$$X(s) = \frac{1}{s + a}. \tag{9.18}$$

For convergence in this example, we require that $\mathcal{Re}\{s + a\} < 0$, or $\mathcal{Re}\{s\} < -a$; that is,

$$-e^{-at}u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s + a}, \qquad \mathcal{Re}\{s\} < -a. \tag{9.19}$$

Comparing eqs. (9.14) and (9.19), we see that the algebraic expression for the Laplace transform is identical for both of the signals considered in Examples 9.1 and 9.2. However, from the same equations, we also see that the set of values of s for which the expression is valid is very different in the two examples. This serves to illustrate the fact that, in specifying the Laplace transform of a signal, both the algebraic expression and the range of values of s for which this expression is valid are required. In general, the range of values of s for which the integral in eq.(9.3) converges is referred to as the *region of convergence* (which we abbreviate as ROC) of the Laplace transform. That is, the ROC consists of those values of $s = \sigma + j\omega$ for which the Fourier transform of $x(t)e^{-\sigma t}$ converges. We will have more to say about the ROC as we develop some insight into the properties of the Laplace transform.

A convenient way to display the ROC is shown in Figure 9.1. The variable s is a complex number, and in the figure we display the complex plane, generally referred to as the s-plane, associated with this complex variable. The coordinate axes are $\mathcal{Re}\{s\}$ along the horizontal axis and $\mathcal{Im}\{s\}$ along the vertical axis. The horizontal and vertical axes are sometimes referred to as the σ-axis and the $j\omega$-axis, respectively. The shaded region in Figure 9.1(a) represents the set of points in the s-plane corresponding to the region of convergence for Example 9.1. The shaded region in Figure 9.1(b) indicates the region of convergence for Example 9.2.

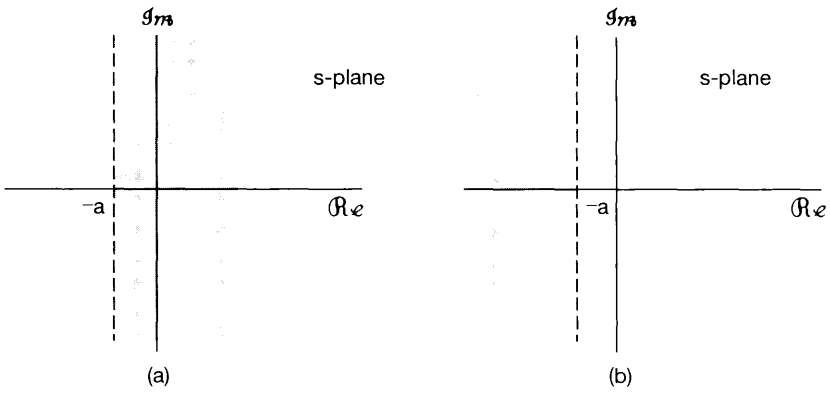


Figure 9.1 (a) ROC for Example 9.1; (b) ROC for Example 9.2.

Example 9.3

In this example, we consider a signal that is the sum of two real exponentials:

$$x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t). \tag{9.20}$$

The algebraic expression for the Laplace transform is then

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} \left[3e^{-2t}u(t) - 2e^{-t}u(t)\right] e^{-st}\,dt \\ &= 3\int_{-\infty}^{\infty} e^{-2t}e^{-st}u(t)\,dt - 2\int_{-\infty}^{\infty} e^{-t}e^{-st}u(t)\,dt. \end{aligned} \tag{9.21}$$

Each of the integrals in eq. (9.21) is of the same form as the integral in eq. (9.10), and consequently, we can use the result in Example 9.1 to obtain

$$X(s) = \frac{3}{s+2} - \frac{2}{s+1}. \tag{9.22}$$

To determine the ROC we note that $x(t)$ is a sum of two real exponentials, and from eq. (9.21) we see that $X(s)$ is the sum of the Laplace transforms of each of the individual terms. The first term is the Laplace transform of $3e^{-2t}u(t)$ and the second term the Laplace transform of $-2e^{-t}u(t)$. From Example 9.1, we know that

$$\begin{aligned} e^{-t}u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+1}, \qquad \mathcal{R}e\{s\} > -1, \\ e^{-2t}u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+2}, \qquad \mathcal{R}e\{s\} > -2. \end{aligned}$$

The set of values of $\mathcal{R}e\{s\}$ for which the Laplace transforms of both terms converge is $\mathcal{R}e\{s\} > -1$, and thus, combining the two terms on the right-hand side of eq. (9.22), we obtain

$$3e^{-2t}u(t) - 2e^{-t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s-1}{s^2+3s+2}, \quad \mathcal{R}e\{s\} > -1. \tag{9.23}$$

Example 9.4

In this example, we consider a signal that is the sum of a real and a complex exponential:

$$x(t) = e^{-2t}u(t) + e^{-t}(\cos 3t)u(t). \tag{9.24}$$

Using Euler's relation, we can write

$$x(t) = \left[e^{-2t} + \frac{1}{2}e^{-(1-3j)t} + \frac{1}{2}e^{-(1+3j)t}\right]u(t),$$

and the Laplace transform of $x(t)$ then can be expressed as

$$\begin{aligned} X(s) = &\int_{-\infty}^{\infty} e^{-2t}u(t)e^{-st}\,dt \\ &+ \frac{1}{2}\int_{-\infty}^{\infty} e^{-(1-3j)t}u(t)e^{-st}\,dt \\ &+ \frac{1}{2}\int_{-\infty}^{\infty} e^{-(1+3j)t}u(t)e^{-st}\,dt. \end{aligned} \tag{9.25}$$

Each of the integrals in eq. (9.25) represents a Laplace transform of the type encountered in Example 9.1. It follows that

$$e^{-2t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+2}, \quad \mathcal{R}e\{s\} > -2, \tag{9.26}$$

$$e^{-(1-3j)t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+(1-3j)}, \quad \mathcal{R}e\{s\} > -1, \tag{9.27}$$

$$e^{-(1+3j)t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+(1+3j)}, \quad \mathcal{R}e\{s\} > -1. \tag{9.28}$$

For all three Laplace transforms to converge simultaneously, we must have $\mathcal{R}e\{s\} > -1$. Consequently, the Laplace transform of $x(t)$ is

$$\frac{1}{s+2} + \frac{1}{2}\left(\frac{1}{s+(1-3j)}\right) + \frac{1}{2}\left(\frac{1}{s+(1+3j)}\right), \quad \mathcal{R}e\{s\} > -1, \tag{9.29}$$

or, with terms combined over a common denominator,

$$e^{-2t}u(t) + e^{-t}(\cos 3t)u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{2s^2+5s+12}{(s^2+2s+10)(s+2)}, \quad \mathcal{R}e\{s\} > -1. \tag{9.30}$$

In each of the four preceding examples, the Laplace transform is rational, i.e., it is a ratio of polynomials in the complex variable s, so that

$$X(s) = \frac{N(s)}{D(s)}, \tag{9.31}$$

where $N(s)$ and $D(s)$ are the numerator polynomial and denominator polynomial, respectively. As suggested by Examples 9.3 and 9.4, $X(s)$ will be rational whenever $x(t)$ is a linear combination of real or complex exponentials. As we will see in Section 9.7, rational

transforms also arise when we consider LTI systems specified in terms of linear constant-coefficient differential equations. Except for a scale factor, the numerator and denominator polynomials in a rational Laplace transform can be specified by their roots; thus, marking the locations of the roots of $N(s)$ and $D(s)$ in the s-plane and indicating the ROC provides a convenient pictorial way of describing the Laplace transform. For example, in Figure 9.2(a) we show the s-plane representation of the Laplace transform of Example 9.3, with the location of each root of the denominator polynomial in eq. (9.23) indicated with "×" and the location of the root of the numerator polynomial in eq. (9.23) indicated with "o." The corresponding plot of the roots of the numerator and denominator polynomials for the Laplace transform in Example 9.4 is given in Figure 9.2(b). The region of convergence for each of these examples is shaded in the corresponding plot.

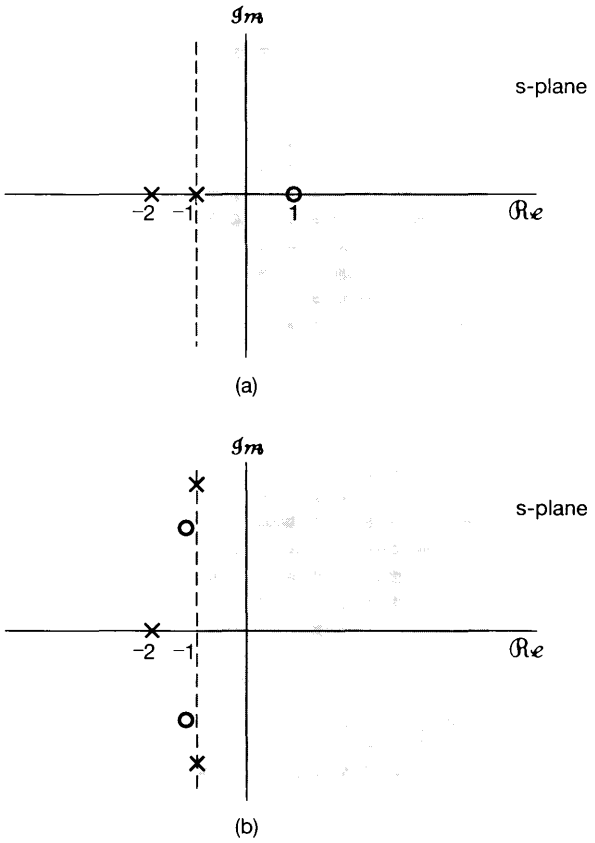


Figure 9.2 s-plane representation of the Laplace transforms for (a) Example 9.3 and (b) Example 9.4. Each × in these figures marks the location of a pole of the corresponding Laplace transform—i.e., a root of the denominator. Similarly, each o marks a zero—i.e., a root of the the numerator. The shaded regions indicate the ROCs.

For rational Laplace transforms, the roots of the numerator polynomial are commonly referred to as the *zeros* of $X(s)$, since, for those values of s, $X(s) = 0$. The roots of the denominator polynomial are referred to as the *poles* of $X(s)$, and for those values of s, $X(s)$ is infinite. The poles and zeros of $X(s)$ in the finite s-plane completely characterize the algebraic expression for $X(s)$ to within a scale factor. The representation of $X(s)$ through its poles and zeros in the s-plane is referred to as the *pole-zero plot* of $X(s)$.

However, as we saw in Examples 9.1 and 9.2, knowledge of the algebraic form of $X(s)$ does not by itself identify the ROC for the Laplace transform. That is, a complete specification, to within a scale factor, of a rational Laplace transform consists of the pole-zero plot of the transform, together with its ROC (which is commonly shown as a shaded region in the s-plane, as in Figures 9.1 and 9.2).

Also, while they are not needed to specify the algebraic form of a rational transform $X(s)$, it is sometimes convenient to refer to poles or zeros of $X(s)$ at infinity. Specifically, if the order of the denominator polynomial is greater than the order of the numerator polynomial, then $X(s)$ will become zero as s approaches infinity. Conversely, if the order of the numerator polynomial is greater than the order of the denominator, then $X(s)$ will become unbounded as s approaches infinity. This behavior can be interpreted as zeros or poles at infinity. For example, the Laplace transform in eq. (9.23) has a denominator of order 2 and a numerator of order only 1, so in this case $X(s)$ has one zero at infinity. The same is true for the transform in eq. (9.30), in which the numerator is of order 2 and the denominator is of order 3. In general, if the order of the denominator exceeds the order of the numerator by k, $X(s)$ will have k zeros at infinity. Similarly, if the order of the numerator exceeds the order of the denominator by k, $X(s)$ will have k poles at infinity.

Example 9.5

Let

$$x(t) = \delta(t) - \frac{4}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t). \tag{9.32}$$

The Laplace transform of the second and third terms on the right-hand side of eq. (9.32) can be evaluated from Example 9.1. The Laplace transform of the unit impulse can be evaluated directly as

$$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{+\infty} \delta(t)e^{-st}\,dt = 1, \tag{9.33}$$

which is valid for any value of s. That is, the ROC of $\mathcal{L}\{\delta(t)\}$ is the entire s-plane. Using this result, together with the Laplace transforms of the other two terms in eq. (9.32), we obtain

$$X(s) = 1 - \frac{4}{3}\frac{1}{s+1} + \frac{1}{3}\frac{1}{s-2}, \quad \mathcal{R}e\{s\} > 2, \tag{9.34}$$

or

$$X(s) = \frac{(s-1)^2}{(s+1)(s-2)}, \quad \mathcal{R}e\{s\} > 2, \tag{9.35}$$

where the ROC is the set of values of s for which the Laplace transforms of all three terms in $x(t)$ converge. The pole-zero plot for this example is shown in Figure 9.3, together with the ROC. Also, since the degrees of the numerator and denominator of $X(s)$ are equal, $X(s)$ has neither poles nor zeros at infinity.

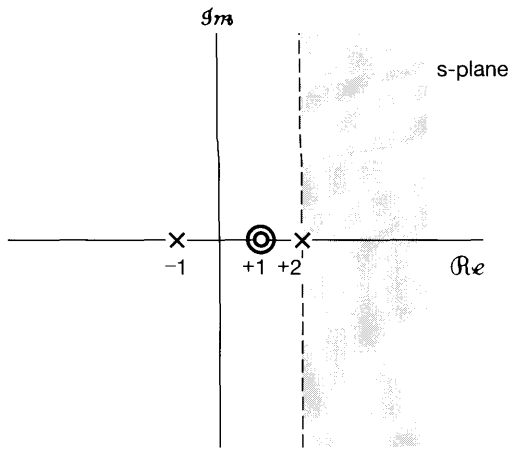


Figure 9.3 Pole-zero plot and ROC for Example 9.5.

Recall from eq. (9.6) that, for $s = j\omega$, the Laplace transform corresponds to the Fourier transform. However, if the ROC of the Laplace transform does not include the $j\omega$-axis, (i.e., if $\mathcal{R}e\{s\} = 0$), then the Fourier transform does not converge. As we see from Figure 9.3, this, in fact, is the case for Example 9.5, which is consistent with the fact that the term $(1/3)e^{2t}u(t)$ in $x(t)$ does not have a Fourier transform. Note also in this example that the two zeros in eq. (9.35) occur at the same value of s. In general, we will refer to the *order* of a pole or zero as the number of times it is repeated at a given location. In Example 9.5 there is a second-order zero at $s = 1$ and two first-order poles, one at $s = -1$, the other at $s = 2$. In this example the ROC lies to the right of the rightmost pole. In general, for rational Laplace transforms, there is a close relationship between the locations of the poles and the possible ROCs that can be associated with a given pole-zero plot. Specific constraints on the ROC are closely associated with time-domain properties of $x(t)$. In the next section, we explore some of these constraints and properties.

9.2 THE REGION OF CONVERGENCE FOR LAPLACE TRANSFORMS

In the preceding section, we saw that a complete specification of the Laplace transform requires not only the algebraic expression for $X(s)$, but also the associated region of convergence. As evidenced by Examples 9.1 and 9.2, two very different signals can have identical algebraic expressions for $X(s)$, so that their Laplace transforms are distinguishable *only* by the region of convergence. In this section, we explore some specific constraints on the ROC for various classes of signals. As we will see, an understanding of these constraints often permits us to specify implicitly or to reconstruct the ROC from knowledge of only the algebraic expression for $X(s)$ and certain general characteristics of $x(t)$ in the time domain.

Property 1: The ROC of $X(s)$ consists of strips parallel to the $j\omega$-axis in the s-plane.

The validity of this property stems from the fact that the ROC of $X(s)$ consists of those values of $s = \sigma + j\omega$ for which the Fourier transform of $x(t)e^{-\sigma t}$ converges. That

is, the ROC of the Laplace transform of $x(t)$ consists of those values of s for which $x(t)e^{-\sigma t}$ is absolutely integrable:[2]

$$\int_{-\infty}^{+\infty} |x(t)|e^{-\sigma t}\,dt < \infty. \tag{9.36}$$

Property 1 then follows, since this condition depends only on σ, the real part of s.

Property 2: For rational Laplace transforms, the ROC does not contain any poles.

Property 2 is easily observed in all the examples studied thus far. Since $X(s)$ is infinite at a pole, the integral in eq. (9.3) clearly does not converge at a pole, and thus the ROC cannot contain values of s that are poles.

Property 3: If $x(t)$ is of finite duration and is absolutely integrable, then the ROC is the entire s-plane.

The intuition behind this result is suggested in Figures 9.4 and 9.5. Specifically, a finite-duration signal has the property that it is zero outside an interval of finite duration, as illustrated in Figure 9.4. In Figure 9.5(a), we have shown $x(t)$ of Figure 9.4 multiplied by a decaying exponential, and in Figure 9.5(b) the same signal multiplied by a growing

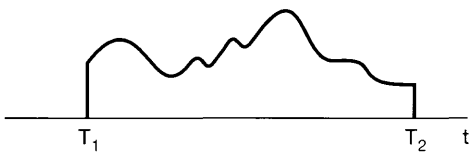


Figure 9.4 Finite-duration signal.

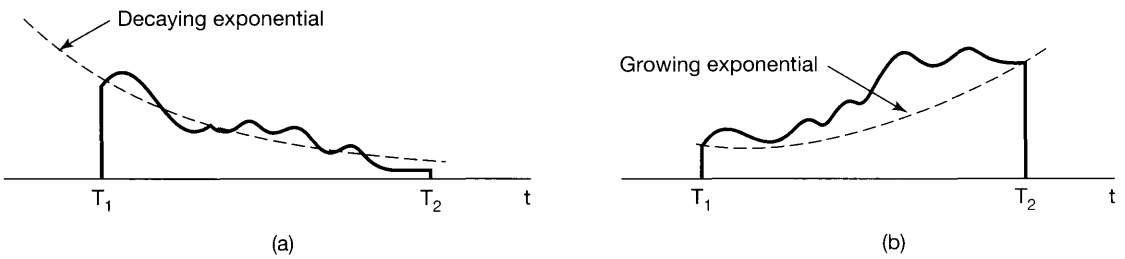


Figure 9.5 (a) Finite-duration signal of Figure 9.4 multiplied by a decaying exponential; (b) finite-duration signal of Figure 9.4 multiplied by a growing exponential.

[2]For a more thorough and formal treatment of Laplace transforms and their mathematical properties, including convergence, see E. D. Rainville, *The Laplace Transform: An Introduction* (New York: Macmillan, 1963), and R. V. Churchill and J. W. Brown, *Complex Variables and Applications* (5th ed.) (New York: McGraw-Hill, 1990). Note that the condition of absolute integrability is one of the Dirichlet conditions introduced in Section 4.1 in the context of our discussion of the convergence of Fourier transforms.

exponential. Since the interval over which $x(t)$ is nonzero is finite, the exponential weighting is never unbounded, and consequently, it is reasonable that the integrability of $x(t)$ not be destroyed by this exponential weighting.

A more formal verification of Property 3 is as follows: Suppose that $x(t)$ is absolutely integrable, so that

$$\int_{T_1}^{T_2} |x(t)|\,dt < \infty. \tag{9.37}$$

For $s = \sigma + j\omega$ to be in the ROC, we require that $x(t)e^{-\sigma t}$ be absolutely integrable, i.e.,

$$\int_{T_1}^{T_2} |x(t)|e^{-\sigma t}\,dt < \infty. \tag{9.38}$$

Eq. (9.37) verifies that s is in the ROC when $\mathcal{R}e\{s\} = \sigma = 0$. For $\sigma > 0$, the maximum value of $e^{-\sigma t}$ over the interval on which $x(t)$ is nonzero is $e^{-\sigma T_1}$, and thus we can write

$$\int_{T_1}^{T_2} |x(t)|e^{-\sigma t}\,dt < e^{-\sigma T_1}\int_{T_1}^{T_2} |x(t)|\,dt. \tag{9.39}$$

Since the right-hand side of eq.(9.39) is bounded, so is the left-hand side; therefore, the s-plane for $\mathcal{R}e\{s\} > 0$ must also be in the ROC. By a similar argument, if $\sigma < 0$, then

$$\int_{T_1}^{T_2} |x(t)|e^{-\sigma t}\,dt < e^{-\sigma T_2}\int_{T_1}^{T_2} |x(t)|\,dt, \tag{9.40}$$

and again, $x(t)e^{-\sigma t}$ is absolutely integrable. Thus, the ROC includes the entire s-plane.

Example 9.6

Let

$$x(t) = \begin{cases} e^{-at}, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}. \tag{9.41}$$

Then

$$X(s) = \int_0^T e^{-at}e^{-st}\,dt = \frac{1}{s+a}[1 - e^{-(s+a)T}]. \tag{9.42}$$

Since in this example $x(t)$ is of finite length, it follows from Property 3 that the ROC is the entire s-plane. In the form of eq. (9.42), $X(s)$ would appear to have a pole at $s = -a$, which, from Property 2, would be inconsistent with an ROC that consists of the entire s-plane. In fact, however, in the algebraic expression in eq. (9.42), both numerator and denominator are zero at $s = -a$, and thus, to determine $X(s)$ at $s = -a$, we can use L'hôpital's rule to obtain

$$\lim_{s\to -a} X(s) = \lim_{s\to -a}\left[\frac{\frac{d}{ds}(1 - e^{-(s+a)T})}{\frac{d}{ds}(s+a)}\right] = \lim_{s\to -a} Te^{-aT}e^{-sT},$$

so that

$$X(-a) = T. \tag{9.43}$$

It is important to recognize that, to ensure that the exponential weighting is bounded over the interval in which $x(t)$ is nonzero, the preceding discussion relies heavily on the fact that $x(t)$ is of finite duration. In the next two properties, we consider modifications of the result in Property 3 when $x(t)$ is of finite extent in only the positive-time or negative-time direction.

Property 4: If $x(t)$ is right sided, and if the line $\mathcal{R}e\{s\} = \sigma_0$ is in the ROC, then all values of s for which $\mathcal{R}e\{s\} > \sigma_0$ will also be in the ROC.

A *right-sided* signal is a signal for which $x(t) = 0$ prior to some finite time T_1, as illustrated in Figure 9.6. It is possible that, for such a signal, there is no value of s for which the Laplace transform will converge. One example is the signal $x(t) = e^{t^2}u(t)$. However, suppose that the Laplace transform converges for some value of σ, which we denote by σ_0. Then

$$\int_{-\infty}^{+\infty} |x(t)|e^{-\sigma_0 t}\,dt < \infty, \tag{9.44}$$

or equivalently, since $x(t)$ is right sided,

$$\int_{T_1}^{+\infty} |x(t)|e^{-\sigma_0 t}\,dt < \infty. \tag{9.45}$$

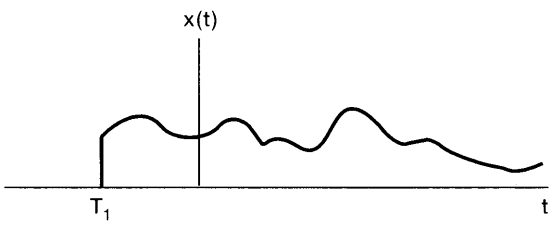


Figure 9.6 Right-sided signal.

Then if $\sigma_1 > \sigma_0$, it must also be true that $x(t)e^{-\sigma_1 t}$ is absolutely integrable, since $e^{-\sigma_1 t}$ decays faster than $e^{-\sigma_0 t}$ as $t \longrightarrow +\infty$, as illustrated in Figure 9.7. Formally, we can say that with $\sigma_1 > \sigma_0$,

$$\begin{aligned}\int_{T_1}^{\infty} |x(t)|e^{-\sigma_1 t}\,dt &= \int_{T_1}^{\infty} |x(t)|e^{-\sigma_0 t}e^{-(\sigma_1-\sigma_0)t}\,dt \\ &\leq e^{-(\sigma_1-\sigma_0)T_1}\int_{T_1}^{\infty} |x(t)|e^{-\sigma_0 t}\,dt.\end{aligned} \tag{9.46}$$

Since T_1 is finite, it follows from eq. (9.45) that the right side of the inequality in eq. (9.46) is finite, and hence, $x(t)e^{-\sigma_1 t}$ is absolutely integrable.

Note that in the preceding argument we explicitly rely on the fact that $x(t)$ is right sided, so that, although with $\sigma_1 > \sigma_0$, $e^{-\sigma_1 t}$ diverges faster than $e^{-\sigma_0 t}$ as $t \to -\infty$, $x(t)e^{-\sigma_1 t}$ cannot grow without bound in the negative-time direction, since $x(t) = 0$ for

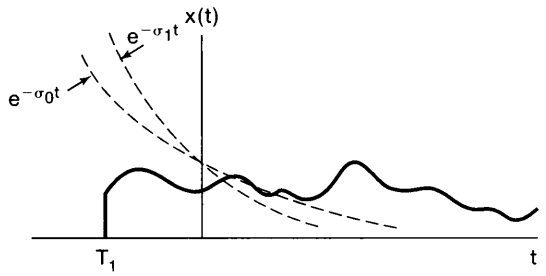


Figure 9.7 If $x(t)$ is right sided and $x(t)e^{-\sigma_0 t}$ is absolutely integrable, then $x(t)e^{-\sigma_1 t}$, $\sigma_1 > \sigma_0$, will also be absolutely integrable.

$t < T_1$. Also, in this case, if a point s is in the ROC, then all the points to the right of s, i.e., all points with larger real parts, are in the ROC. For this reason, the ROC in this case is commonly referred to as a *right-half plane.*

Property 5: If $x(t)$ is left sided, and if the line $\mathcal{R}e\{s\} = \sigma_0$ is in the ROC, then all values of s for which $\mathcal{R}e\{s\} < \sigma_0$ will also be in the ROC.

A *left-sided* signal is a signal for which $x(t) = 0$ after some finite time T_2, as illustrated in Figure 9.8. The argument and intuition behind this property are exactly analogous to the argument and intuition behind Property 4. Also, for a left-sided signal, the ROC is commonly referred to as a *left-half plane,* as if a point s is in the ROC, then all points to the left of s are in the ROC.

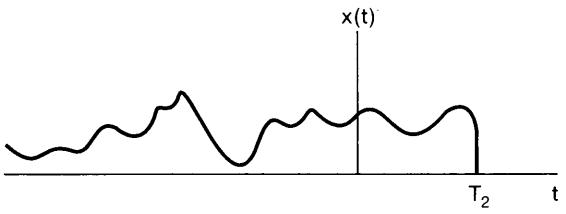


Figure 9.8 Left-sided signal.

Property 6: If $x(t)$ is two sided, and if the line $\mathcal{R}e\{s\} = \sigma_0$ is in the ROC, then the ROC will consist of a strip in the s-plane that includes the line $\mathcal{R}e\{s\} = \sigma_0$.

A *two-sided* signal is a signal that is of infinite extent for both $t > 0$ and $t < 0$, as illustrated in Figure 9.9(a). For such a signal, the ROC can be examined by choosing an arbitrary time T_0 and dividing $x(t)$ into the sum of a right-sided signal $x_R(t)$ and a left-sided signal $x_L(t)$, as indicated in Figures 9.9(b) and 9.9(c). The Laplace transform of $x(t)$ converges for values of s for which the transforms of *both* $x_R(t)$ and $x_L(t)$ converge. From Property 4, the ROC of $\mathcal{L}\{x_R(t)\}$ consists of a half-plane $\mathcal{R}e\{s\} > \sigma_R$ for some value σ_R, and from Property 5, the ROC of $\mathcal{L}\{x_L(t)\}$ consists of a half-plane $\mathcal{R}e\{s\} < \sigma_L$ for some value σ_L. The ROC of $\mathcal{L}\{x(t)\}$ is then the overlap of these two half-planes, as indicated in Figure 9.10. This assumes, of course, that $\sigma_R < \sigma_L$, so that there is some overlap. If this is not the case, then even if the Laplace transforms of $x_R(t)$ and $x_L(t)$ individually exist, the Laplace transform of $x(t)$ does not.

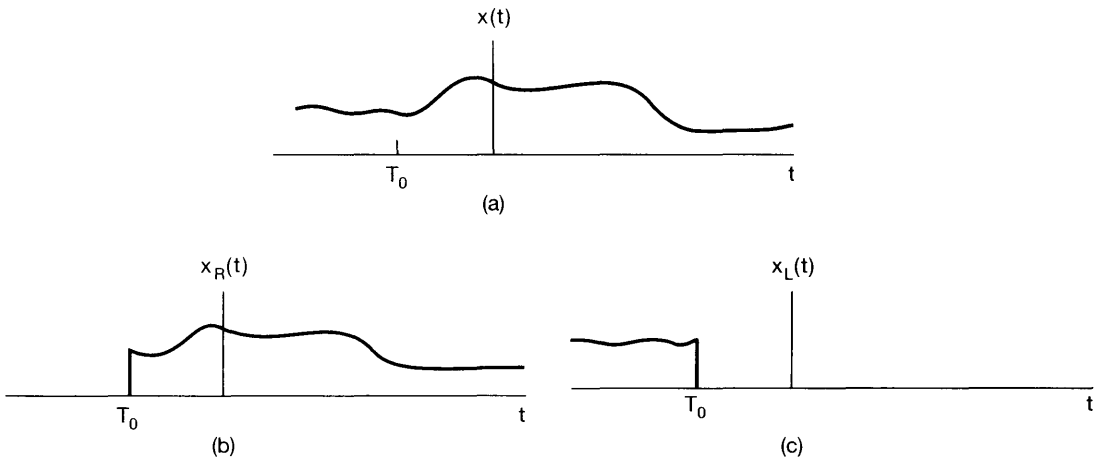


Figure 9.9 Two-sided signal divided into the sum of a right-sided and left-sided signal: (a) two-sided signal $x(t)$; (b) the right-sided signal equal to $x(t)$ for $t > T_0$ and equal to 0 for $t < T_0$; (c) the left-sided signal equal to $x(t)$ for $t < T_0$ and equal to 0 for $t > T_0$.

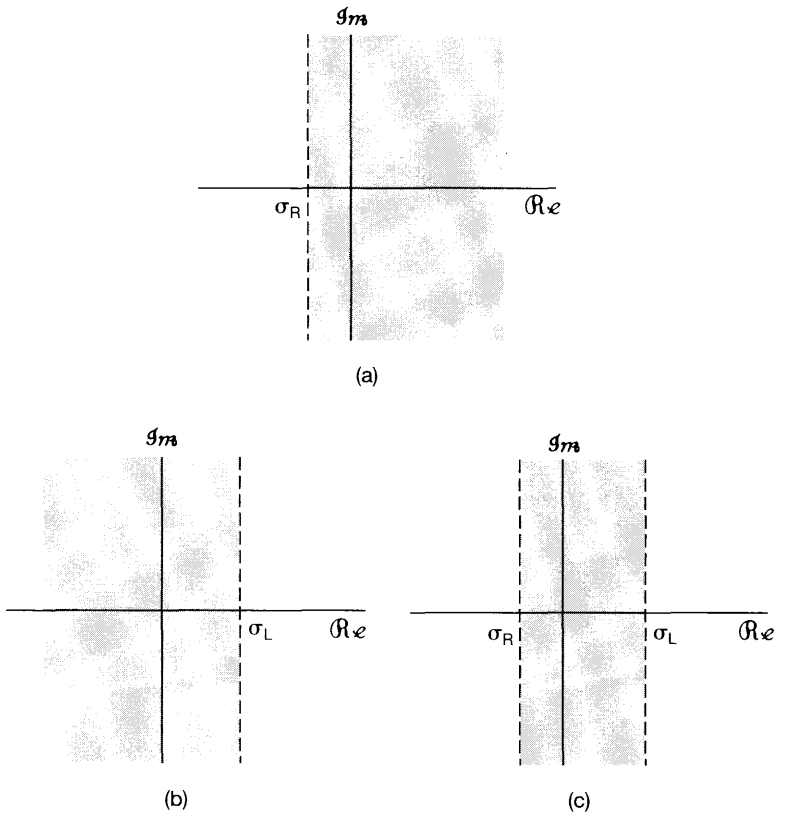


Figure 9.10 (a) ROC for $x_R(t)$ in Figure 9.9; (b) ROC for $x_L(t)$ in Figure 9.9; (c) the ROC for $x(t) = x_R(t) + x_L(t)$, assuming that the ROCs in (a) and (b) overlap.

Example 9.7

Let

$$x(t) = e^{-b|t|}, \tag{9.47}$$

as illustrated in Figure 9.11 for both $b > 0$ and $b < 0$. Since this is a two-sided signal, let us divide it into the sum of a right-sided and left-sided signal; that is,

$$x(t) = e^{-bt}u(t) + e^{+bt}u(-t). \tag{9.48}$$

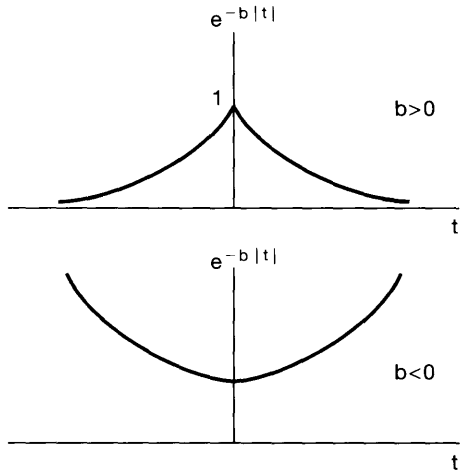


Figure 9.11 Signal $x(t) = e^{-b|t|}$ for both $b > 0$ and $b < 0$.

From Example 9.1,

$$e^{-bt}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+b}, \quad \mathcal{R}e\{s\} > -b, \tag{9.49}$$

and from Example 9.2,

$$e^{+bt}u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{-1}{s-b}, \quad \mathcal{R}e\{s\} < +b. \tag{9.50}$$

Although the Laplace transforms of each of the individual terms in eq. (9.48) have a region of convergence, there is no *common* region of convergence if $b \leq 0$, and thus, for those values of b, $x(t)$ has no Laplace transform. If $b > 0$, the Laplace transform of $x(t)$ is

$$e^{-b|t|} \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+b} - \frac{1}{s-b} = \frac{-2b}{s^2-b^2}, \quad -b < \mathcal{R}e\{s\} < +b. \tag{9.51}$$

The corresponding pole-zero plot is shown in Figure 9.12, with the shading indicating the ROC.

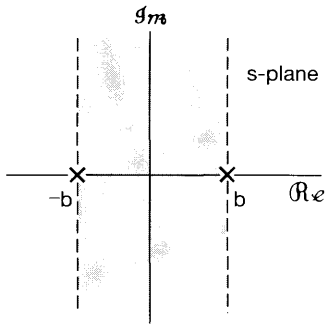


Figure 9.12 Pole-zero plot and ROC for Example 9.7.

A signal either does not have a Laplace transform or falls into one of the four categories covered by Properties 3 through 6. Thus, for any signal with a Laplace transform, the ROC *must* be the entire s-plane (for finite-length signals), a left-half plane (for left-sided signals), a right-half plane (for right-sided signals), or a single strip (for two-sided signals). In all the examples that we have considered, the ROC has the additional property that in each direction (i.e., $\mathcal{R}e\{s\}$ increasing and $\mathcal{R}e\{s\}$ decreasing) it is bounded by poles or extends to infinity. In fact, this is *always* true for rational Laplace transforms:

Property 7: If the Laplace transform $X(s)$ of $x(t)$ is rational, then its ROC is bounded by poles or extends to infinity. In addition, no poles of $X(s)$ are contained in the ROC.

A formal argument establishing this property is somewhat involved, but its validity is essentially a consequence of the facts that a signal with a rational Laplace transform consists of a linear combination of exponentials and, from Examples 9.1 and 9.2, that the ROC for the transform of individual terms in this linear combination must have the property. As a consequence of Property 7, together with Properties 4 and 5, we have

Property 8: If the Laplace transform $X(s)$ of $x(t)$ is rational, then if $x(t)$ is right sided, the ROC is the region in the s-plane to the right of the rightmost pole. If $x(t)$ is left sided, the ROC is the region in the s-plane to the left of the leftmost pole.

To illustrate how different ROCs can be associated with the same pole-zero pattern, let us consider the following example:

Example 9.8

Let

$$X(s) = \frac{1}{(s+1)(s+2)}, \tag{9.52}$$

with the associated pole-zero pattern in Figure 9.13(a). As indicated in Figures 9.13(b)–(d), there are three possible ROCs that can be associated with this algebraic expression, corresponding to three distinct signals. The signal associated with the pole-zero pattern in Figure 9.13(b) is right sided. Since the ROC includes the $j\omega$-axis, the Fourier

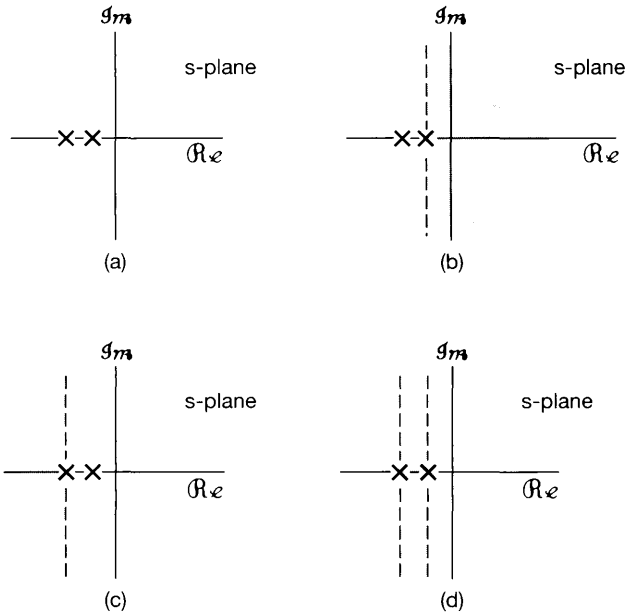


Figure 9.13 (a) Pole-zero pattern for Example 9.8; (b) ROC corresponding to a right-sided sequence; (c) ROC corresponding to a left-sided sequence; (d) ROC corresponding to a two-sided sequence.

transform of this signal converges. Figure 9.13(c) corresponds to a left-sided signal and Figure 9.13(d) to a two-sided signal. Neither of these two signals have Fourier transforms, since their ROCs do not include the $j\omega$-axis.

9.3 THE INVERSE LAPLACE TRANSFORM

In Section 9.1 we discussed the interpretation of the Laplace transform of a signal as the Fourier transform of an exponentially weighted version of the signal; that is, with s expressed as $s = \sigma + j\omega$, the Laplace transform of a signal $x(t)$ is

$$X(\sigma + j\omega) = \mathcal{F}\{x(t)e^{-\sigma t}\} = \int_{-\infty}^{+\infty} x(t)e^{-\sigma t}e^{-j\omega t}\,dt \tag{9.53}$$

for values of $s = \sigma + j\omega$ in the ROC. We can invert this relationship using the inverse Fourier transform as given in eq. (4.9). We have

$$x(t)e^{-\sigma t} = \mathcal{F}^{-1}\{X(\sigma + j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(\sigma + j\omega)e^{j\omega t}\,d\omega, \tag{9.54}$$

or, multiplying both sides by $e^{\sigma t}$, we obtain

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(\sigma + j\omega)e^{(\sigma + j\omega)t}\,d\omega. \tag{9.55}$$

That is, we can recover $x(t)$ from its Laplace transform evaluated for a set of values of $s = \sigma + j\omega$ in the ROC, with σ fixed and ω varying from $-\infty$ to $+\infty$. We can highlight this and gain additional insight into recovering $x(t)$ from $X(s)$ by changing the variable of integration in eq. (9.55) from ω to s and using the fact that σ is constant, so that $ds = j\,d\omega$. The result is the basic inverse Laplace transform equation:

$$\boxed{x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}\,ds.} \tag{9.56}$$

This equation states that $x(t)$ can be represented as a weighted integral of complex exponentials. The contour of integration in eq. (9.56) is the straight line in the s-plane corresponding to all points s satisfying $\mathcal{R}e\{s\} = \sigma$. This line is parallel to the $j\omega$-axis. Furthermore, we can choose any such line in the ROC—i.e., we can choose any value of σ such that $X(\sigma + j\omega)$ converges. The formal evaluation of the integral for a general $X(s)$ requires the use of contour integration in the complex plane, a topic that we will not consider here. However, for the class of rational transforms, the inverse Laplace transform can be determined without directly evaluating eq. (9.56) by using the technique of partial-fraction expansion in a manner similar to that used in Chapter 4 to determine the inverse Fourier transform. Basically, the procedure consists of expanding the rational algebraic expression into a linear combination of lower order terms.

For example, assuming no multiple-order poles, and assuming that the order of the denominator polynomial is greater than the order of the numerator polynomial, we can expand $X(s)$ in the form

$$X(s) = \sum_{i=1}^{m} \frac{A_i}{s + a_i}. \tag{9.57}$$

From the ROC of $X(s)$, the ROC of each of the individual terms in eq. (9.57) can be inferred, and then, from Examples 9.1 and 9.2, the inverse Laplace transform of each of these terms can be determined. There are two possible choices for the inverse transform of each term $A_i/(s + a_i)$ in the equation. If the ROC is to the right of the pole at $s = -a_i$, then the inverse transform of this term is $A_i e^{-a_i t}u(t)$, a right-sided signal. If the ROC is to the left of the pole at $s = -a_i$, then the inverse transform of the term is $-A_i e^{-a_i t}u(-t)$, a left-sided signal. Adding the inverse transforms of the individual terms in eq. (9.57) then yields the inverse transform of $X(s)$. The details of this procedure are best presented through a number of examples.

Example 9.9

Let

$$X(s) = \frac{1}{(s + 1)(s + 2)}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.58}$$

To obtain the inverse Laplace transform, we first perform a partial-fraction expansion to obtain

$$X(s) = \frac{1}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2}. \tag{9.59}$$

As discussed in the appendix, we can evaluate the coefficients A and B by multiplying both sides of eq. (9.59) by $(s + 1)(s + 2)$ and then equating coefficients of equal powers of s on both sides. Alternatively, we can use the relation

$$A = [(s + 1)X(s)]|_{s=-1} = 1, \tag{9.60}$$

$$B = [(s + 2)X(s)]|_{s=-2} = -1. \tag{9.61}$$

Thus, the partial-fraction expansion for $X(s)$ is

$$X(s) = \frac{1}{s + 1} - \frac{1}{s + 2}. \tag{9.62}$$

From Examples 9.1 and 9.2, we know that there are two possible inverse transforms for a transform of the form $1/(s + a)$, depending on whether the ROC is to the left or the right of the pole. Consequently, we need to determine which ROC to associate with each of the individual first-order terms in eq. (9.62). This is done by reference to the properties of the ROC developed in Section 9.2. Since the ROC for $X(s)$ is $\mathcal{R}e\{s\} > -1$, the ROC for the individual terms in the partial-fraction expansion of eq. (9.62) includes $\mathcal{R}e\{s\} > -1$. The ROC for each term can then be extended to the left or right (or both) to be bounded by a pole or infinity. This is illustrated in Figure 9.14. Figure 9.14(a) shows the pole-zero plot and ROC for $X(s)$, as specified in eq. (9.58). Figure 9.14(b) and 9.14(c) represent the individual terms in the partial-fraction expansion in eq. (9.62). The ROC for the sum is indicated with lighter shading. For the term represented by Figure 9.14(c), the ROC for the sum can be extended to the left as shown, so that it is bounded by a pole.

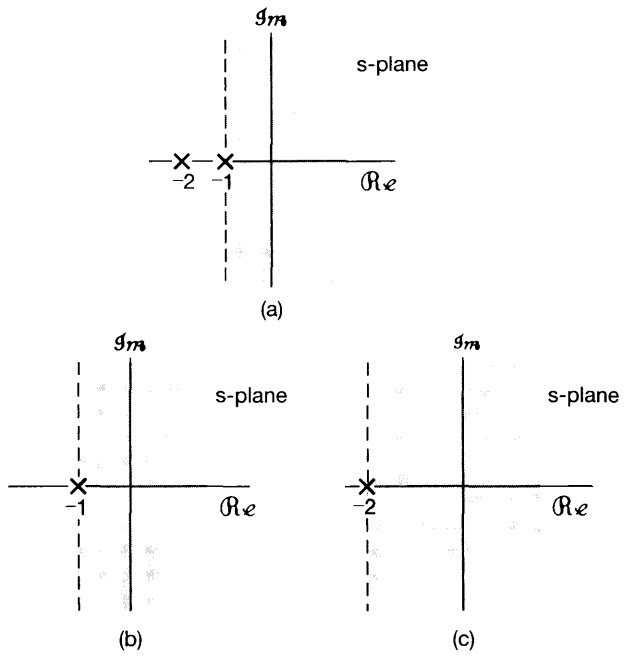


Figure 9.14 Construction of the ROCs for the individual terms in the partial-fraction expansion of $X(s)$ in Example 9.8: (a) pole-zero plot and ROC for $X(s)$; (b) pole at $s = -1$ and its ROC; (c) pole at $s = -2$ and its ROC.

Since the ROC is to the right of both poles, the same is true for each of the individual terms, as can be seen in Figures 9.14(b) and (c). Consequently, from Property 8 in the preceding section, we know that each of these terms corresponds to a right-sided signal. The inverse transform of the individual terms in eq. (9.62) can then be obtained by reference to Example 9.1:

$$e^{-t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+1}, \qquad \mathcal{R}e\{s\} > -1, \tag{9.63}$$

$$e^{-2t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+2}, \qquad \mathcal{R}e\{s\} > -2. \tag{9.64}$$

We thus obtain

$$[e^{-t} - e^{-2t}]u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{(s+1)(s+2)}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.65}$$

Example 9.10

Let us now suppose that the algebraic expression for $X(s)$ is again given by eq. (9.58), but that the ROC is now the left-half plane $\mathcal{R}e\{s\} < -2$. The partial-fraction expansion for $X(s)$ relates only to the algebraic expression, so eq. (9.62) is still valid. With this new ROC, however, the ROC is to the *left* of both poles and thus, the same must be true for each of the two terms in the equation. That is, the ROC for the term corresponding to the pole at $s = -1$ is $\mathcal{R}e\{s\} < -1$, while the ROC for the term with pole at $s = -2$ is $\mathcal{R}e\{s\} < -2$. Then, from Example 9.2,

$$-e^{-t}u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+1}, \qquad \mathcal{R}e\{s\} < -1, \tag{9.66}$$

$$-e^{-2t}u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+2}, \qquad \mathcal{R}e\{s\} < -2, \tag{9.67}$$

so that

$$x(t) = [-e^{-t} + e^{-2t}]u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{(s+1)(s+2)}, \qquad \mathcal{R}e\{s\} < -2. \tag{9.68}$$

Example 9.11

Finally, suppose that the ROC of $X(s)$ in eq. (9.58) is $-2 < \mathcal{R}e\{s\} < -1$. In this case, the ROC is to the left of the pole at $s = -1$ so that this term corresponds to the left-sided signal in eq. (9.66), while the ROC is to the right of the pole at $s = -2$ so that this term corresponds to the right-sided signal in eq. (9.64). Combining these, we find that

$$x(t) = -e^{-t}u(-t) - e^{-2t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{(s+1)(s+2)}, \qquad -2 < \mathcal{R}e\{s\} < -1. \tag{9.69}$$

As discussed in the appendix, when $X(s)$ has multiple-order poles, or when the denominator is not of higher degree than the numerator, the partial-fraction expansion of $X(s)$ will include other terms in addition to the first-order terms considered in Examples 9.9–9.11. In Section 9.5, after discussing properties of the Laplace transform, we develop some other Laplace transform pairs that, in conjunction with the properties, allow us to extend the inverse transform method outlined in Example 9.9 to arbitrary rational transforms.

9.4 GEOMETRIC EVALUATION OF THE FOURIER TRANSFORM FROM THE POLE-ZERO PLOT

As we saw in Section 9.1, the Fourier transform of a signal is the Laplace transform evaluated on the $j\omega$-axis. In this section we discuss a procedure for geometrically evaluating the Fourier transform and, more generally, the Laplace transform at any set of values from the pole-zero pattern associated with a rational Laplace transform. To develop the procedure, let us first consider a Laplace transform with a single zero [i.e., $X(s) = s - a$], which we evaluate at a specific value of s, say, $s = s_1$. The algebraic expression $s_1 - a$ is the sum of two complex numbers, s_1 and $-a$, each of which can be represented as a vector in the complex plane, as illustrated in Figure 9.15. The vector representing the complex number $s_1 - a$ is then the vector sum of $\mathbf{s}_1$ and $-\mathbf{a}$, which we see in the figure to be a vector from the zero at $s = a$ to the point s_1. The value of $X(s_1)$ then has a magnitude that is the length of this vector and an angle that is the angle of the vector relative to the real axis. If $X(s)$ instead has a single pole at $s = a$ [i.e., $X(s) = 1/(s - a)$], then the denominator would be represented by the same vector sum of $\mathbf{s}_1$ and $-\mathbf{a}$, and the value of $X(s_1)$ would have a magnitude that is the *reciprocal* of the length of the vector from the pole to $s = s_1$ and an angle that is the *negative* of the angle of the vector with the real axis.

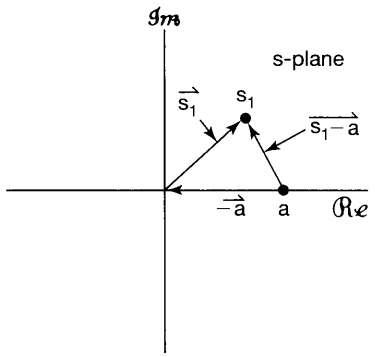


Figure 9.15 Complex plane representation of the vectors $\mathbf{s}_1$, $\mathbf{a}$, and $\mathbf{s}_1 - \mathbf{a}$ representing the complex numbers s_1, a, and $s_1 - a$, respectively.

A more general rational Laplace transform consists of a product of pole and zero terms of the form discussed in the preceding paragraph; that is, it can be factored into the form

$$X(s) = M\frac{\prod_{i=1}^{R}(s - \beta_i)}{\prod_{j=1}^{P}(s - \alpha_j)}. \tag{9.70}$$

To evaluate $X(s)$ at $s = s_1$, each term in the product is represented by a vector from the zero or pole to the point s_1. The magnitude of $X(s_1)$ is then the magnitude of the scale factor M, times the product of the lengths of the zero vectors (i.e., the vectors from the zeros to s_1) divided by the product of the lengths of the pole vectors (i.e., the vectors from the poles to s_1). The angle of the complex number $X(s_1)$ is the sum of the angles of the zero vectors minus the sum of the angles of the pole vectors. If the scale factor M in eq. (9.70) is negative, an additional angle of π would be included. If $X(s)$ has a multiple pole or zero

(or both), corresponding to some of the α_j's being equal to each other or some of the β_i's being equal to each other (or both), the lengths and angles of the vectors from each of these poles or zeros must be included a number of times equal to the order of the pole or zero.

Example 9.12

Let

$$X(s) = \frac{1}{s + \frac{1}{2}}, \qquad \mathcal{R}e\{s\} > -\frac{1}{2}. \tag{9.71}$$

The Fourier transform is $X(s)|_{s = j\omega}$. For this example, then, the Fourier transform is

$$X(j\omega) = \frac{1}{j\omega + 1/2}. \tag{9.72}$$

The pole-zero plot for $X(s)$ is shown in Figure 9.16. To determine the Fourier transform graphically, we construct the pole vector as indicated. The magnitude of the Fourier transform at frequency ω is the reciprocal of the length of the vector from the pole to the point $j\omega$ on the imaginary axis. The phase of the Fourier transform is the negative of the angle of the vector. Geometrically, from Figure 9.16, we can write

$$|X(j\omega)|^2 = \frac{1}{\omega^2 + (1/2)^2} \tag{9.73}$$

and

$$\measuredangle X(j\omega) = -\tan^{-1} 2\omega. \tag{9.74}$$

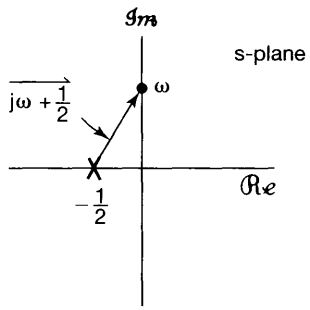


Figure 9.16 Pole-zero plot for Example 9.12. $|X(j\omega)|$ is the reciprocal of the length of the vector shown, and $\measuredangle X(j\omega)$ is the negative of the angle of the vector.

Often, part of the value of the geometric determination of the Fourier transform lies in its usefulness in obtaining an approximate view of the overall characteristics of the transform. For example, in Figure 9.16, it is readily evident that the length of the pole vector monotonically increases with increasing ω, and thus, the magnitude of the Fourier

transform will monotonically *decrease* with increasing ω. The ability to draw general conclusions about the behavior of the Fourier transform from the pole-zero plot is further illustrated by a consideration of general first- and second-order systems.

9.4.1 First-Order Systems

As a generalization of Example 9.12, let us consider the class of first-order systems that was discussed in some detail in Section 6.5.1. The impulse response for such a system is

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t), \tag{9.75}$$

and its Laplace transform is

$$H(s) = \frac{1}{s\tau + 1}, \qquad \mathcal{R}e\{s\} > -\frac{1}{\tau}. \tag{9.76}$$

The pole-zero plot is shown in Figure 9.17. Note from the figure that the length of the pole vector is minimal for $\omega = 0$ and increases monotonically as ω increases. Also, the angle of the pole increases monotonically from 0 to $\pi/2$ as ω increases from 0 to ∞.

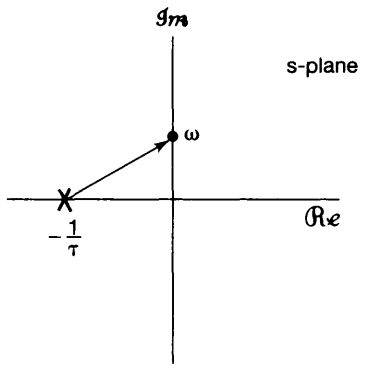


Figure 9.17 Pole-zero plot for first-order system of eq. (9.76).

From the behavior of the pole vector as ω varies, it is clear that the magnitude of the frequency response $H(j\omega)$ monotonically decreases as ω increases, while $\measuredangle H(j\omega)$ monotonically decreases from 0 to $-\pi/2$, as shown in the Bode plots for this system in Figure 9.18. Note also that when $\omega = 1/\tau$, the real and imaginary parts of the pole vector are equal, yielding a value of the magnitude of the frequency response that is reduced by a factor of $\sqrt{2}$, or approximately 3 dB, from its maximum at $\omega = 0$ and a value of $\pi/4$ for the angle of the frequency response. This is consistent with our examination of first-order systems in Section 6.5.1, where we noted that $\omega = 1/\tau$ is often referred to as the 3-dB point or the break frequency—i.e., the frequency at which the straight-line approximation of the Bode plot of $|H(j\omega)|$ has a break in its slope. As we also saw in Section 6.5.1, the time constant τ controls the speed of response of first-order systems, and we now see that

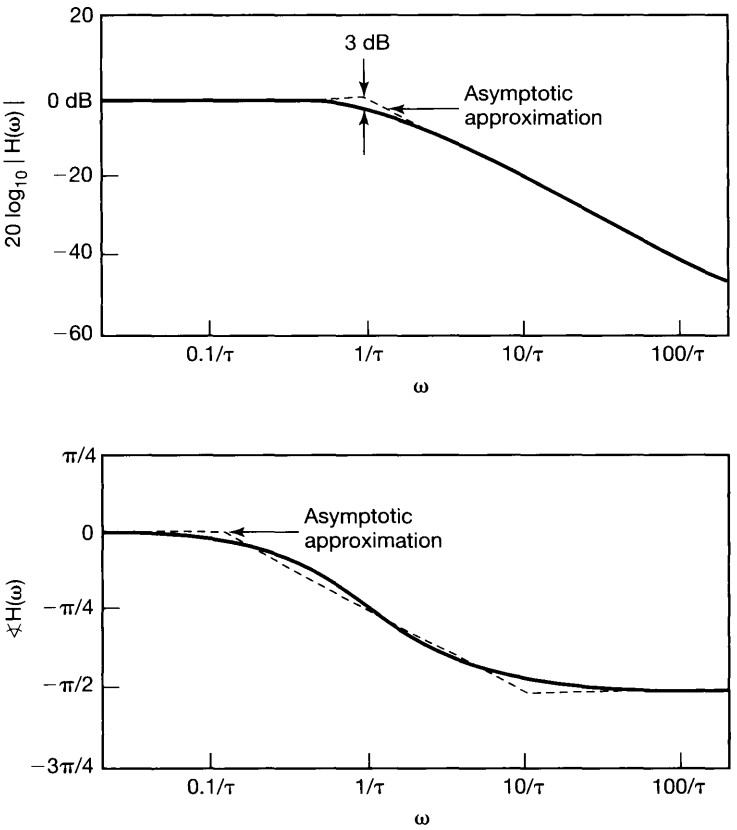


Figure 9.18 Frequency response for a first-order system.

the pole of such a system at $s = -1/\tau$ is on the negative real axis, at a distance to the origin that is the reciprocal of the time constant.

From our graphical interpretation, we can also see how changing the time constant or, equivalently, the position of the pole of $H(s)$ changes the characteristics of a first-order system. In particular, as the pole moves farther into the left-half plane, the break frequency and, hence, the effective cutoff frequency of the system increases. Also, from eq. (9.75) and from Figure 6.19, we see that this same movement of the pole to the left corresponds to a decrease in the time constant τ, resulting in a faster decay of the impulse response and a correspondingly faster rise time in the step response. This relationship between the real part of the pole locations and the speed of the system response holds more generally; that is, poles farther away from the $j\omega$-axis are associated with faster response terms in the impulse response.

9.4.2 Second-Order Systems

Let us next consider the class of second-order systems, which was discussed in some detail in Section 6.5.2. The impulse response and frequency response for the system, originally

given in eqs. (6.37) and (6.33), respectively, are

$$h(t) = M[e^{c_1 t} - e^{c_2 t}]u(t), \tag{9.77}$$

where

$$\begin{aligned} c_1 &= -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}, \\ c_2 &= -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}, \\ M &= \frac{\omega_n}{2\sqrt{\zeta^2 - 1}}, \end{aligned}$$

and

$$H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}. \tag{9.78}$$

The Laplace transform of the impulse response is

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s - c_1)(s - c_2)}. \tag{9.79}$$

For $\zeta > 1$, c_1 and c_2 are real and thus both poles lie on the real axis, as indicated in Figure 9.19(a). The case of $\zeta > 1$ is essentially a product of two first-order terms, as in Section 9.4.1. Consequently, in this case $|H(j\omega)|$ decreases monotonically as $|\omega|$ increases, while $\measuredangle H(j\omega)$ varies from 0 at $\omega = 0$ to $-\pi$ as $\omega \to \infty$. This can be verified from Figure 9.19(a) by observing that the length of the vector from each of the two poles to the point $s = j\omega$ increases monotonically as ω increases from 0, and the angle of each of these vectors increases from 0 to $\pi/2$ as ω increases from 0 to ∞. Note also that as ζ increases, one pole moves closer to the $j\omega$-axis, indicative of a term in the impulse response that decays more slowly, and the other pole moves farther into the left-half plane, indicative of a term in the impulse response that decays more rapidly. Thus, for large values of ζ, it is the pole close to the $j\omega$-axis that dominates the system response for large time. Similarly, from a consideration of the pole vectors for $\zeta \gg 1$, as indicated in Figure 9.19(b), for low frequencies the length and angle of the vector for the pole close to the $j\omega$-axis are much more sensitive to changes in ω than the length and angle of the vector for the pole far from the $j\omega$-axis. Hence, we see that for low frequencies, the characteristics of the frequency response are influenced principally by the pole close to the $j\omega$-axis.

For $0 < \zeta < 1$, c_1 and c_2 are complex, so that the pole-zero plot is that shown in Figure 9.19(c). Correspondingly, the impulse response and step response have oscillatory parts. We note that the two poles occur in complex conjugate locations. In fact, as we discuss in Section 9.5.5, the complex poles (and zeros) for a real-valued signal always occur in complex conjugate pairs. From the figure—particularly when ζ is small, so that the poles are close to the $j\omega$-axis—as ω approaches $\omega_n\sqrt{1 - \zeta^2}$, the behavior of the frequency response is dominated by the pole vector in the second quadrant, and in

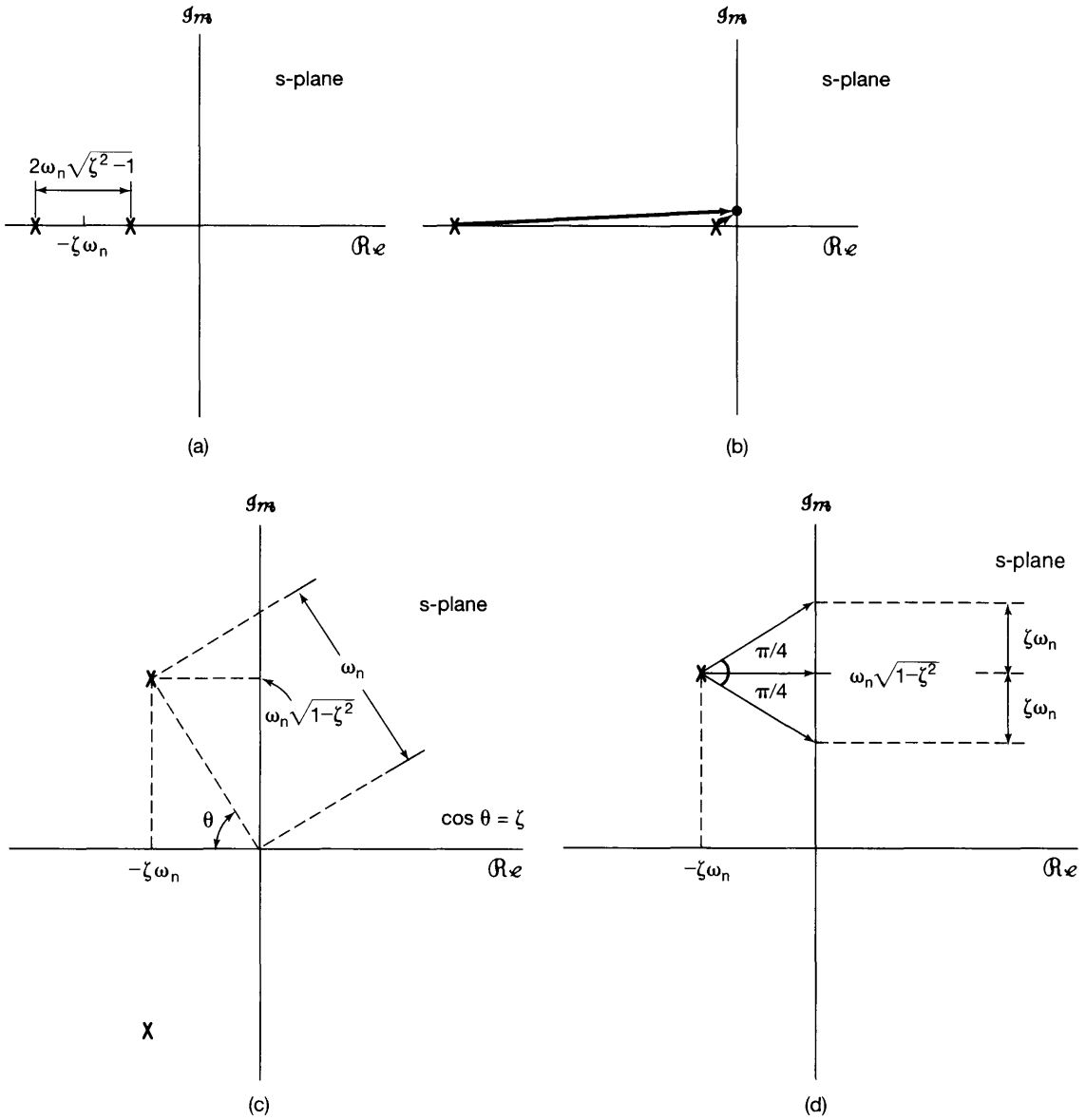


Figure 9.19 (a) Pole-zero plot for a second-order system with $\zeta > 1$; (b) pole vectors for $\zeta \gg 1$; (c) pole-zero plot for a second-order system with $0 < \zeta < 1$; (d) pole vectors for $0 < \zeta < 1$ and for $\omega = \omega_n\sqrt{1-\zeta^2}$ and $\omega = \omega_n\sqrt{1-\zeta^2} \pm \zeta\omega_n$.

particular, the length of that pole vector has a minimum at $\omega = \omega_n\sqrt{1-\zeta^2}$. Thus, qualitatively, we would expect the magnitude of the frequency response to exhibit a peak in the vicinity of that frequency. Because of the presence of the other pole, the peak will occur not exactly at $\omega = \omega_n\sqrt{1-\zeta^2}$, but at a frequency slightly less than this. A careful sketch of the magnitude of the frequency response is shown in Figure 9.20(a) for $\omega_n = 1$ and several values of ζ where the expected behavior in the vicinity of the poles is clearly evident. This is consistent with our analysis of second-order systems in Section 6.5.2.

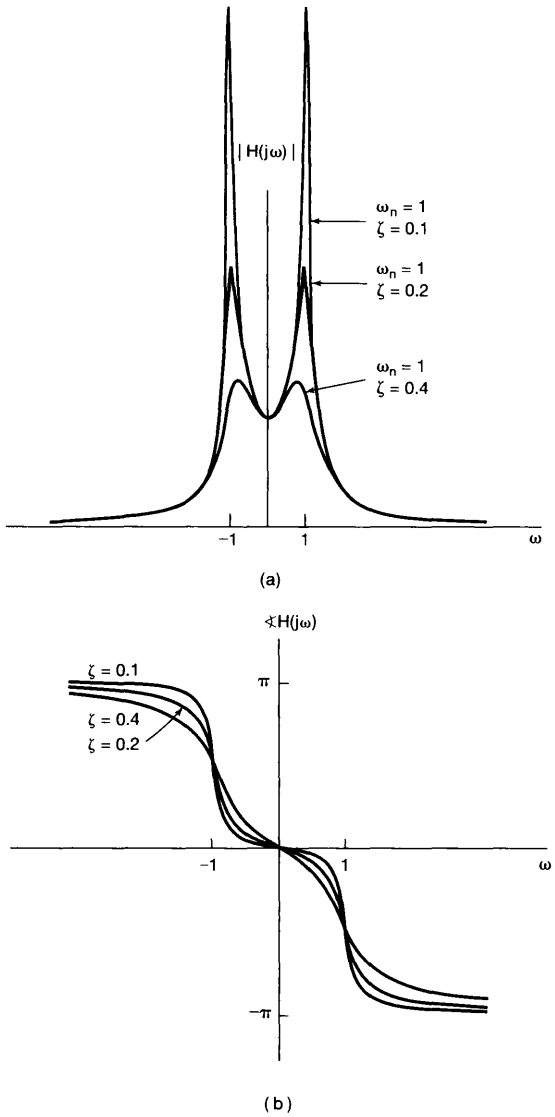


Figure 9.20 (a) Magnitude and (b) phase of the frequency response for a second-order system with $0 < \zeta < 1$.

Thus, for $0 < \zeta < 1$, the second-order system is a nonideal bandpass filter, with the parameter ζ controlling the sharpness and width of the peak in the frequency response. In particular, from the geometry in Figure 9.19(d), we see that the length of the pole vector from the second-quadrant pole increases by a factor of $\sqrt{2}$ from its minimum at $\omega = \omega_n\sqrt{1-\zeta^2}$ when ω increases or decreases from this value by $\zeta\omega_n$. Consequently, for small ζ, and neglecting the effect of the distant third-quadrant pole, $|H(j\omega)|$ is within a factor of $\sqrt{2}$ of its peak value over the frequency range

$$\omega_n\sqrt{1-\zeta^2} - \zeta\omega_n < \omega < \omega_n\sqrt{1-\zeta^2} + \zeta\omega_n.$$

If we define the relative bandwidth B as the length of this frequency interval divided by the undamped natural frequency ω_n, we see that

$$B = 2\zeta.$$

Thus, the closer ζ is to zero, the sharper and narrower the peak in the frequency response is. Note also that B is the reciprocal of the quality measure Q for second-order systems defined in Section 6.5.2. Thus, as the quality increases, the relative bandwidth decreases and the filter becomes increasingly frequency selective.

An analogous picture can be developed for $\measuredangle H(\omega)$, which is plotted in Figure 9.20(b) for $\omega_n = 1$ and several values of ζ. As can be seen from Figure 9.19(d), the angle of the second-quadrant pole vector changes from $-\pi/4$ to 0 to $\pi/4$ as ω changes from $\omega_n\sqrt{1-\zeta^2} - \zeta\omega_n$ to $\omega_n\sqrt{1-\zeta^2}$ to $\omega_n\sqrt{1-\zeta^2} + \zeta\omega_n$. For small values of ζ, the angle for the third-quadrant pole changes very little over this frequency interval, resulting in a rapid change in $\measuredangle H(j\omega)$ of approximately $\pi/2$ over the interval, as captured in the figure.

Varying ω_n with ζ fixed only changes the frequency scale in the preceding discussion—i.e., $|H(\omega)|$ and $\measuredangle H(\omega)$ depend only on ω/ω_n. From Figure 9.19(c), we also can readily determine how the poles and system characteristics change as we vary ζ, keeping ω_n constant. Since $\cos\theta = \zeta$, the poles move along a semicircle with fixed radius ω_n. For $\zeta = 0$, the two poles are on the imaginary axis. Correspondingly, in the time domain, the impulse response is sinusoidal with no damping. As ζ increases from 0 to 1, the two poles remain complex and move into the left-half plane, and the vectors from the origin to the poles maintain a constant overall magnitude ω_n. As the real part of the poles becomes more negative, the associated time response will decay more quickly as $t \to \infty$. Also, as we have seen, as ζ increases from 0 toward 1, the relative bandwidth of the frequency response increases, and the frequency response becomes less sharp and less frequency selective.

9.4.3 All-Pass Systems

As a final illustration of the geometric evaluation of the frequency response, let us consider a system for which the Laplace transform of the impulse response has the pole-zero plot shown in Figure 9.21(a). From this figure, it is evident that for any point along the $j\omega$-axis, the pole and zero vectors have equal length, and consequently, the magnitude of the frequency response is constant and independent of frequency. Such a system is com-

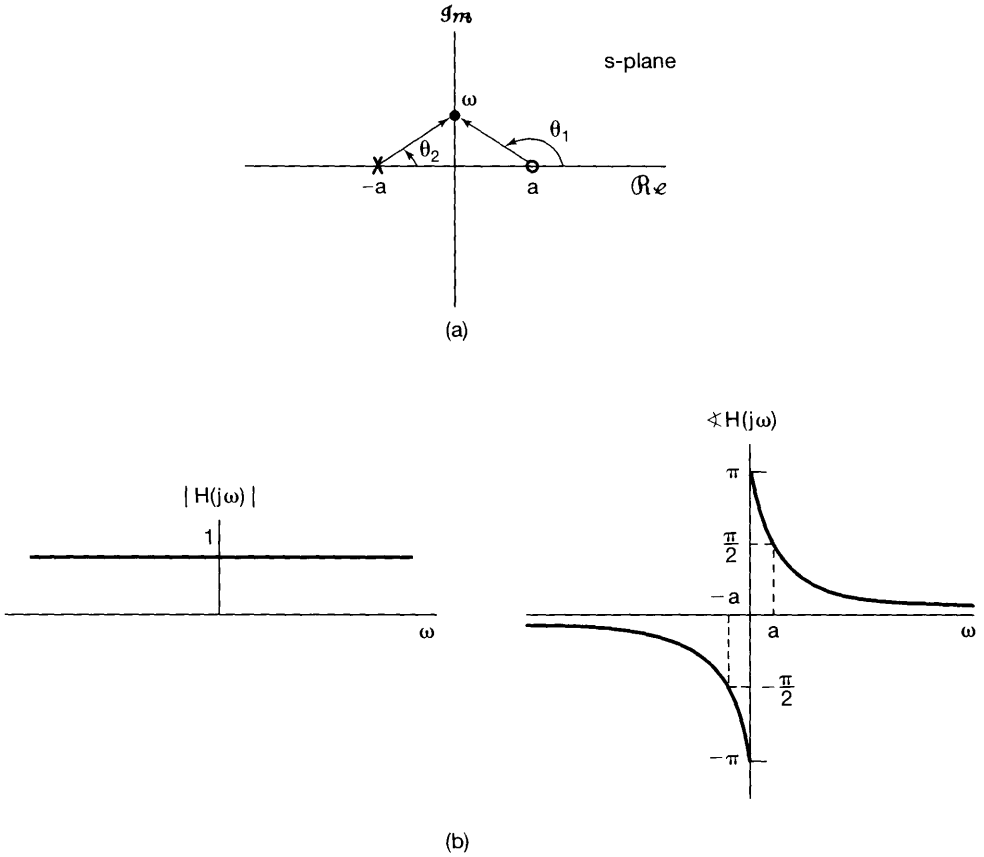


Figure 9.21 (a) Pole-zero plot for an all-pass system; (b) magnitude and phase of an all-pass frequency response.

monly referred to as an *all-pass system,* since it passes all frequencies with equal gain (or attenuation). The phase of the frequency response is $\theta_1 - \theta_2$, or, since $\theta_1 = \pi - \theta_2$,

$$\measuredangle H(j\omega) = \pi - 2\theta_2. \tag{9.80}$$

From Figure 9.21(a), $\theta_2 = \tan^{-1}(\omega/a)$, and thus,

$$\measuredangle H(j\omega) = \pi - 2\tan^{-1}\left(\frac{\omega}{a}\right). \tag{9.81}$$

The magnitude and phase of $H(j\omega)$ are illustrated in Figure 9.21(b).

9.5 PROPERTIES OF THE LAPLACE TRANSFORM

In exploiting the Fourier transform, we relied heavily on the set of properties developed in Section 4.3. In the current section, we consider the corresponding set of properties for

the Laplace transform. The derivations of many of these results are analogous to those of the corresponding properties for the Fourier transform. Consequently, we will not present the derivations in detail, some of which are left as exercises at the end of the chapter. (See Problems 9.52–9.54.)

9.5.1 Linearity of the Laplace Transform

If

$$x_1(t) \overset{\mathcal{L}}{\longleftrightarrow} X_1(s) \quad \text{with a region of convergence that will be denoted as } R_1$$

and

$$x_2(t) \overset{\mathcal{L}}{\longleftrightarrow} X_2(s) \quad \text{with a region of convergence that will be denoted as } R_2,$$

then

$$\boxed{ax_1(t) + bx_2(t) \overset{\mathcal{L}}{\longleftrightarrow} aX_1(s) + bX_2(s), \text{ with ROC containing } R_1 \cap R_2.} \tag{9.82}$$

As indicated, the region of convergence of $X(s)$ is at least the intersection of R_1 and R_2, which could be empty, in which case $X(s)$ has no region of convergence—i.e., $x(t)$ has no Laplace transform. For example, for $x(t)$ as in eq. (9.47) of Example 9.7, with $b > 0$ the ROC for $X(s)$ is the intersection of the ROCs for the two terms in the sum. If $b < 0$, there are no common points in R_1 and R_2; that is, the intersection is empty, and thus, $x(t)$ has no Laplace transform. The ROC can also be larger than the intersection. As a simple example, for $x_1(t) = x_2(t)$ and $a = -b$ in eq. (9.82), $x(t) = 0$, and thus, $X(s) = 0$. The ROC of $X(s)$ is then the entire s-plane.

The ROC associated with a linear combination of terms can always be constructed by using the properties of the ROC developed in Section 9.2. Specifically, from the intersection of the ROCs for the individual terms (assuming that it is not empty), we can find a line or strip that is in the ROC of the linear combination. We then extend this to the right ($\mathcal{Re}\{s\}$ increasing) and to the left ($\mathcal{Re}\{s\}$ decreasing) to the nearest poles (which may be at infinity).

Example 9.13

In this example, we illustrate the fact that the ROC for the Laplace transform of a linear combination of signals can sometimes extend beyond the intersection of the ROCs for the individual terms. Consider

$$x(t) = x_1(t) - x_2(t), \tag{9.83}$$

where the Laplace transforms of $x_1(t)$ and $x_2(t)$ are, respectively,

$$X_1(s) = \frac{1}{s+1}, \qquad \mathcal{R}e\{s\} > -1, \tag{9.84}$$

and

$$X_2(s) = \frac{1}{(s+1)(s+2)}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.85}$$

The pole-zero plot, including the ROCs for $X_1(s)$ and $X_2(s)$, is shown in Figures 9.22(a) and (b). From eq. (9.82),

$$X(s) = \frac{1}{s+1} - \frac{1}{(s+1)(s+2)} = \frac{s+1}{(s+1)(s+2)} = \frac{1}{s+2}. \tag{9.86}$$

Thus, in the linear combination of $x_1(t)$ and $x_2(t)$, the pole at $s = -1$ is canceled by a zero at $s = -1$. The pole-zero plot for $X(s) = X_1(s) - X_2(s)$ is shown in Figure 9.22(c). The intersection of the ROCs for $X_1(s)$ and $X_2(s)$ is $\mathcal{R}e\{s\} > -1$. However, since the ROC is always bounded by a pole or infinity, for this example the ROC for $X(s)$ can be extended to the left to be bounded by the pole at $s = -2$, as a result of the pole-zero cancellation at $s = -1$.

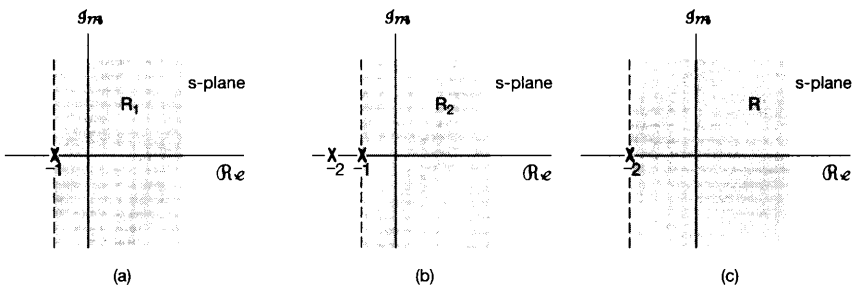


Figure 9.22 Pole-zero plots and ROCs for Example 9.13: (a) $X_1(s)$; (b) $X_2(s)$; (c) $X_1(s) - X_2(s)$. The ROC for $X_1(s) - X_2(s)$ includes the intersection of R_1 and R_2, which can then be extended to be bounded by the pole at $s = -2$.

9.5.2 Time Shifting

If

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \qquad \text{with ROC} = R,$$

then

$$\boxed{x(t - t_0) \overset{\mathcal{L}}{\longleftrightarrow} e^{-st_0}X(s), \quad \text{with ROC} = R.} \tag{9.87}$$

9.5.3 Shifting in the s-Domain

If

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \qquad \text{with ROC} = R,$$

then

$$\boxed{e^{s_0 t}x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s - s_0), \quad \text{with ROC} = R + \mathcal{R}e\{s_0\}.} \tag{9.88}$$

That is, the ROC associated with $X(s - s_0)$ is that of $X(s)$, shifted by $\mathcal{R}e\{s_0\}$. Thus, for any value s that is in R, the value $s + \mathcal{R}e\{s_0\}$ will be in R_1. This is illustrated in Figure 9.23. Note that if $X(s)$ has a pole or zero at $s = a$, then $X(s - s_0)$ has a pole or zero at $s - s_0 = a$—i.e., $s = a + s_0$.

An important special case of eq. (9.88) is when $s_0 = j\omega_0$—i.e., when a signal $x(t)$ is used to modulate a periodic complex exponential $e^{j\omega_0 t}$. In this case, eq. (9.88) becomes

$$e^{j\omega_0 t}x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s - j\omega_0), \quad \text{with ROC} = R. \tag{9.89}$$

The right-hand side of eq. (9.89) can be interpreted as a shift in the s-plane parallel to the $j\omega$-axis. That is, if the Laplace transform of $x(t)$ has a pole or zero at $s = a$, then the Laplace transform of $e^{j\omega_0 t}x(t)$ has a pole or zero at $s = a + j\omega_0$.

9.5.4 Time Scaling

If

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \quad \text{with ROC} = R,$$

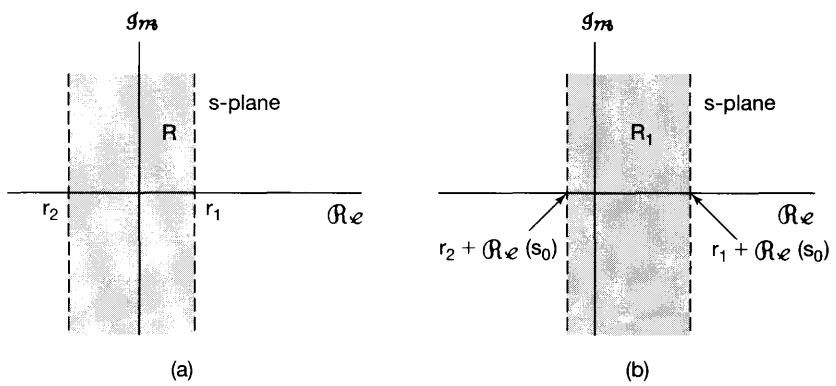


Figure 9.23 Effect on the ROC of shifting in the s-domain: (a) the ROC of $X(s)$; (b) the ROC of $X(s - s_0)$.

then

$$x(at) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{|a|} X\left(\frac{s}{a}\right), \quad \text{with ROC } R_1 = aR. \tag{9.90}$$

That is, for any value s in R [which is illustrated in Figure 9.24(a)], the value a/s will be in R_1, as illustrated in Figure 9.24(b) for a positive value of $a < 1$. Note that, for $0 < a < 1$, there is a compression in the size of the ROC of $X(s)$ by a factor of a, as depicted in Figure 9.24(b), while for a > 1, the ROC is expanded by a factor of a. Also, eq. (9.90) implies that if a is negative, the ROC undergoes a reversal plus a scaling. In particular, as depicted in Figure 9.24(c), the ROC of $1/|a|X(s/a)$ for $0 > a > -1$ involves

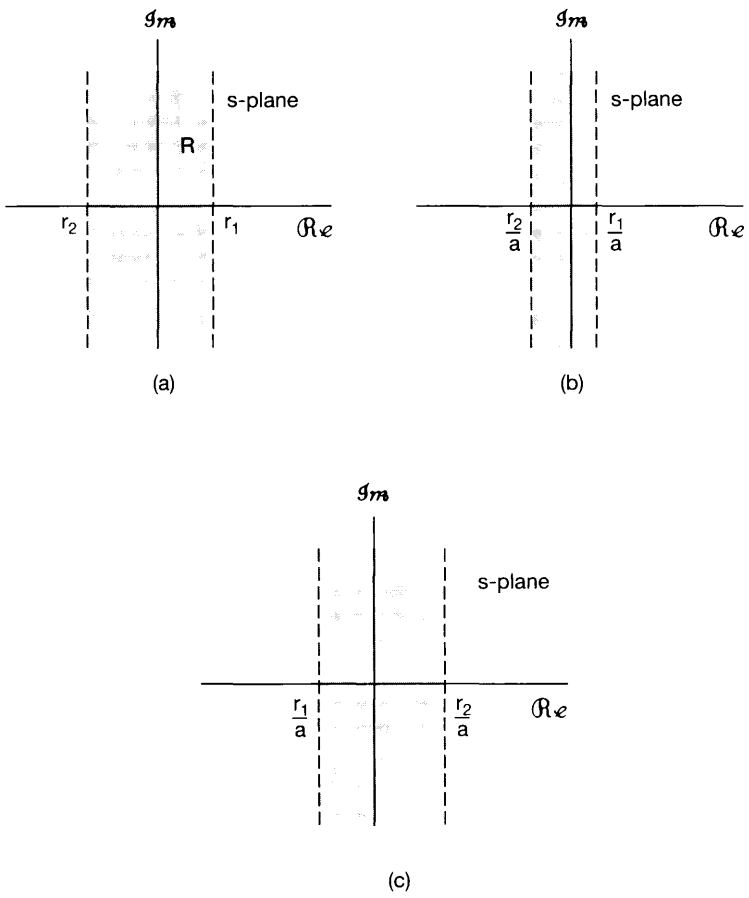


Figure 9.24 Effect on the ROC of time scaling: (a) ROC of $X(s)$; (b) ROC of $(1/|a|)X(s/a)$ for $0 < a < 1$; (c) ROC of $(1/|a|)X(s/a)$ for $0 > a > -1$.

a reversal about the $j\omega$-axis, together with a change in the size of the ROC by a factor of $|a|$. Thus, time reversal of $x(t)$ results in a reversal of the ROC. That is,

$$x(-t) \overset{\mathcal{L}}{\longleftrightarrow} X(-s), \quad \text{with ROC} = -R. \tag{9.91}$$

9.5.5 Conjugation

If

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \quad \text{with ROC} = R, \tag{9.92}$$

then

$$\boxed{x^*(t) \overset{\mathcal{L}}{\longleftrightarrow} X^*(s^*), \quad \text{with ROC} = R.} \tag{9.93}$$

Therefore,

$$X(s) = X^*(s^*) \quad \text{when } x(t) \text{ is real.} \tag{9.94}$$

Consequently, if $x(t)$ is real and if $X(s)$ has a pole or zero at $s = s_0$ (i.e., if $X(s)$ is unbounded or zero at $s = s_0$), then $X(s)$ also has a pole or zero at the complex conjugate point $s = s_0^*$. For example, the transform $X(s)$ for the real signal $x(t)$ in Example 9.4 has poles at $s = 1 \pm 3j$ and zeros at $s = (-5 \pm j\sqrt{71})/2$.

9.5.6 Convolution Property

If

$$x_1(t) \overset{\mathcal{L}}{\longleftrightarrow} X_1(s), \quad \text{with } ROC = R_1,$$

and

$$x_2(t) \overset{\mathcal{L}}{\longleftrightarrow} X_2(s), \quad \text{with } ROC = R_2,$$

then

$$\boxed{x_1(t) * x_2(t) \overset{\mathcal{L}}{\longleftrightarrow} X_1(s)X_2(s), \quad \text{with ROC containing } R_1 \cap R_2.} \tag{9.95}$$

In a manner similar to the linearity property set forth in Section 9.5.1, the ROC of $X_1(s)X_2(s)$ includes the intersection of the ROCs of $X_1(s)$ and $X_2(s)$ and may be larger if pole-zero cancellation occurs in the product. For example, if

$$X_1(s) = \frac{s+1}{s+2}, \quad \mathcal{R}e\{s\} > -2, \tag{9.96}$$

and

$$X_2(s) = \frac{s+2}{s+1}, \qquad \mathcal{R}e\{s\} > -1, \tag{9.97}$$

then $X_1(s)X_2(s) = 1$, and its ROC is the entire s-plane.

As we saw in Chapter 4, the convolution property in the context of the Fourier transform plays an important role in the analysis of linear time-invariant systems. In Sections 9.7 and 9.8 we will exploit in some detail the convolution property for Laplace transforms for the analysis of LTI systems in general and, more specifically, for the class of systems represented by linear constant-coefficient differential equations.

9.5.7 Differentiation in the Time Domain

If

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \qquad \text{with ROC} = R,$$

then

$$\boxed{\frac{dx(t)}{dt} \overset{\mathcal{L}}{\longleftrightarrow} sX(s), \qquad \text{with ROC containing } R.} \tag{9.98}$$

This property follows by differentiating both sides of the inverse Laplace transform as expressed in equation (9.56). Specifically, let

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}ds.$$

Then

$$\frac{dx(t)}{dt} = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} sX(s)e^{st}ds. \tag{9.99}$$

Consequently, $dx(t)/dt$ is the inverse Laplace transform of $sX(s)$. The ROC of $sX(s)$ includes the ROC of $X(s)$ and may be larger if $X(s)$ has a first-order pole at $s = 0$ that is canceled by the multiplication by s. For example, if $x(t) = u(t)$, then $X(s) = 1/s$, with an ROC that is $\mathcal{R}e\{s\} > 0$. The derivative of $x(t)$ is an impulse with an associated Laplace transform that is unity and an ROC that is the entire s-plane.

9.5.8 Differentiation in the s-Domain

Differentiating both sides of the Laplace transform equation (9.3), i.e.,

$$X(s) = \int_{-\infty}^{+\infty} x(t)e^{-st}dt,$$

we obtain

$$\frac{dX(s)}{ds} = \int_{-\infty}^{+\infty} (-t)x(t)e^{-st}dt.$$

Consequently, if

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \qquad \text{with ROC} = R,$$

then

$$\boxed{-tx(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{dX(s)}{ds}, \qquad \text{with ROC} = R.} \tag{9.100}$$

The next two examples illustrate the use of this property.

Example 9.14

Let us find the Laplace transform of

$$x(t) = te^{-at}u(t). \tag{9.101}$$

Since

$$e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \qquad \mathcal{R}e\{s\} > -a,$$

it follows from eq. (9.100) that

$$te^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} -\frac{d}{ds}\left[\frac{1}{s+a}\right] = \frac{1}{(s+a)^2}, \qquad \mathcal{R}e\{s\} > -a. \tag{9.102}$$

In fact, by repeated application of eq. (9.100), we obtain

$$\frac{t^2}{2}e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{(s+a)^3}, \qquad \mathcal{R}e\{s\} > -a, \tag{9.103}$$

and, more generally,

$$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{(s+a)^n}, \qquad \mathcal{R}e\{s\} > -a. \tag{9.104}$$

As the next example illustrates, this specific Laplace transform pair is particularly useful when applying partial-fraction expansion to the determination of the inverse Laplace transform of a rational function with multiple-order poles.

Example 9.15

Consider the Laplace transform

$$X(s) = \frac{2s^2 + 5s + 5}{(s+1)^2(s+2)}, \qquad \mathcal{R}e\{s\} > -1.$$

Applying the partial-fraction expansion method described in the appendix, we can write

$$X(s) = \frac{2}{(s+1)^2} - \frac{1}{(s+1)} + \frac{3}{s+2}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.105}$$

Since the ROC is to the right of the poles at $s = -1$ and -2, the inverse transform of each of the terms is a right-sided signal, and, applying eqs. (9.14) and (9.104), we obtain the inverse transform

$$x(t) = [2te^{-t} - e^{-t} + 3e^{-2t}]u(t).$$

9.5.9 Integration in the Time Domain

If

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s), \qquad \text{with ROC} = R,$$

then

$$\boxed{\int_{-\infty}^{t} x(\tau)d\tau \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}X(s), \text{ with ROC containing } R \cap \{\mathcal{R}e\{s\} > 0\}.} \tag{9.106}$$

This property is the inverse of the differentiation property set forth in Section 9.5.7. It can be derived using the convolution property presented in Section 9.5.6. Specifically,

$$\int_{-\infty}^{t} x(\tau)d\tau = u(t) * x(t). \tag{9.107}$$

From Example 9.1, with $a = 0$,

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}, \qquad \mathcal{R}e\{s\} > 0, \tag{9.108}$$

and thus, from the convolution property,

$$u(t) * x(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}X(s), \tag{9.109}$$

with an ROC that contains the intersection of the ROC of $X(s)$ and the ROC of the Laplace transform of $u(t)$ in eq. (9.108), which results in the ROC given in eq. (9.106).

9.5.10 The Initial- and Final-Value Theorems

Under the specific constraints that $x(t) = 0$ for $t < 0$ and that $x(t)$ contains no impulses or higher order singularities at the origin, one can directly calculate, from the Laplace transform, the initial value $x(0^+)$—i.e., $x(t)$ as t approaches zero from positive values of t. Specifically the *initial-value theorem* states that

$$x(0^+) = \lim_{s \to \infty} sX(s), \tag{9.110}$$

Also, if $x(t) = 0$ for $t < 0$ and, in addition, $x(t)$ has a finite limit as $t \to \infty$, then the *final-value theorem* says that

$$\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s). \tag{9.111}$$

The derivation of these results is considered in Problem 9.53.

Example 9.16

The initial- and final-value theorems can be useful in checking the correctness of the Laplace transform calculations for a signal. For example, consider the signal $x(t)$ in Example 9.4. From eq. (9.24), we see that $x(0^+) = 2$. Also, using eq. (9.29), we find that

$$\lim_{s\to\infty} sX(s) = \lim_{s\to\infty} \frac{2s^3 + 5s^2 + 12s}{s^3 + 4s^2 + 14s + 20} = 2,$$

which is consistent with the initial-value theorem in eq. (9.110).

9.5.11 Table of Properties

In Table 9.1, we summarize the properties developed in this section. In Section 9.7, many of these properties are used in applying the Laplace transform to the analysis and characterization of linear time-invariant systems. As we have illustrated in several examples, the various properties of Laplace transforms and their ROCs can provide us with

TABLE 9.1 PROPERTIES OF THE LAPLACE TRANSFORM

Section	Property	Signal	Laplace Transform	ROC
		$x(t)$	$X(s)$	R
		$x_1(t)$	$X_1(s)$	R_1
		$x_2(t)$	$X_2(s)$	R_2
9.5.1	Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	At least $R_1 \cap R_2$
9.5.2	Time shifting	$x(t - t_0)$	$e^{-st_0}X(s)$	R
9.5.3	Shifting in the s-Domain	$e^{s_0 t}x(t)$	$X(s - s_0)$	Shifted version of R (i.e., s is in the ROC if $s - s_0$ is in R)
9.5.4	Time scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$	Scaled ROC (i.e., s is in the ROC if s/a is in R)
9.5.5	Conjugation	$x^*(t)$	$X^*(s^*)$	R
9.5.6	Convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$	At least $R_1 \cap R_2$
9.5.7	Differentiation in the Time Domain	$\frac{d}{dt}x(t)$	$sX(s)$	At least R
9.5.8	Differentiation in the s-Domain	$-tx(t)$	$\frac{d}{ds}X(s)$	R
9.5.9	Integration in the Time Domain	$\int_{-\infty}^{t} x(\tau)d(\tau)$	$\frac{1}{s}X(s)$	At least $R \cap \{\Re e\{s\} > 0\}$
		Initial- and Final-Value Theorems		
9.5.10	If $x(t) = 0$ for $t < 0$ and $x(t)$ contains no impulses or higher-order singularities at $t = 0$, then $x(0^+) = \lim_{s\to\infty} sX(s)$. If $x(t) = 0$ for $t < 0$ and $x(t)$ has a finite limit as $t \longrightarrow \infty$, then $\lim_{t\to\infty} x(t) = \lim_{s\to\infty} sX(s)$			

considerable information about a signal and its transform that can be useful either in characterizing the signal or in checking a calculation. In Sections 9.7 and 9.8 and in some of the problems at the end of this chapter, we give several other examples of the uses of these properties.

9.6 SOME LAPLACE TRANSFORM PAIRS

As we indicated in Section 9.3, the inverse Laplace transform can often be easily evaluated by decomposing $X(s)$ into a linear combination of simpler terms, the inverse transform of each of which can be recognized. Listed in Table 9.2 are a number of useful Laplace

TABLE 9.2 LAPLACE TRANSFORMS OF ELEMENTARY FUNCTIONS

Transform pair	Signal	Transform	ROC
1	$\delta(t)$	1	All s
2	$u(t)$	$\frac{1}{s}$	$\mathcal{R}e\{s\} > 0$
3	$-u(-t)$	$\frac{1}{s}$	$\mathcal{R}e\{s\} < 0$
4	$\frac{t^{n-1}}{(n-1)!}u(t)$	$\frac{1}{s^n}$	$\mathcal{R}e\{s\} > 0$
5	$-\frac{t^{n-1}}{(n-1)!}u(-t)$	$\frac{1}{s^n}$	$\mathcal{R}e\{s\} < 0$
6	$e^{-\alpha t}u(t)$	$\frac{1}{s+\alpha}$	$\mathcal{R}e\{s\} > -\alpha$
7	$-e^{-\alpha t}u(-t)$	$\frac{1}{s+\alpha}$	$\mathcal{R}e\{s\} < -\alpha$
8	$\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(t)$	$\frac{1}{(s+\alpha)^n}$	$\mathcal{R}e\{s\} > -\alpha$
9	$-\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(-t)$	$\frac{1}{(s+\alpha)^n}$	$\mathcal{R}e\{s\} < -\alpha$
10	$\delta(t-T)$	e^{-sT}	All s
11	$[\cos\omega_0 t]u(t)$	$\frac{s}{s^2+\omega_0^2}$	$\mathcal{R}e\{s\} > 0$
12	$[\sin\omega_0 t]u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\mathcal{R}e\{s\} > 0$
13	$[e^{-\alpha t}\cos\omega_0 t]u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$	$\mathcal{R}e\{s\} > -\alpha$
14	$[e^{-\alpha t}\sin\omega_0 t]u(t)$	$\frac{\omega_0}{(s+\alpha)^2+\omega_0^2}$	$\mathcal{R}e\{s\} > -\alpha$
15	$u_n(t) = \frac{d^n\delta(t)}{dt^n}$	s^n	All s
16	$u_{-n}(t) = \underbrace{u(t) * \cdots * u(t)}_{n \text{ times}}$	$\frac{1}{s^n}$	$\mathcal{R}e\{s\} > 0$

transform pairs. Transform pair 1 follows directly from eq. (9.3). Transform pairs 2 and 6 follow directly from Example 9.1 with $a = 0$ and $a = \alpha$, respectively. Transform pair 4 was developed in Example 9.14 using the differentiation property. Transform pair 8 follows from transform pair 4 using the property set forth in Section 9.5.3. Transform pairs 3, 5, 7, and 9 are based on transform pairs 2, 4, 6 and 8, respectively, together with the time-scaling property of section 9.5.4 with $a = -1$. Similarly, transform pairs 10 through 16 can all be obtained from earlier ones in the table using appropriate properties in Table 9.1 (see Problem 9.55).

9.7 ANALYSIS AND CHARACTERIZATION OF LTI SYSTEMS USING THE LAPLACE TRANSFORM

One of the important applications of the Laplace transform is in the analysis and characterization of LTI systems. Its role for this class of systems stems directly from the convolution property (Section 9.5.6). Specifically, the Laplace transforms of the input and output of an LTI system are related through multiplication by the Laplace transform of the impulse response of the system. Thus,

$$Y(s) = H(s)X(s). \tag{9.112}$$

where $X(s)$, $Y(s)$, and $H(s)$ are the Laplace transforms of the input, output, and impulse response of the system, respectively. Equation (9.112) is the counterpart, in the context of Laplace transforms, of eq. (4.56) for Fourier transform. Also, from our discussion in Section 3.2 on the response of LTI systems to complex exponentials, if the input to an LTI system is $x(t) = e^{st}$, with s in the ROC of $H(s)$, then the output will be $H(s)e^{st}$; i.e., e^{st} is an eigenfunction of the system with eigenvalue equal to the Laplace transform of the impulse response.

If the ROC of $H(s)$ includes the imaginary axis, then for $s = j\omega$, $H(s)$ is the frequency response of the LTI system. In the broader context of the Laplace transform, $H(s)$ is commonly referred to as the system function or, alternatively, the *transfer function*. Many properties of LTI systems can be closely associated with the characteristics of the *system function* in the s-plane. We illustrate this next by examining several important properties and classes of systems.

9.7.1 Causality

For a causal LTI system, the impulse response is zero for $t < 0$ and thus is right sided. Consequently, from the discussion in Section 9.2, we see that

> The ROC associated with the system function for a causal system is a right-half plane.

It should be stressed, however, that the converse of this statement is not necessarily true. That is, as illustrated in Example 9.19 to follow, an ROC to the right of the rightmost

pole does not guarantee that a system is causal; rather, it guarantees only that the impulse response is right sided. However, if $H(s)$ is *rational*, then, as illustrated in Examples 9.17 and 9.18 to follow, we can determine whether the system is causal simply by checking to see if its ROC is a right-half plane. Specifically,

> For a system with a rational system function, causality of the system is equivalent to the ROC being the right-half plane to the right of the rightmost pole.

Example 9.17

Consider a system with impulse response

$$h(t) = e^{-t}u(t). \tag{9.113}$$

Since $h(t) = 0$ for $t < 0$, this system is causal. Also, the system function can be obtained from Example 9.1:

$$H(s) = \frac{1}{s+1}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.114}$$

In this case, the system function is rational and the ROC in eq. (9.114) is to the right of the rightmost pole, consistent with our statement that causality for systems with rational system functions is equivalent to the ROC being to the right of the rightmost pole.

Example 9.18

Consider a system with impulse response

$$h(t) = e^{-|t|}.$$

Since $h(t) \neq 0$ for $t < 0$, this system is not causal. Also, from Example 9.7, the system function is

$$H(s) = \frac{-2}{s^2 - 1}, \qquad -1 < \mathcal{R}e\{s\} < +1.$$

Thus, $H(s)$ is rational and has an ROC that is *not* to the right of the the rightmost pole, consistent with the fact that the system is not causal.

Example 9.19

Consider the system function

$$H(s) = \frac{e^s}{s+1}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.115}$$

For this system, the ROC is to the right of the rightmost pole. Therefore, the impulse response must be right sided. To determine the impulse response, we first use the result

of Example 9.1:

$$e^{-t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+1}, \qquad \mathcal{R}e\{s\} > -1. \tag{9.116}$$

Next, from the time-shifting property of Section 9.5.2 [eq. (9.87)], the factor e^{s} in eq. (9.115) can be accounted for by a shift in the time function in eq. (9.116). Then

$$e^{-(t+1)}u(t+1) \overset{\mathcal{L}}{\longleftrightarrow} \frac{e^{s}}{s+1}, \qquad \mathcal{R}e\{s\} > -1, \tag{9.117}$$

so that the impulse response associated with the system is

$$h(t) = e^{-(t+1)}u(t+1), \tag{9.118}$$

which is nonzero for $-1 < t < 0$. Hence, the system is not causal. This example serves as a reminder that causality implies that the ROC is to the right of the rightmost pole, but the converse is not in general true, unless the system function is rational.

In an exactly analogous manner, we can deal with the concept of anticausality. A system is *anticausal* if its impulse response $h(t) = 0$ for $t > 0$. Since in that case $h(t)$ would be left sided, we know from Section 9.2 that the ROC of the system function $H(s)$ would have to be a left-half plane. Again, in general, the converse is not true. That is, if the ROC of $H(s)$ is a left-half plane, all we know is that $h(t)$ is left sided. However, if $H(s)$ is rational, then having an ROC to the left of the leftmost pole is equivalent to the system being anticausal.

9.7.2 Stability

The ROC of $H(s)$ can also be related to the stability of a system. As mentioned in Section 2.3.7, the stability of an LTI system is equivalent to its impulse response being absolutely integrable, in which case (Section 4.4) the Fourier transform of the impulse response converges. Since the Fourier transform of a signal equals the Laplace transform evaluated along the $j\omega$-axis, we have the following:

> An LTI system is stable if and only if the ROC of its system function H(s) includes the entire $j\omega$-axis [i.e., $\mathcal{R}e\{s\} = 0$].

Example 9.20

Let us consider an LTI system with system function

$$H(s) = \frac{s-1}{(s+1)(s-2)}. \tag{9.119}$$

Since the ROC has not been specified, we know from our discussion in Section 9.2 that there are several different ROCs and, consequently, several different system impulse responses that can be associated with the algebraic expression for $H(s)$ given in eq. (9.119).

If, however, we have information about the causality or stability of the system, the appropriate ROC can be identified. For example, if the system is known to be *causal*, the ROC will be that indicated in Figure 9.25(a), with impulse response

$$h(t) = \left(\frac{2}{3}e^{-t} + \frac{1}{3}e^{2t}\right)u(t). \tag{9.120}$$

Note that this particular choice of ROC does not include the $j\omega$-axis, and consequently, the corresponding system is unstable (as can be checked by observing that $h(t)$ is not absolutely integrable). On the other hand, if the system is known to be *stable*, the ROC is that given in Figure 9.25(b), and the corresponding impulse response is

$$h(t) = \frac{2}{3}e^{-t}u(t) - \frac{1}{3}e^{2t}u(-t),$$

which is absolutely integrable. Finally, for the ROC in Figure 9.25(c), the system is anticausal and unstable, with

$$h(t) = -\left(\frac{2}{3}e^{-t} + \frac{1}{3}e^{2t}\right)u(-t).$$

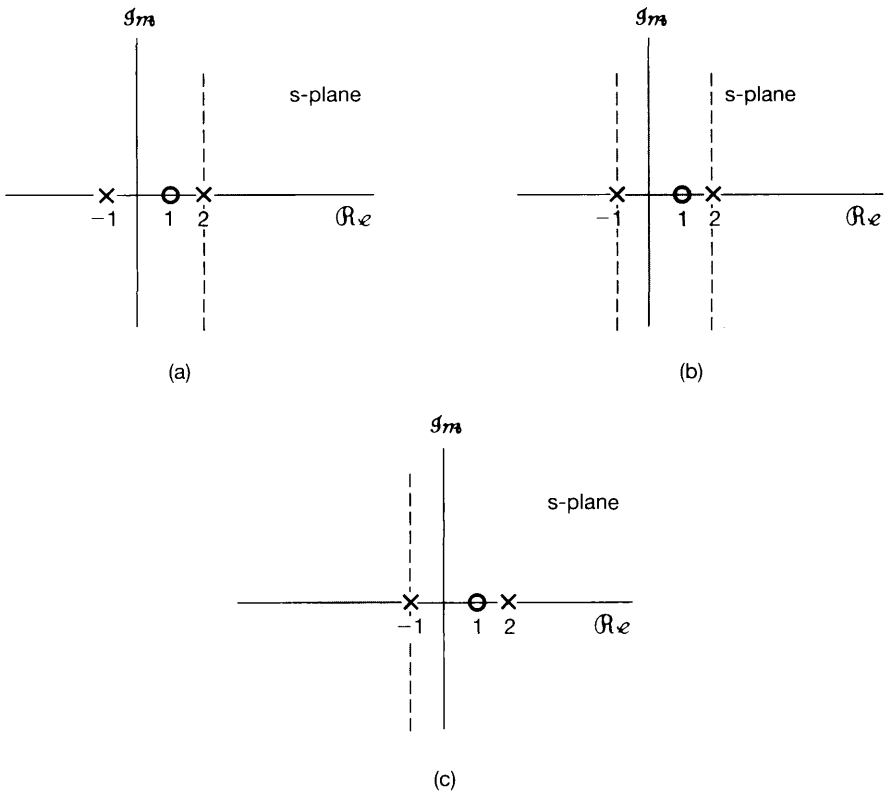


Figure 9.25 Possible ROCs for the system function of Example 9.20 with poles at $s = -1$ and $s = 2$ and a zero at $s = 1$: (a) causal, unstable system; (b) noncausal, stable system; (c) anticausal, unstable system.

It is perfectly possible, of course, for a system to be stable (or unstable) and have a system function that is not rational. For example, the system function in eq. (9.115) is not rational, and its impulse response in eq. (9.118) is absolutely integrable, indicating that the system is stable. However, for systems with rational system functions, stability is easily interpreted in terms of the poles of the system. For example, for the pole-zero plot in Figure 9.25, stability corresponds to the choice of an ROC that is between the two poles, so that the $j\omega$-axis is contained in the ROC.

For one particular and very important class of systems, stability can be characterized very simply in terms of the locations of the poles. Specifically, consider a causal LTI system with a rational system function $H(s)$. Since the system is causal, the ROC is to the right of the rightmost pole. Consequently, for this system to be stable (i.e., for the ROC to include the $j\omega$-axis), the rightmost pole of $H(s)$ must be to the *left* of the $j\omega$-axis. That is,

> A causal system with rational system function $H(s)$ is stable if and only if all of the poles of $H(s)$ lie in the left-half of the s-plane—i.e., all of the poles have negative real parts.

Example 9.21

Consider again the causal system in Example 9.17. The impulse response in eq. (9.113) is absolutely integrable, and thus the system is stable. Consistent with this, we see that the pole of $H(s)$ in eq. (9.114) is at $s = -1$, which is in the left-half of the s-plane. In contrast, the causal system with impulse response

$$h(t) = e^{2t}u(t)$$

is unstable, since $h(t)$ is not absolutely integrable. Also, in this case

$$H(s) = \frac{1}{s-2}, \qquad \mathcal{R}e\{s\} > 2,$$

so the system has a pole at $s = 2$ in the right half of the s-plane.

Example 9.22

Let us consider the class of causal second-order systems previously discussed in Sections 9.4.2 and 6.5.2. The impulse response and system function are, respectively,

$$h(t) = M[e^{c_1 t} - e^{c_2 t}]u(t) \tag{9.121}$$

and

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s-c_1)(s-c_2)}, \tag{9.122}$$

where

$$c_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}, \tag{9.123}$$

$$c_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}, \tag{9.124}$$

$$M = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}}. \tag{9.125}$$

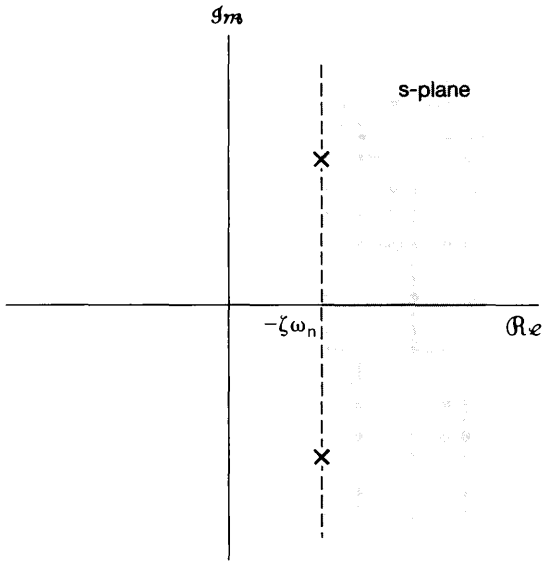


Figure 9.26 Pole locations and ROC for a causal second-order system with $\zeta < 0$.

In Figure 9.19, we illustrated the pole locations for $\zeta > 0$. In Figure 9.26, we illustrate the pole locations for $\zeta < 0$. As is evident from the latter figure and from eqs. (9.124) and (9.125), for $\zeta < 0$ both poles have positive real parts. Consequently, for $\zeta < 0$, the causal second-order system cannot be stable. This is also evident in eq. (9.121), since, with $\mathcal{R}e\{c_1\} > 0$ and $\mathcal{R}e\{c_2\} > 0$, each term grows exponentially as t increases, and thus $h(t)$ cannot be absolutely integrable.

9.7.3 LTI Systems Characterized by Linear Constant-Coefficient Differential Equations

In Section 4.7, we discussed the use of the Fourier transform to obtain the frequency response of an LTI system characterized by a linear constant-coefficient differential equation without first obtaining the impulse response or time-domain solution. In an exactly analogous manner, the properties of the Laplace transform can be exploited to directly obtain the system function for an LTI system characterized by a linear constant-coefficient differential equation. We illustrate this procedure in the next example.

Example 9.23

Consider an LTI system for which the input $x(t)$ and output $y(t)$ satisfy the linear constant-coefficient differential equation

$$\frac{dy(t)}{dt} + 3y(t) = x(t). \tag{9.126}$$

Applying the Laplace transform to both sides of eq. (9.126), and using the linearity and differentiation properties set forth in Sections 9.5.1 and 9.5.7, respectively [(eqs. (9.82) and (9.98)], we obtain the algebraic equation

$$sY(s) + 3Y(s) = X(s). \tag{9.127}$$

Since, from eq. (9.112), the system function is

$$H(s) = \frac{Y(s)}{X(s)},$$

we obtain, for this system,

$$H(s) = \frac{1}{s+3}. \tag{9.128}$$

This, then, provides the algebraic expression for the system function, but not the region of convergence. In fact, as we discussed in Section 2.4, the differential equation itself is not a complete specification of the LTI system, and there are, in general, different impulse responses, all consistent with the differential equation. If, in addition to the differential equation, we know that the system is causal, then the ROC can be inferred to be to the right of the rightmost pole, which in this case corresponds to $\mathcal{R}e\{s\} > -3$. If the system were known to be anticausal, then the ROC associated with $H(s)$ would be $\mathcal{R}e\{s\} < -3$. The corresponding impulse response in the causal case is

$$h(t) = e^{-3t}u(t), \tag{9.129}$$

whereas in the anticausal case it is

$$h(t) = -e^{-3t}u(-t). \tag{9.130}$$

The same procedure used to obtain $H(s)$ from the differential equation in Example 9.23 can be applied more generally. Consider a general linear constant-coefficient differential equation of the form

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}. \tag{9.131}$$

Applying the Laplace transform to both sides and using the linearity and differentiation properties repeatedly, we obtain

$$\left(\sum_{k=0}^{N} a_k s^k\right) Y(s) = \left(\sum_{k=0}^{M} b_k s^k\right) X(s), \tag{9.132}$$

or

$$H(s) = \frac{\left\{\sum_{k=0}^{M} b_k s^k\right\}}{\left\{\sum_{k=0}^{N} a_k s^k\right\}}. \tag{9.133}$$

Thus, the system function for a system specified by a differential equation is always rational, with zeros at the solutions of

$$\sum_{k=0}^{M} b_k s^k = 0 \tag{9.134}$$

and poles at the solutions of

$$\sum_{k=0}^{N} a_k s^k = 0. \tag{9.135}$$

Consistently with our previous discussion, eq. (9.133) does not include a specification of the region of convergence of $H(s)$, since the linear constant-coefficient differential equation by itself does not constrain the region of convergence. However, with additional information, such as knowledge about the stability or causality of the system, the region of convergence can be inferred. For example, if we impose the condition of initial rest on the system, so that it is causal, the ROC will be to the right of the rightmost pole.

Example 9.24

An *RLC* circuit whose capacitor voltage and inductor current are initially zero constitutes an LTI system describable by a linear constant-coefficient differential equation. Consider the series *RLC* circuit in Figure 9.27. Let the voltage across the voltage source be the input signal $x(t)$, and let the voltage measured across the capacitor be the output signal $y(t)$. Equating the sum of the voltages across the resistor, inductor, and capacitor with the source voltage, we obtain

$$RC\frac{dy(t)}{dt} + LC\frac{d^2y(t)}{dt^2} + y(t) = x(t). \tag{9.136}$$

Applying eq. (9.133), we obtain

$$H(s) = \frac{1/LC}{s^2 + (R/L)s + (1/LC)}. \tag{9.137}$$

As shown in Problem 9.64, if the values of R, L, and C are all positive, the poles of this system function will have negative real parts, and consequently, the system will be stable.

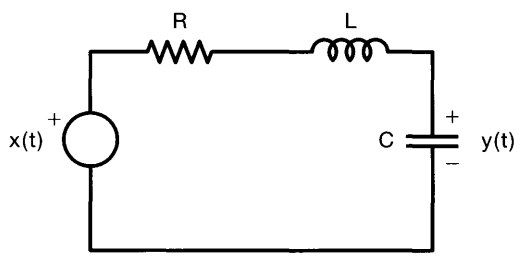


Figure 9.27 A series *RLC* circuit.

9.7.4 Examples Relating System Behavior to the System Function

As we have seen, system properties such as causality and stability can be directly related to the system function and its characteristics. In fact, each of the properties of Laplace transforms that we have described can be used in this way to relate the behavior of the system to the system function. In this section, we give several examples illustrating this.

Example 9.25

Suppose we know that if the input to an LTI system is

$$x(t) = e^{-3t}u(t),$$

then the output is

$$y(t) = [e^{-t} - e^{-2t}]u(t).$$

As we now show, from this knowledge we can determine the system function for this system and from this can immediately deduce a number of other properties of the system.

Taking Laplace transforms of $x(t)$ and $y(t)$, we get

$$X(s) = \frac{1}{s+3}, \qquad \mathcal{R}e\{s\} > -3,$$

and

$$Y(s) = \frac{1}{(s+1)(s+2)}, \qquad \mathcal{R}e\{s\} > -1.$$

From eq. (9.112), we can then conclude that

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s+3}{(s+1)(s+2)} = \frac{s+3}{s^2+3s+2}.$$

Furthermore, we can also determine the ROC for this system. In particular, we know from the convolution property set forth in Section 9.5.6 that the ROC of $Y(s)$ must include at least the intersections of the ROCs of $X(s)$ and $H(s)$. Examining the three possible choices for the ROC of $H(s)$ (i.e., to the left of the pole at $s = -2$, between the poles at -2 and -1, and to the right of the pole at $s = -1$), we see that the only choice that is consistent with the ROCs of $X(s)$ and $Y(s)$ is $\mathcal{R}e\{s\} > -1$. Since this is to the right of the rightmost pole of $H(s)$, we conclude that $H(s)$ is causal, and since both poles of $H(s)$ have negative real parts, it follows that the system is stable. Moreover, from the relationship between eqs. (9.131) and (9.133), we can specify the differential equation that, together with the condition of initial rest, characterizes the system:

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 3x(t).$$

Example 9.26

Suppose that we are given the following information about an LTI system:

1. The system is causal.
2. The system function is rational and has only two poles, at $s = -2$ and $s = 4$.

3. If $x(t) = 1$, then $y(t) = 0$.
4. The value of the impulse response at $t = 0^+$ is 4.

From this information we would like to determine the system function of the system.

From the first two facts, we know that the system is unstable (since it is causal and has a pole at $s = 4$ with positive real part) and that the system function is of the form

$$H(s) = \frac{p(s)}{(s+2)(s-4)} = \frac{p(s)}{s^2 - 2s - 8},$$

where $p(s)$ is a polynomial in s. Because the response $y(t)$ to the input $x(t) = 1 = e^{0 \cdot t}$ must equal $H(0) \cdot e^{0 \cdot t} = H(0)$, we conclude, from fact 3, that $p(0) = 0$—i.e., that $p(s)$ must have a root at $s = 0$ and thus is of the form

$$p(s) = sq(s),$$

where $q(s)$ is another polynomial in s.

Finally, from fact 4 and the initial-value theorem in Section 9.5.10, we see that

$$\lim_{s \to \infty} sH(s) = \lim_{s \to \infty} \frac{s^2 q(s)}{s^2 - 2s - 8} = 4. \tag{9.138}$$

As $s \to \infty$, the terms of highest power in s in both the numerator and the denominator of $sH(s)$ dominate and thus are the only ones of importance in evaluating eq. (9.138). Furthermore, if the numerator has higher degree than the denominator, the limit will diverge. Consequently, we can obtain a finite nonzero value for the limit only if the degree of the numerator of $sH(s)$ is the same as the degree of the denominator. Since the degree of the denominator is 2, we conclude that, for eq. (9.138) to hold, $q(s)$ must be a constant—i.e., $q(s) = K$. We can evaluate this constant by evaluating

$$\lim_{s \to \infty} \frac{Ks^2}{s^2 - 2s - 8} = \lim_{s \to \infty} \frac{Ks^2}{s^2} = K. \tag{9.139}$$

Equating eqs. (9.138) and (9.139), we see that $K = 4$, and thus,

$$H(s) = \frac{4s}{(s+2)(s-4)}.$$

Example 9.27

Consider a stable and causal system with impulse response $h(t)$ and system function $H(s)$. Suppose $H(s)$ is rational, contains a pole at $s = -2$, and does not have a zero at the origin. The location of all other poles and zeros is unknown. For each of the following statements let us determine whether we can definitely say that it is true, whether we can definitely say that it is false, or whether there is insufficient information to ascertain the statement's truth:

(a) $\mathcal{F}\{h(t)e^{3t}\}$ converges.

(b) $\int_{-\infty}^{+\infty} h(t)\,dt = 0$.

(c) $th(t)$ is the impulse response of a causal and stable system.

(d) $dh(t)/dt$ contains at least one pole in its Laplace transform.

(e) $h(t)$ has finite duration.

(f) $H(s) = H(-s)$.

(g) $\lim_{s\to\infty} H(s) = 2$.

Statement (a) is false, since $\mathfrak{F}\{h(t)e^{3t}\}$ corresponds to the value of the Laplace transform of $h(t)$ at $s = -3$. If this converges, it implies that $s = -3$ is in the ROC. A causal and stable system must always have its ROC to the *right* of all of its poles. However, $s = -3$ is not to the right of the pole at $s = -2$.

Statement (b) is false, because it is equivalent to stating that $H(0) = 0$. This contradicts the fact that $H(s)$ does not have a zero at the origin.

Statement (c) is true. According to Table 9.1, the property set forth in Section 9.5.8, the Laplace transform of $th(t)$ has the same ROC as that of $H(s)$. This ROC includes the $j\omega$-axis, and therefore, the corresponding system is stable. Also, $h(t) = 0$ for $t < 0$ implies that $th(t) = 0$ for $t < 0$. Thus, $th(t)$ represents the impulse response of a causal system.

Statement (d) is true. According to Table 9.1, $dh(t)/dt$ has the Laplace transform $sH(s)$. The multiplication by s does not eliminate the pole at $s = -2$.

Statement (e) is false. If $h(t)$ is of finite duration, then if its Laplace transform has any points in its ROC, the ROC must be the entire s-plane. However, this is not consistent with $H(s)$ having a pole at $s = -2$.

Statement (f) is false. If it were true, then, since $H(s)$ has a pole at $s = -2$, it must also have a pole at $s = 2$. This is inconsistent with the fact that all the poles of a causal and stable system must be in the left half of the s-plane.

The truth of statement (g) cannot be ascertained with the information given. The statement requires that the degree of the numerator and denominator of $H(s)$ be equal, and we have insufficient information about $H(s)$ to determine whether this is the case.

9.7.5 Butterworth Filters

In Example 6.3 we briefly introduced the widely-used class of LTI systems known as Butterworth filters. The filters in this class have a number of properties, including the characteristics of the magnitude of the frequency response of each of these filters in the passband, that make them attractive for practical implementation. As a further illustration of the usefulness of Laplace transforms, in this section we use Laplace transform techniques to determine the system function of a Butterworth filter from the specification of its frequency response magnitude.

An Nth-order lowpass Butterworth filter has a frequency response the square of whose magnitude is given by

$$|B(j\omega)|^2 = \frac{1}{1 + (j\omega/j\omega_c)^{2N}}, \tag{9.140}$$

where N is the order of the filter. From eq. (9.140), we would like to determine the system function $B(s)$ that gives rise to $|B(j\omega)|^2$. We first note that, by definition,

$$|B(j\omega)|^2 = B(j\omega)B^*(j\omega). \tag{9.141}$$

If we restrict the impulse response of the Butterworth filter to be real, then from the property of conjugate symmetry for Fourier transforms,

$$B^*(j\omega) = B(-j\omega), \tag{9.142}$$

so that

$$B(j\omega)B(-j\omega) = \frac{1}{1 + (j\omega/j\omega_c)^{2N}}. \tag{9.143}$$

Next, we note that $B(s)|_{s=j\omega} = B(j\omega)$, and consequently, from eq. (9.143),

$$B(s)B(-s) = \frac{1}{1 + (s/j\omega_c)^{2N}}. \tag{9.144}$$

The roots of the denominator polynomial corresponding to the combined poles of $B(s)B(-s)$ are at

$$s = (-1)^{1/2N}(j\omega_c). \tag{9.145}$$

Equation (9.145) is satisfied for any value $s = s_p$ for which

$$|s_p| = \omega_c \tag{9.146}$$

and

$$\measuredangle s_p = \frac{\pi(2k + 1)}{2N} + \frac{\pi}{2}, \qquad k \text{ an integer;} \tag{9.147}$$

that is,

$$s_p = \omega_c \exp\left(j\left[\frac{\pi(2k + 1)}{2N} + \pi/2\right]\right). \tag{9.148}$$

In Figure 9.28 we illustrate the positions of the poles of $B(s)B(-s)$ for $N = 1, 2, 3,$ and 6. In general, the following observations can be made about these poles:

1. There are $2N$ poles equally spaced in angle on a circle of radius ω_c in the s-plane.
2. A pole never lies on the $j\omega$-axis and occurs on the σ-axis for N odd, but not for N even.
3. The angular spacing between the poles of $B(s)B(-s)$ is π/N radians.

To determine the poles of $B(s)$ given the poles of $B(s)B(-s)$, we observe that the poles of $B(s)B(-s)$ occur in pairs, so that if there is a pole at $s = s_p$, then there is also a pole at $s = -s_p$. Consequently, to construct $B(s)$, we choose one pole from each pair. If we restrict the system to be stable and causal, then the poles that we associate with $B(s)$ are the poles along the semicircle in the left-half plane. The pole locations specify $B(s)$ only to within a scale factor. However, from eq. (9.144), we see that $B^2(s)|_{s=0} = 1$, or equivalently, from eq. (9.140), the scale factor is chosen so that the square of the magnitude of the frequency response has unity gain at $\omega = 0$.

To illustrate the determination of $B(s)$, let us consider the cases $N = 1$, $N = 2$, and $N = 3$. In Figure 9.28 we showed the poles of $B(s)B(-s)$, as obtained from eq. (9.148).

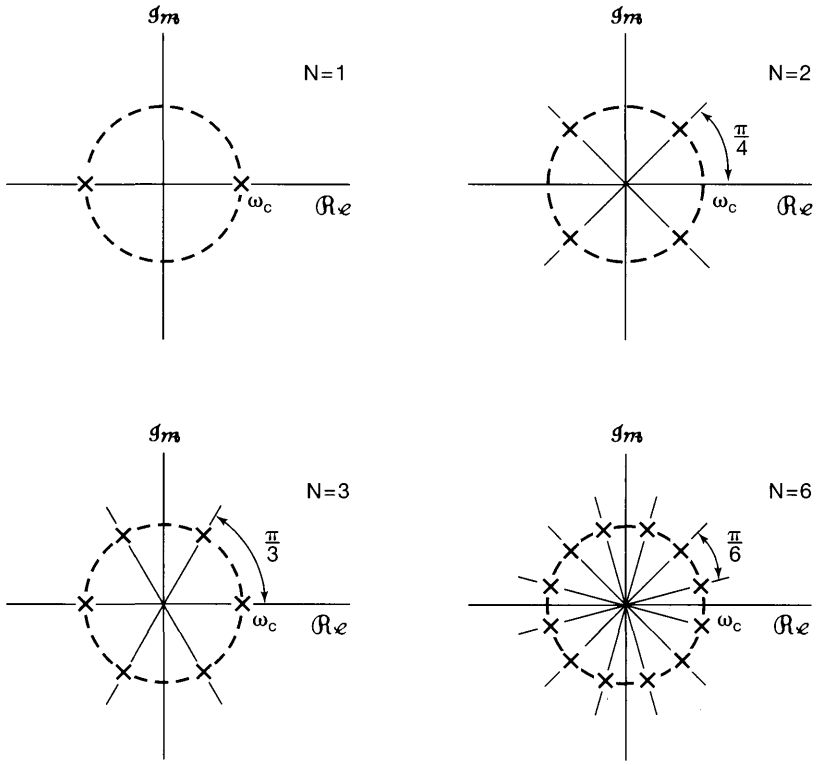


Figure 9.28 Position of the poles of $B(s)B(-s)$ for $N = 1, 2, 3$, and 6.

In Figure 9.29 we show the poles associated with $B(s)$ for each of these values of N. The corresponding transfer functions are:

$$N = 1: \qquad B(s) = \frac{\omega_c}{s + \omega_c}; \tag{9.149}$$

$$N = 2: \qquad B(s) = \frac{\omega_c^2}{(s + \omega_c e^{j(\pi/4)})(s + \omega_c e^{-j(\pi/4)})}$$

$$= \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}; \tag{9.150}$$

$$N = 3: \qquad B(s) = \frac{\omega_c^3}{(s + \omega_c)(s + \omega_c e^{j(\pi/3)})(s + \omega_c e^{-j(\pi/3)})}$$

$$= \frac{\omega_c^3}{(s + \omega_c)(s^2 + \omega_c s + \omega_c^2)} \tag{9.151}$$

$$= \frac{\omega_c^3}{s^3 + 2\omega_c s^2 + 2\omega_c^2 s + \omega_c^3}.$$

Based on the discussion in Section 9.7.3, from $B(s)$ we can determine the associated linear constant-coefficient differential equation. Specifically, for the foregoing three values

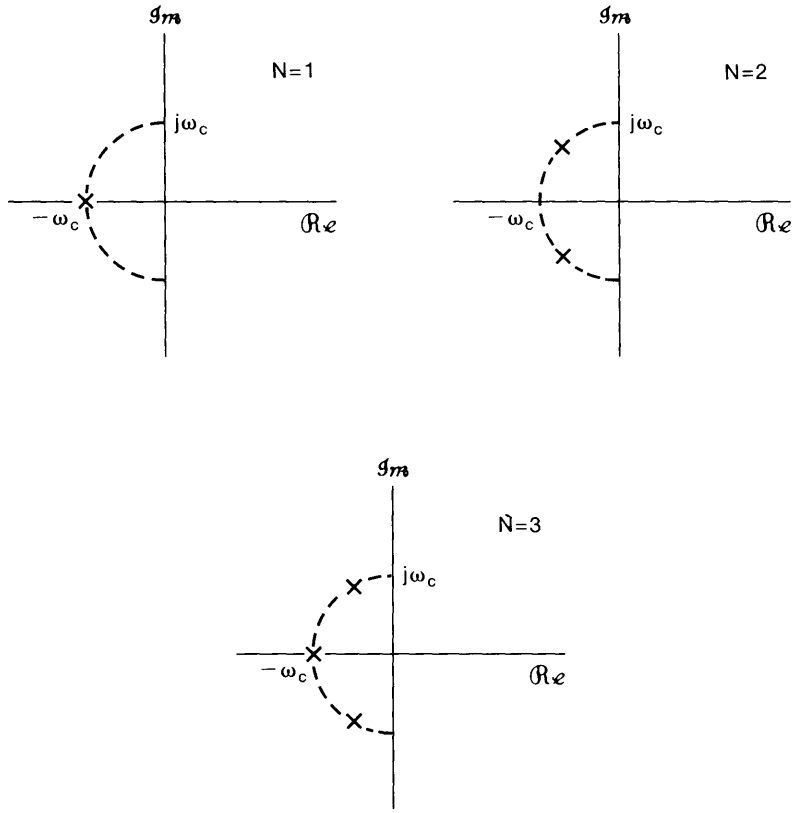


Figure 9.29 Position of the poles of $B(s)$ for $N = 1$, 2, and 3.

of N, the corresponding differential equations are:

$$N = 1: \quad \frac{dy(t)}{dt} + \omega_c y(t) = \omega_c x(t); \tag{9.152}$$

$$N = 2: \quad \frac{d^2y(t)}{dt^2} + \sqrt{2}\omega_c \frac{dy(t)}{dt} + \omega_c^2 y(t) = \omega_c^2 x(t); \tag{9.153}$$

$$N = 3: \quad \frac{d^3y(t)}{dt^3} + 2\omega_c \frac{d^2y(t)}{dt^2} + 2\omega_c^2 \frac{dy(t)}{dt} + \omega_c^3 y(t) = \omega_c^3 x(t). \tag{9.154}$$

9.8 SYSTEM FUNCTION ALGEBRA AND BLOCK DIAGRAM REPRESENTATIONS

The use of the Laplace transform allows us to replace time-domain operations such as differentiation, convolution, time shifting, and so on, with algebraic operations. We have already seen many of the benefits of this in terms of analyzing LTI systems, and in this section we take a look at another important use of system function algebra, namely, in analyzing interconnections of LTI systems and synthesizing systems as interconnections of elementary system building blocks.

9.8.1 System Functions for Interconnections of LTI Systems

Consider the parallel interconnection of two systems, as shown in Figure 9.30(a). The impulse response of the overall system is

$$h(t) = h_1(t) + h_2(t), \tag{9.155}$$

and from the linearity of the Laplace transform,

$$H(s) = H_1(s) + H_2(s). \tag{9.156}$$

Similarly, the impulse response of the series interconnection in Figure 9.30(b) is

$$h(t) = h_1(t) * h_2(t), \tag{9.157}$$

and the associated system function is

$$H(s) = H_1(s)H_2(s). \tag{9.158}$$

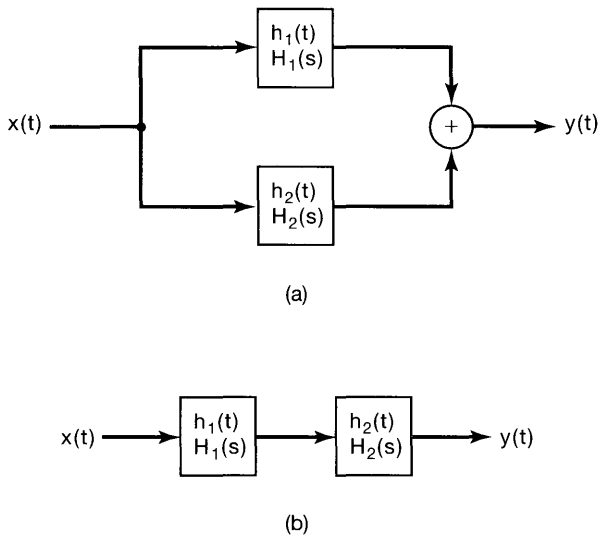


Figure 9.30 (a) Parallel interconnection of two LTI systems; (b) series combination of two LTI systems.

The utility of the Laplace transform in representing combinations of linear systems through algebraic operations extends to far more complex interconnections than the simple parallel and series combinations in Figure 9.30. To illustrate this, consider the feedback interconnection of two systems, as indicated in Figure 9.31. The design, applications, and analysis of such interconnections are treated in detail in Chapter 11. While analysis of the system in the time domain is not particularly simple, determining the overall system function from input $x(t)$ to output $y(t)$ is a straightforward algebraic manipulation. Specifically, from Figure 9.31,

$$Y(s) = H_1(s)E(s), \tag{9.159}$$

$$E(s) = X(s) - Z(s), \tag{9.160}$$

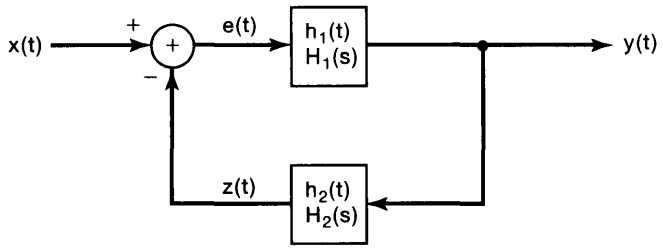


Figure 9.31 Feedback interconnection of two LTI systems.

and

$$Z(s) = H_2(s)Y(s), \tag{9.161}$$

from which we obtain the relation

$$Y(s) = H_1(s)[X(s) - H_2(s)Y(s)], \tag{9.162}$$

or

$$\frac{Y(s)}{X(s)} = H(s) = \frac{H_1(s)}{1 + H_1(s)H_2(s)}. \tag{9.163}$$

9.8.2 Block Diagram Representations for Causal LTI Systems Described by Differential Equations and Rational System Functions

In Section 2.4.3, we illustrated the block diagram representation of an LTI system described by a first-order differential equation using the basic operations of addition, multiplication by a coefficient, and integration. These same three operations can also be used to build block diagrams for higher order systems, and in this section we illustrate this in several examples.

Example 9.28

Consider the causal LTI system with system function

$$H(s) = \frac{1}{s+3}.$$

From Section 9.7.3, we know that this system can also be described by the differential equation

$$\frac{dy(t)}{dt} + 3y(t) = x(t),$$

together with the condition of initial rest. In Section 2.4.3 we constructed a block diagram representation, shown in Figure 2.32, for a first-order system such as this. An equivalent block diagram (corresponding to Figure 2.32 with $a = 3$ and $b = 1$) is shown in

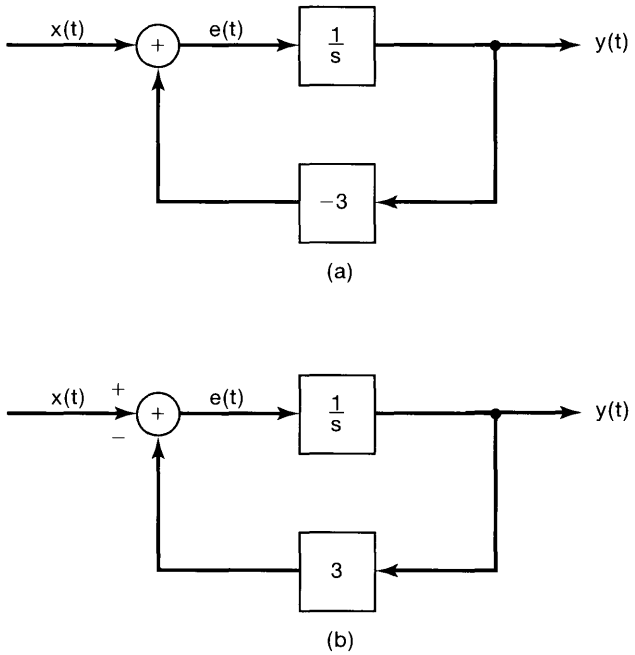


Figure 9.32 (a) Block diagram representation of the causal LTI system in Example 9.28; (b) equivalent block diagram representation.

Figure 9.32(a). Here, $1/s$ is the system function of a system with impulse response $u(t)$, i.e., it is the system function of an integrator. Also, the system function -3 in the feedback path in Figure 9.32(a) corresponds to multiplication by the coefficient -3. The block diagram in the figure involves a feedback loop much as we considered in the previous subsection and as pictured in Figure 9.31, the sole difference being that the two signals that are the inputs to the adder in Figure 9.32(a) are added, rather than subtracted as in Figure 9.31. However, as illustrated in Figure 9.32(b), by changing the sign of the coefficient in the multiplication in the feedback path, we obtain a block diagram representation of exactly the same form as Figure 9.31. Consequently, we can apply eq. (9.163) to verify that

$$H(s) = \frac{1/s}{1 + 3/s} = \frac{1}{s+3}.$$

Example 9.29

Consider now the causal LTI system with system function

$$H(s) = \frac{s+2}{s+3} = \left(\frac{1}{s+3}\right)(s+2). \tag{9.164}$$

As suggested by eq. (9.164), this system can be thought of as a cascade of a system with system function $1/(s+3)$ followed by a system with system function $s+2$, and

we have illustrated this in Figure 9.33(a), in which we have used the block diagram in Figure 9.32(a) to represent $1/(s + 3)$.

It is also possible to obtain an alternative block diagram representation for the system in eq. (9.164). Using the linearity and differentiation properties of the Laplace transform, we know that $y(t)$ and $z(t)$ in Figure 9.33 (a) are related by

$$y(t) = \frac{dz(t)}{dt} + 2z(t).$$

However, the input $e(t)$ to the integrator is exactly the derivative of the output $z(t)$, so that

$$y(t) = e(t) + 2z(t),$$

which leads directly to the alternative block diagram representation shown in Figure 9.33(b). Note that the block diagram in Figure 9.33(a) requires the differentiation of $z(t)$, since

$$y(t) = \frac{dz(t)}{dt} + 2z(t)$$

In contrast, the block diagram in Figure 9.33(b) does not involve the explicit differentiation of any signal.

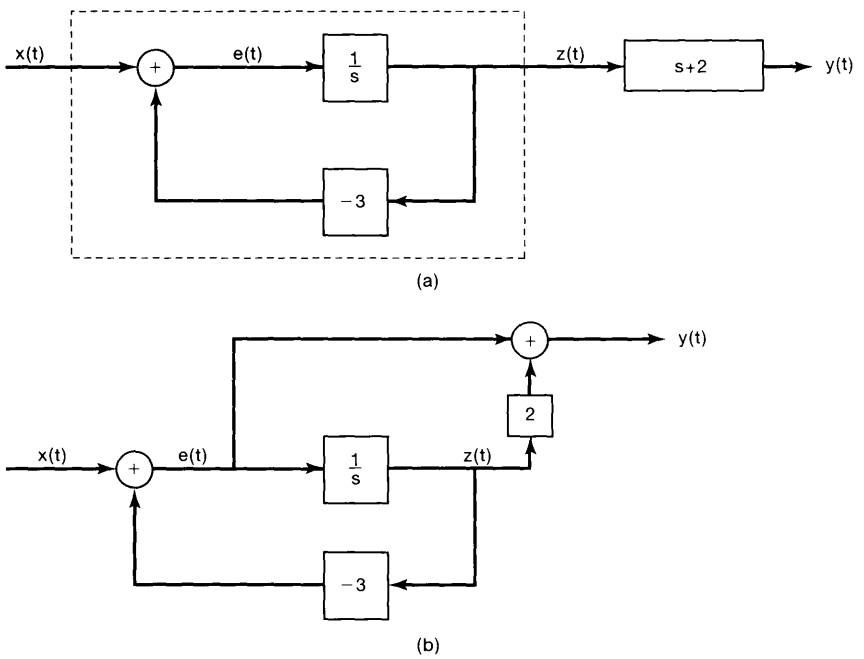


Figure 9.33 (a) Block diagram representations for the system in Example 9.29; (b) equivalent block diagram representation.

Example 9.30

Consider next a causal second-order system with system function

$$H(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s^2+3s+2}. \tag{9.165}$$

The input $x(t)$ and output $y(t)$ for this system satisfy the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = x(t). \tag{9.166}$$

By employing similar ideas to those used in the preceding examples, we obtain the block diagram representation for this system shown in Figure 9.34(a). Specifically, since the

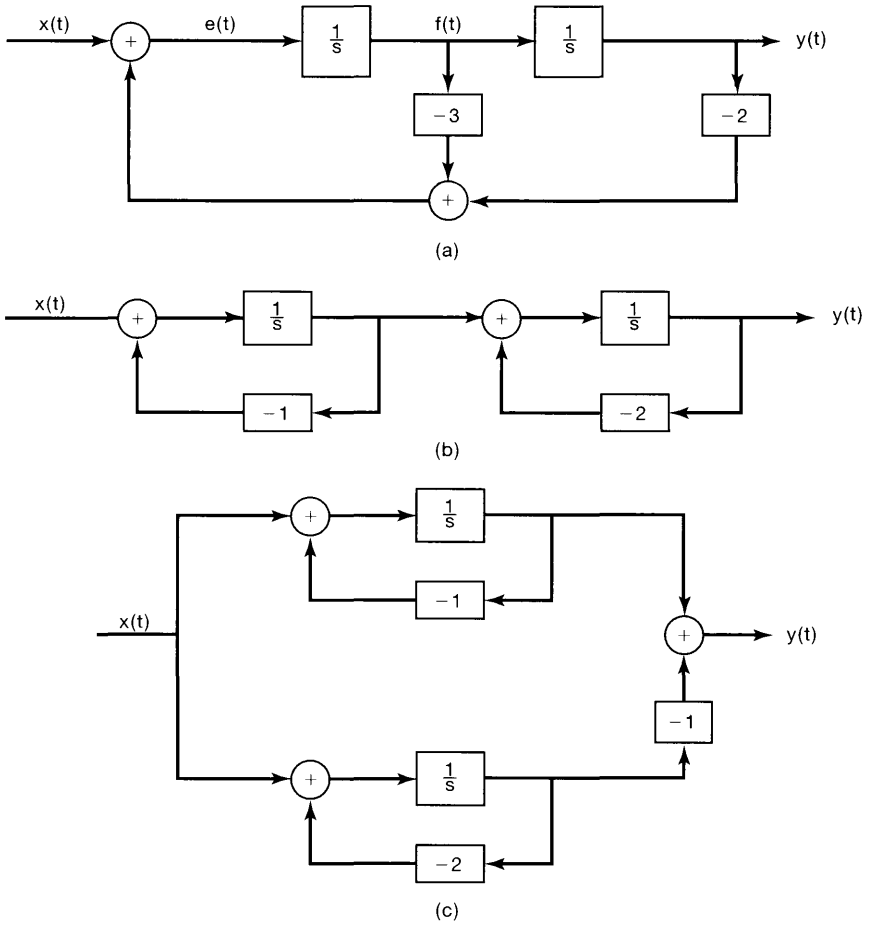


Figure 9.34 Block diagram representations for the system in Example 9.30: (a) direct form; (b) cascade form; (c) parallel form.

input to an integrator is the derivative of the output of the integrator, the signals in the block diagram are related by

$$f(t) = \frac{dy(t)}{dt},$$
$$e(t) = \frac{df(t)}{dt} = \frac{d^2y(t)}{dt}.$$

Also, eq. (9.166) can be rewritten as

$$\frac{d^2y(t)}{dt^2} = -3\frac{dy(t)}{dt} - 2y(t) + x(t),$$

or

$$e(t) = -3f(t) - 2y(t) + x(t),$$

which is exactly what is represented in Figure 9.34(a).

The block diagram in this figure is sometimes referred to as a *direct-form* representation, since the coefficients appearing in the diagram can be directly identified with the coefficients appearing in the system function or, equivalently, the differential equation. Other block diagram representations of practical importance also can be obtained after a modest amount of system function algebra. Specifically, $H(s)$ in eq. (9.165) can be rewritten as

$$H(s) = \left(\frac{1}{s+1}\right)\left(\frac{1}{s+2}\right),$$

which suggests that this system can be represented as the cascade of two first-order systems. The *cascade-form* representation corresponding to $H(s)$ is shown in Figure 9.34(b).

Alternatively, by performing a partial-fraction expansion of $H(s)$, we obtain

$$H(s) = \frac{1}{s+1} - \frac{1}{s+2},$$

which leads to the *parallel-form* representation depicted in Figure 9.34(c).

Example 9.31

As a final example, consider the system function

$$H(s) = \frac{2s^2 + 4s - 6}{s^2 + 3s + 2}. \tag{9.167}$$

Once again, using system function algebra, we can write $H(s)$ in several different forms, each of which suggests a block diagram representation. In particular, we can write

$$H(s) = \left(\frac{1}{s^2 + 3s + 2}\right)(2s^2 + 4s - 6),$$

which suggests the representation of $H(s)$ as the cascade of the system depicted in Figure 9.34(a) and the system with system function $2s^2 + 4s - 6$. However, exactly as we

did in Example 9.29, we can extract the derivatives required for this second system by "tapping" the signals appearing as the inputs to the integrators in the first system. The details of this construction are examined in Problem 9.36, and the result is the direct-form block diagram shown in Figure 9.35. Once again, in the direct-form representation the coefficients appearing in the block diagram can be determined by inspection from the coefficients in the system function in eq. (9.167).

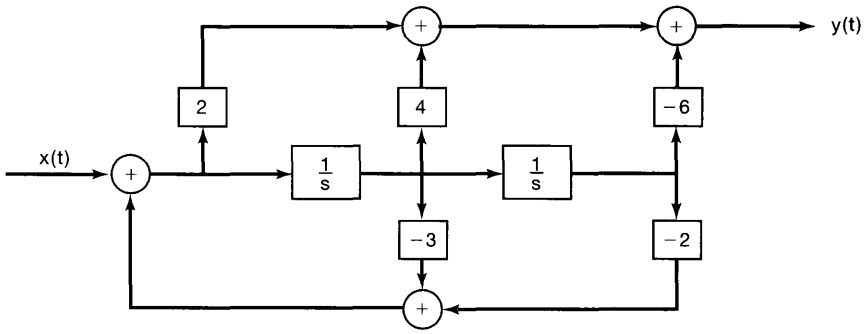


Figure 9.35 Direct-form representation for the system in Example 9.31.

Alternatively, we can write $H(s)$ in the form

$$H(s) = \left(\frac{2(s-1)}{s+2}\right)\left(\frac{s+3}{s+1}\right) \tag{9.168}$$

or

$$H(s) = 2 + \frac{6}{s+2} - \frac{8}{s+1}. \tag{9.169}$$

The first of these suggests a cascade-form representation, while the second leads to a parallel-form block diagram. These are also considered in Problem 9.36.

The methods for constructing block diagram representations for causal LTI systems described by differential equations and rational system functions can be applied equally well to higher order systems. In addition, there is often considerable flexibility in how this is done. For example, by reversing the numerators in eq. (9.168), we can write

$$H(s) = \left(\frac{s+3}{s+2}\right)\left(\frac{2(s-1)}{s+2}\right),$$

which suggests a different cascade form. Also, as illustrated in Problem 9.38, a fourth-order system function can be written as the product of two second-order system functions, each of which can be represented in a number of ways (e.g., direct form, cascade, or parallel), and it can also be written as the sum of lower order terms, each of which has several different representations. In this way, simple low-order systems can serve as building blocks for the implementation of more complex, higher order systems.

9.9 THE UNILATERAL LAPLACE TRANSFORM

In the preceding sections of this chapter, we have dealt with what is commonly called the bilateral Laplace transform. In this section, we introduce and examine a somewhat different transform, the *unilateral Laplace transform*, which is of considerable value in analyzing causal systems and, particularly, systems specified by linear constant-coefficient differential equations with nonzero initial conditions (i.e., systems that are not initially at rest).

The unilateral Laplace transform of a continuous-time signal $x(t)$ is defined as

$$\mathfrak{X}(s) \triangleq \int_{0^-}^{\infty} x(t)e^{-st}\,dt, \tag{9.170}$$

where the lower limit of integration, 0^-, signifies that we include in the interval of integration any impulses or higher order singularity functions concentrated at $t = 0$. Once again we adopt a convenient shorthand notation for a signal and its unilateral Laplace transform:

$$x(t) \overset{\mathcal{UL}}{\longleftrightarrow} \mathfrak{X}(s) = \mathcal{UL}\{x(t)\}. \tag{9.171}$$

Comparing eqs. (9.170) and (9.3), we see that the difference in the definitions of the unilateral and bilateral Laplace transform lies in the lower limit on the integral. The bilateral transform depends on the entire signal from $t = -\infty$ to $t = +\infty$, whereas the unilateral transform depends only on the signal from $t = 0^-$ to ∞. Consequently, two signals that differ for $t < 0$, but that are identical for $t \geq 0$, will have different bilateral Laplace transforms, but identical unilateral transforms. Similarly, any signal that is identically zero for $t < 0$ has identical bilateral and unilateral transforms.

Since the unilateral transform of $x(t)$ is identical to the bilateral transform of the signal obtained from $x(t)$ by setting its value to 0 for all $t < 0$, many of the insights, concepts, and results pertaining to bilateral transforms can be directly adapted to the unilateral case. For example, using Property 4 in Section 9.2 for right-sided signals, we see that the ROC for eq. (9.170) is always a right-half plane. The evaluation of the inverse unilateral Laplace transforms is also the same as for bilateral transforms, with the constraint that the ROC for a unilateral transform must always be a right-half plane.

9.9.1 Examples of Unilateral Laplace Transforms

To illustrate the unilateral Laplace transform, let us consider the following examples:

Example 9.32

Consider the signal

$$x(t) = \frac{t^{n-1}}{(n-1)!}e^{-at}u(t). \tag{9.172}$$

Since $x(t) = 0$ for $t < 0$, the unilateral and bilateral transforms are identical. Thus, from Table 9.2,

$$\mathfrak{X}(s) = \frac{1}{(s+a)^n}, \qquad \mathcal{R}e\{s\} > -a. \tag{9.173}$$

Example 9.33

Consider next

$$x(t) = e^{-a(t+1)}u(t+1). \tag{9.174}$$

The *bilateral* transform $X(s)$ for this example can be obtained from Example 9.1 and the time-shifting property (Section 9.5.2):

$$X(s) = \frac{e^s}{s+a}, \qquad \mathcal{R}e\{s\} > -a. \tag{9.175}$$

By contrast, the unilateral transform is

$$\begin{aligned}\mathfrak{X}(s) &= \int_{0^-}^{\infty} e^{-a(t+1)}u(t+1)e^{-st}\,dt \\ &= \int_{0^-}^{\infty} e^{-a}e^{-t(s+a)}\,dt \\ &= e^{-a}\frac{1}{s+a}, \qquad \mathcal{R}e\{s\} > -a.\end{aligned} \tag{9.176}$$

Thus, in this example, the unilateral and bilateral Laplace transforms are clearly different. In fact, we should recognize $\mathfrak{X}(s)$ as the bilateral transform not of $x(t)$, but of $x(t)u(t)$, consistent with our earlier comment that the unilateral transform is the bilateral transform of a signal whose values for $t < 0^-$ have been set to zero.

Example 9.34

Consider the signal

$$x(t) = \delta(t) + 2u_1(t) + e^t u(t). \tag{9.177}$$

Since $x(t) = 0$ for $t < 0$, and since singularities at the origin are included in the interval of integration, the unilateral transform for $x(t)$ is the same as the bilateral transform. Specifically, using the fact (transform pair 15 in Table 9.2) that the bilateral transform of $u_n(t)$ is s^n, we have

$$\mathfrak{X}(s) = X(s) = 1 + 2s + \frac{1}{s-1} = \frac{s(2s-1)}{s-1}, \qquad \mathcal{R}e\{s\} > 1. \tag{9.178}$$

Example 9.35

Consider the unilateral Laplace transform

$$\mathfrak{X}(s) = \frac{1}{(s+1)(s+2)}. \tag{9.179}$$

In Example 9.9, we considered the inverse transform for a bilateral Laplace transform of the exact form as that in eq. (9.179) and for several ROCs. For the unilateral transform, the ROC must be the right-half plane to the right of the rightmost pole of $\mathcal{X}(s)$; i.e., in this case, the ROC consists of all points s with $\mathcal{R}e\{s\} > -1$. We can then invert this unilateral transform exactly as in Example 9.9 to obtain

$$x(t) = [e^{-t} - e^{-2t}]u(t) \quad \text{for} \quad t > 0^-, \tag{9.180}$$

where we have emphasized the fact that unilateral Laplace transforms provide us with information about signals only for $t > 0^-$.

Example 9.36

Consider the unilateral transform

$$\mathcal{X}(s) = \frac{s^2 - 3}{s + 2}. \tag{9.181}$$

Since the degree of the numerator of $\mathcal{X}(s)$ is not strictly less than the degree of the denominator, we expand $\mathcal{X}(s)$ as

$$\mathcal{X}(s) = A + Bs + \frac{C}{s + 2}. \tag{9.182}$$

Equating eqs. (9.181) and (9.182), and clearing denominators, we obtain

$$s^2 - 3 = (A + Bs)(s + 2) + C, \tag{9.183}$$

and equating coefficients for each power of s yields

$$\mathcal{X}(s) = -2 + s + \frac{1}{s + 2}, \tag{9.184}$$

with an ROC of $\mathcal{R}e\{s\} > -2$. Taking inverse transforms of each term results in

$$x(t) = -2\delta(t) + u_1(t) + e^{-2t}u(t) \quad \text{for} \quad t > 0^-. \tag{9.185}$$

9.9.2 Properties of the Unilateral Laplace Transform

As with the bilateral Laplace transform, the unilateral Laplace transform has a number of important properties, many of which are the same as their bilateral counterparts and several of which differ in significant ways. Table 9.3 summarizes these properties. Note that we have not included a column explicitly identifying the ROC for the unilateral Laplace transform for each signal, since the ROC of any unilateral Laplace transform is always a right-half plane. For example the ROC for a rational unilateral Laplace transform is always to the right of the rightmost pole.

Contrasting Table 9.3 with Table 9.1 for the bilateral transform, we see that, with the caveat that ROCs for unilateral Laplace transforms are always right-half planes, the linearity, s-domain shifting, time-scaling, conjugation and differentiation in the s-domain

TABLE 9.3 PROPERTIES OF THE UNILATERAL LAPLACE TRANSFORM

Property	Signal	Unilateral Laplace Transform
	$x(t)$ $x_1(t)$ $x_2(t)$	$\mathfrak{X}(s)$ $\mathfrak{X}_1(s)$ $\mathfrak{X}_2(s)$
Linearity	$ax_1(t) + bx_2(t)$	$a\mathfrak{X}_1(s) + b\mathfrak{X}_2(s)$
Shifting in the s-domain	$e^{s_0 t}x(t)$	$\mathfrak{X}(s - s_0)$
Time scaling	$x(at), \quad a > 0$	$\frac{1}{a}\mathfrak{X}\left(\frac{s}{a}\right)$
Conjugation	$x^*(t)$	$x^*(s)$
Convolution (assuming that $x_1(t)$ and $x_2(t)$ are identically zero for $t < 0$)	$x_1(t) * x_2(t)$	$\mathfrak{X}_1(s)\mathfrak{X}_2(s)$
Differentiation in the time domain	$\frac{d}{dt}x(t)$	$s\mathfrak{X}(s) - x(0^-)$
Differentiation in the s-domain	$-tx(t)$	$\frac{d}{ds}\mathfrak{X}(s)$
Integration in the time domain	$\int_{0^-}^{t} x(\tau)\,d\tau$	$\frac{1}{s}\mathfrak{X}(s)$
Initial- and Final-Value Theorems If $x(t)$ contains no impulses or higher-order singularities at $t = 0$, then $x(0^+) = \lim_{s\to\infty} s\mathfrak{X}(s)$ $\lim_{t\to\infty} x(t) = \lim_{s\to 0} s\mathfrak{X}(s)$		

properties are identical to their bilateral counterparts. Similarly, the initial- and final-value theorems stated in Section 9.5.10 also hold for unilateral Laplace transforms.[3] The derivation of each of these properties is identical to that of its bilateral counterpart.

The convolution property for unilateral transforms also is quite similar to the corresponding property for bilateral transforms. This property states that if

$$x_1(t) = x_2(t) = 0 \quad \text{for all} \quad t < 0, \tag{9.186}$$

[3]In fact, the initial- and final-value theorems are basically unilateral transform properties, as they apply only to signals $x(t)$ that are identically 0 for $t < 0$.

then

$$x_1(t) * x_2(t) \overset{\mathcal{UL}}{\longleftrightarrow} \mathfrak{X}_1(s)\mathfrak{X}_2(s). \tag{9.187}$$

Equation (9.187) follows immediately from the bilateral convolution property, since, under the conditions of eq. (9.186), the unilateral and bilateral transforms are identical for each of the signals $x_1(t)$ and $x_2(t)$. Thus, the system analysis tools and system function algebra developed and used in this chapter apply without change to unilateral transforms, as long as we deal with causal LTI systems (for which the system function is *both* the bilateral *and* the unilateral transform of the impulse response) with inputs that are identically zero for $t < 0$. An example of this is the integration property in Table 9.3: If $x(t) = 0$ for $t < 0$, then

$$\int_{0^-}^{t} x(\tau)\,d\tau = x(t) * u(t) \overset{\mathcal{UL}}{\longleftrightarrow} \mathfrak{X}(s)\mathcal{U}(s) = \frac{1}{s}\mathfrak{X}(s) \tag{9.188}$$

As a second case in point, consider the following example:

Example 9.37

Suppose a causal LTI system is described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = x(t), \tag{9.189}$$

together with the condition of initial rest. Using eq. (9.133), we find that the system function for this system is

$$\mathcal{H}(s) = \frac{1}{s^2 + 3s + 2}. \tag{9.190}$$

Let the input to this system be $x(t) = \alpha u(t)$. In this case, the unilateral (and bilateral) Laplace transform of the output $y(t)$ is

$$\begin{aligned} \mathcal{Y}(s) &= \mathcal{H}(s)\mathfrak{X}(s) = \frac{\alpha}{s(s+1)(s+2)} \\ &= \frac{\alpha/2}{s} - \frac{\alpha}{s+1} + \frac{\alpha/2}{s+2}. \end{aligned} \tag{9.191}$$

Applying Example 9.32 to each term of eq. (9.191) yields

$$y(t) = \alpha\left[\frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}\right]u(t). \tag{9.192}$$

It is important to note that the convolution property for unilateral Laplace transforms applies only if the signals $x_1(t)$ and $x_2(t)$ in eq. (9.187) are both zero for $t < 0$. That is, while we have seen that the bilateral Laplace transform of $x_1(t) * x_2(t)$ always equals the product of the bilateral transforms of $x_1(t)$ and $x_2(t)$, the unilateral transform of $x_1(t) * x_2(t)$ in general does *not* equal the product of the unilateral transforms if either $x_1(t)$ or $x_2(t)$ is nonzero for $t < 0$. (See, for example, Problem 9.39).

A particularly important difference between the properties of the unilateral and bilateral transforms is the differentiation property. Consider a signal $x(t)$ with unilateral Laplace transform $\mathfrak{X}(s)$. Then, integrating by parts, we find that the unilateral transform of $dx(t)/dt$ is given by

$$\begin{aligned}\int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st}\,dt &= x(t)e^{-st}\Big|_{0^-}^{\infty} + s\int_{0^-}^{\infty} x(t)e^{-st}\,dt \\ &= s\mathfrak{X}(s) - x(0^-).\end{aligned} \tag{9.193}$$

Similarly, a second application of this would yield the unilateral Laplace transform of $d^2x(t)/dt^2$, i.e.,

$$s^2\mathfrak{X}(s) - sx(0^-) - x'(0^-), \tag{9.194}$$

where $x'(0^-)$ denotes the derivative of $x(t)$ evaluated at $t = 0^-$. Clearly, we can continue the procedure to obtain the unilateral transform of higher derivatives.

9.9.3 Solving Differential Equations Using the Unilateral Laplace Transform

A primary use of the unilateral Laplace transform is in obtaining the solution of linear constant-coefficient differential equations with nonzero initial conditions. We illustrate this in the following example:

Example 9.38

Consider the system characterized by the differential equation (9.189) with initial conditions

$$y(0^-) = \beta, \qquad y'(0^-) = \gamma. \tag{9.195}$$

Let $x(t) = \alpha u(t)$. Then, applying the unilateral transform to both sides of eq. (9.189), we obtain

$$s^2\mathcal{Y}(s) - \beta s - \gamma + 3s\mathcal{Y}(s) - 3\beta + 2\mathcal{Y}(s) = \frac{\alpha}{s}, \tag{9.196}$$

or

$$\mathcal{Y}(s) = \frac{\beta(s+3)}{(s+1)(s+2)} + \frac{\gamma}{(s+1)(s+2)} + \frac{\alpha}{s(s+1)(s+2)}, \tag{9.197}$$

where $\mathcal{Y}(s)$ is the unilateral Laplace transform of $y(t)$.

Referring to Example 9.37 and, in particular, to eq. (9.191), we see that the last term on the right-hand side of eq. (9.197) is precisely the unilateral Laplace transform of the response of the system when the initial conditions in eq. (9.195) are both zero ($\beta = \gamma = 0$). That is, the last term represents the response of the causal LTI system described by eq. (9.189) and the condition of initial rest. This response is often referred

to as the *zero-state response*—i.e., the response when the initial state (the set of initial conditions in eq. (9.195)) is zero.

An analogous interpretation applies to the first two terms on the right-hand side of eq. (9.197). These terms represent the unilateral transform of the response of the system when the input is zero ($\alpha = 0$). This response is commonly referred to as the *zero-input response*. Note that the zero-input response is a linear function of the values of the initial conditions (e.g., doubling the values of both β and γ doubles the zero-input response). Furthermore, eq. (9.197) illustrates an important fact about the solution of linear constant-coefficient differential equations with nonzero initial conditions, namely, that the overall response is simply the superposition of the zero-state and the zero-input responses. The zero-state response is the response obtained by setting the initial conditions to zero—i.e., it is the response of an LTI system defined by the differential equation and the condition of initial rest. The zero-input response is the response to the initial conditions with the input set to zero. Other examples illustrating this can be found in Problems 9.20, 9.40, and 9.66.

Finally, for any values of α, β, and γ, we can, of course, expand $\mathcal{Y}(s)$ in eq. (9.197) in a partial-fraction expansion and invert to obtain $y(t)$. For example, if $\alpha = 2$, $\beta = 3$, and $\gamma = -5$, then performing a partial-fraction expansion for eq. (9.197) we find that

$$\mathcal{Y}(s) = \frac{1}{s} - \frac{1}{s+1} + \frac{3}{s+2}. \tag{9.198}$$

Applying Example 9.32 to each term then yields

$$y(t) = [1 - e^{-t} + 3e^{-2t}]u(t) \quad \text{for} \quad t > 0. \tag{9.199}$$

9.10 SUMMARY

In this chapter, we have developed and studied the Laplace transform, which can be viewed as a generalization of the Fourier transform. It is particularly useful as an analytical tool in the analysis and study of LTI systems. Because of the properties of Laplace transforms, LTI systems, including those represented by linear constant-coefficient differential equations, can be characterized and analyzed in the transform domain by algebraic manipulations. In addition, system function algebra provides a convenient tool both for analyzing interconnections of LTI systems and for constructing block diagram representations of LTI systems described by differential equations.

For signals and systems with rational Laplace transforms, the transform is often conveniently represented in the complex plane (s-plane) by marking the locations of the poles and zeros and indicating the region of convergence. From the pole-zero plot, the Fourier transform can be geometrically obtained, within a scaling factor. Causality, stability, and other characteristics are also easily identified from knowledge of the pole locations and the region of convergence.

In this chapter, we have been concerned primarily with the bilateral Laplace transform. However, we also introduced a somewhat different form of the Laplace transform known as the unilateral Laplace transform. The unilateral transform can be interpreted as the bilateral transform of a signal whose values prior to $t = 0^-$ have been set to zero. This form of the Laplace transform is especially useful for analyzing systems described by linear constant-coefficient differential equations with nonzero initial conditions.

Chapter 9 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining three sections contain problems belonging to the basic, advanced, and extension categories, respectively.

BASIC PROBLEMS WITH ANSWERS

9.1. For each of the following integrals, specify the values of the real parameter σ which ensure that the integral converges:

(a) $\int_0^{\infty} e^{-5t} e^{-(\sigma+j\omega)t}\,dt$ **(b)** $\int_{-\infty}^{0} e^{-5t} e^{-(\sigma+j\omega)t}\,dt$

(c) $\int_{-5}^{5} e^{-5t} e^{-(\sigma+j\omega)t}\,dt$ **(d)** $\int_{-\infty}^{\infty} e^{-5t} e^{-(\sigma+j\omega)t}\,dt$

(e) $\int_{-\infty}^{\infty} e^{-5|t|} e^{-(\sigma+j\omega)t}\,dt$ **(f)** $\int_{-\infty}^{0} e^{-5|t|} e^{-(\sigma+j\omega)t}\,dt$

9.2. Consider the signal

$$x(t) = e^{-5t}u(t-1),$$

and denote its Laplace transform by $X(s)$.

(a) Using eq. (9.3), evaluate $X(s)$ and specify its region of convergence.

(b) Determine the values of the finite numbers A and t_0 such that the Laplace transform $G(s)$ of

$$g(t) = Ae^{-5t}u(-t-t_0)$$

has the same algebraic form as $X(s)$. What is the region of convergence corresponding to $G(s)$?

9.3. Consider the signal

$$x(t) = e^{-5t}u(t) + e^{-\beta t}u(t),$$

and denote its Laplace transform by $X(s)$. What are the constraints placed on the real and imaginary parts of β if the region of convergence of $X(s)$ is $\mathcal{R}e\{s\} > -3$?

9.4. For the Laplace transform of

$$x(t) = \begin{cases} e^t \sin 2t, & t \leq 0 \\ 0, & t > 0 \end{cases},$$

indicate the location of its poles and its region of convergence.

9.5. For each of the following algebraic expressions for the Laplace transform of a signal, determine the number of zeros located in the finite s-plane and the number of zeros located at infinity:

(a) $\dfrac{1}{s+1} + \dfrac{1}{s+3}$

(b) $\dfrac{s+1}{s^2-1}$

(c) $\dfrac{s^3-1}{s^2+s+1}$

9.6. An absolutely integrable signal $x(t)$ is known to have a pole at $s = 2$. Answer the following questions:
(a) Could $x(t)$ be of finite duration?
(b) Could $x(t)$ be left sided?
(c) Could $x(t)$ be right sided?
(d) Could $x(t)$ be two sided?

9.7. How many signals have a Laplace transform that may be expressed as

$$\frac{(s-1)}{(s+2)(s+3)(s^2+s+1)}$$

in its region of convergence?

9.8. Let $x(t)$ be a signal that has a rational Laplace transform with exactly two poles, located at $s = -1$ and $s = -3$. If $g(t) = e^{2t}x(t)$ and $G(j\omega)$ [the Fourier transform of $g(t)$] converges, determine whether $x(t)$ is left sided, right sided, or two sided.

9.9. Given that

$$e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \qquad \mathcal{R}e\{s\} > \mathcal{R}e\{-a\},$$

determine the inverse Laplace transform of

$$X(s) = \frac{2(s+2)}{s^2+7s+12}, \qquad \mathcal{R}e\{s\} > -3.$$

9.10. Using geometric evaluation of the magnitude of the Fourier transform from the corresponding pole-zero plot, determine, for each of the following Laplace transforms, whether the magnitude of the corresponding Fourier transform is approximately lowpass, highpass, or bandpass:

(a) $H_1(s) = \dfrac{1}{(s+1)(s+3)}, \quad \mathcal{R}e\{s\} > -1$

(b) $H_2(s) = \dfrac{s}{s^2+s+1}, \quad \mathcal{R}e\{s\} > -\frac{1}{2}$

(c) $H_3(s) = \dfrac{s^2}{s^2+2s+1}, \quad \mathcal{R}e\{s\} > -1$

9.11. Use geometric evaluation from the pole-zero plot to determine the magnitude of the Fourier transform of the signal whose Laplace transform is specified as

$$X(s) = \frac{s^2-s+1}{s^2+s+1}, \qquad \mathcal{R}e\{s\} > -\frac{1}{2}.$$

9.12. Suppose we are given the following three facts about the signal $x(t)$:
1. $x(t) = 0$ for $t < 0$.
2. $x(k/80) = 0$ for $k = 1, 2, 3, \ldots$.
3. $x(1/160) = e^{-120}$.
Let $X(s)$ denote the Laplace transform of $x(t)$, and determine which of the following statements is consistent with the given information about $x(t)$:
(a) $X(s)$ has only one pole in the finite s-plane.
(b) $X(s)$ has only two poles in the finite s-plane.
(c) $X(s)$ has more than two poles in the finite s-plane.

9.13. Let

$$g(t) = x(t) + \alpha x(-t),$$

where

$$x(t) = \beta e^{-t} u(t)$$

and the Laplace transform of $g(t)$ is

$$G(s) = \frac{s}{s^2 - 1}, \qquad -1 < \mathcal{R}e\{s\} < 1.$$

Determine the values of the constants α and β.

9.14. Suppose the following facts are given about the signal $x(t)$ with Laplace transform $X(s)$:

1. $x(t)$ is real and even.
2. $X(s)$ has four poles and no zeros in the finite s-plane.
3. $X(s)$ has a pole at $s = (1/2)e^{j\pi/4}$.
4. $\int_{-\infty}^{\infty} x(t)\,dt = 4$.

Determine $X(s)$ and its ROC.

9.15. Consider two right-sided signals $x(t)$ and $y(t)$ related through the differential equations

$$\frac{dx(t)}{dt} = -2y(t) + \delta(t)$$

and

$$\frac{dy(t)}{dt} = 2x(t).$$

Determine $Y(s)$ and $X(s)$, along with their regions of convergence.

9.16. A causal LTI system S with impulse response $h(t)$ has its input $x(t)$ and output $y(t)$ related through a linear constant-coefficient differential equation of the form

$$\frac{d^3y(t)}{dt^3} + (1 + \alpha)\frac{d^2y(t)}{dt^2} + \alpha(\alpha + 1)\frac{dy(t)}{dt} + \alpha^2 y(t) = x(t).$$

(a) If

$$g(t) = \frac{dh(t)}{dt} + h(t),$$

how many poles does $G(s)$ have?

(b) For what real values of the parameter α is S guaranteed to be stable?

9.17. A causal LTI system S has the block diagram representation shown in Figure P9.17. Determine a differential equation relating the input $x(t)$ to the output $y(t)$ of this system.

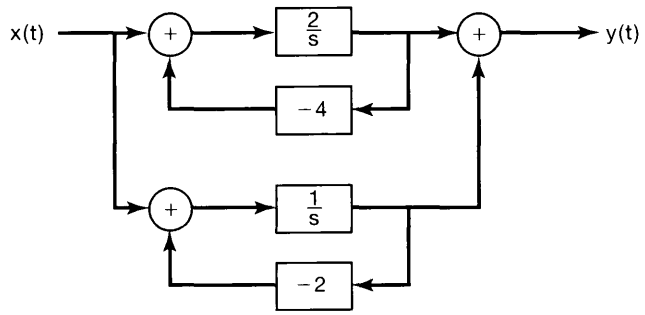


Figure P9.17

9.18. Consider the causal LTI system represented by the *RLC* circuit examined in Problem 3.20.

(a) Determine $H(s)$ and specify its region of convergence. Your answer should be consistent with the fact that the system is causal and stable.

(b) Using the pole-zero plot of $H(s)$ and geometric evaluation of the magnitude of the Fourier transform, determine whether the magnitude of the corresponding Fourier transform has an approximately lowpass, highpass, or bandpass characteristic.

(c) If the value of R is now changed to $10^{-3}\ \Omega$, determine $H(s)$ and specify its region of convergence.

(d) Using the pole-zero plot of $H(s)$ obtained in part (c) and geometric evaluation of the magnitude of the Fourier transform, determine whether the magnitude of the corresponding Fourier transform has an approximately lowpass, highpass, or bandpass characteristic.

9.19. Determine the unilateral Laplace transform of each of the following signals, and specify the corresponding regions of convergence:

(a) $x(t) = e^{-2t}u(t+1)$

(b) $x(t) = \delta(t+1) + \delta(t) + e^{-2(t+3)}u(t+1)$

(c) $x(t) = e^{-2t}u(t) + e^{-4t}u(t)$

9.20. Consider the *RL* circuit of Problem 3.19.

(a) Determine the zero-state response of this circuit when the input current is $x(t) = e^{-2t}u(t)$.

(b) Determine the zero-input response of the circuit for $t > 0^-$, given that

$$y(0^-) = 1.$$

(c) Determine the output of the circuit when the input current is $x(t) = e^{-2t}u(t)$ and the initial condition is the same as the one specified in part (b).

BASIC PROBLEMS

9.21. Determine the Laplace transform and the associated region of convergence and pole-zero plot for each of the following functions of time:

(a) $x(t) = e^{-2t}u(t) + e^{-3t}u(t)$ **(b)** $x(t) = e^{-4t}u(t) + e^{-5t}(\sin 5t)u(t)$

(c) $x(t) = e^{2t}u(-t) + e^{3t}u(-t)$ **(d)** $x(t) = te^{-2|t|}$

(e) $x(t) = |t|e^{-2|t|}$ **(f)** $x(t) = |t|e^{2t}u(-t)$

(g) $x(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 0, & \text{elsewhere} \end{cases}$ **(h)** $x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2 - t, & 1 \le t \le 2 \end{cases}$

(i) $x(t) = \delta(t) + u(t)$ **(j)** $x(t) = \delta(3t) + u(3t)$

9.22. Determine the function of time, $x(t)$, for each of the following Laplace transforms and their associated regions of convergence:

(a) $\frac{1}{s^2+9}$, $\mathcal{R}e\{s\} > 0$

(b) $\frac{s}{s^2+9}$, $\mathcal{R}e\{s\} < 0$

(c) $\frac{s+1}{(s+1)^2+9}$, $\mathcal{R}e\{s\} < -1$

(d) $\frac{s+2}{s^2+7s+12}$, $-4 < \mathcal{R}e\{s\} < -3$

(e) $\frac{s+1}{s^2+5s+6}$, $-3 < \mathcal{R}e\{s\} < -2$

(f) $\frac{(s+1)^2}{s^2-s+1}$, $\mathcal{R}e\{s\} > \frac{1}{2}$

(g) $\frac{s^2-s+1}{(s+1)^2}$, $\mathcal{R}e\{s\} > -1$

9.23. For each of the following statements about $x(t)$, and for each of the four pole-zero plots in Figure P9.23, determine the corresponding constraint on the ROC:

1. $x(t)e^{-3t}$ is absolutely integrable.
2. $x(t) * (e^{-t}u(t))$ is absolutely integrable.
3. $x(t) = 0$, $t > 1$.
4. $x(t) = 0$, $t < -1$.

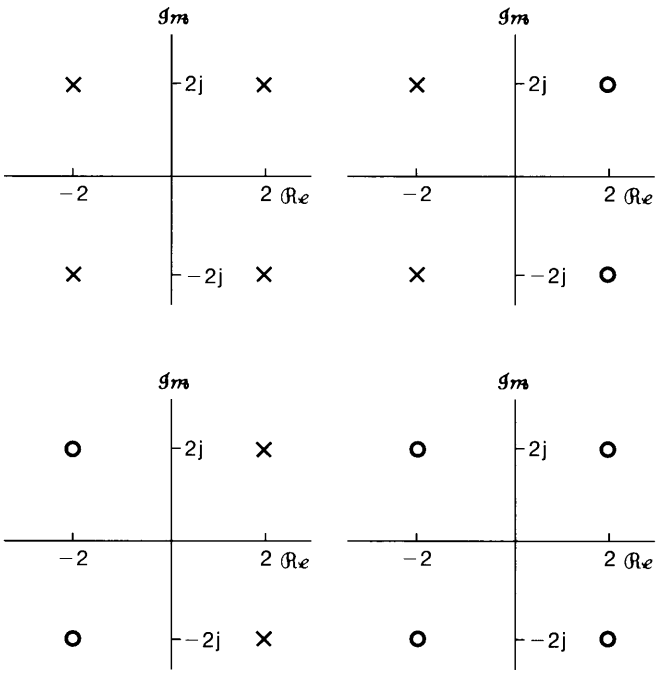


Figure P9.23

9.24. Throughout this problem, we will consider the region of convergence of the Laplace transforms always to include the $j\omega$-axis.

(a) Consider a signal $x(t)$ with Fourier transform $X(j\omega)$ and Laplace transform $X(s) = s + 1/2$. Draw the pole-zero plot for $X(s)$. Also, draw the vector whose length represents $|X(j\omega)|$ and whose angle with respect to the real axis represents $\measuredangle X(j\omega)$ for a given ω.

(b) By examining the pole-zero plot and vector diagram in part (a), determine a different Laplace transform $X_1(s)$ corresponding to the function of time, $x_1(t)$, so that

$$|X_1(j\omega)| = |X(j\omega)|,$$

but

$$x_1(t) \neq x(t).$$

Show the pole-zero plot and associated vectors that represent $X_1(j\omega)$.

(c) For your answer in part (b), determine, again by examining the related vector diagrams, the relationship between $\measuredangle X(j\omega)$ and $\measuredangle X_1(j\omega)$.

(d) Determine a Laplace transform $X_2(s)$ such that

$$\measuredangle X_2(j\omega) = \measuredangle X(j\omega),$$

but $x_2(t)$ is not proportional to $x(t)$. Show the pole-zero plot for $X_2(s)$ and the associated vectors that represent $X_2(j\omega)$.

(e) For your answer in part (d), determine the relationship between $|X_2(j\omega)|$ and $|X(j\omega)|$.

(f) Consider a signal $x(t)$ with Laplace transform $X(s)$ for which the pole-zero plot is shown in Figure P9.24. Determine $X_1(s)$ such that $|X(j\omega)| = |X_1(j\omega)|$ and all poles and zeros of $X_1(s)$ are in the left-half of the s-plane [i.e., $\mathcal{R}e\{s\} < 0$]. Also, determine $X_2(s)$ such that $\measuredangle X(j\omega) = \measuredangle X_2(j\omega)$ and all poles and zeros of $X_2(s)$ are in the left-half of the s-plane.

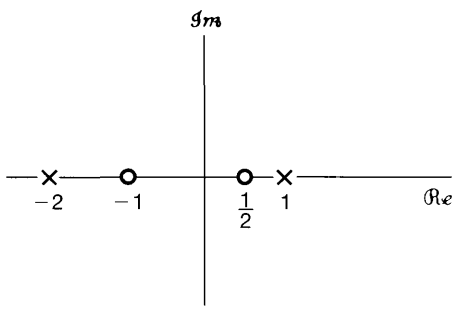


Figure P9.24

9.25. By considering the geometric determination of the Fourier transform, as developed in Section 9.4, sketch, for each of the pole-zero plots in Figure P9.25, the magnitude of the associated Fourier transform.

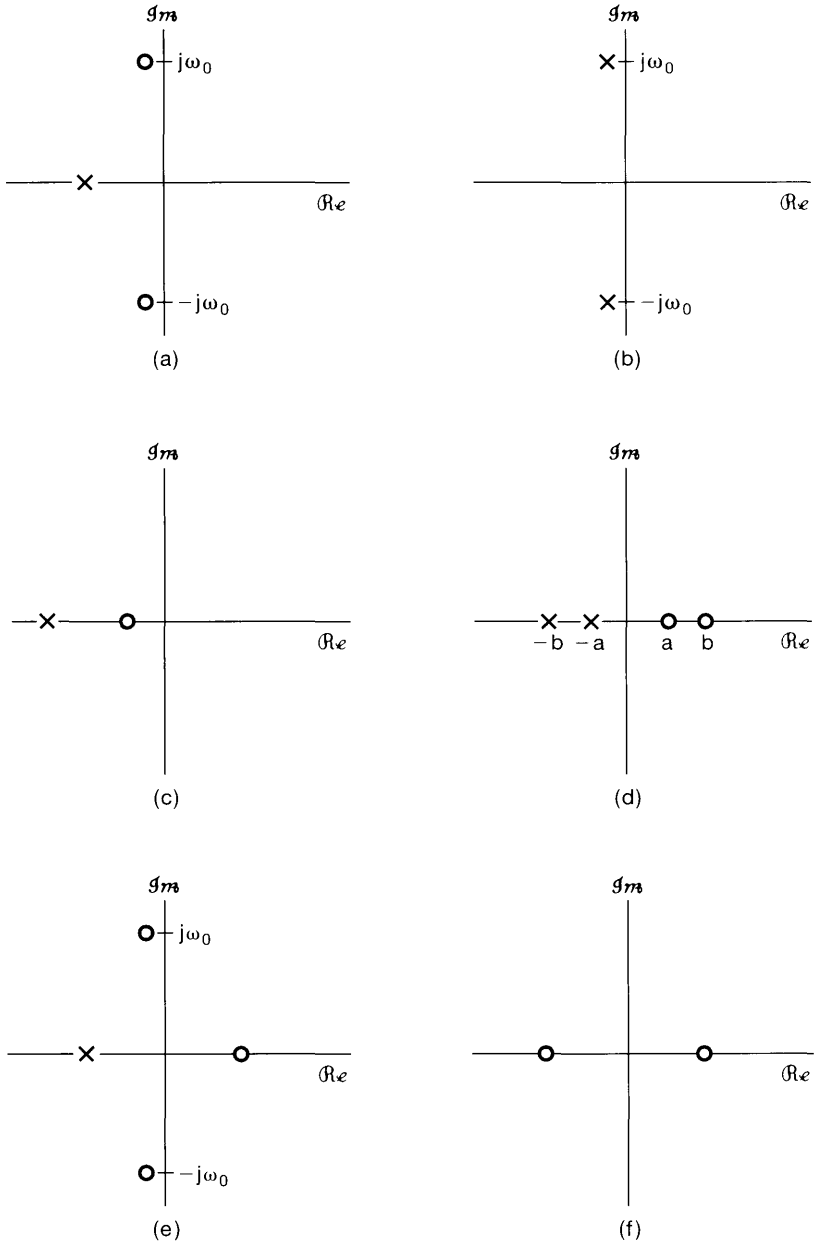


Figure P9.25

9.26. Consider a signal $y(t)$ which is related to two signals $x_1(t)$ and $x_2(t)$ by

$$y(t) = x_1(t-2) * x_2(-t+3)$$

where

$$x_1(t) = e^{-2t}u(t) \qquad \text{and} \qquad x_2(t) = e^{-3t}u(t).$$

Given that

$$e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \quad \mathcal{R}e\{s\} > a,$$

use properties of the Laplace transform to determine the Laplace transform $Y(s)$ of $y(t)$.

9.27. We are given the following five facts about a real signal $x(t)$ with Laplace transform $X(s)$:

1. $X(s)$ has exactly two poles.
2. $X(s)$ has no zeros in the finite s-plane.
3. $X(s)$ has a pole at $s = -1 + j$.
4. $e^{2t}x(t)$ is not absolutely integrable.
5. $X(0) = 8$.

Determine $X(s)$ and specify its region of convergence.

9.28. Consider an LTI system for which the system function $H(s)$ has the pole-zero pattern shown in Figure P9.28.

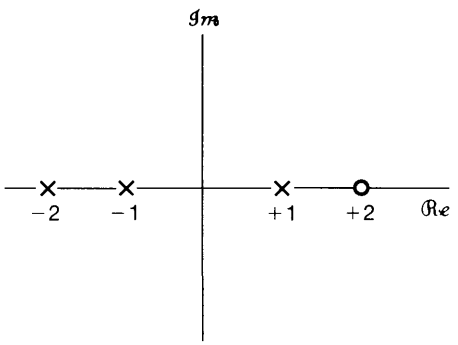


Figure P9.28

(a) Indicate all possible ROCs that can be associated with this pole-zero pattern.

(b) For each ROC identified in part (a), specify whether the associated system is stable and/or causal.

9.29. Consider an LTI system with input $x(t) = e^{-t}u(t)$ and impulse response $h(t) = e^{-2t}u(t)$.

(a) Determine the Laplace transforms of $x(t)$ and $h(t)$.

(b) Using the convolution property, determine the Laplace transform $Y(s)$ of the output $y(t)$.

(c) From the Laplace transform of $y(t)$ as obtained in part (b), determine $y(t)$.

(d) Verify your result in part (c) by explicitly convolving $x(t)$ and $h(t)$.

9.30. A pressure gauge that can be modeled as an LTI system has a time response to a unit step input given by $(1 - e^{-t} - te^{-t})u(t)$. For a certain input $x(t)$, the output is observed to be $(2 - 3e^{-t} + e^{-3t})u(t)$.

For this observed measurement, determine the true pressure input to the gauge as a function of time.

9.31. Consider a continuous-time LTI system for which the input $x(t)$ and output $y(t)$ are related by the differential equation

$$\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t).$$

Let $X(s)$ and $Y(s)$ denote Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of $h(t)$, the system impulse response.

(a) Determine $H(s)$ as a ratio of two polynomials in s. Sketch the pole-zero pattern of $H(s)$.

(b) Determine $h(t)$ for each of the following cases:

1. The system is stable.
2. The system is causal.
3. The system is *neither* stable *nor* causal.

9.32. A causal LTI system with impulse response $h(t)$ has the following properties:

1. When the input to the system is $x(t) = e^{2t}$ for all t, the output is $y(t) = (1/6)e^{2t}$ for all t.
2. The impulse response $h(t)$ satisfies the differential equation

$$\frac{dh(t)}{dt} + 2h(t) = (e^{-4t})u(t) + bu(t),$$

where b is an unknown constant.

Determine the system function $H(s)$ of the system, consistent with the information above. There should be no unknown constants in your answer; that is, the constant b should *not* appear in the answer.

9.33. The system function of a causal LTI system is

$$H(s) = \frac{s+1}{s^2 + 2s + 2}.$$

Determine and sketch the response $y(t)$ when the input is

$$x(t) = e^{-|t|}, \qquad -\infty < t < \infty.$$

9.34. Suppose we are given the following information about a causal and stable LTI system S with impulse response $h(t)$ and a rational system function $H(s)$:

1. $H(1) = 0.2$.
2. When the input is $u(t)$, the output is absolutely integrable.
3. When the input is $tu(t)$, the output is not absolutely integrable.
4. The signal $d^2h(t)/dt^2 + 2\,dh(t)/dt + 2h(t)$ is of finite duration.
5. $H(s)$ has exactly one zero at infinity.

Determine $H(s)$ and its region of convergence.

9.35. The input $x(t)$ and output $y(t)$ of a causal LTI system are related through the block-diagram representation shown in Figure P9.35.

(a) Determine a differential equation relating $y(t)$ and $x(t)$.

(b) Is this system stable?

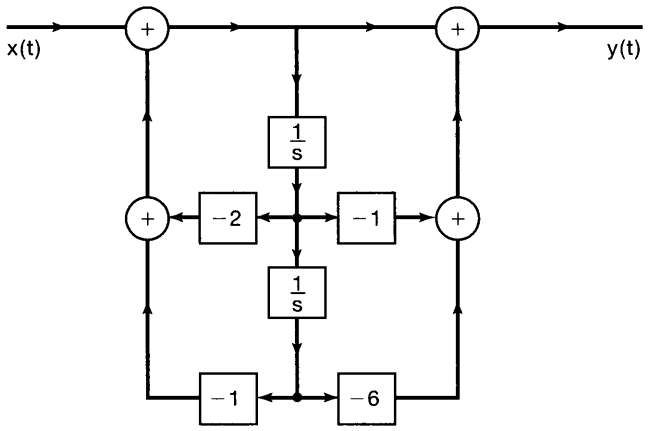


Figure P9.35

9.36. In this problem, we consider the construction of various types of block diagram representations for a causal LTI system S with input $x(t)$, output $y(t)$, and system function

$$H(s) = \frac{2s^2 + 4s - 6}{s^2 + 3s + 2}.$$

To derive the direct-form block diagram representation of S, we first consider a causal LTI system S_1 that has the same input $x(t)$ as S, but whose system function is

$$H_1(s) = \frac{1}{s^2 + 3s + 2}.$$

With the output of S_1 denoted by $y_1(t)$, the direct-form block diagram representation of S_1 is shown in Figure P9.36. The signals $e(t)$ and $f(t)$ indicated in the figure represent respective inputs into the two integrators.

(a) Express $y(t)$ (the output of S) as a linear combination of $y_1(t)$, $dy_1(t)/dt$, and $d^2y_1(t)/dt^2$.

(b) How is $dy_1(t)/dt$ related to $f(t)$?

(c) How is $d^2y_1(t)/dt^2$ related to $e(t)$?

(d) Express $y(t)$ as a linear combination of $e(t)$, $f(t)$, and $y_1(t)$.

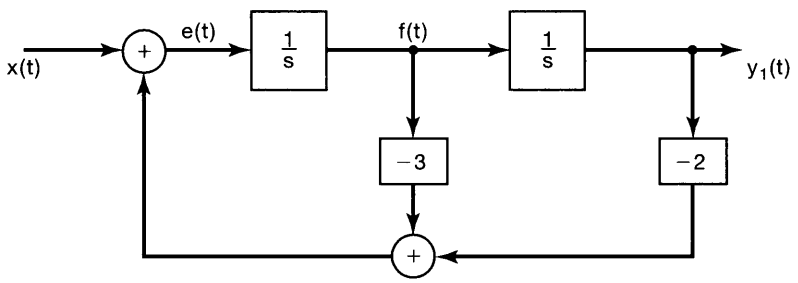


Figure P9.36

(e) Use the result from the previous part to extend the direct-form block diagram representation of S_1 and create a block diagram representation of S.

(f) Observing that

$$H(s) = \left(\frac{2(s-1)}{s+2}\right)\left(\frac{s+3}{s+1}\right),$$

draw a block diagram representation for S as a cascade combination of two subsystems.

(g) Observing that

$$H(s) = 2 + \frac{6}{s+2} - \frac{8}{s+1},$$

draw a block-diagram representation for S as a parallel combination of three subsystems.

9.37. Draw a direct-form representation for the causal LTI systems with the following system functions:

(a) $H_1(s) = \frac{s+1}{s^2+5s+6}$ **(b)** $H_2(s) = \frac{s^2-5s+6}{s^2+7s+10}$ **(c)** $H_3(s) = \frac{s}{(s+2)^2}$

9.38. Consider a fourth-order causal LTI system S whose system function is specified as

$$H(s) = \frac{1}{(s^2 - s + 1)(s^2 + 2s + 1)}.$$

(a) Show that a block diagram representation for S consisting of a cascade of four first-order sections will contain multiplications by coefficients that are not purely real.

(b) Draw a block diagram representation for S as a *cascade* interconnection of two second-order systems, each of which is represented in direct form. There should be no multiplications by nonreal coefficients in the resulting block diagram.

(c) Draw a block diagram representation for S as a *parallel* interconnection of two second-order systems, each of which is represented in direct form. There should be no multiplications by nonreal coefficients in the resulting block diagram.

9.39. Let

$$x_1(t) = e^{-2t}u(t) \quad \text{and} \quad x_2(t) = e^{-3(t+1)}u(t+1).$$

(a) Determine the unilateral Laplace transform $\mathfrak{X}_1(s)$ and the bilateral Laplace transform $X_1(s)$ for the signal $x_1(t)$.

(b) Determine the unilateral Laplace transform $\mathfrak{X}_2(s)$ and the bilateral Laplace transform $X_2(s)$ for the signal $x_2(t)$.

(c) Take the inverse bilateral Laplace transform of the product $X_1(s)X_2(s)$ to determine the signal $g(t) = x_1(t) * x_2(t)$.

(d) Show that the inverse unilateral Laplace transform of the product $\mathfrak{X}_1(s)\mathfrak{X}_2(s)$ is not the same as $g(t)$ for $t > 0^-$.

9.40. Consider the system S characterized by the differential equation

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t).$$

(a) Determine the zero-state response of this system for the input $x(t) = e^{-4t}u(t)$.
(b) Determine the zero-input response of the system for $t > 0^-$, given that

$$y(0^-) = 1, \qquad \left.\frac{dy(t)}{dt}\right|_{t=0^-} = -1, \qquad \left.\frac{d^2y(t)}{dt^2}\right|_{t=0^-} = 1.$$

(c) Determine the output of S when the input is $x(t) = e^{-4t}u(t)$ and the initial conditions are the same as those specified in part (b).

ADVANCED PROBLEMS

9.41. **(a)** Show that, if $x(t)$ is an even function, so that $x(t) = x(-t)$, then $X(s) = X(-s)$.
(b) Show that, if $x(t)$ is an odd function, so that $x(t) = -x(-t)$, then $X(s) = -X(-s)$.
(c) Determine which, if any, of the pole-zero plots in Figure P9.41 could correspond to an even function of time. For those that could, indicate the required ROC.

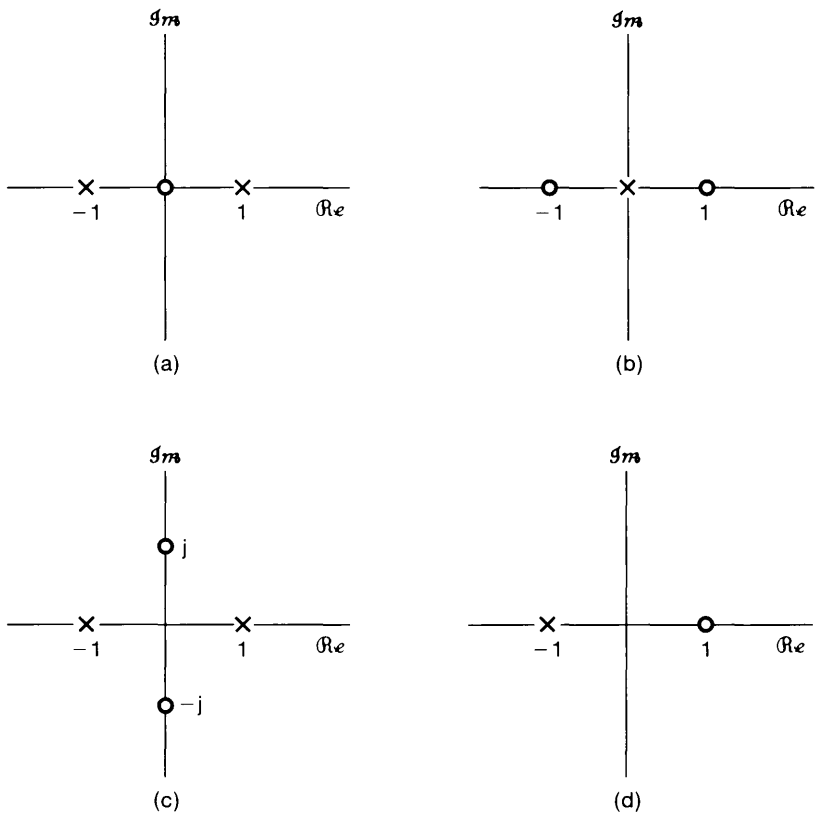


Figure P9.41

9.42. Determine whether each of the following statements is true or false. If a statement is true, construct a convincing argument for it. If it is false, give a counterexample.
(a) The Laplace transform of $t^2u(t)$ does not converge anywhere on the s-plane.
(b) The Laplace transform of $e^{t^2}u(t)$ does not converge anywhere on the s-plane.
(c) The Laplace transform of $e^{j\omega_0 t}$ does not converge anywhere on the s-plane.
(d) The Laplace transform of $e^{j\omega_0 t}u(t)$ does not converge anywhere on the s-plane.
(e) The Laplace transform of $|t|$ does not converge anywhere on the s-plane.

9.43. Let $h(t)$ be the impulse response of a causal and stable LTI system with a rational system function.
(a) Is the system with impulse response $dh(t)/dt$ guaranteed to be causal and stable?
(b) Is the system with impulse response $\int_{-\infty}^{t} h(\tau)\,d\tau$ guaranteed to be causal and unstable?

9.44. Let $x(t)$ be the sampled signal specified as

$$x(t) = \sum_{n=0}^{\infty} e^{-nT}\,\delta(t - nT),$$

where $T > 0$.
(a) Determine $X(s)$, including its region of convergence.
(b) Sketch the pole-zero plot for $X(s)$.
(c) Use geometric interpretation of the pole-zero plot to argue that $X(j\omega)$ is periodic.

9.45. Consider the LTI system shown in Figure P9.45(a) for which we are given the following information:

$$X(s) = \frac{s+2}{s-2},$$

$$x(t) = 0, \qquad t > 0,$$

and

$$y(t) = -\frac{2}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t). \qquad \text{[See Figure P9.45(b).]}$$

(a) Determine $H(s)$ and its region of convergence.
(b) Determine $h(t)$.

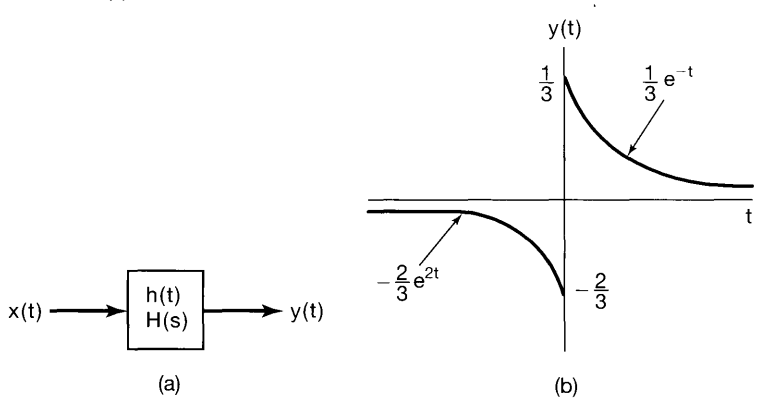


Figure P9.45

(c) Using the system function $H(s)$ found in part (a), determine the output $y(t)$ if the input is

$$x(t) = e^{3t}, \qquad -\infty < t < +\infty.$$

9.46. Let $H(s)$ represent the system function for a causal, stable system. The input to the system consists of the sum of three terms, one of which is an impulse $\delta(t)$ and another a complex exponential of the form $e^{s_0 t}$, where s_0 is a complex constant. The output is

$$y(t) = -6e^{-t}u(t) + \frac{4}{34}e^{4t}\cos 3t + \frac{18}{34}e^{4t}\sin 3t + \delta(t).$$

Determine $H(s)$, consistently with this information.

9.47. The signal

$$y(t) = e^{-2t}u(t)$$

is the output of a causal all-pass system for which the system function is

$$H(s) = \frac{s-1}{s+1}.$$

(a) Find and sketch at least two possible inputs $x(t)$ that could produce $y(t)$.

(b) What is the input $x(t)$ if it is known that

$$\int_{-\infty}^{\infty} |x(t)|\,dt < \infty?$$

(c) What is the input $x(t)$ if it is known that a stable (but not necessarily causal) system exists that will have $x(t)$ as an output if $y(t)$ is the input? Find the impulse response $h(t)$ of this filter, and show by direct convolution that it has the property claimed [i.e., that $y(t) * h(t) = x(t)$].

9.48. The *inverse* of an LTI system $H(s)$ is defined as a system that, when cascaded with $H(s)$, results in an overall transfer function of unity or, equivalently, an overall impulse response that is an impulse.

(a) If $H_1(s)$ denotes the transfer function of an inverse system for $H(s)$, determine the general algebraic relationship between $H(s)$ and $H_1(s)$.

(b) Shown in Figure P9.48 is the pole-zero plot for a stable, causal system $H(s)$. Determine the pole-zero plot for the associated inverse system.

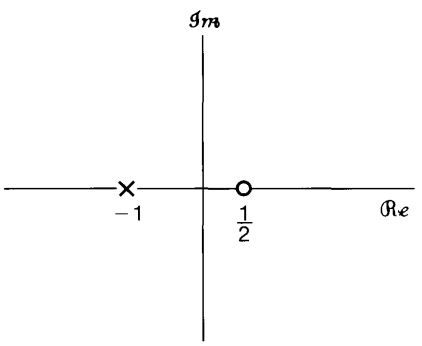


Figure P9.48

9.49. A class of systems, referred to as minimum-delay or minimum-phase systems, is sometimes defined through the statement that these systems are causal and stable and that the inverse systems are also causal and stable.

Based on the preceding definition, develop an argument to demonstrate that all poles *and* zeros of the transfer function of a minimum-delay system must be in the left half of the s-plane [i.e., $\mathcal{R}e\{s\} < 0$].

9.50. Determine whether or not each of the following statements about LTI systems is true. If a statement is true, construct a convincing argument for it. If it is false, give a counterexample.

(a) A stable continuous-time system must have all its poles in the left half of the s-plane [i.e., $\mathcal{R}e\{s\} < 0$].

(b) If a system function has more poles than zeros, and the system is causal, the step response will be continuous at $t = 0$.

(c) If a system function has more poles than zeros, and the system is not restricted to be causal, the step response can be discontinuous at $t = 0$.

(d) A stable, causal system must have all its poles and zeros in the left half of the s-plane.

9.51. Consider a stable and causal system with a real impulse response $h(t)$ and system function $H(s)$. It is known that $H(s)$ is rational, one of its poles is at $-1 + j$, one of its zeros is at $3 + j$, and it has exactly two zeros at infinity. For each of the following statements, determine whether it is true, whether it is false, or whether there is insufficient information to determine the statement's truth.

(a) $h(t)e^{-3t}$ is absolutely integrable.

(b) The ROC for $H(s)$ is $\mathcal{R}e\{s\} > -1$.

(c) The differential equation relating inputs $x(t)$ and outputs $y(t)$ for S may be written in a form having only real coefficients.

(d) $\lim_{s\to\infty} H(s) = 1$.

(e) $H(s)$ does not have fewer than four poles.

(f) $H(j\omega) = 0$ for at least one finite value of ω.

(g) If the input to S is $e^{3t}\sin t$, the output is $e^{3t}\cos t$.

9.52. As indicated in Section 9.5, many of the properties of the Laplace transform and their derivation are analogous to corresponding properties of the Fourier transform and their derivation, as developed in Chapter 4. In this problem, you are asked to outline the derivation of a number of the Laplace transform properties.

Observing the derivation for the corresponding property in Chapter 4 for the Fourier transform, derive each of the following Laplace transform properties. Your derivation must include a consideration of the region of convergence.

(a) Time shifting (Section 9.5.2)

(b) Shifting in the s-domain (Section 9.5.3)

(c) Time scaling (Section 9.5.4)

(d) Convolution property (Section 9.5.6)

9.53. As presented in Section 9.5.10, the initial-value theorem states that, for a signal $x(t)$ with Laplace transform $X(s)$ and for which $x(t) = 0$ for $t < 0$, the initial value of $x(t)$ [i.e., $x(0+)$] can be obtained from $X(s)$ through the relation

$$x(0+) = \lim_{s\to\infty} sX(s). \qquad \text{[eq. (9.110)]}$$

First, we note that, since $x(t) = 0$ for $t < 0$, $x(t) = x(t)u(t)$. Next, expanding $x(t)$ as a Taylor series at $t = 0+$, we obtain

$$x(t) = \left[x(0+) + x^{(1)}(0+)t + \cdots + x^{(n)}(0+)\frac{t^n}{n!} + \cdots\right]u(t), \tag{P9.53–1}$$

where $x^{(n)}(0+)$ denotes the nth derivative of $x(t)$ evaluated at $t = 0+$.

(a) Determine the Laplace transform of an arbitrary term $x^{(n)}(0+)(t^n/n!)u(t)$ on the right-hand side of eq. (P9.53–1). (You may find it helpful to review Example 9.14.)

(b) From your result in part (a) and the expansion in eq. (P9.53–1), show that $X(s)$ can be expressed as

$$X(s) = \sum_{n=0}^{\infty} x^{(n)}(0+)\frac{1}{s^{n+1}}.$$

(c) Demonstrate that eq. (9.110) follows from the result of part (b).

(d) By first determining $x(t)$, verify the initial-value theorem for each of the following examples:

(1) $X(s) = \frac{1}{s+2}$

(2) $X(s) = \frac{s+1}{(s+2)(s+3)}$

(e) A more general form of the initial-value theorem states that if $x^{(n)}(0+) = 0$ for $n < N$, then $x^{(N)}(0+) = \lim_{s\to\infty} s^{N+1}X(s)$. Demonstrate that this more general statement also follows from the result in part (b).

9.54. Consider a real-valued signal $x(t)$ with Laplace transform $X(s)$.

(a) By applying complex conjugation to both sides of eq. (9.56), show that $X(s) = X^*(s^*)$.

(b) From your result in (a), show that if $X(s)$ has a pole (zero) at $s = s_0$, it must also have a pole (zero) at $s = s_0^*$; i.e., for $x(t)$ real, the poles and zeros of $X(s)$ that are not on the real axis must occur in complex conjugate pairs.

9.55. In Section 9.6, Table 9.2, we listed a number of Laplace transform pairs, and we indicated specifically how transform pairs 1 through 9 follow from Examples 9.1 and 9.14, together with various properties from Table 9.1.

By exploiting properties from Table 9.1, show how transform pairs 10 through 16 follow from transform pairs 1 through 9 in Table 9.2.

9.56. The Laplace transform is said to exist for a specific complex s if the magnitude of the transform is finite—that is, if $|X(s)| < \infty$.

Show that a *sufficient* condition for the existence of the transform $X(s)$ at $s = s_0 = \sigma_0 + j\omega_0$ is that

$$\int_{-\infty}^{+\infty} |x(t)|e^{-\sigma_0 t}\,dt < \infty.$$

In other words, show that $x(t)$ exponentially weighted by $e^{-\sigma_0 t}$ is absolutely integrable. You will need to use the result that, for a complex function $f(t)$,

$$\left| \int_a^b f(t)\,dt \right| \leq \int_a^b |f(t)|\,dt. \tag{P9.56–1}$$

Without rigorously proving eq. (P9.56–1), argue its plausibility.

9.57. The Laplace transform $X(s)$ of a signal $x(t)$ has four poles and an unknown number of zeros. The signal $x(t)$ is known to have an impulse at $t = 0$. Determine what information, if any, this provides about the number of zeros and their locations.

9.58. Let $h(t)$ be the impulse response of a causal and stable LTI system with rational system function $H(s)$. Show that $g(t) = \mathcal{R}e\{h(t)\}$ is also the impulse response of a causal and stable system.

9.59. If $\mathfrak{X}(s)$ denotes the unilateral Laplace transform of $x(t)$, determine, in terms of $\mathfrak{X}(s)$, the unilateral Laplace transform of:

(a) $x(t-1)$ **(b)** $x(t+1)$

(c) $\int_{-\infty}^{\infty} x(\tau)\,d\tau$ **(d)** $\frac{d^3x(t)}{dt^3}$

EXTENSION PROBLEMS

9.60. In long-distance telephone communication, an echo is sometimes encountered due to the transmitted signal being reflected at the receiver, sent back down the line, reflected again at the transmitter, and returned to the receiver. The impulse response for a system that models this effect is shown in Figure P9.60, where we have assumed that only one echo is received. The parameter T corresponds to the one-way travel time along the communication channel, and the parameter α represents the attenuation in amplitude between transmitter and receiver.

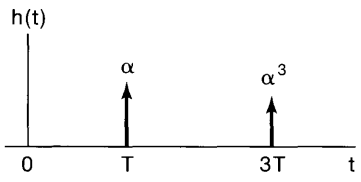


Figure P9.60

(a) Determine the system function $H(s)$ and associated region of convergence for the system.

(b) From your result in part (a), you should observe that $H(s)$ does not consist of a ratio of polynomials. Nevertheless, it is useful to represent it in terms of poles and zeros, where, as usual, the zeros are the values of s for which $H(s) = 0$

and the poles are the values of s for which $1/H(s) = 0$. For the system function found in part (a), determine the zeros and demonstrate that there are no poles.

(c) From your result in part (b), sketch the pole-zero plot for $H(s)$.

(d) By considering the appropriate vectors in the s-plane, sketch the magnitude of the frequency response of the system.

9.61. The autocorrelation function of a signal $x(t)$ is defined as

$$\phi_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)\,dt.$$

(a) Determine, in terms of $x(t)$, the impulse response $h(t)$ of an LTI system for which, when the input is $x(t)$, the output is $\phi_{xx}(t)$ [Figure P9.61(a)].

(b) From your answer in part (a), determine $\Phi_{xx}(s)$, the Laplace transform of $\phi_{xx}(\tau)$ in terms of $X(s)$. Also, express $\Phi_{xx}(j\omega)$, the Fourier transform of $\phi_{xx}(\tau)$, in terms of $X(j\omega)$.

(c) If $X(s)$ has the pole-zero pattern and ROC shown in Figure P9.61(b), sketch the pole-zero pattern and indicate the ROC for $\Phi_{xx}(s)$.

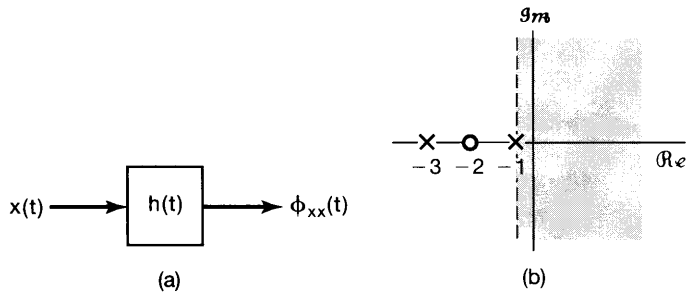


Figure P9.61

9.62. In a number of applications in signal design and analysis, the class of signals

$$\phi_n(t) = e^{-t/2}L_n(t)u(t), \quad n = 0, 1, 2, \ldots, \tag{P9.62-1}$$

where

$$L_n(t) = \frac{e^t}{n!}\frac{d^n}{dt^n}(t^n e^{-t}), \tag{P9.62-2}$$

is encountered.

(a) The functions $L_n(t)$ are referred to as *Laguerre polynomials*. To verify that they in fact have the form of polynomials, determine $L_0(t)$, $L_1(t)$, and $L_2(t)$ explicitly.

(b) Using the properties of the Laplace transform in Table 9.1 and Laplace transform pairs in Table 9.2, determine the Laplace transform $\Phi_n(s)$ of $\phi_n(t)$.

(c) The set of signals $\phi_n(t)$ can be generated by exciting a network of the form in Figure P9.62 with an impulse. From your result in part (b), determine $H_1(s)$ and $H_2(s)$ so that the impulse responses along the cascade chain are the signals $\phi_n(t)$ as indicated.

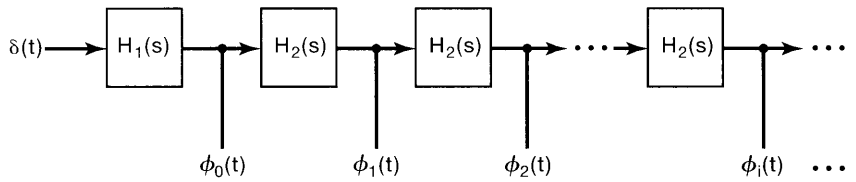


Figure P9.62

9.63. In filter design, it is often possible and convenient to transform a lowpass filter to a highpass filter and vice versa. With $H(s)$ denoting the transfer function of the original filter and $G(s)$ that of the transformed filter, one such commonly used transformation consists of replacing s by $1/s$; that is,

$$G(s) = H\left(\frac{1}{s}\right).$$

(a) For $H(s) = 1/(s + 1/2)$, sketch $|H(j\omega)|$ and $|G(j\omega)|$.

(b) Determine the linear constant-coefficient differential equation associated with $H(s)$ and with $G(s)$.

(c) Now consider a more general case in which $H(s)$ is the transfer function associated with the linear constant-coefficient differential equation in the general form

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{N} b_k \frac{d^k x(t)}{dt^k}. \qquad \text{(P9.63–1)}$$

Without any loss of generality, we have assumed that the number of derivatives N is the same on both sides of the equation, although in any particular case, some of the coefficients may be zero. Determine $H(s)$ and $G(s)$.

(d) From your result in part (c), determine, in terms of the coefficients in eq. (P9.63–1), the linear constant-coefficient differential equation associated with $G(s)$.

9.64. Consider the *RLC* circuit shown in Figure 9.27 with input $x(t)$ and output $y(t)$.

(a) Show that if R, L, and C are all positive, then this LTI system is stable.

(b) How should R, L, and C be related to each other so that the system represents a second-order Butterworth filter?

9.65. **(a)** Determine the differential equation relating $v_i(t)$ and $v_0(t)$ for the *RLC* circuit of Figure P9.65.

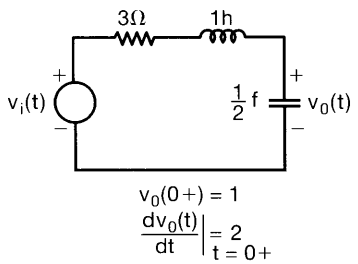


Figure P9.65

(b) Suppose that $v_i(t) = e^{-3t}u(t)$. Using the unilateral Laplace transform, determine $v_o(t)$ for $t > 0$.

9.66. Consider the *RL* circuit shown in Figure P9.66. Assume that the current $i(t)$ has reached a steady state with the switch at position *A*. At time $t = 0$, the switch is moved from position *A* to position *B*.

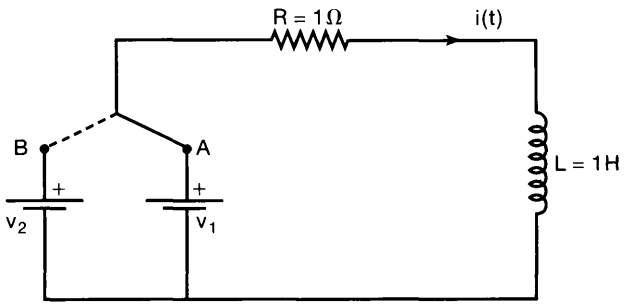


Figure P9.66

(a) Find the differential equation relating $i(t)$ and v_2 for $t > 0^-$. Specify the initial condition (i.e., the value of $i(0^-)$) for this differential equation in terms of v_1.

(b) Using the properties of the unilateral Laplace transform in Table 9.3, determine and plot the current $i(t)$ for each of the following values of v_1 and v_2:

(i) $v_1 = 0\ V, v_2 = 2\ V$

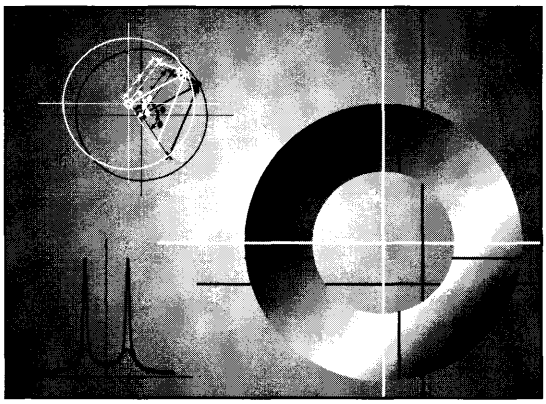
(ii) $v_1 = 4\ V, v_2 = 0\ V$

(iii) $v_1 = 4\ V, v_2 = 2\ V$

Using your answers for (i), (ii), and (iii), argue that the current $i(t)$ may be expressed as a sum of the circuit's zero-state response and zero-input response.

10

THE z-TRANSFORM



10.0 INTRODUCTION

In Chapter 9, we developed the Laplace transform as an extension of the continuous-time Fourier transform. This extension was motivated in part by the fact that it can be applied to a broader class of signals than the Fourier transform can, since there are many signals for which the Fourier transform does not converge but the Laplace transform does. The Laplace transform allowed us, for example, to perform transform analysis of unstable systems and to develop additional insights and tools for LTI system analysis.

In this chapter, we use the same approach for discrete time as we develop the z-transform, which is the discrete-time counterpart of the Laplace transform. As we will see, the motivations for and properties of the z-transform closely parallel those of the Laplace transform. Just as with the relationship between continuous-time and discrete-time Fourier transforms, however, we will encounter some important distinctions between the z-transform and the Laplace transform that arise from the fundamental differences between continuous-time and discrete-time signals and systems.

10.1 THE z-TRANSFORM

As we saw in Section 3.2, for a discrete-time linear time-invariant system with impulse response $h[n]$, the response $y[n]$ of the system to a complex exponential input of the form z^n is

$$y[n] = H(z)z^n, \tag{10.1}$$

where

$$H(z) = \sum_{n=-\infty}^{+\infty} h[n]z^{-n}. \tag{10.2}$$

For $z = e^{j\omega}$ with ω real (i.e., with $|z| = 1$), the summation in eq. (10.2) corresponds to the discrete-time Fourier transform of $h[n]$. More generally, when $|z|$ is not restricted to unity, the summation is referred to as the *z-transform* of $h[n]$.

The z-transform of a general discrete-time signal $x[n]$ is defined as[1]

$$\boxed{X(z) \triangleq \sum_{n=-\infty}^{+\infty} x[n]z^{-n},} \tag{10.3}$$

where z is a complex variable. For convenience, the z-transform of $x[n]$ will sometimes be denoted as $\mathcal{Z}\{x[n]\}$ and the relationship between $x[n]$ and its z-transform indicated as

$$x[n] \overset{z}{\longleftrightarrow} X(z). \tag{10.4}$$

In Chapter 9, we considered a number of important relationships between the Laplace transform and the Fourier transform for continuous-time signals. In a similar, but not identical, way, there are a number of important relationships between the z-transform and the discrete-time Fourier transform. To explore these relationships, we express the complex variable z in polar form as

$$z = re^{j\omega}, \tag{10.5}$$

with r as the magnitude of z and ω as the angle of z. In terms of r and ω, eq. (10.3) becomes

$$X(re^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n](re^{j\omega})^{-n},$$

or equivalently,

$$X(re^{j\omega}) = \sum_{n=-\infty}^{+\infty} \{x[n]r^{-n}\}e^{-j\omega n}. \tag{10.6}$$

From eq. (10.6), we see that $X(re^{j\omega})$ is the Fourier transform of the sequence $x[n]$ multiplied by a real exponential r^{-n}; that is,

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}. \tag{10.7}$$

The exponential weighting r^{-n} may be decaying or growing with increasing n, depending on whether r is greater than or less than unity. We note in particular that, for $r = 1$, or

[1]The z-transform defined in eq. (10.3) is often referred to as the *bilateral* z-transform, to distinguish it from the *unilateral* z-transform, which we develop in Section 10.9. The bilateral z-transform involves a summation from $-\infty$ to $+\infty$, while the unilateral transform has a form similar to eq. (10.3), but with summation limits from 0 to $+\infty$. Since we are mostly concerned with the bilateral z-transform, we will refer to $X(z)$ as defined in eq. (10.3) simply as the z-transform, except in Section 10.9, in which we use the words "unilateral" and "bilateral" to avoid ambiguity.

equivalently, $|z| = 1$, eq. (10.3) reduces to the Fourier transform; that is,

$$X(z)\Big|_{z=e^{j\omega}} = X(e^{j\omega}) = \mathcal{F}\{x[n]\}. \tag{10.8}$$

The relationship between the z-transform and Fourier transform for discrete-time signals parallels closely the corresponding discussion in Section 9.1 for continuous-time signals, but with some important differences. In the continuous-time case, the Laplace transform reduces to the Fourier transform when the real part of the transform variable is zero. Interpreted in terms of the s-plane, this means that the Laplace transform reduces to the Fourier transform on the imaginary axis (i.e., for $s = j\omega$). In contrast, the z-transform reduces to the Fourier transform when the magnitude of the transform variable z is *unity* (i.e, for $z = e^{j\omega}$). Thus, the z-transform reduces to the Fourier transform on the contour in the complex z-plane corresponding to a circle with a radius of unity, as indicated in Figure 10.1. This circle in the z-plane is referred to as the *unit circle* and plays a role in the discussion of the z-transform similar to the role of the imaginary axis in the s-plane for the Laplace transform.

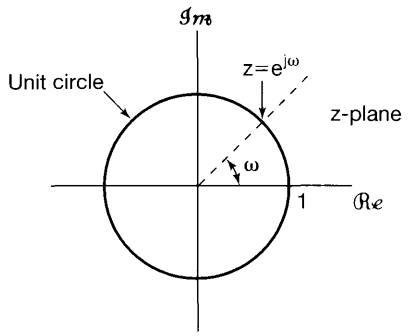


Figure 10.1 Complex z-plane. The z-transform reduces to the Fourier transform for values of z on the unit circle.

From eq. (10.7), we see that, for convergence of the z-transform, we require that the Fourier transform of $x[n]r^{-n}$ converge. For any specific sequence $x[n]$, we would expect this convergence for some values of r and not for others. In general, the z-transform of a sequence has associated with it a range of values of z for which $X(z)$ converges. As with the Laplace transform, this range of values is referred to as the *region of convergence* (ROC). If the ROC includes the unit circle, then the Fourier transform also converges. To illustrate the z-transform and the associated region of convergence, let us consider several examples.

Example 10.1

Consider the signal $x[n] = a^n u[n]$. Then, from eq. (10.3),

$$X(z) = \sum_{n=-\infty}^{+\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n.$$

For convergence of $X(z)$, we require that $\sum_{n=0}^{\infty} |az^{-1}|^n < \infty$. Consequently, the region of convergence is the range of values of z for which $|az^{-1}| < 1$, or equivalently, $|z| > |a|$.

Then

$$X(z) = \sum_{n=0}^{\infty}(az^{-1})^n = \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |z| > |a|. \tag{10.9}$$

Thus, the z-transform for this signal is well-defined for any value of a, with an ROC determined by the magnitude of a according to eq. (10.9). For example, for $a = 1$, $x[n]$ is the unit step sequence with z-transform

$$X(z) = \frac{1}{1-z^{-1}}, \qquad |z| > 1.$$

We see that the z-transform in eq. (10.9) is a rational function. Consequently, just as with rational Laplace transforms, the z-transform can be characterized by its zeros (the roots of the numerator polynomial) and its poles (the roots of the denominator polynomial). For this example, there is one zero, at $z = 0$, and one pole, at $z = a$. The pole-zero plot and the region of convergence for Example 10.1 are shown in Figure 10.2 for a value of a between 0 and 1. For $|a| > 1$, the ROC does not include the unit circle, consistent with the fact that, for these values of a, the Fourier transform of $a^n u[n]$ does not converge.

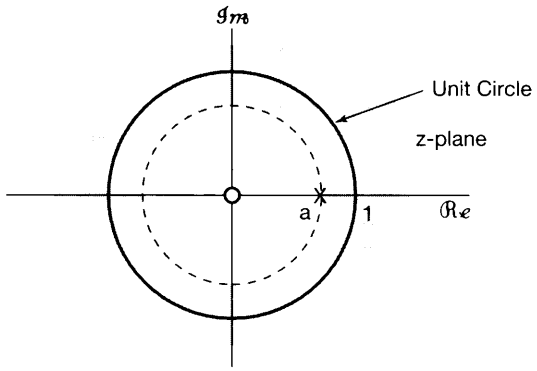


Figure 10.2 Pole-zero plot and region of convergence for Example 10.1 for $0 < a < 1$.

Example 10.2

Now let $x[n] = -a^n u[-n-1]$. Then

$$\begin{aligned} X(z) &= -\sum_{n=-\infty}^{+\infty} a^n u[-n-1]z^{-n} = -\sum_{n=-\infty}^{-1} a^n z^{-n} \\ &= -\sum_{n=1}^{\infty} a^{-n}z^n = 1 - \sum_{n=0}^{\infty}(a^{-1}z)^n. \end{aligned} \tag{10.10}$$

If $|a^{-1}z| < 1$, or equivalently, $|z| < |a|$, the sum in eq. (10.10) converges and

$$X(z) = 1 - \frac{1}{1-a^{-1}z} = \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |z| < |a|. \tag{10.11}$$

The pole-zero plot and region of convergence for this example are shown in Figure 10.3 for a value of a between 0 and 1.

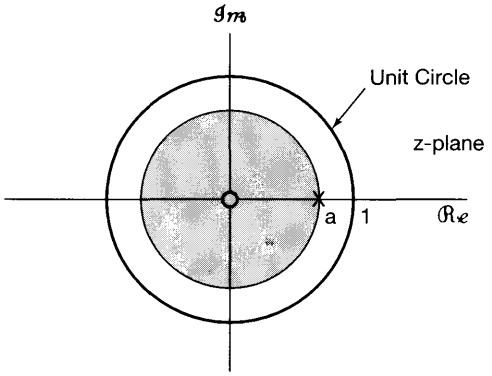


Figure 10.3 Pole-zero plot and region of convergence for Example 10.2 for $0 < a < 1$.

Comparing eqs. (10.9) and (10.11), and Figures 10.2 and 10.3, we see that the algebraic expression for $X(z)$ and the corresponding pole-zero plot are identical in Examples 10.1 and 10.2, and the z-transforms differ only in their regions of convergence. Thus, as with the Laplace transform, specification of the z-transform requires both the algebraic expression and the region of convergence. Also, in both examples, the sequences were exponentials and the resulting z-transforms were rational. In fact, as further suggested by the following examples, $X(z)$ will be rational whenever $x[n]$ is a linear combination of real or complex exponentials:

Example 10.3

Let us consider a signal that is the sum of two real exponentials:

$$x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]. \tag{10.12}$$

The z-transform is then

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{+\infty} \left\{ 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n] \right\} z^{-n} \\ &= 7\sum_{n=-\infty}^{+\infty} \left(\frac{1}{3}\right)^n u[n] z^{-n} - 6\sum_{n=-\infty}^{+\infty} \left(\frac{1}{2}\right)^n u[n] z^{-n} \\ &= 7\sum_{n=0}^{\infty} \left(\frac{1}{3}z^{-1}\right)^n - 6\sum_{n=0}^{+\infty} \left(\frac{1}{2}z^{-1}\right)^n \qquad (10.13) \\ &= \frac{7}{1-\frac{1}{3}z^{-1}} - \frac{6}{1-\frac{1}{2}z^{-1}} = \frac{1-\frac{3}{2}z^{-1}}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})} \qquad (10.14) \\ &= \frac{z(z-\frac{3}{2})}{(z-\frac{1}{3})(z-\frac{1}{2})}. \qquad (10.15) \end{aligned}$$

For convergence of $X(z)$, both sums in eq. (10.13) must converge, which requires that both $|(1/3)z^{-1}| < 1$ and $|(1/2)z^{-1}| < 1$, or equivalently, $|z| > 1/3$ and $|z| > 1/2$. Thus, the region of convergence is $|z| > 1/2$.

The z-transform for this example can also be obtained using the results of Example 10.1. Specifically, from the definition of the z-transform in eq. (10.3), we see that the z-transform is linear; that is, if $x[n]$ is the sum of two terms, then $X(z)$ will be the sum of the z-transforms of the individual terms and will converge when both z-transforms converge. From Example 10.1,

$$\left(\frac{1}{3}\right)^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-\frac{1}{3}z^{-1}}, \quad |z| > \frac{1}{3} \tag{10.16}$$

and

$$\left(\frac{1}{2}\right)^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-\frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}, \tag{10.17}$$

and consequently,

$$7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n] \overset{z}{\longleftrightarrow} \frac{7}{1-\frac{1}{3}z^{-1}} - \frac{6}{1-\frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}, \tag{10.18}$$

as we determined before. The pole-zero plot and ROC for the z-transform of each of the individual terms and for the combined signal are shown in Figure 10.4.

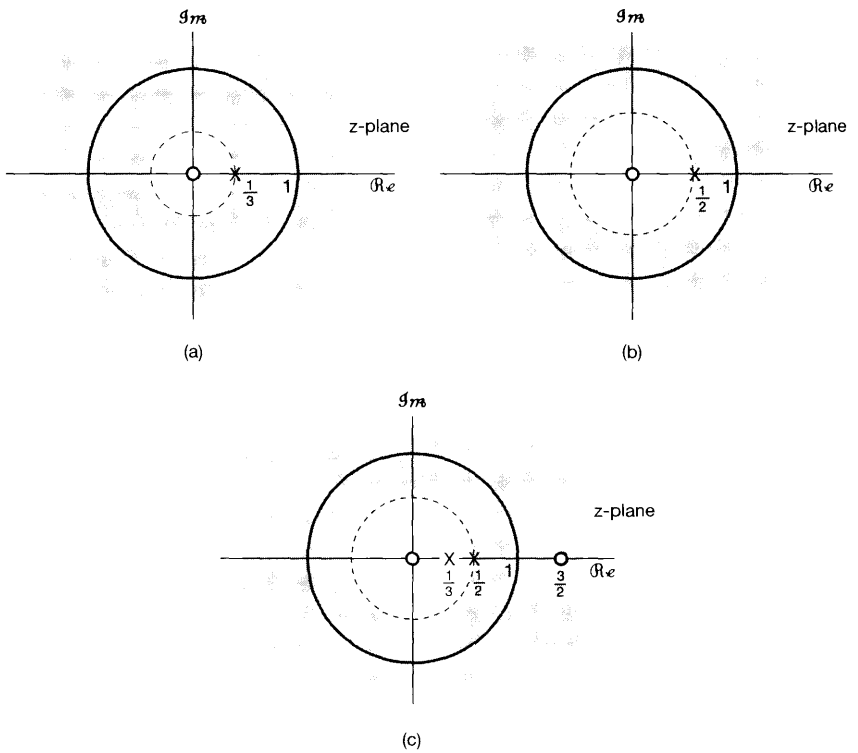


Figure 10.4 Pole-zero plot and region of convergence for the individual terms and the sum in Example 10.3: (a)$1/(1-\frac{1}{3}z^{-1})$, $|z| > \frac{1}{3}$; (b)$1/(1-\frac{1}{2}z^{-1})$, $|z| > \frac{1}{2}$; (c)$7/(1-\frac{1}{3}z^{-1}) - 6/(1-\frac{1}{2}z^{-1})$, $|z| > \frac{1}{2}$.

Example 10.4

Let us consider the signal

$$\begin{aligned} x[n] &= (\tfrac{1}{3})^n \sin(\tfrac{\pi}{4}n)u[n] \\ &= \tfrac{1}{2j}(\tfrac{1}{3}e^{j\pi/4})^n u[n] - \tfrac{1}{2j}(\tfrac{1}{3}e^{-j\pi/4})^n u[n]. \end{aligned}$$

The z-transform of this signal is

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} \left\{ \frac{1}{2j}\left(\frac{1}{3}e^{j\pi/4}\right)^n u[n] - \frac{1}{2j}\left(\frac{1}{3}e^{-j\pi/4}\right)^n u[n] \right\} z^{-n} \\ &= \frac{1}{2j}\sum_{n=0}^{\infty}\left(\frac{1}{3}e^{j\pi/4}z^{-1}\right)^n - \frac{1}{2j}\sum_{n=0}^{\infty}\left(\frac{1}{3}e^{-j\pi/4}z^{-1}\right)^n \qquad (10.19) \\ &= \frac{1}{2j}\frac{1}{1-\frac{1}{3}e^{j\pi/4}z^{-1}} - \frac{1}{2j}\frac{1}{1-\frac{1}{3}e^{-j\pi/4}z^{-1}}, \end{aligned}$$

or equivalently,

$$X(z) = \frac{\frac{1}{3\sqrt{2}}z}{(z-\frac{1}{3}e^{j\pi/4})(z-\frac{1}{3}e^{-j\pi/4})} \qquad (10.20)$$

For convergence of $X(z)$, both sums in eq. (10.19) must converge, which requires that $|(1/3)e^{j\pi/4}z^{-1}| < 1$ and $|(1/3)e^{-j\pi/4}z^{-1}| < 1$, or equivalently, $|z| > 1/3$. The pole-zero plot and ROC for this example are shown in Figure 10.5.

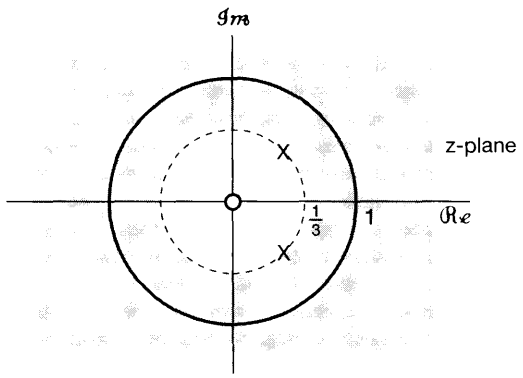


Figure 10.5 Pole-zero plot and ROC for the z-transform in Example 10.4.

In each of the preceding four examples, we expressed the z-transform both as a ratio of polynomials in z and as a ratio of polynomials in z^{-1}. From the definition of the z-transform as given in eq. (10.3), we see that, for sequences which are zero for $n < 0$, $X(z)$ involves only negative powers of z. Thus, for this class of signals, it is particularly convenient for $X(z)$ to be expressed in terms of polynomials in z^{-1} rather than z, and

when appropriate, we will use that form in our discussion. However, reference to the poles and zeros is always in terms of the roots of the numerator and denominator expressed as polynomials in z. Also, it is sometimes convenient to refer to $X(z)$, written as a ratio of polynomials in z, as having poles at infinity if the degree of the numerator exceeds the degree of the denominator or zeros at infinity if the numerator is of smaller degree than the denominator.

10.2 THE REGION OF CONVERGENCE FOR THE z-TRANSFORM

In Chapter 9, we saw that there were specific properties of the region of convergence of the Laplace transform for different classes of signals and that understanding these properties led to further insights about the transform. In a similar manner, we explore a number of properties of the region of convergence for the z-transform. Each of the following properties and its justification closely parallel the corresponding property in Section 9.2.

Property 1: The ROC of $X(z)$ consists of a ring in the z-plane centered about the origin.

This property is illustrated in Figure 10.6 and follows from the fact that the ROC consists of those values of $z = re^{j\omega}$ for which $x[n]r^{-n}$ has a Fourier transform that converges. That is, the ROC of the z-transform of $x[n]$ consists of the values of z for which $x[n]r^{-n}$ is absolutely summable:[2]

$$\sum_{n=-\infty}^{+\infty} |x[n]|r^{-n} < \infty. \tag{10.21}$$

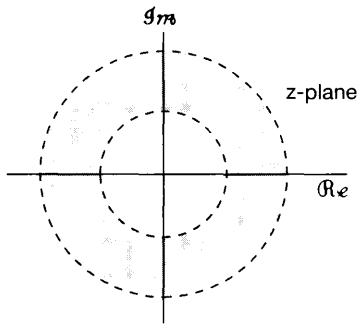


Figure 10.6 ROC as a ring in the z-plane. In some cases, the inner boundary can extend inward to the origin, in which case the ROC becomes a disc. In other cases, the outer boundary can extend outward to infinity.

[2]For a thorough treatment of the mathematical properties of z-transforms, see R.V. Churchill and J.W. Brown, *Complex Variables and Applications* (5th ed.) (New York: McGraw–Hill, 1990), and E. I. Jury, *Theory and Application of the z-Transform Method* (Malabar, FL: R. E. Krieger Pub. Co., 1982).

Thus, convergence is dependent only on $r = |z|$ and not on ω. Consequently, if a specific value of z is in the ROC, then all values of z on the same circle (i.e., with the same magnitude) will be in the ROC. This by itself guarantees that the ROC will consist of concentric rings. As we will see when we discuss Property 6, the ROC must in fact consist of only a single ring. In some cases the inner boundary of the ROC may extend inward to the origin, and in some cases the outer boundary may extend outward to infinity.

Property 2: The ROC does not contain any poles.

As with the Laplace transform, this property is simply a consequence of the fact that at a pole $X(z)$ is infinite and therefore, by definition, does not converge.

Property 3: If $x[n]$ is of finite duration, then the ROC is the entire z-plane, except possibly $z = 0$ and/or $z = \infty$.

A finite-duration sequence has only a finite number of nonzero values, extending, say, from $n = N_1$ to $n = N_2$, where N_1 and N_2 are finite. Thus, the z-transform is the sum of a finite number of terms; that is,

$$X(z) = \sum_{n=N_1}^{N_2} x[n]z^{-n}. \tag{10.22}$$

For z not equal to zero or infinity, each term in the sum will be finite, and consequently, $X(z)$ will converge. If N_1 is negative and N_2 positive, so that $x[n]$ has nonzero values both for $n < 0$ and $n > 0$, then the summation includes terms with both positive powers of z and negative powers of z. As $|z| \to 0$, terms involving negative powers of z become unbounded, and as $|z| \to \infty$, terms involving positive powers of z become unbounded. Consequently, for N_1 negative and N_2 positive, the ROC does not include $z = 0$ or $z = \infty$. If N_1 is zero or positive, there are only negative powers of z in eq. (10.22), and consequently, the ROC includes $z = \infty$. If N_2 is zero or negative, there are only positive powers of z in eq. (10.22), and consequently, the ROC includes $z = 0$.

Example 10.5

Consider the unit impulse signal $\delta[n]$. Its z-transform is given by

$$\delta[n] \overset{z}{\longleftrightarrow} \sum_{n=-\infty}^{+\infty} \delta[n]z^{-n} = 1, \tag{10.23}$$

with an ROC consisting of the entire z-plane, including $z = 0$ and $z = \infty$. On the other hand, consider the delayed unit impulse $\delta[n - 1]$, for which

$$\delta[n - 1] \overset{z}{\longleftrightarrow} \sum_{n=-\infty}^{+\infty} \delta[n - 1]z^{-n} = z^{-1}. \tag{10.24}$$

This z-transform is well defined except at $z = 0$, where there is a pole. Thus, the ROC consists of the entire z-plane, including $z = \infty$ but excluding $z = 0$. Similarly, consider an impulse advanced in time, namely, $\delta[n + 1]$. In this case,

$$\delta[n + 1] \overset{z}{\longleftrightarrow} \sum_{n=-\infty}^{+\infty} \delta[n + 1]z^{-n} = z, \tag{10.25}$$

which is well defined for all finite values of z. Thus, the ROC consists of the entire finite z-plane (including $z = 0$), but there is a pole at infinity.

Property 4: If $x[n]$ is a right-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all finite values of z for which $|z| > r_0$ will also be in the ROC.

The justification for this property follows in a manner identical to that of Property 4 in Section 9.2. A right-sided sequence is zero prior to some value of n, say, N_1. If the circle $|z| = r_0$ is in the ROC, then $x[n]r_0^{-n}$ is absolutely summable. Now consider $|z| = r_1$ with $r_1 > r_0$, so that r_1^{-n} decays more quickly than r_0^{-n} for increasing n. As illustrated in Figure 10.7, this more rapid exponential decay will further attenuate sequence values

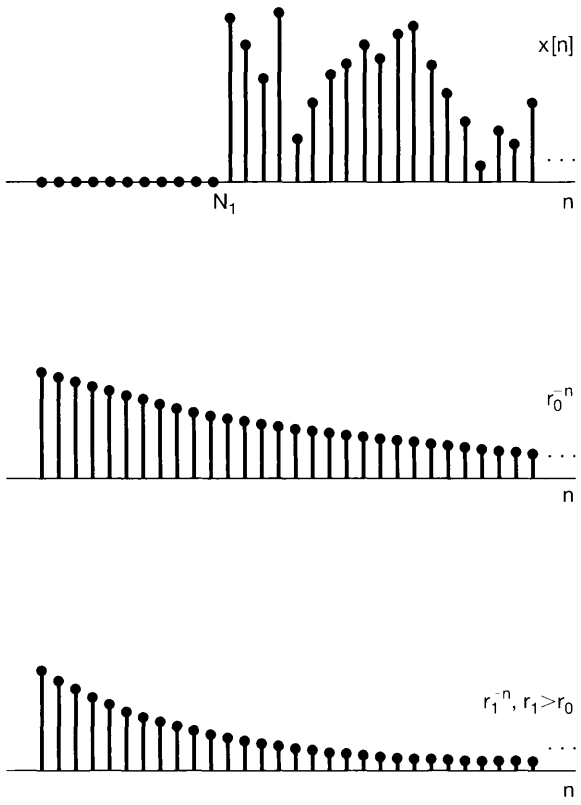


Figure 10.7 With $r_1 > r_0$, $x[n]r_1^{-n}$ decays faster with increasing n than does $x[n]r_0^{-n}$. Since $x[n] = 0, n < N_1$, this implies that if $x[n]r_0^{-n}$ is absolutely summable, then $x[n]r_1^{-n}$ will be also.

for positive values of n and cannot cause sequence values for negative values of n to become unbounded, since $x[n]$ is right sided and, in particular, $x[n]z^{-n} = 0$ for $n < N_1$. Consequently, $x[n]r_1^{-n}$ is absolutely summable.

For right-sided sequences in general, eq. (10.3) takes the form

$$X(z) = \sum_{n=N_1}^{\infty} x[n]z^{-n}, \tag{10.26}$$

where N_1 is finite and may be positive or negative. If N_1 is negative, then the summation in eq. (10.26) includes terms with positive powers of z, which become unbounded as $|z| \to \infty$. Consequently, for right-sided sequences in general, the ROC will not include infinity. However, for the particular class of causal sequences, i.e., sequences that are zero for $n < 0$, N_1 will be nonnegative, and consequently, the ROC will include $z = \infty$.

Property 5: If $x[n]$ is a left-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all values of z for which $0 < |z| < r_0$ will also be in the ROC.

Again, this property closely parallels the corresponding property for Laplace transforms, and the proof of it and its basis in intuition are similar to the proof and intuition for Property 4. In general, for left-sided sequences, from eq. (10.3), the summation for the z-transform will be of the form

$$X(z) = \sum_{n=-\infty}^{N_2} x[n]z^{-n}, \tag{10.27}$$

where N_2 may be positive or negative. If N_2 is positive, then eq. (10.27) includes negative powers of z, which become unbounded as $|z| \to 0$. Consequently, for left-sided sequences, the ROC will not in general include $z = 0$. However, if $N_2 \leq 0$ (so that $x[n] = 0$ for all $n > 0$), the ROC will include $z = 0$.

Property 6: If $x[n]$ is two sided, and if the circle $|z| = r_0$ is in the ROC, then the ROC will consist of a ring in the z-plane that includes the circle $|z| = r_0$.

As with Property 6 in Section 9.2, the ROC for a two-sided signal can be examined by expressing $x[n]$ as the sum of a right-sided and a left-sided signal. The ROC for the right-sided component is a region bounded on the inside by a circle and extending outward to (and possibly including) infinity. The ROC for the left-sided component is a region bounded on the outside by a circle and extending inward to, and possibly including, the origin. The ROC for the composite signal includes the intersection of the ROCs of the components. As illustrated in Figure 10.8, the overlap (assuming that there is one) is a ring in the z-plane.

We illustrate the foregoing properties with examples that closely parallel Examples 9.6 and 9.7.

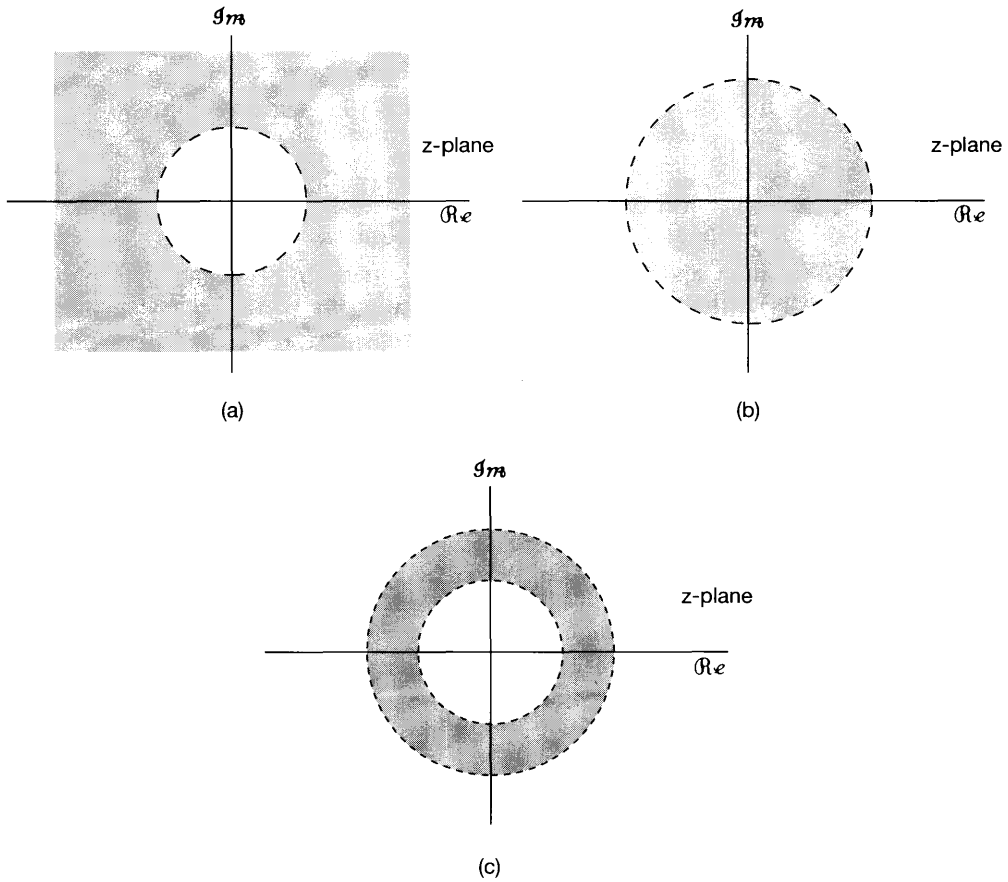


Figure 10.8 (a) ROC for right-sided sequence; (b) ROC for left-sided sequence; (c) intersection of the ROCs in (a) and (b), representing the ROC for a two-sided sequence that is the sum of the right-sided and the left-sided sequence.

Example 10.6

Consider the signal

$$x[n] = \begin{cases} a^n, & 0 \le n \le N-1, \ a > 0 \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$\begin{aligned} X(z) &= \sum_{n=0}^{N-1} a^n z^{-n} \\ &= \sum_{n=0}^{N-1} (az^{-1})^n \\ &= \frac{1-(az^{-1})^N}{1-az^{-1}} = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z-a}. \end{aligned} \tag{10.28}$$

Since $x[n]$ is of finite length, it follows from Property 3 that the ROC includes the entire z-plane except possibly the origin and/or infinity. In fact, from our discussion of Property 3, since $x[n]$ is zero for $n < 0$, the ROC will extend to infinity. However, since $x[n]$ is nonzero for some positive values of n, the ROC will not include the origin. This is evident from eq. (10.28), from which we see that there is a pole of order $N - 1$ at $z = 0$. The N roots of the numerator polynomial are at

$$z_k = ae^{j(2\pi k/N)}, \quad k = 0, 1, \ldots, N-1. \tag{10.29}$$

The root for $k = 0$ cancels the pole at $z = a$. Consequently, there are no poles other than at the origin. The remaining zeros are at

$$z_k = ae^{j(2\pi k/N)}, \quad k = 1, \ldots, N-1. \tag{10.30}$$

The pole-zero pattern is shown in Figure 10.9.

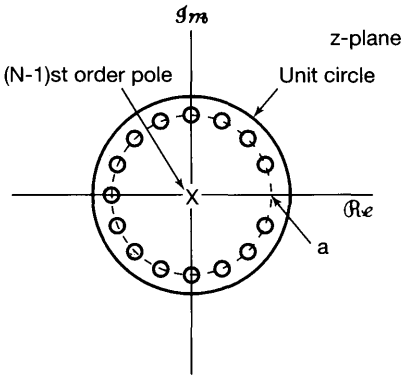


Figure 10.9 Pole-zero pattern for Example 10.6 with $N = 16$ and $0 < a < 1$. The region of convergence for this example consists of all values of z except $z = 0$.

Example 10.7

Let

$$x[n] = b^{|n|}, \quad b > 0. \tag{10.31}$$

This two-sided sequence is illustrated in Figure 10.10, for both $b < 1$ and $b > 1$. The z-transform for the sequence can be obtained by expressing it as the sum of a right-sided and a left-sided sequence. We have

$$x[n] = b^n u[n] + b^{-n} u[-n-1]. \tag{10.32}$$

From Example 10.1,

$$b^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1 - bz^{-1}}, \quad |z| > b, \tag{10.33}$$

and from Example 10.2,

$$b^{-n} u[-n-1] \overset{z}{\longleftrightarrow} \frac{-1}{1 - b^{-1}z^{-1}}, \quad |z| < \frac{1}{b}. \tag{10.34}$$

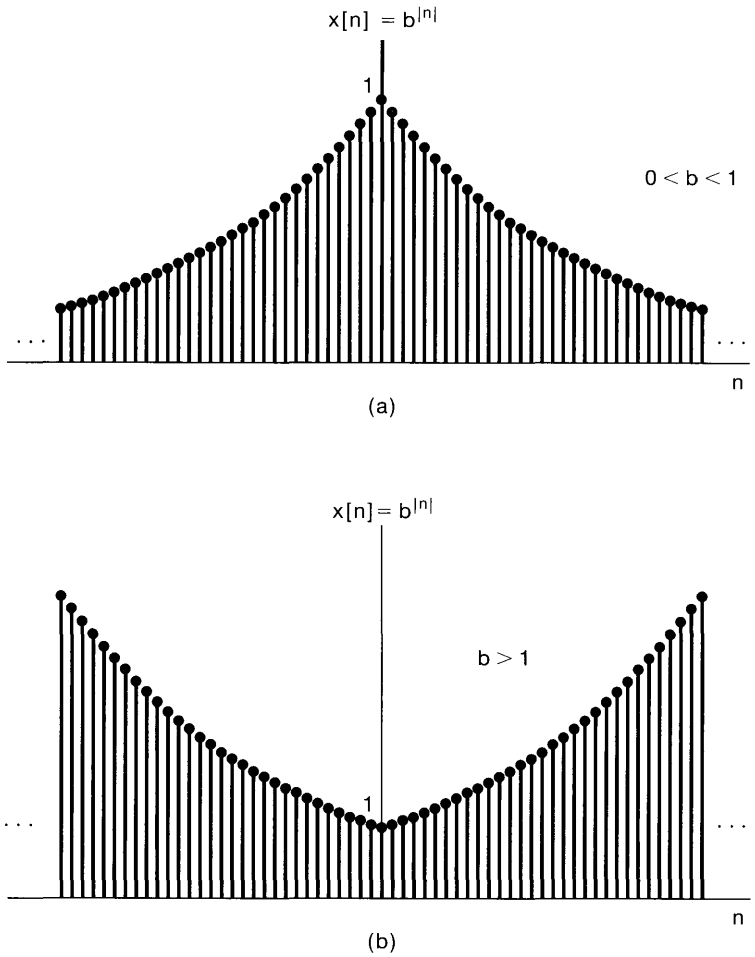


Figure 10.10 Sequence $x[n] = b^{|n|}$ for $0 < b < 1$ and for $b > 1$: (a) $b = 0.95$; (b) $b = 1.05$.

In Figures 10.11(a)–(d) we show the pole-zero pattern and ROC for eqs. (10.33) and (10.34), for values of $b > 1$ and $0 < b < 1$. For $b > 1$, there is no common ROC, and thus the sequence in eq. (10.31) will not have a z-transform, even though the right-sided and left-sided components do individually. For $b < 1$, the ROCs in eqs. (10.33) and (10.34) overlap, and thus the z-transform for the composite sequence is

$$X(z) = \frac{1}{1 - bz^{-1}} - \frac{1}{1 - b^{-1}z^{-1}}, \quad b < |z| < \frac{1}{b}, \tag{10.35}$$

or equivalently,

$$X(z) = \frac{b^2 - 1}{b} \frac{z}{(z - b)(z - b^{-1})}, \quad b < |z| < \frac{1}{b}. \tag{10.36}$$

The corresponding pole-zero pattern and ROC are shown in Figure 10.11(e).

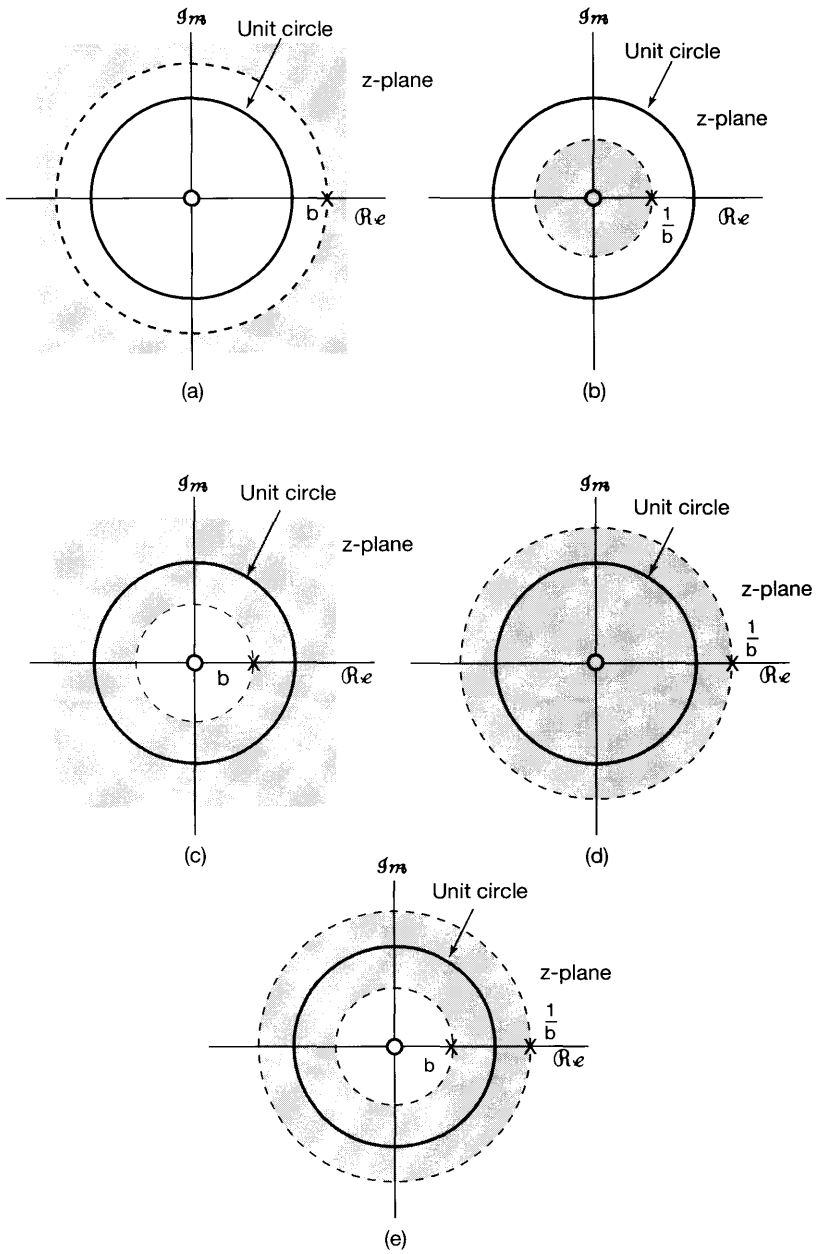


Figure 10.11 Pole-zero plots and ROCs for Example 10.7: (a) eq. (10.33) for $b > 1$; (b) eq. (10.34) for $b > 1$; (c) eq. (10.33) for $0 < b < 1$; (d) eq. (10.34) for $0 < b < 1$; (e) pole-zero plot and ROC for eq. (10.36) with $0 < b < 1$. For $b > 1$, the z-transform of $x[n]$ in eq. (10.31) does not converge for any value of z.

In discussing the Laplace transform in Chapter 9, we remarked that for a rational Laplace transform, the ROC is always bounded by poles or infinity. We observe that in the foregoing examples a similar statement applies to the z-transform, and in fact, this is always true:

Property 7: If the z-transform $X(z)$ of $x[n]$ is rational, then its ROC is bounded by poles or extends to infinity.

Combining Property 7 with Properties 4 and 5, we have

Property 8: If the z-transform $X(z)$ of $x[n]$ is rational, and if $x[n]$ is right sided, then the ROC is the region in the z-plane outside the outermost pole—i.e., outside the circle of radius equal to the largest magnitude of the poles of $X(z)$. Furthermore, if $x[n]$ is causal (i.e., if it is right sided and equal to 0 for $n < 0$), then the ROC also includes $z = \infty$.

Thus, for right-sided sequences with rational transforms, the poles are all closer to the origin than is any point in the ROC.

Property 9: If the z-transform $X(z)$ of $x[n]$ is rational, and if $x[n]$ is left sided, then the ROC is the region in the z-plane inside the innermost nonzero pole—i.e., inside the circle of radius equal to the smallest magnitude of the poles of $X(z)$ other than any at $z = 0$ and extending inward to and possibly including $z = 0$. In particular, if $x[n]$ is anticausal (i.e., if it is left sided and equal to 0 for $n > 0$), then the ROC also includes $z = 0$.

Thus, for left-sided sequences, the poles of $X(z)$ other than any at $z = 0$ are farther from the origin than is any point in the ROC.

For a given pole-zero pattern, or equivalently, a given rational algebraic expression $X(z)$, there are a limited number of different ROCs that are consistent with the preceding properties. To illustrate how different ROCs can be associated with the same pole-zero pattern, we present the following example, which closely parallels Example 9.8.

Example 10.8

Let us consider all of the possible ROCs that can be connected with the function

$$X(z) = \frac{1}{(1 - \frac{1}{3}z^{-1})(1 - 2z^{-1})}. \tag{10.37}$$

The associated pole-zero pattern is shown in Figure 10.12(a). Based on our discussion in this section, there are three possible ROCs that can be associated with this algebraic expression for the z-transform. These ROCs are indicated in Figure 10.12(b)–(d). Each corresponds to a different sequence. Figure 10.12(b) is associated with a right-sided sequence, Figure 10.12(c) with a left-sided sequence, and Figure 10.12(d) with a two-sided sequence. Since Figure 10.12(d) is the only one for which the ROC includes the unit circle, the sequence corresponding to this choice of ROC is the only one of the three for which the Fourier transform converges.

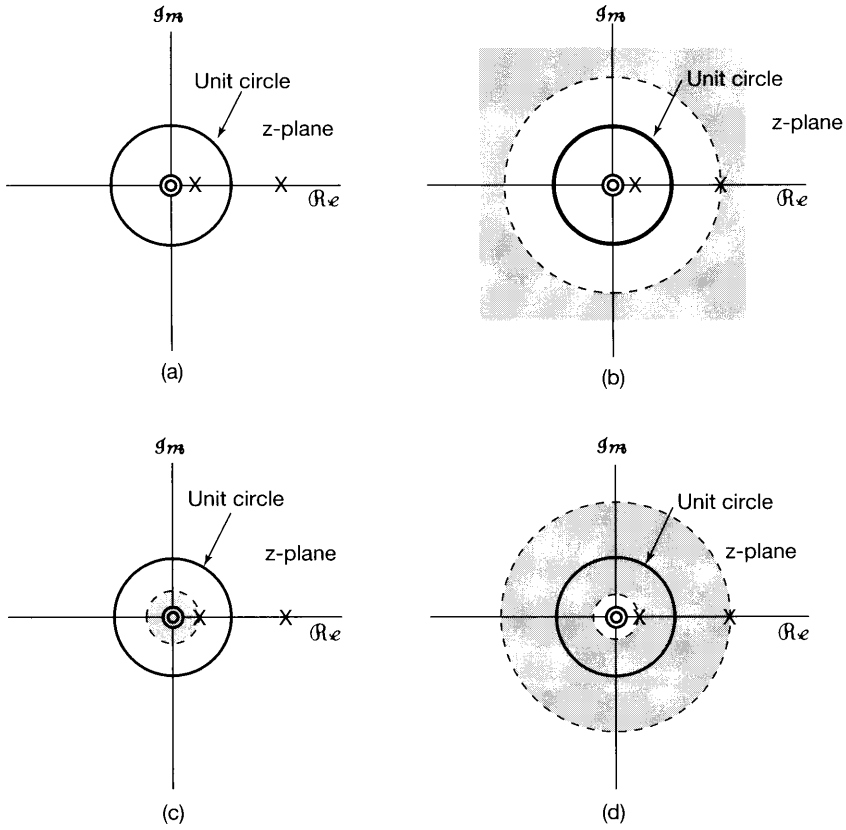


Figure 10.12 The three possible ROCs that can be connected with the expression for the z-transform in Example 10.8: (a) pole-zero pattern for $X(z)$; (b) pole-zero pattern and ROC if $x[n]$ is right sided; (c) pole-zero pattern and ROC if $x[n]$ is left sided; (d) pole-zero pattern and ROC if $x[n]$ is two sided. In each case, the zero at the origin is a second-order zero.

10.3 THE INVERSE z-TRANSFORM

In this section, we consider several procedures for determining a sequence when its z-transform is known. To begin, let us consider the formal relation expressing a sequence in terms of its z-transform. This expression can be obtained on the basis of the interpretation, developed in Section 10.1, of the z-transform as the Fourier transform of an exponentially weighted sequence. Specifically, as expressed in eq. (10.7),

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}, \tag{10.38}$$

for any value of r so that $z = re^{j\omega}$ is inside the ROC. Applying the inverse Fourier transform to both sides of eq. (10.38) yields

$$x[n]r^{-n} = \mathcal{F}^{-1}\{X(re^{j\omega})\},$$

or

$$x[n] = r^n \mathcal{F}^{-1}[X(re^{j\omega})]. \tag{10.39}$$

Using the inverse Fourier transform expression in eq. (5.8), we have

$$x[n] = r^n \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) e^{j\omega n} d\omega,$$

or, moving the exponential factor r^n inside the integral and combining it with the term $e^{j\omega n}$,

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega})(re^{j\omega})^n d\omega. \tag{10.40}$$

That is, we can recover $x[n]$ from its z-transform evaluated along a contour $z = re^{j\omega}$ in the ROC, with r fixed and ω varying over a 2π interval. Let us now change the variable of integration from ω to z. With $z = re^{j\omega}$ and r fixed, $dz = jre^{j\omega} d\omega = jz d\omega$, or $d\omega = (1/j)z^{-1} dz$. The integration in eq. (10.40) is over a 2π interval in ω, which, in terms of z, corresponds to one traversal around the circle $|z| = r$. Consequently, in terms of an integration in the z-plane, eq. (10.40) can be rewritten as

$$\boxed{x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz,} \tag{10.41}$$

where the symbol $\oint$ denotes integration around a counterclockwise closed circular contour centered at the origin and with radius r. The value of r can be chosen as any value for which $X(z)$ converges—i.e., any value such that the circular contour of integration $|z| = r$ is in the ROC. Equation (10.41) is the formal expression for the inverse z-transform and is the discrete-time counterpart of eq. (9.56) for the inverse Laplace transform. As with eq. (9.56), formal evaluation of the inverse transform equation (10.41) requires the use of contour integration in the complex plane. There are, however, a number of alternative procedures for obtaining a sequence from its z-transform. As with Laplace transforms, one particularly useful procedure for rational z-transforms consists of expanding the algebraic expression into a partial-fraction expansion and recognizing the sequences associated with the individual terms. In the following examples, we illustrate the procedure.

Example 10.9

Consider the z-transform

$$X(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad |z| > \tfrac{1}{3}. \tag{10.42}$$

There are two poles, one at $z = 1/3$ and one at $z = 1/4$, and the ROC lies outside the outermost pole. That is, the ROC consists of all points with magnitude greater than that of the pole with the larger magnitude, namely the pole at $z = 1/3$. From Property 4 in Section 10.2, we then know that the inverse transform is a right-sided sequence. As described in the appendix, $X(z)$ can be expanded by the method of partial fractions. For

this example, the partial-fraction expansion, expressed in polynomials in z^{-1}, is

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}. \tag{10.43}$$

Thus, $x[n]$ is the sum of two terms, one with z-transform $1/[1 - (1/4)z^{-1}]$ and the other with z-transform $2/[1 - (1/3)z^{-1}]$. In order to determine the inverse z-transform of each of these individual terms, we must specify the ROC associated with each. Since the ROC for $X(z)$ is outside the outermost pole, the ROC for each individual term in eq. (10.43) must also be outside the pole associated with that term. That is, the ROC for each term consists of all points with magnitude greater than the magnitude of the corresponding pole. Thus,

$$x[n] = x_1[n] + x_2[n], \tag{10.44}$$

where

$$x_1[n] \overset{z}{\longleftrightarrow} \frac{1}{1 - \frac{1}{4}z^{-1}}, \quad |z| > \tfrac{1}{4}, \tag{10.45}$$

$$x_2[n] \overset{z}{\longleftrightarrow} \frac{2}{1 - \frac{1}{3}z^{-1}}, \quad |z| > \tfrac{1}{3}. \tag{10.46}$$

From Example 10.1, we can identify by inspection that

$$x_1[n] = \left(\frac{1}{4}\right)^n u[n] \tag{10.47}$$

and

$$x_2[n] = 2\left(\frac{1}{3}\right)^n u[n], \tag{10.48}$$

and thus,

$$x[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n]. \tag{10.49}$$

Example 10.10

Now let us consider the same algebraic expression for $X(z)$ as in eq. (10.42), but with the ROC for $X(z)$ as $1/4 < |z| < 1/3$. Equation (10.43) is still a valid partial-fraction expansion of the algebraic expression for $X(z)$, but the ROC associated with the individual terms will change. In particular, since the ROC for $X(z)$ is outside the pole at $z = 1/4$, the ROC corresponding to this term in eq. (10.43) is also outside the pole and consists of all points with magnitude greater than 1/4, as it did in the previous example. However, since in this example the ROC for $X(z)$ is inside the pole at $z = 1/3$, that is, since the points in the ROC all have magnitude less than 1/3, the ROC corresponding to this term must also lie inside this pole. Thus, the z-transform pairs for the individual components in eq. (10.44) are

$$x_1[n] \overset{z}{\longleftrightarrow} \frac{1}{1 - \frac{1}{4}z^{-1}}, \quad |z| > \tfrac{1}{4}, \tag{10.50}$$

and

$$x_2[n] \overset{z}{\longleftrightarrow} \frac{2}{1-\frac{1}{3}z^{-1}}, \quad |z| < \tfrac{1}{3}. \tag{10.51}$$

The signal $x_1[n]$ remains as in eq. (10.47), while from Example 10.2, we can identify

$$x_2[n] = -2\left(\frac{1}{3}\right)^n u[-n-1], \tag{10.52}$$

so that

$$x[n] = \left(\frac{1}{4}\right)^n u[n] - 2\left(\frac{1}{3}\right)^n u[-n-1]. \tag{10.53}$$

Example 10.11

Finally, consider $X(z)$ as in eq. (10.42), but now with the ROC $|z| < 1/4$. In this case the ROC is inside both poles, i.e., the points in the ROC all have magnitude smaller than either of the poles at $z = 1/3$ or $z = 1/4$. Consequently the ROC for each term in the partial-fraction expansion in eq. (10.43) must also lie inside the corresponding pole. As a result, the z-transform pair for $x_1[n]$ is given by

$$x_1[n] \overset{z}{\longleftrightarrow} \frac{1}{1-\frac{1}{4}z^{-1}}, \quad |z| < \frac{1}{4}, \tag{10.54}$$

while the z-transform pair for $x_2[n]$ is given by eq. (10.51). Applying the result of Example 10.2 to eq. (10.54), we find that

$$x_1[n] = -\left(\frac{1}{4}\right)^n u[-n-1],$$

so that

$$x[n] = -\left(\frac{1}{4}\right)^n u[-n-1] - 2\left(\frac{1}{3}\right)^n u[-n-1].$$

The foregoing examples illustrate the basic procedure of using partial-fraction expansions to determine inverse z-transforms. As with the corresponding method for the Laplace transform, the procedure relies on expressing the z-transform as a linear combination of simpler terms. The inverse transform of each term can then be obtained by inspection. In particular, suppose that the partial-fraction expansion of $X(z)$ is of the form

$$X(z) = \sum_{i=1}^{m} \frac{A_i}{1-a_i z^{-1}}, \tag{10.55}$$

so that the inverse transform of $X(z)$ equals the sum of the inverse transforms of the individual terms in the equation. If the ROC of $X(z)$ is outside the pole at $z = a_i$, the inverse transform of the corresponding term in eq. (10.55) is $A_i a_i^n u[n]$. On the other hand, if the ROC of $X(z)$ is inside the pole at $z = a_i$, the inverse transform of this term is $-A_i a_i^n u[-n-1]$. In general, the partial-fraction expansion of a rational transform may include terms in

addition to the first-order terms in eq. (10.55). In Section 10.6, we list a number of other z-transform pairs that can be used in conjunction with the z-transform properties to be developed in Section 10.5 to extend the inverse transform method outlined in the preceding example to arbitrary rational z-transforms.

Another very useful procedure for determining the inverse z-transform relies on a power-series expansion of $X(z)$. This procedure is motivated by the observation that the definition of the z-transform given in eq. (10.3) can be interpreted as a power series involving both positive and negative powers of z. The coefficients in this power series are, in fact, the sequence values $x[n]$. To illustrate how a power-series expansion can be used to obtain the inverse z-transform, let us consider three examples.

Example 10.12

Consider the z-transform

$$X(z) = 4z^2 + 2 + 3z^{-1}, \quad 0 < |z| < \infty. \tag{10.56}$$

From the power-series definition of the z-transform in eq. (10.3), we can determine the inverse transform of $X(z)$ by inspection:

$$x[n] = \begin{cases} 4, & n = -2 \\ 2, & n = 0 \\ 3, & n = 1 \\ 0, & \text{otherwise} \end{cases}.$$

That is,

$$x[n] = 4\delta[n + 2] + 2\delta[n] + 3\delta[n - 1]. \tag{10.57}$$

Comparing eqs. (10.56) and (10.57), we see that different powers of z serve as placeholders for sequence values at different points in time; i.e., if we simply use the transform pair

$$\delta[n + n_0] \overset{z}{\longleftrightarrow} z^{n_0},$$

we can immediately pass from eq. (10.56) to (10.57) and vice versa.

Example 10.13

Consider

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|.$$

This expression can be expanded in a power series by long division:

$$\begin{array}{r} 1 + az^{-1} + a^2z^{-2} + \cdots \\ 1 - az^{-1} \overline{)\,1} \\ \underline{1 - az^{-1}} \\ az^{-1} \\ \underline{az^{-1} - a^2z^{-2}} \\ a^2z^{-2} \end{array},$$

or

$$\frac{1}{1 - az^{-1}} = 1 + az^{-1} + a^2z^{-2} + \dots. \tag{10.58}$$

The series expansion of eq. (10.58) converges, since $|z| > |a|$, or equivalently, $|az^{-1}| < 1$. Comparing this equation with the definition of the z-transform in equation (10.3), we see, by matching terms in powers of z, that $x[n] = 0, n < 0$; $x[0] = 1$; $x[1] = a$; $x[2] = a^2$; and in general, $x[n] = a^n u[n]$, which is consistent with Example 10.1.

If, instead, the ROC of $X(z)$ is specified as $|z| < |a|$ or, equivalently, $|az^{-1}| > 1$, then the power-series expansion for $1/(1 - az^{-1})$ in eq. (10.58) does not converge. However, we can obtain a convergent power series by long division again:

$$\begin{array}{r} -a^{-1}z - a^{-2}z^2 - \cdots \\ -az^{-1} + 1 \overline{)\ 1} \\ \underline{1 - a^{-1}z} \\ a^{-1}z \end{array},$$

or

$$\frac{1}{1 - az^{-1}} = -a^{-1}z - a^{-2}z^2 - \dots. \tag{10.59}$$

In this case, then, $x[n] = 0, n \geq 0$; and $x[-1] = -a^{-1}$, $x[-2] = -a^{-2}, \dots$; that is, $x[n] = -a^n u[-n - 1]$. This is consistent with Example 10.2.

The power-series expansion method for obtaining the inverse z-transform is particularly useful for nonrational z-transforms, which we illustrate with the following example:

Example 10.14

Consider the z-transform

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|. \tag{10.60}$$

With $|z| > |a|$, or, equivalently, $|az^{-1}| < 1$, eq. (10.60) can be expanded in a power series using the Taylor's series expansion

$$\log(1 + v) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}v^n}{n}, \quad |v| < 1. \tag{10.61}$$

Applying this to eq. (10.60), we have

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}a^n z^{-n}}{n}, \tag{10.62}$$

from which we can identify

$$x[n] = \begin{cases} (-1)^{n+1}\dfrac{a^n}{n}, & n \geq 1 \\ 0, & n \leq 0 \end{cases}, \tag{10.63}$$

or equivalently,

$$x[n] = \frac{-(-a)^n}{n} u[n-1].$$

In Problem 10.63 we consider a related example with region of convergence $|z| < |a|$.

10.4 GEOMETRIC EVALUATION OF THE FOURIER TRANSFORM FROM THE POLE-ZERO PLOT

In Section 10.1 we noted that the z-transform reduces to the Fourier transform for $|z| = 1$ (i.e., for the contour in the z-plane corresponding to the unit circle), provided that the ROC of the z-transform includes the unit circle, so that the Fourier transform converges. In a similar manner, we saw in Chapter 9 that, for continuous-time signals, the Laplace transform reduces to the Fourier transform on the $j\omega$-axis in the s-plane. In Section 9.4, we also discussed the geometric evaluation of the continuous-time Fourier transform from the pole-zero plot. In the discrete-time case, the Fourier transform can again be evaluated geometrically by considering the pole and zero vectors in the z-plane. However, since in this case the rational function is to be evaluated on the contour $|z| = 1$, we consider the vectors from the poles and zeros to the unit circle rather than to the imaginary axis. To illustrate the procedure, let us consider first-order and second-order systems, as discussed in Section 6.6.

10.4.1 First-Order Systems

The impulse response of a first-order causal discrete-time system is of the general form

$$h[n] = a^n u[n], \tag{10.64}$$

and from Example 10.1, its z-transform is

$$H(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|. \tag{10.65}$$

For $|a| < 1$, the ROC includes the unit circle, and consequently, the Fourier transform of $h[n]$ converges and is equal to $H(z)$ for $z = e^{j\omega}$. Thus, the frequency response for the first-order system is

$$H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}. \tag{10.66}$$

Figure 10.13(a) depicts the pole-zero plot for $H(z)$ in eq. (10.65), including the vectors from the pole (at $z = a$) and zero (at $z = 0$) to the unit circle. With this plot, the geometric evaluation of $H(z)$ can be carried out using the same procedure as described in Section 9.4. In particular, if we wish to evaluate the frequency response in eq. (10.65), we perform the evaluation for values of z of the form $z = e^{j\omega}$. The magnitude of the frequency response at frequency ω is the ratio of the length of the vector $\mathbf{v}_1$ to the length of the vector $\mathbf{v}_2$ shown in Figure 10.13(a). The phase of the frequency response is the angle of $\mathbf{v}_1$ with respect to the real axis minus the angle of $\mathbf{v}_2$. Furthermore, the vector $\mathbf{v}_1$ from

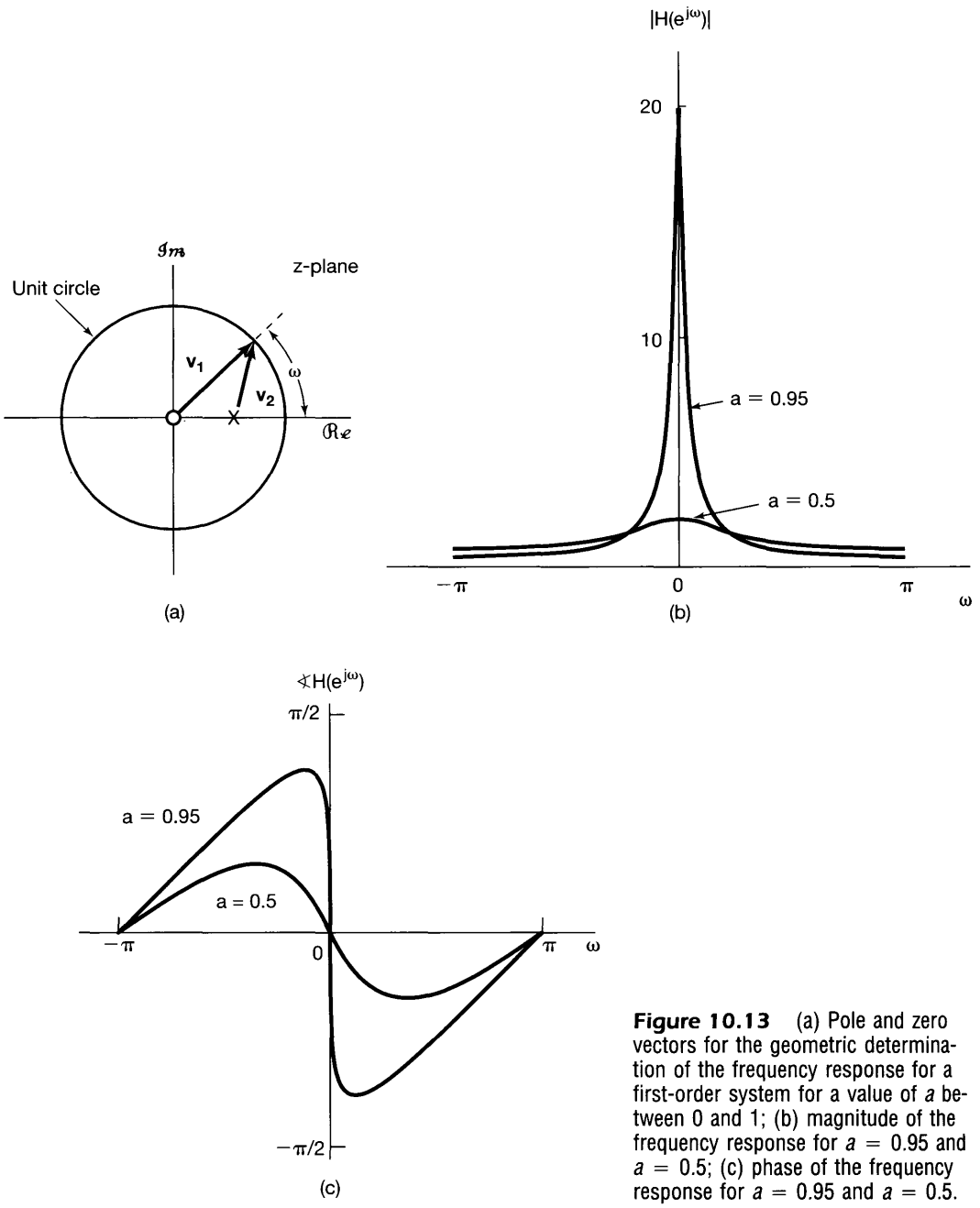


Figure 10.13 (a) Pole and zero vectors for the geometric determination of the frequency response for a first-order system for a value of a between 0 and 1; (b) magnitude of the frequency response for $a = 0.95$ and $a = 0.5$; (c) phase of the frequency response for $a = 0.95$ and $a = 0.5$.

the zero at the origin to the unit circle has a constant length of unity and thus has no effect on the magnitude of $H(e^{j\omega})$. The phase contributed to $H(e^{j\omega})$ by the zero is the angle of the zero vector with respect to the real axis, which we see is equal to ω. For $0 < a < 1$, the pole vector has minimum length at $\omega = 0$ and monotonically increases in length as ω increases from zero to π. Thus, the magnitude of the frequency response will be

maximum at $\omega = 0$ and will decrease monotonically as ω increases from 0 to π. The angle of the pole vector begins at zero and increases monotonically as ω increases from zero to π. The resulting magnitude and phase of $H(e^{j\omega})$ are shown in Figures 10.13(b) and (c), respectively, for two values of a.

The magnitude of the parameter a in the discrete-time first-order system plays a role similar to that of the time constant τ for the continuous-time first-order system of Section 9.4.1. Note first that, as illustrated in Figure 10.13, the magnitude of the peak of $H(e^{j\omega})$ at $\omega = 0$ decreases as $|a|$ decreases toward 0. Also, as was discussed in Section 6.6.1 and illustrated in Figures 6.26 and 6.27, as $|a|$ decreases, the impulse response decays more sharply and the step response settles more quickly. With multiple poles, the speed of response associated with each pole is related to its distance from the origin, with those closest to the origin contributing the most rapidly decaying terms in the impulse response. This is further illustrated in the case of second-order systems, which we consider next.

10.4.2 Second-Order Systems

Next, let us consider the class of second-order systems as discussed in Section 6.6.2, with impulse response and frequency response given in eqs. (6.64) and (6.60), which we respectively repeat here as

$$h[n] = r^n \frac{\sin(n+1)\theta}{\sin\theta} u[n] \tag{10.67}$$

and

$$H(e^{j\omega}) = \frac{1}{1 - 2r\cos\theta e^{-j\omega} + r^2 e^{-j2\omega}}, \tag{10.68}$$

where $0 < r < 1$ and $0 \leq \theta \leq \pi$. Since $H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$, we can infer from eq. (10.68) that the system function, corresponding to the z-transform of the system impulse response, is

$$H(z) = \frac{1}{1 - (2r\cos\theta)z^{-1} + r^2 z^{-2}}. \tag{10.69}$$

The poles of $H(z)$ are located at

$$z_1 = re^{j\theta}, \quad z_2 = re^{-j\theta}, \tag{10.70}$$

and there is a double zero at $z = 0$. The pole-zero plot and the pole and zero vectors with $0 < \theta < \pi/2$ are illustrated in Figure 10.14(a). In this case, the magnitude of the frequency response equals the square of the magnitude of $\mathbf{v}_1$ (since there is a double zero at the origin) divided by the product of the magnitudes of $\mathbf{v}_2$ and $\mathbf{v}_3$. Because the length of the vector $\mathbf{v}_1$ from the zero at the origin is 1 for all values of ω, the magnitude of the frequency response equals the reciprocal of the product of the lengths of the two pole vectors $\mathbf{v}_2$ and $\mathbf{v}_3$. Also, the phase of the frequency response equals twice the angle of $\mathbf{v}_1$ with respect to the real axis minus the sums of the angles of $\mathbf{v}_2$ and $\mathbf{v}_3$. In Figure 10.14(b) we show the magnitude of the frequency response for $r = 0.95$ and $r = 0.75$, while in Figure 10.14(c) we display the phase of $H(e^{j\omega})$ for the same two values of r. We note in particular that, as we move

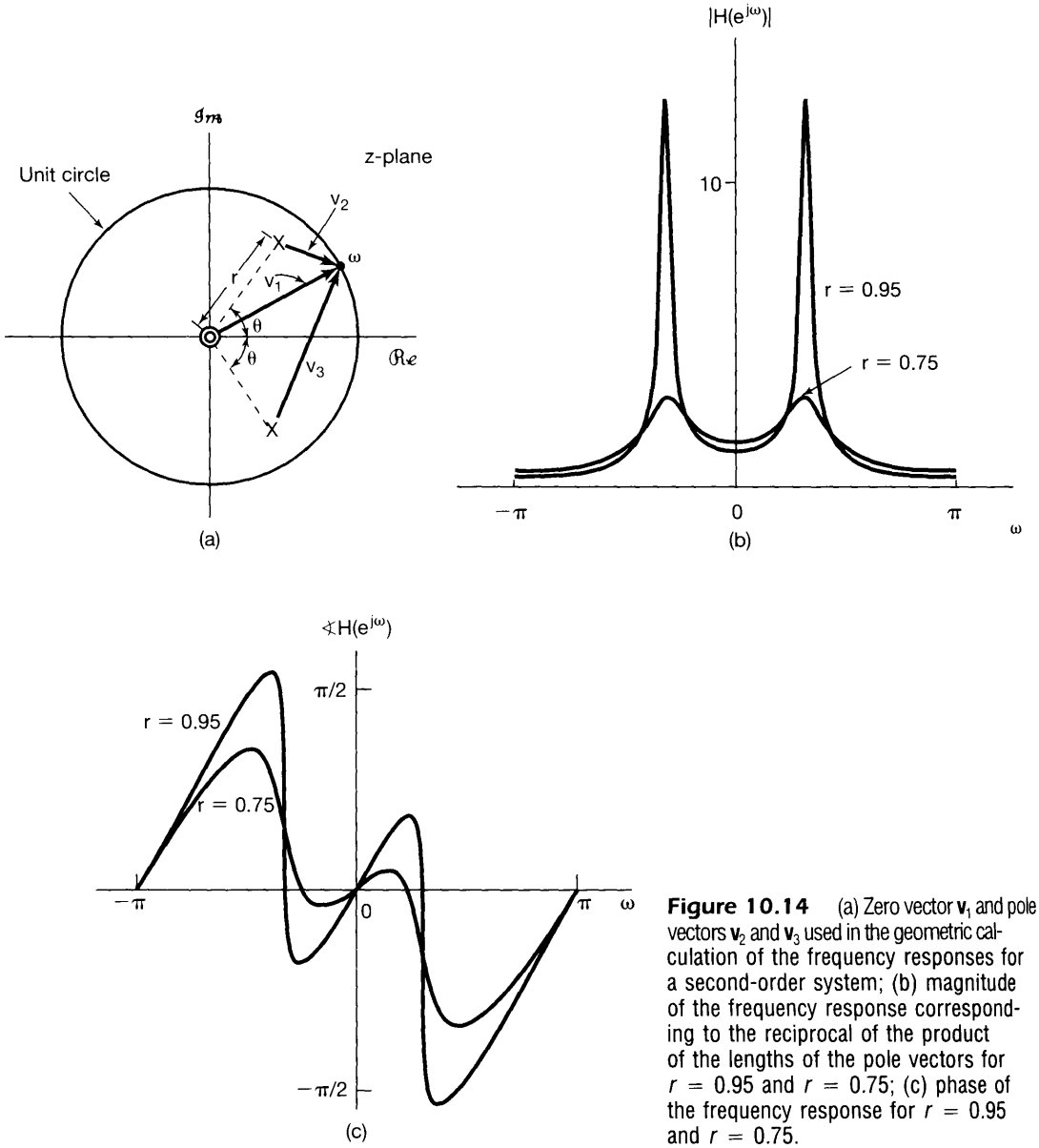


Figure 10.14 (a) Zero vector $\mathbf{v}_1$ and pole vectors $\mathbf{v}_2$ and $\mathbf{v}_3$ used in the geometric calculation of the frequency responses for a second-order system; (b) magnitude of the frequency response corresponding to the reciprocal of the product of the lengths of the pole vectors for $r = 0.95$ and $r = 0.75$; (c) phase of the frequency response for $r = 0.95$ and $r = 0.75$.

along the unit circle from $\omega = 0$ toward $\omega = \pi$, the length of the vector $\mathbf{v}_2$ first decreases and then increases, with a minimum length in the vicinity of the pole location, at $\omega = \theta$. This is consistent with the fact that the magnitude of the frequency response peaks for ω near θ when the length of the vector $\mathbf{v}_2$ is small. Based on the behavior of the pole vectors, it is also evident that as r increases toward unity, the minimum length of the pole vectors will decrease, causing the frequency response to peak more sharply with increasing r. Also, for r near unity, the angle of the vector $\mathbf{v}_2$ changes sharply for ω in the vicinity of θ. Furthermore, from the form of the impulse response [eq. (10.67) and Figure 6.29] or the

step response [eq. (6.67) and Figure 6.30], we see, as we did with the first-order system, that as the poles move closer to the origin, corresponding to r decreasing, the impulse response decays more rapidly and the step response settles more quickly.

10.5 PROPERTIES OF THE z-TRANSFORM

As with the other transforms we have developed, the z-transform possesses a number of properties that make it an extremely valuable tool in the study of discrete-time signals and systems. In this section, we summarize many of these properties. Their derivations are analogous to the derivations of properties for the other transforms, and thus, many are left as exercises at the end of the chapter. (See Problems 10.43 and 10.51–10.54.)

10.5.1 Linearity

If

$$x_1[n] \overset{z}{\longleftrightarrow} X_1(z), \quad \text{with ROC} = R_1,$$

and

$$x_2[n] \overset{z}{\longleftrightarrow} X_2(z), \quad \text{with ROC} = R_2,$$

then

$$\boxed{ax_1[n] + bx_2[n] \overset{z}{\longleftrightarrow} aX_1(z) + bX_2(z), \quad \text{with ROC containing } R_1 \cap R_2.} \tag{10.71}$$

As indicated, the ROC of the linear combination is at least the intersection of R_1 and R_2. For sequences with rational z-transforms, if the poles of $aX_1(z) + bX_2(z)$ consist of all of the poles of $X_1(z)$ and $X_2(z)$ (i.e., if there is no pole-zero cancellation), then the region of convergence will be exactly equal to the overlap of the individual regions of convergence. If the linear combination is such that some zeros are introduced that cancel poles, then the region of convergence may be larger. A simple example of this occurs when $x_1[n]$ and $x_2[n]$ are both of infinite duration, but the linear combination is of finite duration. In this case the region of convergence of the linear combination is the entire z-plane, with the possible exception of zero and/or infinity. For example, the sequences $a^n u[n]$ and $a^n u[n-1]$ both have a region of convergence defined by $|z| > |a|$, but the sequence corresponding to the difference $(a^n u[n] - a^n u[n-1]) = \delta[n]$ has a region of convergence that is the entire z-plane.

10.5.2 Time Shifting

If

$$x[n] \overset{z}{\longleftrightarrow} X(z), \quad \text{with ROC} = R,$$

then

$$\boxed{x[n-n_0] \overset{\mathcal{Z}}{\longleftrightarrow} z^{-n_0}X(z), \quad \text{with ROC} = R, \text{ except for the possible addition or deletion of the origin or infinity.}} \tag{10.72}$$

Because of the multiplication by z^{-n_0}, for $n_0 > 0$ poles will be introduced at $z = 0$, which may cancel corresponding zeros of $X(z)$ at $z = 0$. Consequently, $z = 0$ may be a pole of $z^{-n_0}X(z)$ while it may not be a pole of $X(z)$. In this case the ROC for $z^{-n_0}X(z)$ equals the ROC of $X(z)$ but with the origin deleted. Similarly, if $n_0 < 0$, zeros will be introduced at $z = 0$, which may cancel corresponding poles of $X(z)$ at $z = 0$. Consequently, $z = 0$ may be a zero of $z^{-n_0}X(z)$ while it may not be a pole of $X(z)$. In this case $z = \infty$ is a pole of $z^{-n_0}X(z)$, and thus the ROC for $z^{-n_0}X(z)$ equals the ROC of $X(z)$ but with the $z = \infty$ deleted.

10.5.3 Scaling in the z-Domain

If

$$x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z), \quad \text{with ROC} = R,$$

then

$$\boxed{z_0^n x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X\left(\frac{z}{z_0}\right), \text{ with ROC} = |z_0|R,} \tag{10.73}$$

where $|z_0|R$ is the scaled version of R. That is, if z is a point in the ROC of $X(z)$, then the point $|z_0|z$ is in the ROC of $X(z/z_0)$. Also, if $X(z)$ has a pole (or zero) at $z = a$, then $X(z/z_0)$ has a pole (or zero) at $z = z_0 a$.

An important special case of eq. (10.73) is when $z_0 = e^{j\omega_0}$. In this case, $|z_0|R = R$ and

$$e^{j\omega_0 n}x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(e^{-j\omega_0}z). \tag{10.74}$$

The left-hand side of eq. (10.74) corresponds to multiplication by a complex exponential sequence. The right-hand side can be interpreted as a rotation in the z-plane; that is, all pole-zero locations rotate in the z-plane by an angle of ω_0, as illustrated in Figure 10.15. This can be seen by noting that if $X(z)$ has a factor of the form $1 - az^{-1}$, then $X(e^{-j\omega_0}z)$ will have a factor $1 - ae^{j\omega_0}z^{-1}$, and thus, a pole or zero at $z = a$ in $X(z)$ will become a pole or zero at $z = ae^{j\omega_0}$ in $X(e^{-j\omega_0}z)$. The behavior of the z-transform on the unit circle will then also shift by an angle of ω_0. This is consistent with the frequency-shifting property set forth in Section 5.3.3, where multiplication with a complex exponential in the time domain was shown to correspond to a shift in frequency of the Fourier transform. Also, in the more general case when $z_0 = r_0 e^{j\omega_0}$ in eq. (10.73), the pole and zero locations are rotated by ω_0 and scaled in magnitude by a factor of r_0.

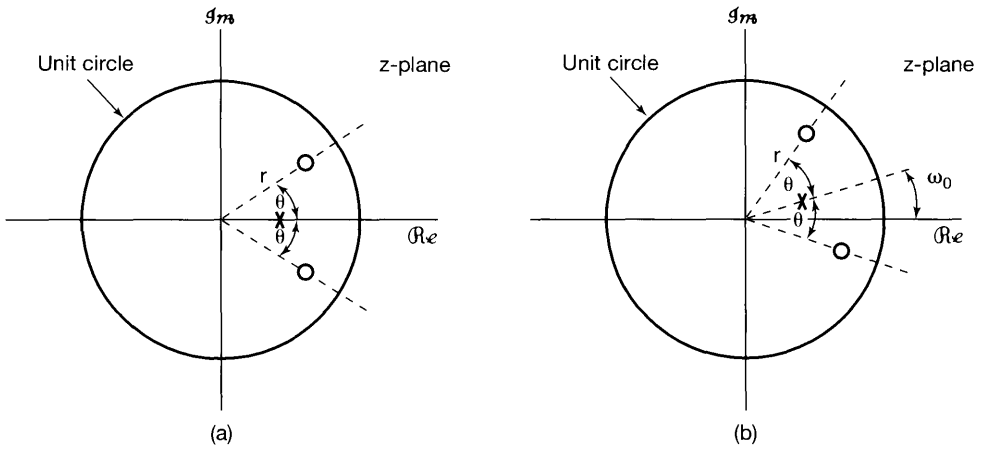


Figure 10.15 Effect on the pole-zero plot of time-domain multiplication by a complex exponential sequence $e^{j\omega_0 n}$: (a) pole-zero pattern for the z-transform for a signal $x[n]$; (b) pole-zero pattern for the z-transform of $x[n]e^{j\omega_0 n}$.

10.5.4 Time Reversal

If

$$x[n] \overset{z}{\longleftrightarrow} X(z), \quad \text{with ROC} = R,$$

then

$$\boxed{x[-n] \overset{z}{\longleftrightarrow} X(\tfrac{1}{z}), \quad \text{with ROC} = \tfrac{1}{R}.} \tag{10.75}$$

That is, if z_0 is in the ROC for $x[n]$, then $1/z_0$ is in the ROC for $x[-n]$.

10.5.5 Time Expansion

As we discussed in Section 5.3.7, the continuous-time concept of time scaling does not directly extend to discrete time, since the discrete-time index is defined only for integer values. However, the discrete-time concept of time expansion—i.e., of inserting a number of zeros between successive values of a discrete-time sequence $x[n]$—can be defined and does play an important role in discrete-time signal and system analysis. Specifically, the sequence $x_{(k)}[n]$, introduced in Section 5.3.7 and defined as

$$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{if } n \text{ is not a multiple of } k \end{cases} \tag{10.76}$$

has $k - 1$ zeros inserted between successive values of the original signal. In this case, if

$$x[n] \overset{z}{\longleftrightarrow} X(z), \quad \text{with ROC} = R,$$

then

$$\boxed{x_{(k)}[n] \overset{z}{\longleftrightarrow} X(z^k), \quad \text{with ROC} = R^{1/k}.} \tag{10.77}$$

That is, if z is in the ROC of $X(z)$, then the point $z^{1/k}$ is in the ROC of $X(z^k)$. Also, if $X(z)$ has a pole (or zero) at $z = a$, then $X(z^k)$ has a pole (or zero) at $z = a^{1/k}$.

The interpretation of this result follows from the power-series form of the z-transform, from which we see that the coefficient of the term z^{-n} equals the value of the signal at time n. That is, with

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n},$$

it follows that

$$X(z^k) = \sum_{n=-\infty}^{+\infty} x[n](z^k)^{-n} = \sum_{n=-\infty}^{+\infty} x[n]z^{-kn}. \tag{10.78}$$

Examining the right-hand side of eq. (10.78), we see that the only terms that appear are of the form z^{-kn}. In other words, the coefficient of the term z^{-m} in this power series equals 0 if m is not a multiple of k and equals $x[m/k]$ if m is a multiple of k. Thus, the inverse transform of eq. (10.78) is $x_{(k)}[n]$.

10.5.6 Conjugation

If

$$x[n] \overset{z}{\longleftrightarrow} X(z), \quad \text{with ROC} = R, \tag{10.79}$$

then

$$\boxed{x^*[n] \overset{z}{\longleftrightarrow} X^*(z^*), \quad \text{with ROC} = R.} \tag{10.80}$$

Consequently, if $x[n]$ is real, we can conclude from eq. (10.80) that

$$X(z) = X^*(z^*).$$

Thus, if $X(z)$ has a pole (or zero) at $z = z_0$, it must also have a pole (or zero) at the complex conjugate point $z = z_0^*$. For example, the transform $X(z)$ for the real signal $x[n]$ in Example 10.4 has poles at $z = (1/3)e^{\pm j\pi/4}$.

10.5.7 The Convolution Property

If

$$x_1[n] \overset{z}{\longleftrightarrow} X_1(z), \quad \text{with ROC} = R_1,$$

and

$$x_2[n] \overset{z}{\longleftrightarrow} X_2(z), \quad \text{with ROC} = R_2,$$

then

$$\boxed{x_1[n] * x_2[n] \overset{z}{\longleftrightarrow} X_1(z)X_2(z), \quad \text{with ROC containing } R_1 \cap R_2.} \tag{10.81}$$

Just as with the convolution property for the Laplace transform, the ROC of $X_1(z)X_2(z)$ includes the intersection of R_1 and R_2 and may be larger if pole-zero cancellation occurs in the product. The convolution property for the z-transform can be derived in a variety of different ways. A formal derivation is developed in Problem 10.56. A derivation can also be carried out analogous to that used for the convolution property for the continuous-time Fourier transform in Section 4.4, which relied on the interpretation of the Fourier transform as the change in amplitude of a complex exponential through an LTI system.

For the z-transform, there is another often useful interpretation of the convolution property. From the definition in eq. (10.3), we recognize the z-transform as a series in z^{-1} where the coefficient of z^{-n} is the sequence value $x[n]$. In essence, the convolution property equation (10.81) states that when two polynomials or power series $X_1(z)$ and $X_2(z)$ are multiplied, the coefficients in the polynomial representing the product are the convolution of the coefficients in the polynomials $X_1(z)$ and $X_2(z)$. (See Problem 10.57).

Example 10.15

Consider an LTI system for which

$$y[n] = h[n] * x[n], \tag{10.82}$$

where

$$h[n] = \delta[n] - \delta[n-1].$$

Note that

$$\delta[n] - \delta[n-1] \overset{z}{\longleftrightarrow} 1 - z^{-1}, \tag{10.83}$$

with ROC equal to the entire z-plane except the origin. Also, the z-transform in eq. (10.83) has a zero at $z = 1$. From eq. (10.81), we see that if

$$x[n] \overset{z}{\longleftrightarrow} X(z), \quad \text{with ROC} = R,$$

then

$$y[n] \overset{z}{\longleftrightarrow} (1 - z^{-1})X(z), \tag{10.84}$$

with ROC equal to R, with the possible deletion of $z = 0$ and/or addition of $z = 1$.

Note that for this system

$$y[n] = [\delta[n] - \delta[n-1]] * x[n] = x[n] - x[n-1].$$

That is, $y[n]$ is the first difference of the sequence $x[n]$. Since the first-difference operation is commonly thought of as a discrete-time counterpart to differentiation, eq. (10.83) can be thought of as the z-transform counterpart of the Laplace transform differentiation property presented in Section 9.5.7.

Example 10.16

Suppose we now consider the inverse of first differencing, namely, accumulation or summation. Specifically, let $w[n]$ be the running sum of $x[n]$:

$$w[n] = \sum_{k=-\infty}^{n} x[k] = u[n] * x[n]. \tag{10.85}$$

Then, using eq. (10.81) together with the z-transform of the unit step in Example 10.1, we see that

$$w[n] = \sum_{k=-\infty}^{n} x[k] \overset{z}{\longleftrightarrow} \frac{1}{1 - z^{-1}} X(z), \tag{10.86}$$

with ROC including at least the intersection of R with $|z| > 1$. Eq. (10.86) is the discrete-time z-transform counterpart of the integration property in Section 9.5.9.

10.5.8 Differentiation in the z-Domain

If

$$x[n] \overset{z}{\longleftrightarrow} X(z), \quad \text{with ROC} = R,$$

then

$$\boxed{nx[n] \overset{z}{\longleftrightarrow} -z\frac{dX(z)}{dz}, \quad \text{with ROC} = R.} \tag{10.87}$$

This property follows in a straightforward manner by differentiating both sides of the expression for the z-transform given in eq. (10.3). As an example of the use of this property, let us apply it to determining the inverse z-transform considered in Example 10.14.

Example 10.17

If

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|, \tag{10.88}$$

then

$$nx[n] \overset{z}{\longleftrightarrow} -z\frac{dX(z)}{dz} = \frac{az^{-1}}{1 + az^{-1}}, \quad |z| > |a|. \tag{10.89}$$

By differentiating, we have converted the z-transform to a rational expression. The inverse z-transform of the right-hand side of eq. (10.89) can be obtained by using Example 10.1 together with the time-shifting property, eq. (10.72), set forth in Section 10.5.2.

Specifically, from Example 10.1 and the linearity property,

$$a(-a)^n u[n] \overset{z}{\longleftrightarrow} \frac{a}{1+az^{-1}}, \quad |z| > |a|. \tag{10.90}$$

Combining this with the time-shifting property yields

$$a(-a)^{n-1} u[n-1] \overset{z}{\longleftrightarrow} \frac{az^{-1}}{1+az^{-1}}, \quad |z| > |a|.$$

Consequently,

$$x[n] = \frac{-(-a)^n}{n} u[n-1]. \tag{10.91}$$

Example 10.18

As another example of the use of the differentiation property, consider determining the inverse z-transform for

$$X(z) = \frac{az^{-1}}{(1-az^{-1})^2}, \quad |z| > |a|. \tag{10.92}$$

From Example 10.1,

$$a^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|, \tag{10.93}$$

and hence,

$$na^n u[n] \overset{z}{\longleftrightarrow} -z\frac{d}{dz}\left(\frac{1}{1-az^{-1}}\right) = \frac{az^{-1}}{(1-az^{-1})^2}, \quad |z| > |a|. \tag{10.94}$$

10.5.9 The Initial-Value Theorem

If $x[n] = 0, n < 0$, then

$$x[0] = \lim_{z\to\infty} X(z). \tag{10.95}$$

This property follows by considering the limit of each term individually in the expression for the z-transform, with $x[n]$ zero for $n < 0$. With this constraint,

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}.$$

As $z \to \infty$, $z^{-n} \to 0$ for $n > 0$, whereas for $n = 0$, $z^{-n} = 1$. Thus, eq. (10.95) follows.

As one consequence of the initial-value theorem, for a causal sequence, if $x[0]$ is finite, then $\lim_{z\to\infty} X(z)$ is finite. Consequently, with $X(z)$ expressed as a ratio of polynomials in z, the order of the numerator polynomial cannot be greater than the order of the denominator polynomial; or, equivalently, the number of finite zeros of $X(z)$ cannot be greater than the number of finite poles.

Example 10.19

The initial-value theorem can also be useful in checking the correctness of the z-transform calculation for a signal. For example, consider the signal $x[n]$ in Example 10.3. From eq. (10.12), we see that $x[0] = 1$. Also, from eq. (10.14),

$$\lim_{z\to\infty} X(z) = \lim_{z\to\infty} \frac{1 - \frac{3}{2}z^{-1}}{(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{2}z^{-1})} = 1,$$

which is consistent with the initial-value theorem.

10.5.10 Summary of Properties

In Table 10.1, we summarize the properties of the z-transform.

10.6 SOME COMMON z-TRANSFORM PAIRS

As with the inverse Laplace transform, the inverse z-transform can often be easily evaluated by expressing $X(z)$ as a linear combination of simpler terms, the inverse transforms of which are recognizable. In Table 10.2, we have listed a number of useful z-transform pairs. Each of these can be developed from previous examples in combination with the properties of the z-transform listed in Table 10.1. For example, transform pairs 2 and 5 follow directly from Example 10.1, and transform pair 7 is developed in Example 10.18. These, together with the time-reversal and time-shifting properties set forth in Sections 10.5.4 and 10.5.2, respectively, then lead to transform pairs 3, 6, and 8. Transform pairs 9 and 10 can be developed using transform pair 2 together with the linearity and scaling properties developed in Sections 10.5.1 and 10.5.3, respectively.

10.7 ANALYSIS AND CHARACTERIZATION OF LTI SYSTEMS USING z-TRANSFORMS

The z-transform plays a particularly important role in the analysis and representation of discrete-time LTI systems. From the convolution property presented in Section 10.5.7,

$$Y(z) = H(z)X(z), \tag{10.96}$$

where $X(z)$, $Y(z)$, and $H(z)$ are the z-transforms of the system input, output, and impulse response, respectively. $H(z)$ is referred to as the *system function* or *transfer function* of the system. For z evaluated on the unit circle (i.e., for $z = e^{j\omega}$), $H(z)$ reduces to the frequency response of the system, provided that the unit circle is in the ROC for $H(z)$. Also, from our discussion in Section 3.2, we know that if the input to an LTI system is the complex exponential signal $x[n] = z^n$, then the output will be $H(z)z^n$. That is, z^n is an eigenfunction of the system with eigenvalue given by $H(z)$, the z-transform of the impulse response.

Many properties of a system can be tied directly to characteristics of the poles, zeros, and region of convergence of the system function, and in this section we illustrate some of these relationships by examining several important system properties and an important class of systems.

TABLE 10.1 PROPERTIES OF THE z-TRANSFORM

Section	Property	Signal	z-Transform	ROC
		$x[n]$	$X(z)$	R
		$x_1[n]$	$X_1(z)$	R_1
		$x_2[n]$	$X_2(z)$	R_2
10.5.1	Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	At least the intersection of R_1 and R_2
10.5.2	Time shifting	$x[n - n_0]$	$z^{-n_0}X(z)$	R, except for the possible addition or deletion of the origin
10.5.3	Scaling in the z-domain	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$	R
		$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$z_0 R$
		$a^n x[n]$	$X(a^{-1}z)$	Scaled version of R (i.e., $\lvert a\rvert R$ = the set of points $\{\lvert a\rvert z\}$ for z in R)
10.5.4	Time reversal	$x[-n]$	$X(z^{-1})$	Inverted R (i.e., R^{-1} = the set of points z^{-1}, where z is in R)
10.5.5	Time expansion	$x_{(k)}[n] = \begin{cases} x[r], & n = rk \\ 0, & n \neq rk \end{cases}$ for some integer r	$X(z^k)$	$R^{1/k}$ (i.e., the set of points $z^{1/k}$, where z is in R)
10.5.6	Conjugation	$x^*[n]$	$X^*(z^*)$	R
10.5.7	Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least the intersection of R_1 and R_2
10.5.7	First difference	$x[n] - x[n-1]$	$(1 - z^{-1})X(z)$	At least the intersection of R and $\lvert z\rvert > 0$
10.5.7	Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1 - z^{-1}}X(z)$	At least the intersection of R and $\lvert z\rvert > 1$
10.5.8	Differentiation in the z-domain	$nx[n]$	$-z\frac{dX(z)}{dz}$	R
10.5.9		Initial Value Theorem If $x[n] = 0$ for $n < 0$, then $x[0] = \lim_{z\to\infty} X(z)$		

TABLE 10.2 SOME COMMON z-TRANSFORM PAIRS

Signal	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
3. $-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert < 1$
4. $\delta[n-m]$	z^{-m}	All z, except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$
6. $-\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert < \lvert\alpha\rvert$
7. $n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert > \lvert\alpha\rvert$
8. $-n\alpha^n u[-n-1]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert < \lvert\alpha\rvert$
9. $[\cos\omega_0 n]u[n]$	$\frac{1-[\cos\omega_0]z^{-1}}{1-[2\cos\omega_0]z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
10. $[\sin\omega_0 n]u[n]$	$\frac{[\sin\omega_0]z^{-1}}{1-[2\cos\omega_0]z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
11. $[r^n\cos\omega_0 n]u[n]$	$\frac{1-[r\cos\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
12. $[r^n\sin\omega_0 n]u[n]$	$\frac{[r\sin\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$

10.7.1 Causality

A causal LTI system has an impulse response $h[n]$ that is zero for $n < 0$, and therefore is right-sided. From Property 4 in Section 10.2 we then know that the ROC of $H(z)$ is the exterior of a circle in the z-plane. For some systems, e.g., if $h[n] = \delta[n]$, so that $H(z) = 1$, the ROC can extend all the way in to and possibly include the origin. Also, in general, for a right-sided impulse response, the ROC may or may not include infinity. For example, if $h[n] = \delta[n+1]$, then $H(z) = z$, which has a pole at infinity. However, as we saw in Property 8 in Section 10.2, for a causal system the power series

$$H(z) = \sum_{n=0}^{\infty} h[n]z^{-n}$$

does not include any positive powers of z. Consequently, the ROC includes infinity. Summarizing, we have the follow principle:

> A discrete-time LTI system is causal if and only if the ROC of its system function is the exterior of a circle, including infinity.

If $H(z)$ is rational, then, from Property 8 in Section 10.2, for the system to be causal, the ROC must be outside the outermost pole and infinity must be in the ROC. Equivalently, the limit of $H(z)$ as $z \longrightarrow \infty$ must be finite. As we discussed in Section 10.5.9, this is equivalent to the numerator of $H(z)$ having degree no larger than the denominator when both are expressed as polynomials in z. That is:

> A discrete-time LTI system with rational system function $H(z)$ is causal if and only if: (a) the ROC is the exterior of a circle outside the outermost pole; and (b) with $H(z)$ expressed as a ratio of polynomials in z, the order of the numerator cannot be greater than the order of the denominator.

Example 10.20

Consider a system with system function whose algebraic expression is

$$H(z) = \frac{z^3 - 2z^2 + z}{z^2 + \frac{1}{4}z + \frac{1}{8}}.$$

Without even knowing the ROC for this system, we can conclude that the system is not causal, because the numerator of $H(z)$ is of higher order than the denominator.

Example 10.21

Consider a system with system function

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}, \quad |z| > 2 \tag{10.97}$$

Since the ROC for this system function is the exterior of a circle outside the outermost pole, we know that the impulse response is right-sided. To determine if the system is causal, we then need only check the other condition required for causality, namely that $H(z)$, when expressed as a ratio of polynomials in z, has numerator degree no larger than the denominator. For this example,

$$H(z) = \frac{2 - \frac{5}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})} = \frac{2z^2 - \frac{5}{2}z}{z^2 - \frac{5}{2}z + 1}, \tag{10.98}$$

so that the numerator and denominator of $H(z)$ are both of degree two, and consequently we can conclude that the system is causal. This can also be verified by calculating the inverse transform of $H(z)$. In particular, using transform pair 5 in Table 10.2, we find that the impulse response of this system is

$$h[n] = \left[\left(\frac{1}{2}\right)^n + 2^n\right]u[n]. \tag{10.99}$$

Since $h[n] = 0$ for $n < 0$, we can confirm that the system is causal.

10.7.2 Stability

As we discussed in Section 2.3.7, the stability of a discrete-time LTI system is equivalent to its impulse response being absolutely summable. In this case the Fourier transform of $h[n]$

converges, and consequently, the ROC of $H(z)$ must include the unit circle. Summarizing, we obtain the following result:

> An LTI system is stable if and only if the ROC of its system function $H(z)$ includes the unit circle, $|z| = 1$.

Example 10.22

Consider again the system function in eq. (10.97). Since the associated ROC is the region $|z| > 2$, which does not include the unit circle, the system is not stable. This can also be seen by noting that the impulse response in eq. (10.99) is not absolutely summable. If, however, we consider a system whose system function has the same algebraic expression as in eq. (10.97) but whose ROC is the region $1/2 < |z| < 2$, then the ROC does contain the unit circle, so that the corresponding system is noncausal but stable. In this case, using transform pairs 5 and 6 from Table 10.2, we find that the corresponding impulse response is

$$h[n] = \left(\frac{1}{2}\right)^n u[n] - 2^n u[-n-1], \tag{10.100}$$

which is absolutely summable.

Also, for the third possible choice of ROC associated with the algebraic expression for $H(z)$ in eq. (10.97), namely, $|z| < 1/2$, the corresponding system is neither causal (since the ROC is not outside the outermost pole) nor stable (since the ROC does not include the unit circle). This can also be seen from the impulse response, which (using transform pair 6 in Table 10.2) is

$$h[n] = -\left[\left(\frac{1}{2}\right)^n + 2^n\right] u[-n-1].$$

As Example 10.22 illustrates, it is perfectly possible for a system to be stable but not causal. However, if we focus on causal systems, stability can easily be checked by examining the locations of the poles. Specifically, for a causal system with rational system function, the ROC is outside the outermost pole. For this ROC to include the unit circle, $|z| = 1$, all of the poles of the system must be inside the unit circle. That is:

> A causal LTI system with rational system function $H(z)$ is stable if and only if all of the poles of $H(z)$ lie inside the unit circle—i.e., they must all have magnitude smaller than 1.

Example 10.23

Consider a causal system with system function

$$H(z) = \frac{1}{1 - az^{-1}},$$

which has a pole at $z = a$. For this system to be stable, its pole must be inside the unit circle, i.e., we must have $|a| < 1$. This is consistent with the condition for the absolute summability of the corresponding impulse response $h[n] = a^n u[n]$.

Example 10.24

The system function for a second-order system with complex poles was given in eq. (10.69), specifically,

$$H(z) = \frac{1}{1 - (2r\cos\theta)z^{-1} + r^2z^{-2}}, \tag{10.101}$$

with poles located at $z_1 = re^{j\theta}$ and $z_2 = re^{-j\theta}$. Assuming causality, we see that the ROC is outside the outermost pole (i.e., $|z| > |r|$). The pole-zero plot and ROC for this system are shown in Figure 10.16 for $r < 1$ and $r > 1$. For $r < 1$, the poles are inside the unit circle, the ROC includes the unit circle, and therefore, the system is stable. For $r > 1$, the poles are outside the unit circle, the ROC does not include the unit circle, and the system is unstable.

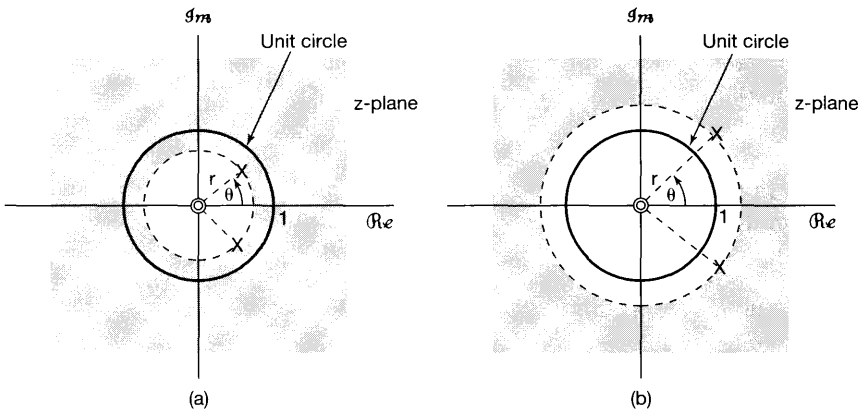


Figure 10.16 Pole-zero plot for a second-order system with complex poles: (a) $r < 1$; (b) $r > 1$.

10.7.3 LTI Systems Characterized by Linear Constant-Coefficient Difference Equations

For systems characterized by linear constant-coefficient difference equations, the properties of the z-transform provide a particularly convenient procedure for obtaining the system function, frequency response, or time-domain response of the system. Let us illustrate this with an example.

Example 10.25

Consider an LTI system for which the input $x[n]$ and output $y[n]$ satisfy the linear constant-coefficient difference equation

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]. \tag{10.102}$$

Applying the z-transform to both sides of eq. (10.102), and using the linearity property set forth in Section 10.5.1 and the time-shifting property presented in Section 10.5.2, we obtain

$$Y(z) - \frac{1}{2}z^{-1}Y(z) = X(z) + \frac{1}{3}z^{-1}X(z),$$

or

$$Y(z) = X(z)\left[\frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}\right]. \tag{10.103}$$

From eq. (10.96), then,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}. \tag{10.104}$$

This provides the algebraic expression for $H(z)$, but not the region of convergence. In fact, there are two distinct impulse responses that are consistent with the difference equation (10.102), one right sided and the other left sided. Correspondingly, there are two different choices for the ROC associated with the algebraic expression (10.104). One, $|z| > 1/2$, is associated with the assumption that $h[n]$ is right sided, and the other, $|z| < 1/2$, is associated with the assumption that $h[n]$ is left sided.

Consider first the choice of ROC equal to $|z| > 1$. Writing

$$H(z) = \left(1 + \frac{1}{3}z^{-1}\right)\frac{1}{1 - \frac{1}{2}z^{-1}},$$

we can use transform pair 5 in Table 10.2, together with the linearity and time-shifting properties, to find the corresponding impulse response

$$h[n] = \left(\frac{1}{2}\right)^n u[n] + \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[n-1].$$

For the other choice of ROC, namely, $|z| < 1$, we can use transform pair 6 in Table 10.2 and the linearity and time-shifting properties, yielding

$$h[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[-n].$$

In this case, the system is anticausal ($h[n] = 0$ for $n > 0$) and unstable.

For the more general case of an Nth-order difference equation, we proceed in a manner similar to that in Example 10.25, applying the z-transform to both sides of the equation and using the linearity and time-shifting properties. In particular, consider an LTI system for which the input and output satisfy a linear constant-coefficient difference equation of the form

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]. \tag{10.105}$$

Then taking z-transforms of both sides of eq. (10.105) and using the linearity and time-shifting properties, we obtain

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z),$$

or

$$Y(z) \sum_{k=0}^{N} a_k z^{-k} = X(z) \sum_{k=0}^{M} b_k z^{-k},$$

so that

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}. \tag{10.106}$$

We note in particular that the system function for a system satisfying a linear constant-coefficient difference equation is always rational. Consistent with our previous example and with the related discussion for the Laplace transform, the difference equation by itself does not provide information about which ROC to associate with the algebraic expression $H(z)$. An additional constraint, such as the causality or stability of the system, however, serves to specify the region of convergence. For example, if we know in addition that the system is causal, the ROC will be outside the outermost pole. If the system is stable, the ROC must include the unit circle.

10.7.4 Examples Relating System Behavior to the System Function

As the previous subsections illustrate, many properties of discrete-time LTI systems can be directly related to the system function and its characteristics. In this section, we give several additional examples to show how z-transform properties can be used in analyzing systems.

Example 10.26

Suppose that we are given the following information about an LTI system:

1. If the input to the system is $x_1[n] = (1/6)^n u[n]$, then the output is

$$y_1[n] = \left[a\left(\frac{1}{2}\right)^n + 10\left(\frac{1}{3}\right)^n\right] u[n],$$

where a is a real number.

2. If $x_2[n] = (-1)^n$, then the output is $y_2[n] = \frac{7}{4}(-1)^n$. As we now show, from these two pieces of information, we can determine the system function $H(z)$ for this system, including the value of the number a, and can also immediately deduce a number of other properties of the system.

The z-transforms of the signals specified in the first piece of information are

$$X_1(z) = \frac{1}{1 - \frac{1}{6}z^{-1}}, \quad |z| > \frac{1}{6}, \tag{10.107}$$

$$\begin{aligned} Y_1(z) &= \frac{a}{1 - \frac{1}{2}z^{-1}} + \frac{10}{1 - \frac{1}{3}z^{-1}} \\ &= \frac{(a + 10) - (5 + \frac{a}{3})z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{2}. \end{aligned} \tag{10.108}$$

From eq. (10.96), it follows that the algebraic expression for the system function is

$$H(z) = \frac{Y_1(z)}{X_1(z)} = \frac{[(a + 10) - (5 + \frac{a}{3})z^{-1}][1 - \frac{1}{6}z^{-1}]}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}. \tag{10.109}$$

Furthermore, we know that the response to $x_2[n] = (-1)^n$ must equal $(-1)^n$ multiplied by the system function $H(z)$ evaluated at $z = -1$. Thus from the second piece of information given, we see that

$$\frac{7}{4} = H(-1) = \frac{[(a + 10) + 5 + \frac{a}{3}][\frac{7}{6}]}{(\frac{3}{2})(\frac{4}{3})}. \tag{10.110}$$

Solving eq. (10.110), we find that $a = -9$, so that

$$H(z) = \frac{(1 - 2z^{-1})(1 - \frac{1}{6}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}, \tag{10.111}$$

or

$$H(z) = \frac{1 - \frac{13}{6}z^{-1} + \frac{1}{3}z^{-2}}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}, \tag{10.112}$$

or, finally,

$$H(z) = \frac{z^2 - \frac{13}{6}z + \frac{1}{3}}{z^2 - \frac{5}{6}z + \frac{1}{6}}. \tag{10.113}$$

Also, from the convolution property, we know that the ROC of $Y_1(z)$ must include at least the intersections of the ROCs of $X_1(z)$ and $H(z)$. Examining the three possible ROCs for $H(z)$ (namely, $|z| < 1/3$, $1/3 < |z| < 1/2$, and $|z| > 1/2$), we find that the only choice that is consistent with the ROCs of $X_1(z)$ and $Y_1(z)$ is $|z| > 1/2$.

Since the ROC for the system includes the unit circle, we know that the system is stable. Furthermore, from eq. (10.113) with $H(z)$ viewed as a ratio of polynomials in z, the order of the numerator does not exceed that of the denominator, and thus we can conclude that the LTI system is causal. Also, using eqs. (10.112) and (10.106), we can write the difference equation that, together with the condition of initial rest, characterizes the system:

$$y[n] - \frac{5}{6}y[n - 1] + \frac{1}{6}y[n - 2] = x[n] - \frac{13}{6}x[n - 1] + \frac{1}{3}x[n - 2].$$

Example 10.27

Consider a stable and causal system with impulse response $h[n]$ and rational system function $H(z)$. Suppose it is known that $H(z)$ contains a pole at $z = 1/2$ and a zero somewhere on the unit circle. The precise number and locations of all of the other poles

and zeros are unknown. For each of the following statements, let us determine whether we can definitely say that it is true, whether we can definitely say that it is false, or whether there is insufficient information given to determine if it is true or not:

(a) $\mathcal{F}\{(1/2)^n h[n]\}$ converges.

(b) $H(e^{j\omega}) = 0$ for some ω.

(c) $h[n]$ has finite duration.

(d) $h[n]$ is real.

(e) $g[n] = n[h[n] * h[n]]$ is the impulse response of a stable system.

Statement (a) is true. $\mathcal{F}\{(1/2)^n h[n]\}$ corresponds to the value of the z-transform of $h[n]$ at $z = 2$. Thus, its convergence is equivalent to the point $z = 2$ being in the ROC. Since the system is stable and causal, all of the poles of $H(z)$ are inside the unit circle, and the ROC includes all the points outside the unit circle, including $z = 2$.

Statement (b) is true because there is a zero on the unit circle.

Statement (c) is false because a finite-duration sequence must have an ROC that includes the entire z-plane, except possibly $z = 0$ and/or $z = \infty$. This is not consistent with having a pole at $z = 1/2$.

Statement (d) requires that $H(z) = H^*(z^*)$. This in turn implies that if there is a pole (zero) at a nonreal location $z = z_0$, there must also be a pole (zero) at $z = z_0^*$. Insufficient information is given to validate such a conclusion.

Statement (e) is true. Since the system is causal, $h[n] = 0$ for $n < 0$. Consequently, $h[n] * h[n] = 0$ for $n < 0$; i.e., the system with $h[n] * h[n]$ as its impulse response is causal. The same is then true for $g[n] = n[h[n] * h[n]]$. Furthermore, by the convolution property set forth in Section 10.5.7, the system function corresponding to the impulse response $h[n] * h[n]$ is $H^2(z)$, and by the differentiation property presented in Section 10.5.8, the system function corresponding to $g[n]$ is

$$G(z) = -z\frac{d}{dz}H^2(z) = -2zH(z)\left[\frac{d}{dz}H(z)\right]. \tag{10.114}$$

From eq. (10.114), we can conclude that the poles of $G(z)$ are at the same locations as those of $H(z)$, with the possible exception of the origin. Therefore, since $H(z)$ has all its poles inside the unit circle, so must $G(z)$. It follows that $g[n]$ is the impulse response of a causal and stable system.

10.8 SYSTEM FUNCTION ALGEBRA AND BLOCK DIAGRAM REPRESENTATIONS

Just as with the Laplace transform in continuous time, the z-transform in discrete time allows us to replace time-domain operations such as convolution and time shifting with algebraic operations. This was exploited in Section 10.7.3, where we were able to replace the difference-equation description of an LTI system with an algebraic description. The use of the z-transform to convert system descriptions to algebraic equations is also helpful in analyzing interconnections of LTI systems and in representing and synthesizing systems as interconnections of basic system building blocks.

10.8.1 System Functions for Interconnections of LTI Systems

The system function algebra for analyzing discrete-time block diagrams such as series, parallel, and feedback interconnections is exactly the same as that for the corresponding continuous-time systems in Section 9.8.1. For example, the system function for the cascade of two discrete-time LTI systems is the product of the system functions for the individual systems in the cascade. Also, consider the feedback interconnection of two systems, as shown in Figure 10.17. It is relatively involved to determine the difference equation or impulse response for the overall system working directly in the time domain. However, with the systems and sequences expressed in terms of their z-transforms, the analysis involves only algebraic equations. The specific equations for the interconnection of Figure 10.17 exactly parallel eqs. (9.159)–(9.163), with the final result that the overall system function for the feedback system of Figure 10.17 is

$$\frac{Y(z)}{X(z)} = H(z) = \frac{H_1(z)}{1 + H_1(z)H_2(z)}. \tag{10.115}$$

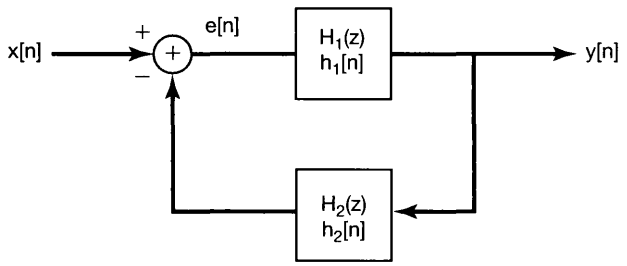


Figure 10.17 Feedback interconnection of two systems.

10.8.2 Block Diagram Representations for Causal LTI Systems Described by Difference Equations and Rational System Functions

As in Section 9.8.2, we can represent causal LTI systems described by difference equations using block diagrams involving three basic operations—in this case, addition, multiplication by a coefficient, and a unit delay. In Section 2.4.3, we described such a block diagram for a first-order difference equation. We first revisit that example, this time using system function algebra, and then consider several slightly more complex examples to illustrate the basic ideas in constructing block diagram representations.

Example 10.28

Consider the causal LTI system with system function

$$H(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}. \tag{10.116}$$

Using the results in Section 10.7.3, we find that this system can also be described by the difference equation

$$y[n] - \frac{1}{4}y[n-1] = x[n],$$

together with the condition of initial rest. In Section 2.4.3 we constructed a block diagram representation for a first-order system of this form, and an equivalent block diagram (corresponding to Figure 2.28 with $a = -1/4$ and $b = 1$) is shown in Figure 10.18(a). Here, z^{-1} is the system function of a unit delay. That is, from the time-shifting property, the input and output of this system are related by

$$w[n] = y[n-1].$$

The block diagram in Figure 10.18(a) contains a feedback loop much as for the system considered in the previous subsection and pictured in Figure 10.17. In fact, with some minor modifications, we can obtain the equivalent block diagram shown in Figure 10.18(b), which is exactly in the form shown in Figure 10.17, with $H_1(z) = 1$ and $H_2(z) = -1/4z^{-1}$. Then, applying eq. (10.115), we can verify that the system function of the system in Figure 10.18 is given by eq. (10.116).

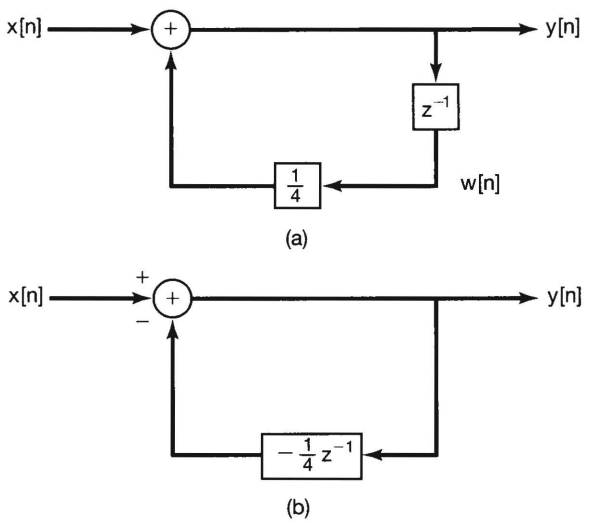


Figure 10.18 (a) Block diagram representations of the causal LTI system in Example 10.28; (b) equivalent block diagram representation.

Example 10.29

Suppose we now consider the causal LTI system with system function

$$H(z) = \frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}} = \left(\frac{1}{1 - \frac{1}{4}z^{-1}}\right)(1 - 2z^{-1}). \tag{10.117}$$

As eq. (10.117) suggests, we can think of this system as the cascade of a system with system function $1/[1 - (1/4)z^{-1}]$ and one with system function $1 - 2z^{-1}$. We have illustrated the cascade in Figure 10.19(a), in which we have used the block diagram in Figure 10.18(a) to represent $1/[1 - (1/4)z^{-1}]$. We have also represented $1 - 2z^{-1}$ using a unit delay, an adder, and a coefficient multiplier. Using the time-shifting property, we then see that the input $v[n]$ and output $y[n]$ of the system with the system function $1 - 2z^{-1}$

are related by

$$y[n] = v[n] - 2v[n-1].$$

While the block diagram in Figure 10.19(a) is certainly a valid representation of the system in eq. (10.117), it has an inefficiency whose elimination leads to an alternative block-diagram representation. To see this, note that the input to both unit delay elements in Figure 10.19(a) is $v[n]$, so that the outputs of these elements are identical; i.e.,

$$w[n] = s[n] = v[n-1].$$

Consequently, we need not keep both of these delay elements, and we can simply use the output of one of them as the signal to be fed to both coefficient multipliers. The result is the block diagram representation in Figure 10.19(b). Since each unit delay element requires a memory register to store the preceding value of its input, the representation in Figure 10.19(b) requires less memory than that in Figure 10.19(a).

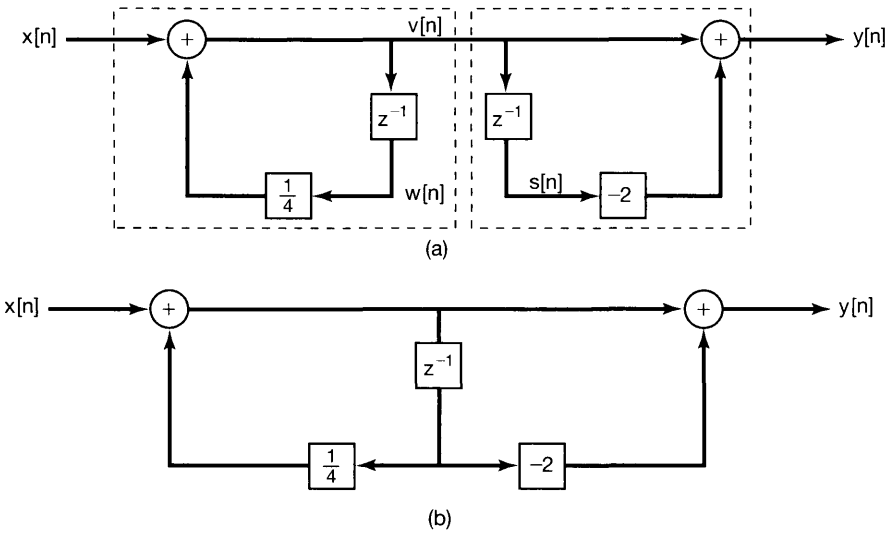


Figure 10.19 (a) Block-diagram representations for the system in Example 10.29; (b) equivalent block-diagram representation using only one unit delay element.

Example 10.30

Next, consider the second-order system function

$$H(z) = \frac{1}{(1 + \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} = \frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}, \tag{10.118}$$

which is also described by the difference equation

$$y[n] + \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n]. \tag{10.119}$$

Using the same ideas as in Example 10.28, we obtain the block-diagram representation for this system shown in Figure 10.20(a). Specifically, since the two system function blocks in this figure with system function z^{-1} are unit delays, we have

$$f[n] = y[n-1],$$
$$e[n] = f[n-1] = y[n-2],$$

so that eq. (10.119) can be rewritten as

$$y[n] = -\frac{1}{4}y[n-1] + \frac{1}{8}y[n-2] + x[n],$$

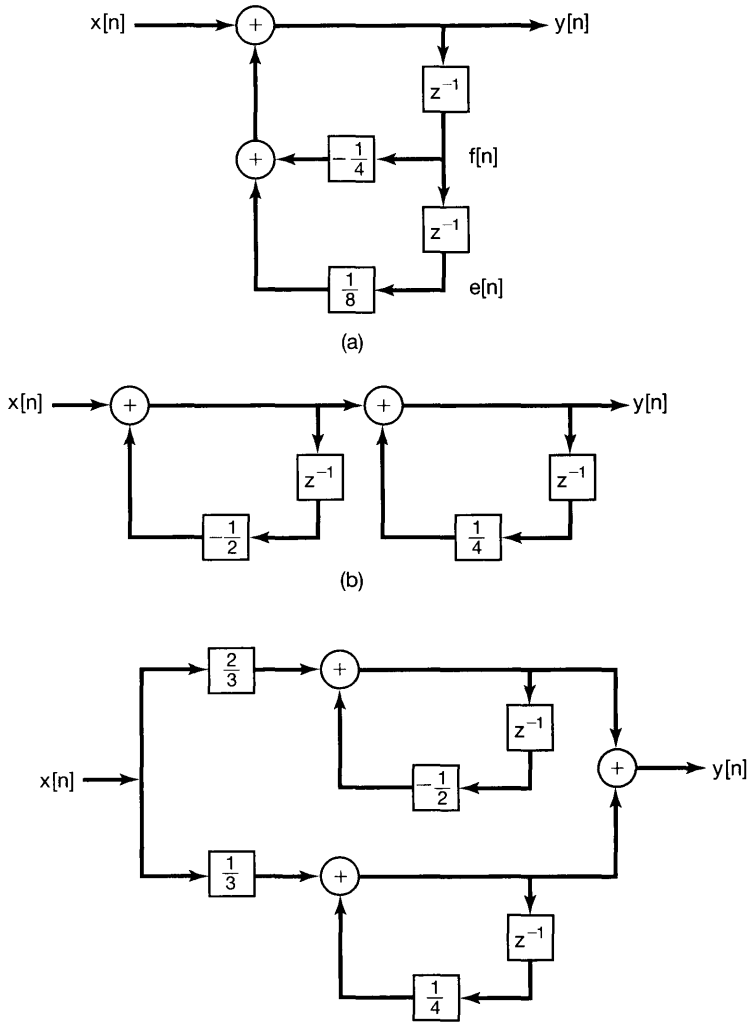


Figure 10.20 Block-diagram representations for the system in Example 10.30: (a) direct form; (b) cascade form; (c) parallel form.

or

$$y[n] = -\frac{1}{4}f[n] + \frac{1}{8}e[n] + x[n],$$

which is exactly what the figure represents.

The block diagram in Figure 10.20(a) is commonly referred to as a *direct-form* representation, since the coefficients appearing in the diagram can be determined by inspection from the coefficients appearing in the difference equation or, equivalently, the system function. Alternatively, as in continuous time, we can obtain both *cascade-form* and *parallel-form* block diagrams with the aid of a bit of system function algebra. Specifically, we can rewrite eq. (10.118) as

$$H(z) = \left(\frac{1}{1 + \frac{1}{2}z^{-1}}\right)\left(\frac{1}{1 - \frac{1}{4}z^{-1}}\right), \tag{10.120}$$

which suggests the cascade-form representation depicted in Figure 10.20(b) in which the system is represented as the cascade of two systems corresponding to the two factors in eq. (10.120).

Also, by performing a partial-fraction expansion, we obtain

$$H(z) = \frac{\frac{2}{3}}{1 + \frac{1}{2}z^{-1}} + \frac{\frac{1}{3}}{1 - \frac{1}{4}z^{-1}},$$

which leads to the parallel-form representation depicted in Figure 10.20(c).

Example 10.31

Finally, consider the system function

$$H(z) = \frac{1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}. \tag{10.121}$$

Writing

$$H(z) = \left(\frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}\right)\left(1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}\right) \tag{10.122}$$

suggests representing the system as the cascade of the system in Figure 10.20(a) and the system with system function $1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}$. However, as in Example 10.29, the unit delay elements needed to implement the first term in eq. (10.122) also produce the delayed signals needed in computing the output of the second system. The result is the direct-form block diagram shown in Figure 10.21, the details of the construction of which are examined in Problem 10.38. The coefficients in the direct-form representation can be determined by inspection from the coefficients in the system function of eq. (10.121).

We can also write $H(z)$ in the forms

$$H(z) = \left(\frac{1 + \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1}}\right)\left(\frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}}\right) \tag{10.123}$$

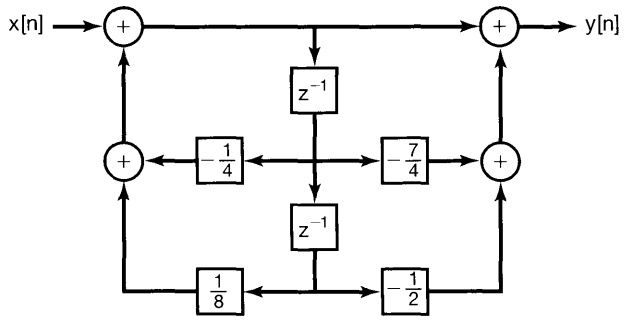


Figure 10.21 Direct-form representation for the system in Example 10.31.

and

$$H(z) = 4 + \frac{5/3}{1 + \frac{1}{2}z^{-1}} - \frac{14/3}{1 - \frac{1}{4}z^{-1}}. \tag{10.124}$$

Eq. (10.123) suggests a cascade-form representation, while eq. (10.124) leads to a parallel-form block diagram. These are also considered in Problem 10.38.

The concepts used in constructing block-diagram representations in the preceding examples can be applied directly to higher order systems, and several examples are considered in Problem 10.39. As in continuous time, there is typically considerable flexibility in doing this—e.g., in how numerator and denominator factors are paired in a product representation as in eq. (10.123), in the way in which each factor is implemented, and in the order in which the factors are cascaded. While all of these variations lead to representations of the same system, in practice there are differences in the behavior of the different block diagrams. Specifically, each block-diagram representation of a system can be translated directly into a computer algorithm for the implementation of the system. However, because the finite word length of a computer necessitates quantizing the coefficients in the block diagram and because there is numerical roundoff as the algorithm operates, each of these representations will lead to an algorithm that only approximates the behavior of the original system. Moreover, the errors in each of these approximations will be somewhat different. Because of these differences, considerable effort has been put into examining the relative merits of the various block-diagram representations in terms of their accuracy and sensitivity to quantization effects. For discussions of this subject, the reader may turn to the references on digital signal processing in the bibliography at the end of the book.

10.9 THE UNILATERAL z-TRANSFORM

The form of the z-transform considered thus far in this chapter is often referred to as the *bilateral z-transform.* As was the case with the Laplace transform, there is an alternative form, referred to as the *unilateral z-transform,* that is particularly useful in analyzing causal systems specified by linear constant-coefficient difference equations with nonzero initial conditions (i.e., systems that are not initially at rest). In this section, we introduce the unilateral z-transform and illustrate some of its properties and uses, paralleling our discussion of the unilateral Laplace transform in Section 9.9.

The unilateral z-transform of a sequence $x[n]$ is defined as

$$\mathcal{X}(z) = \sum_{n=0}^{\infty} x[n]z^{-n}. \tag{10.125}$$

As in previous chapters, we adopt a convenient shorthand notation for a signal and its unilateral z-transform:

$$x[n] \overset{\mathcal{UZ}}{\longleftrightarrow} \mathcal{X}(z) = \mathcal{UZ}\{x[n]\}. \tag{10.126}$$

The unilateral z-transform differs from the bilateral transform in that the summation is carried out only over nonnegative values of n, whether or not $x[n]$ is zero for $n < 0$. Thus the unilateral z-transform of $x[n]$ can be thought of as the bilateral transform of $x[n]u[n]$ (i.e., $x[n]$ multiplied by a unit step). In particular, then, for any sequence that is zero for $n < 0$, the unilateral and bilateral z-transforms will be identical. Referring to the discussion of regions of convergence in Section 10.2, we also see that, since $x[n]u[n]$ is always a right-sided sequence, the region of convergence of $\mathcal{X}(z)$ is always the exterior of a circle.

Because of the close connection between bilateral and unilateral z-transforms, the calculation of unilateral transforms proceeds much as for bilateral transforms, with the caveat that we must take care to limit the range of summation in the transform to $n \geq 0$. Similarly, the calculation of inverse unilateral transforms is basically the same as for bilateral transforms, once we take into account the fact that the ROC for a unilateral transform is *always* the exterior of a circle.

10.9.1 Examples of Unilateral z-Transforms and Inverse Transforms

Example 10.32

Consider the signal

$$x[n] = a^n u[n]. \tag{10.127}$$

Since $x[n] = 0, n < 0$, the unilateral and bilateral transforms are equal for this example, and thus, in particular,

$$\mathcal{X}(z) = \frac{1}{1 - az^{-1}}, \qquad |z| > |a|. \tag{10.128}$$

Example 10.33

Let

$$x[n] = a^{n+1}u[n + 1]. \tag{10.129}$$

In this case the unilateral and bilateral transforms are *not* equal, since $x[-1] = 1 \neq 0$. The bilateral transform is obtained from Example 10.1 and the time-shifting property set forth in Section 10.5.2. Specifically,

$$X(z) = \frac{z}{1 - az^{-1}}, \qquad |z| > |a|. \tag{10.130}$$

In contrast, the unilateral transform is

$$\begin{aligned}\mathfrak{X}(z) &= \sum_{n=0}^{\infty} x[n]z^{-n} \\ &= \sum_{n=0}^{\infty} a^{n+1}z^{-n},\end{aligned}$$

or

$$\mathfrak{X}(z) = \frac{a}{1-az^{-1}}, \qquad |z| > |a|. \tag{10.131}$$

Example 10.34

Consider the unilateral z-transform

$$\mathfrak{X}(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1-\frac{1}{4}z^{-1})(1-\frac{1}{3}z^{-1})}. \tag{10.132}$$

In Example 10.9, we considered the inverse transform for a bilateral z-transform $X(z)$ of the same form as in eq. (10.132) and for several different ROCs. In the case of the unilateral transform, the ROC must be the exterior of the circle of radius equal to the largest magnitude of the poles of $\mathfrak{X}(z)$—in this instance, all points z with $|z| > 1/3$. We can then invert the unilateral transform exactly as in Example 10.9, yielding

$$x[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n] \quad \text{for} \quad n \geq 0. \tag{10.133}$$

In eq. (10.133), we have emphasized the fact that inverse unilateral z-transforms provide us with information about $x[n]$ *only* for $n \geq 0$.

Another approach to inverse transforms introduced in Section 10.3, namely, identifying the inverse transforms from the coefficients in the power-series expansion of the z-transform, also can be used for unilateral transforms. However, in the unilateral case, a constraint which must be satisfied is that, as a consequence of eq. (10.125), the power-series expansion for the transform cannot contain terms with positive powers of z. For instance, in Example 10.13 we performed long division on the bilateral transform

$$X(z) = \frac{1}{1-az^{-1}} \tag{10.134}$$

in two ways, corresponding to the two possible ROCs for $X(z)$. Only one of these choices, namely, that corresponding to the ROC $|z| > |a|$, led to a series expansion without positive powers of z, i.e.,

$$\frac{1}{1-az^{-1}} = 1 + az^{-1} + a^2z^{-2} + \ldots, \tag{10.135}$$

and this is the *only* choice for the expansion if eq. (10.134) represents a unilateral transform.

Note that the requirement that $\mathcal{X}(z)$ have a power-series expansion with no terms with positive powers of z implies that not every function of z can *be* a unilateral z-transform. In particular, if we consider a rational function of z written as a ratio of polynomials in z (not in z^{-1}), i.e.,

$$\frac{p(z)}{q(z)}, \tag{10.136}$$

then for this to be a unilateral transform (with the appropriately chosen ROC as the exterior of a circle), the degree of the numerator must be no bigger than the degree of the denominator.

Example 10.35

A simple example illustrating the preceding point is given by the rational function in eq. (10.130), which we can write as a ratio of polynomials in z:

$$\frac{z^2}{z-a}. \tag{10.137}$$

There are two possible bilateral transforms that can be associated with this function, namely those corresponding to the two possible ROCs, $|z| < |a|$ and $|z| > |a|$. The choice $|z| > |a|$ corresponds to a right-sided sequence, but *not* to a signal that is zero for all $n < 0$, since its inverse transform, which is given by eq. (10.129), is nonzero for $n = -1$.

More generally, if we associate eq. (10.136) with the bilateral transform with the ROC that is the exterior of the circle with radius given by the magnitude of the largest root of $q(z)$, then the inverse transform will certainly be right sided. However, for it to be zero for *all* $n < 0$, it must also be the case that degree($p(z)$) $\leq$ degree($q(z)$).

10.9.2 Properties of the Unilateral z-Transform

The unilateral z-transform has many important properties, some of which are identical to their bilateral counterparts and several of which differ in significant ways. Table 10.3 summarizes these properties. Note that we have not included a column explicitly identifying the ROC for the unilateral z-transform for each signal, since the ROC of any unilateral z-transform is always the exterior of a circle. For example, the ROC for a rational unilateral z-transform is always outside the outermost pole.

By contrasting this table with the corresponding Table 10.1 for bilateral z-transforms, we can gain considerable insight into the nature of the unilateral transform. In particular, several properties—namely, linearity, scaling in the z-domain, time expansion, conjugation, and differentiation in the z-domain—are identical to their bilateral counterparts, as is the initial-value theorem stated in Section 10.5.9, which is fundamentally a unilateral transform property, since it requires $x[n] = 0$ for $n < 0$. One bilateral property, namely, the time-reversal property set forth in Section 10.5.4, obviously has no meaningful counterpart for the unilateral transform, while the remaining properties differ in important ways between the bilateral and unilateral cases.

TABLE 10.3 PROPERTIES OF THE UNILATERAL z-TRANSFORM

Property	Signal	Unilateral z-Transform
—	$x[n]$	$\mathcal{X}(z)$
—	$x_1[n]$	$\mathcal{X}_1(z)$
—	$x_2[n]$	$\mathcal{X}_2(z)$
Linearity	$ax_1[n] + bx_2[n]$	$a\mathcal{X}_1(z) + b\mathcal{X}_2(z)$
Time delay	$x[n-1]$	$z^{-1}\mathcal{X}(z) + x[-1]$
Time advance	$x[n+1]$	$z\mathcal{X}(z) - zx[0]$
Scaling in the z-domain	$e^{j\omega_0 n}x[n]$	$\mathcal{X}(e^{-j\omega_0}z)$
	$z_0^n x[n]$	$\mathcal{X}(z/z_0)$
	$a^n x[n]$	$\mathcal{X}(a^{-1}z)$
Time expansion	$x_k[n] = \begin{cases} x[m], & n = mk \\ 0, & n \neq mk \quad \text{for any } m \end{cases}$	$\mathcal{X}(z^k)$
Conjugation	$x^*[n]$	$\mathcal{X}^*(z^*)$
Convolution (assuming that $x_1[n]$ and $x_2[n]$ are identically zero for $n < 0$)	$x_1[n] * x_2[n]$	$\mathcal{X}_1(z)\mathcal{X}_2(z)$
First difference	$x[n] - x[n-1]$	$(1 - z^{-1})\mathcal{X}(z) - x[-1]$
Accumulation	$\sum_{k=0}^{n} x[k]$	$\frac{1}{1-z^{-1}}\mathcal{X}(z)$
Differentiation in the z-domain	$nx[n]$	$-z\frac{d\mathcal{X}(z)}{dz}$
	Initial Value Theorem $x[0] = \lim_{z\to\infty}\mathcal{X}(z)$	

Let us examine the difference in the convolution property first. Table 10.3 states that if $x_1[n] = x_2[n] = 0$ for all $n < 0$, then

$$x_1[n] * x_2[n] \overset{uz}{\longleftrightarrow} \mathcal{X}_1(z)\mathcal{X}_2(z). \tag{10.138}$$

Since in this case the unilateral and bilateral transforms are identical for each of these signals, eq. (10.138) follows from the bilateral convolution property. Thus, the system analysis and system function algebra developed and used in this chapter apply without change to unilateral transforms, as long as we are considering causal LTI systems (for which the system function is both the bilateral *and* the unilateral transform of the impulse response) with inputs that are identically zero for $n < 0$. An example of such application is to the accumulation or summation property in Table 10.3. Specifically, if $x[n] = 0$ for $n < 0$, then

$$\sum_{k=0}^{n} x[k] = x[n] * u[n] \overset{uz}{\longleftrightarrow} \mathcal{X}(z)\mathcal{U}(z) = \mathcal{X}(z)\frac{1}{1-z^{-1}}. \tag{10.139}$$

As a second example, consider the following:

Example 10.36

Consider the causal LTI system described by the difference equation

$$y[n] + 3y[n-1] = x[n], \tag{10.140}$$

together with the condition of initial rest. The system function for this system is

$$\mathcal{H}(z) = \frac{1}{1+3z^{-1}}. \tag{10.141}$$

Suppose that the input to the system is $x[n] = \alpha u[n]$, where α is a given constant. In this case, the unilateral (and bilateral) z-transform of the output $y[n]$ is

$$\begin{aligned} \mathcal{Y}(z) &= \mathcal{H}(z)\mathcal{X}(z) = \frac{\alpha}{(1+3z^{-1})(1-z^{-1})} \\ &= \frac{(3/4)\alpha}{1+3z^{-1}} + \frac{(1/4)\alpha}{1-z^{-1}}. \end{aligned} \tag{10.142}$$

Applying Example 10.32 to each term of eq. (10.142) yields

$$y[n] = \alpha\left[\frac{1}{4} + \left(\frac{3}{4}\right)(-3)^n\right]u[n]. \tag{10.143}$$

An important point to note here is that the convolution property for unilateral z-transforms applies *only* if the signals $x_1[n]$ and $x_2[n]$ in eq. (10.138) are both identically zero for $n < 0$. While it is generally true that the bilateral transform of $x_1[n] * x_2[n]$ equals the product of the bilateral transforms of $x_1[n]$ and $x_2[n]$, the unilateral transform of $x_1[n] * x_2[n]$ in general does not equal the product of the unilateral transforms if $x_1[n]$ or $x_2[n]$ is nonzero for $n < 0$. This point is explored further in Problem 10.41.

Much of the importance of the unilateral z-transform lies in its application to analyzing causal systems and, in particular, systems characterized by linear constant-coefficient difference equations with possibly nonzero initial conditions. In Section 10.7 we saw how the bilateral transform—particularly the shifting property for bilateral z-transforms—could be used to analyze and compute solutions for LTI systems characterized by such difference equations, together with the assumption of initial rest. As we will now see, the shifting property for unilateral transforms, which differs from its bilateral counterpart, plays an analogous role for initialized systems.

To develop the shifting property for the unilateral transform, consider the signal

$$y[n] = x[n-1]. \tag{10.144}$$

Then

$$\begin{aligned} \mathcal{Y}(z) &= \sum_{n=0}^{\infty} x[n-1]z^{-n} \\ &= x[-1] + \sum_{n=1}^{\infty} x[n-1]z^{-n} \\ &= x[-1] + \sum_{n=0}^{\infty} x[n]z^{-(n+1)}, \end{aligned}$$

or

$$\mathcal{Y}(z) = x[-1] + z^{-1}\sum_{n=0}^{\infty} x[n]z^{-n}, \tag{10.145}$$

so that

$$\mathcal{Y}(z) = x[-1] + z^{-1}\mathcal{X}(z). \tag{10.146}$$

By repeated application of eq. (10.146), the unilateral transform of

$$w[n] = y[n-1] = x[n-2] \tag{10.147}$$

is

$$\mathcal{W}(z) = x[-2] + x[-1]z^{-1} + z^{-2}\mathcal{X}(z). \tag{10.148}$$

Continuing this iterative procedure, we can also determine the unilateral transform of $x[n-m]$ for any positive value of m.

Eq. (10.146) is sometimes referred to as the time delay property, since $y[n]$ in eq. (10.144) is a delayed version of $x[n]$. There is also a time advance property for unilateral transforms that relates the transform of an advanced version of $x[n]$ to $\mathcal{X}(z)$. Specifically, as shown in Problem 10.60,

$$x[n+1] \overset{uz}{\longleftrightarrow} z\mathcal{X}(z) - zx[0]. \tag{10.149}$$

10.9.3 Solving Difference Equations Using the Unilateral z-Transform

The following example illustrates the use of unilateral z-transforms and the time delay property to solve linear constant-coefficient difference equations with nonzero initial conditions:

Example 10.37

Consider again the difference equation (10.140) with $x[n] = \alpha u[n]$ and with the initial condition

$$y[-1] = \beta. \tag{10.150}$$

Applying the unilateral transform to both sides of eq. (10.140) and using the linearity and time delay properties, we obtain

$$\mathcal{Y}(z) + 3\beta + 3z^{-1}\mathcal{Y}(z) = \frac{\alpha}{1-z^{-1}}. \tag{10.151}$$

Solving for $\mathcal{Y}(z)$ yields

$$\mathcal{Y}(z) = -\frac{3\beta}{1+3z^{-1}} + \frac{\alpha}{(1+3z^{-1})(1-z^{-1})}. \tag{10.152}$$

Referring to Example 10.36 and, in particular, eq. (10.142), we see that the second term on the right-hand side of eq. (10.152) equals the unilateral z-transform of the response of the system when the initial condition in eq. (10.150) is zero ($\beta = 0$). That is, this term represents the response of the causal LTI system described by eq. (10.140), together with the condition of initial rest. As in continuous-time, this response is frequently referred to as the zero-state response, i.e., the response when the initial condition or state is zero.

The first term on the right-hand side of eq. (10.152) is interpreted as the unilateral transform of the zero-input response—i.e., the response of the system when the input is zero ($\alpha = 0$). The zero-input response is a linear function of the value β of the initial condition. Moreover, eq. (10.152) illustrates the fact that the solution of a linear constant-coefficient difference equation with nonzero initial state is the superposition of the zero-state and zero-input responses. The zero-state response, obtained by setting the initial condition to zero, corresponds to the response of the causal LTI system defined by the difference equation and the condition of initial rest. The zero-input response is the response to the initial condition alone with the input set to zero. Problems 10.20 and 10.42 provide other examples illustrating the use of unilateral transforms to solve difference equations with nonzero initial conditions.

Finally, for any values of α and β, we can expand $\mathcal{Y}(z)$ in eq. (10.152) by the method of partial fractions and invert the result to obtain $y[n]$. For example, if $\alpha = 8$ and $\beta = 1$,

$$\mathcal{Y}(z) = \frac{3}{1 + 3z^{-1}} + \frac{2}{1 - z^{-1}}, \tag{10.153}$$

and applying the unilateral transform pair in Example 10.32 to each term yields

$$y[n] = [3(-3)^n + 2]u[n], \quad \text{for } n \geq 0. \tag{10.154}$$

10.10 SUMMARY

In this chapter, we have developed the z-transform for discrete-time signals and systems. The discussion and development closely paralleled the corresponding treatment of the Laplace transform for continuous-time signals, but with some important differences. For example, in the complex s-plane the Laplace transform reduces to the Fourier transform on the imaginary axis, whereas in the complex z-plane the z-transform reduces to the Fourier transform on the unit circle. For the Laplace transform the ROC consists of a strip or half-plane (i.e., a strip extending to infinity in one direction), whereas for the z-transform the ROC is a ring, perhaps extending outward to infinity or inward to include the origin. As with the Laplace transform, time-domain characteristics such as the right-sided, left-sided, or two-sided nature of a sequence and the causality or stability of an LTI system can be associated with properties of the region of convergence. In particular, for rational z-transforms, these time-domain characteristics can be associated with the pole locations in relation to the region of convergence.

Because of the properties of z-transforms, LTI systems, including those described by linear constant-coefficient difference equations, can be analyzed in the transform domain by algebraic manipulations. System function algebra also is a very useful tool for the analysis of interconnections of LTI systems and for the construction of block diagram representations of LTI systems described by difference equations.

For the most part in this chapter, we have focused on bilateral z-transforms. However, as with the Laplace transform, we have also introduced a second form of the z-transform known as the unilateral z-transform. The unilateral transform, which can be viewed as the bilateral transform of a signal whose values for $n < 0$ have been set to zero, is particularly useful for analyzing systems described by linear constant-coefficient difference equations with nonzero initial conditions.

Chapter 10 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining three sections contain problems belonging to the basic, advanced, and extension categories, respectively.

BASIC PROBLEMS WITH ANSWERS

10.1. Determine the constraint on $r = |z|$ for each of the following sums to converge:

(a) $\sum_{n=-1}^{\infty} (\frac{1}{2})^{n+1} z^{-n}$ **(b)** $\sum_{n=1}^{\infty} (\frac{1}{2})^{-n+1} z^{n}$

(c) $\sum_{n=0}^{\infty} \{\frac{1+(-1)^n}{2}\} z^{-n}$ **(d)** $\sum_{n=-\infty}^{\infty} (\frac{1}{2})^{|n|} \cos(\frac{\pi}{4} n) z^{-n}$

10.2. Consider the signal

$$x[n] = \left(\frac{1}{5}\right)^n u[n-3].$$

Use eq. (10.3) to evaluate the z-transform of this signal, and specify the corresponding region of convergence.

10.3. Let

$$x[n] = (-1)^n u[n] + \alpha^n u[-n-n_0].$$

Determine the constraints on the complex number α and the integer n_0, given that the ROC of $X(z)$ is

$$1 < |z| < 2.$$

10.4. Consider the signal

$$x[n] = \begin{cases} (\frac{1}{3})^n \cos(\frac{\pi}{4} n), & n \leq 0 \\ 0, & n > 0 \end{cases}.$$

Determine the poles and ROC for $X(z)$.

10.5. For each of the following algebraic expressions for the z-transform of a signal, determine the number of zeros in the finite z-plane and the number of zeros at infinity.

(a) $\dfrac{z^{-1}(1-\frac{1}{2}z^{-1})}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{4}z^{-1})}$

(b) $\dfrac{(1-z^{-1})(1-2z^{-1})}{(1-3z^{-1})(1-4z^{-1})}$

(c) $\dfrac{z^{-2}(1-z^{-1})}{(1-\frac{1}{4}z^{-1})(1+\frac{1}{4}z^{-1})}$

10.6. Let $x[n]$ be an absolutely summable signal with rational z-transform $X(z)$. If $X(z)$ is known to have a pole at $z = 1/2$, could $x[n]$ be
(a) a finite-duration signal?
(b) a left-sided signal?
(c) a right-sided signal?
(d) a two-sided signal?

10.7. Suppose that the algebraic expression for the z-transform of $x[n]$ is

$$X(z) = \frac{1-\frac{1}{4}z^{-2}}{(1+\frac{1}{4}z^{-2})(1+\frac{5}{4}z^{-1}+\frac{3}{8}z^{-2})}.$$

How many different regions of convergence could correspond to $X(z)$?

10.8. Let $x[n]$ be a signal whose rational z-transform $X(z)$ contains a pole at $z = 1/2$. Given that

$$x_1[n] = \left(\frac{1}{4}\right)^n x[n]$$

is absolutely summable and

$$x_2[n] = \left(\frac{1}{8}\right)^n x[n]$$

is not absolutely summable, determine whether $x[n]$ is left sided, right sided, or two sided.

10.9. Using partial-fraction expansion and the fact that

$$a^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|,$$

find the inverse z-transform of

$$X(z) = \frac{1-\frac{1}{3}z^{-1}}{(1-z^{-1})(1+2z^{-1})}, \quad |z| > 2.$$

10.10. Consider the following algebraic expression for the z-transform $X(z)$ of a signal $x[n]$:

$$X(z) = \frac{1+z^{-1}}{1+\frac{1}{3}z^{-1}}.$$

(a) Assuming the ROC to be $|z| > 1/3$, use long division to determine the values of $x[0]$, $x[1]$, and $x[2]$.

(b) Assuming the ROC to be $|z| < 1/3$, use long division to determine the values of $x[0]$, $x[-1]$, and $x[-2]$.

10.11. Find the inverse z-transform of

$$X(z) = \frac{1}{1{,}024}\left[\frac{1{,}024 - z^{-10}}{1 - \frac{1}{2}z^{-1}}\right], \quad |z| > 0.$$

10.12. By considering the geometric interpretation of the magnitude of the Fourier transform from the pole-zero plot, determine, for each of the following z-transforms, whether the corresponding signal has an approximately lowpass, bandpass, or highpass characteristic:

(a) $X(z) = \dfrac{z^{-1}}{1 + \frac{8}{9}z^{-1}}, \quad |z| > \frac{8}{9}$

(b) $X(z) = \dfrac{1 + \frac{8}{9}z^{-1}}{1 - \frac{16}{9}z^{-1} + \frac{64}{81}z^{-2}}, \quad |z| > \frac{8}{9}$

(c) $X(z) = \dfrac{1}{1 + \frac{64}{81}z^{-2}}, \quad |z| > \frac{8}{9}$

10.13. Consider the rectangular signal

$$x[n] = \begin{cases} 1, & 0 \le n \le 5 \\ 0, & \text{otherwise} \end{cases}.$$

Let

$$g[n] = x[n] - x[n-1].$$

(a) Find the signal $g[n]$ and directly evaluate its z-transform.

(b) Noting that

$$x[n] = \sum_{k=-\infty}^{n} g[k],$$

use Table 10.1 to determine the z-transform of $x[n]$.

10.14. Consider the triangular signal

$$g[n] = \begin{cases} n - 1, & 2 \le n \le 7 \\ 13 - n, & 8 \le n \le 12 \\ 0, & \text{otherwise} \end{cases}.$$

(a) Determine the value of n_0 such that

$$g[n] = x[n] * x[n - n_0],$$

where $x[n]$ is the rectangular signal considered in Problem 10.13.

(b) Use the convolution and shift properties in conjunction with $X(z)$ found in Problem 10.13 to determine $G(z)$. Verify that your answer satisfies the initial-value theorem.

10.15. Let

$$y[n] = \left(\frac{1}{9}\right)^n u[n].$$

Determine two distinct signals such that each has a z-transform $X(z)$ which satisfies both of the following conditions:

1. $[X(z) + X(-z)]/2 = Y(z^2)$.
2. $X(z)$ has only one pole and only one zero in the z-plane.

10.16. Consider the following system functions for stable LTI systems. Without utilizing the inverse z-transform, determine in each case whether or not the corresponding system is causal.

(a) $\dfrac{1 - \frac{4}{3}z^{-1} + \frac{1}{2}z^{-2}}{z^{-1}(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$

(b) $\dfrac{z - \frac{1}{2}}{z^2 + \frac{1}{2}z - \frac{3}{16}}$

(c) $\dfrac{z + 1}{z + \frac{4}{3} - \frac{1}{2}z^{-2} - \frac{2}{3}z^{-3}}$

10.17. Suppose we are given the following five facts about a particular LTI system S with impulse response $h[n]$ and z-transform $H(z)$:

1. $h[n]$ is real.
2. $h[n]$ is right sided.
3. $\lim_{z \to \infty} H(z) = 1$.
4. $H(z)$ has two zeros.
5. $H(z)$ has one of its poles at a nonreal location on the circle defined by $|z| = 3/4$.

Answer the following two questions:

(a) Is S causal? **(b)** Is S stable?

10.18. Consider a causal LTI system whose input $x[n]$ and output $y[n]$ are related through the block diagram representation shown in Figure P10.18.

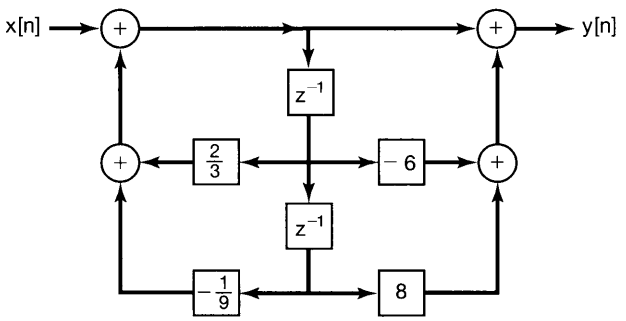


Figure P10.18

(a) Determine a difference equation relating $y[n]$ and $x[n]$.

(b) Is this system stable?

10.19. Determine the unilateral z-transform of each of the following signals, and specify the corresponding regions of convergence:

(a) $x_1[n] = (\frac{1}{4})^n u[n+5]$
(b) $x_2[n] = \delta[n+3] + \delta[n] + 2^n u[-n]$
(c) $x_3[n] = (\frac{1}{2})^{|n|}$

10.20. Consider a system whose input $x[n]$ and output $y[n]$ are related by

$$y[n-1] + 2y[n] = x[n].$$

(a) Determine the zero-input response of this system if $y[-1] = 2$.
(b) Determine the zero-state response of the system to the input $x[n] = (1/4)^n u[n]$.
(c) Determine the output of the system for $n \geq 0$ when $x[n] = (1/4)^n u[n]$ and $y[-1] = 2$.

BASIC PROBLEMS

10.21. Determine the z-transform for each of the following sequences. Sketch the pole-zero plot and indicate the region of convergence. Indicate whether or not the Fourier transform of the sequence exists.

(a) $\delta[n+5]$ **(b)** $\delta[n-5]$
(c) $(-1)^n u[n]$ **(d)** $(\frac{1}{2})^{n+1} u[n+3]$
(e) $(-\frac{1}{3})^n u[-n-2]$ **(f)** $(\frac{1}{4})^n u[3-n]$
(g) $2^n u[-n] + (\frac{1}{4})^n u[n-1]$ **(h)** $(\frac{1}{3})^{n-2} u[n-2]$

10.22. Determine the z-transform for the following sequences. Express all sums in closed form. Sketch the pole-zero plot and indicate the region of convergence. Indicate whether the Fourier transform of the sequence exists.

(a) $(\frac{1}{2})^n\{u[n+4] - u[n-5]\}$ **(b)** $n(\frac{1}{2})^{|n|}$
(c) $|n|(\frac{1}{2})^{|n|}$ **(d)** $4^n \cos[\frac{2\pi}{6}n + \frac{\pi}{4}]u[-n-1]$

10.23. Following are several z-transforms. For each one, determine the inverse z-transform using both the method based on the partial-fraction expansion and the Taylor's series method based on the use of long division.

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{4}z^{-2}}, \quad |z| > \frac{1}{2}.$$

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{4}z^{-2}}, \quad |z| < \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{(1 - \frac{1}{2}z^{-1})^2}, \quad |z| > \frac{1}{2}.$$

$$X(z) = \frac{z^{-1} - \frac{1}{2}}{(1 - \frac{1}{2}z^{-1})^2}, \quad |z| < \frac{1}{2}.$$

10.24. Using the method indicated, determine the sequence that goes with each of the following z-transforms:

(a) Partial fractions:

$$X(z) = \frac{1 - 2z^{-1}}{1 + \frac{5}{2}z^{-1} + z^{-2}}, \text{ and } x[n] \text{ is absolutely summable.}$$

(b) Long division:

$$X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{2}z^{-1}}, \text{ and } x[n] \text{ is right sided.}$$

(c) Partial fractions:

$$X(z) = \frac{3}{z - \frac{1}{4} - \frac{1}{8}z^{-1}}, \text{ and } x[n] \text{ is absolutely summable.}$$

10.25. Consider a right-sided sequence $x[n]$ with z-transform

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}. \tag{P10.25–1}$$

(a) Carry out a partial-fraction expansion of eq. (P10.25–1) expressed as a ratio of polynomials in z^{-1}, and from this expansion, determine $x[n]$.

(b) Rewrite eq. (P10.25–1) as a ratio of polynomials in z, and carry out a partial-fraction expansion of $X(z)$ expressed in terms of polynomials in z. From this expansion, determine $x[n]$, and demonstrate that the sequence obtained is identical to that obtained in part (a).

10.26. Consider a left-sided sequence $x[n]$ with z-transform

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}.$$

(a) Write $X(z)$ as a ratio of polynomials in z instead of z^{-1}.

(b) Using a partial-fraction expression, express $X(z)$ as a sum of terms, where each term represents a pole from your answer in part (a).

(c) Determine $x[n]$.

10.27. A right-sided sequence $x[n]$ has z-transform

$$X(z) = \frac{3z^{-10} + z^{-7} - 5z^{-2} + 4z^{-1} + 1}{z^{-10} - 5z^{-7} + z^{-3}}.$$

Determine $x[n]$ for $n < 0$.

10.28. **(a)** Determine the z-transform of the sequence

$$x[n] = \delta[n] - 0.95\,\delta[n - 6].$$

(b) Sketch the pole-zero pattern for the sequence in part (a).

(c) By considering the behavior of the pole and zero vectors as the unit circle is traversed, develop an approximate sketch of the magnitude of the Fourier transform of $x[n]$.

10.29. By considering the geometric determination of the frequency response as discussed in Section 10.4, sketch, for each of the pole-zero plots in Figure P10.29, the magnitude of the associated Fourier transform.

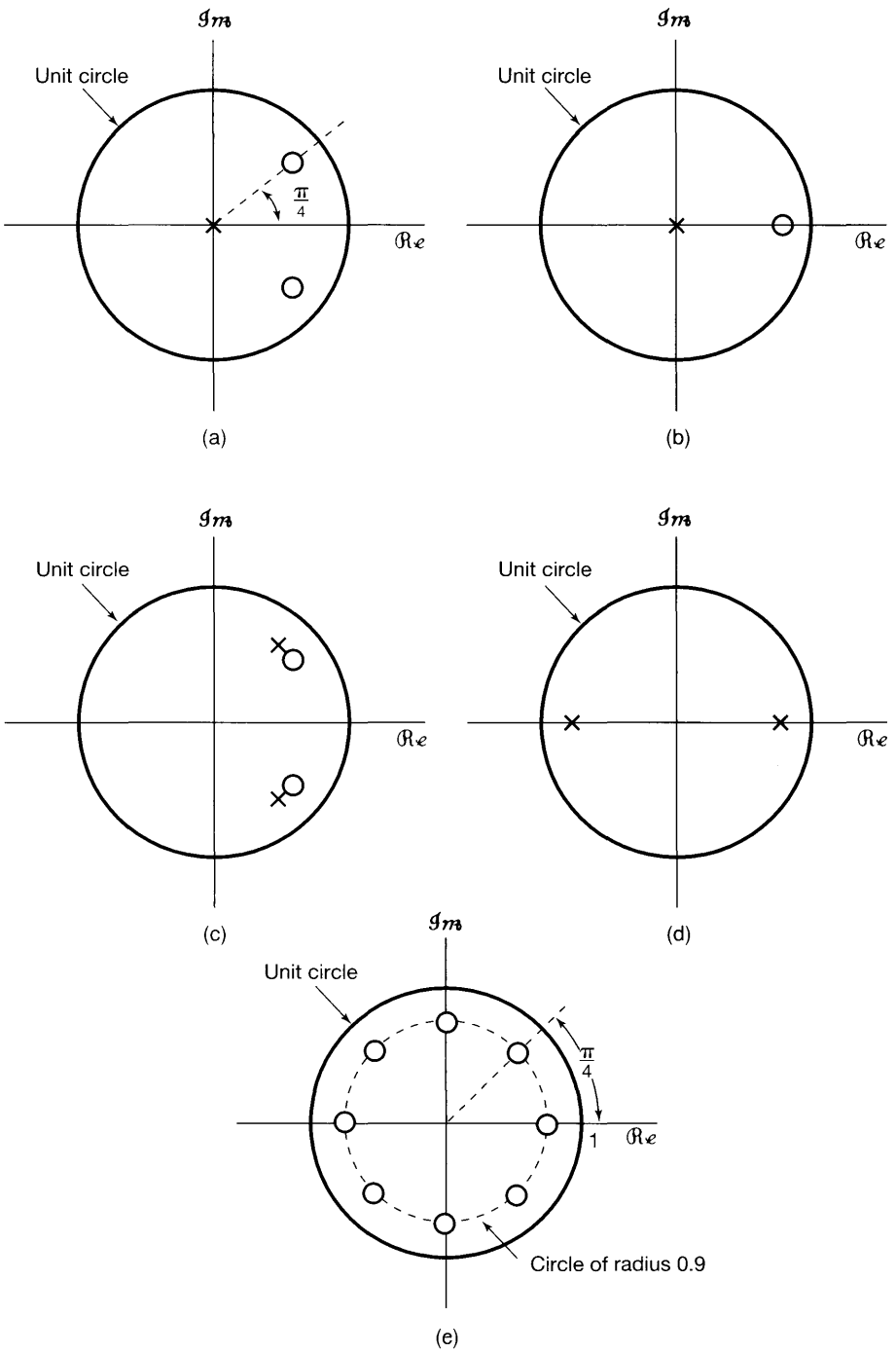


Figure P10.29

10.30. Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n+3] * x_2[-n+1]$$

where

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n] \quad \text{and} \quad x_2[n] = \left(\frac{1}{3}\right)^n u[n].$$

Given that

$$a^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|,$$

use properties of the z-transform to determine the z-transform $Y(z)$ of $y[n]$.

10.31. We are given the following five facts about a discrete-time signal $x[n]$ with z-transform $X(z)$:

1. $x[n]$ is real and right-sided.
2. $X(z)$ has exactly two poles.
3. $X(z)$ has two zeros at the origin.
4. $X(z)$ has a pole at $z = \frac{1}{2}e^{j\pi/3}$.
5. $X(1) = \frac{8}{3}$.

Determine $X(z)$ and specify its region of convergence.

10.32. Consider an LTI system with impulse response

$$h[n] = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

and input

$$x[n] = \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}.$$

(a) Determine the output $y[n]$ by explicitly evaluating the discrete convolution of $x[n]$ and $h[n]$.

(b) Determine the output $y[n]$ by computing the inverse z-transform of the product of the z-transforms of the input and the unit sample response.

10.33. **(a)** Determine the system function for the causal LTI system with difference equation

$$y[n] - \frac{1}{2}y[n-1] + \frac{1}{4}y[n-2] = x[n].$$

(b) Using z-transforms, determine $y[n]$ if

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$

10.34. A causal LTI system is described by the difference equation

$$y[n] = y[n-1] + y[n-2] + x[n-1].$$

(a) Find the system function $H(z) = Y(z)/X(z)$ for this system. Plot the poles and zeros of $H(z)$ and indicate the region of convergence.
(b) Find the unit sample response of the system.
(c) You should have found the system to be unstable. Find a stable (noncausal) unit sample response that satisfies the difference equation.

10.35. Consider an LTI system with input $x[n]$ and output $y[n]$ for which

$$y[n-1] - \frac{5}{2}y[n] + y[n+1] = x[n].$$

The system may or may not be stable or causal.

By considering the pole-zero pattern associated with the preceding difference equation, determine three possible choices for the unit sample response of the system. Show that each choice satisfies the difference equation.

10.36. Consider the linear, discrete-time, shift-invariant system with input $x[n]$ and output $y[n]$ for which

$$y[n-1] - \frac{10}{3}y[n] + y[n+1] = x[n].$$

The system is stable. Determine the unit sample response.

10.37. The input $x[n]$ and output $y[n]$ of a causal LTI system are related through the block-diagram representation shown in Figure P10.37.

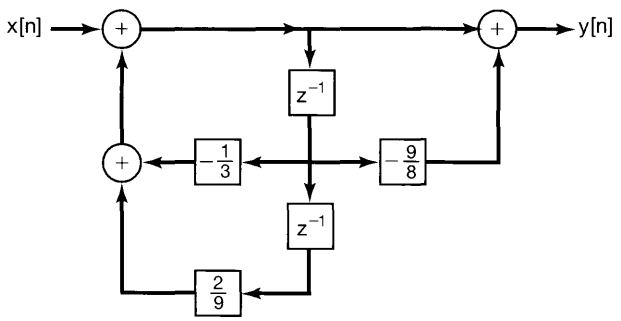


Figure P10.37

(a) Determine a difference equation relating $y[n]$ and $x[n]$.
(b) Is this system stable?

10.38. Consider a causal LTI system S with input $x[n]$ and a system function specified as

$$H(z) = H_1(z)H_2(z),$$

where

$$H_1(z) = \frac{1}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

and

$$H_2(z) = 1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}.$$

A block diagram corresponding to $H(z)$ may be obtained as a cascade connection of a block diagram for $H_1(z)$ followed by a block diagram for $H_2(z)$. The result is shown in Figure P10.38, in which we have also labeled the intermediate signals $e_1[n]$, $e_2[n]$, $f_1[n]$, and $f_2[n]$.

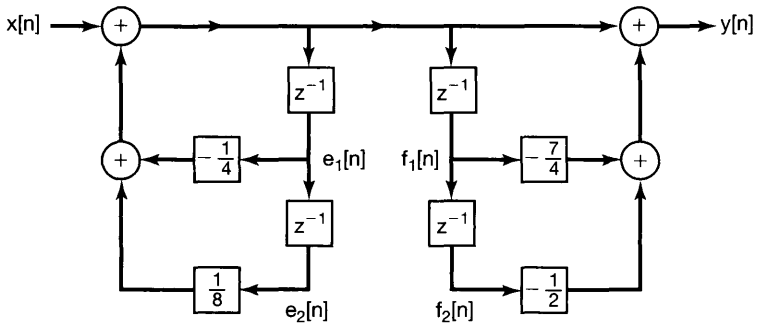


Figure P10.38

(a) How is $e_1[n]$ related to $f_1[n]$?
(b) How is $e_2[n]$ related to $f_2[n]$?
(c) Using your answers to the previous two parts as a guide, construct a direct-form block diagram for S that contains only two delay elements.
(d) Draw a cascade-form block diagram representation for S based on the observation that

$$H(z) = \left(\frac{1 + \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1}}\right)\left(\frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}}\right).$$

(e) Draw a parallel-form block diagram representation for S based on the observation that

$$H(z) = 4 + \frac{5/3}{1 + \frac{1}{2}z^{-1}} - \frac{14/3}{1 - \frac{1}{4}z^{-1}}.$$

10.39. Consider the following three system functions corresponding to causal LTI systems:

$$H_1(z) = \frac{1}{(1 - z^{-1} + \frac{1}{4}z^{-2})(1 - \frac{2}{3}z^{-1} + \frac{1}{9}z^{-2})},$$

$$H_2(z) = \frac{1}{(1 - z^{-1} + \frac{1}{2}z^{-2})(1 - \frac{1}{2}z^{-1} + z^{-2})},$$

$$H_3(z) = \frac{1}{(1 - z^{-1} + \frac{1}{2}z^{-2})(1 - z^{-1} + \frac{1}{4}z^{-2})}.$$

(a) For each system function, draw a direct-form block diagram.
(b) For each system function, draw a block diagram that corresponds to the cascade connection of two second-order block diagrams. Each second-order block diagram should be in direct form.
(c) For each system function, determine whether there exists a block diagram representation which is the cascade of four first-order block diagrams with the constraint that all the coefficient multipliers must be real.

10.40. Determine the unilateral z-transform for each of the sequences in Problem 10.21.

10.41. Consider the following two signals:

$$x_1[n] = \left(\frac{1}{2}\right)^{n+1} u[n+1],$$

$$x_2[n] = \left(\frac{1}{4}\right)^{n} u[n].$$

Let $\mathfrak{X}_1(z)$ and $X_1(z)$ respectively be the unilateral and bilateral z-transforms of $x_1[n]$, and let $\mathfrak{X}_2(z)$ and $X_2(z)$ respectively be the unilateral and bilateral z-transforms of $x_2[n]$.

(a) Take the inverse bilateral z-transform of $X_1(z)X_2(z)$ to determine $g[n] = x_1[n] * x_2[n]$.
(b) Take the inverse unilateral z-transform of $\mathfrak{X}_1(z)\mathfrak{X}_2(z)$ to obtain a signal $q[n]$ for $n \geq 0$. Observe that $q[n]$ and $g[n]$ are not identical for $n \geq 0$.

10.42. For each of the following difference equations and associated input and initial conditions, determine the zero-input and zero-state responses by using the unilateral z-transform:

(a) $y[n] + 3y[n-1] = x[n]$,
$$x[n] = \left(\frac{1}{2}\right)^{n} u[n],$$
$$y[-1] = 1.$$

(b) $y[n] - \frac{1}{2}y[n-1] = x[n] - \frac{1}{2}x[n-1]$,
$$x[n] = u[n],$$
$$y[-1] = 0.$$

(c) $y[n] - \frac{1}{2}y[n-1] = x[n] - \frac{1}{2}x[n-1]$,
$$x[n] = u[n],$$
$$y[-1] = 1.$$

ADVANCED PROBLEMS

10.43. Consider an even sequence $x[n]$ (i.e., $x[n] = x[-n]$) with rational z-transform $X(z)$.

(a) From the definition of the z-transform, show that

$$X(z) = X\left(\frac{1}{z}\right).$$

(b) From your results in part (a), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = 1/z_0$.

(c) Verify the result in part (b) for each of the following sequences:

(1) $\delta[n+1] + \delta[n-1]$

(2) $\delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$

10.44. Let $x[n]$ be a discrete-time signal with z-transform $X(z)$. For each of the following signals, determine the z-transform in terms of $X(z)$:

(a) $\Delta x[n]$, where Δ is the first-difference operator defined by

$$\Delta x[n] = x[n] - x[n-1]$$

(b) $x_1[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

(c) $x_1[n] = x[2n]$

10.45. Determine which of the following z-transforms could be the transfer function of a discrete-time linear system that is not necessarily stable, but for which the unit sample response is zero for $n < 0$. State your reasons clearly.

(a) $\dfrac{(1 - z^{-1})^2}{1 - \frac{1}{2}z^{-1}}$ **(b)** $\dfrac{(z-1)^2}{z - \frac{1}{2}}$

(c) $\dfrac{(z - \frac{1}{4})^5}{(z - \frac{1}{2})^6}$ **(d)** $\dfrac{(z - \frac{1}{4})^6}{(z - \frac{1}{2})^5}$

10.46. A sequence $x[n]$ is the output of an LTI system whose input is $s[n]$. The system is described by the difference equation

$$x[n] = s[n] - e^{8\alpha}s[n-8],$$

where $0 < \alpha < 1$.

(a) Find the system function

$$H_1(z) = \frac{X(z)}{S(z)},$$

and plot its poles and zeros in the z-plane. Indicate the region of convergence.

(b) We wish to recover $s[n]$ from $x[n]$ with an LTI system. Find the system function

$$H_2(z) = \frac{Y(z)}{X(z)}$$

such that $y[n] = s[n]$. Find all possible regions of convergence for $H_2(z)$, and for each, tell whether or not the system is causal or stable.

(c) Find all possible choices for the unit impulse response $h_2[n]$ such that

$$y[n] = h_2[n] * x[n] = s[n].$$

10.47. The following is known about a discrete-time LTI system with input $x[n]$ and output $y[n]$:

1. If $x[n] = (-2)^n$ for all n, then $y[n] = 0$ for all n.

2. If $x[n] = (1/2)^n u[n]$ for all n, then $y[n]$ for all n is of the form

$$y[n] = \delta[n] + a\left(\frac{1}{4}\right)^n u[n],$$

where a is a constant.

(a) Determine the value of the constant a.

(b) Determine the response $y[n]$ if the input $x[n]$ is

$$x[n] = 1, \text{ for all } n.$$

10.48. Suppose a second-order causal LTI system has been designed with a real impulse response $h_1[n]$ and a rational system function $H_1(z)$. The pole-zero plot for $H_1(z)$ is shown in Figure P10.48(a). Now consider another causal second-order system with impulse response $h_2[n]$ and rational system function $H_2(z)$. The pole-zero plot for $H_2(z)$ is shown in Figure P10.48(b). Determine a sequence $g[n]$ such that the following three conditions hold:

(1) $h_2[n] = g[n]h_1[n]$

(2) $g[n] = 0$ for $n < 0$

(3) $\sum_{k=0}^{\infty} |g[k]| = 3$

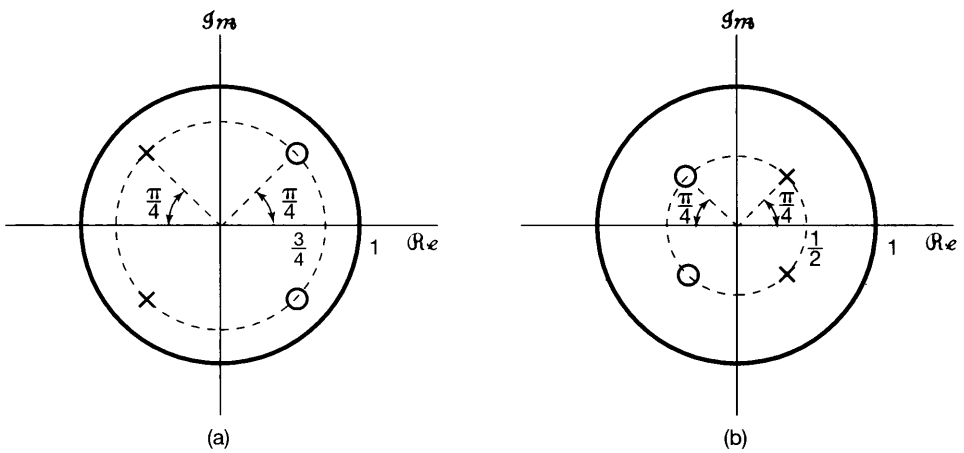


Figure P10.48

10.49. In Property 4 of Section 10.2, it was stated that if $x[n]$ is a right-sided sequence and if the circle $|z| = r_0$ is in the ROC, then all finite values of z for which $|z| > r_0$ will also be in the ROC. In this discussion an intuitive explanation was given. A more formal argument parallels closely that used for Property 4 of Section 9.2, relating to the Laplace transform. Specifically, consider a right-sided sequence

$$x[n] = 0, \quad n < N_1,$$

and for which

$$\sum_{n=-\infty}^{\infty} |x[n]|r_0^{-n} = \sum_{n=N_1}^{\infty} |x[n]|r_0^{-n} < \infty.$$

Then if $r_0 \leq r_1$,

$$\sum_{n=N_1}^{\infty} |x[n]|r_1^{-n} \leq A \sum_{n=N_1}^{\infty} |x[n]|r_0^{-n}, \tag{P10.49–1}$$

where A is a positive constant.

(a) Show that eq. (P10.49–1) is true, and determine the constant A in terms of r_0, r_1, and N_1.

(b) From your result in part (a), show that Property 4 of Section 10.2 follows.

(c) Develop an argument similar to the foregoing one to demonstrate the validity of Property 5 of Section 10.2.

10.50. A discrete-time system with the pole-zero pattern shown in Figure P10.50(a) is referred to as a first-order all-pass system, since the magnitude of the frequency response is constant regardless of frequency.

(a) Demonstrate algebraically that $|H(e^{j\omega})|$ is constant.

To demonstrate the same property geometrically, consider the vector diagram in Figure P10.50(b). We wish to show that the length of v_2 is proportional to the length of v_1 independently of the frequency ω.

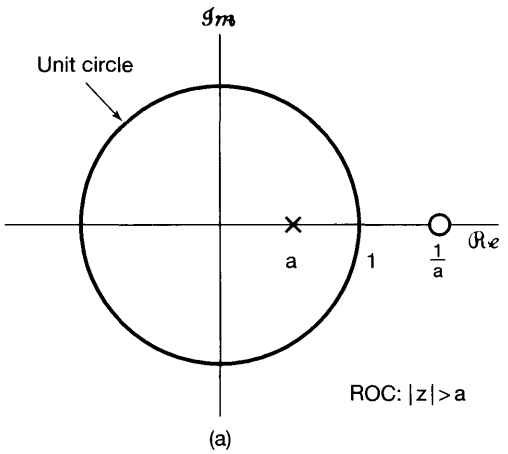


Figure P10.50a

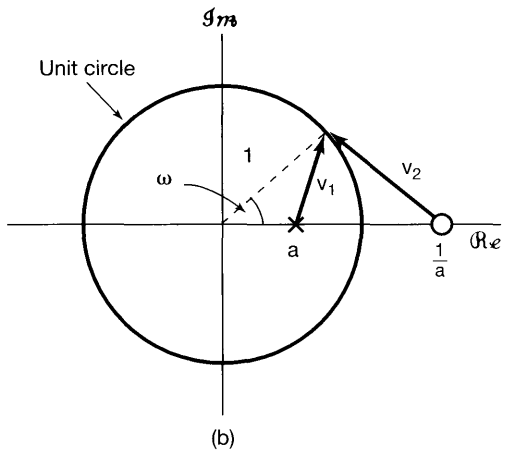


Figure P10.50b

(b) Express the length of v_1 using the law of cosines and the fact that v_1 is one leg of a triangle for which the other two legs are the unit vector and a vector of length a.

(c) In a manner similar to that in part (b), determine the length of v_2 and show that it is proportional in length to v_1 independently of ω.

10.51. Consider a real-valued sequence $x[n]$ with rational z-transform $X(z)$.

(a) From the definition of the z-transform, show that

$$X(z) = X^*(z^*).$$

(b) From your result in part (a), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = z_0^*$.

(c) Verify the result in part (b) for each of the following sequences:

(1) $x[n] = (\frac{1}{2})^n u[n]$

(2) $x[n] = \delta[n] - \frac{1}{2}\delta[n-1] + \frac{1}{4}\delta[n-2]$

(d) By combining your results in part (b) with the result of Problem 10.43(b), show that for a real, even sequence, if there is a pole (zero) of $H(z)$ at $z = \rho e^{j\theta}$, then there is also a pole (zero) of $H(z)$ at $z = (1/\rho)e^{j\theta}$ and at $z = (1/\rho)e^{-j\theta}$.

10.52. Consider a sequence $x_1[n]$ with z-transform $X_1(z)$ and a sequence $x_2[n]$ with z-transform $X_2(z)$, where

$$x_2[n] = x_1[-n].$$

Show that $X_2(z) = X_1(1/z)$, and from this, show that if $X_1(z)$ has a pole (or zero) at $z = z_0$, then $X_2(z)$ has a pole (or zero) at $z = 1/z_0$.

10.53. **(a)** Carry out the proof for each of the following properties in Table 10.1:

(1) Property set forth in Section 10.5.2

(2) Property set forth in Section 10.5.3

(3) Property set forth in Section 10.5.4

(b) With $X(z)$ denoting the z-transform of $x[n]$ and R_x the ROC of $X(z)$, determine, in terms of $X(z)$ and R_x, the z-transform and associated ROC for each of the following sequences:
(1) $x^*[n]$
(2) $z_0^n x[n]$, where z_0 is a complex number

10.54. In Section 10.5.9, we stated and proved the initial-value theorem for causal sequences.
(a) State and prove the corresponding theorem if $x[n]$ is anticausal (i.e., if $x[n] = 0, n > 0$).
(b) Show that if $x[n] = 0, n < 0$, then

$$x[1] = \lim_{z \to \infty} z(X(z) - x[0]).$$

10.55. Let $x[n]$ denote a causal sequence (i.e., if $x[n] = 0, n < 0$) for which $x[0]$ is nonzero and finite.
(a) Using the initial-value theorem, show that there are no poles or zeros of $X(z)$ at $z = \infty$.
(b) Show that, as a consequence of your result in part (a), the number of poles of $X(z)$ in the finite z-plane equals the number of zeros of $X(z)$ in the finite z-plane. (The finite z-plane excludes $z = \infty$.)

10.56. In Section 10.5.7, we stated the convolution property for the z-transform. To show that this property holds, we begin with the convolution sum expressed as

$$x_3[n] = x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]. \qquad \text{(P10.56–1)}$$

(a) By taking the z-transform of eq. (P10.56–1) and using eq. (10.3), show that

$$X_3(z) = \sum_{k=-\infty}^{\infty} x_1[k]\hat{X}_2(z),$$

where $\hat{X}_2(z)$ is the transform of $x_2[n-k]$.
(b) Using your result in part (a) and property 10.5.2 in Table 10.1, show that

$$X_3(z) = X_2(z) \sum_{k=-\infty}^{\infty} x_1[k]z^{-k}.$$

(c) From part (b), show that

$$X_3(z) = X_1(z)X_2(z)$$

as stated in eq. (10.81).

10.57. Let

$$X_1(z) = x_1[0] + x_1[1]z^{-1} + \cdots + x_1[N_1]z^{-N_1},$$
$$X_2(z) = x_2[0] + x_2[1]z^{-1} + \cdots + x_2[N_2]z^{-N_2}.$$

Define

$$Y(z) = X_1(z)X_2(z),$$

and let

$$Y(z) = \sum_{k=0}^{M} y[k]z^{-k}.$$

(a) Express M in terms of N_1 and N_2.
(b) Use polynomial multiplication to determine $y[0]$, $y[1]$, and $y[2]$.
(c) Use polynomial multiplication to show that, for $0 \leq k \leq M$,

$$y[k] = \sum_{m=-\infty}^{\infty} x_1[m]x_2[k-m].$$

10.58. A minimum-phase system is a system that is causal and stable and for which the inverse system is also causal and stable. Determine the necessary constraints on the location in the z-plane of the poles and zeros of the system function of a minimum-phase system.

10.59. Consider the digital filter structure shown in Figure P10.59.

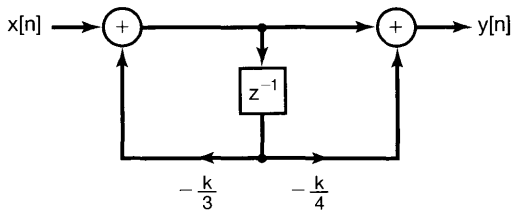


Figure P10.59

(a) Find $H(z)$ for this causal filter. Plot the pole-zero pattern and indicate the region of convergence.
(b) For what values of the k is the system stable?
(c) Determine $y[n]$ if $k = 1$ and $x[n] = (2/3)^n$ for all n.

10.60. Consider a signal $x[n]$ whose unilateral z-transform is $\mathfrak{X}(z)$. Show that the unilateral z-transform of $y[n] = x[n+1]$ may be specified as

$$\mathfrak{Y}(z) = z\mathfrak{X}(z) - zx[0].$$

10.61. If $\mathfrak{X}(z)$ denotes the unilateral z-transform of $x[n]$, determine, in terms of $\mathfrak{X}(z)$, the unilateral z-transform of:
(a) $x[n+3]$ **(b)** $x[n-3]$ **(c)** $\sum_{k=-\infty}^{n} x[k]$

EXTENSION PROBLEMS

10.62. The autocorrelation sequence of a sequence $x[n]$ is defined as

$$\phi_{xx}[n] = \sum_{k=-\infty}^{\infty} x[k]x[n+k].$$

Determine the z-transform of $\phi_{xx}[n]$ in terms of the z-transform of $x[n]$.

10.63. By using the power-series expansion

$$\log(1 - w) = -\sum_{i=1}^{\infty} \frac{w^i}{i}, \quad |w| < 1,$$

determine the inverse of each of the following two z-transforms:

(a) $X(z) = \log(1 - 2z)$, $|z| < \frac{1}{2}$

(b) $X(z) = \log(1 - \frac{1}{2}z^{-1})$, $|z| > \frac{1}{2}$

10.64. By first differentiating $X(z)$ and using the appropriate properties of the z-transform, determine the sequence for which the z-transform is each of the following:

(a) $X(z) = \log(1 - 2z)$, $|z| < \frac{1}{2}$

(b) $X(z) = \log(1 - \frac{1}{2}z^{-1})$, $|z| > \frac{1}{2}$

Compare your results for (a) and (b) with the results obtained in Problem 10.63, in which the power-series expansion was used.

10.65. The *bilinear transformation* is a mapping for obtaining a rational z-transform $H_d(z)$ from a rational Laplace transform $H_c(s)$. This mapping has two important properties:

1. If $H_c(s)$ is the Laplace transform of a causal and stable LTI system, then $H_d(z)$ is the z-transform of a causal and stable LTI system.
2. Certain important characteristics of $|H_c(j\omega)|$ are preserved in $|H_d(e^{j\omega})|$.

In this problem, we illustrate the second of these properties for the case of all-pass filters.

(a) Let

$$H_c(s) = \frac{a - s}{s + a},$$

where a is real and positive. Show that

$$|H_c(j\omega)| = 1.$$

(b) Let us now apply the bilinear transformation to $H_c(s)$ in order to obtain $H_d(z)$. That is,

$$H_d(z) = H_c(s)\Big|_{s = \frac{1 - z^{-1}}{1 + z^{-1}}}.$$

Show that $H_d(z)$ has one pole (which is inside the unit circle) and one zero (which is outside the unit circle).

(c) For the system function $H_d(z)$ derived in part (b), show that $|H_d(e^{j\omega})| = 1$.

10.66. The bilinear transformation, introduced in the previous problem, may also be used to obtain a discrete-time filter, the magnitude of whose frequency response is similar to the magnitude of the frequency response of a given continuous-time low-pass filter. In this problem, we illustrate the similarity through the example of a continuous-time second-order Butterworth filter with system function $H_c(s)$.

(a) Let

$$H_d(z) = H_c(s)\Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}}.$$

Show that

$$H_d(e^{j\omega}) = H_c\left(j\tan\frac{\omega}{2}\right).$$

(b) Given that

$$H_c(s) = \frac{1}{(s + e^{j\pi/4})(s + e^{-j\pi/4})}$$

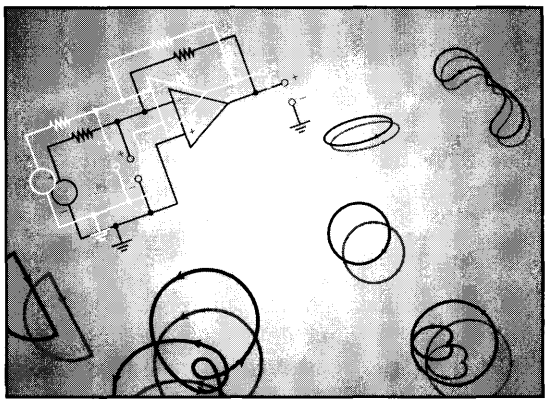
and that the corresponding filter is causal, verify that $H_c(0) = 1$, that $|H_c(j\omega)|$ decreases monotonically with increasing positive values of ω, that $|H_c(j)|^2 = 1/2$ (i.e., that $\omega_c = 1$ is the half-power frequency), and that $H_c(\infty) = 0$.

(c) Show that if the bilinear transformation is applied to $H_c(s)$ of part (b) in order to obtain $H_d(z)$, then the following may be asserted about $H_d(z)$ and $H_d(e^{j\omega})$:

1. $H_d(z)$ has only two poles, both of which are *inside* the unit circle.
2. $H_d(e^{j0}) = 1$.
3. $|H_d(e^{j\omega})|$ decreases monotonically as ω goes from 0 to π.
4. The half-power frequency of $H_d(e^{j\omega})$ is $\pi/2$.

11

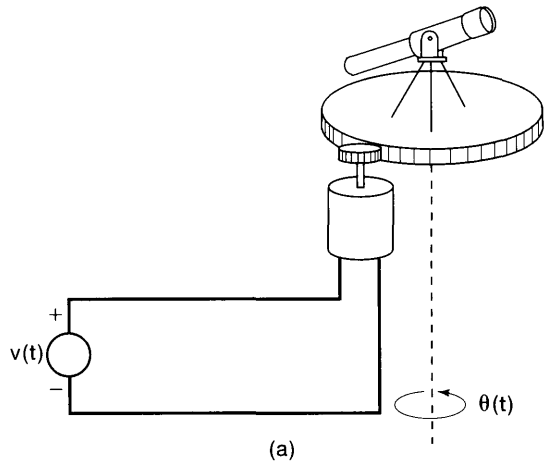
LINEAR FEEDBACK SYSTEMS



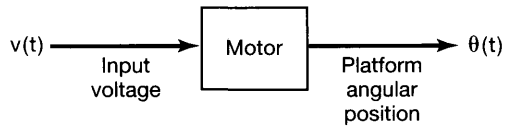
11.0 INTRODUCTION

It has long been recognized that in many situations there are particular advantages to be gained by using feedback—that is, by using the output of a system to control or modify the input. For example, it is common in electromechanical systems, such as a motor whose shaft position is to be maintained at a constant angle, to measure the error between the desired and the true position and to use this error in the form of a signal to turn the shaft in the appropriate direction. This is illustrated in Figure 11.1, where we have depicted the use of a dc motor for the accurate pointing of a telescope. In Figure 11.1(a) we have indicated pictorially what such a system would look like, where $v(t)$ is the input voltage to the motor and $\theta(t)$ is the angular position of the telescope platform. The block diagram for the motor-driven pointing system is shown in Figure 11.1(b). A feedback system for controlling the position of the telescope is illustrated in Figure 11.1(c), and a block diagram equivalent to this system is shown in Figure 11.1(d). The external, or *reference,* input to this feedback system is the desired shaft angle θ_D. A potentiometer is used to convert the angle into a voltage $K_1\theta_D$ proportional to θ_D. Similarly, a second potentiometer produces a voltage $K_1\theta(t)$ proportional to the actual platform angle. These two voltages are compared, producing an error voltage $K_1\big(\theta_D - \theta(t)\big)$, which is amplified and then used to drive the electric motor.

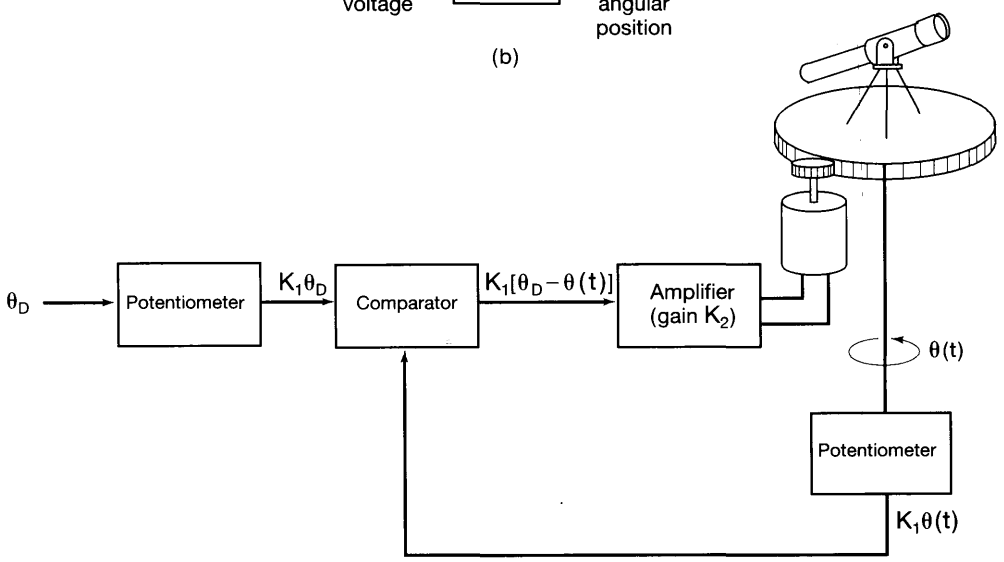
Figure 11.1 suggests two different methods for pointing the telescope. One of these is the feedback system of Figures 11.1(c) and (d). Here, the input that we must provide is the desired reference angle θ_D. Alternatively, if the initial angle, the desired angle, and the detailed electrical and mechanical characteristics of the motor-shaft assembly were known exactly, we could specify the precise history of the input voltage $v(t)$ that would first accelerate and then decelerate the shaft, bringing the platform to a stop at the desired



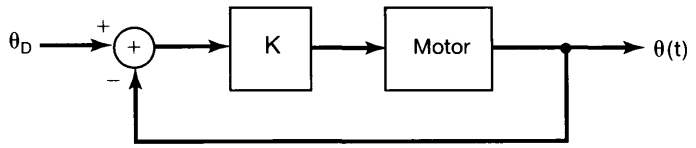
(a)



(b)



(c)



(d)

Figure 11.1 Use of feedback to control the angular position of a telescope: (a) dc motor-driven telescope platform; (b) block diagram of the system in (a); (c) feedback system for pointing the telescope; (d) block diagram of the system in (c) (here, $K = K_1K_2$).

position without the use of feedback, as in Figures 11.1(a) and (b). A system operating in accordance with Figures 11.1(a) and (b) is typically referred to as an *open-loop system,* in contrast to the *closed-loop system* of Figures 11.1(c) and (d). In a practical environment, there are clear advantages to controlling the motor-shaft angle with the closed-loop system rather than with the open-loop system. For example, in the closed-loop system, when the shaft has been rotated to the correct position, any disturbance from this position will be sensed, and the resulting error will be used to provide a correction. In the open-loop system, there is no mechanism for providing a correction. As another advantage of the closed-loop system, consider the effect of errors in modeling the characteristics of the motor-shaft assembly. In the open-loop system, a precise characterization of the system is required to design the correct input. In the closed-loop system, the input is simply the desired shaft angle and does not require precise knowledge of the system. This insensitivity of the closed-loop system to disturbances and to imprecise knowledge of the system are two important advantages of feedback.

The control of an electric motor is just one of a great many examples in which feedback plays an important role. Similar uses of feedback can be found in a wide variety of applications, such as chemical process control, automotive fuel systems, household heating systems, and aerospace systems, to name just a few. In addition, feedback is also present in many biological processes and in the control of human motion. For example, when a person reaches for an object, it is usual during the reaching process to monitor visually the distance between the hand and the object so that the velocity of the hand can be smoothly decreased as the distance (i.e., the error) between the hand and the object decreases. The effectiveness of using the system output (hand position) to control the input is clearly demonstrated by alternatively reaching with and without the use of visual feedback.

In addition to its use in providing an error-correcting mechanism that can reduce sensitivity to disturbances and to errors in the modeling of the system that is to be controlled, another important characteristic of feedback is its potential for stabilizing a system that is inherently unstable. Consider the problem of trying to balance a broomstick in the palm of the hand. If the hand is held stationary, small disturbances (such as a slight breeze or inadvertent motion of the hand) will cause the broom to fall over. Of course, if one knows exactly what disturbances will occur, and if one can control the motion of the hand perfectly, it is possible to determine in advance how to move the hand to balance the broom. This is clearly unrealistic; however, by always moving the hand in the direction in which the broom is falling, the broom can be balanced. This, of course, *requires* feedback in order to sense the direction in which the broom is falling. A second example that is closely related to the balancing of a broom is the problem of controlling a so-called inverted pendulum, which is illustrated in Figure 11.2. As shown, an inverted pendulum consists of a thin rod with a weight at the top. The bottom of the rod is mounted on a cart that can move in either direction along a track. Again, if the cart is kept stationary, the inverted pendulum

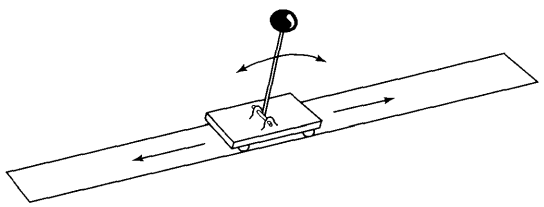


Figure 11.2 An inverted pendulum.

will topple over. The problem of stabilizing the pendulum is one of designing a feedback system that will move the cart to keep the pendulum vertical. This example is examined in Problem 11.56. A third example, which again bears some similarity to the balancing of a broom, is the problem of controlling the trajectory of a rocket. In this case, much as the movement of the hand is used to compensate for disturbances in the position of the broom, the direction of the thrust of the rocket is used to correct for changes in aerodynamic forces and wind disturbances that would otherwise cause the rocket to deviate from its course. Again, feedback is important, because these forces and disturbances are never precisely known in advance.

The preceding examples provide some indication of why feedback may be useful. In the next two sections we introduce the basic block diagrams and equations for linear feedback systems and discuss in more detail a number of applications of feedback and control, both in continuous time and in discrete time. We also point out how feedback can have harmful as well as useful effects. These examples of the uses and effects of feedback will give us some insight into how changes in the parameters in a feedback control system lead to changes in the behavior of the system. Understanding this relationship is essential in designing feedback systems that have certain desirable characteristics. With this material as background, we will then develop, in the remaining sections of the chapter, several specific techniques that are of significant value in the analysis and design of continuous-time and discrete-time feedback systems.

11.1 LINEAR FEEDBACK SYSTEMS

The general configuration of a continuous-time LTI feedback system is shown in Figure 11.3(a) and that of a discrete-time LTI feedback system in Figure 11.3(b). Because of

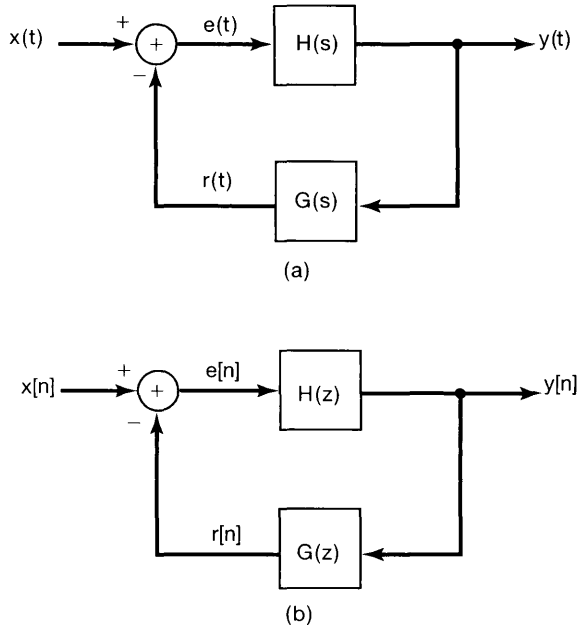


Figure 11.3 Basic feedback system configurations in (a) continuous time and (b) discrete time.

the typical applications in which feedback is utilized, it is natural to restrict the systems in these figures to be causal. This will be our assumption throughout the chapter. In that case, the system functions in Figure 11.3 can be interpreted either as unilateral or as bilateral transforms, and, as a consequence of causality, the ROC's associated with them will always be to the right of the rightmost pole for Laplace transforms and outside the outermost pole for z-transforms.

It should also be noted that the convention used in Figure 11.3(a) is that $r(t)$, the signal fed back, is subtracted from the input $x(t)$ to form $e(t)$. The identical convention is adopted in discrete time. Historically, this convention arose in tracking-system applications, where $x(t)$ represented a desired command and $e(t)$ represented the error between the command and the actual response $r(t)$. This was the case, for example, in our discussion of the pointing of a telescope. In more general feedback systems, $e(t)$ and $e[n]$, the discrete-time counterpart of $e(t)$, may not correspond to or be directly interpretable as error signals.

The system function $H(s)$ in Figure 11.3(a) or $H(z)$ in Figure 11.3(b) is referred to as the *system function of the forward path* and $G(s)$ or $G(z)$ as the *system function of the feedback path.* The system function of the overall system of Figure 11.3(a) or (b) is referred to as the *closed-loop system function* and will be denoted by $Q(s)$ or $Q(z)$. In Sections 9.8.1 and 10.8.1, we derived expressions for the system functions of feedback interconnections of LTI systems. Applying these results to the feedback systems of Figure 11.3, we obtain

$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}, \tag{11.1}$$

$$Q(z) = \frac{Y(z)}{X(z)} = \frac{H(z)}{1 + G(z)H(z)}. \tag{11.2}$$

Equations (11.1) and (11.2) represent the fundamental equations for the study of LTI feedback systems. In the following sections, we use these equations as the basis for gaining insight into the properties of feedback systems and for developing several tools for their analysis.

11.2 SOME APPLICATIONS AND CONSEQUENCES OF FEEDBACK

In the introduction, we provided a brief, intuitive look at some of the properties and uses of feedback systems. In this section, we examine a number of the characteristics and applications of feedback in somewhat more quantitative terms, using the basic feedback equations (11.1) and (11.2) as a starting point. Our purpose is to provide an introduction to and an appreciation for the applications of feedback, rather than to develop any of these applications in detail. In the sections that follow, we focus in more depth on several specific techniques for analyzing feedback systems that are useful in a wide range of problems, including many of the applications that we are about to describe.

11.2.1 Inverse System Design

In some applications, one would like to synthesize the inverse of a given continuous-time system. Suppose that this system has system function $P(s)$, and consider the feedback system shown in Figure 11.4. Applying equation (11.1) with $H(s) = K$ and $G(s) = P(s)$,

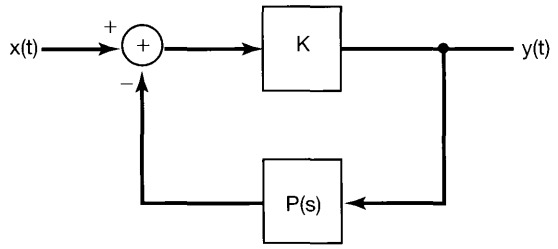


Figure 11.4 Form of a feedback system used in implementing the inverse of the system with system function $P(s)$.

we find that the closed-loop system function is

$$Q(s) = \frac{K}{1 + KP(s)}. \tag{11.3}$$

If the gain K is sufficiently large so that $KP(s) \gg 1$, then

$$Q(s) \simeq \frac{1}{P(s)}, \tag{11.4}$$

in which case the feedback system approximates the inverse of the system with system function $P(s)$.

It is important to note that the result in eq. (11.4) requires that the gain K be sufficiently high, but is otherwise not dependent on the precise value of the gain. Operational amplifiers are one class of devices that provide this kind of gain and are widely used in feedback systems. One common application of the inversion inherent in eq. (11.4) is in the implementation of integrators. A capacitor has the property that its current is proportional to the derivative of the voltage. By inserting a capacitor in the feedback path around an operational amplifier, the differentiation property of the capacitor is inverted to provide integration. This specific application is explored in more detail in Problems 11.50–11.52.

Although our discussion is for the most part restricted to linear systems, it is worth pointing out that this same basic approach is commonly used in inverting a nonlinearity. For example, systems for which the output is the logarithm of the input are commonly implemented by utilizing the exponential current-voltage characteristics of a diode as feedback around an operational amplifier. This is explored in more detail in Problem 11.53.

11.2.2 Compensation for Nonideal Elements

Another common use of feedback is to correct for some of the nonideal properties of the open-loop system. For example, feedback is often used in the design of amplifiers to provide constant-gain amplification in a given frequency band, and in fact, it is this application, pioneered by H. S. Black at Bell Telephone Laboratories in the 1920s, that is generally considered to have been the catalyst for the development of feedback control as a practical and useful system design methodology.

Specifically, consider an open-loop frequency response $H(j\omega)$ which provides amplification over the specified frequency band, but which is not constant over that range. For example, operational amplifiers or the vacuum tube amplifiers of concern to Black and his colleagues typically provide considerable, but not precisely controlled, amplification.

While such devices can provide raw amplification levels of several orders of magnitude, the price one pays for this includes uncertain levels of amplification that can fluctuate with frequency, time, temperature, etc., and that can also introduce unwanted phase and nonlinear distortions. What Black proposed was placing such a powerful, but uncertain and erratic, amplifier in a feedback loop as in Figure 11.3(a) with $G(s)$ chosen to be constant, i.e., $G(s) = K$. In this case, assuming the closed-loop system is stable, its frequency response is

$$Q(j\omega) = \frac{H(j\omega)}{1 + KH(j\omega)}. \tag{11.5}$$

If, over the specified frequency range,

$$|KH(j\omega)| \gg 1, \tag{11.6}$$

then

$$Q(j\omega) \simeq \frac{1}{K}. \tag{11.7}$$

That is, the closed-loop frequency response is constant, as desired. This of course assumes that the system in the feedback path can be designed so that its frequency response $G(j\omega)$ has a constant gain K over the desired frequency band, which is precisely what we assumed we could *not* ensure for $H(j\omega)$. The difference between the requirement on $H(j\omega)$ and that on $G(j\omega)$, however, is that $H(j\omega)$ must provide amplification, whereas, from eq. (11.7), we see that for the overall closed-loop system to provide a gain greater than unity, K must be less than 1. That is, $G(j\omega)$ must be an attenuator over the specified range of frequencies. In general, an attenuator with approximately flat frequency characteristics is considerably easier to realize than an amplifier with approximately flat frequency response (since an attenuator can be constructed from passive elements).

The use of feedback to flatten the frequency response incurs some cost, however, and it is this fact that led to the considerable skepticism with which Black's idea was met. In particular, from eqs. (11.6) and (11.7), we see that

$$|H(j\omega)| \gg \frac{1}{K} \simeq Q(j\omega), \tag{11.8}$$

so that the closed-loop gain $1/K$ will be substantially less than the open-loop gain $|H(j\omega)|$. This apparently significant loss of gain, attributable to what Black referred to as *degenerative* or negative feedback, was initially viewed as a serious weakness in his negative-feedback amplifier. Indeed, the effect had been known for many years and had led to the conviction that negative feedback was not a particularly useful mechanism. However, Black pointed out that what one gave up in overall gain was often more than offset by the reduced sensitivity of the overall closed-loop amplifier: The closed-loop system function is essentially equal to eq. (11.7), independently of variations in $H(j\omega)$, as long as $|H(j\omega)|$ is large enough. Thus, if the open-loop amplifier is initially designed with considerably more gain than is actually needed, the closed-loop amplifier will provide the desired levels of amplification with greatly reduced sensitivity. This concept and its application to extending the bandwidth of an amplifier are explored in Problem 11.49.

11.2.3 Stabilization of Unstable Systems

As mentioned in the introduction, one use of feedback systems is to stabilize systems that, without feedback, are unstable. Examples of this kind of application include the control of the trajectory of a rocket, the regulation of nuclear reactions in a nuclear power plant, the stabilization of an aircraft, and the natural and regulatory control of animal populations.

To illustrate how feedback can be used to stabilize an unstable system, let us consider a simple first-order continuous-time system with

$$H(s) = \frac{b}{s - a}. \tag{11.9}$$

With $a > 0$, the system is unstable. Choosing the system function $G(s)$ to be a constant gain K, we see that the closed-loop system function in eq. (11.1) becomes

$$\begin{aligned} Q(s) &= \frac{H(s)}{1 + KH(s)} \\ &= \frac{b}{s - a + Kb}. \end{aligned} \tag{11.10}$$

The closed-loop system will be stable if the pole is moved into the left half of the s-plane. This will be the case if

$$Kb > a. \tag{11.11}$$

Thus, we can stabilize the system with a constant gain in the feedback loop if that gain is chosen to satisfy eq. (11.11). This type of feedback system is referred to as a *proportional feedback system,* since the signal that is fed back is proportional to the output of the system.

As another example, consider the second-order system

$$H(s) = \frac{b}{s^2 + a}. \tag{11.12}$$

If $a > 0$, the system is an oscillator (i.e., $H(s)$ has its poles on the $j\omega$-axis), and the impulse response of the system is sinusoidal. If $a < 0$, $H(s)$ has one pole in the left-half plane and one in the right-half plane. Thus, in either case, the system is unstable. In fact, as considered in Problem 11.56, the system function given in eq. (11.12) with $a < 0$ can be used to model the dynamics of the inverted pendulum described in the introduction.

Let us first consider the use of proportional feedback for this second-order system; that is, we take

$$G(s) = K. \tag{11.13}$$

Substituting into eq. (11.1), we obtain

$$Q(s) = \frac{b}{s^2 + (a + Kb)}. \tag{11.14}$$

In our discussion of second-order systems in Chapter 6, we considered a transfer function of the form

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \tag{11.15}$$

For such a system to be stable, ω_n must be real and positive (i.e., $\omega_n^2 > 0$), and ζ must be positive (corresponding to positive damping). From eqs. (11.14) and (11.15), it follows that with proportional feedback we can only influence the value of ω_n^2, and consequently, we cannot stabilize the system because we cannot introduce any damping. To suggest a type of feedback that can be used to stabilize this system, recall the mass-spring-dashpot mechanical system described in our examination of second-order systems in Section 6.5.2. We saw that damping in that system was the result of the inclusion of a dashpot, which provided a restoring force proportional to the *velocity* of the mass. This suggests that we consider *proportional-plus-derivative* feedback, that is, a $G(s)$ of the form

$$G(s) = K_1 + K_2 s, \tag{11.16}$$

which yields

$$Q(s) = \frac{b}{s^2 + bK_2 s + (a + K_1 b)}. \tag{11.17}$$

The closed-loop poles will be in the left-half plane, and hence, the closed-loop system will be stable as long as we choose K_1 and K_2 to guarantee that

$$bK_2 > 0, \qquad a + K_1 b > 0. \tag{11.18}$$

The preceding discussion illustrates how feedback can be used to stabilize continuous-time systems. The stabilization of unstable systems is an important application of feedback for discrete-time systems as well. Examples of discrete-time systems that are unstable in the absence of feedback are models of population growth. To illustrate how feedback can prevent the unimpeded growth of populations, let us consider a simple model for the evolution of the population of a single species of animal. Let $y[n]$ denote the number of animals in the nth generation, and assume that without the presence of any impeding influences, the birthrate is such that the population would double each generation. In this case, the basic equation for the population dynamics of the species is

$$y[n] = 2y[n-1] + e[n], \tag{11.19}$$

where $e[n]$ represents any additions to or deletions from the population that are caused by external influences.

This population model is obviously unstable, with an impulse response that grows exponentially. However, in any ecological system, there are a number of factors that will inhibit the growth of a population. For example, limits on the food supply for the species will manifest themselves through a reduction in population growth when the number of animals becomes large. Similarly, if the species has natural enemies, it is often reasonable to assume that the population of the predators will grow when the population of the prey increases and, consequently, that the presence of natural enemies will retard population growth. In addition to natural influences such as these, there may be effects introduced by

humans that are aimed at population control. For example, the food supply or the predator population may fall under human regulation. In addition, stocking lakes with fish or importing animals from other areas can be used to promote growth, and the control of hunting or fishing can also provide a regulative effect. Because all of these influences depend on the size of the population (either naturally or by design), they represent feedback effects.

Based on the preceding discussion, we can separate $e[n]$ into two parts by means of the equation

$$e[n] = x[n] - r[n], \tag{11.20}$$

where $r[n]$ represents the effect of the regulative influences described in the previous paragraph and $x[n]$ incorporates any other external effects, such as the migration of animals or natural disasters or disease. Note that we have included a minus sign in eq. (11.20). This is consistent with our convention of using negative feedback, and here it also has the physical interpretation that, since the uninhibited growth of the population is unstable, the feedback term plays the role of a *retarding* influence. To see how the population can be controlled by the presence of this feedback term, suppose that the regulative influences account for the depletion of a fixed proportion β of the population in each generation. Since, according to our model, the surviving fraction of each generation will double in size, it follows that

$$y[n] = 2(1 - \beta)y[n - 1] + x[n]. \tag{11.21}$$

Comparing eq. (11.21) with eqs. (11.19) and (11.20), we see that

$$r[n] = 2\beta y[n - 1]. \tag{11.22}$$

The factor of 2 here represents the fact that the depletion of the present population decreases the number of births in the next generation.

This example of the use of feedback is illustrated in Figure 11.5. Here, the system function of the forward path is obtained from eq. (11.19) as

$$H(z) = \frac{1}{1 - 2z^{-1}}, \tag{11.23}$$

while from eq. (11.22) the system function of the feedback path is

$$G(z) = 2\beta z^{-1}. \tag{11.24}$$

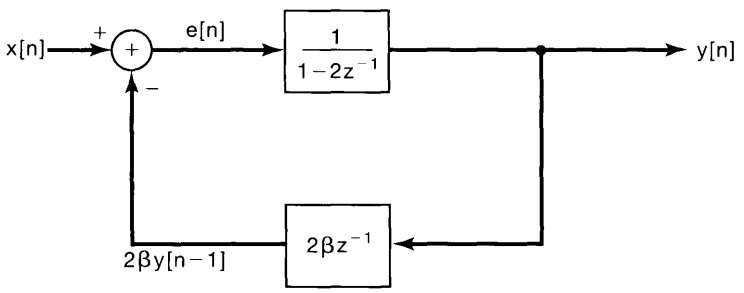


Figure 11.5 Block diagram of a simple feedback model of population dynamics.

Consequently, the closed-loop system function is

$$Q(z) = \frac{H(z)}{1 + G(z)H(z)} = \frac{1}{1 - 2(1 - \beta)z^{-1}}. \tag{11.25}$$

If $\beta < 1/2$, the closed-loop system is still unstable, whereas it is stable[1] if $1/2 < \beta < 3/2$.

Clearly, this example of population growth and control is extremely simplified. For instance, the feedback model of eq. (11.22) does not account for the fact that the part of $r[n]$ which is due to the presence of natural enemies depends upon the population of the predators, which in turn has its own growth dynamics. Such effects can be incorporated by making the feedback model more complex to reflect the presence of other dynamics in an ecological system, and the resulting models for the evolution of interacting species are extremely important in ecological studies. However, even without the incorporation of these effects, the simple model that we have described here does illustrate the basic ideas of how feedback can prevent both the unlimited proliferation of a species and its extinction. In particular, we can see at an elementary level how human-induced factors can be used. For example, if a natural disaster or an increase in the population of natural enemies causes a drastic decrease in the population of a species, a tightening of limits on hunting or fishing and accelerated efforts to increase the population can be used to decrease β in order to *destabilize* the system to allow for rapid growth, until a normal-size population is again attained.

Note also that for this type of problem, it is not usually the case that one wants strict stability. If the regulating influences are such that $\beta = 1/2$, and if all other external influences are zero (i.e., if $x[n] = 0$), then $y[n] = y[n - 1]$. Therefore, as long as $x[n]$ is small and averages to zero over several generations, a value of $\beta = 1/2$ will result in an essentially constant population. However, for this value of β the system is unstable, since eq. (11.21) then reduces to

$$y[n] = y[n - 1] + x[n]. \tag{11.26}$$

That is, the system is equivalent to an accumulator. Thus, if $x[n]$ is a unit step, the output grows without bound. Consequently, if a steady trend is expected in $x[n]$, caused, for example, by a migration of animals into a region, a value of $\beta > 1/2$ would need to be used to stabilize the system and thus to keep the population within bounds and maintain an ecological balance.

11.2.4 Sampled-Data Feedback Systems

In addition to dealing with problems such as the one just described, discrete-time feedback techniques are of great importance in a wide variety of applications involving continuous-time systems. The flexibility of digital systems has made the implementation of *sampled-data feedback systems* an extremely attractive option. In such a system, the output of a continuous-time system is sampled, some processing is done on the resulting sequence of samples, and a discrete sequence of feedback commands is generated. This sequence

[1]Although, in the context of our population example, β could never exceed unity, since $\beta > 1$ corresponds to removing more than 100% of the population.

is then converted to a continuous-time signal that is fed back to and subtracted from the external input to produce the actual input to the continuous-time system.

Clearly, the constraint of causality on feedback systems imposes a restriction on the process of converting the discrete-time feedback signal to a continuous-time signal (e.g., ideal lowpass filtering or any noncausal approximation of it is not allowed). One of the most widely used conversion systems is the zero-order hold (introduced in Section 7.1.2). The structure of a sampled-data feedback system involving a zero-order hold is depicted in Figure 11.6(a). In the figure, we have a continuous-time LTI system with system function $H(s)$ that is sampled to produce a discrete-time sequence

$$p[n] = y(nT). \tag{11.27}$$

The sequence $p[n]$ is then processed by a discrete-time LTI system with system function $G(z)$, and the resulting output is put through a zero-order hold to produce the continuous-time signal

$$z(t) = d[n] \qquad \text{for} \quad nT \leq t < (n+1)T. \tag{11.28}$$

This signal is subtracted from the external input $x(t)$ to produce $e(t)$.

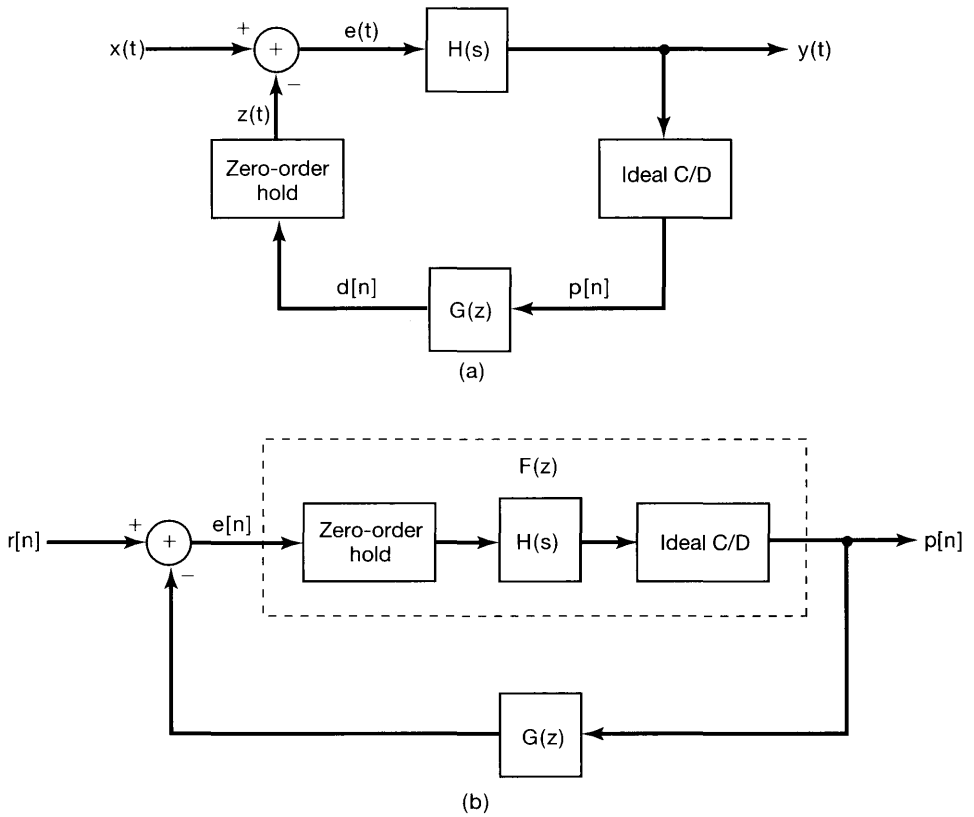


Figure 11.6 (a) A sampled-data feedback system using a zero-order hold; (b) equivalent discrete-time system.

Suppose also that $x(t)$ is constant over intervals of length T. That is,

$$x(t) = r[n] \qquad \text{for} \quad nT \leq t < (n+1)T, \tag{11.29}$$

where $r[n]$ is a discrete-time sequence. This is an approximation that is usually valid in practice, as the sampling rate is typically fast enough so that $x(t)$ does not change appreciably over intervals of length T. Furthermore, in many applications, the external input is itself actually generated by applying a zero-order hold operation to a discrete sequence. For example, in systems such as advanced aircraft, the external inputs represent human operator commands that are themselves first processed digitally and then converted back to continuous-time input signals. Because the zero-order hold is a linear operation, the feedback system of Figure 11.6(a) when $x(t)$ is given by eq. (11.29) is equivalent to the system of Figure 11.6(b).

As shown in Problem 11.60, the discrete-time system with input $e[n]$ and output $p[n]$ is an LTI system with system function $F(z)$ that is related to the continuous-time system function $H(s)$ by means of a *step-invariant* transformation. That is, if $s(t)$ is the step response of the continuous-time system, then the step response $q[n]$ of the discrete-time system consists of equally spaced samples of $s(t)$. Mathematically,

$$q[n] = s(nT) \qquad \text{for all } n. \tag{11.30}$$

Once we have determined $F(z)$, we have a completely discrete-time feedback system model (Figure 11.6(b)) exactly capturing the behavior of the continuous-time feedback system (Figure 11.6(a)) at the sampling instants $t = nT$, and we can then consider designing the feedback system function $G(z)$ to achieve our desired objectives. An example of designing such a sampled-data feedback system to stabilize an unstable continuous-time system is examined in detail in Problem 11.60.

11.2.5 Tracking Systems

As mentioned in Section 11.0, one of the important applications of feedback is in the design of systems in which the objective is to have the output track or follow the input. There is a broad range of problems in which tracking is an important component. For example, the telescope-pointing problem discussed in Section 11.0 is a tracking problem: The feedback system of Figures 11.1(c) and (d) has as its input the desired pointing angle, and the purpose of the feedback loop is to provide a mechanism for driving the telescope to follow the input. In airplane autopilots the input is the desired flight path of the vehicle, and the autopilot feedback system uses the aircraft control surfaces (rudder, ailerons, and elevator) and thrust control in order to keep the aircraft on the prescribed course.

To illustrate some of the issues that arise in the design of tracking systems, consider the discrete-time feedback system depicted in Figure 11.7(a). The examination of discrete-time tracking systems of this form often arises in analyzing the characteristics of sampled-data tracking systems for continuous-time applications. One example of such a system is a digital autopilot. In Figure 11.7(a), $H_p(z)$ denotes the system function of the system whose output is to be controlled. This system is often referred to as the *plant,* a term that can be traced to applications such as the control of power plants, heating systems, and chemical-processing plants. The system function $H_c(z)$ represents a compensator, which is the element to be designed. Here, the input to the compensator is the tracking error—

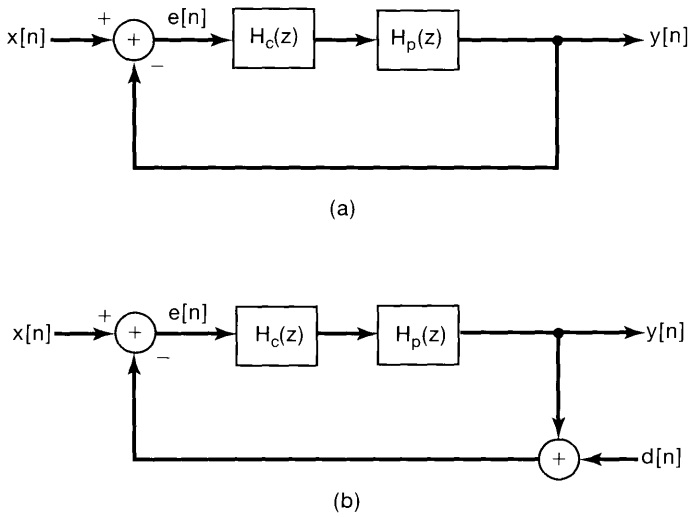


Figure 11.7 (a) Discrete-time tracking system; (b) tracking system of (a) with a disturbance $d[n]$ in the feedback path accounting for the presence of measurement errors.

that is, the difference $e[n]$ between the input $x[n]$ and the output $y[n]$. The output of the compensator is the input to the plant (for example, the actual voltage applied to the motor in the feedback system of Figures 11.1(c) and (d) or the actual physical input to the drive system of the rudder of an aircraft).

To simplify notation, let $H(z) = H_c(z)H_p(z)$. In this case, the application of eq. (11.2) yields the relationship

$$Y(z) = \frac{H(z)}{1 + H(z)}X(z). \tag{11.31}$$

Also, since $Y(z) = H(z)E(z)$, it follows that

$$E(z) = \frac{1}{1 + H(z)}X(z), \tag{11.32}$$

or, specializing to $z = e^{j\omega}$, we obtain

$$E(e^{j\omega}) = \frac{1}{1 + H(e^{j\omega})}X(e^{j\omega}). \tag{11.33}$$

Equation (11.33) provides us with some insight into the design of tracking systems. Specifically, for good tracking performance, we would like $e[n]$ or, equivalently, $E(e^{j\omega})$ to be small. That is,

$$\frac{1}{1 + H(e^{j\omega})}X(e^{j\omega}) \approx 0. \tag{11.34}$$

Consequently, for that range of frequencies for which $X(e^{j\omega})$ is nonzero, we would like $|H(e^{j\omega})|$ to be large. Thus, we have one of the fundamental principles of feedback system design: Good tracking performance requires a large gain. This desire for a large gain, however, must typically be tempered, for several reasons. One reason is that if the gain is *too* large, the closed-loop system may have undesirable characteristics (such as too little

damping) or might in fact become unstable. This possibility is discussed in the next section and is also addressed by the methods developed in subsequent sections.

In addition to the issue of stability, there are other reasons for wanting to limit the gain in a tracking system. For example, in implementing such a system, we must measure the output $y[n]$ in order to compare it to the command input $x[n]$, and any measuring device used will have inaccuracies and error sources (such as thermal noise in the electronics of the device). In Figure 11.7(b), we have included these error sources in the form of a disturbance input $d[n]$ in the feedback loop. Some simple system function algebra yields the following relationship between $Y(z)$ and the transforms $X(z)$ and $D(z)$ of $x[n]$ and $d[n]$:

$$Y(z) = \left[\frac{H(z)}{1 + H(z)}X(z)\right] - \left[\frac{H(z)}{1 + H(z)}D(z)\right]. \tag{11.35}$$

From this expression, we see that in order to minimize the influence of $d[n]$ on $y[n]$, we would like $H(z)$ to be small so that the second term on the right-hand side of eq. (11.35) is small.

From the preceding development, we see that the goals of tracking and of minimizing the effect of measurement errors are conflicting, and one must take this into account in coming up with an acceptable system design. In general, the design depends on more detailed information concerning the characteristics of the input $x[n]$ and the disturbance $d[n]$. For example, in many applications $x[n]$ has a significant amount of its energy concentrated at low frequencies, while measurement error sources such as thermal noise have a great deal of energy at high frequencies. Consequently, one usually designs the compensator $H_c(z)$ so that $|H(e^{j\omega})|$ is large at low frequencies and is small for ω near $\pm\pi$.

There are a variety of other issues that one must consider in designing tracking systems, such as the presence of disturbances at other points in the feedback loop. (For example, the effect of wind on the motion of an aircraft must be taken into account in designing an autopilot.) The methods of feedback system analysis introduced in this chapter provide the necessary tools for examining each of these issues. In Problem 11.57, we use some of these tools to investigate several other aspects of the problem of designing tracking systems.

11.2.6 Destabilization Caused by Feedback

As well as having many applications, feedback can have undesirable effects and can in fact cause instability. For example, consider the telescope-pointing system illustrated in Figure 11.1. From the discussion in the preceding section, we know that it would be desirable to have a large amplifier gain in order to achieve good performance in tracking the desired pointing angle. On the other hand, as we increase the gain, we are likely to obtain faster tracking response at the expense of a reduction in system damping, resulting in significant overshoot and ringing in response to changes in the desired angle. Furthermore, instability can result if the gain is increased too much.

Another common example of the possible destabilizing effect of feedback is feedback in audio systems. Consider the situation depicted in Figure 11.8(a). Here, a loudspeaker produces an audio signal that is an amplified version of the sounds picked up by a microphone. Note that in addition to other audio inputs, the sound coming from the speaker itself may be sensed by the microphone. How strong this particular signal is depends upon

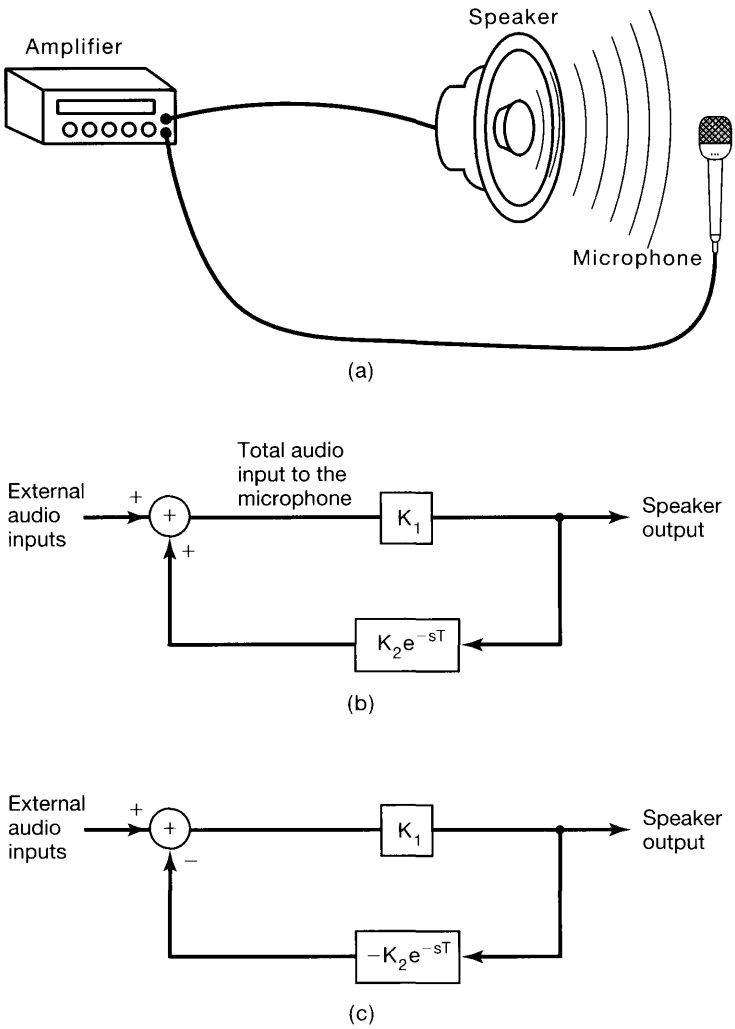


Figure 11.8 (a) Pictorial representation of the phenomenon of audio feedback; (b) block diagram representation of (a); (c) block diagram in (b) redrawn as a negative feedback system. (*Note:* e^{-sT} is the system function of a T-second time delay.)

the distance between the speaker and the microphone. Specifically, because of the attenuating properties of air, the strength of the signal reaching the microphone from the speaker decreases as the distance between the speaker and the microphone increases. In addition, due to the finite speed of propagation of sound waves, there is time delay between the signal produced by the speaker and that sensed by the microphone.

This audio feedback system is represented in block diagram form in Figure 11.8(b). Here, the constant K_2 in the feedback path represents the attenuation, and T is the propagation delay. The constant K_1 is the amplifier gain. Also, note that the output from the feedback path is *added* to the external input. This is an example of *positive feedback.* As discussed at the beginning of the section, the use of a negative sign in the definition of the basic feedback system of Figure 11.3 is purely conventional, and positive and negative feedback systems can be analyzed using the same tools. For example, as illustrated in Figure 11.8(c), the feedback system of Figure 11.8(b) can be written as a negative feedback

system by adding a minus sign to the feedback-path system function. From this figure and from eq. (11.1), we can determine the closed-loop system function:

$$Q(s) = \frac{K_1}{1 - K_1 K_2 e^{-sT}}. \tag{11.36}$$

Later we will return to this example, and, using a technique that we will develop in Section 11.3, we will show that the system of Figure 11.8 is unstable if

$$K_1 K_2 > 1. \tag{11.37}$$

Since the attenuation due to the propagation of sound through the air decreases (i.e., K_2 *increases*) as the distance between the speaker and the microphone decreases, if the microphone is placed too close to the speaker, so that eq. (11.37) is satisfied, the system will be unstable. The result of this instability is an excessive amplification and distortion of audio signals.

It is interesting to note that positive, or what Black referred to as *regenerative,* feedback had also been known for some time before he invented his negative feedback amplifier and, ironically, had been viewed as a very useful mechanism (in contrast to the skeptical view of negative feedback). Indeed, positive feedback can be useful. For example, it was already known in the 1920s that the destabilizing influence of positive feedback could be used to generate oscillating signals. This use of positive feedback is illustrated in Problem 11.54.

In this section, we have described a number of the applications of feedback. These and others, such as the use of feedback in the implementation of recursive discrete-time filters (see Problem 11.55), are considered in more detail in the problems at the end of the chapter. From our examination of the uses of feedback and the possible stabilizing and destabilizing effects that it can have, it is clear that some care must be taken in designing and analyzing feedback systems to ensure that the closed-loop system behaves in a desirable fashion. Specifically, in Sections 11.2.3 and 11.2.6, we have seen several examples of feedback systems in which the characteristics of the closed-loop system can be significantly altered by changing the values of one or two parameters in the feedback system. In the remaining sections of this chapter, we develop several techniques for analyzing the effect of changes in such parameters on the closed-loop system and for designing systems to meet desired objectives such as stability, adequate damping, etc.

11.3 ROOT-LOCUS ANALYSIS OF LINEAR FEEDBACK SYSTEMS

As we have seen in a number of the examples and applications we have discussed, a useful type of feedback system is that in which the system has an adjustable gain K associated with it. As this gain is varied, it is of interest to examine how the poles of the closed-loop system change, since the locations of these poles tell us a great deal about the behavior of the system. For example, in stabilizing an unstable system, the adjustable gain is used to move the poles into the left-half plane for a continuous-time system or inside the unit circle for a discrete-time system. In addition, in Problem 11.49, we show that feedback can be used to broaden the bandwidth of a first-order system by moving the pole so as to decrease the time constant of the system. Furthermore, just as feedback can be used

to relocate the poles to improve system performance, as we saw in Section 11.2.6, there is the potential danger that with an improper choice of feedback a stable system can be destabilized, which is usually undesirable.

In this section, we discuss a particular method for examining the locus (i.e., the path) in the complex plane of the poles of the closed-loop system as an adjustable gain is varied. The procedure, referred to as the *root-locus method,* is a graphical technique for plotting the closed-loop poles of a rational system function $Q(s)$ or $Q(z)$ as a function of the value of the gain. The technique works in an identical manner for both continuous-time and discrete-time systems.

11.3.1 An Introductory Example

To illustrate the basic nature of the root-locus method for analyzing a feedback system, let us reexamine the discrete-time example considered in the preceding section and specified by the system functions

[eq. (11.23)] $$H(z) = \frac{1}{1 - 2z^{-1}} = \frac{z}{z - 2} \tag{11.38}$$

and

[eq. (11.24)] $$G(z) = 2\beta z^{-1} = \frac{2\beta}{z}, \tag{11.39}$$

where β now is viewed as an adjustable gain. Then, as we noted earlier, the closed-loop system function is

[eq. (11.25)] $$Q(z) = \frac{1}{1 - 2(1 - \beta)z^{-1}} = \frac{z}{z - 2(1 - \beta)}. \tag{11.40}$$

In this example, it is straightforward to identify the closed-loop pole as being located at $z = 2(1 - \beta)$. In Figure 11.9(a), we have plotted the locus of the pole for the system as β varies from 0 to $+\infty$. In part (b) of the figure, we have plotted the locus as β varies from 0 to $-\infty$. In each plot, we have indicated the point $z = 2$, which is the open-loop pole [i.e., it is the pole of $Q(z)$ for $\beta = 0$]. As β increases from 0, the pole moves to the left of the point $z = 2$ along the real axis, and we have indicated this by including an arrow on the thick line to show how the pole changes as β is increased. Similarly, for $\beta < 0$, the pole of $Q(z)$ moves to the *right* of $z = 2$, and the direction of the arrow in Figure 11.9(b) indicates how the pole changes as the magnitude of β increases. For $1/2 < \beta < 3/2$, the pole lies inside the unit circle, and thus, the system is stable.

As a second example, consider a continuous-time feedback system with

$$H(s) = \frac{s}{s - 2} \tag{11.41}$$

and

$$G(s) = \frac{2\beta}{s}, \tag{11.42}$$

where β again represents the adjustable gain. Since $H(s)$ and $G(s)$ in this example are algebraically identical to $H(z)$ and $G(z)$, respectively, in the preceding example, the same

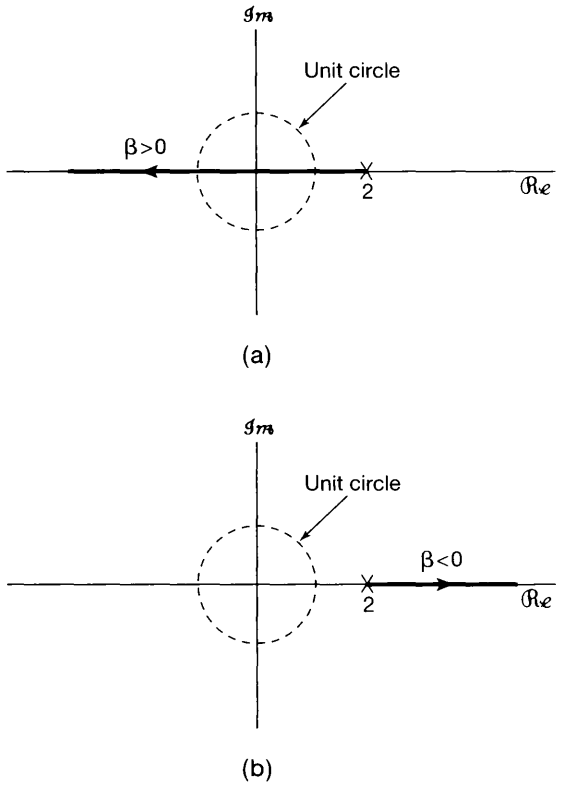


Figure 11.9 Root locus for the closed-loop system of eq. (11.40) as the value of β is varied: (a) $\beta > 0$; (b) $\beta < 0$. Note that we have marked the point $z = 2$ that corresponds to the pole location when $\beta = 0$.

will be true for the closed-loop system function

$$Q(s) = \frac{s}{s - 2(1 - \beta)} \tag{11.43}$$

vis-à-vis $Q(z)$, and the locus of the pole as a function of β will be identical to the locus in that example.

The relationship between these two examples stresses the fact that the locus of the poles is determined by the algebraic expressions for the system functions of the forward and feedback paths and is not inherently associated with whether the system is a continuous-time or discrete-time system. However, the interpretation of the result is intimately connected with its continuous-time or discrete-time context. In the discrete-time case it is the location of the poles in relation to the unit circle that is important, whereas in the continuous-time case it is their location in relation to the imaginary axis. Thus, as we have seen for the discrete-time example in eq. (11.40), the system is stable for $1/2 < \beta < 3/2$, while the continuous-time system of eq. (11.43) is stable for $\beta > 1$.

11.3.2 Equation for the Closed-Loop Poles

In the simple example considered in the previous section the root locus was easy to plot, since we could first explicitly determine the closed-loop pole as a function of the gain parameter and then plot the location of the pole as we changed the gain. For more complex

systems, one cannot expect to find such simple closed-form expressions for the closed-loop poles. However, it is still possible to sketch accurately the locus of the poles as the value of the gain parameter is varied from $-\infty$ to $+\infty$, without actually solving for the location of the poles for any specific value of the gain. This technique for determining the root locus is extremely useful in gaining insight into the characteristics of a feedback system. Also, as we develop the method, we will see that once we have determined the root locus, there is a relatively straightforward procedure for determining the value of the gain parameter that produces a closed-loop pole at any specified location along the root locus. We will phrase our discussion in terms of the Laplace transform variable s, with the understanding that it applies equally well to the discrete-time case.

Consider a modification of the basic feedback system of Figure 11.3(a), where either $G(s)$ or $H(s)$ is cascaded with an adjustable gain K. This is illustrated in Figure 11.10. In either of these cases, the denominator of the closed-loop system function is $1 + KG(s)H(s)$.[2] Therefore, the equation for the poles of the closed-loop system are the solutions of the equation

$$1 + KG(s)H(s) = 0. \tag{11.44}$$

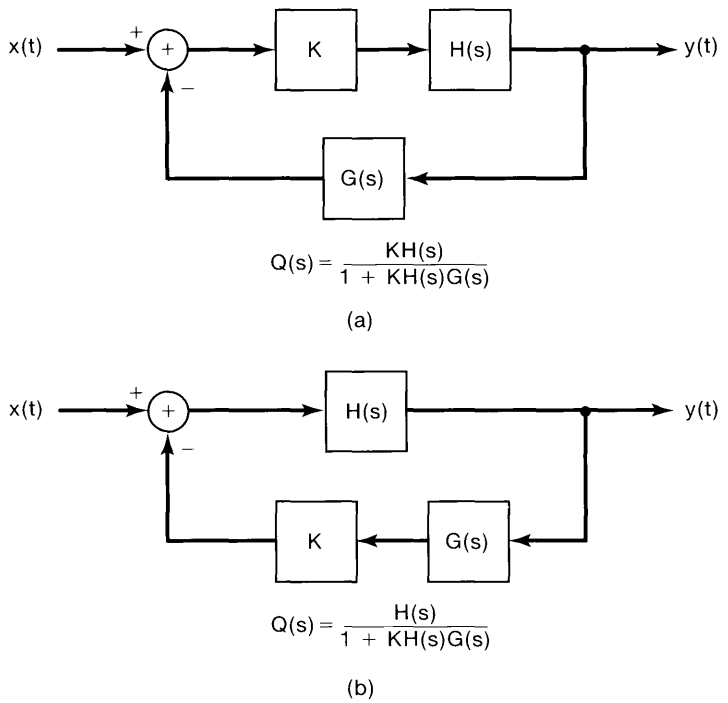


Figure 11.10 Feedback systems containing an adjustable gain: (a) system in which the gain is located in the forward path; (b) system with the gain in the feedback path.

[2]In the following discussion, we assume for simplicity that there is no pole-zero cancellation in the product $G(s)H(s)$. The presence of such pole-zero cancellations does not cause any real difficulties, and the procedure we will outline here is easily extended to that case (Problem 11.32). In fact, the simple example at the start of this section [eqs. (11.41) and (11.42)] *does* involve a pole-zero cancellation, at $s = 0$.

Rewriting eq. (11.44), we obtain the basic equation determining the closed-loop poles:

$$G(s)H(s) = -\frac{1}{K}. \tag{11.45}$$

The technique for plotting the root locus is based on the properties of this equation and its solutions. In the remainder of this section, we will discuss some of these properties and indicate how they can be exploited in determining the root locus.

11.3.3 The End Points of the Root Locus: The Closed-Loop Poles for K = 0 and |K| = +∞

Perhaps the most immediate observation that one can make about the root locus is that obtained by examining eq. (11.45) for $K = 0$ and $|K| = \infty$. For $K = 0$, the solution of this equation must yield the poles of $G(s)H(s)$, since $1/K = \infty$. To illustrate, recall the example given by eqs. (11.41) and (11.42). If we let β play the role of K, we see that eq. (11.45) becomes

$$\frac{2}{s-2} = -\frac{1}{\beta}. \tag{11.46}$$

Therefore, for $\beta = 0$, the pole of the system will be located at the pole of $2/(s-2)$ (i.e., at $s = 2$), which agrees with what we depicted in Figure 11.9.

Suppose now that $|K| = \infty$. Then $1/K = 0$, so that the solutions of eq. (11.45) must approach the *zeros* of $G(s)H(s)$. If the order of the numerator of $G(s)H(s)$ is smaller than that of the denominator, then some of these zeros, equal in number to the difference in order between the denominator and numerator, will be at infinity.

Referring again to eq. (11.46), since the order of the denominator of $2/(s-2)$ is 1, while the order of the numerator is zero, we conclude that in this example there is one zero at infinity and no zeros in the finite s-plane. Thus, as $|\beta| \to \infty$, the closed-loop pole approaches infinity. Again, this agrees with Figure 11.9, in which the magnitude of the pole increases without bound as $|\beta| \to \infty$ for either $\beta > 0$ or $\beta < 0$.

While the foregoing observations provide us with basic information as to the closed-loop pole locations for the extreme values of K, the following result is the key to our being able to plot the root locus without actually solving for the closed-loop poles as explicit functions of the gain.

11.3.4 The Angle Criterion

Consider again eq. (11.45). Since the right-hand side of this equation is real, a point s_0 can be a closed-loop pole only if the left-hand side of the equation, i.e., $G(s_0)H(s_0)$, is also real. Writing

$$G(s_0)H(s_0) = \left|G(s_0)H(s_0)\right| e^{j\measuredangle G(s_0)H(s_0)}, \tag{11.47}$$

we see that, for $G(s_0)H(s_0)$ to be real, it must be true that

$$e^{j\measuredangle G(s_0)H(s_0)} = \pm 1. \tag{11.48}$$

That is, for s_0 to be a closed-loop pole, we must have

$$\measuredangle G(s_0)H(s_0) = \text{integer multiple of } \pi. \tag{11.49}$$

Returning to eq. (11.46), we see immediately that in order for $2/(s_0 - 2)$ to be real, it is necessary that s_0 be real. For more complex system functions, it is not as easy to determine the values of s_0 for which $G(s_0)H(s_0)$ is real. However, as we will see, the use of the angle criterion given by eq. (11.49), together with the geometric method described in Chapter 9 for evaluating $\measuredangle G(s_0)H(s_0)$, greatly facilitates the determination of the root locus.

The angle criterion given by eq. (11.49) provides us with a direct method for determining whether a point s_0 could be a closed-loop pole for *some* value of the gain K. A further examination of eq. (11.45) gives us a way in which to calculate the value of the gain corresponding to any point on the root locus. Specifically, suppose that s_0 satisfies

$$\measuredangle G(s_0)H(s_0) = \text{odd multiple of } \pi. \tag{11.50}$$

Then $e^{j\measuredangle G(s_0)H(s_0)} = -1$, and from eq. (11.47) we see that

$$G(s_0)H(s_0) = -\left|G(s_0)H(s_0)\right|. \tag{11.51}$$

Substituting eq. (11.51) into eq. (11.45), we find that if

$$K = \frac{1}{\left|G(s_0)H(s_0)\right|}, \tag{11.52}$$

then s_0 is a solution of the equation and hence a closed-loop pole.

Similarly, if s_0 satisfies the condition

$$\measuredangle G(s_0)H(s_0) = \text{even multiple of } \pi, \tag{11.53}$$

then

$$G(s_0)H(s_0) = \left|G(s_0)H(s_0)\right|. \tag{11.54}$$

Thus, if

$$K = -\frac{1}{\left|G(s_0)H(s_0)\right|}, \tag{11.55}$$

then s_0 is a solution of eq. (11.45) and hence a closed-loop pole.

For the example given in eq. (11.46), if s_0 is on the real line and $s_0 < 2$, then

$$\measuredangle\left(\frac{2}{s_0 - 2}\right) = -\pi, \tag{11.56}$$

and from eq. (11.52), the value of β for which s_0 is the closed-loop pole is

$$\beta = \frac{1}{\left|\frac{2}{s_0 - 2}\right|} = \frac{2 - s_0}{2}. \tag{11.57}$$

That is,

$$s_0 = 2(1 - \beta), \tag{11.58}$$

which agrees with eq. (11.43).

Summarizing the last two observations that we have made, we see that the *root locus* for the closed-loop system, that is, the set of points in the complex s-plane that are closed-loop poles for *some* value of K as K varies from $-\infty$ to $+\infty$, are precisely those points that satisfy the angle condition of eq. (11.49). Furthermore:

1. A point s_0 for which

$$\sphericalangle G(s_0)H(s_0) = \text{odd multiple of } \pi \tag{11.59}$$

is on the root locus and is a closed-loop pole for some value of $K > 0$. The value of the gain that makes s_0 a closed-loop pole is given by eq. (11.52).

2. A point s_0 for which

$$\sphericalangle G(s_0)H(s_0) = \text{even multiple of } \pi \tag{11.60}$$

is on the root locus and is a closed-loop pole for some value of $K < 0$. The value of the gain that makes s_0 a closed-loop pole is given by eq. (11.55).

Therefore, we have now reduced the problem of determining the root locus to that of searching for points that satisfy the angle requirements given by eqs. (11.59) and (11.60). These equations can be refined further to a set of properties that aid in sketching the root locus. Before discussing these properties, however, let us consider a simple example.

Example 11.1

Let

$$H(s) = \frac{1}{s+1}, \qquad G(s) = \frac{1}{s+2}. \tag{11.61}$$

Recall that in Section 9.4 we discussed the geometric evaluation of Laplace transforms. In that section, we saw that the angle of the rational Laplace transform

$$\frac{\prod_{k=1}^{m}(s - \beta_k)}{\prod_{k=1}^{n}(s - \alpha_k)} \tag{11.62}$$

evaluated at some point s_0 in the complex plane equals the sum of the angles of the vectors from each of the zeros to s_0 minus the sum of the angles from each of the poles to s_0. Applying this to the product of $G(s)H(s)$, where $G(s)$ and $H(s)$ are as given in eq. (11.61), we can determine geometrically those points in the s-plane that satisfy the angle criteria, eqs. (11.59) and (11.60), and therefore can sketch the root locus.

In Figure 11.11, we have plotted the poles of $G(s)H(s)$ and have denoted by θ and ϕ the angles from each of the poles to the point s_0. Let us first test the angle criterion

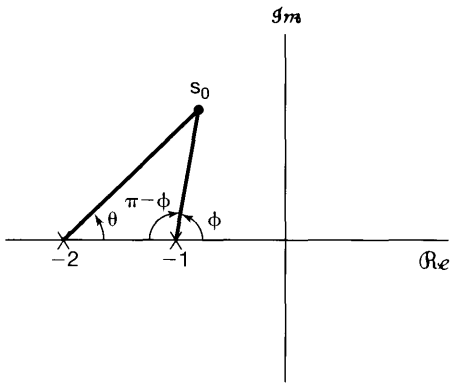


Figure 11.11 Geometric procedure for evaluating angle criterion in Example 11.1.

for points s_0 on the real axis. To begin with, the angle contribution from both poles is zero when s_0 is on the real axis to the right of -1. Thus,

$$\measuredangle G(s_0)H(s_0) = 0 = 0 \cdot \pi, \qquad s_0 \text{ real and greater than } -1, \tag{11.63}$$

and by eq. (11.60), these points are on the root locus for $K < 0$. For points between the two poles, the pole at -1 contributes an angle of $-\pi$, and the pole at -2 contributes 0. Thus,

$$\measuredangle G(s_0)H(s_0) = -\pi, \qquad s_0 \text{ real}, -2 < s_0 < -1. \tag{11.64}$$

These points are on the locus for $K > 0$. Finally, each pole contributes an angle of $-\pi$ when s_0 is real and less than -2, so that

$$\measuredangle G(s_0)H(s_0) = -2\pi, \qquad s_0 \text{ real and less than } -2.$$

Therefore, these points are on the locus for $K < 0$.

Let us now examine points in the upper half of the s-plane. (Since the impulse responses are real-valued, the complex poles occur in conjugate pairs. Therefore, we can immediately determine the poles in the lower half-plane after we have examined the upper half.) From Figure 11.11, the angle of $G(s_0)H(s_0)$ at the point s_0 is

$$\measuredangle G(s_0)H(s_0) = -(\theta + \phi). \tag{11.65}$$

Also, it is clear that as s_0 ranges over the upper half-plane (but not the real axis), we have

$$0 < \theta < \pi, \qquad 0 < \phi < \pi. \tag{11.66}$$

Thus,

$$-2\pi < \measuredangle G(s)H(s) < 0, \tag{11.67}$$

Therefore, we see immediately that *no* point in the upper half-plane can be on the locus for $K < 0$ [since $\measuredangle G(s)H(s)$ never equals an even multiple of π]. In addition, if s_0 is to be on the locus for $K > 0$, we must have

$$\measuredangle G(s_0)H(s_0) = -(\theta + \phi) = -\pi, \tag{11.68}$$

or

$$\theta = \pi - \phi. \tag{11.69}$$

Examining the geometry of Figure 11.11, we see that this occurs only for those points located on the straight line that is parallel to the imaginary axis and that bisects the line joining the poles at -1 and -2. We have now examined the entire s-plane and have determined all those points on the root locus. In addition, we know that for $K = 0$, the closed-loop poles equal the poles of $G(s)H(s)$, and as $|K| \to \infty$, the closed-loop poles go to the zeros of $G(s)H(s)$, which in this case are both at infinity. Putting these results together, we can draw the entire root locus, depicted in Figure 11.12, in which we have indicated the direction of increasing $|K|$, both for $K > 0$ and for $K < 0$.

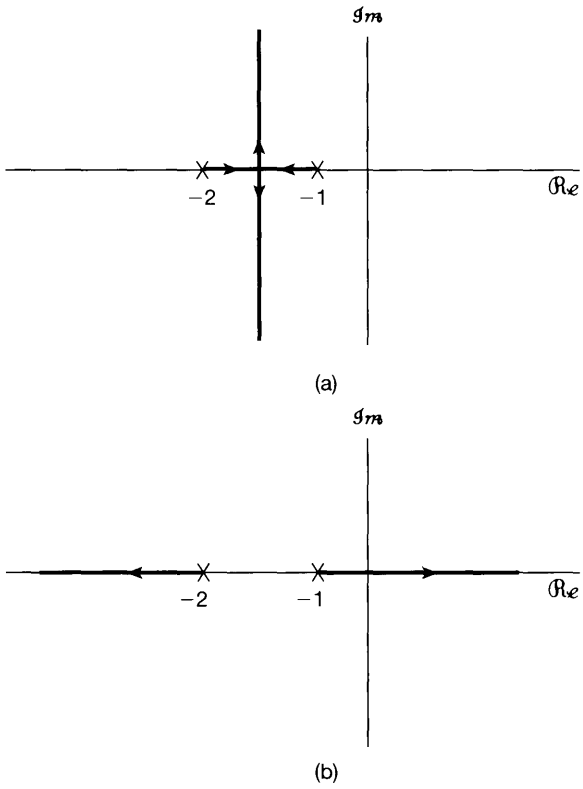


Figure 11.12 Root locus for Example 11.1: (a) $K > 0$; (b) $K < 0$. The poles of $G(s)H(s)$, which are located at $s = -1$ and $s = -2$, are indicated.

Note from the figure that for $K > 0$ there are two branches of the root locus and that the same is true for $K < 0$. The reason for the existence of two branches is that in this example the closed-loop system is a second-order system and consequently has two poles for any specified value of K. Therefore, the root locus has two branches, each of which traces the location of one of the closed-loop poles as K is varied, and for any particular value of K, there is one closed-loop pole on each branch. Again, if we wish to calculate the value of K for which a specific point s_0 on the locus is a closed-loop pole, we can use eqs. (11.52) and (11.55).

11.3.5 Properties of the Root Locus

The procedure outlined in the preceding subsection provides us, in principle, with a method for determining the root locus for any continuous-time or discrete-time LTI feedback system. That is, we simply determine, graphically or otherwise, all those points that satisfy eq. (11.59) or eq. (11.60). Fortunately, there are a number of other geometric properties concerning root loci that make the sketching of a locus far less tedious. To begin our discussion of these properties, let us assume that we have placed $G(s)H(s)$ in the standard form

$$G(s)H(s) = \frac{s^m + b_{m-1}s^{m-1} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} = \frac{\prod_{k=1}^{m}(s - \beta_k)}{\prod_{k=1}^{n}(s - \alpha_k)}, \tag{11.70}$$

where the β_k's denote the zeros and the α_k's denote the poles. In general, these may be complex. Also, from eq. (11.70), we see that we are assuming that the leading coefficient in both the numerator and the denominator of $G(s)H(s)$ is $+1$. This can always be achieved by dividing the numerator and denominator by the denominator coefficient of s^n and absorbing the resulting numerator coefficient of s^m into the gain K. For example,

$$K\frac{2s+1}{3s^2+5s+2} = K\frac{\frac{2}{3}s+\frac{1}{3}}{s^2+\frac{5}{3}s+\frac{2}{3}} = \left(\frac{2}{3}\right)K\frac{s+\frac{1}{2}}{s^2+\frac{5}{3}s+\frac{2}{3}}, \tag{11.71}$$

and the quantity $(2/3)K$ is then regarded as the overall gain that is varied in determining the root locus.

We assume, in addition, that

$$m \leq n, \tag{11.72}$$

which is the case that is usually encountered in practice. (Problem 11.33 considers the case $m > n$.) The following are some properties that include earlier observations and that aid in sketching the root locus.

Property 1: For $K = 0$, the solutions of eq. (11.45) are the poles of $G(s)H(s)$. Since we are assuming n poles, the root locus has n branches, each one starting (for $K = 0$) at a pole of $G(s)H(s)$.

Property 1 includes the general version of a fact which we noted in Example 11.1: that there is one branch of the root locus for each closed-loop pole. The next property is also simply a restatement of one of our earlier observations.

Property 2: As $|K| \to \infty$, each branch of the root locus approaches a zero of $G(s)H(s)$. Since we are assuming that $m \leq n$, $n - m$ of these zeros are at infinity.

Property 3: Parts of the real s-axis that lie to the left of an *odd* number of real poles and zeros of $G(s)H(s)$ are on the root locus for $K > 0$. Parts of the real s-axis that lie to the left of an *even* number (possibly zero) of poles and zeros of $G(s)H(s)$ are on the root locus for $K < 0$.

We can show that Property 3 is true as follows: From our discussion in Example 11.1 and from Figure 11.13(a), we see that if a point on the real s-axis is to the right of a real pole or zero of $G(s)H(s)$, that pole or zero contributes zero to $\measuredangle G(s_0)H(s_0)$. On the other hand, if s_0 is to the left of a zero, that zero contributes $+\pi$, whereas if s_0 is to the left of a pole, we get a contribution of $-\pi$ (since we subtract the pole angles). Hence, if s_0 is to the left of an odd number of real poles and zeros, the total contribution of these poles and zeros is an odd multiple of π, whereas if s_0 is to the left of an even number of real poles and zeros, the total contribution is an even multiple of π. From eqs. (11.59) and (11.60), we will have the result stated in Property 3 if we can show that the total contribution from all poles and zeros with nonzero imaginary parts is an even multiple of π. The key here is that such poles and zeros occur in complex-conjugate pairs, and we can consider the contribution from each such pair, as illustrated in Figure 11.13(b). The symmetry in the picture clearly indicates that the sum of the angles from this pair to any point s_0 on the real axis is precisely 2π. Summing over all conjugate zero pairs and subtracting the sum over all conjugate pole pairs, we get the desired result. Thus, *any* segment of the real line

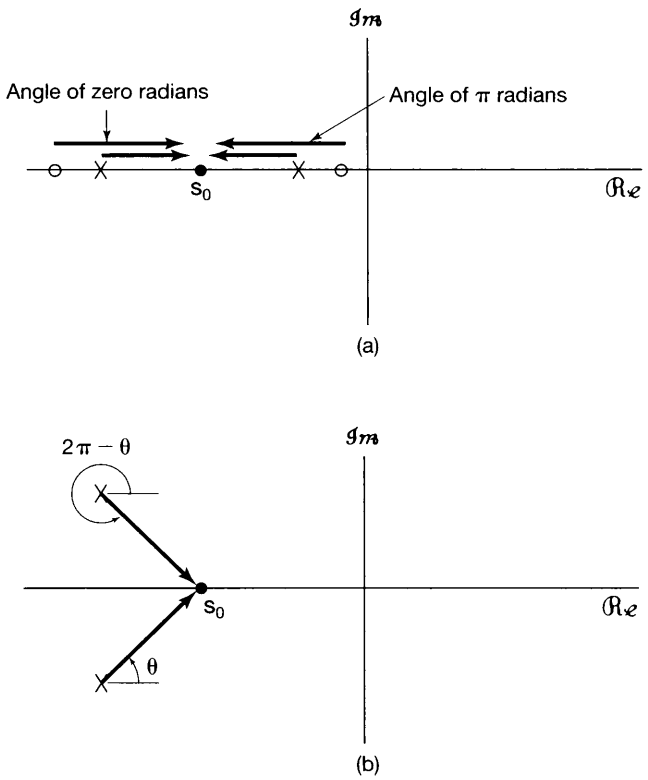


Figure 11.13 (a) Angle contribution from real poles and zeros to a point on the real axis; (b) total angle contribution from a complex-conjugate pole pair to a point on the real axis.

between real poles or zeros is on the root locus either for $K > 0$ or for $K < 0$, depending on whether it lies to the left of an odd or an even number of poles and zeros of $G(s)H(s)$.

As one consequence of Properties 1 through 3, consider a segment of the real axis between two poles of $G(s)H(s)$, with no zeros between these poles. From Property 1 the root locus begins at the poles, and from Property 3 the entire portion of the real axis between the two poles will lie on the root locus for a positive or negative range of values of K. Therefore, as $|K|$ increases from zero, the two branches of the root locus that begin at these poles move toward each other along the segment of the real axis between the poles. From Property 2, as $|K|$ increases toward infinity, each branch of the root locus must approach a zero. Since there are no zeros along that portion of the real axis, the only way that this can happen is if the branches break off into the complex plane for $|K|$ sufficiently large. This is illustrated in Figure 11.12, in which the locus for $K > 0$ has a portion between two real poles. As K is increased, the root locus eventually leaves the real axis, forming two complex-conjugate branches. Summarizing this discussion, we have the following property of the root locus:

Property 4: Branches of the root locus between two real poles must break off into the complex plane for $|K|$ large enough.

Properties 1–4 serve to illustrate how characteristics of the root locus can be deduced from eqs. (11.45), (11.59), and (11.60). In many cases, plotting the poles and zeros of $G(s)H(s)$ and then using these four properties suffices to provide a reasonably accurate sketch of the root locus. (See Examples 11.2 and 11.3.) In addition to these properties, however, there are numerous other characteristics of the root locus that allow one to obtain sketches of increasing accuracy. For example, from Property 2, we know that $n - m$ branches of the root locus approach infinity. In fact, these branches approach infinity at specific angles that can be calculated, and therefore, the branches are asymptotically parallel to lines at these angles. Moreover, it is possible to draw in the asymptotes and then determine the point at which they intersect. These two properties and several others are illustrated in Problems 11.34–11.36 and 11.41–11.42. A more detailed development of the root-locus method can be found in more advanced texts such as those listed in the bibliography at the end of the book.

In the remainder of this section we present two examples, one in continuous time and one in discrete time, that illustrate how the four properties that we have just described allow us to sketch the root locus and to deduce the stability characteristics of a feedback system as the gain K is varied.

Example 11.2

Let

$$G(s)H(s) = \frac{s - 1}{(s + 1)(s + 2)}. \tag{11.73}$$

From Properties 1 and 2, the root locus for either K positive or K negative starts at the points $s = -1$ and $s = -2$. One branch terminates at the zero at $s = 1$ and the other at infinity.

Let us first consider $K > 0$. The root locus in this case is illustrated in Figure 11.14(a). From Property 3, we can identify the regions of the real axis that are on

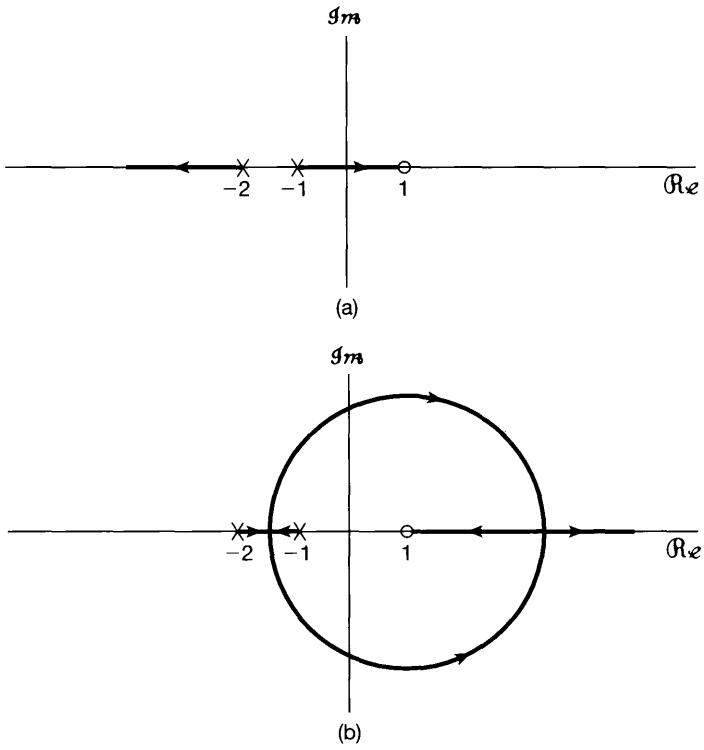


Figure 11.14 Root locus for Example 11.2: (a) $K > 0$; (b) $K < 0$. The poles of $G(s)H(s)$ at $s = -1$ and $s = -2$ and the zero of $G(s)H(s)$ at $s = 1$ are indicated in the figure.

the root locus for $K > 0$—specifically, $\mathcal{R}e\{s\} < -2$ and $-1 < \mathcal{R}e\{s\} < 1$. Therefore, one branch of the root locus for $K > 0$ originates at $s = -1$ and approaches $s = 1$ as $K \to \infty$. The other begins at $s = -2$ and extends to the left toward $\mathcal{R}e\{s\} = -\infty$ as $K \to +\infty$.

Thus, we see that for $K > 0$, if K is sufficiently large, the system will become unstable, as one of the closed-loop poles moves into the right-half plane. The procedure that we have used for sketching the root locus does not, of course, indicate the value of K for which this instability develops. However, for this particular example, we see that the value of K for which the instability occurs corresponds to the root locus passing through $s = 0$. Consequently, from eq. (11.52), the corresponding value of K is

$$K = \frac{1}{|G(0)H(0)|} = 2. \tag{11.74}$$

Thus, the system is stable for $0 \leq K < 2$, but is unstable for $K \geq 2$.

For $K < 0$, the portions of the real axis lying on the root locus are $\mathcal{R}e\{s\} > 1$ and $-2 < \mathcal{R}e\{s\} < -1$. Thus, the root locus again starts at the points $s = -2$ and $s = -1$, moving into the region $-2 < \mathcal{R}e\{s\} < -1$. At some point, it breaks off into the complex plane and follows a trajectory such that it returns to the real axis for $s > 1$. Upon the root locus' return to the real axis, one branch moves to the left toward the zero at $s = 1$ and the other to the right toward $s = \infty$, as indicated in Figure 11.14(b), in which we have displayed an accurate plot of the root locus for $K < 0$.

Rules can also be developed to indicate the locations at which the root locus leaves and enters the real axis. Even without that precise a description, however, we can sketch the general shape of the root locus in Figure 11.14(b) and can therefore deduce that for $K < 0$, the system also becomes unstable for $|K|$ sufficiently large.

Example 11.3

Consider the discrete-time feedback system illustrated in Figure 11.15. In this case,

$$G(z)H(z) = \frac{z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)} = \frac{z}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}. \tag{11.75}$$

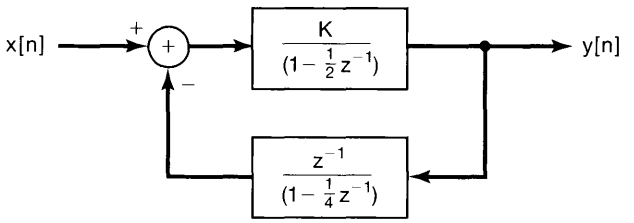


Figure 11.15 Discrete-time feedback system of Example 11.3.

As discussed at the beginning of this section, the techniques for sketching the root locus of a discrete-time feedback system are identical to those used in the continuous-time case. Therefore, in a manner exactly analogous to that of the preceding example, we can deduce the basic form of the root locus for this example, which is illustrated in Figure 11.16. In this case the portion of the real axis between the two poles of $G(z)H(z)$ (at $z = 1/4$ and $z = 1/2$) is on the root locus for $K > 0$, and as K increases the locus breaks off into the complex plane and returns to the axis at some point in the left-half plane. From there, one branch approaches the zero of $G(z)H(z)$ at $z = 0$, and the other approaches infinity as $K \to \infty$. The form of the root locus for $K < 0$ consists of two branches on the real axis, one approaching 0 and the other infinity.

As we remarked earlier, while the form of the root locus does not depend on whether the system is a continuous-time or discrete-time system, any conclusion regarding stability based on examining the locus certainly does. For this example, we can conclude that for $|K|$ sufficiently large the system is unstable, since one of the two poles has magnitude greater than 1. In particular, from the $K > 0$ root locus in Figure 11.16(a), we see that the transition from stability to instability occurs when one of the closed-loop poles is at $z = -1$. From eq. (11.52), the corresponding value of K is

$$K = \frac{1}{|G(-1)H(-1)|} = \frac{15}{8}. \tag{11.76}$$

Similarly, from Figure 11.16(b), the stability-instability transition occurs when one of the closed-loop poles is at $z = 1$, and from eq. (11.55), the corresponding value of K is

$$K = -\frac{1}{|G(1)H(1)|} = -\frac{3}{8}. \tag{11.77}$$

Putting this together, we see that the closed-loop system in Figure 11.16 is stable if

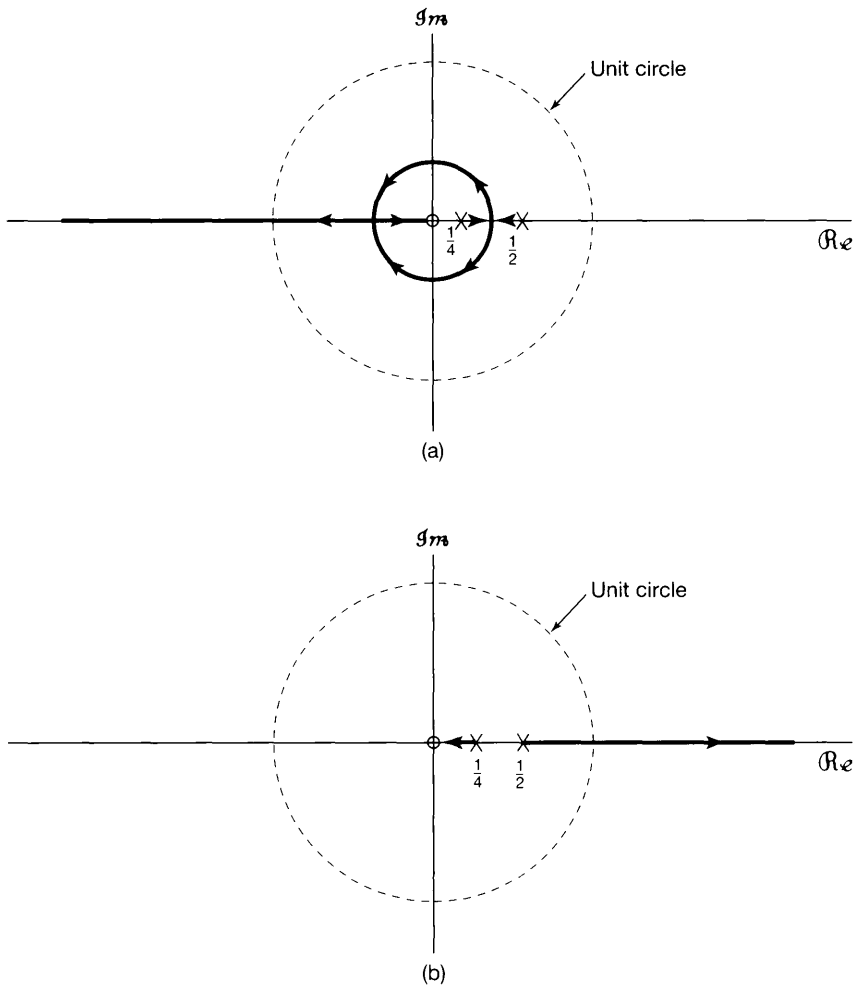


Figure 11.16 Root locus for Example 11.3: (a) $K > 0$; (b) $K < 0$. The poles of $G(z)H(z)$ at $z = 1/4$ and $z = 1/2$ and the zero of $G(z)H(z)$ at $z = 0$ are indicated in the figure.

$$-\frac{3}{8} < K < \frac{15}{8} \tag{11.78}$$

and is unstable for K outside this range.

11.4 THE NYQUIST STABILITY CRITERION

As developed in Section 11.3, the root-locus technique provides detailed information concerning the location of closed-loop poles as the system gain is varied. From root-locus plots, one can determine the damping and the stability characteristics of the system as K is varied. Determination of the root locus requires the analytic description of the system

functions of the forward and feedback paths and is applicable only when these transforms are rational. For example, it cannot be directly applied in situations in which our knowledge of the system functions is obtained purely from experimentation.

In this section, we introduce another method for the determination of the stability of feedback systems as a function of an adjustable gain parameter. This technique, referred to as the *Nyquist criterion,* differs from the root-locus method in two basic ways. Unlike the root-locus method, the Nyquist criterion does *not* provide detailed information concerning the location of the closed-loop poles as a function of K, but rather, simply determines whether or not the system is stable for any specified value of K. On the other hand, the Nyquist criterion can be applied to nonrational system functions and in situations in which no analytic description of the forward and feedback path system functions is available.

Our objective in this section is to outline the basic ideas behind the Nyquist criterion for both continuous-time and discrete-time systems. As we will see, both the discrete-time and continuous-time Nyquist tests are the result of the same fundamental concept, although, as with the root-locus method, the actual criteria for stability differ because of the differences between continuous and discrete time. More detailed developments of the ideas behind the Nyquist criterion and its use in the design of feedback systems can be found in texts on the analysis and synthesis of feedback systems and automatic control systems, including those listed in the bibliography at the end of the book.

To introduce the method, let us recall that the poles of the closed-loop systems of Figure 11.10 and their discrete-time counterparts are the solutions of the equations

$$1 + KG(s)H(s) = 0 \qquad \text{(continuous time)} \tag{11.79}$$

and

$$1 + KG(z)H(z) = 0 \qquad \text{(discrete time).} \tag{11.80}$$

For discrete-time systems, we want to determine whether any of the solutions of eq. (11.80) lie outside the unit circle, and for continuous-time systems whether any of the solutions of eq. (11.79) lie in the right half of the s-plane. The Nyquist criterion fixes this by examination of the values of $G(s)H(s)$ along the $j\omega$-axis and the values of $G(z)H(z)$ along the unit circle. The basis for this is the encirclement property, which we develop in the following subsection.

11.4.1 The Encirclement Property

Consider a general rational function $W(p)$, where p is a complex variable,[3] and suppose that we plot $W(p)$ for values of p along a closed contour in the p-plane, which we traverse in a clockwise direction. This is illustrated in Figure 11.17 for a function $W(p)$ that has two zeros and no poles. In Figure 11.17(a) we show a closed contour C in the p-plane, and in Figure 11.17(b) we plot the closed contour of the values of $W(p)$ as p varies around C.

[3]Because we will use the property we are about to develop for both continuous-time and discrete-time feedback systems, we have chosen to describe it in terms of a general complex variable p. In the next subsection we use this property to analyze continuous-time feedback systems, where the complex variable is s. Following this, in Section 11.4.3 we use the encirclement property for discrete-time feedback systems, in which context the complex variable is z.

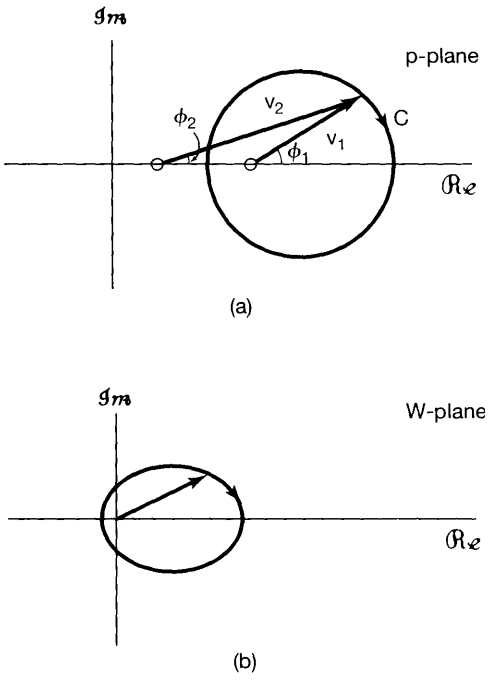


Figure 11.17 Basic encirclement property. The closed curve in (b) represents a plot of the values of $W(p)$ for values of p along the curve C in (a). Here, the arrow on the curve C in (a) indicates the direction in which C is traversed, and the arrow in (b) indicates the corresponding direction along the contour of values of $W(p)$.

In this example, there is one zero of $W(p)$ inside the contour and one zero of $W(p)$ outside the contour. At any point p on C, the angle of $W(p)$ is the sum of the angles of the two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ to the point p. As we traverse the contour once, the angle ϕ_1 of the vector from the zero *inside* the contour encounters a net change of -2π radians, whereas the angle ϕ_2 of the vector from the zero *outside* the contour encounters no net change. Thus, on the plot of $W(p)$, there is a net change in angle of -2π. Said another way, the plot of $W(p)$ in Figure 11.17(b) encircles the origin once in the clockwise direction. More generally, for an arbitrary rational $W(p)$, as we traverse a closed contour in the clockwise direction, any poles and zeros of $W(p)$ outside the contour will contribute no net change to the angle of $W(p)$, whereas each zero inside the contour will contribute a net change of -2π and each pole inside will contribute a net change of $+2\pi$. Since each net change of -2π in $W(p)$ corresponds to one clockwise encirclement of the origin in the plot of $W(p)$, we can state the following basic *encirclement property:*

Encirclement Property: As a closed contour C in the p-plane is traversed once in the clockwise direction, the corresponding plot of $W(p)$ for values of p along the contour encircles the origin in the clockwise direction a net number of times equal to the number of zeros minus the number of poles contained within the contour.

In applying this statement, a counterclockwise encirclement is interpreted as the negative of one clockwise encirclement. For example, if there is one pole and no zeros inside the contour, there will be one counterclockwise, or equivalently, minus-one clockwise, encirclement.

Example 11.4

Consider the function

$$W(p) = \frac{p-1}{(p+1)(p^2+p+1)}. \tag{11.81}$$

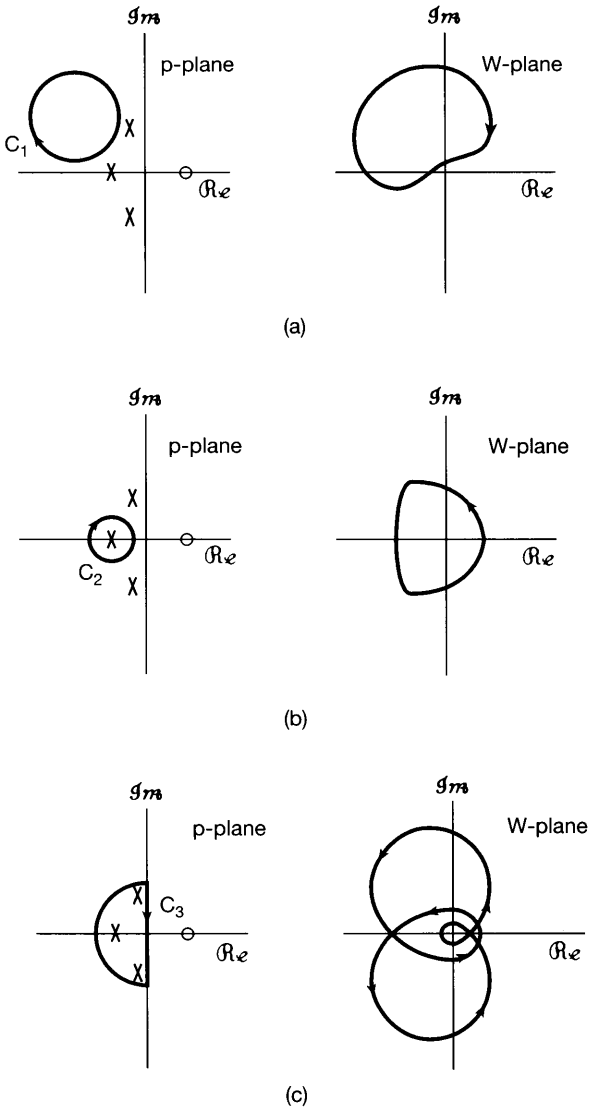
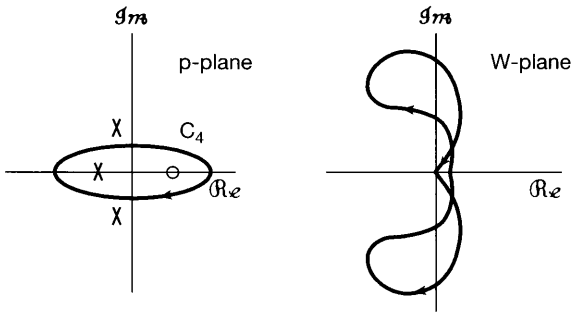
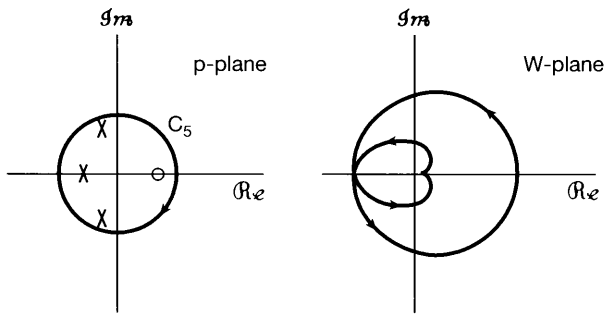


Figure 11.18 Basic encirclement property for Example 11.4: (a) the contour encircles no poles or zeros and consequently $W(p)$ has no encirclements of the origin; (b) the contour encircles one pole and therefore $W(p)$ has one encirclement of the origin; (c) the contour encircles three poles and therefore $W(p)$ has three encirclements of the origin;



(d)



(e)

Figure 11.18 *Continued* (d) the contour encircles one pole and one zero and therefore $W(p)$ has no encirclements of the origin; (e) the contour encircles three poles and one zero. $W(p)$ has two encirclements of the origin.

In Figure 11.18, we depict several closed contours in the complex p-plane and the corresponding plots of $W(p)$ along each of these contours. In Figure 11.18(a), the contour C_1 does not encircle any of the poles or zeros of $W(p)$, and consequently, the plot of $W(p)$ has no net encirclements of zero. In Figure 11.18(b), only the pole at $p = -1$ is contained within the contour C_2, and the plot of $W(p)$ encircles the origin once in the counterclockwise direction (minus-one clockwise encirclements). In Figure 11.18(c), C_3 encircles all three poles, and the plot of $W(p)$ encircles the origin three times in a counterclockwise direction. In Figure 11.18(d), C_4 encircles one pole and one zero, and therefore, the plot of $W(p)$ has no net encirclements of the origin. Finally, in Figure 11.18(e), all of the poles and the one zero of $W(p)$ are contained within C_5, and thus, the plot of $W(p)$ along this contour has two net counterclockwise encirclements of the origin.

11.4.2 The Nyquist Criterion for Continuous-Time LTI Feedback Systems

In this section, we exploit the encirclement property in examining the stability of the continuous-time feedback system of Figure 11.10. Stability of this system requires that

no zeros of $1 + KG(s)H(s)$, or equivalently, of the function

$$R(s) = \frac{1}{K} + G(s)H(s) \tag{11.82}$$

lie in the right half of the s-plane. Thus, in applying the general result developed in the preceding subsection, we can consider the contour indicated in Figure 11.19. From the plot of $R(s)$, as s traverses the contour C we can obtain a count of the number of zeros minus the number of poles of $R(s)$ contained within the contour by counting the number of clockwise encirclements of the origin. As M increases to infinity, this then corresponds to the number of zeros minus the number of poles of $R(s)$ in the right half of the s-plane.

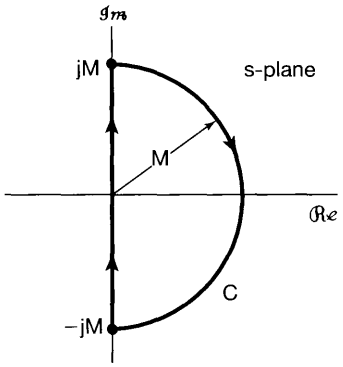


Figure 11.19 Closed contour containing a portion of the right-half plane; as $M \to \infty$, the contour encloses the entire right-half plane.

Let us examine the evaluation of $R(s)$ along the contour in Figure 11.19 as M increases to infinity. Along the semicircular portion of the contour extending into the right-half plane, we must ensure that $R(s)$ remains bounded as M increases. Accordingly, we will assume that $R(s)$ has at least as many poles as zeros. In that case,

$$R(s) = \frac{b_n s^n + b_{n-1}s^{n-1} + \cdots + b_0}{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_0} \tag{11.83}$$

and

$$\lim_{|s|\to\infty} R(s) = \frac{b_n}{a_n} = \text{constant}. \tag{11.84}$$

Therefore, as M increases to infinity, the value of $R(s)$ does not change as we traverse the semicircular part of the contour, and consequently, the constant value along this part is equal to the value of $R(s)$ at the end points [i.e., $R(j\omega)$ at $\omega = \pm\infty$].

Therefore, the plot of $R(s)$ along the contour of Figure 11.19 can be obtained by plotting $R(s)$ along the part of the contour that coincides with the imaginary axis—that is, the plot of $R(j\omega)$ as ω varies from $-\infty$ to $+\infty$. Since $R(j\omega)$ equals $1/K + G(j\omega)H(j\omega)$, $R(s)$ along the contour can be drawn from knowledge of $G(j\omega)$ and $H(j\omega)$. If both the forward- and feedback-path systems are stable, $G(j\omega)$ and $H(j\omega)$ are simply the frequency-response functions of these systems. However, the encirclement property for the general function $W(p)$ is simply a property of complex functions; it has nothing to

do with whether $W(p)$ arose as the Laplace or z-transform of any signal and, consequently, has nothing to do with regions of convergence. Thus, even if the forward- and feedback-path systems are unstable, if we examine the plot of the *function* $R(j\omega) = 1/K + G(j\omega)H(j\omega)$ for $-\infty < \omega < \infty$, we know that the net number of clockwise encirclements of the origin will equal the number of zeros minus the number of poles of $R(s)$ that lie in the right-half plane.

Furthermore, from eq. (11.82), we see that the poles of $R(s)$ are simply the poles of $G(s)H(s)$, while the zeros of $R(s)$ are the closed-loop poles. In addition, since $G(j\omega)H(j\omega) = R(j\omega) - 1/K$, it follows that the plot of $G(j\omega)H(j\omega)$ encircles the point $-1/K$ *exactly* as many times as $R(j\omega)$ encircles the origin. The plot of $G(j\omega)H(j\omega)$ as ω varies from $-\infty$ to $+\infty$ is called the *Nyquist plot.* From the encirclement property, we then see that

$$\begin{array}{l}\text{The number of clockwise}\\ \text{encirclements of the point}\\ -1/K \text{ by the Nyquist plot}\end{array} = \begin{array}{l}\text{The number of right-half}\\ \text{plane closed-loop poles}\\ \text{minus the number of right-}\\ \text{half plane poles of } G(s)H(s).\end{array} \tag{11.85}$$

While the open-loop system $G(s)H(s)$ may have unstable poles, for the closed-loop system to be stable we require no right-half plane closed-loop poles. This yields the *continuous-time Nyquist stability criterion:*

Continuous-Time Nyquist Stability Criterion: For the closed-loop system to be stable, the net number of *clockwise* encirclements of the point $-1/K$ by the Nyquist plot of $G(j\omega)H(j\omega)$ must equal *minus* the number of right-half-plane poles of $G(s)H(s)$. Equivalently, the net number of *counterclockwise* encirclements of the point $-1/K$ by the Nyquist plot of $G(j\omega)H(j\omega)$ must *equal* the number of right-half-plane poles of $G(s)H(s)$.

For example, if the forward- and feedback-path systems are stable, then the Nyquist plot is simply the plot of the frequency response of the cascade of these two systems. In this case, since there are no poles of $G(s)H(s)$ in the right-half plane, the Nyquist criterion requires that, for stability, the net number of encirclements of the point $-1/K$ must be zero.

Example 11.5

Let

$$G(s) = \frac{1}{s+1}, \qquad H(s) = \frac{1}{\frac{1}{2}s+1}. \tag{11.86}$$

The Bode plot for $G(j\omega)H(j\omega)$ is shown in Figure 11.20. The Nyquist plot depicted in Figure 11.21 is constructed directly from the plots of the log magnitude and phase of $G(j\omega)H(j\omega)$. That is, each point on the Nyquist plot has polar coordinates consisting of the magnitude $|G(j\omega)H(j\omega)|$ and angle $\measuredangle G(j\omega)H(j\omega)$ for some value of ω. The coordinates of $G(j\omega)H(j\omega)$ for $\omega < 0$ are obtained from the values for $\omega > 0$ through the use of the conjugate symmetry property of $G(j\omega)H(j\omega)$. This property manifests itself geometrically in a very simple way, which facilitates the sketching of the Nyquist plot for any feedback system composed of systems with

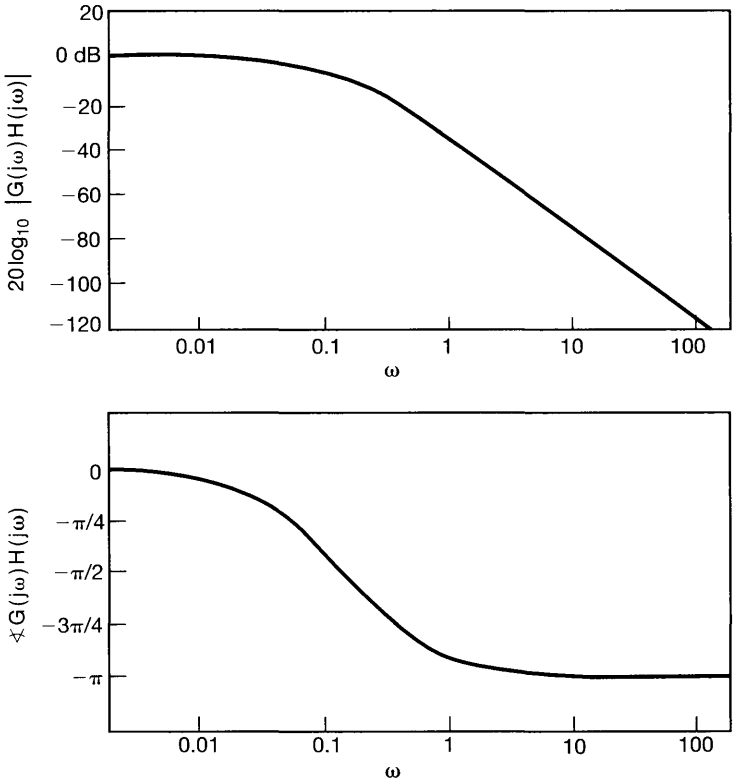


Figure 11.20 Bode plot for $G(j\omega)H(j\omega)$ in Example 11.5.

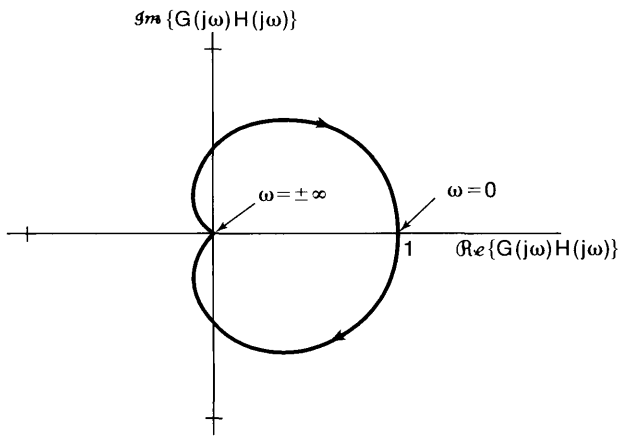


Figure 11.21 Nyquist plot of $G(j\omega)H(j\omega)$ for Example 11.5. The arrow on the curve indicates the direction of increasing ω.

real impulse responses. Specifically, since $|G(-j\omega)H(-j\omega)| = |G(j\omega)H(j\omega)|$ and $\sphericalangle G(-j\omega)H(-j\omega) = -\sphericalangle G(j\omega)H(j\omega)$, the Nyquist plot of $G(j\omega)H(j\omega)$ for $\omega \leq 0$ is a reflection about the real axis of the plot for $\omega \geq 0$. Note also that we have included an arrow on the Nyquist plot in Figure 11.21. This arrow indicates the direction of increasing ω. That is, it indicates the direction in which the Nyquist plot is traversed (as ω varies from $-\infty$ to $+\infty$) for the counting of encirclements in the application of the Nyquist criterion.

In this example there are no right-half-plane open-loop poles, and consequently, the Nyquist criterion requires that, for stability, there be no net encirclements of the point $-1/K$. Thus, by inspection of Figure 11.21, we see that the closed-loop system will be stable if the point $-1/K$ falls outside the Nyquist contour—that is, if

$$-\frac{1}{K} \leq 0 \quad \text{or} \quad -\frac{1}{K} > 1, \tag{11.87}$$

which is equivalent to

$$K \geq 0 \quad \text{or} \quad 0 > K > -1. \tag{11.88}$$

Combining these two conditions, we obtain the result that the closed-loop system will be stable for any choice of K greater than -1.

Example 11.6

Consider now

$$G(s)H(s) = \frac{s+1}{(s-1)(\frac{1}{2}s+1)}. \tag{11.89}$$

The Nyquist plot for this system is indicated in Figure 11.22. For this example, $G(s)H(s)$ has one right-half-plane pole. Thus, for stability we require one counterclockwise encirclement of the point $-1/K$, which in turn requires that the point $-1/K$ fall inside the contour. Hence, we will have stability if and only if $-1 < -1/K < 0$, that is, if $K > 1$.

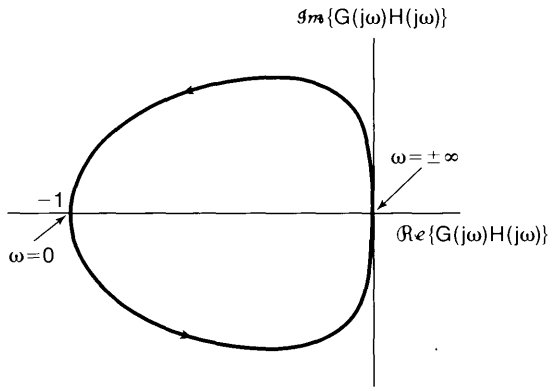


Figure 11.22 Nyquist plot for Example 11.6. The arrow on the curve indicates the direction of increasing ω.

In the foregoing discussion, we have introduced and illustrated a form for the Nyquist stability criterion that applies to an extremely large class of feedback systems. In addition,

there are a number of refinements and extensions of the criterion that allow it to be used for many other feedback systems as well. For example, as we have developed it, the Nyquist plot can be drawn without any difficulties for stable or unstable $G(s)H(s)$, as long as there are no poles of $G(s)H(s)$ exactly *on* the $j\omega$-axis. When such poles do occur, the value of $G(j\omega)H(j\omega)$ is infinite at those points. However, as considered in Problem 11.44, the Nyquist criterion can be modified to allow for poles of $G(s)H(s)$ on the $j\omega$-axis. In addition, as mentioned at the beginning of this section, the Nyquist criterion can also be extended to the case in which $G(s)$ and $H(s)$ are not rational. For example, it can be shown that if the forward- and feedback-path systems are both stable, the Nyquist criterion is the same when the system functions are nonrational as it is when they are rational. That is, the closed-loop system is stable if there are no net encirclements of the point $-1/K$. To illustrate the application of the Nyquist criterion for nonrational system functions, we present the following example:

Example 11.7

Consider the acoustic feedback example discussed in Section 11.2.6. Referring to Figure 11.8(b), let $K = K_1K_2$ and

$$G(s)H(s) = -e^{-sT} = e^{-(sT+j\pi)}, \tag{11.90}$$

where we have used the fact that $e^{-j\pi} = -1$. In this case,

$$G(j\omega)H(j\omega) = e^{-j(\omega T+\pi)}, \tag{11.91}$$

and as ω varies from $-\infty$ to ∞, $G(j\omega)H(j\omega)$ traces out a circle of radius 1 in the clockwise direction, with one full revolution for every change of $2\pi/T$ in ω. This is illustrated in Figure 11.23. Since the forward- and feedback-path systems are stable [the cascade $G(s)H(s)$ is simply a time delay], the Nyquist stability criterion indicates that the closed-loop system will be stable if and only if $-1/K$ does not fall inside the unit circle. Equivalently, we require for stability that

$$|K| < 1. \tag{11.92}$$

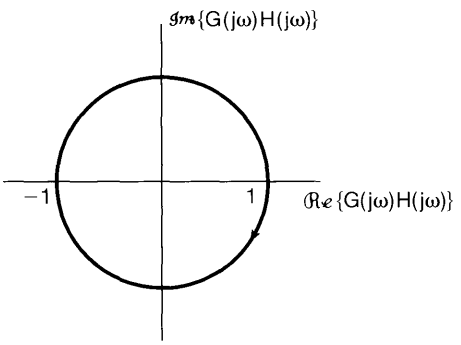


Figure 11.23 Nyquist plot for Example 11.7.

Since K_1 and K_2 represent an acoustic gain and attenuation, respectively, they are both positive, which yields the stability criterion

$$K_1K_2 < 1. \tag{11.93}$$

11.4.3 The Nyquist Criterion for Discrete-Time LTI Feedback Systems

As in the continuous-time case, the Nyquist stability criterion for discrete-time systems is based on the fact that the difference in the number of poles and zeros inside a contour, for a rational function, can be determined by examining a plot of the value of the function along the contour. The difference between the continuous-time and discrete-time cases is the choice of the contour. For the discrete-time case, stability of the closed-loop feedback system requires that no zeros of

$$R(z) = \frac{1}{K} + G(z)H(z) \tag{11.94}$$

lie outside the unit circle.

Recall that the encirclement property relates to poles and zeros *inside* any specified contour. On the other hand, in examining the stability of a discrete-time system, we are concerned with the zeros of $R(z)$ *outside* the unit circle. Therefore, in order to make use of the encirclement property, we first make a simple modification. Let us consider the rational function

$$\hat{R}(z) = R\left(\frac{1}{z}\right) \tag{11.95}$$

obtained by replacing z by its reciprocal. As seen in Problem 10.43, if z_0 is a zero (pole) of $R(z)$, then $1/z_0$ is a zero (pole) of $\hat{R}(z)$. Since $1/|z_0|$ is less than 1 if $|z_0| > 1$, any zero or pole of $R(z)$ *outside* the unit circle corresponds to a zero or pole of $\hat{R}(z)$ *inside* the unit circle.

From the basic encirclement property, we know that as z traverses the unit circle in a clockwise direction, the net number of clockwise encirclements of the origin by $\hat{R}(z)$ equals the difference between the number of its zeros and poles inside the unit circle. However, from the previous paragraph, this equals the difference between the number of zeros and poles of $R(z)$ *outside* the unit circle. Furthermore, on the unit circle, $z = e^{j\omega}$ and $1/z = e^{-j\omega}$. Therefore,

$$\hat{R}(e^{j\omega}) = R(e^{-j\omega}). \tag{11.96}$$

From this, we see that evaluating $\hat{R}(z)$ as z traverses the unit circle in the clockwise direction is identical to evaluating $R(z)$ as z traverses the unit circle in the *counterclockwise* direction. In sum, then,

The number of clockwise encirclements of the origin by the plot of $R(e^{j\omega})$ as the unit circle is traversed in the counterclockwise direction (e.g., as ω increases from 0 to 2π)	=	The number of zeros of $R(z)$ outside the unit circle minus the number of poles of $R(z)$ outside the unit circle.

(11.97)

Much as in the continuous-time case, counting the encirclements of the origin by $R(e^{j\omega})$ is equivalent to counting the number of encirclements of the point $-1/K$ by the

plot of $G(e^{j\omega})H(e^{j\omega})$, again referred to as the Nyquist plot, which is graphed as ω varies from 0 to 2π. Also, the poles of $R(z)$ are precisely the poles of $G(z)H(z)$, and the zeros of $R(z)$ are the closed-loop poles. Therefore, the encirclement property stated in the preceding paragraph implies that the net number of clockwise encirclements by the Nyquist plot of the point $-1/K$ equals the number of closed-loop poles outside the unit circle minus the number of poles of $G(z)H(z)$ outside the unit circle. In order that the closed-loop system be stable, we require no closed-loop poles outside the unit circle. This yields the *discrete-time Nyquist stability criterion*:

Discrete-Time Nyquist Stability Criterion: For the closed-loop system to be stable, the net number of *clockwise* encirclements of the point $-1/K$ by the Nyquist plot of $G(e^{j\omega})H(e^{j\omega})$ as ω varies from 0 to 2π must equal *minus* the number of poles of $G(z)H(z)$ that lie outside the unit circle. Equivalently, the net number of *counterclockwise* encirclements of the point $-1/K$ by the Nyquist plot of $G(e^{j\omega})H(e^{j\omega})$ as ω varies from 0 to 2π must *equal* the number of poles of $G(z)H(z)$ outside the unit circle.

Example 11.8

Let

$$G(z)H(z) = \frac{z^{-2}}{1 + \frac{1}{2}z^{-1}} = \frac{1}{z(z + \frac{1}{2})}. \tag{11.98}$$

The Nyquist plot of this curve is shown in Figure 11.24. Since $G(z)H(z)$ has no poles outside the unit circle, for the stability of the closed-loop system there must be no encirclements of the point $-1/K$. From the figure, we see that this will be the case either if $-1/K < -1$ or if $-1/K > 2$. Thus, the system is stable for $-1/2 < K < 1$.

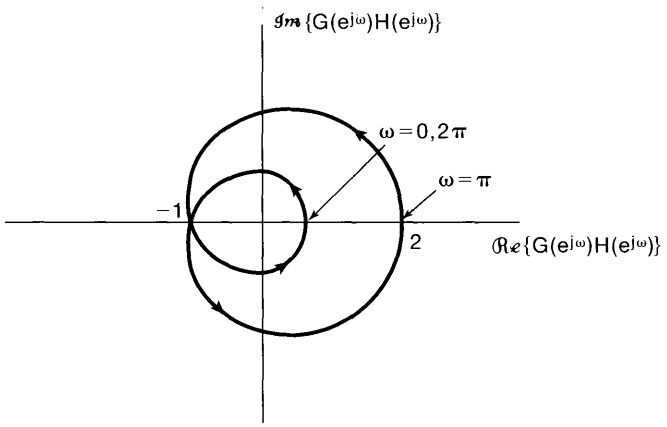


Figure 11.24 Nyquist plot for Example 11.8. The arrow on the curve indicates the direction in which the curve is traversed as ω increases from 0 to 2π.

Just as in continuous time, if the forward and feedback paths are stable, then the Nyquist plot can be obtained from the frequency responses $H(e^{j\omega})$ and $G(e^{j\omega})$ of these

systems. If the forward and feedback paths are unstable, then the frequency responses are not defined. Nevertheless, the *function* $G(z)H(z)$ can still be evaluated on the unit circle, and the Nyquist stability criterion can be applied.

As we have seen in this section, the Nyquist stability criterion provides a useful method for determining the range of values of the gain K for which a continuous-time or discrete-time feedback is stable (or unstable). This criterion and the root-locus method are extremely important tools in the design and implementation of feedback systems, and each has its own uses and limitations. For example, the Nyquist criterion can be applied to nonrational system functions, whereas the root-locus method cannot. On the other hand, root-locus plots allow us to examine not only stability, but also other characteristics of the closed-loop system response, such as damping, oscillation frequency, and so on, which are readily identifiable from the location of the poles of the closed-loop system. In the next section, we introduce an additional tool for the analysis of feedback systems that highlights another important characteristic of closed-loop system behavior.

11.5 GAIN AND PHASE MARGINS

In this section, we introduce and examine the concept of the *margin of stability* in a feedback system. It is often of interest not only to know *whether* a feedback system is stable, but also to determine how much the gain in the system can be perturbed and how much additional phase shift can be added to the system before it becomes unstable. Information such as this is important because in many applications the forward and feedback system functions are known only approximately or may change slightly during operation because of wear, the effect of high temperatures on components, or similar influences.

As an example, consider the telescope-pointing system described in Section 11.0 and illustrated in Figures 11.1(c) and (d). This system consists of a motor, a potentiometer converting the shaft angle to a voltage, and an amplifier that is used to amplify the voltage representing the difference between the desired and the actual shaft angles. Assuming that we have obtained approximate descriptions of each of these components, we can set the amplifier gain so that the system will be stable if these descriptions are accurate. However, the amplifier gain and the constant of proportionality that describes the angle-voltage characteristic of the potentiometer are never known exactly, and therefore, the actual gain in the feedback system may differ from the nominal value assumed in designing the system. Furthermore, the damping characteristics of the motor cannot be determined with absolute precision, and thus, the actual time constant of the motor response may differ from the approximate value in the specification of the system. For example, if the actual motor time constant is larger than the nominal value used in the design, the motor will respond more sluggishly than anticipated, thereby producing an effective time delay in the feedback system. As we have discussed in earlier chapters, and as we will again in Example 11.11, time delays have the effect of increasing the negative phase in the frequency response of a system, and this phase shift can have a destabilizing influence on the system. Because of the possible presence of gain and phase errors such as those we have just described, it is clearly desirable to set the amplifier gain so that there is some margin for error—that is, so that the actual system will remain stable even if it differs somewhat from the approximate model used in the design process.

In this section, we introduce one method for quantifying the margin of stability in a feedback system. To do this, we consider a closed-loop system, depicted in Figure 11.25, that has been designed to be stable based on nominal values for the forward- and feedback-path system functions. For our discussion here, we let $H(s)$ and $G(s)$ denote these nominal values. Also, since the basic concepts are identical for both continuous-time and discrete-time systems, we will again focus our development on the continuous-time case, and at the end of the section we illustrate the application of these ideas to a discrete-time example.

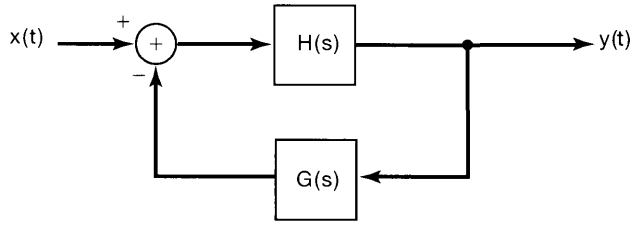


Figure 11.25 Typical feedback system designed to be stable, assuming nominal descriptions for $H(s)$ and $G(s)$.

To assess the margin of stability in our feedback system, suppose that the actual system is as depicted in Figure 11.26, where we have allowed for the possibility of a gain K and phase shift ϕ in the feedback path. In the nominal system K is unity and ϕ is zero, but in the actual system either or both may have a different value. Therefore, it is of interest to know how much variation can be tolerated in these quantities without losing closed-loop system stability. In particular, the *gain margin* of the feedback system is defined as the minimum amount of additional gain K, with $\phi = 0$, that is required so that the closed-loop system becomes unstable. Similarly, the *phase margin* is the additional amount of phase shift, with $K = 1$, that is required for the system to be unstable. By convention, the phase margin is expressed as a positive quantity; that is, it equals the magnitude of the additional negative phase shift at which the feedback system becomes unstable.

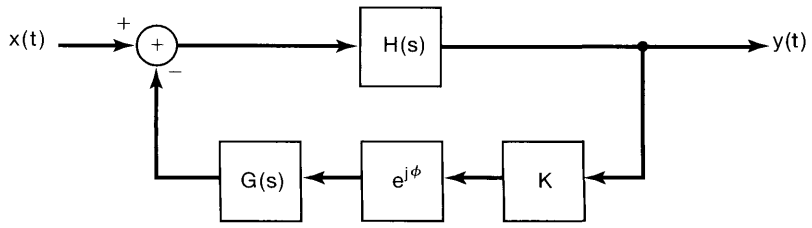


Figure 11.26 Feedback system containing possible gain and phase deviations from the nominal description depicted in Figure 11.25.

Since the closed-loop system of Figure 11.25 is stable, the system of Figure 11.26 can become unstable if, as K and ϕ are varied, at least one pole of the closed-loop system crosses the $j\omega$-axis. If a pole of the closed-loop system is on the $j\omega$-axis at, say, $\omega = \omega_0$, then at this frequency

$$1 + Ke^{-j\phi}G(j\omega_0)H(j\omega_0) = 0, \tag{11.99}$$

or

$$Ke^{-j\phi}G(j\omega_0)H(j\omega_0) = -1. \tag{11.100}$$

Note that with $K = 1$ and $\phi = 0$, by our assumption of stability for the nominal feedback system of Figure 11.25, there is no value of ω_0 for which eq. (11.100) is satisfied. The gain margin of this system is the minimum value of $K > 1$ for which eq. (11.100) has a solution for *some* ω_0 with $\phi = 0$. That is, the gain margin is the smallest value of K for which the equation

$$KG(j\omega_0)H(j\omega_0) = -1 \tag{11.101}$$

has a solution ω_0. Similarly, the phase margin is the minimum value of ϕ for which eq. (11.100) has a solution for some ω_0 when $K = 1$. In other words, the phase margin is the smallest value of $\phi > 0$ for which the equation

$$e^{-j\phi}G(j\omega_0)H(j\omega_0) = -1 \tag{11.102}$$

has a solution.

To illustrate the calculation and graphical interpretation of gain and phase margins, we consider the following example.

Example 11.9

Let

$$G(s)H(s) = \frac{4(1 + \frac{1}{2}s)}{s(1 + 2s)(1 + 0.05s + (0.125s)^2)}. \tag{11.103}$$

The Bode plot for this example is shown in Figure 11.27. Note that, as discussed in Problem 6.31, the factor of $1/j\omega$ in $G(j\omega)H(j\omega)$ contributes $-90°$ ($-\pi/2$ radians) of phase shift and a 20-dB-per-decade increase in $|G(j\omega)H(j\omega)|$. To determine the gain margin, we observe that, with $\phi = 0$, the only frequency at which eq. (11.101) can be satisfied is that for which $\measuredangle G(j\omega_0)H(j\omega_0) = -\pi$. At this frequency, the gain margin in decibels can be identified by inspection of Figure 11.27. We first examine Figure 11.27(b) to determine the frequency ω_1 at which the angle curve crosses the line $-\pi$ radians. Locating the point at this same frequency in Figure 11.27(a) provides us with the value of $|G(j\omega_1)H(j\omega_1)|$. For eq. (11.101) to be satisfied for $\omega_0 = \omega_1$, K must equal $1/|G(j\omega_1)H(j\omega_1)|$. This value is the gain margin. As illustrated in Figure 11.27(a), the gain margin expressed in decibels can be identified as the amount the log-magnitude curve would have to be shifted up so that the curve intersects the 0-dB line at the frequency ω_1.

In a similar fashion, we can determine the phase margin. Note first that the only frequency at which eq. (11.102) can be satisfied is that for which $|G(j\omega_0)H(j\omega_0)| = 1$, or equivalently, $20 \log_{10} |G(j\omega_0)H(j\omega_0)| = 0$. To determine the phase margin, we first find the frequency ω_2 in Figure 11.27(a) at which the log-magnitude curve crosses the 0-dB line. Locating the point at this same frequency in Figure 11.27(b) then provides us with the value of $\measuredangle G(j\omega_2)H(j\omega_2)$. For eq. (11.102) to be satisfied for $\omega_0 = \omega_2$, the angle of the left-hand side of this equation must be $-\pi$. The value of ϕ for which this is true is the phase margin. As illustrated in Figure 11.27(b), the phase margin can be identified as the amount the angle curve would have to be lowered so that the curve intersects the line $-\pi$ at the frequency ω_2.

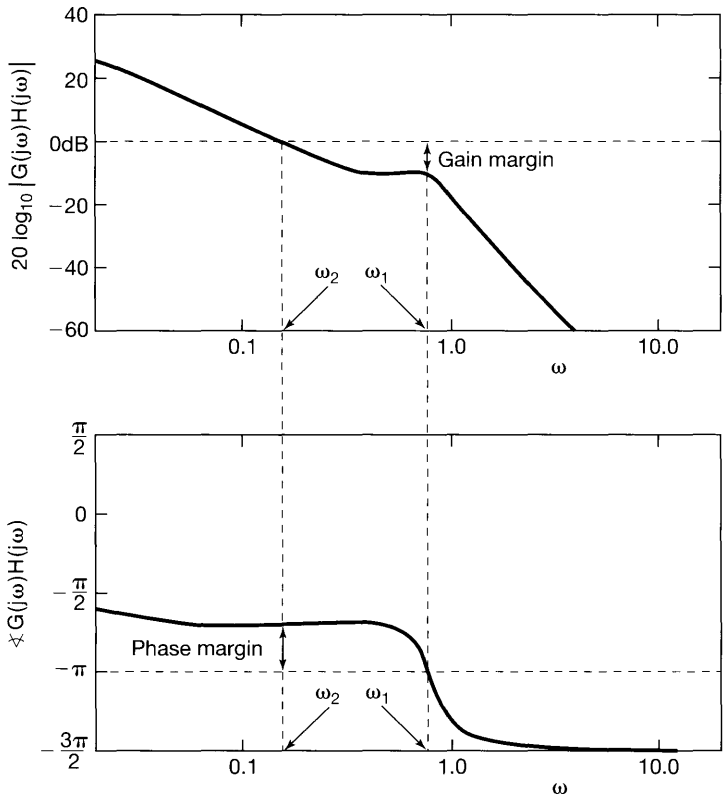


Figure 11.27 Use of Bode plots to calculate gain and phase margins for the system of Example 11.9.

In determining gain and phase margins, it is not always of interest to identify explicitly the *frequency* at which the poles will cross the $j\omega$-axis. As an alternative, we can identify the gain and phase margins from a *log magnitude-phase diagram.* For example, the log magnitude-phase diagram for the system of Figure 11.27 is shown in Figure 11.28. In this figure, we plot $20 \log_{10} |G(j\omega)H(j\omega)|$ versus $\angle G(j\omega)H(j\omega)$ as ω varies from 0 to $+\infty$. Therefore, because of the conjugate symmetry of $G(j\omega)H(j\omega)$, the plot contains the same information as the Nyquist plot, in which $\mathcal{R}e\{G(j\omega)H(j\omega)\}$ is plotted versus $\mathcal{I}m\{G(j\omega)H(j\omega)\}$ for $-\infty < \omega < \infty$. As we have indicated, the phase margin can be read off by locating the intersection of the log magnitude-phase plot with the 0-dB line. That is, the phase margin is the amount of additional negative phase shift required to shift the log magnitude-phase curve so that it intersects the 0-dB line with exactly 180° (π radians) of phase shift. Similarly, the gain margin is directly obtained from the intersection of the log magnitude-phase curve with the line $-\pi$ radians, and this represents the amount of additional gain needed so that the curve crosses the line $-\pi$ with a magnitude of 0 dB.

The following examples provide several other elementary illustrations of log magnitude-phase diagrams:

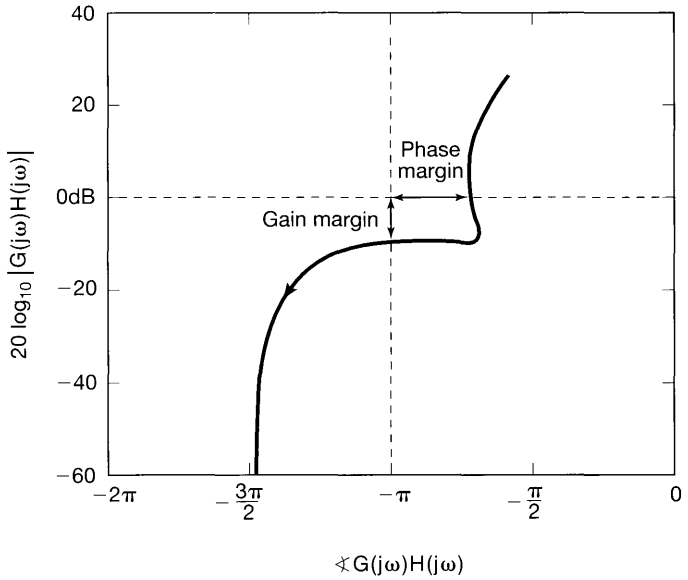


Figure 11.28 Log magnitude-phase plot for the system of Example 11.9.

Example 11.10

Let

$$G(s)H(s) = \frac{1}{\tau s + 1}, \qquad \tau > 0. \tag{11.104}$$

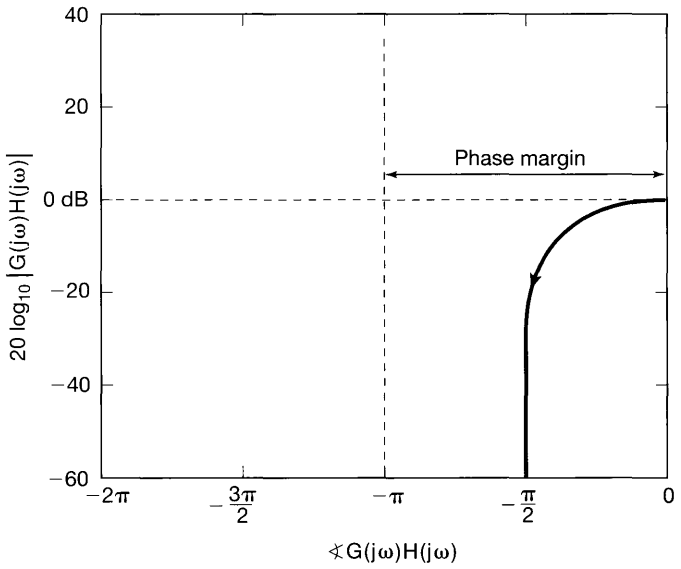


Figure 11.29 Log magnitude-phase plot for the first-order system of Example 11.10.

In this case, we obtain the log magnitude-phase plot depicted in Figure 11.29. This has a phase margin of π, and since the curve does not intersect the line $-\pi$, the system has infinite gain margin (i.e., we can increase the gain as much as we like and maintain stability). This is consistent with the conclusion that we can draw by examining the system illustrated in Figure 11.30(a). In Figure 11.30(b), we have depicted the root locus for this system with $\phi = 0$ and $K > 0$. From the figure, it is evident that the system is stable for any positive value of K. In addition, if $K = 1$ and $\phi = \pi$, so that $e^{j\phi} = -1$, the closed-loop system function for the system of Figure 11.30(a) is $1/\tau s$, which has a pole at $s = 0$, so that the system is unstable.

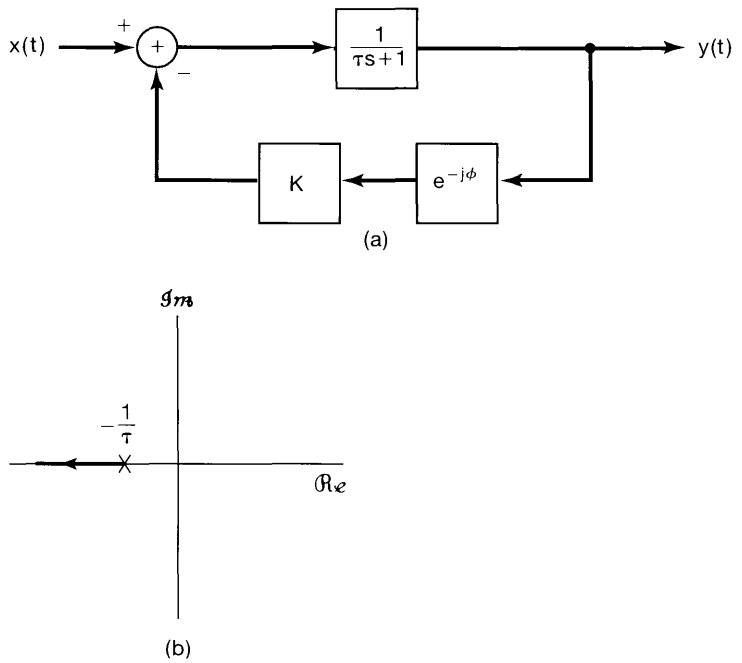


Figure 11.30 (a) First-order feedback system with possible gain and phase variations in the feedback path; (b) root locus for this system with $\phi = 0$, $K > 0$.

Example 11.11

Suppose we now consider the second-order system

$$H(s) = \frac{1}{s^2 + s + 1}, \qquad G(s) = 1. \tag{11.105}$$

The system $H(s)$ has an undamped natural frequency of 1 and a damping ratio of 0.5. The log magnitude-phase plot for this system is illustrated in Figure 11.31. Again we have infinite gain margin, but a phase margin of only $\pi/2$, since it can be shown by a straightforward calculation that $|H(j\omega)| = 1$ for $\omega = 1$, and at this frequency $\measuredangle H(j\omega) = -\pi/2$.

We can now illustrate the type of problem that can be solved using the concepts of gain and phase margins. Suppose that the feedback system specified by eq. (11.105) cannot be realized. Rather, some unavoidable time delay is introduced into the feedback

path. That is,

$$G(s) = e^{-s\tau}, \tag{11.106}$$

where τ is the time delay. What we would like to know is how small this delay must be to ensure the stability of the closed-loop system.

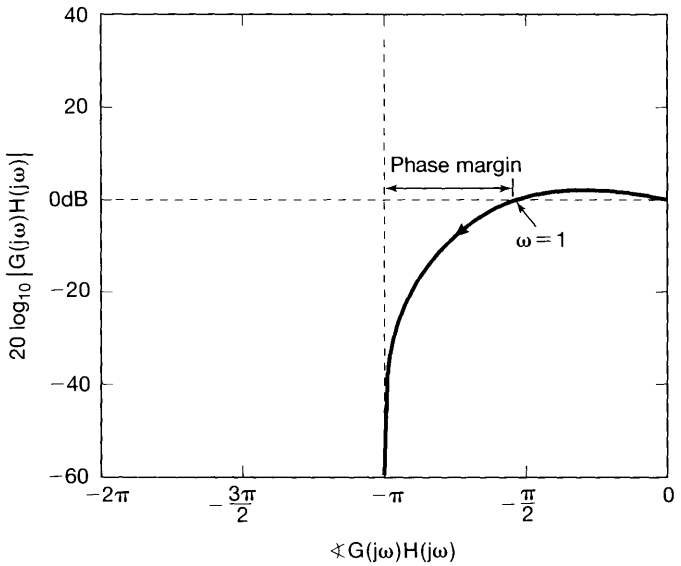


Figure 11.31 Log magnitude-phase plot for the second-order system of Example 11.11.

The first point to note is that

$$|e^{-j\omega\tau}| = 1, \tag{11.107}$$

so the delay does not change the magnitude of $H(j\omega)G(j\omega)$. On the other hand,

$$\angle e^{-j\omega\tau} = -\omega\tau \text{ radians.} \tag{11.108}$$

Thus, every point on the curve in Figure 11.31 is shifted to the *left*. The amount of the shift is proportional to the value of ω for each point on the log magnitude-phase curve.

From this discussion, we see that instability will occur once the phase margin is reduced to zero, and this will happen when the phase shift introduced by the delay is equal to $-\pi/2$ at $\omega = 1$. That is, the critical value τ^* of the time delay satisfies

$$\angle e^{-j\tau^*} = -\tau^* = -\frac{\pi}{2}, \tag{11.109}$$

or (assuming that the units of ω are radians/second)

$$\tau^* \approx 1.57 \text{ seconds.} \tag{11.110}$$

Thus, for any time delay $\tau < \tau^*$, the system remains stable.

Example 11.12

Consider again the acoustic feedback system discussed in Section 11.2.6 and Example 11.7. Here, we assume that the system of Figure 11.8 has been designed with $K_1K_2 < 1$, so that the closed-loop system is stable. In this case, the log magnitude-phase plot for

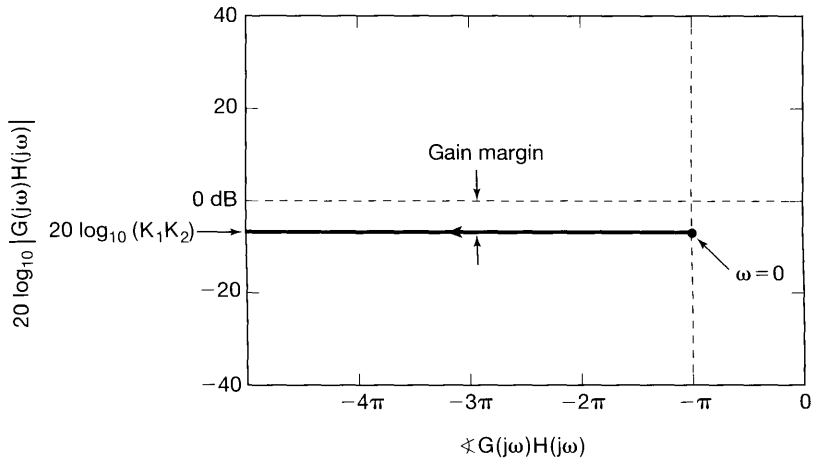


Figure 11.32 Log magnitude-phase plot for Example 11.12.

$G(s)H(s) = K_1K_2e^{-(sT+j\pi)}$ is illustrated in Figure 11.32. From the figure, we see that the system has infinite phase margin and a gain margin in decibels of $-20\log_{10}(K_1K_2)$ (i.e., this is precisely the gain factor that, when multiplied by K_1K_2, equals 1).

As indicated at the start of the section, the definitions of the gain and phase margin are the same for discrete-time feedback systems as for continuous-time systems. Specifically, if we have a stable discrete-time feedback system, the gain margin is the minimum amount of additional gain required in the feedback system such that the closed-loop system becomes unstable. Similarly, the phase margin is the minimum amount of additional negative phase shift required for the feedback system to be unstable. The following example illustrates the graphical calculation of phase and gain margins for a discrete-time feedback system; the procedure is essentially the same as for continuous-time systems.

Example 11.13

In this example, we illustrate the concept of gain and phase margin for the discrete-time feedback system shown in Figure 11.33. Here,

$$G(z)H(z) = \frac{\frac{7\sqrt{2}}{4}z^{-1}}{1 - \frac{7\sqrt{2}}{8}z^{-1} + \frac{49}{64}z^{-2}}, \tag{11.111}$$

and by direct calculation we can check that the feedback system is stable for $K = 1$ and $\phi = 0$. In Figure 11.34, we have displayed the log magnitude-phase diagram for

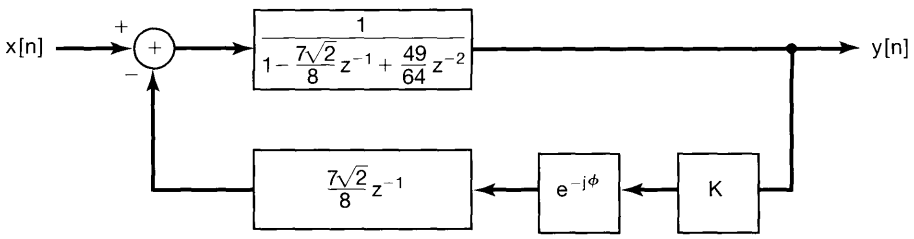


Figure 11.33 Discrete-time feedback system of Example 11.13.

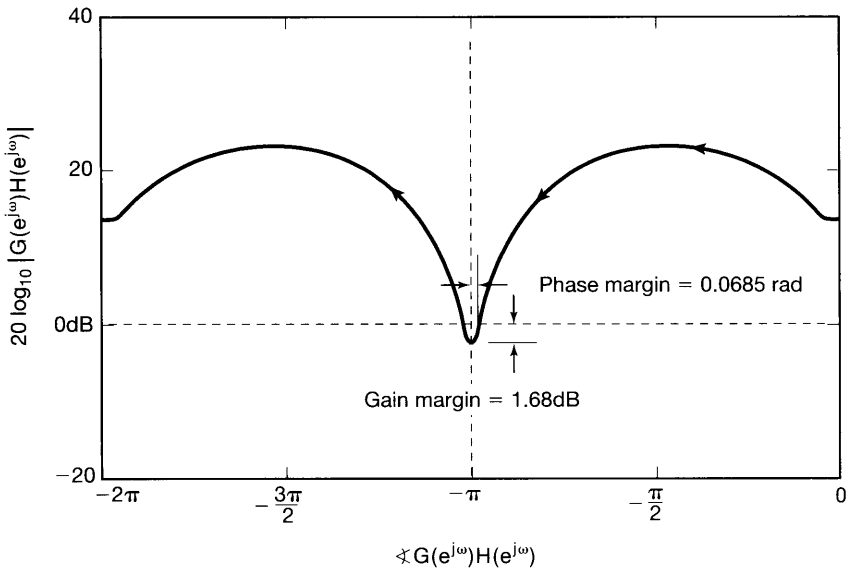


Figure 11.34 Log magnitude-phase diagram for the discrete-time feedback system of Example 11.13.

the system; that is, we have plotted $20 \log_{10} |G(e^{j\omega})H(e^{j\omega})|$ versus $\measuredangle G(e^{j\omega})H(e^{j\omega})$ as ω varies from 0 to 2π. The system has a gain margin of 1.68 dB and a phase margin of 0.0685 radians (3.93°).

In concluding this section, it should be stressed that the gain margin is the *minimum* value of gain that moves one or more of the closed-loop poles onto the $j\omega$-axis in continuous time or the unit circle in discrete time and, consequently, causes the system to become unstable. It is important to note, however, that this does *not* imply that the system is unstable for *all* values of gain above the value specified by the gain margin. For example, as illustrated in Problem 11.47, as K increases, the root locus may move from the left-half plane into the right-half plane and then cross back into the left-half plane. The gain margin provides us with the information about how much the gain can be increased until the poles *first* reach the $j\omega$-axis, but it tells us nothing about the possibility that the system may again be stable for even larger values of the gain. To obtain such information, we must either refer to the root locus or use the Nyquist stability criterion. (See Problem 11.47.).[4]

11.6 SUMMARY

In this chapter, we have examined a number of the applications and several techniques for the analysis of feedback systems. We have seen how the use of Laplace and z-transforms allows us to analyze these systems algebraically and graphically. In Section 11.2 we indicated several of the applications of feedback, including the design of inverse systems,

[4]For detailed discussions of this point and also of gain and phase margins and log magnitude-phase diagrams in general, see the texts on feedback listed in the bibliography at the end of the book.

the stabilization of unstable systems, and the design of tracking systems. We also saw that feedback can destabilize, as well as stabilize, a system.

In Section 11.3, we described the root-locus method for plotting the poles of the closed-loop system as a function of a gain parameter. Here, we found that the geometric evaluation of the phase of a rational Laplace transform or z-transform allowed us to gain a significant amount of insight into the properties of the root locus. These properties often permit us to obtain a reasonably accurate sketch of the root locus without performing complex calculations.

In contrast to the root-locus method, the Nyquist criterion of Section 11.4 is a technique for determining the stability of a feedback system, again as a function of a variable gain, *without* obtaining a detailed description of the location of the closed-loop poles. The Nyquist criterion is applicable to nonrational system functions and thus can be used when all that is available are experimentally determined frequency responses. The same is true of the gain and phase margins described in Section 11.5. These quantities provide a measure of the margin of stability in a feedback system and therefore are of importance to designers in that they allow them to determine how robust a feedback system is to discrepancies between estimates of the forward- and feedback-path system functions and their actual values.

Chapter 11 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining three sections contain problems belonging to the basic, advanced, and extension categories, respectively.

BASIC PROBLEMS WITH ANSWERS

11.1. Consider the interconnection of discrete-time LTI systems shown in Figure P11.1. Express the overall system function for this interconnection in terms of $H_0(z)$, $H_1(z)$, and $G(z)$.

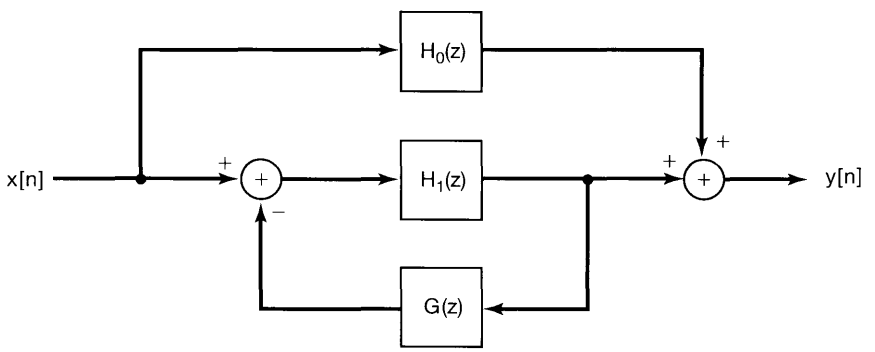


Figure P11.1

11.2. Consider the interconnection of discrete-time LTI systems shown in Figure P11.2. Express the overall system function for this interconnection in terms of $H_1(s)$, $H_2(s)$, $G_1(s)$, and $G_2(s)$.

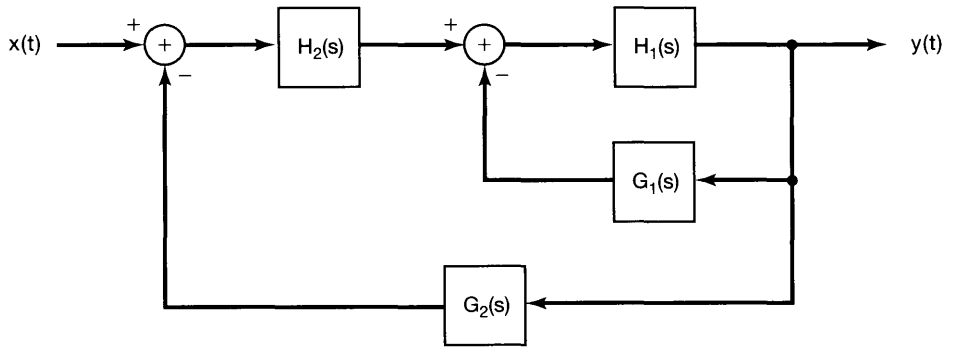


Figure P11.2

11.3. Consider the continuous-time feedback system depicted in Figure 11.3(a) with

$$H(s) = \frac{1}{s-1} \qquad \text{and} \qquad G(s) = s - b.$$

For what real values of b is the feedback system stable?

11.4. A causal LTI system S with input $x(t)$ and output $y(t)$ is represented by the differential equation

$$\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt}.$$

S is to be implemented using the feedback configuration of Figure 11.3(a) with $H(s) = 1/(s+1)$. Determine $G(s)$.

11.5. Consider the discrete-time feedback system depicted in Figure 11.3(b) with

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \qquad \text{and} \qquad G(z) = 1 - bz^{-1}.$$

For what real values of b is the feedback system stable?

11.6. Consider the discrete-time feedback system depicted in Figure 11.3(b) with

$$H(z) = 1 - z^{-N} \qquad \text{and} \qquad G(z) = \frac{z^{-1}}{1 - z^{-N}}.$$

Is this system IIR or FIR?

11.7. Suppose the closed-loop poles of a feedback system satisfy

$$\frac{1}{(s+2)(s+3)} = -\frac{1}{K}.$$

Use the root-locus method to determine the values of K for which the feedback system is guaranteed to be stable.

11.8. Suppose the closed-loop poles of a feedback system satisfy

$$\frac{s-1}{(s+1)(s+2)} = -\frac{1}{K}.$$

Use the root-locus method to determine the negative values of K for which the feedback system is guaranteed to be stable.

11.9. Suppose the closed-loop poles of a feedback system satisfy

$$\frac{(s+1)(s+3)}{(s+2)(s+4)} = -\frac{1}{K}.$$

Use the root-locus method to determine whether there are any values of the adjustable gain K for which the system's impulse response has an oscillatory component of the form $e^{-at}\cos(\omega_0 t + \phi)$, where $\omega_0 \neq 0$.

11.10. The root locus corresponding to $G(s)H(s) = -1/K$ is illustrated in Figure P11.10. In this figure, the start ($K = 0$) and end of each branch of the root locus are marked by a '•' symbol. Specify the poles and zeros of $G(s)H(s)$.

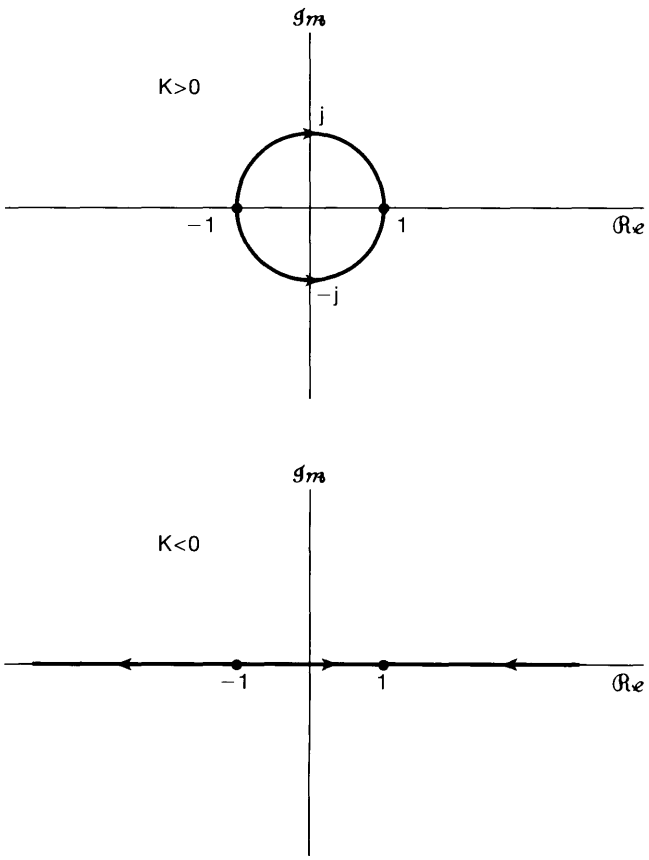


Figure P11.10

11.11. Suppose the closed-loop poles of a discrete-time feedback system satisfy

$$\frac{z^{-2}}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{2}z^{-1})} = -\frac{1}{K}.$$

Using the root-locus method, determine the positive values of K for which this system is stable.

11.12. Each of the four locations $z = 1/2$, $z = 1/4$, $z = 0$, and $z = -1/2$ is a single-order pole or zero of $G(z)H(z)$. Furthermore, $G(z)H(z)$ is known to have only two poles. What information can you deduce about the poles and zeros of $G(z)H(z)$ from the fact that for all K, the root locus corresponding to

$$G(z)H(z) = -\frac{1}{K}$$

is on the real axis.

11.13. Consider the block diagram of Figure P11.13 for a discrete-time system. Use the root-locus method to determine the values of K for which the system is guaranteed to be stable.

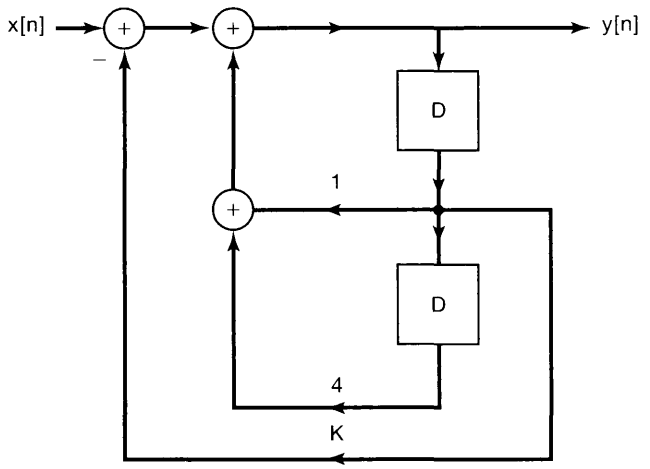


Figure P11.13

11.14. Let C be a closed path that lies on the unit circle in the p-plane and that is traversed in the clockwise direction in order to evaluate $W(p)$. For each of the following expressions for $W(p)$, determine the net number of times the plot of $W(p)$ encircles the origin in a clockwise direction:

(a) $W(p) = \frac{(1-\frac{1}{2}p^{-1})}{(1-\frac{1}{4}p^{-1})}$

(b) $W(p) = \frac{(1-2p^{-1})}{(1-\frac{1}{2}p^{-1})(1-2p^{-1}+4p^{-2})}$

11.15. Consider a continuous-time feedback system whose closed-loop poles satisfy

$$G(s)H(s) = \frac{1}{(s+1)} = -\frac{1}{K}.$$

Use the Nyquist plot and the Nyquist stability criterion to determine the range of values of K for which the closed-loop system is stable. *Hint:* In sketching the Nyquist plot, you may find it useful to sketch the corresponding Bode plot first. It also is helpful to determine the values of ω for which $G(j\omega)H(j\omega)$ is real.

11.16. Consider a continuous-time feedback system whose closed-loop poles satisfy

$$G(s)H(s) = \frac{1}{(s+1)(s/10+1)} = -\frac{1}{K}.$$

Use the Nyquist plot and the Nyquist stability criterion to determine the range of values of K for which the closed-loop system is stable.

11.17. Consider a continuous-time feedback system whose closed-loop poles satisfy

$$G(s)H(s) = \frac{1}{(s+1)^4} = -\frac{1}{K}.$$

Use the Nyquist plot and the Nyquist stability criterion to determine the range of values of K for which the closed-loop system is stable.

11.18. Consider a discrete-time feedback system whose closed-loop poles satisfy

$$G(z)H(z) = z^{-3} = -\frac{1}{K}.$$

Use the Nyquist plot and the Nyquist stability criterion to determine the range of values of K for which the closed-loop system is stable.

11.19. Consider a feedback system, either in continuous-time or discrete-time, and suppose that the Nyquist plot for the system passes through the point $-1/K$. Is the feedback system stable or unstable for this value of the gain? Explain your answer.

11.20. Consider the basic continuous-time feedback system of Figure 11.3(a). Determine the phase and gain margin for the following specification of $H(s)$ and $G(s)$:

$$H(s) = \frac{s+1}{s^2+s+1}, \quad G(s) = 1.$$

BASIC PROBLEMS

11.21. Consider the feedback system of Figure P11.21. Find the closed-loop poles and zeros of this system for the following values of K:

(i) $K = 0.1$
(ii) $K = 1$
(iii) $K = 10$
(iv) $K = 100$

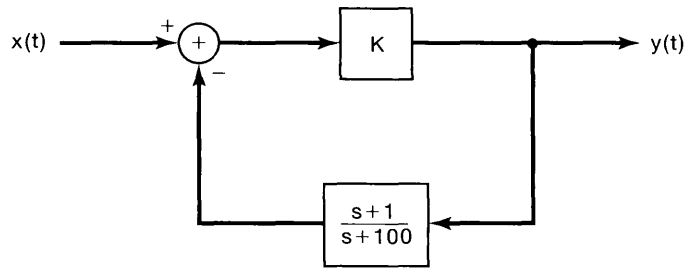


Figure P11.21

11.22. Consider the basic feedback system of Figure 11.3(a). Determine the closed-loop system impulse response for each of the following specifications of the system functions in the forward and feedback paths:

(a) $H(s) = \frac{1}{(s+1)(s+3)}$, $G(s) = 1$

(b) $H(s) = \frac{1}{s+3}$, $G(s) = \frac{1}{s+1}$

(c) $H(s) = \frac{1}{2}$, $G(s) = e^{-s/3}$

11.23. Consider the basic feedback systems of Figure 11.3(b). Determine the closed-loop system impulse response for each of the following specifications of the system functions in the forward and feedback paths:

(a) $H(z) = \frac{z^{-1}}{1-\frac{1}{2}z^{-1}}$, $G(z) = \frac{2}{3} - \frac{1}{6}z^{-1}$

(b) $H(z) = \frac{2}{3} - \frac{1}{6}z^{-1}$, $G(z) = \frac{z^{-1}}{1-\frac{1}{2}z^{-1}}$

11.24. Sketch the root loci for $K > 0$ and $K < 0$ for each of the following:

(a) $G(s)H(s) = \frac{1}{s+1}$

(b) $G(s)H(s) = \frac{1}{(s-1)(s+3)}$

(c) $G(s)H(s) = \frac{1}{s^2+s+1}$

(d) $G(s)H(s) = \frac{s+1}{s^2}$

(e) $G(s)H(s) = \frac{(s+1)^2}{s^3}$

(f) $G(s)H(s) = \frac{s^2+2s+2}{s^2(s-1)}$

(g) $G(s)H(s) = \frac{(s+1)(s-1)}{s(s^2+2s+2)}$

(h) $G(s)H(s) = \frac{(1-s)}{(s+2)(s+3)}$

11.25. Sketch the root loci for $K > 0$ and $K < 0$ for each of the following:

(a) $G(z)H(z) = \frac{z-1}{z^2-\frac{1}{4}}$

(b) $G(z)H(z) = \frac{2}{z^2-\frac{1}{4}}$

(c) $G(z)H(z) = \frac{z^{-1}(1+z^{-1})}{1-\frac{1}{4}z^{-2}}$

(d) $G(z)H(z) = z^{-1} - z^{-2}$

(e) $G(z)H(z)$ is the system function of the causal LTI system described by the difference equation

$$y[n] - 2y[n-1] = x[n-1] - x[n-2].$$

11.26. Consider a feedback system with

$$G(s)H(s) = \frac{(s-a)(s-b)}{s(s+3)(s+6)}.$$

Sketch the root locus for $K > 0$ and $K < 0$ for the following values of a and b:

(a) $a = 1, \quad b = 2$ **(b)** $a = -2, \quad b = 2$ **(c)** $a = -4, \quad b = 2$
(d) $a = -7, \quad b = 2$ **(e)** $a = -1, \quad b = -2$ **(f)** $a = -4, \quad b = -2$
(g) $a = -7, \quad b = -2$ **(h)** $a = -5, \quad b = -4$ **(i)** $a = -7, \quad b = -4$
(j) $a = -7, \quad b = -8$

11.27. Consider a feedback system with

$$H(s) = \frac{s+2}{s^2+2s+4}, \quad G(s) = K.$$

(a) Sketch the root locus for $K > 0$.
(b) Sketch the root locus for $K < 0$.
(c) Find the smallest positive value of K for which the closed-loop impulse response does not exhibit any oscillatory behavior.

11.28. Sketch the Nyquist plot for each of the following specifications of $G(s)H(s)$, and use the continuous-time Nyquist criterion to determine the range of values of K (if any such range exists) for which the closed-loop system is stable. *Note:* In sketching the Nyquist plots, you may find it useful to sketch the corresponding Bode plots first. It also is helpful to determine the values of ω for which $G(j\omega)H(j\omega)$ is real.

(a) $G(s)H(s) = \frac{1}{s-1}$ **(b)** $G(s)H(s) = \frac{1}{s^2-1}$
(c) $G(s)H(s) = \frac{1}{(s+1)^2}$ **(d)** $G(s)H(s) = \frac{1}{(s+1)^3}$
(e) $G(s)H(s) = \frac{1-s}{(s+1)^2}$ **(f)** $G(s)H(s) = \frac{s+1}{(s-1)^2}$
(g) $G(s)H(s) = \frac{s+1}{s^2-4}$ **(h)** $G(s)H(s) = \frac{1}{s^2+2s+2}$
(i) $G(s)H(s) = \frac{s+1}{s^2-2s+2}$ **(j)** $G(s)H(s) = \frac{s+1}{(s+100)(s-1)^2}$
(k) $G(s)H(s) = \frac{s^2}{(s+1)^3}$

11.29. Consider the basic continuous-time feedback system of Figure 11.3(a). Sketch the log magnitude-phase diagram, and roughly determine the phase and gain margin, for each of the following choices of $G(s)$ and $H(s)$. You may find it useful to use the straight-line approximations to the Bode plots developed in Chapter 6 to aid you in sketching the log magnitude-phase diagrams. Be careful, however, to take into account how the actual frequency response deviates from its approximation near break frequencies when there are underdamped second-order terms present. (See Section 6.5.2.)

(a) $H(s) = \frac{10s+1}{s^2+s+1}$, $G(s) = 1$
(b) $H(s) = \frac{s/10+1}{s^2+s+1}$, $G(s) = 1$
(c) $H(s) = \frac{1}{(s+1)^2(s+10)}$, $G(s) = 100$
(d) $H(s) = \frac{1}{(s+1)^3}$, $G(s) = \frac{1}{s+1}$
(e) $H(s) = \frac{1-s}{(s+1)(s+10)}$, $G(s) = 1$

(f) $H(s) = \frac{1-s/100}{(s+1)^2}$, $G(s) = \frac{10s+1}{s/10+1}$

(g) $H(s) = \frac{1}{s(s+1)}$, $G(s) = \frac{1}{s+1}$

Note: Your sketch for part (g) should reflect the fact that for this feedback system $|G(j\omega)H(j\omega)| \to \infty$ as $\omega \to 0$; what is the phase of $G(j\omega)H(j\omega)$ for $\omega = 0^+$, i.e., for ω an infinitesimal amount larger than 0?

11.30. Sketch the Nyquist plot for each of the following specifications of $G(z)H(z)$, and use the discrete-time Nyquist criterion to determine the range of values of K (if any such range exists) for which the closed-loop system is stable. [*Note:* In sketching the Nyquist plots, you may find it useful to first sketch the magnitude and phase plots as a function of frequency or at least calculate $|G(e^{j\omega})H(e^{j\omega})|$ and $\measuredangle G(e^{j\omega})H(e^{j\omega})$ at several points. Also, it is helpful to determine the values of ω for which $G(e^{j\omega})H(e^{j\omega})$ is real.]

(a) $G(z)H(z) = \frac{1}{z-\frac{1}{2}}$ **(b)** $G(z)H(z) = \frac{1}{z-2}$

(c) $G(z)H(z) = z^{-1}$ **(d)** $G(z)H(z) = z^{-2}$

(e) $G(z)H(z) = \frac{1}{(z+\frac{1}{2})(z-\frac{3}{2})}$ **(f)** $G(z)H(z) = \frac{z-\sqrt{3}}{z(z+1/\sqrt{3})}$

(g) $G(z)H(z) = \frac{1}{z^2-z+\frac{1}{3}}$ **(h)** $G(z)H(z) = \frac{z-\frac{1}{2}}{z(z-2)}$

(i) $G(z)H(z) = \frac{(z+1)^2}{z^3}$

11.31. Consider the basic discrete-time system in Figure 11.3(b). Sketch the log magnitude-phase diagram, and roughly determine the phase and gain margin, for each of the following choices of $G(z)$ and $H(z)$. You may find it useful to determine the values of ω for which either $|G(e^{j\omega})H(e^{j\omega})| = 1$ or $\measuredangle G(e^{j\omega}) = -\pi$.

(a) $H(z) = z^{-1}$, $G(z) = \frac{1}{2}$

(b) $H(z) = \frac{z^{-1}}{1-\frac{1}{2}z^{-1}}$, $G(z) = \frac{1}{2}$

(c) $H(z) = \frac{1}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{2}z^{-1})}$, $G(z) = z^{-2}$

(d) $H(z) = \frac{2}{z-2}$, $G(z) = 1$

(e) $H(z) = \frac{1}{z+\frac{1}{2}}$, $G(z) = \frac{1}{z-\frac{3}{2}}$

(f) $H(z) = \frac{1}{z+\frac{1}{2}}$, $G(z) = 1 - \frac{3}{2}z^{-1}$

(g) $H(z) = \frac{\frac{1}{2}}{z^2-z+\frac{1}{3}}$, $G(z) = 1$

(h) $H(z) = \frac{1}{z-1}$, $G(z) = \frac{1}{4}z^{-1}$

(*Note:* Your sketch for part (h) should reflect the fact that, for this feedback system, $G(z)H(z)$ has a pole at $z = 1$; what are the values of $\measuredangle G(e^{j\omega})H(e^{j\omega})$ for $e^{j\omega}$ just on either side of the point $z = 1$?)

ADVANCED PROBLEMS

11.32. **(a)** Consider the feedback system of Figure 11.10(b) with

$$H(s) = \frac{N_1(s)}{D_1(s)}, \quad G(s) = \frac{N_2(s)}{D_2(s)}. \tag{P11.32-1}$$

Assume that there is no pole-zero cancellation in the product $G(s)H(s)$. Show that the zeros of the closed-loop system function consist of the zeros of $H(s)$ and the poles of $G(s)$.

(b) Use the result of part (a) together with the appropriate property of the root locus to confirm that, with $K = 0$, the closed-loop system zeros are the zeros of $H(s)$ and the closed-loop poles are the poles of $H(s)$.

(c) While it is usual for $H(s)$ and $G(s)$ in eq. (P11.32–1) to be in reduced form [i.e., the polynomials $N_1(s)$ and $D_1(s)$ have no common factors, and the same is true of $N_2(s)$ and $D_2(s)$], it may happen that $N_1(s)$ and $D_2(s)$ have common factors or $N_2(s)$ and $D_1(s)$ have common factors. To see what occurs when such common factors are present, let $p(s)$ denote the greatest common factor of $N_2(s)$ and $D_1(s)$. That is,

$$\frac{N_1(s)}{p(s)} \quad \text{and} \quad \frac{D_2(s)}{p(s)}$$

are both polynomials and have *no* common factors. Similarly,

$$\frac{N_2(s)}{q(s)} \quad \text{and} \quad \frac{D_1(s)}{q(s)}$$

are polynomials and have no common factors. Show that the closed-loop system function can be written as

$$Q(s) = \frac{p(s)}{q(s)}\left[\frac{\hat{H}(s)}{1 + K\hat{G}(s)\hat{H}(s)}\right], \tag{P11.32–2}$$

where

$$\hat{H}(s) = \frac{N_1(s)/p(s)}{D_1(s)/q(s)}$$

and

$$\hat{G}(s) = \frac{N_2(s)/q(s)}{D_2(s)/p(s)}.$$

Therefore, from eq. (P11.32–2) and part (a), we see that the zeros of $Q(s)$ are the zeros of $p(s)$, the zeros of $\hat{H}(s)$, and the poles of $\hat{G}(s)$, while the poles of $Q(s)$ are the zeros of $q(s)$ and the solutions of

$$1 + K\hat{G}(s)\hat{H}(s) = 0. \tag{P11.32–3}$$

By construction, there is no pole-zero cancellation in the product $\hat{G}(s)\hat{H}(s)$, and thus, we can apply the root-locus method described in Section 11.3 to sketch the locations of the solutions of eq. (P11.32–3) as K is varied.

(d) Use the procedure outlined in part (c) to determine the closed-loop zeros, any closed-loop poles whose locations are independent of K, and the locus of the remaining closed-loop poles for $K > 0$ when

$$H(s) = \frac{s + 1}{(s + 4)(s + 2)}, \quad G(s) = \frac{s + 2}{s + 1}.$$

(e) Repeat part (d) for

$$H(z) = \frac{1 + z^{-1}}{1 - \frac{1}{2}z^{-1}}, \quad G(z) = \frac{z^{-1}}{1 + z^{-1}}.$$

(f) Let

$$H(z) = \frac{z^2}{(z-2)(z+2)}, \quad G(z) = \frac{1}{z^2}.$$

(i) Sketch the root locus for $K > 0$ and for $K < 0$.
(ii) Find all the values of K for which the overall system is stable.
(iii) Find the impulse response of the closed-loop system when $K = 4$.

11.33. Consider the feedback system of Figure 11.10(a), and suppose that

$$G(s)H(s) = \frac{\prod_{k=1}^{m}(s - \beta_k)}{\prod_{k=1}^{n}(s - \alpha_k)},$$

where $m > n$.[5] In this case $G(s)H(s)$ has $m - n$ poles at infinity (see Chapter 9), and we can adapt the root-locus rules given in the text by noting that (1) there are m branches of the root locus and (2) for $K = 0$, all branches of the root locus begin at poles of $G(s)H(s)$, $m - n$ of which are at infinity. Furthermore, as $|K| \to \infty$, these branches converge to the m zeros of $G(s)H(s)$, namely, $\beta_1, \beta_2, \ldots, \beta_m$. Use these facts to assist you in sketching the root locus (for $K > 0$ and for $K < 0$) for each of the following:

(a) $G(s)H(s) = s - 1$
(b) $G(s)H(s) = (s + 1)(s + 2)$
(c) $G(s)H(s) = \frac{(s+1)(s+2)}{s-1}$

11.34. In Section 11.3, we derived a number of properties that can be of value in determining the root locus for a feedback system. In this problem, we develop several additional properties. We derive these properties in terms of continuous-time systems, but, as with all root-locus properties, they hold as well for discrete-time root loci. For our discussion of these properties, we refer to the basic equation satisfied by the closed-loop poles, namely,

$$G(s)H(s) = -\frac{1}{K}, \tag{P11.34-1}$$

[5]Note that for a continuous-time system, the condition $m > n$ implies that the system with system function $G(s)H(s)$ involves differentiation of the input. [In fact, the inverse transform of $G(s)H(s)$ includes singularity functions up to the order $m - n$.] In discrete time, if $G(z)H(z)$, written as a ratio of polynomials in z, has $m > n$, it is necessarily the system function of a noncausal system. [In fact, the inverse transform of $G(z)H(z)$ has a nonzero value at time $n - m < 0$.] Thus, the case considered in this problem is actually of interest only for continuous-time systems.

where

$$G(s)H(s) = \frac{\prod_{k=1}^{m}(s-\beta_k)}{\prod_{k=1}^{n}(s-\alpha_k)} = \frac{\sum_{k=0}^{m} b_k s^k}{\sum_{k=0}^{n} a_k s^k}. \qquad \text{(P11.34–2)}$$

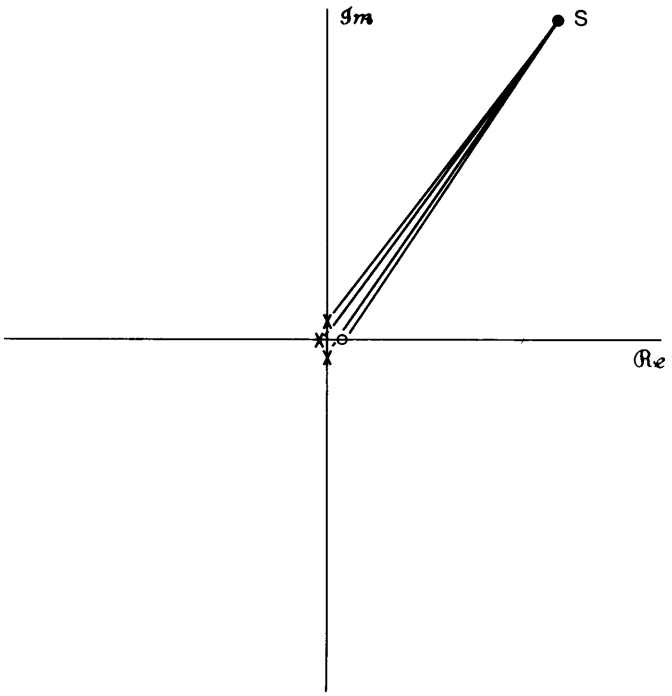


Figure P11.34

Throughout this problem, we assume that $m \leq n$.

(a) From Property 2, we know that $n - m$ branches of the root locus go to zeros of $G(s)H(s)$ located at infinity. In this first part, we demonstrate that it is straightforward to determine the angles at which these branches approach infinity. Specifically, consider searching the remote part of the s-plane [i.e., the region where $|s|$ is extremely large and far from any of the poles and zeros of $G(s)H(s)$]. This region is illustrated in Figure P11.34. Use the geometry of the picture, together with the angle criterion for $K > 0$ and for $K < 0$, to deduce that:

- For $K > 0$, the $n - m$ branches of the root locus that approach infinity do so at the angles

$$\frac{(2k+1)\pi}{n-m}, \qquad k = 0, 1, \ldots, n-m-1.$$

- For $K < 0$, the $n - m$ branches of the root locus that approach infinity do so at the angles

$$\frac{2k\pi}{n-m}, \qquad k = 0, 1, \ldots, n-m-1.$$

Thus, the branches of the root locus that approach infinity do so at specified angles that are arranged symmetrically. For example, for $n - m = 3$ and $K > 0$, we see that the asymptotic angles are $\pi/3$, π, and $5\pi/3$. The result of part (a), together with one additional fact, allows us to draw in the asymptotes for the branches of the root locus that approach infinity. Specifically, all of the $n - m$ asymptotes intersect at a single point on the real axis. This is derived in the next part of the problem.

(b) (i) As a first step, consider a general polynomial equation

$$s^r + f_{r-1}s^{r-1} + \cdots + f_0 = (s - \xi_1)(s - \xi_2)\cdots(s - \xi_r) = 0.$$

Show that

$$f_{r-1} = -\sum_{i=1}^{r} \xi_i.$$

(ii) Perform long division on $1/G(s)H(s)$ to write

$$\frac{1}{G(s)H(s)} = s^{n-m} + \gamma_{n-m-1}s^{n-m-1} + \ldots. \qquad \text{(P11.34–3)}$$

Show that

$$\gamma_{n-m-1} = a_{n-1} - b_{m-1} = \sum_{k=1}^{m} \beta_k - \sum_{k=1}^{n} \alpha_k.$$

[See eq. (P11.34–2).]

(iii) Argue that the solution of eq. (P11.34–1) for *large* s is an approximate solution of the equation

$$s^{n-m} + \gamma_{n-m-1}s^{n-m-1} + \gamma_{n-m-2}s^{n-m-2} + \cdots + \gamma_0 + K = 0.$$

(iv) Use the results of (i)–(iii) to deduce that the sum of the $n - m$ closed-loop poles that approach infinity is asymptotically equal to

$$b_{m-1} - a_{n-1}.$$

Thus, the center of gravity of these $n - m$ poles is

$$\frac{b_{m-1} - a_{n-1}}{n-m},$$

which does not depend on K. Consequently, we have $n - m$ closed-loop poles that approach $|s| = \infty$ at evenly spaced angles and that have a center of gravity that is independent of K. From this, we can deduce that:

The asymptotes of the $n-m$ branches of the root locus that approach infinity intersect at the point

$$\frac{b_{m-1}-a_{n-1}}{n-m}=\frac{\sum_{k=1}^{n}\alpha_k-\sum_{k=1}^{m}\beta_k}{n-m}.$$

This point of intersection of the asymptotes is the same for $K>0$ and $K<0$.

(c) Suppose that

$$G(s)H(s)=\frac{1}{(s+1)(s+3)(s+5)}.$$

(i) What are the asymptotic angles for the closed-loop poles that approach infinity for $K>0$ and for $K<0$?
(ii) What is the point of intersection of the asymptotes?
(iii) Draw in the asymptotes, and use them to help you sketch the root locus for $K>0$ and for $K<0$.

(d) Repeat part (c) for each of the following:
(i) $G(s)H(s)=\frac{s+1}{s(s+2)(s+4)}$
(ii) $G(s)H(s)=\frac{1}{s^4}$
(iii) $G(s)H(s)=\frac{1}{s(s+1)(s+5)(s+6)}$
(iv) $G(s)H(s)=\frac{1}{(s+2)^2(s-1)^2}$
(v) $G(s)H(s)=\frac{s+3}{(s+1)(s^2+2s+2)}$
(vi) $G(s)H(s)=\frac{s+1}{(s+2)^2(s^2+2s+2)}$
(vii) $G(s)H(s)=\frac{s+1}{(s+100)(s-1)(s-2)}$

(e) Use the result of part (a) to explain why the following statement is true: For any continuous-time feedback system, with $G(s)H(s)$ given by eq. (P11.34–2), if $n-m\geq 3$, we can make the closed-loop system unstable by choosing $|K|$ large enough.

(f) Repeat part (c) for the discrete-time feedback system specified by

$$G(z)H(z)=\frac{z^{-3}}{(1-z^{-1})(1+\frac{1}{2}z^{-1})}.$$

(g) Explain why the following statement is true: For any discrete-time feedback system with

$$G(z)H(z)=\frac{z^m+b_{m-1}z^{m-1}+\cdots+b_0}{z^n+a_{n-1}z^{n-1}+\cdots+a_0},$$

if $n>m$, we can make the closed-loop system unstable by choosing $|K|$ large enough.

11.35. (a) Consider again the feedback system of Example 11.2:

$$G(s)H(s)=\frac{s-1}{(s+1)(s+2)}.$$

The root locus for $K < 0$ is plotted in Figure 11.14(b). For some value of K, the closed-loop poles are on the $j\omega$-axis. Determine this value of K and the corresponding locations of the closed-loop poles by examining the real and imaginary parts of the equation

$$G(j\omega)H(j\omega) = -\frac{1}{K},$$

which must be satisfied if the point $s = j\omega$ is on the root locus for any given values of K. Use this result plus the analysis in Example 11.2 to find the full range of values of K (positive and negative) for which the closed-loop system is stable.

(b) Note that the feedback system is unstable for $|K|$ sufficiently large. Explain why this is true in general for continuous-time feedback systems for which $G(s)H(s)$ has a zero in the right-half plane and for discrete-time feedback systems for which $G(z)H(z)$ has a zero outside the unit circle.

11.36. Consider a continuous-time feedback system with

$$G(s)H(s) = \frac{1}{s(s+1)(s+2)}. \tag{P11.36–1}$$

(a) Sketch the root locus for $K > 0$ and for $K < 0$. (*Hint:* The results of Problem 11.34 are useful here.)

(b) If you have sketched the locus correctly, you will see that for $K > 0$, two branches of the root locus cross the $j\omega$-axis, passing from the left-half plane into the right-half plane. Consequently, we can conclude that the closed-loop system is stable for $0 < K < K_0$, where K_0 is the value of the gain for which the two branches of the root locus intersect the $j\omega$-axis. Note that the sketch of the root locus does not by itself tell us what the value of K_0 is or the exact point on the $j\omega$-axis where the branches cross. As in Problem 11.35, determine K_0 by solving the pair of equations obtained as the real and imaginary parts of

$$G(j\omega)H(j\omega) = -\frac{1}{K_0}. \tag{P11.36–2}$$

Determine the corresponding two values of ω (which are the negatives of each other, since poles occur in complex-conjugate pairs).

From your root-locus sketches in part (a), note that there is a segment of the real axis between two poles which is on the root locus for $K > 0$, and a different segment is on the locus for $K < 0$. In both cases, the root locus breaks off from the real axis at some point. In the next part of this problem, we illustrate how one can calculate these breakaway points.

(c) Consider the equation denoting the closed-loop poles:

$$G(s)H(s) = -\frac{1}{K}. \tag{P11.36–3}$$

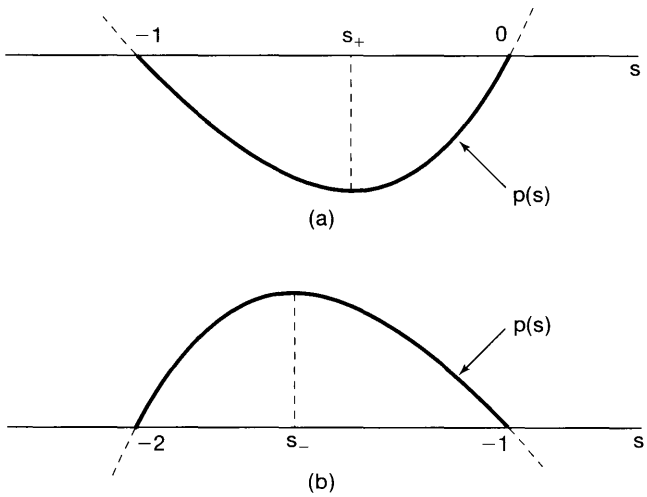


Figure P11.36

Using eq. (P11.36–1), show that an equivalent equation for the closed loop poles is

$$p(s) = s^3 + 3s^2 + 2s = -K. \qquad \text{(P11.36–4)}$$

Consider the segment of the real axis between 0 and -1. This segment is on the root locus for $K \geq 0$. For $K = 0$, two branches of the locus begin at 0 and -1 and approach each other as K is increased.

(i) Use the facts stated, together with eq. (P11.36–4), to explain why the function $p(s)$ has the form shown in Figure P11.36(a) for $-1 \leq s \leq 0$ and why the point s_+ where the minimum occurs is the breakaway point (i.e., it is the point where the two branches of the $K > 0$ locus break from the segment of the real axis between -1 and 0).

Similarly, consider the root locus for $K < 0$ and, more specifically, the segment of the real axis between -1 and -2 that is part of this locus. For $K = 0$, two branches of the root locus begin at -1 and -2, and as K is decreased, these poles approach each other.

(ii) In an analogous fashion to that used in part (i), explain why the function $p(s)$ has the form shown in Figure P11.36(b) and why the point s_- where the maximum occurs is the breakaway point for $K < 0$.

Thus, the breakaway points correspond to the the maxima and minima of $p(s)$ as s ranges over the negative real line.

(iii) The points at which $p(s)$ has a maximum or minimum are the solutions of the equation

$$\frac{dp(s)}{ds} = 0.$$

Use this fact to find the breakaway points s_+ and s_-, and then use eq. (P11.36–4) to find the gains at which these points are closed-loop poles.

In addition to the method illustrated in part (c), there are other, partially analytical, partially graphical methods for determining breakaway points. It is also possible to use a procedure similar to the one just illustrated in part (c) to find the

"break-in" points, where two branches of the root locus merge onto the real axis. These methods plus the one illustrated are described in advanced texts such as those listed in the bibliography at the end of the book.

11.37. One issue that must always be taken into account by the system designer is the possible effect of unmodeled aspects of the system one is attempting to stabilize or modify through feedback. In this problem, we provide an illustration of why this is the case. Consider a continuous-time feedback system, and suppose that

$$H(s) = \frac{1}{(s+10)(s-2)} \tag{P11.37-1}$$

and

$$G(s) = K. \tag{P11.37-2}$$

(a) Use root-locus techniques to show that the closed-loop system will be stable if K is chosen large enough.

(b) Suppose that the system we are trying to stabilize by feedback actually has a system function

$$H(s) = \frac{1}{(s+10)(s-2)(10^{-3}s+1)}. \tag{P11.37-3}$$

The added factor can be thought of as representing a first-order system in cascade with the system of eq. (P11.37–1). Note that the time constant of the added first order system is extremely small and thus will appear to have a step response that is almost instantaneous. For this reason, one often neglects such factors in order to obtain simpler and more tractable models that capture all of the important characteristics of the system. However, one must still keep these neglected dynamics in mind in obtaining a useful feedback design. To see why this is the case, show that if $G(s)$ is given by eq. (P11.37–2) and $H(s)$ is as in eq. (P11.37–3), then the closed-loop system will be unstable if K is chosen *too* large. *Hint:* See Problem 11.34.

(c) Use root-locus techniques to show that if

$$G(s) = K(s+100),$$

then the feedback system will be stable for all values of K sufficiently large if $H(s)$ is given by eq. (P11.37–1) *or* eq. (P11.37–3).

11.38. Consider the feedback system of Figure 11.3(b) with

$$H(z) = \frac{Kz^{-1}}{1-z^{-1}}$$

and

$$G(z) = 1 - az^{-1}.$$

(a) Sketch the root locus for $K > 0$ and $K < 0$ when $a = 1/2$.

(b) Repeat part (a) when $a = -1/2$.

(c) With $a = -1/2$, find a value of K for which the closed-loop impulse response is of the form

$$(A + Bn)\alpha^n$$

for some values of the constants A, B, and α, with $|\alpha| < 1$. (*Hint:* What must the denominator of the closed-loop system function look like in this case?)

11.39. Consider the feedback system of Figure P11.39 with

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad G(z) = K. \tag{P11.39–1}$$

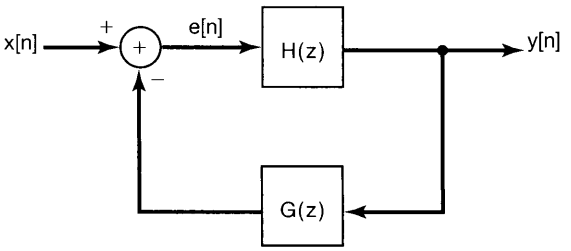


Figure P11.39

(a) Plot the root locus for $K > 0$.

(b) Plot the root locus for $K < 0$. (*Note:* Be careful with this root locus. By applying the angle criterion on the real axis, you will find that as K is decreased from zero, the closed loop approaches $z = +\infty$ along the positive real axis and then returns along the negative real axis from $z = -\infty$. Check that this is in fact the case by explicitly solving for the closed-loop pole as a function of K. At what value of K is the pole at $|z| = \infty$?)

(c) Find the full range of values of K for which the closed-loop system is stable.

(d) The phenomenon observed in part (b) is a direct consequence of the fact that in this example the numerator and denominator of $G(z)H(z)$ have the same degree. When this occurs in a discrete-time feedback system, it means that there is a delay-free loop in the system. That is, the output at a given point in time is being fed back into the system and in turn affects its own value at the same point in time. To see that this is the case in the system we are considering here, write the difference equation relating $y[n]$ and $e[n]$. Then write $e[n]$ in terms of the input and output for the feedback system. Contrast this result with that of the feedback system with

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad G(z) = Kz^{-1}. \tag{P11.39–2}$$

The primary consequence of having delay-free loops is that such feedback systems cannot be implemented in the form depicted. For example, for the system of eq. (P11.39–1), we cannot first calculate $e[n]$ and then $y[n]$,

because $e[n]$ depends on $y[n]$. Note that we *can* perform this type of calculation for the system of eq. (P11.39–2), since $e[n]$ depends on $y[n-1]$.

(e) Show that the feedback system of eq. (P11.39–1) represents a causal system, except for the value of K for which the closed-loop pole is at $|z| = \infty$.

11.40. Consider the discrete-time feedback system depicted in Figure P11.40. The system in the forward path is not very well damped, and we would like to choose the feedback system function so as to improve the overall damping. By using the root-locus method, show that this can be done with

$$G(z) = 1 - \frac{1}{2}z^{-1}.$$

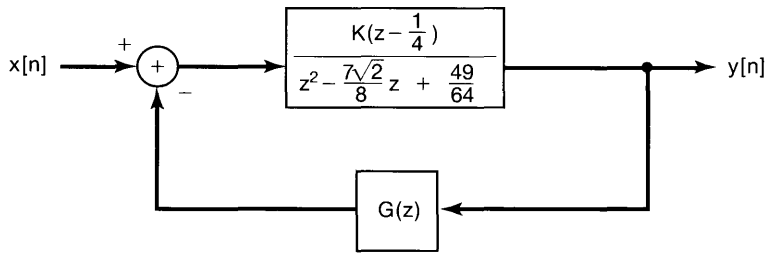


Figure P11.40

Specifically, sketch the root locus for $K > 0$, and specify the value of the gain K for which a significant improvement in damping is obtained.

11.41. **(a)** Consider a feedback system with

$$H(z) = \frac{z+1}{z^2 + z + \frac{1}{4}}, \quad G(z) = \frac{K}{z-1}.$$

(i) Write the closed-loop system function explicitly as a ratio of two polynomials. (The denominator polynomial will have coefficients that depend on K.)

(ii) Show that the sum of the closed-loop poles is independent of K.

(b) More generally, consider a feedback system with system function

$$G(z)H(z) = K\frac{z^m + b_{m-1}z^{m-1} + \cdots + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_0}.$$

Show that if $m \leq n-2$, the sum of the closed-loop poles is independent of K.

11.42. Consider again the discrete-time feedback system of Example 11.3:

$$G(z)H(z) = \frac{z}{(z-\frac{1}{2})(z-\frac{1}{4})}.$$

The root loci for $K > 0$ and $K < 0$ are depicted in Figure 11.16.

(a) Consider the root locus for $K > 0$. In this case, the system becomes unstable when one of the closed-loop poles is less than or equal to -1. Find the value of K for which $z = -1$ is a closed-loop pole.

(b) Consider the root locus for $K < 0$. In this case, the system becomes unstable when one of the closed-loop poles is greater than or equal to 1. Find the value of K for which $z = 1$ is a closed-loop pole.

(c) What is the full range of values of K for which the closed-loop system is stable?

11.43. Consider a discrete-time feedback system with

$$G(z)H(z) = \frac{1}{z(z-1)}.$$

(a) Sketch the root locus for $K > 0$ and for $K < 0$.

(b) If you have sketched the root locus correctly for $K > 0$, you will see that the two branches of the root locus cross and exit from the unit circle. Consequently, we can conclude that the closed-loop system is stable for $0 < K < K_0$, where K_0 is the value of the gain for which the two branches intersect the unit circle. At what points on the unit circle do the branches exit from it? What is the value of K_0?

11.44. As mentioned in Section 11.4, the continuous-time Nyquist criterion can be extended to allow for poles of $G(s)H(s)$ on the $j\omega$-axis. In this problem, we will illustrate the general technique for doing this by means of several examples. Consider a continuous-time feedback system with

$$G(s)H(s) = \frac{1}{s(s+1)}. \tag{P11.44–1}$$

When $G(s)H(s)$ has a pole at $s = 0$, we modify the contour of Figure 11.19 by avoiding the origin. To do this, we indent the contour by adding a semicircle of infinitesimal radius ϵ into the right-half plane. [See Figure P11.44(a).] Thus, only a small part of the right-half plane is not enclosed by the modified contour, and its area goes to zero as we let $\epsilon \to 0$. Consequently, as $M \to \infty$, the contour will enclose the entire right-half plane. As in the text, $G(s)H(s)$ is a constant (in this case zero) along the circle of infinite radius. Thus, to plot $G(s)H(s)$ along the contour, we need only plot it for the portion of the contour consisting of the $j\omega$-axis and the infinitesimal circle.

(a) Show that

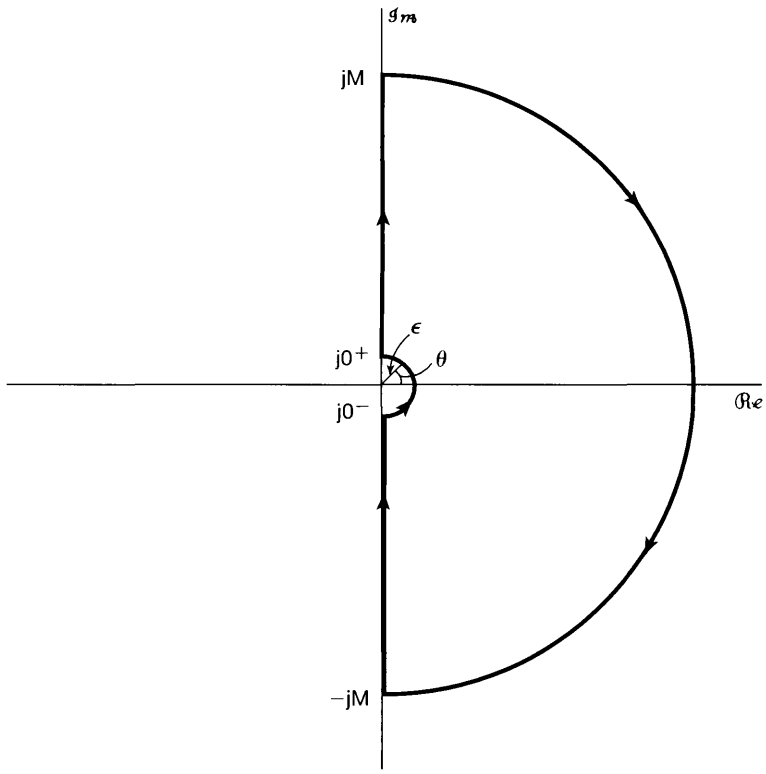
$$\measuredangle G(j0^+)H(j0^+) = -\frac{\pi}{2}$$

and

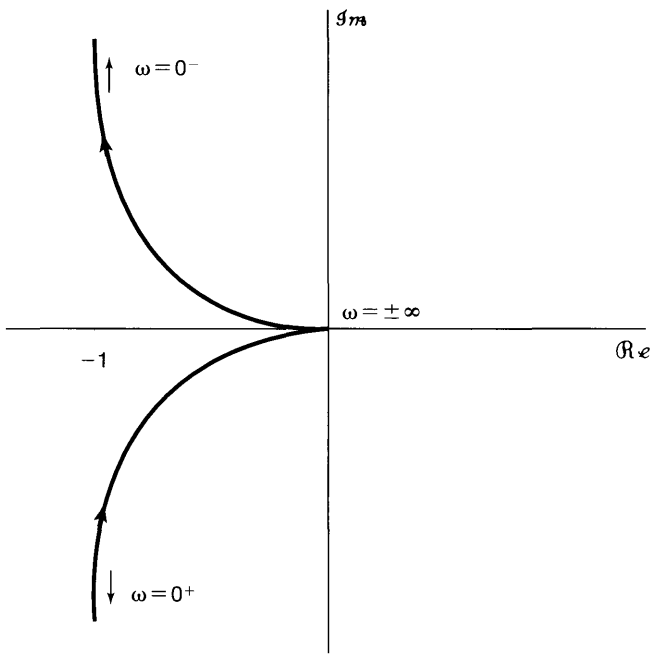
$$\measuredangle G(j0^-)H(j0^-) = \frac{\pi}{2},$$

where $s = j0^-$ is the point where the infinitesimal semicircle meets the $j\omega$-axis just below the origin and $s = j0^+$ is the corresponding point just above the origin.

(b) Use the result of part (a) together with eq. (P11.44–1) to verify that Figure P11.44(b) is an accurate sketch of $G(s)H(s)$ along the portions of the contour



(a)



(b)

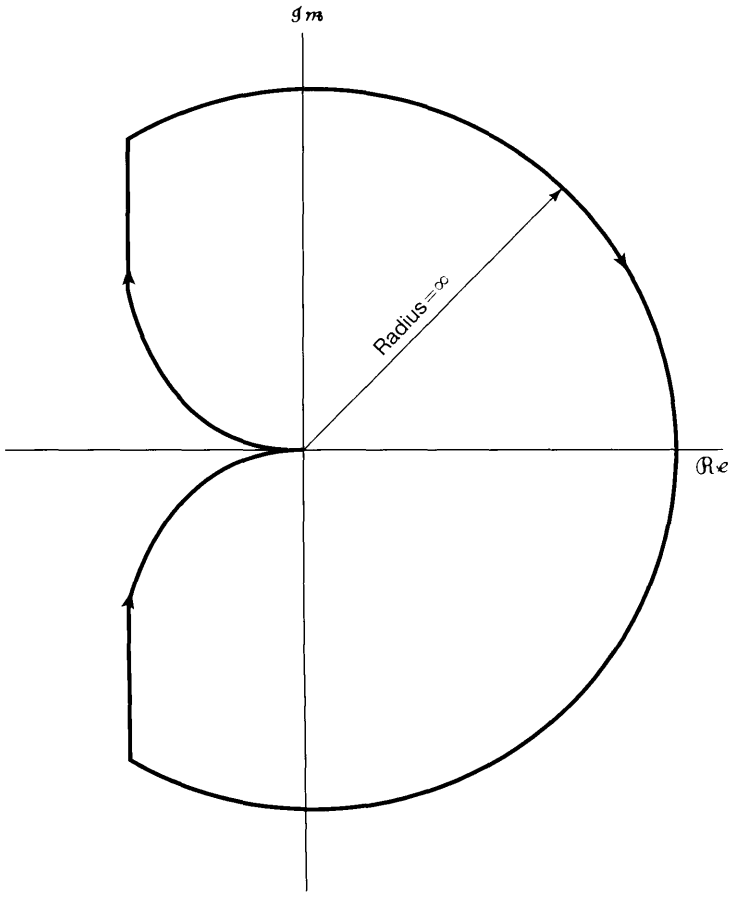
Figure P11.44

from $-j\infty$ to $j0^-$ and $j0^+$ to $j\infty$. In particular, check that $\measuredangle G(j\omega)H(j\omega)$ and $|G(j\omega)H(j\omega)|$ behave in the manner depicted in the figure.

(c) All that remains to be done is to determine the plot of $G(s)H(s)$ along the small semicircle about $s = 0$. Note that as $\epsilon \to 0$, the magnitude of $G(s)H(s)$ along this contour goes to infinity. Show that as $\epsilon \to 0$, the contribution of the pole at $s = -1$ to $\measuredangle G(s)H(s)$ along the semicircle is zero. Then show that as $\epsilon \to 0$,

$$\measuredangle G(s)H(s) = -\theta,$$

where θ is as defined in Figure P11.44(a). Thus, since θ varies from $-\pi/2$ at $s = j0^-$ to $+\pi/2$ at $s = j0^-$ in the counterclockwise direction, $\measuredangle G(s)H(s)$ must go from $+\pi/2$ at $s = j0^+$ to $-\pi/2$ at $s = j0^+$ in the clockwise direction. The result is the complete Nyquist plot depicted in Figure P11.44(c).



(c)

Figure P11.44 *Continued*

(d) Using the Nyquist plot of Figure P11.44(c), find the range of values of K for which the closed-loop feedback system is stable. (*Note:* As presented in the text, the continuous-time Nyquist criterion states that, for closed-loop system stability, the net number of clockwise encirclements of the point $-1/K$ must equal minus the net number of right-half plane poles of $G(s)H(s)$. In the present example, note that the pole of $G(s)H(s)$ at $s = 0$ is *outside* the modified contour. Consequently, it is *not* included in counting the poles of $G(s)H(s)$ in the right-half plane [i.e., only poles of $G(s)H(s)$ strictly *inside* the right-half plane are counted in applying the Nyquist criterion]. Thus, in this case, since $G(s)H(s)$ has no poles strictly inside the right-half plane, we must have *no* encirclements of the point $s = -1/K$ for closed-loop system stability.)

(e) Follow the steps outlined in parts (a)–(c) to sketch the Nyquist plots for each of the following:

(i) $G(s)H(s) = \frac{(s/10)+1}{s(s+1)}$

(ii) $G(s)H(s) = \frac{1}{s(s+1)^2}$

(iii) $G(s)H(s) = \frac{1}{s^2}$ [be careful in calculating $\measuredangle G(s)H(s)$ along the infinitesimal semicircle]

(iv) $G(s)H(s) = \frac{s+1}{s(1-s)}$ [be careful in calculating $\measuredangle G(j\omega)H(j\omega)$ as ω is varied; make sure to take the minus sign in the denominator into account]

(v) $G(s)H(s) = \frac{s+1}{s^2}$ [same remark as for (iii)]

In each case, use the Nyquist criterion to determine the range of values of K (if any such range exists) for which the closed-loop system is stable. Also, use another method (root locus or direct calculation of the closed-loop poles as a function of K) to provide a partial check of the correctness of your Nyquist plot. [*Note:* In sketching the Nyquist plots, you may find it useful to sketch the Bode plots of $G(s)H(s)$ first. It may also be helpful to determine the values of ω for which $G(j\omega)H(j\omega)$ is real.]

(f) Repeat part (e) for:

(i) $G(s)H(s) = \frac{1}{s^2+1}$

(ii) $G(s)H(s) = \frac{s+1}{s^2+1}$

Note: In these cases there are *two* poles on the imaginary axis; accordingly, you will need to modify the contour of Figure 11.19 to avoid each of them. Use infinitesimal semicircles, as in Figure P11.44(a).

11.45. Consider a system with system function

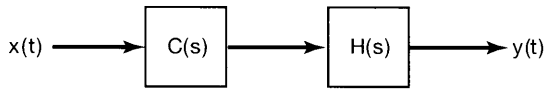
$$H(s) = \frac{1}{(s+1)(s-2)}. \tag{P11.45–1}$$

Because this system is unstable, we would like to devise some method for its stabilization.

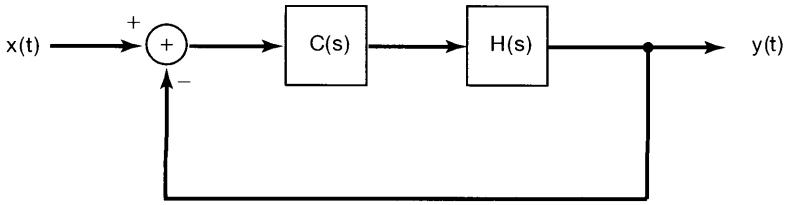
(a) Consider first a series compensation scheme as illustrated in Figure P11.45(a). Show that the overall system of this figure is stable if the system function

$$C(s) = \frac{s-2}{s+3}.$$

In practice, this is *not* considered to be a particularly useful way to attempt to stabilize a system. Explain why.



(a)



(b)

Figure P11.45

(b) Suppose that instead we use a feedback system, as depicted in Figure P11.45(b). Is it possible to stabilize this system using a constant gain, that is,

$$C(s) = K,$$

for the stabilizing element? Justify your answer using Nyquist techniques.

(c) Show that the system of Figure P11.45(b) can be stabilized if $C(s)$ is a proportional plus derivative system—that is, if

$$C(s) = K(s + a).$$

Consider both the case $0 < a < 1$ and the case $a > 1$.

(d) Suppose that

$$C(s) = K(s + 2).$$

Choose the value of K such that the closed-loop system has a pair of complex poles with a damping ratio $\zeta = 1/2$. (*Hint:* In this case, the denominator of the closed-loop system must have the form

$$s^2 + \omega_n s + \omega_n^2$$

for some value of $\omega_n > 0$.)

(e) Pure derivative compensation is both impossible to obtain and undesirable in practice. This is because the required amplification of arbitrarily high frequencies neither can be obtained nor is advisable, as all real systems are subject to some level of high-frequency disturbances. Thus, suppose that we consider a compensator of the form

$$C(s) = K\left(\frac{s + a}{s + b}\right), \quad a, b > 0. \tag{P11.45–2}$$

If $b < a$, this is a *lag network*: $\sphericalangle C(j\omega) < 0$ for all $\omega > 0$, so that the phase of the output of the system lags the phase of the input. If $b > a$, $\sphericalangle C(j\omega) > 0$ for all $\omega > 0$, and the system is then called a *lead network*.

(i) Show that it is possible to stabilize the system with the lead compensator

$$C(s) = K\frac{s + \frac{1}{2}}{s + 2} \qquad \text{(P11.45–3)}$$

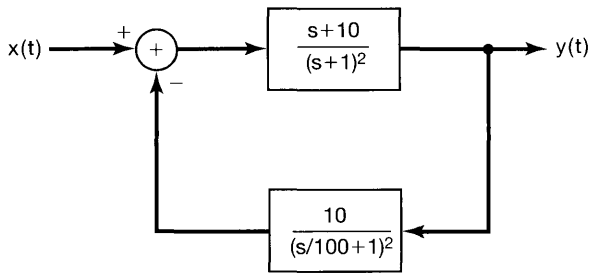
if K is chosen large enough.

(ii) Show that it is not possible to stabilize the feedback system of Figure P11.45(b) using the lag network

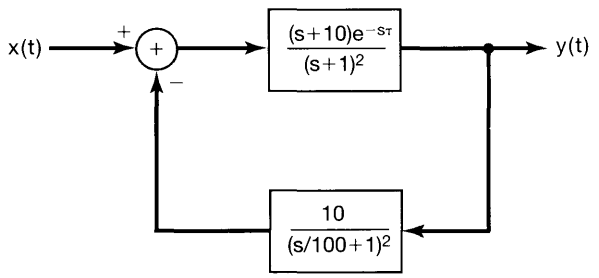
$$C(s) = K\frac{s + 3}{s + 2}.$$

Hint: Use the results of Problem 11.34 in sketching the root locus. Then determine the points on the $j\omega$-axis that are on the root locus and the values of K for which each of these points is a closed-loop pole. Use this information to prove that for no value of K are *all* of the closed-loop poles in the left-half plane.

11.46. Consider the continuous-time feedback system depicted in Figure P11.46(a).



(a)



(b)

Figure P11.46

(a) Use the straight-line approximations to Bode plots developed in Chapter 6 to obtain a sketch of the log magnitude-phase plot of this system. Estimate the phase and gain margins from your plot.

(b) Suppose that there is an unknown delay within the feedback system, so that the actual feedback system is as shown in Figure P11.46(b). Approximately what is the largest delay τ that can be tolerated before the feedback system becomes unstable? Use your results from part (a) for this calculation.

(c) Calculate more precise values of the phase and gain margins, and compare these to your results in part (a). This should give you some idea of the size of the errors that are incurred in using the approximate Bode plots.

11.47. As mentioned at the end of Section 11.5, the phase and gain margins may provide *sufficient* conditions to ensure that a stable feedback system remains stable. For example, we showed that a stable feedback system will remain stable as the gain is increased, until we reach a limit specified by the gain margin. This does *not* imply (a) that the feedback system cannot be made unstable by *decreasing* the gain or (b) that the system will be unstable for *all* values of gain greater than the gain margin limit. In this problem, we illustrate these two points.

(a) Consider a continuous-time feedback system with

$$G(s)H(s) = \frac{1}{(s-1)(s+2)(s+3)}.$$

Sketch the root locus for this system for $K > 0$. Use the properties of the root locus described in the text and in Problem 11.34 to help you draw the locus accurately. Once you do so, you should see that for small values of the gain K the system is unstable, for larger values of K the system is stable, while for still larger values of K the system again becomes unstable. Find the range of values of K for which the system is stable. *Hint:* Use the same method as is employed in Example 11.2 and Problem 11.35 to determine the values of K at which branches of the root locus pass through the origin and cross the $j\omega$-axis.

If we set our gain somewhere within the stable range that you have just found, we can increase the gain somewhat and maintain stability, but a large enough increase in gain causes the system to become unstable. This maximum amount of increase in gain at which the closed-loop system just becomes unstable is the gain margin. Note that if we *decrease* the gain too much, we can also cause instability.

(b) Consider the feedback system of part (a) with the gain K set at a value of 7. Show that the closed-loop system is stable. Sketch the log magnitude-phase plot of this system, and show that there are two nonnegative values of ω for which $\measuredangle G(j\omega)H(j\omega) = -\pi$. Further, show that, for one of these values $7|G(j\omega)H(j\omega)| < 1$, and for the other $7|G(j\omega)H(j\omega)| > 1$. The first value provides us with the usual gain margin—that is, the factor $1/|7G(j\omega)H(j\omega)|$ by which we can increase the gain and cause instability. The second provides us with the factor $1/|7G(j\omega)H(j\omega)|$ by which we can decrease the gain and just cause instability.

(c) Consider a feedback system with

$$G(s)H(s) = \frac{(s/100 + 1)^2}{(s + 1)^3}.$$

Sketch the root locus for $K > 0$. Show that two branches of the root locus begin in the left-half plane and, as K is increased, move into the right-half plane and then back into the left-half plane. Do this by examining the equation

$$G(j\omega)H(j\omega) = -\frac{1}{K}.$$

Specifically, by equating the real and imaginary parts of this equation, show that there are two values of $K \geq 0$ for which the closed-loop poles lie on the $j\omega$-axis.

Thus, if we set the gain at a small enough value so that the system is stable, then we can increase the gain up until the point at which the two branches of the root locus intersect the $j\omega$-axis. For a range of values of gain beyond this point, the closed-loop system is unstable. However, if we *continue* to increase the gain, the system will again become stable for K large enough.

(d) Sketch the Nyquist plot for the system of part (c), and confirm the conclusions reached in part (c) by applying the Nyquist criterion. (Make sure to count the *net* number of encirclements of $-1/K$.)

Systems such as that considered in parts (c) and (d) of this problem are often referred to as being *conditionally stable* systems, because their stability properties may change several times as the gain is varied.

11.48. In this problem, we illustrate the discrete-time counterpart of the technique described in Problem 11.44. Specifically, the discrete-time Nyquist criterion can be extended to allow for poles of $G(z)H(z)$ on the unit circle.

Consider a discrete-time feedback system with

$$G(z)H(z) = \frac{z^{-2}}{1 - z^{-1}} = \frac{1}{z(z - 1)}. \qquad \text{(P11.48–1)}$$

In this case, we modify the contour on which we evaluate $G(z)H(z)$, as illustrated in Figure P11.48(a).

(a) Show that

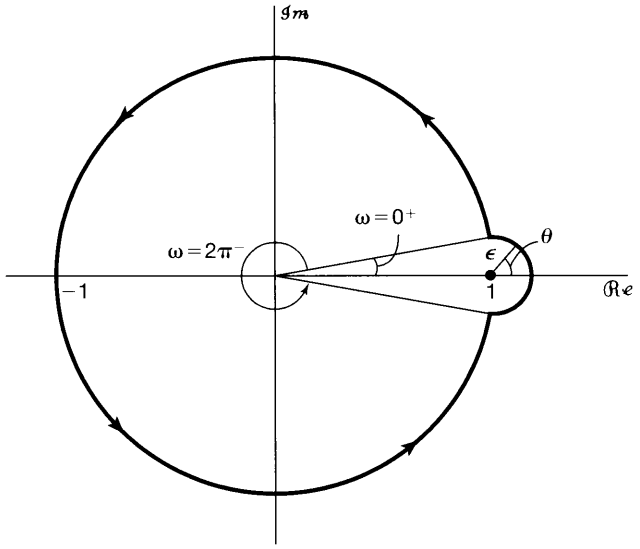
$$\measuredangle G(e^{j0^+})H(e^{j0^+}) = -\frac{\pi}{2}$$

and

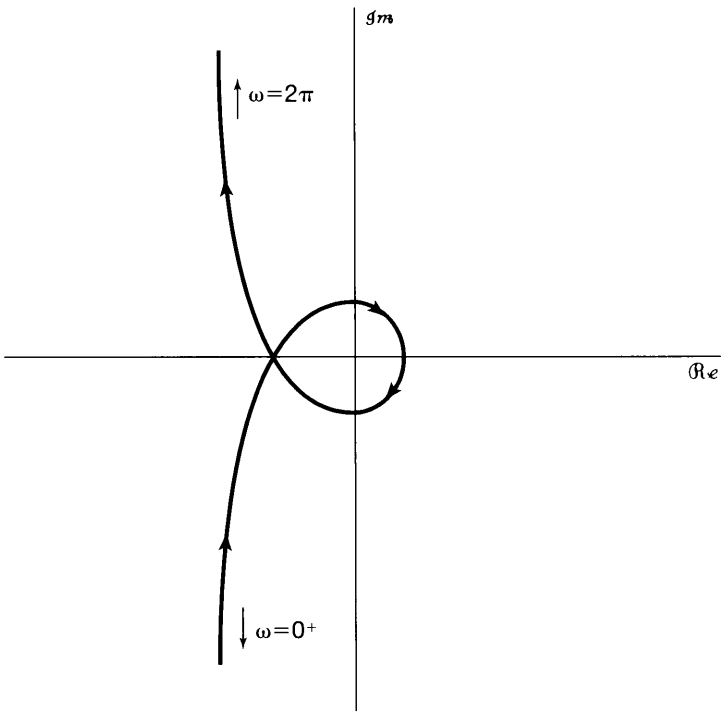
$$\measuredangle G(e^{j2\pi^-})H(e^{j2\pi^-}) = \frac{\pi}{2},$$

where $z = e^{j2\pi}$ is the point below the real axis at which the small semicircle intersects the unit circle and $z = e^{j0^+}$ is the corresponding point above the real axis.

(b) Use the results of part (a) together with eq. (P11.48–1) to verify that Figure P11.48(b) is an accurate sketch of $G(z)H(z)$ along the portion of the



(a)



(b)

Figure P11.48

contour $z = e^{j\omega}$ as ω varies from 0^+ to $2\pi^-$ in a counterclockwise direction. In particular, verify that the angular variation of $G(e^{j\omega})H(e^{j\omega})$ is as indicated.

(c) Find the value of ω for which $\measuredangle G(e^{j\omega})H(e^{j\omega}) = -\pi$, and verify that

$$|G(e^{j\omega})H(e^{j\omega})| = 1$$

at this point. [*Hint:* Use the geometrical method for evaluating $\measuredangle G(e^{j\omega})H(e^{j\omega})$ together with some elementary geometry to determine the value of ω.]

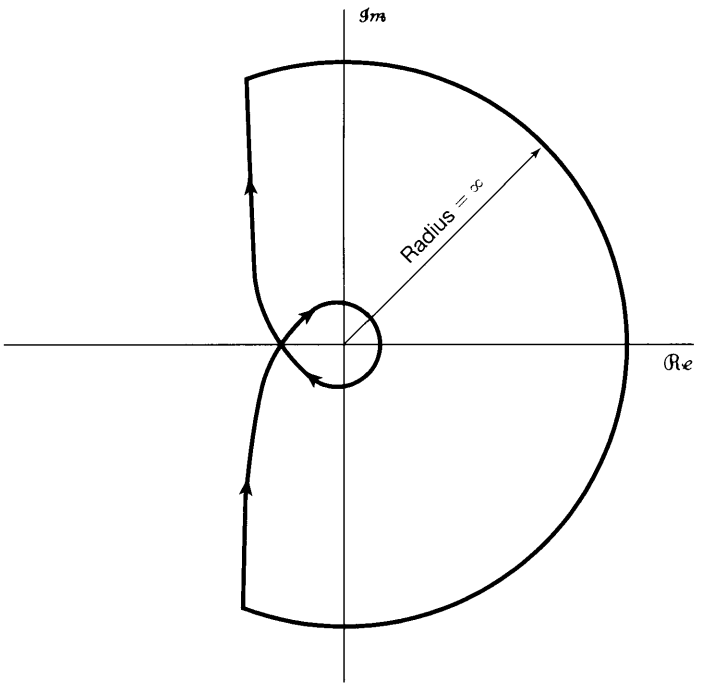
(d) Consider next the plot of $G(z)H(z)$ along the small semicircle about $z = 1$. Note that as $\epsilon \to 0$, the magnitude of $G(z)H(z)$ along this contour goes to infinity. Show that as $\epsilon \to 0$, the contribution of the pole at $z = 0$ to $\measuredangle G(z)H(z)$ along the semicircle is zero. Then show that as $\epsilon \to 0$,

$$\measuredangle G(z)H(z) = -\theta,$$

where θ is as defined in Figure P11.48(a).

Thus, since θ varies from $-\pi/2$ to $+\pi/2$ in the counterclockwise direction, $\measuredangle G(z)H(z)$ varies from $+\pi/2$ to $-\pi/2$ in the clockwise direction. The result is the complete Nyquist plot of Figure P11.48(c).

(e) Using the Nyquist plot, find the range of values of K for which the closed-loop feedback system is stable. [*Note:* Since the pole of $G(z)H(z)$ at $z = 1$ is



(c)

Figure P11.48 *Continued*

inside the modified contour, it is not included in counting the poles of $G(z)H(z)$ outside the unit circle. That is, only poles *strictly outside* the unit circle are counted in applying the Nyquist criterion. Thus, in this case, since $G(z)H(z)$ has no poles strictly outside the unit circle, we must have no encirclements of the point $z = -1/K$ for closed-loop stability.]

(f) Follow the steps outlined in parts (a), (b), and (d) to sketch the Nyquist plots for each of the following:

(i) $\frac{z+\frac{1}{2}+\sqrt{3}}{z-1}$

(ii) $\frac{1}{(z-1)(z+\frac{1}{2}+\sqrt{3})}$

(iii) $\frac{z+1}{z(z-1)}$

(iv) $\frac{z-1/\sqrt{3}}{(z-1)^2}$ [be careful in calculating $\sphericalangle G(z)H(z)$ along the infinitesimal semicircle]

For each of the preceding, use the Nyquist criterion to determine the range of values of K (if any such range exists) for which the closed-loop system is stable. Also, use another method (root locus or direct calculation of the closed-loop poles as a function of K) to provide a partial check of the correctness of your Nyquist plot. *Note:* In sketching the Nyquist plots, you may find it useful to first sketch the magnitude and phase plots as a function of frequency or at least calculate $|G(e^{j\omega})H(e^{j\omega})|$ and $\sphericalangle G(e^{j\omega})H(e^{j\omega})$ at several points. Also, it is helpful to determine the values of ω for which $G(e^{j\omega})H(e^{j\omega})$ is real.

(g) Repeat part (f) for

$$G(z)H(z) = \frac{1}{z^2 - 1}.$$

In this case there are two poles on the unit circle, and thus, you must modify the contour around each of these by including an infinitesimal semicircle that extends outside the unit circle, thereby placing the pole inside the contour.

EXTENSION PROBLEMS

11.49. In this problem, we provide an illustration of how feedback can be used to increase the bandwidth of an amplifier. Consider an amplifier whose gain falls off at high frequencies. That is, suppose the system function of this amplifier is

$$H(s) = \frac{Ga}{s + a}.$$

(a) What is the dc gain of the amplifier (i.e., the magnitude of its frequency response at 0 frequency)?

(b) What is the system time constant?

(c) Suppose we define the bandwidth of the system as the frequency at which the magnitude of the amplifier frequency response is $1/\sqrt{2}$ times its magnitude at dc. What is the bandwidth of the amplifier?

(d) Suppose we place the amplifier in a feedback loop as depicted in Figure P11.49. What is the dc gain of the closed-loop system? What are the time constant and the bandwidth of the closed-loop system?

(e) Find the value of K that leads to a closed-loop bandwidth that is exactly double the bandwidth of the open-loop amplifier. What are the corresponding closed-loop system time constant and dc gain?

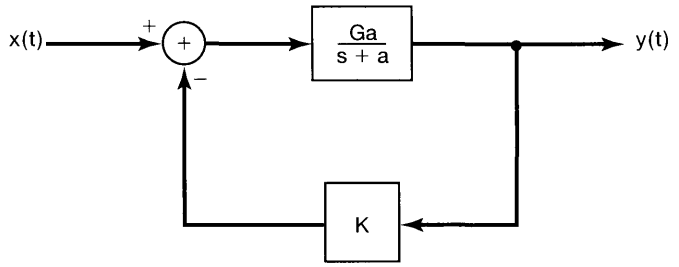


Figure P11.49

11.50. As mentioned in the text, an important class of devices used in the implementation of feedback systems is the class of operational amplifiers. A model for such an amplifier is depicted in Figure P11.50(a). The amplifier's input is the difference

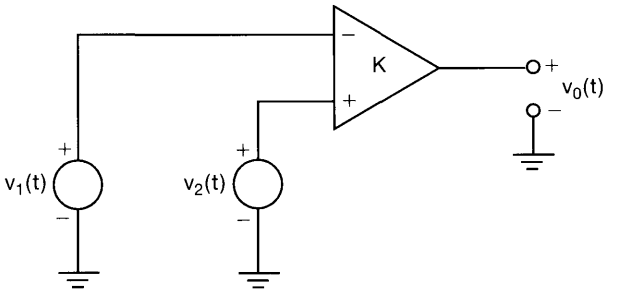


Figure P11.50a

between two voltages $v_2(t)$ and $v_1(t)$, and the output voltage is an amplified version of the input; that is,

$$v_o(t) = K[v_2(t) - v_1(t)]. \qquad \text{(P11.50–1)}$$

Consider an operational amplifier connection shown in Figure P11.50(b). In this figure, $Z_1(s)$ and $Z_2(s)$ are impedances. (That is, each is the system function of an LTI system whose input is the current flowing through the impedance element and whose output is the voltage across the element.) Making the approximation that the input impedance of the operational amplifier is infinite and that its output impedance is zero, we obtain the following relationship between $V_1(s)$, $V_i(s)$, and $V_o(s)$, the Laplace transforms of $v_1(t)$, $v_i(t)$, and $v_o(t)$, respectively:

$$V_1 = \left[\frac{Z_2(s)}{Z_1(s) + Z_2(s)}\right] V_i(s) + \left[\frac{Z_1(s)}{Z_1(s) + Z_2(s)}\right] V_o(s). \qquad \text{(P11.50–2)}$$

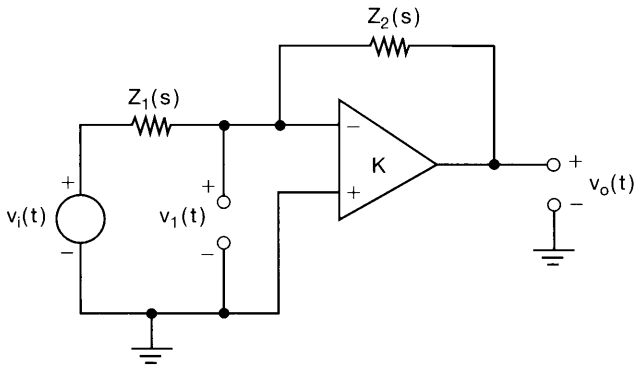


Figure P11.50b

Also, from eq. (P11.50–1) and Figure P11.50(b), we see that

$$V_o(s) = -KV_1(s). \tag{P11.50–3}$$

(a) Show that the system function

$$H(s) = \frac{V_o(s)}{V_i(s)}$$

for the interconnection of Figure P11.50(b) is identical to the overall closed-loop system function for the system of Figure P11.50(c).

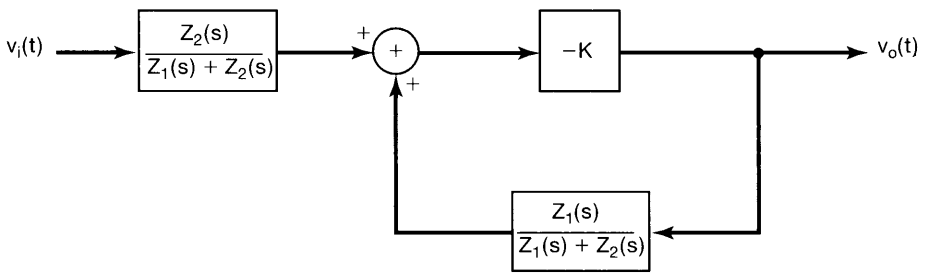


Figure P11.50c

(b) Show that if $K >> 1$, then

$$H(s) \approx -\frac{Z_2(s)}{Z_1(s)}.$$

11.51. (a) Suppose that in Figure P11.50(b) $Z_1(s)$ and $Z_2(s)$ are both pure resistances, say, R_1 and R_2, respectively. A typical value for R_2/R_1 is in the range 1 to 10^3, while a typical value for K is 10^6. Using the results of Problem 11.50(a), calculate the actual system function for this value of K and for R_2/R_1 equal to 1 and then to 10^3, and compare each resulting value to $-R_2/R_1$. This should give you some idea of how good the approximation of Problem 11.50(b) typically is.

(b) One of the important uses of feedback is in the reduction of system sensitivity to variations in parameters. This is particularly important for circuits involving operational amplifiers, which have high gains that may be known only approximately.

(i) Consider the circuit discussed in part (a), with $R_2/R_1 = 10^2$. What is the percentage change in the closed-loop gain of the system if K changes from 10^6 to 5×10^5?

(ii) How large must K be so that a 50% reduction in its value results in only a 1% reduction in the closed-loop gain? Again, take $R_2/R_1 = 10^2$.

11.52. Consider the circuit of Figure P11.52. This circuit is obtained by using

$$Z_1(s) = R, \qquad Z_2(s) = \frac{1}{CS}$$

in Figure P11.50(b). Using the results from Problem 11.50, show that the system behaves approximately like an integrator. In what frequency range (expressed in terms of K, R, and C) does this approximation break down?

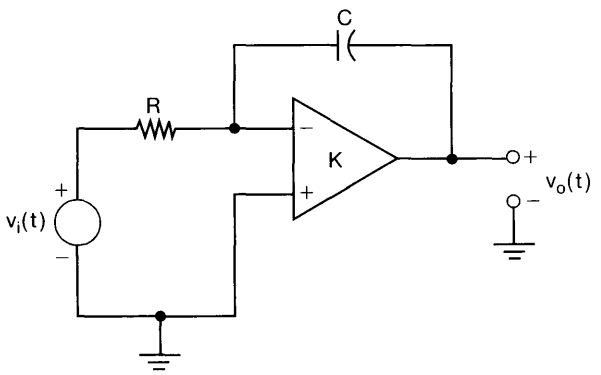


Figure P11.52

11.53. Consider the circuit depicted in Figure P11.53(a), which is obtained from the circuit of Figure P11.50(b) by using $Z_1(s) = R$ and by replacing $Z_2(s)$ with a diode that has an exponential current-voltage relationship. Assume that this relationship is of the form

$$i_d(t) = Me^{qv_d(t)/kT}, \tag{P11.53–1}$$

where M is a constant that depends upon the construction of the diode, q is the charge of an electron, k is Boltzmann's constant, and T is absolute temperature. Note that the idealized relationship of eq. (P11.53–1) assumes that there is no possibility of a negative diode current. Usually, there *is* some small maximum negative value of diode current, but we will neglect this possibility in our analysis.

(a) Assuming that the input impedance of the operational amplifier is infinite and that its output impedance is zero, show that the following relations hold:

$$v_o(t) = v_d(t) + Ri_d(t) + v_i(t), \tag{P11.53–2}$$

$$v_o(t) = -K[v_o(t) - v_d(t)]. \tag{P11.53–3}$$

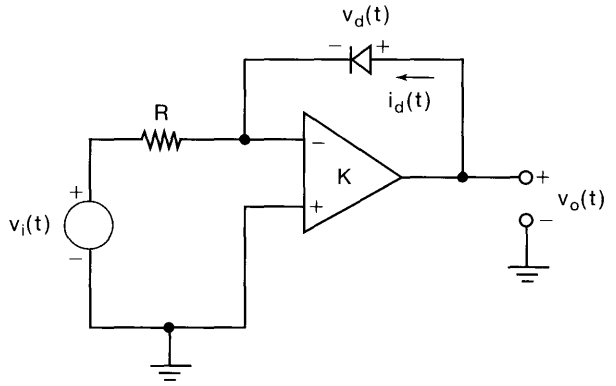


Figure P11.53a

(b) Show that for K large, the relationship between $v_o(t)$ and $v_i(t)$ is essentially the same as in the feedback system of Figure P11.53(b), in which the system in the feedback path is a nonlinear memoryless system with input $v_o(t)$ and output

$$w(t) = RMe^{qv_o(t)/kT}.$$

(c) Show that for K large,

$$v_o(t) \approx \frac{kT}{q} \ln\left(-\frac{v_i(t)}{RM}\right). \qquad \text{(P11.53–4)}$$

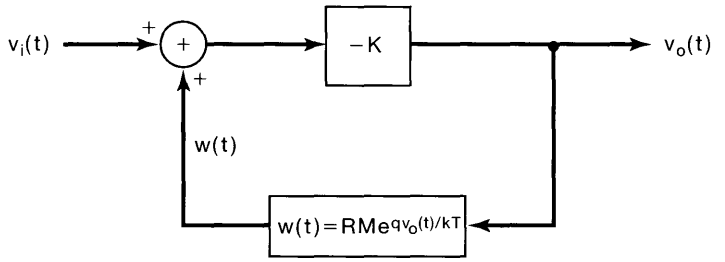


Figure P11.53b

Note that eq. (P11.53–4) makes sense only for a negative $v_i(t)$, which is consistent with the requirement that the diode current cannot be negative. If a positive $v_i(t)$ is applied, the current $i_d(t)$ cannot balance the current through the resistor. Thus, a nonnegligible current is fed into the amplifier, causing it to saturate.

11.54. In this problem, we explore the use of positive feedback for generating oscillating signals.

(a) Consider the system illustrated in Figure P11.54(a). Show that $x_f(t) = x_i(t)$ if

$$G(s)H(s) = -1. \qquad \text{(P11.54–1)}$$

Suppose that we connect terminals 1 and 2 in Figure P11.54(a) and make $x_i(t) = 0$. Then the output of the system should remain unchanged if we

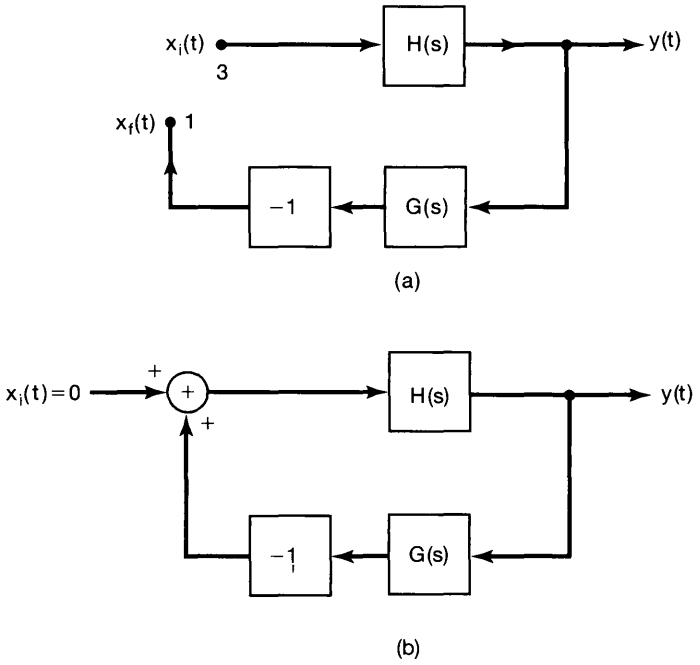


Figure P11.54

satisfy eq. (P11.54–1). The system now produces an output without any input. Therefore, the system shown in Figure P11.54(b) is an oscillator, provided that eq. (P11.54–1) is satisfied.

(b) A commonly used oscillator in practice is the sinusoidal oscillator. For such an oscillator, we may rewrite the condition of eq. (P11.54–1) as

$$G(j\omega_0)H(j\omega_0) = -1. \qquad \text{(P11.54–2)}$$

What is the value of the closed-loop gain for the system shown in Figure P11.54(b) at ω_0 when eq. (P11.54–2) is satisfied?

(c) A sinusoidal oscillator may be constructed on the basis of the principle outlined above by using the circuit shown in Figure P11.54(c). The input to the

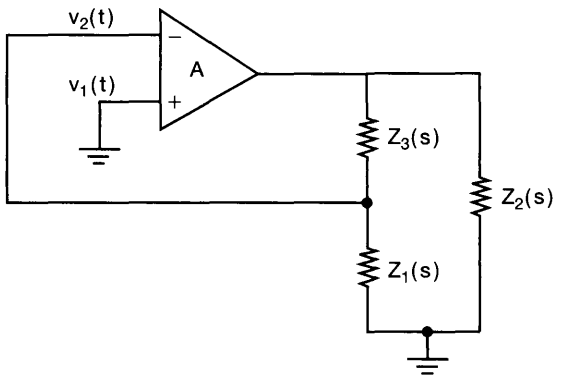


Figure P11.54c

amplifier is the difference between the voltages $v_1(t)$ and $v_2(t)$. In this circuit, the amplifier has a gain of A and an output resistance of R_0. $Z_1(s)$, $Z_2(s)$, and $Z_3(s)$ are impedances. (That is, each is the system function of an LTI system whose input is the current flowing through the impedance element and whose output is the voltage across the element.) It can be shown that, for this circuit,

$$H(s) = \frac{-AZ_L(s)}{Z_L(s) + R_0},$$

where

$$Z_L = \frac{Z_2(s)(Z_1(s) + Z_3(s))}{Z_1(s) + Z_2(s) + Z_3(s)}.$$

Also, we can show that

$$G(s) = \frac{-Z_1(s)}{Z_1(s) + Z_3(s)}.$$

(i) Show that

$$G(s)H(s) = \frac{AZ_1(s)Z_2(s)}{R_0(Z_1(s) + Z_2(s) + Z_3(s)) + Z_2(s)(Z_1(s) + Z_3(s))}.$$

(ii) If $Z_1(s)$, $Z_2(s)$, and $Z_3(s)$ are pure reactances (i.e., inductances or capacitances), we can write $Z_1(j\omega) = jX_1(j\omega)$, $Z_2(j\omega) = jX_2(j\omega)$, and $Z_3(j\omega) = jX_3(j\omega)$, where $X_i(j\omega)$, $i = 1, 2, 3$, are all real. Using the results of parts (b) and (i), show that a necessary condition for the circuit to produce oscillations is

$$X_1(j\omega) + X_2(j\omega) + X_3(j\omega) = 0.$$

(iii) Show also that, in addition to the constraint of part (ii), the constraint $AX_1(j\omega) = X_2(j\omega)$ has to be satisfied for the circuit to produce oscillations. [Since $X_i(j\omega)$ is positive for inductances and negative for capacitances, the latter constraint requires that $Z_1(s)$ and $Z_2(s)$ be reactances of the same type (i.e., both should be inductances or both should be capacitances).]

(iv) Let us assume that $Z_1(s)$ and $Z_2(s)$ are both inductances such that

$$X_1(j\omega) = X_2(j\omega) = \omega L.$$

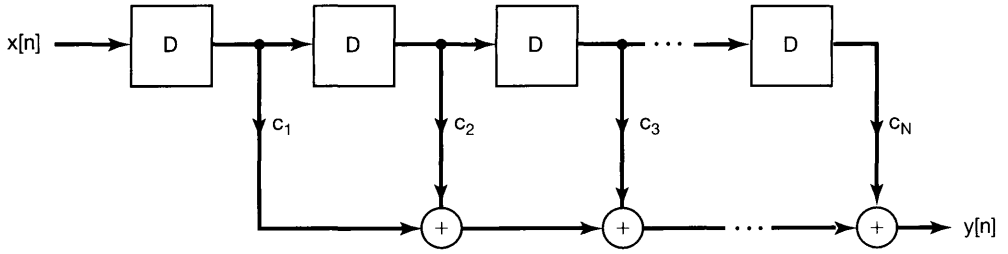
Let us also assume that

$$X_3(j\omega) = -1/(\omega C)$$

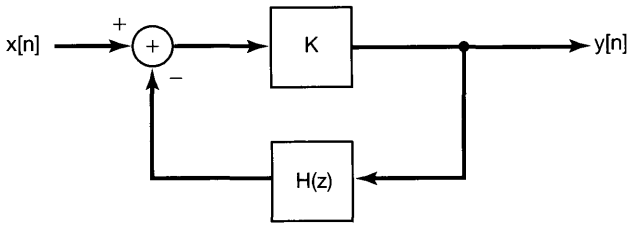
is a capacitance. Use the condition derived in (ii) to determine the frequency (in terms of L and C) at which the circuit oscillates.

11.55. (a) Consider the nonrecursive discrete-time LTI filter depicted in Figure P11.55(a). Through the use of feedback around this nonrecursive system, a recursive filter can be implemented. To do so, consider the configuration shown in Figure P11.55(b), in which $H(z)$ is the system function of the nonrecursive LTI system of Figure P11.55(a). Determine the overall system function of this feedback

system, and find the difference equation relating the input to the output of the overall system.



(a)



(b)

Figure P11.55

(b) Now suppose that $H(z)$ in Figure P11.55(b) is the system function of a recursive LTI system. Specifically, suppose that

$$H(z) = \frac{\sum_{i=1}^{N} c_i z^{-i}}{\sum_{i=1}^{N} d_i z^{-i}}.$$

Show how one can find values of the coefficients $K, c_1, \ldots, c_N$, and $d_0, \ldots, d_N$, such that the closed-loop system function is

$$Q(z) = \frac{\sum_{i=0}^{N} b_i z^{-i}}{\sum_{i=0}^{N} a_i z^{-i}},$$

where the a_i and b_i are specified coefficients.

In this problem, we have seen that the use of feedback provides us with alternative implementations of LTI systems specified by linear constant-coefficient difference equations. The implementation in part (a), consisting of feedback around a nonrecursive system, is particularly interesting, as some technologies are ideally suited to implementing tapped delay-line structures (i.e., systems consisting of chains of delays with taps at each delay whose outputs are weighted and then summed).

11.56. Consider an inverted pendulum mounted on a movable cart, as depicted in Figure P11.56. Here, we have modeled the pendulum as consisting of a massless rod of length L with a mass m attached at the end. The variable $\theta(t)$ denotes the pendulum's angular deflection from the vertical, g is gravitational acceleration, $s(t)$ is the position of the cart with respect to some reference point, $a(t)$ is the acceleration of the cart, and $x(t)$ represents the angular acceleration resulting from any disturbances, such as gusts of wind.

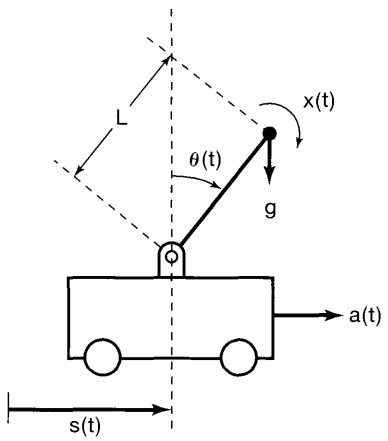


Figure P11.56

Our goal in this problem is to analyze the dynamics of the inverted pendulum and, more specifically, to investigate the problem of balancing the pendulum by a judicious choice of the acceleration $a(t)$ of the cart. The differential equation relating $\theta(t)$, $a(t)$, and $x(t)$ is

$$L\frac{d^2\theta(t)}{dt^2} = g\sin[\theta(t)] - a(t)\cos[\theta(t)] + Lx(t). \qquad \text{(P11.56–1)}$$

This relation merely equates the actual acceleration of the mass along a direction perpendicular to the rod to the applied accelerations [gravity, the disturbance acceleration due to $x(t)$, and the cart's acceleration] along this direction.

Note that eq. (P11.56–1) is a nonlinear differential equation. The detailed, exact analysis of the behavior of the pendulum requires that we examine this equation; however, we can obtain a great deal of insight into the dynamics of the pendulum by performing a linearized analysis. Specifically, let us examine the dynamics of the pendulum when it is nearly vertical [i.e., when $\theta(t)$ is small]. In this case, we

can make the approximations

$$\sin[\theta(t)] \approx \theta(t), \cos[\theta(t)] \approx 1. \tag{P11.56–2}$$

(a) Suppose that the cart is stationary [i.e., $a(t) = 0$], and consider the causal LTI system with input $x(t)$ and output $\theta(t)$ described by eq. (P11.56–1), together with the approximations given in eq. (P11.56–2). Find the system function for this system, and show that it has a pole in the right-half of the plane, implying that the system is unstable.

(b) The result of part (a) indicates that if the cart is stationary, any minor angular disturbance caused by $x(t)$ will lead to growing angular deviations from the vertical. Clearly, at some point, these deviations will become sufficiently large so that the approximations of eq. (P11.56–2) will no longer be valid. At this point the linearized analysis is no longer accurate, but the fact that it is accurate for small angular displacements allows us to conclude that the vertical equilibrium position is unstable, since small angular displacements will grow rather than diminish.

We now wish to consider the problem of stabilizing the vertical position of the pendulum by moving the cart in an appropriate fashion. Suppose we try proportional feedback—that is,

$$a(t) = K\theta(t).$$

Assume that $\theta(t)$ is small, so that the approximations in eq. (P11.56–2) are valid. Draw a block diagram of the linearized system with $\theta(t)$ as the output, $x(t)$ as the external input, and $a(t)$ as the signal that is fed back. Show that the resulting closed-loop system is unstable. Find a value of K such that if $x(t) = \delta(t)$, the pendulum will sway back and forth in an undamped oscillatory fashion.

(c) Consider using the proportional-plus-derivative (PD) feedback,

$$a(t) = K_1\theta(t) + K_2\frac{d\theta(t)}{dt}.$$

Show that one can find values of K_1 and K_2 that stabilize the pendulum. In fact, using

$$g = 9.8 \text{ m/sec}^2$$

and (P11.56–3)

$$L = 0.5 \text{ m},$$

choose values of K_1 and K_2 so that the damping ratio of the closed loop system is 1 and the natural frequency is 3 rad/sec.

11.57. In this problem, we consider several examples of the design of tracking systems. Consider the system depicted in Figure P11.57. Here, $H_p(s)$ is the system whose output is to be controlled, and $H_c(s)$ is the compensator to be designed. Our objective in choosing $H_c(s)$ is that we would like the output $y(t)$ to follow the input $x(t)$. In particular, in addition to stabilizing the system, we would also like to design the system so that the error $e(t)$ decays to zero for certain specified inputs.

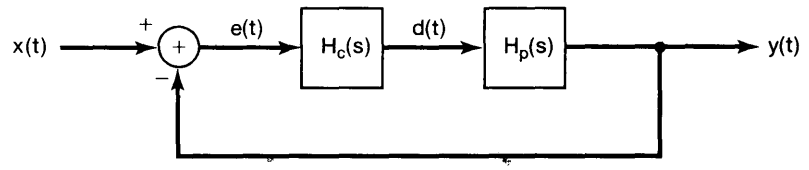


Figure P11.57

(a) Suppose that

$$H_p(s) = \frac{\alpha}{s + \alpha}, \quad \alpha \neq 0. \tag{P11.57–1}$$

Show that if $H_c(s) = K$ (which is known as *proportional* or *P* control), we can choose K so as to stabilize the system and so that $e(t) \to 0$ if $x(t) = \delta(t)$. Show that we *cannot* get $e(t) \to 0$ if $x(t) = u(t)$.

(b) Again let $H_p(s)$ be as in eq. (P11.57–1), and suppose that we use *proportional-plus-integral* (PI) control—that is,

$$H_c(s) = K_1 + \frac{K_2}{s}.$$

Show that we can choose K_1 and K_2 so as to stabilize the system, and we can also get $e(t) \to 0$ if $x(t) = u(t)$. Thus, the system can track a step. In fact, this illustrates a basic and important principle in feedback system design: To track a step [$X(s) = 1/s$], we need an integrator ($1/s$) in the feedback system. An extension of this principle is considered in the next problem.

(c) Suppose that

$$H_p(s) = \frac{1}{(s-1)^2}.$$

Show that we cannot stabilize this system with a PI controller, but that we can stabilize it and have it track a step if we use *proportional-plus-integral-plus-differential* (PID) control, i.e.,

$$H_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

11.58. In Problem 11.57, we discussed how the presence of an integrator in a feedback system can make it possible for the system to track a step input with zero error in the steady state. In this problem, we extend the idea. Consider the feedback system depicted in Figure P11.58, and suppose that the overall closed-loop system is stable. Suppose also that

$$H(s) = \frac{K\prod_{k=1}^{m}(s - \beta_k)}{s^l\prod_{k=1}^{n-l}(s - \alpha_k)},$$

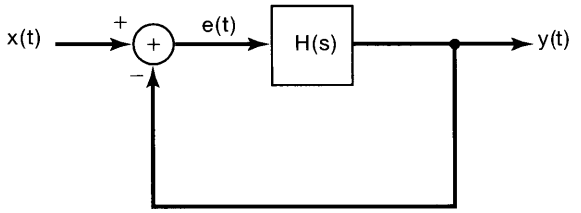


Figure P11.58

where the α_k and β_k are given nonzero numbers and l is a positive integer. The feedback system of Figure P11.58 is often referred to as a *Type l* feedback system.

(a) Use the final-value theorem (Section 9.5.10) to show that a Type 1 feedback system can track a step—that is, that

$$e(t) \to 0 \qquad \text{if} \quad x(t) = u(t).$$

(b) Similarly, show that a Type 1 system cannot track a ramp, but rather, that

$$e(t) \to \text{a finite constant if} \qquad x(t) = u_{-2}(t).$$

(c) Show that, for a Type 1 system, unbounded results ensue if

$$x(t) = u_{-k}(t)$$

with $k > 2$.

(d) More generally, show that, for a Type l system:
(i) $e(t) \to 0$ if $x(t) = u_{-k}(t)$ with $k \leq l$
(ii) $e(t) \to$ a finite constant if $x(t) = u_{(-l+1)}(t)$
(iii) $e(t) \to \infty$ if $x(t) = u_{-k}(t)$ with $k > l + 1$

11.59. (a) Consider the discrete-time feedback system of Figure P11.59. Suppose that

$$H(z) = \frac{1}{(z-1)(z+\frac{1}{2})}.$$

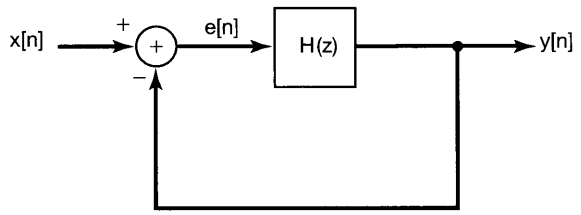


Figure P11.59

Show that this system can track a unit step in the sense that if $x[n] = u[n]$, then

$$\lim_{n\to\infty} e[n] = 0. \qquad \text{(P11.59–1)}$$

(b) More generally, consider the feedback system of Figure P11.59, and assume that the closed-loop system is stable. Suppose that $H(z)$ has a pole at $z = 1$.

Show that the system can track a unit step. [*Hint:* Express the transform $E(z)$ of $e[n]$ in terms of $H(z)$ and the transform of $u[n]$; explain why all the poles of $E(z)$ are inside the unit circle.]

(c) The results of parts (a) and (b) are discrete-time counterparts of the results for continuous-time systems discussed in Problems 11.57 and 11.58. In discrete time, we can also consider the design of the systems that track specified inputs *perfectly* after a finite number of steps. Such systems are known as *deadbeat feedback systems.*

Consider the discrete-time system of Figure P11.59 with

$$H(z) = \frac{z^{-1}}{1 - z^{-1}}.$$

Show that the overall closed-loop system is a deadbeat feedback system with the property that it tracks a step input exactly after one step: that is, if $x[n] = u[n]$, then $e[n] = 0$, $n \geq 1$.

(d) Show that the feedback system of Figure P11.59 with

$$H(z) = \frac{\frac{3}{4}z^{-1} + \frac{1}{4}z^{-2}}{(1 + \frac{1}{4}z^{-1})(1 - z^{-1})}$$

is a deadbeat system with the property that the output tracks a unit step perfectly after a finite number of steps. At what time step does the error $e[n]$ first settle to zero?

(e) More generally, for the feedback system of Figure P11.59, find $H(z)$ so that $y[n]$ perfectly tracks a unit step for $n \geq N$ and, in fact, so that

$$e[n] = \sum_{k=0}^{N-1} a_k \delta[n - k], \tag{P11.59–2}$$

where the a_i are specified constants. *Hint:* Use the relationship between $H(z)$ and $E(z)$ when the input is a unit step and $e[n]$ is given by eq. (P11.59–2).

(f) Consider the system of Figure P11.59 with

$$H(z) = \frac{z^{-1} + z^{-2} - z^{-3}}{(1 + z^{-1})(1 - z^{-1})^2}.$$

Show that this system tracks a ramp $x[n] = (n+1)u[n]$ exactly after two time steps.

11.60. In this problem, we investigate some of the properties of sampled-data feedback systems and illustrate the use of such systems. Recall from Section 11.2.4 that in a sampled-data feedback system the output of a continuous-time system is sampled. The resulting sequence of samples is processed by a discrete-time system, the output of which is converted to a continuous-time signal that in turn is fed back and subtracted from the external input to produce the actual input to the continuous-time system.

(a) Consider the system within dashed lines in Figure 11.6(b). This is a discrete-time system with input $e[n]$ and output $p[n]$. Show that it is an LTI system. As we have indicated in the figure, we will let $F(z)$ denote the system function of this system.

(b) Show that in Figure 11.6(b) the discrete-time system with system function $F(z)$ is related to the continuous-time system with system function $H(s)$ by means of a *step-invariant* transformation. That is, if $s(t)$ is the step response of the continuous-time system and $q[n]$ is the step response of the discrete-time system, then

$$q[n] = s(nT) \qquad \text{for all } n.$$

(c) Suppose that

$$H(s) = \frac{1}{s-1}, \mathcal{R}e\{s\} > 1.$$

Show that

$$F(z) = \frac{(e^T - 1)z^{-1}}{1 - e^T z^{-1}}, \quad |z| > e^T.$$

(d) Suppose that $H(s)$ is as in part (c) and that $G(z) = K$. Find the range of values of K for which the closed-loop discrete-time system of Figure 11.6(b) is stable.

(e) Suppose that

$$G(z) = \frac{K}{1 + \frac{1}{2}z^{-1}}.$$

Under what conditions on T can we find a value of K that stabilizes the overall system? Find a particular pair of values for K and T that yield a stable closed-loop system. *Hint:* Examine the root locus, and find the values for which the poles enter or leave the unit circle.

APPENDIX
PARTIAL-FRACTION EXPANSION

A.1 INTRODUCTION

The purpose of this appendix is to describe the technique of partial-fraction expansion. This tool is of great value in the study of signals and systems; in particular, it is very useful in inverting Fourier, Laplace, or z-transforms and in analyzing LTI systems described by linear constant-coefficient differential or difference equations. The method of partial-fraction expansion consists of taking a function that is the ratio of polynomials and expanding it as a linear combination of simpler terms of the same type. The determination of the coefficients in the linear combination is the basic problem to be solved in obtaining the expansion. As we will see, this is a relatively straightforward problem in algebra that can be solved very efficiently with a bit of "bookkeeping."

To illustrate the basic idea behind and role of partial-fraction expansion, consider the analysis developed in Section 6.5.2 for a second-order continuous-time LTI system specified by the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 x(t). \tag{A.1}$$

The frequency response of this system is

$$H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}, \tag{A.2}$$

or, if we factor the denominator,

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_1)(j\omega - c_2)}, \tag{A.3}$$

where

$$\begin{aligned} c_1 &= -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}, \\ c_2 &= -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}. \end{aligned} \tag{A.4}$$

Having $H(j\omega)$, we are in a position to answer a variety of questions related to the system. For example, to determine the impulse response of the system, recall that for any number α with $\mathcal{R}e\{s\} < 0$, the Fourier transform of

$$x_1(t) = e^{\alpha t}u(t) \tag{A.5}$$

is

$$X_1(j\omega) = \frac{1}{j\omega - \alpha}, \tag{A.6}$$

while if

$$x_2(t) = te^{\alpha t}u(t), \tag{A.7}$$

then

$$X_2(j\omega) = \frac{1}{(j\omega - \alpha)^2}. \tag{A.8}$$

Therefore, if we can expand $H(j\omega)$ as a sum of terms of the form of eq. (A.6) or (A.8), we can determine the inverse transform of $H(j\omega)$ by inspection. For example, in Section 6.5.2 we noted that when $c_1 \neq c_2$, $H(j\omega)$ in eq. (A.3) could be rewritten in the form

$$H(j\omega) = \left(\frac{\omega_n^2}{c_1 - c_2}\right)\frac{1}{j\omega - c_1} + \left(\frac{\omega_n^2}{c_2 - c_1}\right)\frac{1}{j\omega - c_2}. \tag{A.9}$$

In this case, the Fourier transform pair of eqs. (A.5) and (A.6) allows us to write down immediately the inverse transform of $H(j\omega)$ as

$$h(t) = \left[\frac{\omega_n^2}{c_1 - c_2}e^{c_1 t} + \frac{\omega_n^2}{c_2 - c_1}e^{c_2 t}\right]u(t). \tag{A.10}$$

While we have phrased the preceding discussion in terms of continuous-time Fourier transforms, similar concepts also arise in discrete-time Fourier analysis and in the use of Laplace and z-transforms. In all of these cases, we encounter the important class of *rational transforms*—that is, transforms that are ratios of polynomials in some variable. Also, in each of these contexts, we find reasons for expanding these transforms as sums of simpler terms such as in eq. (A.9). In this section, in order to develop a general procedure for calculating the expansions, we consider rational functions of a general variable v; that is, we examine functions of the form

$$H(v) = \frac{\beta_m v^m + \beta_{m-1}v^{m-1} + \ldots + \beta_1 v + \beta_0}{\alpha_n v^n + \alpha_{n-1}v^{n-1} + \ldots + \alpha_1 v + \alpha_0}. \tag{A.11}$$

For continuous-time Fourier analysis ($j\omega$) plays the role of v, while for Laplace transforms that role is played by the complex variable s. In discrete-time Fourier analysis, v is usually taken to be $e^{-j\omega}$, while for z-transforms, we can use either z^{-1} or z. After we have developed the basic techniques of partial-fraction expansion, we will illustrate their application to the analysis of both continuous-time and discrete-time LTI systems.

A.2 PARTIAL-FRACTION EXPANSION AND CONTINOUS-TIME SIGNALS AND SYSTEMS

For our purposes, it is convenient to consider rational functions in one of two standard forms. The second of these, which is often useful in the analysis of discrete-time signals and systems, will be discussed shortly. The first of the standard forms is

$$G(v) = \frac{b_{n-1}v^{n-1} + b_{n-2}v^{n-2} + \ldots + b_1 v + b_0}{v^n + a_{n-1}v^{n-1} + \ldots + a_1 v + a_0}. \tag{A.12}$$

In this form the coefficient of the highest order term in the denominator is 1, and the order of the numerator is at least one less than the order of the denominator. (The order of the numerator will be less than $n-1$ if $b_{n-1}=0$.)

If we are given $H(v)$ in the form of eq. (A.11), we can obtain a rational function of the form of eq. (A.12) by performing two straightforward calculations. First, we divide both the numerator and the denominator of $H(v)$ by α_n. This yields

$$H(v)=\frac{\gamma_m v^m+\gamma_{m-1}v^{m-1}+\ldots+\gamma_1 v+\gamma_0}{v^n+a_{n-1}v^{n-1}+\ldots+a_1 v+a_0}, \tag{A.13}$$

where

$$\gamma_m=\frac{\beta_m}{\alpha_n}, \qquad \gamma_{m-1}=\frac{\beta_{m-1}}{\alpha_n}, \qquad \ldots,$$
$$a_{n-1}=\frac{\alpha_{n-1}}{\alpha_n}, \qquad a_{n-2}=\frac{\alpha_{n-2}}{\alpha_n}, \qquad \ldots.$$

If $m<n$, $H(v)$ is called a *strictly proper* rational function, and in this case, letting $b_0=\gamma_0, b_1=\gamma_1, \ldots, b_m=\gamma_m$, and setting any remaining b's equal to zero, we see that $H(v)$ in eq. (A.13) is already of the form of eq. (A.12). In most of the discussions in this book in which rational functions are considered, we are concerned primarily with strictly proper rational functions. However, if $H(v)$ is not proper (i.e., if $m \geq n$), we can perform a preliminary calculation that allows us to write $H(v)$ as the sum of a polynomial in v and a strictly proper rational function. That is,

$$\begin{aligned} H(v)=&\, c_{m-n}v^{m-n}+c_{m-n-1}v^{m-n-1}+\ldots+c_1 v+c_0 \\ &+\frac{b_{n-1}v^{n-1}+b_{n-2}v^{n-2}+\ldots+b_1 v+b_0}{v^n+a_{n-1}v^{n-1}+\ldots+a_1 v+a_0}. \end{aligned} \tag{A.14}$$

The coefficients $c_0, c_1, \ldots, c_{m-n}$ and $b_0, b_1, \ldots, b_{n-1}$ can be obtained by equating eqs. (A.13) and (A.14) and then multiplying through by the denominator. This yields

$$\begin{aligned} \gamma_m v^m+\ldots+\gamma_1 v+\gamma_0 =&\, b_{n-1}v^{n-1}+\ldots+b_1 v+b_0 \\ &+(c_{m-n}v^{m-n}+\ldots+c_0)(v^n+a_{n-1}v^{n-1}+\ldots+a_0). \end{aligned} \tag{A.15}$$

By equating the coefficients of equal powers of v on both sides of eq. (A.15), we can determine the c's and b's in terms of the a's and γ's. For example, if $m=2$ and $n=1$, so that

$$H(v)=\frac{\gamma_2 v^2+\gamma_1 v+\gamma_0}{v+a_1}=c_1 v+c_0+\frac{b_0}{v+a_1}, \tag{A.16}$$

then eq. (A.15) becomes

$$\begin{aligned} \gamma_2 v^2+\gamma_1 v+\gamma_0 &= b_0+(c_1 v+c_0)(v+a_1) \\ &= b_0+c_1 v^2+(c_0+a_1 c_1)v+a_1 c_0. \end{aligned}$$

Equating the coefficients of equal powers of v, we obtain the equations

$$\begin{aligned} \gamma_2 &= c_1, \\ \gamma_1 &= c_0+a_1 c_1, \\ \gamma_0 &= b_0+a_1 c_0. \end{aligned}$$

The first equation yields the value of c_1, which can then be used in the second to solve for c_0, which in turn can be used in the third to solve for b_0. The result is

$$\begin{aligned} c_1 &= \gamma_2, \\ c_0 &= \gamma_1 - a_1\gamma_2, \\ b_0 &= \gamma_0 - a_1(\gamma_1 - a_1\gamma_2). \end{aligned}$$

The general case of eq. (A.15) can be solved in an analogous fashion.

Our goal now is to focus on the proper rational function $G(v)$ in eq. (A.12) and to expand it into a sum of simpler proper rational functions. To see how this can be done, consider the case of $n = 3$, so that eq. (A.12) reduces to

$$G(v) = \frac{b_2v^2 + b_1v + b_0}{v^3 + a_2v^2 + a_1v + a_0}. \tag{A.17}$$

As a first step, we factor the denominator of $G(v)$ in order to write it in the form

$$G(v) = \frac{b_2v^2 + b_1v + b_0}{(v - \rho_1)(v - \rho_2)(v - \rho_3)}. \tag{A.18}$$

Assuming for the moment that the roots ρ_1, ρ_2, and ρ_3 of the denominator are all distinct, we would like to expand $G(v)$ into a sum of the form

$$G(v) = \frac{A_1}{v - \rho_1} + \frac{A_2}{v - \rho_2} + \frac{A_3}{v - \rho_3}. \tag{A.19}$$

The problem, then, is to determine the constants A_1, A_2, and A_3. One approach is to equate eqs. (A.18) and (A.19) and to multiply through the denominator. In this case, we obtain the equation

$$\begin{aligned} b_2v^2 + b_1v + b_0 = {} & A_1(v - \rho_2)(v - \rho_3) \\ & + A_2(v - \rho_1)(v - \rho_3) \\ & + A_3(v - \rho_1)(v - \rho_2). \end{aligned} \tag{A.20}$$

By expanding the right-hand side of eq. (A.20) and then equating coefficients of equal powers of v, we obtain a set of linear equations that can be solved for A_1, A_2, and A_3.

Although this approach always works, there is a much easier method. Consider eq. (A.19), and suppose that we would like to calculate A_1. Then, multiplying through by $v - \rho_1$, we obtain

$$(v - \rho_1)G(v) = A_1 + \frac{A_2(v - \rho_1)}{v - \rho_2} + \frac{A_3(v - \rho_1)}{v - \rho_3}. \tag{A.21}$$

Since ρ_1, ρ_2, and ρ_3 are distinct, the last two terms on the right-hand side of eq. (A.21) are zero for $v = \rho_1$. Therefore,

$$A_1 = [(v - \rho_1)G(v)]|_{v=\rho_1}, \tag{A.22}$$

or, using eq. (A.18),

$$A_1 = \frac{b_2\rho_1^2 + b_1\rho_1 + b_0}{(\rho_1 - \rho_2)(\rho_1 - \rho_3)}. \tag{A.23}$$

Similarly,

$$A_2 = [(v - \rho_2)G(v)]|_{v=\rho_2} = \frac{b_2\rho_2^2 + b_1\rho_2 + b_0}{(\rho_2 - \rho_1)(\rho_2 - \rho_3)}, \tag{A.24}$$

$$A_3 = [(v - \rho_3)G(v)]|_{v=\rho_3} = \frac{b_2\rho_3^2 + b_1\rho_3 + b_0}{(\rho_3 - \rho_1)(\rho_3 - \rho_2)}. \tag{A.25}$$

Suppose now that $\rho_1 = \rho_3 \neq \rho_2$; that is,

$$G(v) = \frac{b_2v^2 + b_1v + b_0}{(v - \rho_1)^2(v - \rho_2)}. \tag{A.26}$$

In this case, we look for an expansion of the form

$$G(v) = \frac{A_{11}}{v - \rho_1} + \frac{A_{12}}{(v - \rho_1)^2} + \frac{A_{21}}{v - \rho_2}. \tag{A.27}$$

Here, we need the $1/(v - \rho_1)^2$ term in order to obtain the correct denominator in eq. (A.26) when we collect terms over a least common denominator. We also need to include the $1/(v - \rho_1)$ term in general. To see why this is so, consider equating eqs. (A.26) and (A.27) and multiplying them through by the denominator of eq. (A.26):

$$\begin{aligned} b_2v^2 + b_1v + b_0 = {} & A_{11}(v - \rho_1)(v - \rho_2) \\ & + A_{12}(v - \rho_2) + A_{21}(v - \rho_1)^2. \end{aligned} \tag{A.28}$$

Again, if we equate coefficients of equal powers of v, we obtain three equations (for the coefficients of the v^0, v^1, and v^2 terms). If we omit the A_{11} term in eq. (A.27), we will then have three equations in two unknowns, which in general will not have a solution. By including this term, we can always find a solution. In this case also, however, there is a much simpler method. Consider eq. (A.27) and multiply through by $(v - \rho_1)^2$:

$$(v - \rho_1)^2G(v) = A_{11}(v - \rho_1) + A_{12} + \frac{A_{21}(v - \rho_1)^2}{v - \rho_2}. \tag{A.29}$$

From the preceding example, we see immediately how to determine A_{12}:

$$A_{12} = [(v - \rho_1)^2G(v)]|_{v=\rho_1} = \frac{b_2\rho_1^2 + b_1\rho_1 + b_0}{\rho_1 - \rho_2}. \tag{A.30}$$

As for A_{11}, suppose that we differentiate eq. (A.29) with respect to v:

$$\frac{d}{dv}[(v - \rho_1)^2G(v)] = A_{11} + A_{21}\left[\frac{2(v - \rho_1)}{v - \rho_2} - \frac{(v - \rho_1)^2}{(v - \rho_2)^2}\right]. \tag{A.31}$$

It is then apparent that the final term in eq. (A.31) is zero for $v = \rho_1$, and therefore,

$$\begin{aligned} A_{11} &= \left[\frac{d}{dv}(v - \rho_1)^2G(v)\right]\Bigg|_{v=\rho_1} \\ &= \frac{2b_2\rho_1 + b_1}{\rho_1 - \rho_2} - \frac{b_2\rho_1^2 + b_1\rho_1 + b_0}{(\rho_1 - \rho_2)^2}. \end{aligned} \tag{A.32}$$

Finally, by multiplying eq. (A.27) by $v - \rho_2$, we find that

$$A_{21} = [(v - \rho_2)G(v)]|_{v=\rho_2} = \frac{b_2\rho_2^2 + b_1\rho_2 + b_0}{(\rho_2 - \rho_1)^2}. \tag{A.33}$$

This example illustrates all of the basic ideas behind partial-fraction expansion in the general case. Specifically, suppose that the denominator of $G(v)$ in eq. (A.12) has distinct roots $\rho_1, \ldots, \rho_\tau$ with *multiplicities* $\sigma_1, \ldots, \sigma_\tau$; that is,

$$G(v) = \frac{b_{n-1}v^{n-1} + \ldots + b_1 v + b_0}{(v - \rho_1)^{\sigma_1}(v - \rho_2)^{\sigma_2} \ldots (v - \rho_r)^{\sigma_r}}. \tag{A.34}$$

In this case, $G(v)$ has a partial-fraction expansion of the form

$$\begin{aligned} G(v) = {} & \frac{A_{11}}{v - \rho_1} + \frac{A_{12}}{(v - \rho_1)^2} + \ldots + \frac{A_{1\sigma_1}}{(v - \rho_1)^{\sigma_1}} \\ & + \frac{A_{21}}{v - \rho_2} + \ldots + \frac{A_{2\sigma_2}}{(v - \rho_2)^{\sigma_2}} \\ & + \ldots + \frac{A_{r1}}{v - \rho_r} + \ldots + \ldots + \frac{A_{r\sigma_r}}{(v - \rho_r)^{\sigma_r}} \\ = {} & \sum_{i=1}^{r} \sum_{k=1}^{\sigma_i} \frac{A_{ik}}{(v - \rho_i)^k}, \end{aligned} \tag{A.35}$$

where the A_{ik} are computed from the equation[1]

$$\boxed{A_{ik} = \frac{1}{(\sigma_i - k)!} \left[\frac{d^{\sigma_i - k}}{dv^{\sigma_i - k}} [(v - \rho_i)^{\sigma_i} G(v)] \right] \Bigg|_{v=\rho_i}} \tag{A.36}$$

This result can be checked much as in the example: Multiply both sides of eq. (A.35) by $(v - \rho_i)^{\sigma_i}$ and differentiate repeatedly, until A_{ik} is no longer multiplied by a power of $v - \rho_i$. Then set $v = \rho_i$.

Example A.1

In Example 4.25, we examine an LTI system described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t). \tag{A.37}$$

The frequency response of this system is

$$H(j\omega) = \frac{j\omega + 2}{(j\omega)^2 + 4j\omega + 3}. \tag{A.38}$$

To determine the impulse response for this system, we expand $H(j\omega)$ into a sum of simpler terms whose inverse transforms can be obtained by inspection. Making the subsitution of v for $j\omega$, we obtain the function

[1]Here, we use the factorial notation $r!$ for the product $r(r-1)(r-2)\ldots 2 \cdot 1$. The quantity 0! is defined to be equal to 1.

$$G(v) = \frac{v+2}{v^2+4v+3} = \frac{v+2}{(v+1)(v+3)}. \tag{A.39}$$

The partial-fraction expansion for $G(v)$ is then

$$G(v) = \frac{A_{11}}{v+1} + \frac{A_{21}}{v+3}, \tag{A.40}$$

where

$$A_{11} = [(v+1)G(v)]\,|_{v=-1} = \frac{-1+2}{-1+3} = \frac{1}{2}, \tag{A.41}$$

$$A_{21} = [(v+3)G(v)]\,|_{v=-3} = \frac{-3+2}{-3+1} = \frac{1}{2}. \tag{A.42}$$

Thus,

$$H(j\omega) = \frac{\frac{1}{2}}{j\omega+1} + \frac{\frac{1}{2}}{j\omega+3}, \tag{A.43}$$

and the impulse response of the system, obtained by inverting eq. (A.43), is

$$h(t) = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-3t}u(t). \tag{A.44}$$

The system described by eq. (A.37) can also be analyzed using the techniques of Laplace transform analysis, as developed in Chapter 9. The system function for this system is

$$H(s) = \frac{s+2}{s^2+4s+3}, \tag{A.45}$$

and if we substitute v for s, we obtain the same $G(v)$ given in eq. (A.39). Thus, the partial-fraction expansion proceeds exactly as in eqs. (A.40)–(A.42), with the result that

$$H(s) = \frac{\frac{1}{2}}{s+1} + \frac{\frac{1}{2}}{s+3}. \tag{A.46}$$

Inverting this transform, we again obtain the impulse response, as given in eq. (A.44).

Example A.2

We now illustrate the method of partial-fraction expansion when there are repeated factors in the denominator. In Example 4.26, we considered the response of the system described in eq. (A.37) when the input was

$$x(t) = e^{-t}u(t). \tag{A.47}$$

From eq. 4.81, the Fourier transform of the output of the system is

$$Y(j\omega) = \frac{j\omega+2}{(j\omega+1)^2(j\omega+3)}. \tag{A.48}$$

Substituting v for $j\omega$, we obtain the rational function

$$G(v) = \frac{v+2}{(v+1)^2(v+3)}. \tag{A.49}$$

The partial-fraction expansion for this function is

$$G(v) = \frac{A_{11}}{v+1} + \frac{A_{12}}{(v+1)^2} + \frac{A_{21}}{v+3}, \tag{A.50}$$

where, from eq. (A.36),

$$A_{11} = \frac{1}{(2-1)!}\frac{d}{dv}[(v+1)^2 G(v)]\,|_{v=-1} = \frac{1}{4}, \tag{A.51}$$

$$A_{12} = [(v+1)^2 G(v)]\,|_{v=-1} = \frac{1}{2}, \tag{A.52}$$

$$A_{12} = [(v+3)G(v)]\,|_{v=-3} = -\frac{1}{4}. \tag{A.53}$$

Therefore,

$$Y(j\omega) = \frac{\frac{1}{4}}{j\omega+1} + \frac{\frac{1}{2}}{(j\omega+1)^2} - \frac{\frac{1}{4}}{j\omega+3}, \tag{A.54}$$

and taking inverse transforms, we get

$$y(t) = \left[\frac{1}{4}e^{-t} + \frac{1}{2}te^{-t} - \frac{1}{4}e^{-3t}\right]u(t). \tag{A.55}$$

Again, this analysis could also have been performed using Laplace transforms, and the algebra would be identical to that given in eqs. (A.49)–(A.55).

A.3 PARTIAL-FRACTION EXPANSION AND DISCRETE-TIME SIGNALS AND SYSTEMS

As mentioned previously, in performing partial-fraction expansions for discrete-time Fourier transforms or for z-transforms, it is often more convenient to deal with a slightly different form for rational functions. Suppose, then, that we have a rational function in the form

$$G(v) = \frac{d_{n-1}v^{n-1} + \ldots + d_1 v + d_0}{f_n v^n + \ldots + f_1 v + 1}. \tag{A.56}$$

This form for $G(v)$ can be obtained from $G(v)$ in eq. (A.12) by dividing the numerator and denominator by a_0.

With $G(v)$ as in eq. (A.56), the corresponding factorization of the denominator is of the form

$$G(v) = \frac{d_{n-1}v^{n-1} + \ldots + d_1 v + d_0}{(1-\rho_1^{-1}v)^{\sigma_1}(1-\rho_2^{-1}v)^{\sigma_2}\ldots(1-\rho_r^{-1}v)^{\sigma_r}}, \tag{A.57}$$

and the form of the partial-fraction expansion that results is

$$G(v) = \sum_{i=1}^{r}\sum_{k=1}^{\sigma_i}\frac{B_{ik}}{(1-\rho_i^{-1}v)^k}. \tag{A.58}$$

The B_{ik} can be calculated in a manner similar to that used earlier:

$$B_{ik} = \frac{1}{(\sigma_i - k)!}(-\rho_i)^{\sigma_i - k}\left[\frac{d^{\sigma_i - k}}{dv^{\sigma_i - k}}[(1 - \rho_i^{-1}v)^{\sigma_i}G(v)]\right]\Bigg|_{v=\rho_i}. \tag{A.59}$$

As before, the validity of eq. (A.59) can be determined by multiplying both sides of eq. (A.58) by $(1 - \rho_i^{-1}v)^{\sigma_i}$, then differentiating repeatedly with respect to v, until B_{ik} is no longer multiplied by a power of $1 - \rho_i^{-1}v$, and finally, setting $v = \rho_i$.

Example A.3

Consider the causal LTI system in Example 5.19 characterized by the difference equation

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]. \tag{A.60}$$

The frequency response of the system is

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-2j\omega}}. \tag{A.61}$$

For discrete-time transforms such as this, it is most convenient to substitute v for $e^{-j\omega}$. Making this substitution, we obtain the rational function

$$G(v) = \frac{2}{1 - \frac{3}{4}v + \frac{1}{8}v^2} = \frac{2}{(1 - \frac{1}{2}v)(1 - \frac{1}{4}v)}. \tag{A.62}$$

Using the partial-fraction expansion specified by eqs. (A.57)–(A.59), we obtain

$$G(v) = \frac{B_{11}}{1 - \frac{1}{2}v} + \frac{B_{21}}{1 - \frac{1}{4}v}, \tag{A.63}$$

$$B_{11} = \left[\left(1 - \frac{1}{2}v\right)G(v)\right]\Bigg|_{v=2} = \frac{2}{1 - \frac{1}{2}} = 4, \tag{A.64}$$

$$B_{21} = \left[\left(1 - \frac{1}{4}v\right)G(v)\right]\Bigg|_{v=4} = \frac{2}{1 - 2} = -2. \tag{A.65}$$

Thus,

$$H(e^{j\omega}) = \frac{4}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{4}e^{-j\omega}}, \tag{A.66}$$

and taking the inverse transform of eq. (A.66), we obtain the unit impulse response:

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n]. \tag{A.67}$$

In Section 10.7, we developed the tools of z-transform analysis for the examination of discrete-time LTI systems specified by linear constant-coefficient difference

equations. Applying those techniques to this example, we find that the system function can be determined by inspection from eq. (A.60) and is

$$H(z) = \frac{2}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}. \tag{A.68}$$

Then, substituting v for z^{-1}, we obtain $G(v)$ as in eq. (A.62). Thus, using the partial-fraction expansion calculations in eqs. (A.63)–(A.65), we find that

$$H(z) = \frac{4}{1 - \frac{1}{2}z^{-1}} - \frac{2}{1 - \frac{1}{4}z^{-1}}, \tag{A.69}$$

which, when inverted, again yields the unit impulse response of eq. (A.67).

Example A.4

Suppose that the input to the system considered in Example A.3 is

$$x[n] = \left(\frac{1}{4}\right)^n u[n]. \tag{A.70}$$

Then from Example 5.20, the Fourier transform of the output is

$$Y(e^{j\omega}) = \frac{2}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{4}e^{-j\omega})^2}. \tag{A.71}$$

Substituting v for $e^{-j\omega}$ yields

$$G(v) = \frac{2}{(1 - \frac{1}{2}v)(1 - \frac{1}{4}v)^2}. \tag{A.72}$$

Thus, using eqs. (A.58) and (A.59), we obtain the partial-fraction expansion

$$G(v) = \frac{B_{11}}{1 - \frac{1}{4}v} + \frac{B_{12}}{(1 - \frac{1}{4}v)^2} + \frac{B_{21}}{1 - \frac{1}{2}v} \tag{A.73}$$

and find

$$B_{11} = (-4)\left[\frac{d}{dv}\left(1 - \frac{1}{4}v\right)^2 G(v)\right]\Bigg|_{v=4} = -4, \tag{A.74}$$

$$B_{12} = \left[\left(1 - \frac{1}{4}v\right)^2 G(v)\right]\Bigg|_{v=4} = -2, \tag{A.75}$$

$$B_{21} = \left[\left(1 - \frac{1}{2}v\right)G(v)\right]\Bigg|_{v=2} = 8. \tag{A.76}$$

Therefore,

$$Y(j\omega) = -\frac{4}{1 - \frac{1}{4}e^{-j\omega}} - \frac{2}{(1 - \frac{1}{4}e^{-j\omega})^2} + \frac{8}{1 - \frac{1}{2}e^{-j\omega}}, \tag{A.77}$$

which can be inverted by inspection as follows, using the Fourier transform pairs in Table 4.2:

$$y[n] = \left\{ -4\left(\frac{1}{4}\right)^n - 2(n+1)\left(\frac{1}{4}\right)^n + 8\left(\frac{1}{2}\right)^n \right\} u[n]. \tag{A.78}$$

Example A.5

Improper rational functions are often encountered in the analysis of discrete-time systems. To illustrate this, and also to show how they can be analyzed using the techniques developed here, consider the causal LTI system characterized by the difference equation

$$y[n] + \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n] + 3x[n-1] + \frac{11}{6}x[n-2] + \frac{1}{3}x[n-3].$$

The frequency response of this system is

$$H(e^{j\omega}) = \frac{1 + 3e^{-j\omega} + \frac{11}{6}e^{-j2\omega} + \frac{1}{3}e^{-j3\omega}}{1 + \frac{5}{6}e^{-j\omega} + \frac{1}{6}e^{-j2\omega}}. \tag{A.79}$$

Substituting v for $e^{-j\omega}$, we obtain

$$G(v) = \frac{1 + 3v + \frac{11}{6}v^2 + \frac{1}{3}v^3}{1 + \frac{5}{6}v + \frac{1}{6}v^2}. \tag{A.80}$$

This rational function can be written as the sum of a polynomial and a proper rational function:

$$G(v) = c_0 + c_1 v + \frac{b_1 v + b_0}{1 + \frac{5}{6}v + \frac{1}{6}v^2}. \tag{A.81}$$

Equating eqs. (A.80) and (A.81), and multiplying by $1 + \frac{5}{6}v + \frac{1}{6}v^2$, we obtain

$$\begin{aligned} 1 + 3v + \frac{11}{6}v^2 + \frac{1}{3}v^3 = (c_0 + b_0) + \left(\frac{5}{6}c_0 + c_1 + b_1\right)v \\ + \left(\frac{1}{6}c_0 + \frac{5}{6}c_1\right)v^2 + \frac{1}{6}c_1 v^3. \end{aligned} \tag{A.82}$$

Equating coefficients, we see that

$$\begin{aligned} \frac{1}{6}c_1 = \frac{1}{3} &\rightarrow c_1 = 2, \\ \frac{1}{6}c_0 + \frac{5}{6}c_1 = \frac{11}{6} &\rightarrow c_0 = 1, \\ \frac{5}{6}c_0 + c_1 + b_1 = 3 &\rightarrow b_1 = \frac{1}{6}, \\ c_0 + b_0 = 1 &\rightarrow b_0 = 0. \end{aligned} \tag{A.83}$$

Thus,

$$H(e^{j\omega}) = 1 + 2e^{-j\omega} + \frac{\frac{1}{6}e^{-j\omega}}{1 + \frac{5}{6}e^{-j\omega} + \frac{1}{6}e^{-j2\omega}}. \tag{A.84}$$

Also, we can use the method developed here to expand the proper rational function in eq. (A.81):

$$\frac{\frac{1}{6}v}{1+\frac{5}{6}v+\frac{1}{6}v^2} = \frac{\frac{1}{6}v}{(1+\frac{1}{3}v)(1+\frac{1}{2}v)} = \frac{B_{11}}{(1+\frac{1}{3}v)} + \frac{B_{21}}{(1+\frac{1}{2}v)}. \tag{A.85}$$

The coefficients are

$$B_{11} = \left.\left(\frac{\frac{1}{6}v}{1+\frac{1}{2}v}\right)\right|_{v=-3} = 1,$$

$$B_{21} = \left.\left(\frac{\frac{1}{6}v}{1+\frac{1}{3}v}\right)\right|_{v=-2} = -1.$$

Therefore, we find that

$$H(e^{j\omega}) = 1 + 2e^{-j\omega} + \frac{1}{1+\frac{1}{3}e^{-j\omega}} - \frac{1}{1+\frac{1}{2}e^{-j\omega}}, \tag{A.86}$$

and by inspection, we can determine the impulse response of this system:

$$h[n] = \delta[n] + 2\delta[n-1] + \left[\left(-\frac{1}{3}\right)^n - \left(-\frac{1}{2}\right)^n\right]u[n]. \tag{A.87}$$

BIBLIOGRAPHY

The purpose of this bibliography is to provide the reader with sources for additional and more advanced treatments of topics in signal and system analysis. This is by no means meant to be an exhaustive list but rather it is intended to indicate directions for further study and several references for each.

We have divided the bibliography into sixteen different subject areas. The first few deal with the mathematical techniques of signal and system analysis including texts on background mathematics (calculus, differential and difference equations, and complex variables), the theory of Fourier series and of Fourier, Laplace, and z-transforms, and additional topics in mathematics that are commonly encountered and used in signal and system analysis. Several of the sections that follow deal with more thorough and specialized treatments of topics in signals and systems introduced in this text, including filtering, sampling and discrete-time signal processing, communications, and feedback and control. We have also provided a list of other basic books on signals and systems as well as several texts on circuit theory. In addition, we have provided lists of references on several topics that represent important subjects for more advanced study for those interested either in expanding their knowledge of the methods of signals and systems or in exploring applications that make use of these advanced techniques. In particular we include sections on state space models and methods, multidimensional signal and image processing, speech processing, multirate and multiresolution signal analysis, random signals and statistical signal processing, and nonlinear systems. Finally we have included a list of references dealing with a sampling of other applications and advanced topics. Together, the references collected in this bibliography should provide the reader with an appreciation for the breadth of topics and applications that comprise the field of signals and systems.

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Answers

Chapter 1 Answers

1.1. $-0.5, -0.5, j, -j, j, 1+j, 1+j, 1-j, 1-j$

1.2. $5e^{j0}, 2e^{j\pi}, 3e^{-j\pi/2}, e^{-j\pi/3}, \sqrt{2}e^{j\pi/4}, 2e^{-j\pi/2}, \sqrt{2}e^{j\pi/4}, e^{j\pi/2}, e^{-j\pi/12}$.

1.3. **(a)** $P_\infty = 0, E_\infty = \frac{1}{4}$ **(b)** $P_\infty = 1, E_\infty = \infty$ **(c)** $P_\infty = \frac{1}{2}, E_\infty = \infty$
(d) $P_\infty = 0, E_\infty = \frac{4}{3}$ **(e)** $P_\infty = 1, E_\infty = \infty$ **(f)** $P_\infty = \frac{1}{2}, E_\infty = \infty$

1.4. **(a)** $n < 1$ and $n > 7$ **(b)** $n < -6$ and $n > 0$ **(c)** $n < -4$ and $n > 2$
(d) $n < -2$ and $n > 4$ **(e)** $n < -6$ and $n > 0$

1.5. **(a)** $t > -2$ **(b)** $t > -1$ **(c)** $t > -2$ **(d)** $t < 1$ **(e)** $t < 9$

1.6. **(a)** No **(b)** No **(c)** Yes

1.7. **(a)** $|n| > 3$ **(b)** all t **(c)** $|n| < 3, |n| \to \infty$ **(d)** $|t| \to \infty$

1.8. **(a)** $A = 2, a = 0, \omega = 0, \phi = \pi$ **(b)** $A = 1, a = 0, \omega = 3, \phi = 0$
(c) $A = 1, a = 1, \omega = 3, \phi = \frac{\pi}{2}$ **(d)** $A = 1, a = 2, \omega = 100, \phi = \frac{\pi}{2}$

1.9. **(a)** $T = \frac{\pi}{5}$ **(b)** Not periodic **(c)** $N = 2$
(d) $N = 10$ **(e)** Not periodic

1.10. π

1.11. 35

1.12. $M = -1, n_0 = -3$

1.13. 4

1.14. $A_1 = 3, t_1 = 0, A_2 = -3, t_2 = 1$

1.15. **(a)** $y[n] = 2x[n-2] + 5x[n-3] + 2x[n-4]$ **(b)** No

1.16. **(a)** No **(b)** 0 **(c)** No

1.17. **(a)** No; e.g., $y(-\pi) = x(0)$ **(b)** Yes

1.18. **(a)** Yes **(b)** Yes **(c)** $C \leq (2n_0 + 1)B$

1.19. **(a)** Linear, not time invariant **(b)** Not linear, time invariant
(c) Linear, time invariant **(d)** Linear, not time invariant

1.20. **(a)** $\cos(3t)$ **(b)** $\cos(3t-1)$

Chapter 2 Answers

2.1. **(a)** $y_1[n] = 2\delta[n+1] + 4\delta[n] + 2\delta[n-1] + 2\delta[n-2] - 2\delta[n-4]$
(b) $y_2[n] = y_1[n+2]$ **(c)** $y_3[n] = y_2[n]$

2.2. $A = n - 9, B = n + 3$

2.3. $2[1 - \frac{1}{2}^{n+1}]u[n]$

2.4. $y[n] = \begin{cases} n-6, & 7 \leq n \leq 11 \\ 6, & 12 \leq n \leq 18 \\ 24-n, & 19 \leq n \leq 23 \\ 0, & \text{otherwise} \end{cases}$

2.5. $N = 4$

2.6. $y[n] = \begin{cases} \frac{3^n}{2}, & n < 0 \\ \frac{1}{2}, & n \geq 0 \end{cases}$

2.7. **(a)** $u[n-2] - u[n-6]$ **(b)** $u[n-4] - u[n-8]$ **(c)** No
(d) $y[n] = 2u[n] - \delta[n] - \delta[n-1]$

2.8. $y(t) = \begin{cases} t+3, & -2 < t \leq -1 \\ t+4, & -1 < t \leq 0 \\ 2-2t, & 0 < t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$

2.9. $A = t - 5, B = t - 4$

2.10. **(a)** $y(t) = \begin{cases} t, & 0 \leq t \leq \alpha \\ \alpha, & \alpha \leq t \leq 1 \\ 1 + \alpha - t, & 1 \leq t \leq 1 + \alpha \\ 0, & \text{otherwise} \end{cases}$ **(b)** $\alpha = 1$

2.11. **(a)** $y(t) = \begin{cases} 0, & -\infty < t \leq 3 \\ \frac{1-e^{-3(t-3)}}{3}, & 3 < t \leq 5 \\ \frac{(1-e^{-6})e^{-3(t-5)}}{3}, & 5 < t \leq \infty \end{cases}$

(b) $g(t) = e^{-3(t-3)}u(t-3) - e^{-3(t-5)}u(t-5)$ **(c)** $g(t) = \frac{dy(t)}{dt}$

2.12. $A = \frac{1}{1-e^{-3}}$

2.13. **(a)** $A = \frac{1}{5}$ **(b)** $g[n] = \delta[n] - \frac{1}{5}\delta[n-1]$

2.14. $h_1(t), h_2(t)$

2.15. $h_2[n]$

2.16. **(a)** True **(b)** False **(c)** True **(d)** True

2.17. **(a)** $y(t) = \frac{1-j}{6}[e^{(-1+3j)t} - e^{-4t}]u(t)$

(b) $y(t) = \frac{1}{6}[e^{-t}(\cos 3t + \sin 3t) - e^{-4t}]u(t)$

2.18. $(1/4)^{n-1}u[n-1]$

2.19. **(a)** $\alpha = \frac{1}{4}, \beta = 1$ **(b)** $[2(\frac{1}{2})^n - (\frac{1}{4})^n]u[n]$

2.20. **(a)** 1 **(b)** 0 **(c)** 0

Chapter 3 Answers

3.1. $x(t) = 4\cos(\frac{\pi}{4}t) + 8\cos(\frac{3\pi}{4}t + \frac{\pi}{2})$

3.2. $x[n] = 1 + 2\sin(\frac{4\pi}{5}n + \frac{3\pi}{4}) + 4\sin(\frac{8\pi}{5}n + \frac{5\pi}{6})$

3.3. $\omega_0 = \frac{\pi}{3}, a_0 = 2, a_2 = a_{-2} = \frac{1}{2}, a_5 = a^*_{-5} = -2j$

3.4. $a_k = \begin{cases} 0, & k = 0 \\ e^{-jk\pi/2}\frac{3\sin(\frac{k\pi}{2})}{k\pi}, & k \neq 0 \end{cases}$

3.5. $\omega_2 = \omega_1, b_k = e^{-jk\omega_1}[a_{-k} + a_k]$

3.6. **(a)** $x_2(t), x_3(t)$ **(b)** $x_2(t)$

3.7. $a_k = \begin{cases} \frac{2}{T}, & k = 0 \\ \frac{b_k}{j\frac{2\pi}{T}k}, & k \neq 0 \end{cases}$

3.8. $x_1(t) = \sqrt{2}\sin(\pi t), x_2(t) = -\sqrt{2}\sin(\pi t)$

3.9. $a_0 = 3, a_1 = 1 - 2j, a_2 = -1, a_3 = 1 + 2j$

3.10. $a_0 = 0, a_{-1} = -j, a_{-2} = -2j, a_{-3} = -3j$

3.11. $A = 10, B = \frac{\pi}{5}, C = 0$

3.12. $c_k = 6$ for all k

3.13. $y(t) = 0$

3.14. $H(e^{j\pi/2}) = H^*(e^{j3\pi/2}) = 2e^{j\pi/4}, H(e^{j0}) = H(e^{j\pi}) = 0$

3.15. $|k| > 8$

3.16. **(a)** 0 **(b)** $\sin(\frac{3\pi}{8}n + \frac{\pi}{4})$ **(c)** 0

3.17. S_1 and S_3 are not LTI.

3.18. S_1 and S_2 are not LTI.

3.19. **(a)** $\frac{dy(t)}{dt} + y(t) = x(t)$ **(b)** $H(j\omega) = (\frac{1}{1+j\omega})$ **(c)** $y(t) = \frac{1}{\sqrt{2}}\cos(t - \frac{\pi}{4})$

3.20. **(a)** $\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + y(t) = x(t)$ **(b)** $H(j\omega) = (\frac{1}{1+j\omega-\omega^2})$ **(c)** $-\cos t$

Chapter 4 Answers

4.1. **(a)** $\frac{e^{-j\omega}}{2+j\omega}$ **(b)** $\frac{4e^{-j\omega}}{4+\omega^2}$

4.2. **(a)** $2\cos\omega$ **(b)** $-2j\sin 2\omega$

4.3. **(a)** $\frac{\pi}{j}[e^{j\pi/4}\delta(\omega - 2\pi) - e^{-j\pi/4}\delta(\omega + 2\pi)]$
(b) $2\pi\delta(\omega) + \pi[e^{j\pi/8}\delta(\omega - 6\pi) + e^{-j\pi/8}\delta(\omega + 6\pi)]$

4.4. **(a)** $1 + \cos 4\pi t$ **(b)** $\frac{4j\sin^2 t}{\pi t}$

4.5. $x(t) = -\frac{2\sin(3(t-3/2))}{\pi(t-3/2)}, t = \frac{k\pi}{3} + \frac{3}{2}$ for nonzero integers k

4.6. **(a)** $X_1(j\omega) = 2X(-j\omega)\cos\omega$ **(b)** $X_2(j\omega) = \frac{1}{3}e^{-j2\omega}X(j\frac{\omega}{3})$
(c) $X_3(j\omega) = -\omega^2 e^{-j\omega}X(j\omega)$

4.7. **(a)** neither, neither **(b)** imaginary, odd **(c)** imaginary, neither
(d) real, even

4.8. **(a)** $\frac{2\sin(\omega/2)}{j\omega^2} + \pi\delta(\omega)$ **(b)** $\frac{2\sin(\omega/2)}{j\omega^2}$

4.9. **(a)** $\frac{\sin\omega}{j\omega^2} - \frac{e^{-j\omega}}{j\omega}$ **(b)** $\frac{\sin\omega}{\omega}$ **(c)** $\frac{\sin\omega}{j\omega^2} - \frac{\cos\omega}{j\omega}$

4.10. **(a)** $X(j\omega) = \begin{cases} j/2\pi, & -2 \le \omega < 0 \\ -j/2\pi, & 0 \le \omega < 2 \\ 0, & \text{otherwise} \end{cases}$ **(b)** $A = \frac{1}{2\pi^3}$

4.11. $A = \frac{1}{3}, B = 3$

4.12. **(a)** $-\frac{4j\omega}{(1+\omega^2)^2}$ **(b)** $-j2\pi\omega e^{-|\omega|}$

4.13. **(a)** No **(b)** Yes **(c)** Yes

4.14. $x(t) = \sqrt{12}[e^{-t} - e^{-2t}]u(t)$

4.15. $x(t) = 2te^{-|t|}u(t)$

4.16. **(a)** $g(t) = \pi \sum_{k=-\infty}^{\infty} \delta(t - \frac{k\pi}{4})$ **(b)** $X(j\omega) = \begin{cases} 4, & |\omega| \le 1 \\ 0, & 1 < |\omega| \le 4 \end{cases}$

4.17. **(a)** False **(b)** True

4.18. $h(t) = \begin{cases} \frac{5}{4}, & |t| < 1 \\ -\frac{|t|}{4} + \frac{3}{2}, & 1 \le |t| \le 5 \\ -\frac{|t|}{8} + \frac{7}{8}, & 5 < |t| < 7 \\ 0, & \text{otherwise} \end{cases}$

4.19. $x(t) = e^{-4t}u(t)$

4.20. $h(t) = \frac{2}{\sqrt{3}}e^{-t/2}\sin(\frac{\sqrt{3}}{2}t)u(t)$

Chapter 5 Answers

5.1. **(a)** $\frac{e^{-j\omega}}{1-\frac{1}{2}e^{-j\omega}}$ **(b)** $\frac{0.75e^{-j\omega}}{1.25-\cos\omega}$

5.2. **(a)** $2\cos\omega$ **(b)** $2j\sin(2\omega)$

5.3. **(a)** $\frac{\pi}{j}\{e^{j\pi/4}\delta(\omega - \frac{\pi}{3}) - e^{-j\pi/4}\delta(\omega + \frac{\pi}{3})\}$
(b) $4\pi\delta(\omega) + \pi\{e^{j\pi/8}\delta(\omega - \frac{\pi}{6}) + e^{-j\pi/8}\delta(\omega + \frac{\pi}{6})\}$

5.4. **(a)** $x_1[n] = 1 + \cos(\frac{\pi}{2}n)$ **(b)** $-4\frac{\sin^2(\frac{\pi}{2}n)}{\pi n}$

5.5. $x[n] = \frac{\sin[\frac{\pi}{4}(n-\frac{3}{2})]}{\pi(n-\frac{3}{2})}$, and $x[n] = 0$ for $n = \pm\infty$

5.6. **(a)** $X_1(e^{j\omega}) = (2\cos\omega)X(e^{-j\omega})$ **(b)** $X_2(e^{j\omega}) = \mathcal{R}e\{X(e^{j\omega})\}$
(c) $X_3(e^{j\omega}) = -\frac{d^2}{d\omega^2}X(e^{j\omega}) - 2j\frac{d}{d\omega}X(e^{j\omega}) + X(e^{j\omega})$

5.7. **(a)** imaginary, neither **(b)** real, odd **(c)** real, neither

5.8. $x[n] = \begin{cases} 1, & n \le -2 \\ n+3, & -1 \le n \le 1 \\ 4, & n \ge 2 \end{cases}$

5.9. $x[n] = -\delta[n+2] + \delta[n+1] + \delta[n]$

5.10. $A = 2$

5.11. $\alpha = \pi$

5.12. $\frac{\pi}{2} \le |\omega_c| \le \pi$

5.13. $h_2[n] = -2(\frac{1}{4})^n u[n]$

5.14. $h[n] = \frac{16}{17}\delta[n] - \frac{1}{17}\delta[n-2]$

5.15. $\omega_c = 3\pi/4$

5.16. **(a)** $\alpha = \frac{1}{4}$ **(b)** $N = 4$ **(c)** No

5.17. $b_k = \frac{1}{2}(-1)^k$

5.18. $a_k = \frac{1}{3}(\frac{1}{2})^{|k|}$

5.19. **(a)** $H(e^{j\omega}) = \frac{1}{(1-\frac{1}{2}e^{-j\omega})(1+\frac{1}{3}e^{-j\omega})}$

(b) $h[n] = \frac{3}{5}(\frac{1}{2})^n u[n] + \frac{2}{5}(-\frac{1}{3})^n u[n]$

5.20. **(a)** $H(e^{j\omega}) = \frac{\frac{4}{5}e^{-j\omega}}{1-\frac{4}{5}e^{-j\omega}}$

(b) $y[n] - \frac{4}{5}y[n-1] = \frac{4}{5}x[n-1]$

Chapter 6 Answers

6.1. **(a)** $A = |H(j\omega_0)|$ **(b)** $t_0 = -\frac{\sphericalangle H(j\omega_0)}{\omega_0}$

6.2. $\sphericalangle H(e^{j\omega_0}) = -n_0(\omega_0) + 2\pi k$ for some integer k.

6.3. **(a)** $A = 1$ **(b)** $\tau(\omega) > 0$ for $\omega > 0$

6.4. **(a)** $2\cos(\frac{\pi}{2}n - \pi)$ **(b)** $2\sin(\frac{7\pi}{2}n - \frac{3\pi}{4})$

6.5. **(a)** $g(t) = 2\cos(2\omega_c t)$ **(b)** more concentrated

6.6. **(a)** $g[n] = (-1)^n$ **(b)** more concentrated

6.7. **(a)** 1,000 Hz and 3,000 Hz **(b)** 800 Hz and 3,200 Hz

6.8. $\pi - \omega_p \leq \omega \leq \pi$

6.9. Final value = 2/5, t_0 = 2/5 sec

6.10. **(a)** $20\log_{10}|H(j\omega)| \approx \begin{cases} -20, & \omega \ll 0.1 \\ 20\log_{10}(\omega), & 0.1 \ll \omega \ll 40 \\ 32, & \omega \gg 40 \end{cases}$

(b) $20\log_{10}|H(j\omega)| \approx \begin{cases} 20, & \omega \ll 0.2 \\ -20\log_{10}(\omega) + 6, & 0.2 \ll \omega \ll 50 \\ -28, & \omega \gg 50 \end{cases}$

6.11. **(a)** $20\log_{10}|H(j\omega)| \approx \begin{cases} 20, & \omega \ll 0.5 \\ -20\log_{10}(\omega) + 14, & 0.5 \ll \omega \ll 50 \\ -40\log_{10}(\omega) + 48, & \omega \gg 50 \end{cases}$

(b) $20\log_{10}|H(j\omega)| \approx \begin{cases} 0, & \omega \ll 1 \\ -40\log_{10}\omega, & 1 \ll \omega \ll 50 \\ -20\log_{10}\omega - 34, & \omega \gg 50 \end{cases}$

6.12. $H_2(j\omega) = \frac{0.01(j\omega+40)}{(j\omega+1)(j\omega+8)}$

6.13. **(a)** not unique **(b)** unique

6.14. $H_I(j\omega) = 0.2 \times 10^{-4}\frac{(j\omega+50)(j\omega+10)}{(j\omega+0.2)^2}$

6.15. **(a)** critically damped **(b)** underdamped
(c) overdamped **(d)** underdamped

6.16. $y[n] + \frac{1}{2}y[n-1] = \frac{3}{2}x[n]$

6.17. **(a)** oscillatory **(b)** nonoscillatory

6.18. No

6.19. $R \geq 2\sqrt{\frac{L}{C}}$

6.20. $\tau(\omega) = 2$

Chapter 7 Answers

7.1. $|\omega| > 5{,}000\pi$

7.2. (a) and (c)

7.3. **(a)** $8{,}000\pi$ **(b)** $8{,}000\pi$ **(c)** $16{,}000\pi$

7.4. **(a)** ω_0 **(b)** ω_0 **(c)** $2\omega_0$ **(d)** $3\omega_0$

7.5. $|H(j\omega)| = \begin{cases} T, & |\omega| \leq \omega_c \\ 0, & \text{otherwise} \end{cases}$, where $\frac{\omega_0}{2} < \omega_c < \frac{2\pi}{T} - \frac{\omega_0}{2}$, $\measuredangle H(j\omega) = 0$

7.6. $T_{\max} = \frac{\pi}{\omega_1 + \omega_2}$

7.7. $H(j\omega) = \frac{2\sin(\omega T/2)}{\omega T} \times e^{j(\omega T/2)}$

7.8. **(a)** Yes

(b) $g(t) = \sum_{k=-4}^{4} a_k e^{jk\pi t}$, where $a_k = \begin{cases} 0, & k = 0 \\ -j(\frac{1}{2})^{k+1}, & 1 \leq k \leq 4 \\ j(\frac{1}{2})^{-k+1}, & -4 \leq k \leq -1 \end{cases}$

7.9. $\omega_0 = 50\pi$

7.10. **(a)** False **(b)** True **(c)** True

7.11. **(a)** $X_c(j\omega)$ is real
(b) $\text{Max}\{X_c(j\omega)\} = 0.5 \times 10^{-3}$
(c) $X_c(j\omega) = 0$ for $|\omega| \geq 1{,}500\pi$
(d) $X_c(j\omega) = X_c(j(\omega - 2{,}000\pi))$ for $0 \leq \omega \leq 2{,}000\pi$

7.12. $|\omega| \geq 750\pi$

7.13. $h[n] = \delta[n-2]$

7.14. $h[n] = -\frac{\sin[\pi(n-\frac{1}{2})]}{T\pi(n-\frac{1}{2})^2}$

7.15. $N = 2$

7.16. $x[n] = 4(\frac{\sin(\pi n/2)}{\pi n})^2$

7.17. Ideal lowpass filter with cutoff frequency $\pi/2$ and passband gain of unity

7.18. Ideal lowpass filter with cutoff frequency $\pi/4$ and passband gain of 2.

7.19. **(a)** $y[n] = \frac{\sin(5\omega_1 n/3)}{5\pi n}$ **(b)** $y[n] = \frac{1}{5}\delta[n]$

7.20. **(a)** Yes **(b)** No

Chapter 8 Answers

8.1. $m(t) = \frac{1}{2}e^{-j\omega_c t}$

8.2. **(a)** No constraint necessary **(b)** $|\omega_c| > 1{,}000\pi$

8.3. $y(t) = 0$

8.4. $y(t) = \sin 200\pi t$

8.5. $m = \frac{3}{2\pi}$

8.6. $A = 4$

8.7. $\omega_0 = 2\omega_c, A = 2$

8.8. **(a)** Yes **(b)** Yes, $x(t) = \{y(t)\sin\omega_c t\} * \frac{2\sin\omega_c t}{\pi t}$

8.9. **(a)** $|\omega| > 2\omega_c$ **(b)** $\omega_0 = \omega_c, A = 2$

8.10. **(a)** $X(j\omega) = 0$ for $|\omega| \geq 1{,}000\pi$ **(b)** $\omega_c = 1{,}000\pi, A = 4$

8.11. **(a)** $\frac{\omega_c}{2} \leq |\omega| \leq \frac{3\omega_c}{2}$, Gain $= 1$ **(b)** $A = 2|a_1|, \phi = \measuredangle a_1$

8.12. $\Delta = 0.5 \times 10^{-4}$

8.13. **(a)** $p(0) = \frac{1}{T_1}$ **(b)** $p(kT_1) = 0$

8.14. $Y(j\omega) = \pi\delta(\omega - \omega_c) - \frac{m\pi}{2j}\delta(\omega - \omega_c - \omega_m) - \frac{m\pi}{2j}\delta(\omega - \omega_c + \omega_m)$

8.15. $\omega_0 = 0$ and $\omega_0 = \pi$

8.16. $0 \leq \omega \leq \frac{3\pi}{8}$ and $\frac{5\pi}{8} \leq \omega \leq \pi$

8.17. $0 \leq |\omega| \leq \frac{\pi}{2}$

8.18. $H(e^{j\omega}) = \begin{cases} j, & 0 < \omega \leq \frac{\pi}{4} \\ -j, & -\frac{\pi}{4} \leq \omega < 0 \end{cases}$

8.19. $N = 20$

8.20. $p[n] = \sum_{k=-\infty}^{\infty} \delta[n - 2k]$

Chapter 9 Answers

9.1. **(a)** $\sigma > -5$ **(b)** $\sigma < -5$ **(c)** $-\infty \leq \sigma \leq \infty$
(d) no value of σ **(e)** $|\sigma| < 5$ **(f)** $\sigma < 5$

9.2. **(a)** $\frac{e^{-(s+5)}}{s+5}, \mathcal{Re}\{s\} > -5$ **(b)** $A = -1, t_0 = -1, \mathcal{Re}\{s\} < -5$

9.3. $\mathcal{Re}\{\beta\} = 3, \mathcal{Im}\{\beta\}$ arbitrary

9.4. $1 + 2j, 1 - 2j, \mathcal{Re}\{s\} < 1$

9.5. **(a)** 1,1 **(b)** 0,1 **(c)** 1,0

9.6. **(a)** no **(b)** yes **(c)** no **(d)** yes

9.7. 4

9.8. two sided

9.9. $x(t) = 4e^{-4t}u(t) - 2e^{-3t}u(t)$

9.10. **(a)** lowpass **(b)** bandpass **(c)** highpass

9.11. $|X(j\omega)| = 1$

9.12. **(a)** not consistent **(b)** consistent **(c)** consistent

9.13. $\alpha = -1, \beta = \frac{1}{2}$

9.14. $X(s) = 1/[4(s^2 - \frac{s}{\sqrt{2}} + \frac{1}{4})(s^2 + \frac{s}{\sqrt{2}} + \frac{1}{4})], -\frac{\sqrt{2}}{4} < \mathcal{Re}\{s\} < \frac{\sqrt{2}}{4}$

9.15. $X(s) = \frac{s}{s^2+4}; \mathcal{Re}\{s\} > 0, \quad Y(s) = \frac{2}{s^2+4}; \mathcal{Re}\{s\} > 0$

9.16. **(a)** 2 **(b)** $\alpha > 0$

9.17. $\frac{d^2y(t)}{dt^2} + 10\frac{dy(t)}{dt} + 16y(t) = 12x(t) + 3\frac{dx(t)}{dt}$

9.18. **(a)** $H(s) = \frac{1}{s^2+s+1}, \mathcal{Re}\{s\} > -\frac{1}{2}$
(b) Lowpass
(c) $H(s) = \frac{1}{s^2+10^{-3}s+1}, \mathcal{Re}\{s\} > -0.0005$
(d) Bandpass

9.19. **(a)** $\frac{1}{s+2}, \mathcal{Re}\{s\} > -2$
(b) $1 + \frac{e^{-6}}{s+2}, \mathcal{Re}\{s\} > -2$
(c) $\frac{1}{s+4} + \frac{1}{s+2}, \mathcal{Re}\{s\} > -2$

9.20. **(a)** $e^{-t}u(t) - e^{-2t}u(t)$
(b) $e^{-t}u(t)$
(c) $2e^{-t}u(t) - e^{-2t}u(t)$

Chapter 10 Answers

10.1. **(a)** $|z| > \frac{1}{2}$ **(b)** $|z| < \frac{1}{2}$ **(c)** $|z| > 1$ **(d)** $\frac{1}{2} < |z| < 2$

10.2. $X(z) = \frac{1}{125}\frac{z^{-3}}{1-\frac{1}{5}z^{-1}}; |z| > \frac{1}{5}$

10.3. $|\alpha| = 2$, n_0 arbitrary

10.4. poles at $z = \frac{1}{3}e^{\pm j\pi/4}$, ROC: $|z| < \frac{1}{3}$

10.5. **(a)** 1,1 **(b)** 2,0 **(c)** 1,2

10.6. **(a)** No **(b)** No **(c)** Yes **(d)** Yes

10.7. 3

10.8. two sided

10.9. $x[n] = \frac{2}{9}u[n] + \frac{7}{9}(-2)^n u[n]$

10.10. **(a)** $x[0] = 1, x[1] = \frac{2}{3}, x[2] = -\frac{2}{9}$ **(b)** $x[0] = 3, x[-1] = -6, x[-2] = 18$

10.11. $x[n] = \begin{cases} (\frac{1}{2})^n, & 0 \le n \le 9 \\ 0, & \text{otherwise} \end{cases}$

10.12. **(a)** highpass **(b)** lowpass **(c)** bandpass

10.13. **(a)** $G(z) = 1 - z^{-6}; |z| > 0$ **(b)** $X(z) = \frac{1-z^{-6}}{1-z^{-1}}; |z| > 0$

10.14. **(a)** $n_0 = 2$ **(b)** $G(z) = (\frac{z^{-1}-z^{-7}}{1-z^{-1}})^2$

10.15. $(\frac{1}{3})^n u[n]$ and $(-\frac{1}{3})^n u[n]$

10.16. **(a)** Not causal **(b)** Causal **(c)** Not causal

10.17. **(a)** Yes **(b)** Yes

10.18. **(a)** $y[n] - \frac{2}{3}y[n-1] + \frac{1}{9}y[n-2] = x[n] - 6x[n-1] + 8x[n-2]$
(b) Yes

10.19. **(a)** $\chi_1(z) = \frac{1}{1-\frac{1}{4}z^{-1}}, |z| > \frac{1}{4}$
(b) $\chi_2(z) = 2$, All z
(c) $\chi_3(z) = \frac{1}{1-\frac{1}{2}z^{-1}}, |z| > \frac{1}{2}$

10.20. **(a)** $-(-\frac{1}{2})^n u[n]$
(b) $\frac{1}{3}(-\frac{1}{2})^n u[n] + \frac{1}{6}(\frac{1}{4})^n u[n]$
(c) $-\frac{2}{3}(-\frac{1}{2})^n u[n] + \frac{1}{6}(\frac{1}{4})^n u[n]$

Chapter 11 Answers

11.1. $H_0(z) + \frac{H_1(z)}{1+G(z)H_1(z)}$

11.2. $\frac{H_1(s)H_2(s)}{1+H_1(s)G_1(s)+H_1(s)H_2(s)G_2(s)}$

11.3. $b < -1$

11.4. $G(s) = \frac{1}{s}$

11.5. $-\frac{5}{2} < b < \frac{3}{2}$

11.6. FIR

11.7. $K > -6$

11.8. $-3 < k < 0$

11.9. No, root locus stays on real axis

11.10. Double pole at $s = -1$, double zero at $s = 1$

11.11. $0 < k < \frac{5}{4}$

11.12. Pole and zero positions alternate on the real axis

11.13. Unstable for all K

11.14. **(a)** 0 **(b)** 1

11.15. $K > -1$

11.16. $K > -1$

11.17. $-1 < K < 4$

11.18. $-1 < K < 1$

11.19. Unstable

11.20. Gain margin is infinite, phase margin is $2\tan^{-1}\sqrt{2}$

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